Bullseye Pistol

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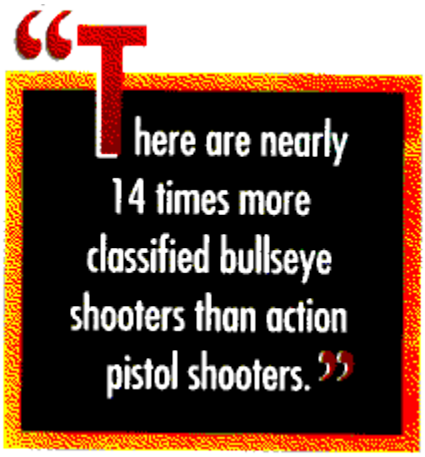
Get Into Bullseye Shooting

By Sherwood Veith

Source: https://www.bullseyepistol.com/getinto.htm

Once you've got the right guns and loads, all you need are steely nerves, a disciplined trigger finger, and a rock-solid hold to compete in this challenging sport.

Here's something new for you to try. Take a one-pound coffee can and spray paint the bottom black. Set the can on its side downrange about chest height with the bottom facing you. Now take your favorite revolver or pistol in one hand, standing up with no rest, and shoot 10 rounds at the bottom of the coffee can. If you need to, take 10 minutes. By the way, set the coffee can 50 yards away.



I've just described one-third of a bullseye handgun match, also known as "conventional pistol" in official NRA terminology. Bullseye is the granddaddy of all popular handgun competition in this country. Its three-gun, 2700-point format goes back to the 1941 National Matches and has been the standard ever since for national and state championships, regional championships, and most outdoor tournaments.

Today the NRA has 42,000 classified bullseye shooters nationwide, and that's not counting the thousands of competitors who compete in unsanctioned matches. An NRA-sanctioned bullseye match is a three-gun affair that requires you to shoot identical 90 shot courses of fire with each gun. Gun one must be a .22 rimfire pistol or revolver, gun two must be a centerfire pistol or revolver of .32 caliber or larger, and gun three must be a .45 caliber pistol or revolver. Most competitors use two guns for an entire match: a .22 rimfire semiautomatic pistol and a 1911-style .45 ACP. Using two guns instead of three saves the expense of a third match-grade pistol, and it requires the shooter to accustom himself to the recoil, grip, and trigger pull of only two guns. Even so, a Smith & Wesson Model 52 in .38 Special or a Walther

in .32 S&W Long is occasionally seen on the centerfire line. Also, optical sights are allowed, including conventional scopes and red dot sights, although many shooters still use open sights.

Slow Fire

The standard 90-shot course of fire begins with 30 shots of slow fire. In the slow-fire stage the target is placed at 50 yards and you're given 10 minutes to fire 10 shots. The targets are scored and repaired after each string of 10 shots. The slow-fire course gives shooters fits. It places more stress on shooter and gun than the other two courses and is often the deciding factor in a match. At 50 yards the 10-ring-only 3.3 inches in diameter-seems minuscule. Inside of the 3.3-inch 10-ring is a 1.7-inch X-ring used to break ties. The mental fortitude required to concentrate on the sight and trigger squeeze while at the same time holding the gun steady with one hand is substantial. Most shooters worry more about their slow-fire scores than any other part of the match. Having a gun that can keep 10 shots out of 10 in a three-inch group at 50 yards is necessary too. Conventional wisdom says that your gun and ammunition combination should be able to hold the X-ring at 50 yards from a machine rest. That's a 1.7-inch group, remember. Bull-barreled target .22s manufactured by Smith & Wesson, Ruger, Browning, High Standard, and many of



the European makers are capable of this level of accuracy right out of the box, assuming the shooter has done his homework with respect to ammunition. The 1.7-inch group with the 1911 .45 is much more elusive; it requires the skills of an above-average pistolsmith and solid load development.

Ammunition used for the .22 stage ranges from standard-velocity target fodder to expensive imported match grade stuff. Most shooters find what works in their gun from an accuracy and reliability standpoint and then stick with it, almost to a fault. We've all read time and time

again how ammunition, powder, and primers can differ from lot to lot. If you are looking for match accuracy of 1.7 inches at 50 yards, then you must be prepared to test each lot of ammo. For the .45, most competitors shoot 200-grain cast-lead SWCs purchased from a commercial bulletmaker over Hercules Bullseye powder and a standard Large Pistol primer. While Bullseye is the powder of choice and has been since FDR was President, my personal favorite is Winchester WST. WST possesses all the best attributes of Bullseye, but it burns cleaner. W231, Solo 1000, and HP38 are also used on the bullseye line. The key to match-accurate ammunition is to launch the bullet at 730 to 780 fps. If you don't have access to a chronograph, try 3.6 to 4.3 grains of Bullseye or WST and vary your charges by .1 grain. Odds are you'll hit a good load. While this isn't intended to be a loading clinic, one final thought is in order: Top shooters weigh their cast bullets, whether homemade or commercially cast. Unseen air bubbles can cause more grief and bad scores at 50 yards than you can imagine. I weigh my slow-fire bullets to .5 grain, but other shooters vary theirs by up to as much as one full grain. The key ingredient is to weed out the really bad bullets that could be two to three grains off the average. Yes, all this emphasis on accuracy is necessary to score well. The best shooters will "clean" slow-fire targets occasionally, and even an average shooter can see the difference between an 8-ring gun and an X-ring gun. If your gun and ammunition can't hold the X- or 10-ring, you don't have a prayer of ever shooting a perfect slow-fire target. Even if you're not capable of shooting a perfect score, you will still score better with topnotch iron and ammo.

Timed Fire

After the slow-fire course is complete, the targets are moved to 25 yards for timed fire. Then 30 shots are fired in five-shot strings, 10 shots to a target, with 20 seconds allowed for each five- shot string. Unlike other handgun sports that use a handheld timer, in bullseye each target rotates to and from the shooter to regulate the timing. As the shooters prepare to fire, only the edge of the target is visible. The target rotates to face the shooter when shooting is to begin. The target rotates away when time has expired. Timed fire is considered the easiest stage of the match. The target is the same one used for slow fire, except only the 9- and 10-rings are black instead of the 8-, 9-, and 10-rings, and it's at half the distance. And 20 seconds to fire five shots is plenty of time. As a friend of mine says, timed fire is sustained slow fire. Top shooters shoot perfect 100's and 99's every time, and even average shooters will clean a timed-fire target on occasion.

Rapid Fire

The third and final stage is rapid fire. It's identical to timed fire: same target at 25 yards with

only 10 seconds allowed for each five-shot string. There is nothing like standing on the line with your .45 in hand and shooting alongside 30 or 40 shooters during a rapid-fire string. The noise, the smoke, the adrenaline- you just have to be there to appreciate it. The top shooters shoot 98s, 99s, and perfect 100's every time in rapid fire. But for the beginner the .45 rapid-fire course is the most challenging. A .45 held in one hand-even with light target-load ammunition-is a handful, and a lapse in concentration during rapid fire is readily apparent on the target (or rather all over the target).

Take The Challenge - You'll Have Fun

Now for those of you with your face twisted up in knots wondering what's the point of shooting a pistol one handed at 50 yards with light-loaded ammunition, let me say I understand completely. After all, you've never shot a pistol with one hand and probably never will, you'll never practice with target loads that don't approximate the recoil of full-power ammo, and only hunters and silhouette shooters with steady rests shoot at 50 yards or more with a handgun and actually expect to hit the target. The point is bullseye is like Indy car racing. Practical? Maybe not. Fun and challenging? Definitely. It's so much fun that there are nearly 14 times more classified bullseye shooters than action pistol shooters. As for challenging, consider that no one has ever fired a perfect 900 out of 900 let alone 2700 out of 2700. To get started in bullseye, contact the NRA's competition department at 800-672-3888 and ask for a bullseye rule book and a sample copy of Shooting Sports USA. Shooting Sports is the NRA's monthly publication dedicated to target shooting. The last 10 pages or so of each issue are dedicated to listing all the NRA-sanctioned events across the country in each discipline. Shooting Sports is a "must have" for the bullseye shooter. So the next time that tin can looks like a boring target, move it out to 50 yards and give it a try with one hand. You might find you have the steely nerves, disciplined trigger finger, and the rock-solid hold of a bullseye shooter. Of course, you might have fun too.

This article is a mirror of one of the same title from the Nov. 1996 issue of HandGunning.

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INTRODUCTORY INFORMATION

Bullseye Pistol Shooting

By Charles E. Petty

Source: http://www.bullseyepistol.com/bullseye.htm

Lots of challenging and entertaining handgun shooting competitions have come along in the recent years, but the demanding old 2700 game grows ever more popular.



In the last 20 years or so, a series of new pistol shooting sports has grabbed the attention of the shooting public. Action pistol, pin shooting, practical competition and silhouette all have had phenomenal growth, and matches like the Bianchi Cup, the Second Chance and the Steel Challenge have become well-publicized big-money events. So a lot of shooters are surprised to find that bullseye, three-gun or conventional pistol, whatever name you choose, is still the most popular of the pistol-shooting games. It's hard to say what prompts the new interest, but I suspect it's a combination of things. Of them, the increasing use of the red-dot sights is an important factor. This type of sight makes it possible for many shooters to continue, or start, shooting after the age when they begin experiencing difficulty seeing iron sights. And, while this may be hard to prove, I think another reason is that bullseye competitors simply get to shoot a lot. Many of the popular speed and practical shooting events feature a little shooting and a lot of waiting. Even though these games are challenging and fun, shooting for 30

seconds and then waiting an hour or two for the next stage in them can be tedious. Conventional pistol is a lot easier for both the competitor and the tournament sponsor-for the Competitor because no holsters and other accessories are required, and for the sponsor because one range officer can control an entire line of shooters. Small indoor ranges throughout the nation host regular leagues and tournaments that require little time and organization. But if there's a primary reason for bullseye's popularity, I believe it's the challenge that appeals to competitors. As far as I'm concerned, bullseye pistol is rivaled only by international free pistol as a test of precision shooting ability. Bullseye is deceptively simple, but it offers challenging competitive opportunities all the way up to the national level.

The classic outdoor pistol match is called a "2700." Shooters fire 270 shots with a maximum value of 10 points each, hence the name. Those 270 shots are divided into three 90-shot events, fired with .22, center-fire and .45 pistols.

This format got its start as a way to combine shooting with the civilian's .22, the police officer's .38 revolver and the military man's .45 autoloader. As .45 accuracy improved, however, shooters began to use the .45 for both center-fire and .45 matches, and today it is rare to see a pure center-fire pistol.

The 90-shot, 900-point aggregate consists of four matches: slow-fire, the National Match Course, timed-fire and rapid-fire. Slow-, timed- and rapid-fire are 20-shot events, but the National Match Course has 30: 10 shots slowfire at 50 yds;. and 10 each timed- (five rounds in 20 seconds) and rapid- (five rounds in 10 seconds) fire at 25 yds. The sustained-fire stages are timed by turning the targets perpendicular to the firing line until the time for shooting begins. They turn to face the shooter, then swivel back to their starting position when time expires. While a full-fledged 2700 is fired at 25 and 50 yds, proportionally reduced targets make it possible to fire all stages at 20 or 25 yds. or indoors at 50 ft. The targets used in outdoor competition have a tie-breaking X-ring of 1.695" diameter and a 10-ring 3.36" in diameter. The 50-yd. target's 8, 9 and 10 rings are black, while only the 9 and 10 rings are black on the 25-yd. sheet, so the sight picture is similar, despite the difference in distance. While a big 2700 match is an all-day affair, NRA recognizes many shorter courses of fire, and many clubs and leagues firing indoor matches use the Gallery Course. It has a single 10 minute, 10-shot slow-fire stage and two five-shot strings each of timed- and rapid-fire. With preparation periods, one relay can take less than 20 minutes. One of the appealing features of conventional competition is the multiple opportunities to win. At the National Matches, and even at large state or regional events, awards are presented for every match and for sub-aggregates. When each of these is multiplied by the many different classes and categories possible at a big match, there can be hundreds of opportunities to win. The NRA classification system groups shooters of similar ability into one of four classes that range from Marksman up to the top-level Master. Shooters may be further grouped into categories--juniors, women, collegians, police and service members. For more information, call NRA's Bob Piccoli at (703) 267-1451.

Time limits are generous and the scoring rings of the targets are, by comparison with the international target, enormous. It isn't hard to put a bullet in the l0-ring. The hard part is trying to put most of them there.

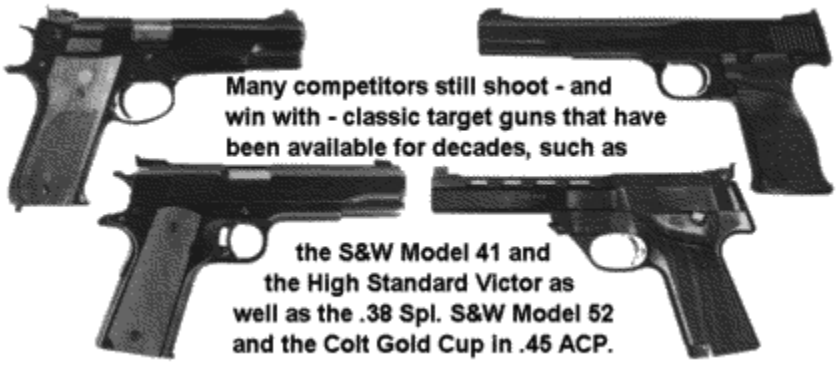
A score of 2600 or more remains, after all these years, the four-minute mile of conventional

pistol shooting. When Hershel Anderson set a national record that has stood for around three decades by firing 2680x2700, he was asked how he did it. His reply, reported as, "I didn't shoot too many nines" was a masterful understatement. For me, slow-fire was always the most difficult event because there was more time for something to go wrong. The basic requirements are to establish sight alignment (the relationship of the front and rear sights), sight picture (the relationship of the aligned sights to the target) and to cause the pistol to fire (trigger control) without disturbing either the sight alignment or sight picture. But a fact of physiology remains to bedevil shooters. Nobody can hold a gun perfectly still. Convincing yourself that there will always be some movement in the sight picture is hard. With red-dot sights, that little red dot literally dances before your eyes, making even a tiny amount of movement more noticeable. This inevitable motion is "minimum arc of movement" (sometimes called wobble area), and it is very much a complex physiological response. The movement must be accepted and concentration directed to keeping the sights aligned and exercising proper trigger control. If we do, the shot will still be good. Building and maintaining concentration is paramount. In slow-fire, where you have the luxury of time, you can go through a little ritual of preparation for each shot. I try to visualize the sight alignment and picture a perfect shot, and then I keep that image in mind as I raise the gun. As the gun is coming up, I try to superimpose the mental image over the real one and establish sight alignment; holding it as the gun rises slightly above the target and gradually settles into a natural position that is, hopefully, a perfect six o'clock hold at the bottom edge of the target. This "natural point of aim" is controlled primarily by the position of your feet. and is determined by extending your shooting hand toward the target, closing your eyes, wobbling your arm around, and letting it settle to a natural position. When you open your eyes, your hand should be pointing squarely at the center of the target. If it isn't, you move your back foot and repeat the exercise until you get it right. With practice, you learn a stance that does this almost automatically. As the gun begins to settle, there is a brief period of time, perhaps five seconds, when things are relatively unstable, followed by another interval of 5-10 seconds when the hold or sight picture is optimum. After that. muscle fatigue causes the hold to deteriorate, and if the shot isn't fired within about 20 seconds, the gun must be put down. Here lies one of the most difficult challenges in shooting. If, after awhile, the shot won't break, there's an almost irresistible urge to get rid of that round so you can go on and do a better job with the next one. It takes a lot of discipline to make yourself bring the gun down and relax a bit instead of fighting awhile longer in hopes of making it come out all right. The inevitable result of this, if you give in, will be a jerked shot. One of the difficult things to learn is to "call" your shots. This is done by relating the shot's point of impact on the target to the face of an imaginary clock. While this is a very basic part of serious shooting, it demands intense concentration to know exactly where the sights are at the moment the shot breaks and to recognize whether there were any shooter errors. Sometimes the first sign of a flinch or trigger jerk is the location of the bullet's impact. The technique for timed- and rapid-fire really

isn't much different, with the obvious exception of the time limits. I use the range officer's commands as cues to indicate where I should be in my preparation for the string. When the command to load is given, I get my feet where they're supposed to be and establish a proper firm grip on the gun. Then I rest the gun on the bench and begin breathing deeply, simply taking a series of slow, deep breaths. When the range officer says "ready on the right," I raise the gun above the target and begin to lower it so that by the time he finishes saying "ready on the firing line" the sights are aligned on the edge of the target right where the bottom of the black center will be when the targets turn into their facing position. My ideal string will have the first shot break just as the target is fully faced, allowing the maximum amount of time for the other four shots. The mental change of gears to go from timed-fire to rapid-fire is hard to analyze, but my observation is that most of us don't use the full 20 seconds of timed-fire. If that's the case, then going to rapid-fire doesn't really require doubling the pace. Good range officers are a gift to shooters, for it really helps if they are consistent and deliver their commands the same way each time . One of the hardest things is dealing with the everyday distractions that attend pistol matches. You know, people talking and prowling around behind the line or even slamming car doors in the nearby parking lot. Little disturbances we'd usually never even notice can achieve monumental proportions when you're trying to shoot. We're told to "ignore them" but a better technique is to analyze the noise and determine whether it's something that should concern you. Same for all the rest of them. This is easier said than done, but learning to deal with these things effectively is the key to real achievement in the shooting world. The same applies to match pressure. Some shooters study the scoreboard intensely, while others won't go near it at gun point. One of the axioms of pistol shooting is that you usually beat yourself. The best way I know to learn to manage these things is to face them often. There is no substitute for shooting under match conditions The things the human brain can do are remarkable, and concentration, to the exclusion of everything around you, is an awesome experience. I've been on the firing line at Camp Perry with hundreds of competitors shooting .45 rapid-fire at the same time and never heard the first bang. Professional athletes have begun to talk about something they call "being in the zone." This is a state of super-concentration where you can sometimes do remarkable things. Shooters report that it seemed as if they were able to guide the bullet to the X-ring by nothing more than willpower. Another analogy might be to describe it as going on autopilot where the proper fundamentals of sight alignment and trigger control are applied almost effortlessly. It's been said often enough to become axiomatic: shooting is 10% physical and 90% mental. Although some may argue about the percentages, 1 don't believe anyone would dispute that the physical skills required for shooting are not that difficult to acquire for most people. The mental management techniques needed to achieve a high level of performance are, however, staggeringly difficult. Dr. Bud Ferrante has served as team psychologist for the U.S. Shooting Team and Olympic athletes as well. "The specialty area of sports psychology has evolved rapidly over the last 10 years," Ferrante explained. "Individuals representing elite world-class

organizations all the way down to youth sports have utilized sports psychology practitioners to improve their athletic performance as well as enhance their capacities." Shooters learn techniques for relaxation, awareness, concentration and visualization. While the basic techniques may be generalized, their application is very much an individual thing, so Ferrante's work has included group seminars on specific topics combined with one-on-one work to develop a method for each individual. "Don't worry about score. Don't take it personally, it's not a measure of your self-worth, it's just your score. Don't compare yourself with other people. That's a great error that can hold you back," he said. Instead, he suggests, concentrate on "performing the shot." We've always been told to take it one shot at a time and try to get all the elements, stance, grip, trigger, sights and follow through to come together for one good shot. Ferrante suggests that we analyze the shot in terms of what felt good and what felt bad and learn from it. Ferrante agreed with my comments about outside distractions. "You can't separate the shooter from the world," he added. Talking with Ferrante and Kimberly Whitchard, another sports psychologist, I asked for a proper name for the "zone" phenomenon. Ferrante used the term "flow" to describe the same thing and likened it to a Zen philosophy of "stilling the mind." Whitchard, who has worked extensively with professional golfers, compares it to a state of self-hypnosis that can be almost trance-like. Her experience is that the condition can be learned and some people can achieve it almost at will. But whether your aspiration is to become a world-class shooter or simply have fun. conventional pistol, bullseye if you prefer, is a great way to improve shooting skills. It is also excellent training for all the other pistol-shooting events. Once you've mastered the basic elements of sight alignment and trigger control they are applicable to every shooting game. Since bullseye pistol is a test of pure accuracy, it's logical to conclude that it requires the most accurate guns. New shooters often believe they don't need the best equipment because they lack the skill to take advantage of it. The truth is that they're the ones who will benefit most from having the best. If a clunker .45 is only capable of shooting 8" groups at 25 yds., what chance does a new shooter have of learning anything? This is not to discourage new shooters from getting started with a gun or guns they already own, but they should realize that a gun's accuracy limitations can lead to frustration when it is time to retrieve It is also imperative to know what sort of accuracy your gun/ammunition combination can deliver. A few groups fired from a good rest will answer the question, but it's important to have a baseline. Buying factory ammunition is expensive these days, but it is no longer safe to assume that good handloads will be better. In fact, it takes a lot of hard work to make handloads that will equal the accuracy of today's factory match loads. So the first step is to find out what the gun will do with good factory ammunition and then perhaps try to duplicate those results with lower cost handloads. The .22s are probably the most ammunition-sensitive, and it's quite remarkable to see how a particular gun will display a preference for one brand or type over another. Once again, testing is the only way to find out. Match-grade .22 ammunition usually does shoot better groups, but the difference between it and regular standard velocity ammo may not be

dramatic. That the cost difference, at retail, can be nearly 10 times higher would seem to be a powerful incentive to try some of the less costly alternatives. The guns that are popular now are little different from those popular 20 years ago and, other than the red-dot sights many of us use, things really haven't changed very much in terms of basic equipment. Even though the game is called three-gun, use their .45s in the center-tire portion of the match. This isn't done to just save the cost of another gun, but because it's the best way. Those who have shot one of the .38 Spl. M1911Al conversions or the S&W Model 52, the usual alternatives to the .45 in center-fire matches, probably loved it, for the soft recoil of the .38 makes the .45 seem almost punishing, but I can tell you from bitter experience that the times the .38 will help you are far outnumbered by the times it will hurt. Using the .45 eliminates having to learn another gun. Even if the .38 is built on the same frame, the different recoil and follow-through requirements complicate things needlessly. A Master-class shooter may benefit from using a .38 Spl. semiauto conversion, but others should evaluate them carefully. Equipment surveys at Camp Perry have changed little over the years. The S&W Model 41 and High Standards dominate the .22 events, with a few Rugers and imports thrown in for variety. The demise of old High Standard was bad news, but there are still plenty of the older guns out there, and they are undeniably popular. Fortunately, High Standard parts and service are still readily available. and High Standard Mfg. and Stoeger are making High Standard target guns again. As far as the .45 is concerned, Colt guns are still the leaders, but the Springfield form the basis of good guns. Most guns, in my opinion, even the Model 41, need a trigger job, and most .45s need a lot more.



For bullseye guns, the accuracy standards must measure up on the most difficult Portion of the course, 50-yd. slow-fire. If the gun/ammo combination isn't capable, at the minimum, of keeping every shot within the l0-ring. a shooter can lose points and never know whether it was his fault or the guns.

A better standard is a maximum of 3" groups at 50 yds. For the good .22s this isn't as much of a problem, but it takes a pretty good .45 to do it. But accuracy is a product of both the gun and the ammo, and each must be proven. The NRA Pistol Rules don't place many limits on the

guns shooters may use. Trigger pull must be 2 lbs. on .22s and 3-1/2 lbs. on .45s. The sight radius of .45s can be no more than 10", while .22s can have a barrel length of no more than 10" The one exception is the service pistol or "hardball" gun. These are used in National Trophy matches at Camp Perry and DCM "leg" matches at the state and regional level. They must generally conform to the outlines of the service M191 1 or Beretta M9 9x19 mm pistols, though adjustable sights. trigger shoes and stops are allowed. Conventional pistol doesn't require much in the way of accessories. Gallery shooters can carry a .22 and a couple of boxes of ammo in an attache case. Veteran pistol shooters almost invariably carry boxes that hold several pistols, ammunition and accessories. These have a top-opening lid that, when opened, serves as a mounting point for a spotting telescope. A carrying strap enables shooters to carry everything to the firing line in one load. Eye and hearing protection are a must. Many shooters use special glasses that block the non-shooting eye, and older competitors have adopted the amplifying electronic ear muffs that help them hear range commands while protecting their remaining hearing. Knowing that your equipment is capable of a high level of performance is reassuring and necessary, but it also eliminates a whole list of possible excuses for poor performance.

Due to the popularity of bullseye pistol, there is a probably a club or a range near you that holds outdoor or indoor matches. A few matches are given in the "Regional Report" and a comprehensive listing is part of Shooting Sports USA (available by calling (703) 267-1583). For more information on the sport, contact: Bob Piccoli, manager of NRA Competitions' Pistol Dept. at 11250 Waples Mill Rd., Fairfax, VA 22030-9400, (703) 267-1451.



American Rifleman

This article is a mirror of one of the same title from the June 1997 issue of the American Rifleman, the Official Journal for members of the National Rifle Association.
This periodical is so good that it alone would justify the annual membership dues if there were no other benefits!

INTRODUCTORY INFORMATION

Introduction to Pistol Competition

By NRA Competitions Division

Source: http://www.bullseyepistol.com/comp.htm

The purpose of this page is to give general information on how to get started in the sport of conventional pistol competition. Items discussed include: How To Get Started, Equipment, Ammunition, Accessories, Eye and Ear Protection, Targets, Course of Fire, NRA Classification System, Tournament Entry and, Other Activities. The information will answer most of the most often asked questions that a beginner will have. The NRA stands ready to assist you and if you have any questions, we hope you will contact us. For more information on conventional pistol competition write to the National Rifle Association, Competitions Division, 11250 Waples Mill Rd., Fairfax, VA 22030. If you wish, you may phone us at (703) 267-1451.

Many individuals become interested in pistol competition; however, unless they start off with the proper information, they find it difficult. The cost of equipment is generally a stumbling block. Many feel that unless they have the best of everything they cannot compete. This is not true. Most start with a minimum investment of a .22 caliber rimfire pistol (autoloader or revolver), spotting scope and stand, and most important, eye and ear protection. It is also advisable to have a copy of the current NRA Rule Book.

Pistol Competition may be fired outdoors or indoors. The course of fire is basically the same for both, but the distance is different.

How To Get Started

If you have an interest in taking up the sport of pistol competition, it is recommended that you check the Coming Events Section of *SHOOTING SPORTS USA*. All upcoming NRA sanctioned tournaments are listed in this section. Find a tournament being conducted near you and contact the listed sponsor to request a program. Attend this tournament as a spectator. This will give you an opportunity to observe how it is conducted and talk to the sponsor and competitors. Be sure you don't disturb the competitors during the match. The time between relays is a good time to talk to them.

You will see a variety of equipment and accessories. Competitors have their own opinion as to what is best. This may sound confusing, but remember, you're there to gather information.

If there is a club in your area, make arrangements to attend one or more if its practice sessions. This will serve the same purpose of attending a tournament except a practice session is not always conducted under match conditions. However, this will give you a better

opportunity to talk about equipment. Also, you may have an opportunity to actually shoot one or more types (brands) of pistol which will help you to decide which seems best for you.

An excellent way for a new shooter to start in competitive shooting is a league. Although NRA rules are used, a league is generally informal. Usually a handicap system is used so all individuals or teams have an equal chance of winning. A Sanctioned League Handbook and application to have a league sanctioned are available at no cost from the NRA Competitions Division.

Equipment

It was stated previously that you don't need the "best of everything" to participate in competition. There are many good values in used equipment. If others know you are "in the market," you will hear of many good deals. Although the question of which is best is asked often, there is no answer. As you will find, each competitor uses his or her favorite brand. This can be related to buying a car. You may like one make and someone else a different make. However, both are satisfied with what they have.

Section 3 of the NRA Pistol Rule Book defines authorized equipment and ammunition. This section is not meant to restrict equipment but to define limitations.

Pistol - Autoloader or revolver? Up until about 30 years ago the revolver was the one to use. Some competitors still use a revolver, but the autoloader is now used almost exclusively. Autoloaders have been developed where they are capable of top-notch accuracy. An autoloader will provide an advantage when firing timed fire (5 shots in 10 seconds) courses.

It should be noted that the standard course of fire is a "3-gun aggregate." This is fired with .22 caliber rimfire, center fire, and .45 caliber pistols. However, it is not necessary that you have 3 different guns. In most tournaments you may enter and fire only one or more states of the aggregate. Many competitors entering the complete aggregate only own a .22 and .45 caliber pistol since the .45 caliber may be used for the center fire stage.

Spotting Scopes

A scope is necessary as this will allow you to see your shots on your target in order to make sight corrections. Scopes need a stand for support or some means to mount on a gun box if used. They come in various price ranges and, as with all optics, you get what you pay for. Good resolution is important as you will need to see a .22 caliber hole on a target at 50 yards if you fire outdoors. A 20X to 30X is generally used.

Ammunition

Not much can be said about ammunition. Obviously, you will need the proper ammunition for the pistol you'll use. Match grade ammunition is available commercially and costs more than "regular." This is manufactured under high standards and is more accurate for competitive shooting. Many competitors hand-load their own ammunition (except .22 rimfire). This is not only cost-effective but allows for loads to be "customized" for a particular gun. In many cases, hand-loaded ammunition is more accurate than commerically produced match grade ammunition. If you use tha hand-load route, be sure to follow all safety precautions.

Accessories

There are many accessories available, and no attempt will be made to mention them all. Some of the most common and useful ones will be discussed.

1. Sights - All target pistols come with an adjustable rear sight. This is a must. However, various brands of adjustable sights are available, some better made than others. Again, experience will tell which one is better for you. Optical and electronic sights are available and are currently allowed in NRA Conventional Pistol competition. Those sights which project an image upon the target (laser) are not permitted. These sights are a help to the shooter whose eyesight is "not what it used to be." They are not recommended for the new shooter who is still mastering the fundamentals of sight alignment, which is absolutely necessary for a champion shooter.
2. Grips - All pistols come with grips. Unfortunately, these are made for a standard size hand. Since no two peopele are identical, it is unlikely that the grips from the factory will fit properly. Custom grips are available as accessories and will vary in cost depending on if you want true custom grips (made exclusively for you) or off the shelf.
3. Shooting Box or Kit - Some means are necessary to transport your pistol and accessories to and from the range. This can be as elaborate as a large box which holds everything to a simple cloth or leather case. The choice will depend on the type and amount of shooting you do. The box has an advantage as it can be used for storage at home.
4. Eye and Ear Protection - These items are a must and should be the first accessories you purchase. If you normally wear glasses and they have hardened lenses, you are covered for eye protection. If not, you should acquire shooting glasses designed for that purpose. Ear plugs or muffs are necessary also. Some shooters wear both.

Targets

Rule 18..15(e) in the NRA Conventional Pistol Rule Book states in part -- "it is the competitors responsibility to frame the correct target for the specific match and distance." As a new pistol competitor, you need to be familiar with what the proper targets are. NRA official targets are described in Section 4 in the NRA Rule Book. Sections 7 and 17 will give the ragets required for various courses fired.

Course of Fire

NRA conventional pistol competition consists of firing slow, timed, and rapid fire. This is done at 50 and 25 yards outdoors and almost exclusively at 50 feet indoors. Generally an outdoor match will consist of 20 shots, slow fire at 50 yards (2 10-shot strings, 10 minutes per string), 20 shots, timed fire at 25 yards (4 5-shot strings, 20 seconds per string), 20 shots, rapid fire at 25 yards (4 5-shot strings, 10 seconds per string), and the National Match Course (10-shots, slow fire at 50 yards, 10-shots timed fire, and 10-shots rapid fire). This match consists of 90-shots for a possible aggregate total of 900 points. For a 2700 aggregate this match is fired once with each gun; .22 caliber rimfire, centerfire, and .45 caliber. Many match programs call for only one or two guns, that is a 900 or 1800 aggregate.

Most indoor tournaments are fired with .22 caliber rimfire only for a 900 aggregate. However, some indoor matches use all guns for a complete 2700 aggregate.

NRA Classification System

Many new shooters hesitate to enter competition because they feel they are not good enough and would not win anything. This is true to some extent as, with most sports, the first time generally does not prove productive as far as awards are concerned. The NRA developed, some years ago, the NRA Classification System to provide an equitable distribution of awards. This places all shooters in a particular class; Marksman, Sharpshooter, Expert, Master, or High Master, depending on their average. Tournament sponsors award prizes in each class and in some tournaments, depending on the number entered, second and third place. A new competitor must enter the first tournament in the Master class. Thereafter, he/she may use a Temporary Score Record Book, which may be obtained from the tournament sponsor or NRA to enter match scores and compute the average for each match fired. The next tournament would be entered in the class which the average covers. After a minimum of 360 shots, fired in NRA Sanctioned Competition, have been reported to NRA by the tournament sponsor, an average is taken and an Official Classification Card is sent to the competitor. The competitor must then compete in that class until a new classification card is sent by NRA. For complete information on the NRA Classification System, see Section 19 in the NRA Pistol Rule Book.

Another system generally used by sponsors to distribute awards is the Category system. Sponsors, using the Category system, will give awards to winners of various categories such as Civilian, Service, Police, Junior, Women, etc. Not all of these are used by all sponsors.

Tournament Entry

When entering a tournament, you will be required to fill out a Registration Entry Card commonly known as an SR-1 card which will be provided by the tournament sponsor. This card provides the information needed to place you in your proper class and category. Part of this SR-1 card is sent by the sponsor to NRA at the end of the tournament with your scores, so they can be posted to the classification system maintained at NRA Headquarters.

It is very important that you put your **NRA membership ID** number (if you are an NRA member) on the SR-1 card. (If you are not currently an NRA member your scores will still be posted for classification purposes, however, you may only compete in "Approved" tournaments. "Registered" tournaments are restricted to members). This will assure that your scores are posted properly and quickly. It is also very important that you always use the same name. For example, if initials are used, such as "J. D. Smith", then continue to use initials, rather than sometimes using "Joe Smith."

Other Activities

Competitive shooting is in itself a great hobby. However, this activity is generally done on weekends with maybe a practice session during the week. NRA has a program whereby the practice session and matches can be used to earn attractive awards. This is the *NRA Qualification Program*. In this, a shooter tries to equal or beat a "par" or "set" score. For complete details on the NRA Qualification Program, write to the NRA Safety and Education Division, 11250 Waples Mill Rd., Fairfax, VA 22030.

NRA also offers membership in three honorary clubs - The "2600, 2650 and 2670 Club." Membership is obtained by individuals firing a score of 2600 or better for the "2600 Club," a score of 2650 or better for the "2650 Club", and a score of 2670 or better for the "2670 Club". Scores must be fired in an NRA Sanctioned Registered 2700 aggregate tournament either indoor or outdoor.

For additional information, or questions relating to competition, you may contact the NRA Competitions Division by E-Mail at competitions@nra.org

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References

- How To Get Started
- Equipment
- Ammunition
- Accessories
- Eye and Ear Protection
- Targets
- Course of Fire
- NRA Classification System
- Tournament Entry
- Other Activities

Further Reading

- competitions@nra.org (mailto:competitions@nra.org)

INTRODUCTORY INFORMATION

Tips for Bullseye Pistol Shooting

By Frank Higginson (as told to Mike LaVoie)

Source: http://bullseyepistol.com/frankhig.htm

Frank Higginson is a top shooter of all time. He still holds records from 1970's. He is a nice guy, easy going, and the finest instructor I have ever met. He has a way of teaching in terms that we can understand.

Scores are the result of groups. Everything we shoot on paper is the result of how we performed on the firing line. We are concerned with how our shots are grouping. Obviously the smaller the group, the better shooting you are doing. I will try to put into writing the proven techniques of a National Champion... -ML

SAFETY

Safety is most important to the shooting sports. Ear protection and Eye protection are mandatory. Commercial glasses from an optometrist are fine. Good glasses are important. If you've bought the standard shooting glasses, hold your glasses out in front of you and look though the glasses at an object. Move the glasses slowly up and down. Is the object clear or distorted? If the object is clear your glasses are okay. If the object is distorted then you'd better buy a good pair of glasses. If you use a "red dot" sight a little darker lens than needed will make the dot look sharper.

There are three types of ear protection; Sponge, Molded, and Outside ear protection.

1. Sponge - the sponge is soft foam that you roll between your fingers to make it thin and then you place the sponge in your ear. Most are disposable, cheap in cost and fit most size ears.
2. Molded ear protection is either a plastic insert that pushes into your ear hat is very inexpensive or a custom fit earpiece. The custom earpiece is a funny looking device that fits your ear perfectly. The "maker" of the molded earpiece takes "putty" and adds hardener to it; next they gently push the "putty" into your ears. The person getting the "putty stuffed into their ears has to not talk or move their jaw for about 10 minutes. The "putty" hardens in your ear. The "maker" than removes the now hardened "putty" and "seals" the earpiece. This earpiece is very comfortable and reduces noise level very good.
3. Outside ear protection is the usual ear protection you normally see shooters wearing. Some sort of plastic "cups" that cover each ear. There are two types of

cushions on the molded ear protectors, Air and Liquid. Air is the best because if you get a leak during a National Competition, liquid can be very distracting.

The Molded earpiece also can have an Amplifier build in with a sound cut off.

Wearing Two ear protectors can help you focus due to the extra noise reduction.

In cold weather warm up your ear protection by body heat. Place the sponge or molded ear protectors in your pocket. Place the outside ear protectors on your leg before the match.

One set of Ears and Eyes are given to most people DON'T LOSE THEM.

STORAGE OF EQUIPMENT

Don't store equipment in hot areas or areas subject to fluctuation of temperature. Near the furnace is not a good idea.

Do not store guns in a gun sock.

Hard plastic covered cases can be a problem if moisture was introduced into the soft foam inside.

A Dehumidifier is perfect to get rid of moisture in a humid area (New England)

Desiccants are very good.

22 ammo is not sealed, and most commercial ammo is not sealed against moisture.

Keep equipment out of direct sunlight, the metal of the gun can get searing hot, and the heat can effect the accuracy of the ammo. Always cover your equipment from the sun.

Cold guns will attract moisture, including inside the gun, i.e., springs, hammer, etc.

Your scope in cold weather to hot temperature will fog scopes, shooting glasses.

Wood also absorbs moisture, grips, wood shooting box.

Always check your guns once a month for rust. Take your grips off to check for rust.

LUBRICATION

Lubrication is very important in a match pistol. A good inexpensive thin oil will do, because you use a lot of it. FP-10 firepower and the other synthetic gun lubes are not necessary. A lot of oil doesn't mean you should dunk the gun into a bucket of oil either. A lot of oil on the rails works well. Oil attracts unburned powder, so it stands to reason that you don't want to get oil into the inner workings of a match firearm.

Breakfree has polymers in it that aren't any good.

Hoppes #9 lube oil is very good.

Remington oil turns to varnish.

Gun Slick isn't any good because it has graphite in it that will wear the parts you put it on, must people put it on the hammer, sear. What a surprise it must be to ruin a good trigger job.

Oiling the case of the top round on the magazine won't hurt unless you are shooting a high power round. The pressures will escape past the case instead of the case stopping the gasses. It can very dangerous.

If it works for you use it, if you are shooting high power the results can be dangerous.

Should you get caught in a rainstorm or high humidity, WD-40 works very well to remove moisture. You must lube the gun after using WD-40.

Don't use Gun Scrubber on wood.

Never vacuum primers and powder as the primers may explode and start a fire. (not to mention they are kinda loud when they explode.)

Always wash out black powder cans when they are empty. Throwing black powder in the trash is very dangerous.

GRIPS

The body is a machine. Your body gets hot and cold. When your body gets hot you perspire, hot weather makes your hands swell. When it's cold weather your hands shrink. The point being if you are using orthopedic grips, the grips may be too big or too small, depending on the weather. I recommend stock grips.

CARE AND CLEANING

You must use a good cleaning solvent. When you are using a good cleaning solvent, use good ventilation so you don't get poisoned.

Don't dip the brush into the solvent jar or you will contaminate the entire bottle. Use a "jag" and dip the cloth wrapped around the "jag" and then dip it into the solvent.

Use a good bronze bore brush. One good trick I use is, take a shotgun 12 gauge cleaning rod and brush and run it up and down the barrel. Of course you will have the barrel out and laying down on a table. Place your hand on top of the barrel and press down. With the other hand run the brush up and back through the barrel.

Don't use a stainless brush.

The best cleaning rod to use is a stainless steel rod with a Teflon coating.

The second best is to use a Polished one piece steel rod.

An Aluminum is good to use as long as you make sure no oxidation has occurred on the rod as aluminum oxide can form on the rod making it an abrasive.

Don't use fiberglass, plastic or wood rods as contaminates can be stuck to them.

Clean your rod every pass and run the rod straight though the barrel.

You don't want to wear the barrel because of hitting the barrel with a cleaning rod.

Don't let Hoppes soak into a stainless steel barrel as it will eat it up.

Sweets solvent is a good gun cleaner, you must use a good oil after cleaning.

Stay away from flashy gimmicks. The Lewis lead remover and the steel wool pad are good if you want to remove some lead, but you will not be able to clean the lead-copper out of the corners of the lands and groves.

Look for a gray steak in a 22 barrel, this means your barrel needs to be cleaned. 22's should be completely cleaned every 6 months.

Any perfectly clean barrel may throw off your first shot. (Not in the usual group.)

Machine rests are very useful but the machine rest may not be as accurate as shooting from your hand.

Use lens tissue for glasses and scopes and use lens cleaner.

The dots on the glass of scopes are oil spots.

Keep your glass carrying case clean so you don't scratch your shooting glasses.

You can use soap to stop your glasses from fogging.

Glasses that change tint should be left on the shooting bench before you go down range to score targets or the tint will be too dark for the next stage. These Glasses aren't any good as rapid weather changes.

GUN BOX

A good designed box fits the equipment you carry. Keep the gun box clean, and always check the hardware on the box. Check the strap, hooks, handles, scope brackets. Always keep a plastic bag that will cover your box during a rainstorm inside the shooting box.

NON-SHOOTING EQUIPMENT

Clothing should be loose fitting and comfortable. You will need rain gear due to the weather. You should shoot in long sleeve shirt and pants. Shoes should be comfortable and should have

flat soles. Never wear new shoes until they are broken in. Never wear sandals, fire ants love sandals.

Bug juice-Suntan lotion-Earplugs- Scope, Tools to adjust sights and tighten scopes. Garbage bag to cover box in case of rain, cut out front and rear of bag so you can see through scope. The garbage bag can also make a poncho for you if necessary.

CLEANING THE BARREL

Due to progress, we don't have to clean barrels as soon or much as Black powder. This is due to smokeless powder. Use a hard lead bullet and use good alox on bullet, 50-50 mix Bees wax and Alox mix.

Clean the barrel after shooting lead than switching to copper. Use a dry brush than follow up with a clean cloth to clean the barrel.

Always shoot though a dry bore

HOW TO SHOOT

Now the stuff you have been waiting so patiently, how to shoot Hi Master scores.

The next paragraph is the MOST important to precision shooting.

1. IDENTIFY YOUR TARGET
2. SEE THE FRONT SIGHT (DOT)
3. MOVE THE TRIGGER STRAIGHT TO THE REAR.
4. BOTH MUST BE DONE AT THE SAME TIME.

Memorize the above paragraph. This is the secret to precision shooting.

#1 states "See the front sight (dot)" I mean don't just look at the front sight (dot), See the front sight (dot). Is there a scratch on the front sight? Does your red dot look sharp, or is there a "tail" off it? Some shooters will purposely put a scratch on the front sight to give themselves a focus point to see, If they see the scratch, they are Seeing the front sight.

#2 states Move the trigger straight to the rear. This seems simple enough, but the hard part is really #3 Both must be done at the same time.

The above is a must do.

O.K. you memorized that part and you will never forget it and do it every time, Right? Well, here are more tips that will move you to a better performance.

The "dot" shows where the barrel is pointed.

The Dot must be centered in the scope as well as centered on the target. The Dot in the scope may look different at different times of the day, it's your eye that's changing during the day.

It doesn't matter if you like the dot bright or light, Whatever works best for you. The Dot may look square or triangle, or round, your eye sees it that way. Another person can look threw the same scope and see a different "dot" than you do.

Magnification causes "parallax" so don't use any magnification. Parallax is the difference in apparent direction of an object from the viewer when looked at from two different positions. In other words if the scope is on an angle and you see the "dot" dead centered, and you break the shot at that instance, the shot will not hit where the dot was centered.

To help your concentration and not be distracted by the shooter next to you, by their movements or their brass whacking you, Blinders on the side of your shooting glasses and an opaque milk color patch on your non- shooting eye helps. Wearing the patch on your shooting eye is not a good idea, and wearing patches on both eyes doesn't help at all except if you want some sleep.

You are the only person on the range!!!! Don't concern yourself with the problems of the other shooters, you are the only person on the range. If you lend your screwdriver to a shooter, what will you be thinking of, Shooting or the screwdriver? You will be thinking about the screwdriver!

You go to a match to win it. Accept your wobble area and shoot within it.

Think positively.

Shooting, You learn more about yourself than any other sport.

CHECK YOUR GEAR

Look for cracks or shiny spots on the magazines near the top of the magazine (see picture). If you see any cracks the magazine is broken. If you see any shiny spots near the top of the magazine, it is worn out.

Check the extractor on a High Standard to be sure it is clean and sharp. If the High Standard barrel is tightened by a screw, make sure the screw is tight. If the barrel is a push button, any time you bang the push button check to make sure the barrel is tight.

Check scope mounts and be sure scope mount screws are tight.

Check your 45 firing pin block for a crack. If you see a crack replace the block.

Look for a gray area on the slide, midway between the ejection port and the front sight.

Check the gun box hooks and latches for damage or if they are loose.

TIPS

Use one magazine

When the command "Is the line ready?" is given, if you are not ready say that you aren't ready very loudly. Yell "NOT READY!" The line will wait for you. Once the command "ready on the right?" is called, it's too late to stop the line.

SLOW FIRE

10 ROUNDS IN 10 MINUTES

Always shoot in a cadence. Slow yourself down and shoot in 1-minute intervals. If everything is working smooth you will be in a cadence. If you run into trouble take a break, there isn't any hurry. Keep your sight alignment. Allow yourself to move. Do not be dead still. The only person who doesn't move is dead. Accept the wobble area. Allow the gun to move around.

GRIPPING THE GUN

How do you hang onto the gun? Pick up the gun in your non-shooting hand and stuff the gun into your shooting hand. Then extend your fingers around he grip. The grip should be natural and comfortable to you. Grip equally with your four fingers. (Your four fingers do not include your trigger finger.)

INTRODUCTORY INFORMATION

The Cause for Competition

By Jake Shevlin

Source: https://www.bullseyepistol.com/oursport.htm

There are lifelong rewarding values and satisfaction from participating in competitive activities. In James A. Michener's "Sports in America", the author analytically dissects modern sport and quickly rejects the spectator variety as an ongoing advantage to anyone. He calls individualistic activities such as golf and shooting, "life-long sports" and says that a person should choose a lifelong program that will enhance his general well being. Michener states that competition is an "extension of our very nature." The NRA Conventional Outdoor Pistol [Bullseye] competitive shooter is participating in one of the three most difficult individual disciplined sports games in the world. The other two, are worldwide Professional Golf and International Trap. It takes a lot of "guts" to participate fully in these activities. Practice and preparation, travel and its side effects, extreme weather, [including wind, rain, cold, and heat], monetary cost, physical and family sacrifice all take their toll, but these are the lot of the golfer or shooter who travels the golf, international trap, or pistol "circuit." Nonetheless, the satisfaction of winning the "Masters" at Augusta, an Olympic Gold Medal, an NRA National Championship at Camp Perry, or a DCM "Distinguished Pistol" medal is beyond belief! To master any of these sports is self-rewarding.

However, author Michener also claims, "We don't have to compete with or compare ourselves against the best." Creative competition [competing with yourself], encourages the human being to be better than he might otherwise have been. He agrees that incentives like moving up in class, the firing of a "Hole in One", 50/50 birds, or Ten "X's" can be as rewarding as going to the winners circle. Therefore, the winning of the Sharpshooter Class at the local monthly 2700 match, the attainment of an NRA Expert Pistol classification, or just shooting a few more X's than usual is also beyond belief! It is the competitive spirit that drives and rewards the person, permeates the balance of his life, and offers him success in his other pursuits.

As Michener says, "For life to be meaningful, there MUST be competition, either external or internal."

You are invited to any local pistol club match or league, and to any NRA Registered or Approved Pistol tournament. Match dates and contact numbers for NRA Registered and Approved tournaments are listed in NRA's "Shooting Sports USA".

Make your life meaningful; first, visit and observe the action, and then, "come out and try it." Prove to yourself that you have the "guts." Note: Match entry requires knowledgeable preparation! Please contact the Match Director or any one of the match shooters.

Jake Shevlin

1999 [Written initially for publication in "The Smoking Gun" at ER&P Club, Inc.]

INTRODUCTORY INFORMATION

Tips on Being a Good Competitor

By NRA Competitions Division

Source: https://www.bullseyepistol.com/nratips.htm

Tournament officials are not the only ones who have duties at a match. The success of any match depends as much on competitors as it does on the proper functioning of the tournament officials. Match personnel and competitors must work together during a tournament.

Here are some general rules which, if practiced consistently, will contribute to the smooth operation of any match as well as making you a welcome addition to that competition.

Know the Program

> The only way to know both what to expect at a match, as well as what is expected of you, is to *get a copy of the program and read it thoroughly*. Saying "I didn't know that" isn't going to make any difference to other competitors or to a jury. The conditions under which the match is going to be fired are listed in the program. Once you've paid your entry fee, you've accepted those conditions. Familiarizing yourself with the program in advance is the only way to be sure that you'll be free to concentrate on your shooting.

Check Your Equipment

> The night before you leave for a match, get *all* your equipment together in one place and make sure you have everything you'll need. Be sure that your ammo is right for the gun you'll be shooting, and that you have enough for the course of fire planned. Take along a screwdriver, pencil or ballpoint pen (fiber tips are terrible in the rain) and your eye and ear protection. Even if you don't normally wear glasses, and you're sure that shooting .22 caliber won't bother you, *you need eye and ear protection*. Many ranges have a mandatory eye and ear protection requirement. Don't forget rain gear - you'd rather have it and not need it than the other way around. Be sure to take your data book, classification card or Silhouette book and your NRA membership card.

Guns and Ammo

Make absolutely certain that the gun or guns you are going to use are clean, in the best of condition, zeroed and legal for the tournament. Where appropriate, you'll want to take along extra magazines or clips. Again, be sure you have enough ammunition to complete the tournament, including extras for possible refires or shootoffs. You'd hate to forfeit a match because you run out of ammo halfway through a shootoff. Make sure to bring the right amount for the gun you will be shooting.

Know the Rules

How well do you know the rules? All competitive shooters, novice or experienced, should have a copy of the current rule book for the competition they're shooting, and should be familiar with it. If a rule is unclear to you, you can ask a tournament official, Official Referee or Match Supervisor for help, or contact the NRA Competitions Division for clarification. There are two important things to remember about the rules:

1. The rules apply to everyone, from a High Master with several National Championships to his credit to a Marksman attending their second match, and
2. You may not agree with all the rules, but you must follow them, both in spirit and in letter.

Key Technicalities

There are a few details not generally covered by rule books or tournament programs, but which are important:

1. Be sure your entry card is filled out completely, correctly and *legibly*. Include current classification and special category, when appropriate. Your NRA ID number is required as it is the key to your records.

2. At any match where competitors score for each other, you must make neat, legible figures. There must be no question whether a figure is a "1" or a "7", and each box on the scorecard must be properly filled in with a figure. For example, a miss is written as an "M", it is not an empty space, a dash, or anything else. In any match, an "X" in the first box, followed by a line through the next nine boxes DOES NOT mean 10 X's, but 1 X and 9 misses. As the scorer, how

would you like to inform the shooter that their first "clean" ever doesn't exist as far as the Stat Office is concerned?

3. To carry our example further, the shooter also has a responsibility to make sure the score fired has been marked on the card properly, and if not, to take the proper steps to change it through the Range Officer. Never sign your scorecard until you have fired the match and have verified the shot values and total score shown on the card. Once you and the scorer have signed the card, you've accepted the shot values indicated there and have no appeal.

4. Know the difference between a "challenge" and a "protest". You challenge the evaluation of a particular shot. You protest a) any injustice you feel has been done to you (except evaluation of a target): b) the conditions under which another shooter has been permitted to fire, or c) the equipment which another competitor has been permitted to use.

Some competitors feel that protesting is "causing trouble" and they "don't want to make waves." These same competitors will then complain "unofficially" about another competitor and everything "that person's allowed to get away with." Don't forget, if you're not part of the solution, you're part of the problem. If you, as a competitor, see valid rule infractions which are not corrected after notifying a match official, protest and get an official from a Referee or Jury.

5. Be punctual: Better than being punctual, get there early. An hour is sometimes not too long to get squadding, get out your gear, and ready to participate. If the program states the match begins at 8:00 am, you can be 99% sure that the first relay will be on the line at 8:00 am ready to shoot. Allow yourself plenty of time to get to the match, especially if the location of the range is unfamiliar to you. Arriving late and rushing about to get yourself and your equipment ready is almost guaranteed to ruin your shooting day, so give yourself plenty of time. If you arrive after the 3 minute preparation period, you might not be able to shoot at all.

6. Don't be afraid to go to your first match. Everyone has to start somewhere. Provided you follow the rules, other shooters are always happy and willing to answer questions and help you along.

7. Offer to help out. The vast majority of tournaments are conducted by just a handful of volunteers. Extra help is always welcome. Granted, you've paid your entry fee and are entitled to devote your attention to your shooting, but you can still offer to police the range after the match, put away equipment or any number of other tasks. Without the people who give up their shooting time in order to provide it for others, there wouldn't be a tournament. So do your part as a competitor.

8. Be sure clothing worn to a tournament reflects your concern for comfort and safety as well as recognition on your part that competitive shooting should be represented in as positive a manner as possible. Especially when TV or newspaper coverage will take place, attire should be in good taste in order to enhance the image of this sport being conveyed to the general public. Items containing controversial or offensive slogans or which, in any other way, could detract from the traditional sporting aspect of competition are unnecessary and strongly discouraged. In some cases, inappropriate clothing could be the basis for a match sponsor not allowing a competitor to participate.

9. Enjoy yourself. Sometimes it's hard to remember to do that, but try to keep in mind that while competitive shooting can be serious, demanding, and nerve-wracking, it is still great fun and is populated by the nicest people in the world - other shooters.

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Further Reading

- NRA Competitions Division (mailto:competitions@nra.org)

INTRODUCTORY INFORMATION

Is There a Secret to Shooting High Scores?

By Jake Shevlin

Source: http://bullseyepistol.com/secret.htm

This treatise is in answer to comments a number of shooters made to me not too long ago: "Is there a secret to shooting high scores?" "What is the secret to shooting high scores?" "I know you know the secret. Tell me the secret!"

In answering, We will emulate, in part, the thoughts in the lead editorial of the New York Sun, Sept. 21, 1897

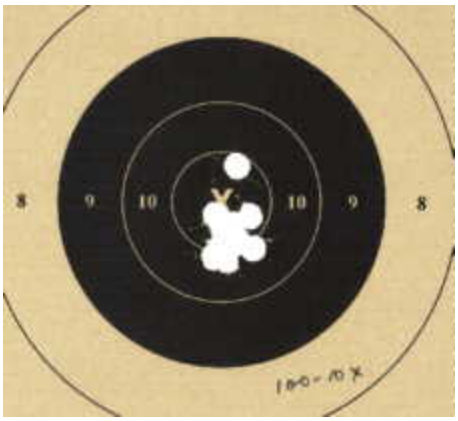
"Yes, Virginia, there is a Santa Claus."

Yes shooter, there IS a secret to shooting high pistol scores. It exists as certainly as love,



generosity, and devotion exist, and you know that they abound and give to your life its highest beauty and joy. It exists as certainly as persistence, hard work, sacrifice, and tenacity exist. You must know also, that when these qualities abound, they give to ones life its highest satisfaction of accomplishment. Alas! How dreary would be the world if there were no God given attributes such as these, and how dreary would be the world if there were no secret to shooting high pistol scores. It would be as dreary as if there were no Hagar the Horrible, no Superman, no James Bond, and no 2700s. And regarding the latter, we would have no "cause for competition."

The secret to shooting high scores is slowly revealed and compounded by the volume and density of the attributes of persistence, hard work, sacrifice, and tenacity that the shooter develops, in a combining overall attitude relative to the goal of high scores. The secret is elusive; it appears and disappears in direct relation to the volume and density of these attributes. The more time and effort the shooter devotes to the goal, the more the secret is revealed.



To have the secret revealed, the shooter must sacrifice. Sacrifice his own, sometimes misguided, conceptions about the vicissitudes of pistol shooting, and of what is required to demand a high score. Sacrifice his time for other endeavors, including family, other hobbies, and leisure vacations. Sacrifice his money for decent equipment. Sacrifice his convenience by traveling long distances, most of the time alone. Sacrifice his comfort by shooting, both match and practice, in the most severe types of weather, and in the most difficult of places. Sacrifice his priorities by shooting ALL the matches he can possibly attend, and shooting ALL the practice he can, in utter disregard of the mental and physical cost.
Regarding sacrifice: "Being a Master shooter is doing all the shooting he loves to do on the days he doesn't feel like shooting!" This sacrifice mandates hard work, with the accompanying persistence and tenacity to carry it out. It must necessarily occur without any encumbrances, mental or physical. The quest must be an unrelenting pursuit of the goal. The shooter has to be thinking of shooting most all of the time. The shooter's focus on shooting must thoroughly possess him. In the morning prior to the match or practice he must visualize the drills involved. In the evening just before going to sleep he has to fire a match in his minds eye, shot by shot! The shooter must also observe the actions and attitudes of High Master shooters, and converse with them so as to learn all he can. These possessive thoughts must include not only the shooting itself, but all the peripherals involved, such as equipment care, ammo supply, travel equipment, and travel planning, etc. In addition, the thoughts require analyzing new shooting techniques and changes in equipment. In other words, the shooter must constantly be concerned with the sum total of the entire drill involved in his endeavor to unlock the secret.



Regarding encumbrances: Hershel Anderson, USAMTU, multiple times National Matches Winner and the shooter holding THE highest 2700 Record of all time [2680-159x], had the least amount of physical encumbrances of any shooter we have ever known. His equipment on the line at the National Matches consisted of the match gun, two magazines, match ammo, and a gun box with spotting scope. The gun box was otherwise empty and unadorned! Simplicity at it's finest! His mental encumbrances were, in all probability minimal or nonexistent. He completed his Army career at the USAMTU as a National Matches Winner, and then returned the following year for another National Matches win as a civilian. Hershel's record score was fired July 24, 1974 in Nashville, TN.
In regards to equipment: Gil Hebard, a National Matches Winner ['50s/'60s], and principal merchant supplier of pistol shooting equipment in that era, remarked that a shooter may be able to "buy his way to the secret". Gil may have been slightly biased, but most knowledgeable shooters will find no real fault in the fact that decent equipment is vital to success.



To have the secret revealed the shooter must constantly demand of himself the possessive thoughts about shooting. The reason for this is to load the sub-conscience, so that many, or even all, parts of the entire shooting drill will allow a robotic reaction on the part of the shooter, guaranteeing the proficiency gain required to master the gun.
How about this scenario? Kim Dyer, USAMTU, Who fired a Women's National 2700 Record at Eustis, Fla. in the '70s [all of 26 years old and 98 lbs. at the time], when asked, "What do you do with an unscheduled afternoon?" Said, "Well, I grab two cans of hardball [ammo] and head to the range." Imagine! 500 or more rounds of hardball shot by a 98 pounder, in one afternoon? Tenacity at it's ultimate! The secret to shooting high pistol scores is self-

rewarding. It is part of the competitive spirit that drives the shooter, permeates the balance of his life, and offers him success in his other pursuits. Once you attain it you will always remember it, but it takes constant vigilance to keep it. The secret to shooting high pistol scores is real and attainable, but intangible, and not always readily explainable. But as sure as there is love, generosity, devotion, persistence, hard work, sacrifice, and tenacity in the human spirit, and as sure as there are leprechauns in Ireland, Santa Claus, and the Easter Bunny, there IS a secret to shooting high pistol scores! "None of us will ever accomplish anything excellent or commanding except when he listens to this whisper which is heard by him alone." Ralph Waldo Emerson, poet.

Jake Shevlin
2000

INTRODUCTORY INFORMATION

Pistol Etiquette

By Ron Porter, CSSA Pistol Executive

Source: https://www.bullseyepistol.com/etiquete.htm

This article was written with the help of Chuck Towne and Fred Crowle, and was published in the CSSA's newletter, Colorado Shooting several years ago.

In all sports there are written and unwritten rules. I would like to discuss some of the unwritten rules or points of etiquette which are applicable to conventional bullseye competition.

The first point of etiquette involves scoring. Scoring should be done efficiently, quickly, and accurately. To accomplish this, each competitor should score the other competitor's target first before looking at his/her own. The scoring is best accomplished by first counting the number of shots to verify the required number (10), writing each shot value on the score card, and then adding the values to get the total score. The totaling is most easily done by adding together the lost points (because these numbers will be the smallest) and then subtracting their total from the possible perfect score of 100. The scorer should write the score on the competitor's target. Only after doing that should the shooter feel free to review his own target and return to the firing line.

A shooter who records his score in a score book should keep the fired target with the score written on it until he has time to transfer the score to his score book. Alternatively, he can write his score on the stub of the score card and keep that for the later transfer to the score book. A shooter should try to avoid being the last one back to the firing line. And he should make all of his preparations (e.g., loading magazines) for the next stage of shooting before the range officer calls shooters to the line for that string. In the scoring, it is the scorer's responsibility to score the target as accurately as he can, and it is the shooter's option to accept the score or contest it with arbitration from the range officer, referee, or scoring jury. In arbitration of a contested score, the scorer is not involved. Each competitor has the responsibility of knowing the rules and exceptions as listed in the program, and of abiding by them. Particularly, he needs to know the rules about alibis, skidders, plugging shots, and challenges.

The second point of etiquette is about behavior of shooters after they have finished shooting. For example, in slow fire a shooter has ten minutes for ten shots. Most shooters finish their ten shots in from three to seven minutes. When they have finished, they should not begin socializing, because talking behind the line is both distracting and impolite. A shooter should have the same consideration for someone who is still shooting as he got: after all, no one was

distracting him with talking while he was shooting. This courtesy applies as well to range officers as to shooters.

The third point of etiquette is about picking up brass. Everyone picks up expended cartridges after shooting their large caliber pistols. But this should be done in a way which will not disturb shooters still in their string of shooting. It is most distracting to have someone reaching between your legs or bumping your feet while you are trying to concentrate on a shot. Incidentally, shooters should mark their brass. Then, a shooter can pick up all the brass in his area and sort it out (his, mine, hers) to minimize confusion and save time.

Now, some advice for shooting better. When there is time to talk, it natural to talk about the shooting you have just done. It is better if you talk about the good shots rather than the bad ones. Lanny Bassham, in his tapes about mental management (shooting is largely a sport of mental discipline), tells the story of shooting wild 8's; he kept a record of each shot. After a while, he said that he got real good at shooting these 8's because he was imprinting them on his subconscious mind. So, it's a good reason why you should talk about your good, not bad, shots.

Eventually, a shooter will realize that he is shooting more against himself than against anyone else in any match. Remember, the actual competition (time when shooters rank against one another) doesn't start until all the scores are in and have been totaled. A good shooter has learned that he performs better when he can shut out all distractions. And since this is easier to do when the distraction are the fewest, don't be hurt if a fellow shooter seems to be ignoring you. He is just trying to improve his concentration by shutting out you and everything that is irrelevant to his shooting.

INTRODUCTORY INFORMATION

The Ultimate Bullseye Primer

By Mike LaVoie

Source: https://www.bullseyepistol.com/mikelav1.htm

Precision Shooting

1. Identify your target
2. Watch the front sight (Red Dot)
3. Press the trigger smoothly straight to the rear

Do #2 and #3 at the same time *[Frank Higginson]*
Sounds simple; doesn't it?

1. Identify your target. Look at your port number and color. Then look down range before you lift your gun and identify the same number and color. It is real easy to cross fire onto someone else's target. You lose 10 points per shot. It is very embarrassing to cross fire. (Believe me I know.) If someone cross fires on your target, be a good sport and ask to refire the target. You have that right. (If you shoot better on the alibi you can only get the highest points on the first target.) The competitor should have a range officer check the target if the competitor disagrees, but don't let him/ her change your mind. Call them as you see them. Do not be intimidated!
   Cross firing- there are 2 kinds of people: those that have and those that will.
2. Watch your front sight. Stare at your front sight or red dot, 100% focus. See the scratches on the front sight. See the imperfections on the red dot.
3. Press the trigger smoothly straight to the rear. Apply even pressure to keep the trigger moving to the rear. In rapid fire you have 2 seconds to move the trigger to the rear smoothly and fire the shot. You have a minute a shot in slow fire. Use the time to focus on the next shot. Don't stop moving the trigger. That is called "playing with the trigger" or "chicken finger". That will get you into trouble.

Scores

Scores are the results of groups. Everything we do on paper targets is the result of groups. Groups show your wobble area. When scoring somebody else's target, score accurately. Use an overlay and don't give points away if the shooter did not shoot them. Let the competitor whine and cry and make such a fuss. I never lower my standards. It's not fair to all the other shooters. Why have a match if everyone scores a 100 on each target? I don't want anyone to score my target higher than what I shot. It isn't sporting or fair to the others. Once your are labeled a cheater, you lose the respect of everyone. If you disagree with the score the person gave you, go first to the person who scored the target and then to a range officer if you still have a problem. Never go to the referee first. Then if your still not satisfied that your target was scored correctly go to the referee. You may have to give him/her $2 to challenge the score, but if you win the challenge you get your $2.00 back.

Wobble Area

Wobble area is the area your barrel is moving around when it is extended out. Pretend to have a zillion power laser on the barrel of your pistol. The wobble area would be the part that is all burnt on the paper target. Put a blank paper behind your target. Never change the blank paper. At the end of the match or training look at the blank paper and you can see your wobble area. Your wobble area is the same size as the groups you have just shot. Wobble area increases or decreases with training, eating, stress, drugs and smoking. You have to accept your wobble area. It is not going to get any better at the given time you are about to shoot. Your wobble area may increase or decrease as the match goes on. I find as my muscles get warmed up and stretched out during the match my wobble area decreases. I like to fire 10 rounds or so of air pistol in my basement before I travel to a match. I set the trigger to the same weight as the firearm I am about to shoot.

Gripping the Gun

To Correctly position the gun in your shooting hand:

1. Hold the gun in your non-shooting hand by the barrel. Don't put your hand in front of the barrel.
2. Observe on your shooting hand the space between your thumb knuckle and pointing finger knuckle- this is the web of your hand.

3. Take your shooting hand and place the web area firmly into the top of the grip, snug against the frame or beaver tail on some 45's.
4. Take your bottom 3 fingers and wrap them around the grip.
5. Take your shooting finger and lay it on the side of the frame.
6. Force your thumb around to the other side of the gun.
7. Then relax your grip when the guns on the bench.
8. Hold the gun like a handshake when your ready to shoot.

Rules

It is the competitors responsibility to know the rules. Here are some rules that are good to know. Whatever classification you have attained, you can shoot that class in ANY Indoor/ Outdoor NRA competition. For example: You have made marksman in indoor pistol shooting therefore you can shoot any NRA match; pistol, small bore, high power etc. in marksman class. Look at each target before you hang them. Look for holes, the proper target for that event; slow, timed, rapid, the correct distance target. Make sure your name or competitor number is on each target, when required. Score your targets as you pull them to verify posted scores. Outdoors make sure the score on the scoresheet is correct. Check addition. Verify your scores in the time allotted to make sure they are correct. Believe me when I tell you how frustrating it is to shoot better than you ever have and find the posted scores are wrong and the time limit is up to challenge it.

Muscle Memory

One can never practice enough. Muscle memory is only accomplished by live firing, dry firing or air pistol practice. Limit your dry firing and air pistol practice to 30 minutes to avoid burnout and boredom. Lifting a weight every day and holding the weight in your shooting position for 90 seconds builds your shooting muscles. You should do the same exercise with your weak hand to balance your muscles. Exercise is very important. Aerobic exercise can be done on the day of a match. Body building, free weights and nautilus type machines should be avoided up to 3 days before a match. Push-ups can be done early in the day, if you have a night match. Advanced athletes should do up to 25 push-ups every day, 5 days a week. Aerobics are great to calm yourself and keep in shape. Bringing your heart rate up twenty minutes a day, three days a week is minimum to keep the heart in good shape. Stair steppers,

walking, running, step aerobics, ski machines, bicycles, and stationary bicycle machines are a few ways to do aerobics. If you build up your health, you will be less stressed under match conditions and you won't fatigue.

Everything Effects Your Shooting

Eat light at least two hours before a match. No Caffeine, sugar, chocolate, fats or smoking. It is hard not to eat or drink the stuff that tastes so good, but think of the good taste of shooting a tight group. 2700's start early and run until mid afternoon. You should eat a light lunch. I eat a banana and a bagel. Also bring plenty of water for summer matches. Stretching your muscles before and during a match helps relax your muscles and mind. Do Not stop your normal routine the day of a match. For example, if you have 3 cups of coffee in the morning continue to have them. You could experience withdrawal symptoms if you decide not to follow your regular routine.

Shooting To Win

One theory is to "see" yourself shooting in "real time". Picture yourself walking up to the line, setting up your gun box, preparing your equipment and mind to shoot. Identify your target. See yourself shooting each perfect shot in slow, timed and rapid fire. Visualize yourself winning the match. Now you just have to go through the routine. Identify your target. Pickup up the gun by the barrel with your non-shooting hand and correctly position it in your shooting hand. Raise the gun slightly above the target with your finger off the trigger. Slowly lower your arm to the black of the target. "See" the front sight (red dot). " See" your wobble area. Accept your wobble area. Don't fight it. If the wobble area has sharp corners or choppy movements, you are fighting your wobble area. Your wobble area should be rounded. (This is easier to notice with a red dot sight.) Feel the gun relaxed in your hand (like a handshake). Stop breathing. Start pressing on the trigger without disturbing the sights. Keep the trigger moving. When the shot breaks you should be surprised. Then follow through. This means watch the movement of the sights or red dot as it moves from the recoil. Call your shot. Good scores aren't hard. Good scores flow free. *[Storrs Dutko]*

Match Jitters

Yes, we all know match jitters. The butterflies in the stomach. The feeling you're going to pee your pants. How do you deal with it? Well everyone gets match jitters. It's normal. It took me 3 years not be nervous during the local Wednesday night matches. I still get them at Camp Perry during the Team matches. Shooting with top shooters like Charlie Gippert and the others on the top team really can put the pressure on you, if you let it. How do you deal with it? Joking around really helps! You have to compete in every match you can. You have to tell yourself, "been there, done that ". You have to tell yourself, "I have done this so many times before, I know how to do this. Open my mind. Set my mind free." Do stretching exercises before and during the match. Breathe slow at least 4 breaths before each shot in slow fire. Breathe in through your nose and slowly blow out through your mouth making a blowing sound you can hear. It's relaxing. Do the best you can. You can't ask for anything more. Good scores are a product of relaxation of muscles and mind! The next Shot is all that matters. The next shot is all that matters. The last shot is history. It doesn't matter anymore, done, finished, gone...The next shot is all that matters!!! Open your mind. Set your mind free.

Comfort Zone

We all know our average. When we start to shoot higher than it, we leave our comfort zone. Our pulse gets heavier, our wobble area gets big. Wammo, the shot gets thrown. Ahhhh, we feel better. We are back to our comfort zone. Ah, Bullshit to the comfort zone. You have practiced long and hard to hit those 10's. You expect to hit them. You can do them all day long. That's what I expect. That's what I trained for. That's what I will do.

Mounting A Scope On A .45

The scope should be mounted as low as possible. Mounting it on the slide is the way to go. Frame mounted scopes will not be accurate when the slide, lug and rails start to wear. The top shooters have them on the slide. Working out, eating right, and dryfiring, a half hour a day, and of course if you can live fire, will get your scores up. When you shoot your 45, say to yourself during timed and rapid, "keep the trigger moving".

Dropping The Slide On A .45

Don't drop the slide on a precision .45 without a magazine and round in it. Let the slide down slowly. If you don't you will elongate the lugholes, which will damage the barrel to frame rail or fracture the lug, which can cause more damage, depending on when??? it fractures.

Tips

Is your dot not as bright as you want it? How old are those batteries? Keep them fresh for outdoor matches where the sun is very bright. Do you put your finger fully into the trigger housing wrapped around the trigger? Use the tip of your finger to press the trigger to the rear. The trigger pull feels lighter as you cam the trigger. You are using leverage when you do this. Muscle memory is a big part of shooting. Your mind and muscles remember what to do at a subconscious level. You think the shots into the center. Follow through is very important. Watching the front site with 100 % focus is the most important, with the proper trigger movement. (Smooth, 8 seconds or less for slow fire, 4 seconds or less for each timed fire shot, 2 seconds for each rapid fire shot.) You must be able to call your shots, meaning where the shot should be on the target after you shoot it, even on rapid fire. I should be able to walk up to you and ask you, "Where did the third shot hit on the first string?" and you should be able to answer me. After slow fire, use a timer to look at seconds to get your mind set for 2 seconds. Tap your finger or foot to the 2 seconds or press against something like your pressing the trigger. You will accidentally fire an early shot. It is part of the learning process. You may see the "hotshots" break an early shot by accident. You have to focus on shooting when the guy next to you shoots early. It's hard to do. Do not use shock buffers in the 1911. It's rubber or plastic and when it falls apart it jams the gun.

How To Analyze Your Performance

1. Did I shoot on the correct target?
2. Did I "see" my front sight?
3. How "focused" was I during the shot?
4. Did I "call" the shot?
5. Did I keep the trigger moving or did I "play" with the trigger?

6. How big was my wobble area?
7. Timed and rapid: how was my timing?
8. Did I keep the trigger moving smoothly?
9. Did I let anything distract me, such as shots in the white, someone else's early shot etc?
10. Did my equipment work okay? Did I have all the accessories I needed?
11. Was I a good sport whether I won or lost?
12. Where were my groups and how big were they?
13. Did I get the first shot off quickly during rapid fire? Meaning, did I have enough trigger pressure applied when the targets turned or when I heard the horn or buzzer.
14. Did I have the right clothing to keep warm or dry?
15. How did my ammo perform?

Equipment

You need a good Bullseye gun. A rimfire firearm is good to start with because you need a rimfire firearm for most matches. The ammo can be inexpensive and you don't have to reload. Buying a used target gun is usually a good idea. Buying equipment is a vicious cycle until you learn from your mistakes or learn from someone that's been there before. This is what usually happens: 1. New shooter buys a $250+ firearm. . Next they need a trigger job $75+ and firearm is at the gunsmith for three months. Grips $50, magazines, $20 each times 4 = $80. So far the total is easily reaching $455. 2. New shooter now looks around and sees the other shooters shooting the "Plasma smorf fantastic racing shooting machines," and the new shooter wants one. 3. New shooter puts their perfectly good firearm on the market so the they can upgrade to the perceived better firearm. 4. Educated new shooter finds "built" firearms for great prices. You can only sell your firearm for blue book or going rates in the area. You might have sunk $200 dollars into a $300 gun. Don't think your going to sell that firearm for $500 dollars. 5. New Shooter finds the new "Plasma smorf fantastic racing shooting machine" shoots as well as their last firearm. Another option is to save and shell out the $475 or less and buy a used Smith and Wesson 41, or a High Standard Victor, Supermatic, or other High Standard 107 Frame built in the Hamden, Connecticut plant or a Ruger Bull barrel firearm that's been already "built". Note: Rugers are hard to disassemble. If you can afford the fancy smancy guns and are serious about Bullseye Shooting then the Pardini, Hammerly 208S and Walther are the usual choices but once you own one you can't use the excuse "must be something wrong with the gun". One mistake that many shooters make is that they need a

long barrel to make the firearm more accurate. The bullet accelerates fastest in the first inch and a half. The longer the bullet stays in the barrel the more likely you can move the firearm away from the point you "broke" the shot. Don't forget the gases leaving the barrel still effect the shot passing around and behind the bullet. This can cause tumbling bullets. Follow through is important. You will need a gun box so you can keep all shooting equipment in one place. Supplies you will need are: Screwdriver Allen wrenches Cleaning rod Shooting diary Pen Extra batteries for red dots scopes Scoring overlays and plugs Rule book Shooting glasses (indoor-yellow tint and outdoor-darker tint) Earmuffs (I use earplugs and earmuffs, called double plugging) Magazines One inch surgical tape Garbage bag (when it rains outside you place the bag over your box) Wooden or brass rod just in case the bullet takes a nap in your barrel (push the bullet out the way it came in) Spare parts I take it back, you don't need a gun box, you need a pick up truck to carry all this stuff around. Did I remember AMMO?

Ammo

Indoors is very forgiving for accuracy because your only shooting 50 or 75 feet. Outdoors at 50 yards you find out which ammo shoots tight. Try different brands.

.22 Caliber

General rule is don't use Remington or Winchester Rimfire ammo. They aren't reliable to go bang each time. I have seen many misfires due to these brand 22's. I use R.W.S. Target ammo from Grice Gun Shop. I pay about $250 per case of 5000 rounds. I like to buy a case of ammo (5,000 rounds) because all the bricks (500 rounds) are from the same lot. Same lot means all the rounds are made the same. This matters at 50 yards.

.38 Caliber

SLOW FIRE 50 YARDS: Bullet head- 148 grain H.B.W.C. (Hollow Base Wad Cutter) from Zero Powder- 3.1 grains W.S.T. (Winchester Super Target) Primers- CCI primers Brass- Winchester NOTE: Wind affects the trajectory of the 38 more than the 45 at 50 yards INDOORS AND 25 YARD TIMED AND RAPID (SHORT LINE) LOAD: Bullet head- Zero in mixed cases Powder- 2.5 grain W.S.T. (Winchester Super Target) Primers- CCI Primers Brass- mixed

.45 Caliber

SLOW FIRE 50 YARDS: Factory Federal Match 185 GRAIN- Semi Wad Cutter INDOORS AND 25 YARD TIMED AND RAPID (SHORT LINE) LOAD: Bullet head -200 lead S.W.C. (Semi-Wad Cutter) from Accucast Powder- 4.1 grains Powder W.S.T. (Winchester Super Target) made by Winchester Primers - CCI primers Brass – mixed for 50 ft, Federal for big matches and 25 yards Note: Never vacuum primers and powder as the primers may explode and start a fire. (Not to mention that they are loud when they explode.)

Supplies

Shooting Supplies & Equipment

Champion's Choice, Inc. 201 International Blvd. LaVergne, TN 37086 615-793-4066 www.champchoice@nashville.com

.22 Ammo

Grice Gun Shop P.O. Box 1028 Clearfield, PA. 16830 814-765-9273

Best Magazine for Competitive Bullseye Shooting

SHOOTING SPORTS USA NRA 11250 Waples Mill RD Fairfax, VA 22030 703-267-1000 *Annual subscriptions for NRA members are $ 5.00 for classified shooters and $ 10.00 for non-classified shooters

Ultradot Sights & Hammerli's and Gunsmithing

Larry's Guns 49 Hawthorne St Portland, ME 04103 207-772-0998 Fax: 207-772-0628 www.larrysguns.com Note: Larry Carter holds many local, state, regional and national pistol championship records. I Buy Most of my Equipment, Firearms and minor Gunsmithing here-

Talk to Moe Mo's Competitior Supplies & Range M.C.S. 34 Delmar Drive Brookfield, CT 06804 203-775-1013

Best Gunsmithing for Major Jobs

George Madore 405 Camelot Dr. Brookhaven, PA. 19015 610-872-3854 Best 45 for You to Buy to Send to George Madore to Accurize (Blued, bottom of the line model) Springfield Armory 420 West Main Geneseo, IL 61254 309-944-5631 Fax: 309-944-3676 www.springfield-armory.com

Best grips

Bowlers Olympic grips 27a High St Little Bytham Grantham Lincs NG334QJ 047684493 (telephone)

Best Grips- Fastest to Acquire

Morini Grips sold by: International Shooting Supply ISS P.O.Box 185234 Fort Worth TX 76181 817-595-2090 877-595-2090 Toll free

Best Loading Press

DILLON 550B Dillon Precision Products, Inc. 8009 E. Dillons Way Scottsdale, AZ 85260 800-762-3845 480-948-8009 480-998-2786 Fax www.dillonprecision.com

INTRODUCTORY INFORMATION

About Mike LaVoie

By Some Accomplishments of Mike LaVoie

Source: http://bullseyepistol.com/mikelav2.htm

- 1999 CT Sectional Open Indoor Pistol Champion
- 1999 CT Sectional Indoor Resident Champion
- 1998 Bronze-Military Leg Match Camp Perry
- 1997 Presidents 100
- 1997 Distinguished Pistol Badge
- Conventional Pistol Team National Champion
- 1997- 2 National Record Champion Civilian and Open team
- 1996 Resident CT State Indoor Champion
- 1995 Resident CT State Indoor Champion
- 1994 Resident CT Outdoor State Champion
- 1994 Bronze Expert Class Camp Perry National Matches
- NRA-USA Civilian Pistol Team 1993
- CT State Gold Pistol Team 1992-Present
- West Haven Police Team & Co 1989-Present (Undefeated)
- Match Winner of the Bridgeport Rifle Club Pistol Matches Since 1992
- Board Member Bridgeport Rifle Club
- Member Ansonia Rod & Gun Club
- Member of Stratford Gun Collectors
- NRA Certified Pistol Instructor
- Instructor for Advanced Precision Pistol
- Life Member of the NRA
- 2622 3 gun score
- 887 One gun

INTRODUCTORY INFORMATION

Outdoor Bullseye Pistol Shooting

By Frank Savino

Source: http://bullseyepistol.com/savino2.htm

This is an article I wrote for the Connecticut Marksman many years ago. Seeing a new group of shooters coming up in the ranks, I decided to dust the article off and update it in May of 2014 – Frank Savino

Here in Connecticut we have a large number of pistol shooters who compete in the various indoor pistol leagues around the state during the winter. When spring comes around, most of these shooters put their guns away and don't touch them until the following September. Those shooters miss out on the outdoor competitive pistol shooting season. I feel a lot of these shooters do not shoot outdoor matches because they are unsure of their ability, or don't know what is involved in shooting this type of match. There is an unnecessary intimidation factor which doesn't need to exist. In this article, I am going to explain what is needed to compete and how to compete in a 2700 outdoor pistol match. Most of this information also applies to other pistol leagues and matches, whether they are fired indoors or outdoors. Outdoor 2700 matches are fired with three guns and are sometimes referred to as three gun matches. The three guns are .22 caliber, centerfire, and .45 caliber pistols. Competitors only need two guns to compete as I will explain later on. There are some shooters who use revolvers for bullseye pistol shooting but the sport is dominated by semiautomatic pistols. The first gun is the .22 caliber pistol or revolver. These consist of Rugers, Smith & Wesson model 41's, High Standards, Hammerlis, and Pardinis. The Feinwerkbau AW93 is beginning to gain in popularity. Some competitors use 1911 pistols with .22 conversion units like the Marvel. The .22 Ruger target pistol with a trigger job and dot sight is an excellent entry level gun. The referee may randomly weigh triggers on competitor's pistols at some of the bigger matches such as Regional Championships, State Championships or the National Championships. The trigger pull on the .22 must be at least two pounds. Next, the Center Fire Pistol is any center fire pistol or revolver .32 caliber or larger. Guns that fall into this category are Hammerli or Pardini .32 cal pistols; these are .32 S&W long caliber. The Smith and Wesson Model 52 in .38 special was popular at one time. Some shooter use custom 1911pistols in .38 special, 38 super, or 9mm. Most shooters use a .45 caliber pistol for their centerfire matches. Using the .45 for centerfire eliminates the need for competitors to have three guns and three different types of ammunition. The trigger pull on centerfire pistols and revolvers must be at least two and a half pounds, however if you are using a .45 caliber semiautomatic pistol for the centerfire portion of the match, the trigger pull must be at least three and a half pounds. The third gun is the .45 caliber pistol or revolver. Most competitors use a 1911 style pistol for the .45 matches. You will need one with a good trigger and adjustable sights or dot sight to get

started. However to be competitive in the upper classes such as Master and High Master you will need a pistol that has been accurized by a pistolsmith. Shooters in these classes have pistols, which will shoot groups under two inches out of a Ransom Rest at fifty yards. The trigger pull on the .45 caliber semiautomatic pistol must be at least three and a half pounds. The trigger pull on a .45 caliber revolver must be at least two and a half pounds. For accuracy work on your target pistols, use a pistolsmith who specializes in pistols for bullseye shooting. Do not go to your local gunsmith, you need a pistolsmith who knows the game of bullseye pistol shooting. A friend of mine brought his Gold Cup to his local gunsmith to be accurized and invested a large amount of money into the gun. As he got to be a better shooter, he realized his gun was not shooting as well as it should. He sent his pistol to a pistolsmith who specialized in bullseye pistols for evaluation and found out the accuracy work was not done correctly. He then had to pay this pistolsmith to redo the work of the first gunsmith. For sights, most shooters use electronic red dot sights such as those available from Ultradot and Aimpoint. If you use iron sights, they will need to be adjustable. It is perfectly acceptable to shoot iron sights, but most shooters use dot sights. I did recently see a shooter break 2600 with iron sights, Walking down the firing line looking at grips you will see everything from custom grips to straight military type grips. I have seen shooters win matches with all types of grips, so use whatever is comfortable. One word of caution, it does occasionally rain during a match and rubber grips get slippery when they are wet, so I recommend staying away from rubber grips. As far as other equipment goes you will need a gun box with a spotting scope. Eye and ear protection are necessary items. Don't forget a screwdriver to adjust your sights and any other small tools you may need for your guns. Bring your cleaning equipment. You will need a light-duty staple gun to staple repair centers to your target. Other helpful items are brass catchers and screens to deflect your neighbor’s brass. You will also need a baseball cap to keep the sun out of your eyes, rain gear, insect repellent, sunscreen and a folding chair. An “Empty Chamber Indicator” is another necessary item, these can be easily made from a piece of orange string from a lawn string trimmer. Some shooters use a small cart or handtruck to get there equipment to the firing line. This is especially handy during the National Championships at Camp Perry where you may need to carry your equipment a long distance. For ammo any good quality standard velocity or match grade .22 caliber ammunition should do. When picking out .22 ammunition you need to buy ammo that works well in your gun. Just because one brand of ammo works well or is super accurate in your friends pistol does not mean it will work in your gun. Buy small amounts of different brands of ammo and test it to see what works best in your gun. Factory loaded .45 caliber match ammo is very expensive. Military and some civilian shooters use factory match ammo made by Federal or Atlanta Arms. The only way most civilian shooters can to afford to shoot the center fire and .45 match is to reload. The most popular loads for the .45 use 185 grain or 200 grain semi wadcutter bullets, or 185 grain Nosler type jacketed hollow point bullets. Consult a reloading manual for proper loads for your firearm. Now that you have your equipment ready, it’s time to discuss

the match. Most outdoor matches are what is called a 2700. Competitors fire 270 rounds, 90 with each gun. Each 90 shot match or 900 as they are called are the same course of fire except they are fired with the different caliber guns. Competitors fire the .22 caliber 900 first, then the center fire 900, and finally the .45 caliber 900. The 900 point matches are made up of four fired matches. First is the Slow Fire Match, which is twenty rounds slow fire. The second match is the National Match Course which is ten rounds slow fire, ten rounds timed fire and ten rounds rapid fire. The third match, the Timed fire match is twenty rounds timed fire. The fourth match is the Rapid Fire Match which is twenty rounds rapid fire. All shooting is done ten rounds per target. After each ten round target, shooters score and repair targets. All slow fire is fired at fifty yards. Timed and rapid fire is fired at twenty-five yards. For slow fire, competitors fire two ten rounds targets with a ten-minute time limit for each target. In timed fire and rapid fire, shooters fire two strings of five rounds per target. Each five round string in timed fire is has a twenty second time limit. The five round strings in rapid fire have a ten second time limit. One aspect of match shooting all competitors should be familiar with is re-fires. A competitor who has a firearm malfunction or ammunition failure in timed or rapid fire may re-fire that string. Shooters are allowed one re-fire in each fired match. In other words, one re-fire in the National Match Course, one re-fire in the timed fire match, and one re-fire in the rapid fire match. If a competitor has another malfunction on a subsequent sting in the same match, he should try to clear the problem and continue to fire if he can do it safely and time allows. A competitor does not have to re-fire a string if he chooses not to, he can accept the score for the number of shots he fired up until the malfunction. It is important for a competitor who has a malfunction not to make an attempt to clear the malfunction. Do not touch the gun with the non-shooting hand. Any attempt to clear the malfunction will result in the re-fire being denied. When a malfunction or ammunition failure occurs in timed or rapid fire the competitor should keep the firearm in his shooting hand and keep it pointed down range. At the end of the string, the range officer will ask if there are any re-fires. The shooter will then raise his non shooting hand. A line officer will verify the problem and determine the number of rounds fired. After all shooters have fired both strings for that target, the range officer will run a re-fire string for the shooters needing to re-fire. The shooter will load five rounds on command and re-fire the complete five round string. In most cases, the shooter will have fired more than ten rounds at the target after firing a re-fire string. The person scoring the target needs to find out how many shots the competitor should have fired and score the ten shots of lowest value. Competitors are required to score each other's targets. The range officer will generally tell competitors to pass their scorecards to the shooter on their right and score the shooter's target on their left. Necessary items for scoring include a pen and small clip board. Shooters should bring a baggie large enough to fit the clipboard and their hand into. If it starts raining, the clipboard with the scorecards is kept in the baggie to keep dry and the scorer can reach in the baggie to write on the scorecard without taking it out and getting it wet. Another good investment is a Fisher space pen, these will write on wet surfaces like

scorecards if they get wet, or wet targets. Another useful item is an overlay for scoring. The scorecard is set up with ten boxes for each target. Scorers should start with the box on the left and place the shot of highest value in that box then work to the right placing the a value for each shot in the box in descending order. The letter M is put in boxes to signify a miss rather than a numeric zero. The shots are then totaled and the score for the target is placed in the proper box. The scorer also needs to write the score on the repair center of the target he scored. At the end of each match, the scorer adds up the total score for the match and signs the score card. The shooter then signs the scorecard, accepting the score. Finally, the shooter turns the scorecard in to the line officer. There is usually a fifteen-minute break between the .22 Match and the Center Fire Match. Shooters get a lunch break between the Center Fire Match and the .45 Match which is usually about forty-five minutes. The 2700 described in this article is the most common type of outdoor match. Some matches may vary from this. Obtain a match program from the match sponsor, as this will spell out the course of fire and match conditions for the tournament. Competitors compete in one of five classes: Marksman, Sharpshooter, Expert, Master and High Master. Shooters are placed in classes according to their averages so they are shooting against other shooters of the same skill level. The first time a competitor shoots a match, he will compete in the Master class unless the match has a special Tyro class. New competitors will put their scores from their matches in a temporary classification book and calculate their average and temporary classification. They do this until the competitor fires 360 rounds in competition and the NRA sends them a classification card. A competitor who holds an NRA Indoor Pistol Classification may use that classification for his first outdoor match. There are also other matches that are fired in conjunction with Regional Championships, State Championships, and the National Matches. These are Distinguished Revolver matches, and CMP leg matches. The scores for these matches are not part of the aggregate score from the 2700. There is an additional fee for shooting these matches; competitors in the 2700 do not need to fire these additional matches if they choose not to. However, don't be surprised if you show up to a match and someone offers to loan you a service pistol or revolver and ammunition to shoot the leg matches. The Distinguished Revolver Match is fired with service type revolvers. Competitors who place in the top 10% of the non-distinguished shooters earn ten, eight, or six leg points. A shooter who earns thirty leg points is awarded the Distinguished Revolver badge by the NRA. Firearms used in this match are unmodified revolvers in .38 special with iron sights (see NRA rule 3.1.4). Ammunition is limited to .38 special with 158 grain lead bullets. The course of fire for this match is a thirty round national match course. Civilian Marksmanship Program (CMP) Service Pistol leg matches are also held in conjunction with Regional Championships, State Championships, and the National Championships. Competitors fire .45 caliber 1911 or 9mm M9 service pistols with iron sights in this match. Allowable alterations to these pistols are limited and listed in the CMP rule book. Ammunition is restricted to round nose, full metal jacket, ball type ammo. Competitors placing in the top 10% of the non-distinguished shooters

competing in the match earn ten, eight, or six leg points. Shooters need to accumulate thirty leg points to be awarded the Distinguished Pistol badge by the CMP. The thirty points accumulated by the shooter must include one eight or ten point leg. Non-distinguished shooters are only eligible to fire four leg matches per year. The course of fire for this match is a thirty shot National Match Course. Competitors may enter partial matches if space is available at the range. Match sponsors give first preference to shooters who enter the full three-gun 2700, but will allow competitors to enter and only shoot one or two of the 900's if space is available. A new shooter who only has a .22 can enter and only shoot the .22 caliber 900 if space is available. Most shooters are willing to help new shooters. If you have questions, contact the match sponsor. If you have questions at a match, ask the line officer or range officer. Tell the match sponsor that you are a new shooter when you are entering a match. Usually they will keep an eye on new shooters to answer any questions and ensure you enjoy the event. Now you are ready to go out and shoot a match. A list of NRA sanctioned matches can be found at http://www.nrapublications.org/wp-content/ComingEvents/CE_Download.pdf. Another good source of upcoming matches in the northeast is the New Jersey Pistol website http://www.njpistol.com/.

Helpful Resources: Shooting Sports USA – a free digital magazine put out by the NRA with competitive shooting news and articles http://www.nrapublications.org/index.php/shooting-sports-usa/

NRA Pistol Rule Book - http://compete.nra.org/documents/pdf/compete/RuleBooks/Pistol/pistol-book.pdf

Civilian Marksmanship Program - http://www.odcmp.com/Competitions.htm

The Encyclopedia of Bullseye Pistol - http://www.bullseyepistol.com/

Bullseye Pistol Gear - http://www.bullseyegear.com/bullseyepistol/home.php

Champion's Choice - http://www.champchoice.com/store/main.aspx

Champion Shooter Supply - http://www.championshooters.com/

Larry's Guns - http://www.larrysguns.com/

Connecticut State Rifle and Revolver Association - http://www.csrra.com/index.html

Bullseye L Internet Forum - http://www.bullseyeforum.net/

Further Reading

- http://www.nrapublications.org/wp-content/ComingEvents/CE_Download.pdf (http://www.nrapublications.org/wp-content/ComingEvents/CE_Download.pdf)
- http://www.njpistol.com/ (http://www.njpistol.com/)
- http://www.nrapublications.org/index.php/shooting-sports-usa/ (http://www.nrapublications.org/index.php/shooting-sports-usa/)
- http://compete.nra.org/documents/pdf/compete/RuleBooks/Pistol/pistol-book.pdf (http://compete.nra.org/documents/pdf/compete/RuleBooks/Pistol/pistol-book.pdf)

- http://www.odcmp.com/Competitions.htm (http://www.odcmp.com/Competitions.htm)
- http://www.bullseyegear.com/bullseyepistol/home.php (http://www.bullseyegear.com/bullseyepistol/home.php)
- http://www.champchoice.com/store/main.aspx (http://www.champchoice.com/store/main.aspx)
- http://www.championshooters.com/ (http://www.championshooters.com/)
- http://www.larrysguns.com/ (http://www.larrysguns.com/)
- http://www.csrra.com/index.html (http://www.csrra.com/index.html)
- http://www.bullseyeforum.net/ (http://www.bullseyeforum.net/)

INTRODUCTORY INFORMATION

Outdoor Bullseye Pistol Shooting (Earlier Version)

By Frank Savino

Source: http://bullseyepistol.com/savino.htm

Here in Connecticut we have a large number of pistol shooters who compete in the various indoor pistol leagues around the state during the winter. When spring comes around most of them put their guns and equipment away and don't touch them until the following September. Those shooters miss the best shooting of the year: Outdoor Bullseye Pistol Shooting. I feel a lot of these shooters do not shoot outdoor matches because they are not sure how to shoot them. I am going to explain in this article what you need and how to compete in an outdoor match.

Outdoor matches are fired with three guns. The first is the .22 caliber pistol or revolver. I have seen shooters do well with Rugers and Browning Buckmarks. Smith & Wesson model 41's and High Standards are the most popular. Some of the top shooters are using firearms made by Hammerli, Walther and an occasional Pardini. The trigger pull on the .22 must be at least two pounds.

Next the Center Fire Pistol is any center fire pistol or revolver .32 caliber or larger. Guns that fall into this category are .32 caliber pistols such as the Walther or the Smith and Wesson Model 52 in .38 special. Clark Custom Guns also makes a 1911 style pistol in .38 special. Most shooters use a .45 caliber pistol for their center fire matches. The trigger pull on center fire guns must be at least two and a half pounds, however if you are using a .45 caliber semiautomatic pistol the trigger pull must be at least three and a half pounds.

The third gun is the .45 caliber pistol or revolver. Most competitors use a 1911 style pistol for the .45 matches. You will need one with a good trigger and adjustable sights to get started. However to be competitive in the upper classes such as Master and High Master you will need one that has been accurized by a pistolsmith. Shooters in these classes have pistols, which will shoot one to two inch groups out of a Ransom Rest at fifty yards. The trigger pull on the .45 caliber semiautomatic pistol must be at least three and a half pounds. The trigger pull on a .45 caliber revolver must be at least two and a half pounds.

There are some shooters who use revolvers for bullseye pistol shooting but the sport is dominated by semiautomatic pistols. There are a few people who have done well with revolvers but if you are going to purchase a firearm for bullseye pistol invest in a semiautomatic pistol.

For accuracy work on your target pistols use a pistolsmith who specializes in pistols for bullseye shooting. He will know what you need as well as how to do it. Do not go to your local

gunsmith. They may be good with your hunting or other sporting firearms but good target guns require a top pistolsmith who knows the game of bullseye pistol shooting. A friend of mine brought his Gold Cup to his local gunsmith to be accurized and invested a large amount of money into the gun. As he got to be a better shooter he realized his gun was not shooting as well as it should. He sent his pistol to one of the top bullseye pistolsmiths in the country for evaluation only to find out the accuracy work was not done correctly. He then had to pay the pistolsmith to redo the work of the first gunsmith.

For sights most shooters use electronic red dot sights such as those available from Aimpoint, Tasco and Ultradot. If you use iron sights they will need to be adjustable. Some good scores are still fired with iron sights. I have seen several matches won in the last year or two by shooters using iron sights. Walking down the firing line looking at grips you will see everything from custom grips made from a drawing of the shooters hand to straight military type grips. I have seen shooters win matches with all types of grips so use whatever is comfortable. One word of caution, it does occasionally rain during a match and rubber grips get slippery when they are wet.

As far as other equipment goes you will need a gun box or case to transport your firearms. A spotting scope is helpful. Eye and ear protection is necessary equipment. Don't forget a screwdriver to adjust your sights and any other small tools you may need for your gun. Bring your cleaning equipment. You will need a light duty staple gun. And don't forget your rain gear.

For ammo any good quality standard velocity or match grade .22 caliber ammo should do. When picking out .22 ammunition you need to buy ammo that works well in your gun. Just because one brand of ammo works well or is super accurate out of your friends pistol does not mean it will work in your gun. Buy small amounts of different brands of ammo and test it to see what works best in your gun.

Factory loaded .45 caliber match ammo will cost about twenty dollars for a box of fifty rounds. The only way for most of us to afford to shoot the center fire and .45 match is to reload. The most popular loads for the .45 use 185 grain and 200 grain semi wad cutter bullets. Consult a reloading manual for proper loads for your firearm. Now that you have your equipment ready we will discuss the match. Most outdoor matches are what is called a 2700. Competitors fire 270 rounds, 90 with each gun. Each 90 shot match or 900 as they are called are of the same course of fire except they are fired with different guns. First competitors fire the .22 caliber 900, then the center fire 900, and lastly the .45 caliber 900. The 900 point matches are made up of four fired matches. First is the Slow Fire Match, which is 20 rounds slow fire. The second match is the National Match Course which is ten rounds slow fire, ten rounds timed fire and ten rounds rapid fire. The third match, the Timed fire match is 20 rounds timed fire. And the fourth match is the Rapid Fire Match, 20 rounds rapid fire.

All shooting is done ten rounds per target. After each ten round target shooters score and repair targets. All slow fire is fired at fifty yards. Timed and rapid fire is fired at twenty-five yards. For slow fire, competitors fire ten rounds with a ten-minute time limit. In timed fire and rapid fire, shooters fire two string of five rounds per target. Each five round string in timed fire is fired in twenty seconds. The five round strings in rapid fire are fired in ten seconds per string This may sound like too much shooting for some people but competitors must go down range to score and repair targets after every ten rounds. So shooters get about a five-minute break after each target while this is going on. There is also a fifteen-minute break between the .22 Match and the Center Fire Match. Shooters get a lunch break between the Center Fire Match and the .45 Match.

The 2700 I described is the most common type of outdoor match. Some matches may vary from this. Obtain a match program from the match sponsor, as this will describe the course of fire and match conditions for the tournament.

Also the Connecticut State Rifle and Revolver Association runs leagues for shooters who only shoot .22's. Also shooters can enter the 2700 and only shoot on or two of the 900's.

Competitors compete in one of five classes Marksman, Sharpshooter, Expert, Master and High Master. Shooters are placed in classes according to their averages so they are shooting against other shooters of the same skill level. The first match you fire you will have to compete in the Master class. Then you will put the scores from your matches in a temporary classification book and calculate your average and classification. You do this until you fire 360 rounds in competition and the NRA sends you a Classification card. If you have an NRA Indoor Pistol Classification you may use that classification for your first outdoor match.

The dates for the C.S.R.&R.A. matches are listed on the back page of the Marksman. Shooters who are interested in traveling can get a complete list of NRA Registered matches in Shooting Sports USA which is put out by the NRA.

TOOLS OF THE TRADE

The Essential Equipment

By John Dreyer

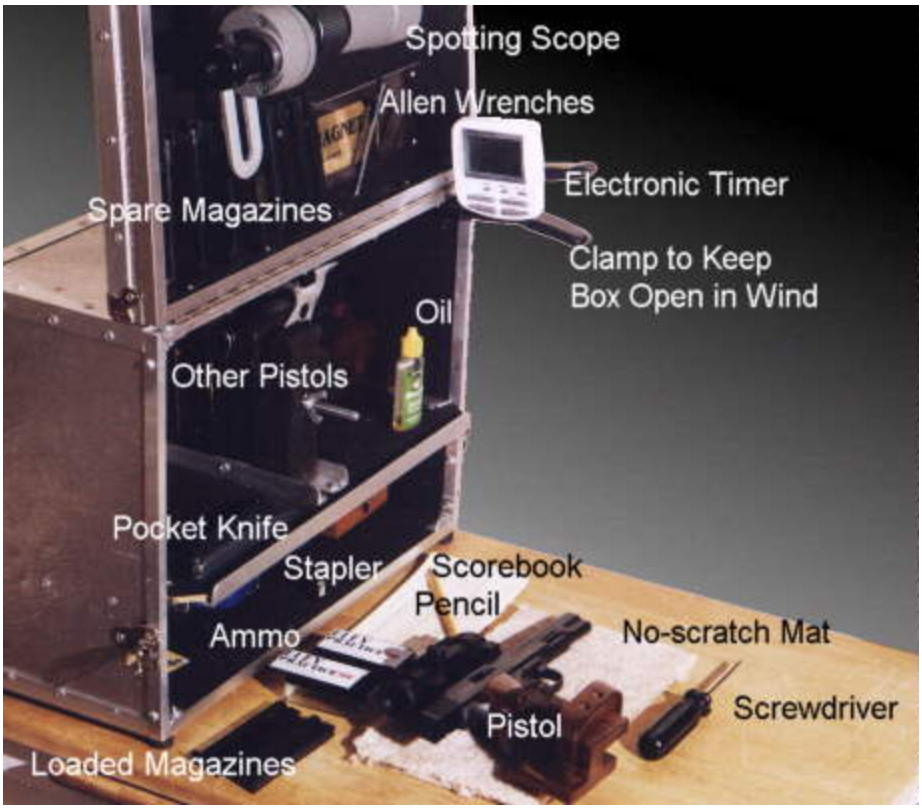
Source: http://www.bullseyepistol.com/equipmnt.htm

Target Pistol | Carrying Case | Spotting Scope | Ear and Eye Protection | Stop Watch | Scoring Aids | Tools | Cleaning Supplies | Targets | Ammunition | Reloading Equipment

When you think of the equipment required for a competitor to excel in Bullseye Shooting, you might compare it to any specialized sport. While you may only need the bare essentials to get started in this sport, to become truly competitive you will have to purchase some special equipment. For example, a single hockey stick and a puck is all that a youngster needs to begin a lifelong affinity with the sport of hockey and to learn the most basic skills. However, to participate in competition and at the same time further develop his skills, he will need skates, pads, gloves, a helmet and more. The same is true with the developing newcomer to the sport of Bullseye Pistol Shooting.
Use this article as a guide to selecting and acquiring equipment to participate in this sport. I encourage you to get involved in the sport with borrowed pistols and a bare minimum of equipment! I am certain, however, that if you truly wish to become a competitor, and possibly a champion, you will eventually obtain just about every item discussed in this article. But a word of caution: gadgetry can easily become an obsession! So, remember that unneeded expenses can be avoided by using a little logic.

Keep it Organized!

If you acquire a lot of "stuff" there is a tendency, especially for beginners, to create a real mess of junk on the firing line. This is not the place to be fumbling through items you probably don't even need during a pistol match! Just remember that Herschel Anderson, one of the greatest shooters of all time, brought an incredibly small number of items to the firing line. Whatever you do decide to cart up to the firing line, keep it organized on the shooting bench. Here is what I bring to a match and how I organize it on the bench, the exact identical way every time...



Selecting a .45 Caliber Target Pistol

If your plans are to shoot only .22 Caliber Gallery events, you will not need to consider the purchase of a .45 caliber pistol. You should skip down to the section titled: Selecting a .22 Target Pistol.

Purchasing a .45 caliber target pistol might be the most expensive investment you make in this sport. So make it count.

The typical pistol for .45 caliber matches is an accurized 1911A1 model or variant with pre-series 80 internals. View this excellent article by Master Pistolsmith Larry Leutenegger covering everything you need to know about accurized pistols.

There are probably only four ways to obtain that competition-ready accurized pistol:

1. Mass-Produced Custom Model Pistol
2. Pistolsmith Hand-Built Model from New Parts
3. Stock Pistol Accurized by a Pistolsmith
4. Stock Pistol with a Do-It-Yourself Job

In the "MPC" group, the manufacturer uses engineering and machining advantages to build the pistol from the ground up. The frame and slide are manufactured in house and tolerances are extremely small. With fewer steps in the accurization process to be done by hand, the manufacturer can produce more pistols annually with reasonable wait times. And while they are extremely accurate and stay that way for a long time, the trigger action from the factory might need a touch up by a qualified pistolsmith.

In the "Pistolsmith Hand-Built" group, the qualified pistolsmith selects a quality frame and slide and all other parts. He assembles the pistol, fitting all of the parts using time-honored methods. The quality of the final product depends on his parts selection and hand-fitting skills. There can be enormous wait times for hand-built pistols, sometimes over 18 months. But depending on your pistolsmith, the wait may be well-worth it for the tremendous quality you receive.

In the "Stock Pistol Accurized" group, the shooter purchases a stock pistol (called a base gun) on his own and sends it to a qualified pistolsmith. At his shop he replaces the barrel and usually all parts of the internal action, closes tolerances, and works it over for function and trigger action. The quality of the final product,and the life-span of the pistol's accuracy not only depend on the parts selection and skills of the pistolsmith, but of the integrity of the materials in the base gun.

And finally, in the "Do-it Yourself" group, the shooter purchases a stock pistol and does his own accuracy job to save money. Oftentimes it is as simple as squeezing the slide in a vise and peening the rails, along with the fitting of a Kart Easy-Fit barrel. Brownell's carries drop-in trigger assemblies that simplify and reduce the time required in improving the trigger action. Results can vary widely and it is definitely harder than it looks, but I have seen some fellows obtain surprising accuracy even on their first attempt.



Pictured on the right is a typical .45 caliber pistol setup that you might find on any firing line. This particular pistol is a (MPC) Les Baer Bullseye model with their 1.5" group guarantee. Shortly after purchase, it was sent to pistolsmith Mike Curtis for a trigger action job. It has a short trigger with the addition of a trigger shoe. It has a slide rail scope mount with an UltraDot 1" model sight. The grips are by Thomas Rink through Larry's Guns.

While mounting the dot sight directly upon the slide is the preferred method, it may also be mounted to a frame (bridge) mount, such as the Clark Scope Doctor mount.

Details such as frontstrap checkering or a beavertail safety are clearly optional, but they might make the pistol easier to shoot. Other additions such as an ambidextrous safety or a flared magwell are unnecessary expenses and no benefit to a bullseye gun.

Selecting a .22 Caliber Target Pistol

Competition-quality .22 caliber pistols come with all different shapes, sizes and price tags. Space limitations allow mention of the more popular options only.

THE ELITE:

These pistols are prized for their tremendous design, light weight, superb and hightly adjustable triggers, outstanding accuracy and flawless reliability. The newcomer, the Feinwerkbau AW93 carries a hefty price tag of $2500 new. Flawless in almost every way except for a best method of attaching the dot sight. The Hmmerli 208s is the final and finest offering in the time-tested Walther/Hmmerli Olympic series. Now discontinued, a nice one can be found second-hand for about $2000. The Knapp scope mount is likely the best method for attaching a dot sight.



Feinwerkbau AW93-L



Hmmerli 208s

HIGH-PERFORMANCE:

These pistols are top performers on the firing line. They are well-made and well-designed. The trigger action of both pistols is very good, but only adjustable for over-travel. The Smith & Wesson Model 41 has not changed since its introduction in 1957 and carries a price tag of $1000 new. Used ones can be found anywhere between $400 and $800. With the Sig/ Hmmerli, modeled after the Olympic series, its Competition model with the 6" barrel sells for $1000 new. Recommended for either model is anatomical grips from Rink or Fung.



Smith & Wesson M41



Sig Trailside/Hmmerli X-esse

CONVERTIBLES:

This category contains .22 caliber setups that are convertible to another caliber and are equally accurate in either. In the tradition of Fred Kart and John Dworak who have made quality .22 conversion units atop the 1911 lower, Bob Marvel stormed the market with his super-accurate offering. For about $450 for a new conversion unit, you can convert a .45 caliber wad gun into a fine .22 with the same trigger action. Similarly, Walther, Hmmerli and Pardini all offer pistols that are convertible between .22 caliber and .32 S&W Long caliber. All three are of exceptional quality and have outstanding triggers that are adjustable in any way thinkable, but they are quite muzzle heavy in their balance. And with all three around $1700, plus another $900 for the conversion to the second caliber, they are not cheap either.



Marvel Conversion on 1911 Frame



Walther GSP Expert



Hmmerli SP20



Pardini SP/MP

MID-LEVEL PISTOLS:

The following pistols are classified together only because they fall in the same price-range. The Walther SP22 series is a relative newcomer with its M4 version best suited for bullseye carrying a price tag of $600 new. While it is definitely odd-looking, it recoils softly, balances nicely and has a trigger adjustable for creep and weight. The High Standard pistols are truly the last of a dying breed, and after the company changed hands three times, their pistols have some real pluses and minuses. On the plus side, they have nice triggers adjustable for weight and a frame that feels exactly like a 1911. On the minus side is finish, reliability, parts availiability and service. But, if you're willing to take a chance, a new High Standard Victor could be yours for $850.



Walther SP22 M4



High Standard Victor

ENTRY-LEVEL MODELS:

If you are new to the sport, there are two models of well-performing pistols that are a small investment to purchase new. Of the two, the Ruger MkI or MkII at $350 has been around a lot longer and its track record of accuracy and reliability is excellent. A trigger action job or a drop-in kit from Volquartsen makes it quite a competitor. The Browning Buckmark at $400 is another option with a variety of barrel options available. A trigger action job will benefit it as well.



Ruger Mark I or II



Browning Buckmark

Extra Magazines

It is a good idea to have extra magazines for several reasons. The obvious one is to have a replacement if one stops functioning during a match. Another is to allow you to preload

several magazines during a target change so that you are not rushed between two strings of timed or rapid fire. Some fellows take this to an expensive extreme and have a magazine preloaded for EVERY string of the match! On the other hand, a large number of top competitors feel more comfortable using a single magazine and loading it before each string; by keeping their empty backup magazines out of view, they avoid the mental error of inserting an empty magazine. *I prefer to reload only two magazines at a time after every stage.*

Shooting Box



Pachmayr Box

By design, competition pistol boxes solve several problems. First, they allow secure storage and transport of your valuable firearms in a lockable case. Second, they provide the space and means to keep all your shooting accessories organized and handy at the firing line.

Pictured on the left is the "classic" design of the Pachmayr competition pistol box. This style remains the most popular among bullseye shooters as it still dominates the firing lines nationwide. In this design, the opened lid serves as a stable and easily accessable holder for your spotting scope. This and other designs of pistol boxes are designed to accomodate the organization needs of the competitive pistol shooter at the firing line. It is simply your decision to find the case that is right for you.

Competition pistol boxes vary greatly in both price and workmanship. The least expensive boxes are made of vinyl-covered plywood such as those by Gun-Ho. At the other end of the price spectrum, you will find durable aluminum boxes that will last a lifetime of abuse.

If the funds are not there for a specialized competition box, a foam-lined gun or photo case, a briefcase, or even a cardboard box may suffice in the meantime!

Suitable Spotting Scope

A spotting scope is an invaluable tool for the target shooter. It allows you to verify where each shot hits in slow fire and where your five shot strings group are in timed and rapid fire. You will want a scope of the 20x or 25x power and the largest objective lens and BEST SHARPNESS you can afford. This is a one time investment that is worth spending a few extra bucks. An inexpensive scope will cause eyestrain and will make seeing .22 holes at 50 yards difficult or impossible.
Gil Hebard manufactures a handy gadget that attaches your scope to the inside of the lid on your pistol box. This gadget allows you to place the scope in just about any position imaginable, accommodating the height of any shooting bench (or shooter for that matter).

Ear and Eye Protection

If you are blessed with good vision and you don't already wear glasses, you will need to buy shooting glasses. Protection from a rare shattered case fragment or bullet splatter from the backstop easily justify the expense of glasses. As another benefit, yellow-tinted lenses will slightly improve contrast outdoors and may seemingly brighten and clarify the target. You may find that you prefer clear lenses on a normal day and grey lenses in extreme bright sunshine. Regardless of color, the QUALITY of the optics is the most important factor. Inexpensive safety googles from the hardware store would not be of high quality.

Ear protection is equally important! Besides preventing the obvious long-term damaging effects of gunfire, maximum ear protection will help eliminate flinching. The unmuffled roar of gunfire during rapid fire would cause ANYONE to get a little shaky! You have several options in this matter: (inside-the-ear) earplugs, (over-the-ears) ear muffs, or as I recommend, BOTH. Don't consider any product with less than 20dB of reduction as ear protection. Hearing protection is inexpensive and is readily available at your local hardware store.

Another item that will assist in optimizing your vision is a cap. The bill of the cap will keep light from reflecting off the inside of your glasses and will also prevent an ejected case from the pistol of the shooter to your left from falling between your glasses and your face. An uncomfortable distraction at the least.

Electronic Stop Watch or Timer



Electronic Timer

For under twenty dollars, you should purchase an electronic stop watch or timer. This little gadget will make sure that you make the best use of the time allotted in slow fire. Without it, you will probably shoot too quickly, wasting the opportunity to wait for the wind to stop blowing or for your arm strength to be restored after a shot. The inverse is a bigger problem, having taken too much time and still having two rounds left unfired at the end of ten minutes. Radio Shack makes an electronic timer that is not only two timers that operate independently in one unit, but they can be set to either count DOWN from a time you preset, or count UP from zero. I prefer to just reset it to ZERO and count up to ten minutes starting at the command "commence firing." Whichever method you choose, keep it simple and do it the same way EACH TIME.

Scoring Aids

A scoring overlay, a clear piece of plastic with "circles" of common bullet sizes, is a valuable tool in scoring targets. It allows you to determine whether those borderline shots are actually touching the scoring ring. Some overlays have a small arc of a few inches of the four innermost rings of the target. Tight groups that make a big hole, thus "erasing" part of the scoring ring, can be scored by reconstructing the scoring ring with the overlay. A second overlay on top shows which shots are touching the "restored" ring. They are cheap. Get a set.

Tools for Repairs or Adjustments

A complete toolkit for your pistol is like the American Express card: Don't leave home without it! Make certain that you have every Allen wrench or screw driver you might need for every adjustable part on your gun. If you need to adjust your sights, you will have the proper size screwdriver for the job. If your optical sight's mount or one of your pistol's grip panels loosens, you can tighten it back up properly. A pocket or utility knife is also a valuable tool that shouldn't be overlooked, with two purposes being opening a new box of cartridges or prying a stuck case out of the chamber.

Another handy gadget to keep in your box is a 12" length of wooden dowel rod for clearing your barrel of a squib round. For the .45 caliber pistol, a 3/8" diameter rod is perfect for the task.

Cleaning Supplies

A properly cleaned and maintained gun will operate more reliably, produce greater accuracy and will wear more slowly than one that is not. Only a fool would risk losing a match with a double-alibi in rapid fire due to a dirty gun! The argument that regular pistol cleaning is unnecessary because the modern smokeless powders and non-corrosive primers are not caustic is really not valid - their residue is ABRASIVE and will cause wear during operation if not removed.

Similarly, a pistol should be lubricated, but never to excess. While the .45 caliber pistol might prefer quite a bit of lubrication, the .22 caliber pistol should be oiled at a bare minimum in only the proper places. Too much oil may attract dirt and residue and actually induce wear.

Here is an extensive list of essential cleaning supplies for the competitive shooter. They are inexpensive and readily available.

Item	Description
Pull-Through	I suggest cleaning a .22 caliber bore with a pull-through and a patch only. I like the PatchWorm brand best.
Cotton Patches	Buy 100% cotton patches only. They clean and absorb better than miserable synthetic. Use the correct size!
Bore Cleaning Rod	A threaded aluminum rod long enough to reach from the breach to and beyond the muzzle. Aluminum is a good choice because it is a soft metal.
Brass Jag	A brass jag screwed onto your rod in conjunction with a cotton patch will do a phenomenal job of cleaning a bore, for any caliber pistol.
Copper Bore Brush	Not for .22 barrels, but you will absolutely have to run a brush with solvent through your centerfire caliber barrels, with extra attention to the rifling near the chamber. As a significant aid to accuracy, I run a brush through my .45's bore periodically during the course of a match as well.
Toothbrush	Your basic straight dental toothbrush will gently clean just about any part of your gun.
Cotton Swabs	Indispensable for cleaning hard to reach places. I buy medical swabs on long sticks from a company called Lab-Safety Supply. They're great!
Tooth Picks	A great no-scratch tool for removing crud in corners and grooves.
Gun Scrubber	

	It ain't cheap, but it does work fast. A few squirts on the inside of the .45 slide and presto! Don't get it in your eyes or on your skin!
Hoppes No. 9 or Kerosene	A super solvent that lifts away powder and lead residue from parts and leaves them clean and rust protected, with no gritty residue. A less expensive and still very effective alternative is kerosene.
Shooter's Choice Lead Remover	An indespensible product for removing stubborn lead buildup near the chamber of centerfire barrels. Dab a little and wait a few minutes, use a brush then a patch. Rinse with a solvent like Gun Scrubber or Hoppes.
Gun Oil	I use a thinner oil (like RemOil) during the colder months and a thicker oil (like Hoppes) during the hotter ones. Otherwise, oil is oil. Go sparingly.

Competition Targets for Practice

While targets are furnished by the officials at a competition or league event, you must provide your own when you practice. You should always practice with the correct targets intended for the event you are shooting. Not doing so may create bad habits and a variety of sighting problems, and it will give incorrect scoring feedback of your shooting. The official targets are not expensive, and you'll appreciate firing on real "tag board" targets as opposed to trying to print your own.

The official NRA targets for conventional pistol are as follows:

Outdoor Competition		
Range	**Match/Stage**	**NRA No.**
50 yd	Slow Fire	B-6
25 yd	Timed & Rapid Fire	B-8
25 yd	Slow Fire (Short Course)	B-16
	Timed & Rapid Fire	B-8
Gallery Competition		
Range	**Match/Stage**	**NRA No.**
50 ft	Slow Fire	B-2
	Timed & Rapid Fire	B-3
20 yd	Slow Fire	B-4
	Timed & Rapid Fire	B-5
25 ft	Slow Fire	TQ-6
	Timed & Rapid Fire	TQ-7

Selecting .22 Caliber Ammunition

The task of selecting ammo for your .22 pistol is an important one. Regardless of the pistol, the following factors must be considered in the selection process:

1. Accuracy (and consistency from lot to lot)
2. Functioning (some types malfunction in certain guns)
3. Cost (the biggest variable)

If a certain ammo does not function well in your pistol, change to another. It is the only of the three factors of which you have no control. Beyond that, selection is really a matter of how much accuracy you can afford. Fortunately, even some of the most inexpensive cartridges still produce satisfactory groups. View this article for the best practices for conducting your own Ammunition Accuracy Test.

Reloading Your Centerfire Ammunition

It is not necessary that you reload your own centerfire cartridges, but chances are good that you will if you shoot bullseye for any length of time. One reason is economic. Match ammunition is expensive; $30 is a pretty good starting point for a box of 50 rounds of .45 ACP match ammo. If you shoot double .45 in a 2700 match, you'll need four boxes to cover the CF and 45 phases, plus a little reserve for alibis. Most shooters quail at the thought of spending $120 on ammunition for a single match. Reloading can help reduce those costs, or at least give you more ammo for the same amount of money. Reloading typically costs from $.12 to $.30 per round (for .45 ACP), depending on component selection and how many times you re-use your brass. This puts the cost of a 50-round box somewhere between $6.00 and $15.00, which looks pretty good compared to factory match ammo.

Reloading can also help you realize your gun's potential. Each match gun is a unique, finely fitted machine with unique tastes in ammunition. Reloaders can vary powder charges and change bullet seating depths and crimp diameters, all of which will affect a load's performance. With enough experimentation you can find the load that extracts maximum accuracy from your pistol.

When selecting a press you need to consider several things. First, think realistically about how much reloading you'll be doing. If you're only going to reload 200-400 rounds a month, you may be able to get by with a single-stage press. If you're planning on reloading much more than that, you may be happier with a progressive. Progressives are much faster - an experienced user can load 500 rounds or more in an hour with a progressive, versus 100-200 rounds with a single-stage press. Progressives are also capable of producing match-quality ammunition, just like a single stage. Some people recommend that beginning reloaders start

with a single-stage press, due to the greater simplicity of the machine. Others started reloading on progressives and have had no problems mastering the process.

References

- Target Pistol
- Carrying Case
- Spotting Scope
- Ear and Eye Protection
- Stop Watch
- Scoring Aids
- Tools
- Cleaning Supplies
- Targets
- Ammunition
- Reloading Equipment
- Selecting a .22 Target Pistol
- accurized pistols
- lubrication
- bare minimum

Further Reading

- PatchWorm (http://20-20.8m.com/)
- Lab-Safety Supply (http://www.labsafety.com/)

TOOLS OF THE TRADE

Don't Compromise on Your Equipment

By Randy Pafford

Source: https://www.bullseyepistol.com/pafford.htm

You know, there is always this tendency to pat people on the head and say "you have to practice." And it's true. Without lots of practice, you cannot do well. Dry fire, get to range, organize your practice sessions, get coaching, etc. The fundamentals are key; I'm having a little trouble getting a clean trigger release in slowfire this year, and it's costing me about 20 points per 2700 so far. So I don't claim to be the best shot around, or know all the answers. Regardless, equipment is very important in bullseye pistol. You can not shoot good scores in an outdoor 2700 without good equipment. Nope, sorry, it cannot be done. That 10 ring is 3 1/3 inches. If your gun is shooting 4, 5, or six inch groups, you are going to have a lot of trouble putting together a good slow fire score. And how are you going to be able to call your shots? "Hey, I thought that shot was good. How come there is a hole down in the 7 ring?" Calling your shots, correcting errors, and re-enforcing things done right is fundamental. But you can't do it if your gun will not shoot to call. A vital consideration is gun fit. Yes, gun fit. Some people have short stocky builds, and forearms the size of telephone poles. They can shoot a heavy gun well. Some of us have longer, rangier builds and don't manage a heavy gun as well. Some people have big hands and need long triggers. Some people have small hands, and with a long trigger, the trigger pull will seem like 10 lb. instead of 3 1/2. Can't see the front sights anymore? Well, do what you like; I'd put a dot sight on rather than screwing around with trick prescriptions. Got a creepy, 5 lb. trigger pull? Go ahead, be a man and keep it that way. The rest of us will go for a smooth 3 1/2 lb. pull on our .45s. And by the way, I have listened to some of the top shooters in the country, and also some coaches, and they will tell you equipment is important. I've never seen anyone improve very fast with lousy equipment. The people I've seen improve fast have -- surprise -- good equipment. Go back to some of the excellent articles in the Gil Hebard book and read carefully. You will note that several of these articles say "you need good equipment." I have never seen anybody shoot a master score with a .45 that wasn't properly accurized, didn't have a trigger job, etc. And that is a little expensive. I'm not saying it can't be done, just that none of the master shooters I know choose to try it. Going a step further I have seen good shooters absolutely held back by their equipment. I know one guy that had a new .45 built and watched his scores move up fifty [that's 50] points in the 2700. From 2550s to around 2600. He later replaced the slightly shot out barrel on his old 41 last year and picked up a few more points. We knew the barrel was shot out because we could see it putting 4 inch groups on the target at 50 yards from the machine rest. So we popped the barrel from my 41 on it and watched about two inches fall off the group. Equipment is important. In this shooter's case, part of the problem was old guns

that literally had worn out barrels. But he got very frustrated sometimes when shooting slowfire and the shots were not to call. This frustration, and lack of confidence in your equipment, is devastating to a shooter. Did I just miss, or did the damn gun throw it out? Ranting aside, I want to stress you don't have to spend an incredible amount of money on your equipment. A Ruger with a trigger job and stocks that fit your hand may work quite nicely, for example. I've seen 880 broken with a Browning Buckmark. But I've also seen sharpshooters and marksmen competing with equipment that would cost me 30 points per 900 aggregate. Some of them are not serious, and don't really mind. I always suggest to people interested in bullseye they start with a .22 and see if they like the sport before investing in a .45. I also think I made better progress just working with a .22 for a year or so than I would have trying to shoot both a .22 and a .45. I don't want folks to feel that they have to spend a fortune to participate in this sport, or that practice [and sometimes natural gifts] do not separate competitors. But equipment is important, and people who say otherwise do not win matches.

TOOLS OF THE TRADE

The Custom-Built Handgun

By Larry Leutenegger

Source: http://bullseyepistol.com/cstomgun.htm

Why a custom Bullseye gun is important.



Figure 1: Competition, even on a casual or local level, is completely different from plinking

First, let's get one thing straight. A custom gun isn't important for Bullseye shooting. An *extremely accurate* gun is. Unfortunately, it seems that the only way to get an extremely accurate gun that suits a Bullseye shooter's needs is to have one built.

Having said that, there's a simple answer with some complicated details to the "Why."

The simple answer is one word: "Competition."

"Competition" means this: You are no longer shooting just for fun (plinking). You are now shooting for score, no matter how casual the league or match may be. Against other shooters' scores and, most importantly, against your own previous scores (see figure 1).

Here are the complicated details:

Lots of folks say, "I'll use my stock gun, and when I can out-shoot *IT*, I'll get a custom." Well there is one thing right with this attitude, and three things wrong. The One Thing Right is, if you have absolutely no budget, at least you're participating in a great sport.

Wrong Thing #1: You will never be sure of how good you are when your equipment is questionable. How do you know when you are out-shooting your stock gun? When I test stock

guns (all at 50 yards), they shoot groups from 4" (very rarely) to 14", most falling between 6" and 12". Let's say we put your stock gun into a Ransom Rest and find out it will group 8" at 50 yards.



Figure 2: The potential area of hits from an 8" gun shot from a Ransom Rest.



Figure 3: The potential area of hits from an 8" gun by a shooter with an 8" wobble area.

Well, that's 8-ring accuracy (see figure 2). Now, in a match, if you shoot an 80, you may have just shot as well as a Ransom Rest (and, in counterpoint, with a good gun, you might have shot a 100-10x), or you may have just gotten lucky. *There is no way to tell.* In addition, most beginners have a decent wobble which, when added to the 8" capability of the gun, gives a very wide hit area. If you have an 8" wobble and a 8" gun, you have a theoretical group sizes of

16," which can put shots in the 5-ring or anything inside it (see figure 3). You can theoretically shoot a 50 or a 100-10x without any difference in your technique.



Figure 4: The potential area of hits from a 2" gun by a shooter with an 8" wobble area.

However, if your gun shoots 2" and you have the same 8" wobble, that's a 10" group and you will score at least a 70 and most likely a 85 when you figure in the law of averages (you'd have to be really unlucky for all your shots to break only at the outside edges of your wobble) (see figure 4).

Wrong Thing #2: You can not develop the core skills you need using the wrong tools. You will never learn to throw a perfect spiral football pass using a tennis ball, no matter how much you practice. In theory, you can learn the perfect golf swing using lousy clubs, but you will never know you have the perfect swing because lousy clubs will spray the hits all over the golf course. The same with an inaccurate gun. With an 8" gun, even if you improve your wobble and reduce it to 6", you still have a theoretical spread of 14" and a score of from 60 to 100-10x. However, with a good gun, reducing your wobble like that would deliver at least a 90, and probably a 95, assuming your core skills are excellent.

That's another very important point to remember: A good gun will give you immediate feedback on your technique, good or bad. If you try, say, a different grip for your next target and shoot a 75 or a 95, that's instant, unquestionable feedback on the value of your change.

Wrong Thing #2a: You will develop bad habits with bad equipment. Only experts can pick up bad equipment and shoot it decently, and that's because they've already developed the core skills. How is a beginner going to learn proper trigger control with a 7-lb. trigger with a 1/4" of notchy travel? He or she will come up with strategies to compensate for bad equipment instead of improving their core skills.

Wrong Thing #3. You will become discouraged. No one can continue to shoot 20s and 30s next to people shooting 90s and 95s without a dip in morale, even with the nicest people next to you (and Bullseye shooters ARE the nicest people). Everyone needs reinforcement, and better scores are the best kind.

Bullseye is a great sport. Sure, go ahead and try it with a stock gun or -- better yet -- borrow a Bullseye gun (I guarantee you, if you show up at a club match and say you're interested in Bullseye but would like to try it first, someone there will lend you a good gun, good ammo and walk you through a match). But as soon as you decide you want to pursue this sport, figure out a way to upgrade your equipment.

Why your custom gun should come from not just any gunsmith, but an experienced BULLSEYE pistolsmith.

The same kind of gun (a Colt 1911) can be totally different when customized for different sports (for example, IPSC vs. Bullseye). Let me use an automotive analogy. You want a great NASCAR stock car. There's a guy in town that builds cars that have won championships - but in drag racing. You wouldn't go to him for a car that you need to run for 500 miles at max RPMs when he's known for building cars that run 5 seconds at max RPMs, even though both are built around V8s. And you shouldn't go to, for example, an IPSC pistolsmith for a Bullseye gun. (I'm comparing to IPSC 'smiths because the guns can seem pretty close -- I presume I don't have to tell you not to go to a riflesmith for a Bullseye pistol.) The different sports require totally different things from a gun, and the guns are used in totally different ways.

This is not to say IPSC 'smiths can't make an accurate 1911 .45. They can and do. But they don't shoot bullseye and don't work every day making Bullseye shooters happy. Often, they don't even understand Bullseye shooters. (I know my ex-wife didn't.) I do a lot of trigger jobs on guns from famous shops. The owners have sometimes sent them back numerous times for the trigger pull. Each time, the famous shop tries, but they don't know what to do because they've never held that gun out, gotten good sight alignment, started the trigger squeeze, watched the wobble slow and waited waited hoped hoped prayed oh please oh please oh please for the trigger to break. I've been shooting Bullseye for over 26 years. As the saying goes, "Been there, shot that." So I redo a lot of triggers on expensive guns. The owners are happy now, but I or any other reputable Bullseye pistolsmith could probably have built those guns right the first time, probably for less money, a shorter wait and less frustration. And they would be just as accurate, if not more so.

There's another aspect folks should know about when deciding between what I call "mass-produced customs" (MPCs) and a real custom Bullseye gun: Guarantees. No, not warrantees. Guarantees. Most MPCs come with a 30-day warranty, which in my mind is no warranty. I give a one-year guarantee, and so do most other small-shop Bullseye 'smiths. The mass

producers are turning out dozens -- maybe hundreds of guns a week. I turn out one or maybe two. Every gun that goes through an MPC has four, five -- who knows how many guys working on it. Every gun that comes from my shop has one -- ME. I know what's inside that gun. I know what's outside that gun. Every gun that goes out of my shop has my name on it, and I know every customer that buys one. So if it's not right, I'll make it right, usually for no extra charge. And most other Bullseye pistolsmiths would do the same. And two years from now, when you call me or any other small-shop Bullseye pistolsmith, we'll remember you and your pistol.

One last reason to get your Bullseye pistol from a Bullseye pistolsmith: support. When you buy a gun from me, it's not just a business transaction. It's the beginning of a relationship. For instance, I and some of the other Bullseye 'smiths attend the large national matches every year, like Canton and Camp Perry. *We're not there to compete or to sell guns*. We're there to support our customers. 90-to-95% of my time at Perry is tuning and fixing guns. (The rest of the time is jawing with tire-kickers.) I sometimes put in 18-hour days there because my customers need their guns to perform there, even the ones I didn't build. The big shops (at least the ones that bother showing up) bring guns you can buy. Some just send shooting teams. Neither does you any good if the gun they sold you starts malfunctioning. I -- and the other Bullseye 'smiths who can make the trip -- bring tools. Think about that when you start shopping for a Bullseye pistol.

What a Bullseye pistolsmith actually does.

(Disclaimer: the opinions expressed here are mine. If other 'smiths agree fine. If other 'smiths disagree, fine too. It takes all kinds to make a horse race.)



Figure 7: The five areas where a Bullseye pistolsmith attains great gains in accuracy. (See text.)

A Bullseye pistolsmith's job is to build the most accurate gun possible. To do that, we employ some tried and true methods and techniques, largely grouped into these five areas (see figure 7):

1. Slide-to-frame fit
2. Barrel-to-slide fit 3. Trigger 4. Functionality/reliability 5. Comfort/personal preferences

When having a target pistol built, remember that when you start with good parts, you end up with a better finished product. Some frames and slides are questionable and it makes the gunsmith's job much more difficult -- trying to make everything work properly and meet their accuracy standards. (You're not going to win the Daytona 500 with the engine block out of your Daddy's pickup.)

For slide-to-frame fitting, I start by measuring all the parts that will mate (see figure 8 and 9). A key fit for accuracy in a 1911 is the bottom slide rail to the frame groove, and this has to be close. I go for a .002" clearance because, over 25 years, I've discovered that this gives the best compromise between tightness and functioning. After all, you need some place for oil and dirt to go when the gun starts to get dirty. I straighten and smooth the area by hand, with a fine-cut Swiss pillar file cut down to fit that area.



Figure 8: Frame-groove-width measurements are taken at several points.



Figure 9: Slide rail width measurements are also taken at several points

To control side-to-side play, I narrow the slide to almost the same width as the frame grove. (I use a personal technique that, while not be a high-tech secret, is something I'd rather not share.) Anyway, then I lap the slide to the frame using very fine abrasive and a lot of elbow grease. This results in a pretty consistent .002" clearance. That's it for the side-to-side movement.

Now I have to get rid of the up-and-down play. This is done by moving down the rails of the frame to fit the bottom rail of the slide, using the same .002" clearance (see figure 10), using judicious application of a hammer and precision gages to keep the bearing area consistent. I lap it once more with very fine abrasive and a lot of elbow grease.



Figure 10: Using a custom-made micrometer to measure slide rail height.



Figure 11: The National Match bushing being machined to .001" of the barrel outside diameter.

As for barrels, I use Kart or Bar-Sto. If a customer wants another brand, that's fine, but I can't guarantee two-inch accuracy at fifty yards (I usually still get it). I can get almost any barrel to shoot, but probably not as well as a Kart or Bar-Sto. More matches -- including National Championships -- have been won with Kart and Bar-Sto barrels than with any other brand, and it's very rare that you get one that won't shoot. They don't cost more than other brands and they're not in short supply, so I can't see why you'd go with another brand. But hey, if you want to, it's your nickel.



Figure 12: The outside of same National Match bushing being machined to fit the slide.

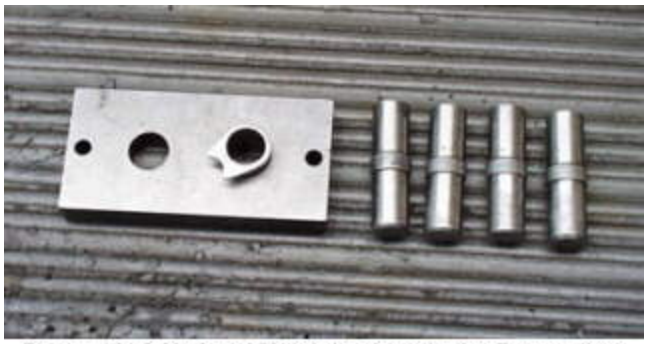


Figure 13: A National Match bushing in the fixture plate and different expanders to fit the bushing to the slide.

Barrel-to-slide fit is probably the most important area for accuracy. If you have a tight barrel-to-slide fit, the slide-to-frame fit can actually loosen and you won't notice a loss in accuracy with open sights or a slide-mounted scope. (If the frame-to-slide fit loosens and you have a frame-mounted scope, you may notice a loss in accuracy.) I fit the barrel in such a way that it will return each and every time to the same place in lock-up. First I fit the bushing to the barrel as close to .001" as possible (see figure 11 and 12). Then I fit the bushing to the slide by expanding the rear of the bushing -- where it's not supporting the barrel -- so you'll need a metal bushing wrench to get it on and off (see figure 13). Then I hand-fit the hood in both length and width to the slide (see figure 14 and 15), then hand-fit the underlugs (see figure 16). All of these areas must be very precise for a tight, consistent, long-lived total lock-up.



Figure 14: The human eye is the best tool to gauge the fit of the barrel hood in the slide.



Figure 15: Very judicious removal of metal with an extremely fine Swiss pillar file for fitting barrel hood.

Next, and probably as important as the fit, is the trigger. A great trigger pull requires fitting of the trigger to the disconnector to the sear to the hammer, plus all the pins and springs that hold them in place. Everything bears on everything else, so a good trigger pull can be like a three-dimensional jigsaw puzzle (see figure 17). I prefer the spur hammers because I can get that beautiful, soft-feeling trigger pulls with them that my guns are known for. I don't know why, but the cool-looking commander hammers never seem to feel soft enough to suit me (they always feel more crisp and harder, even though they are the same weight). But if a customer likes and wants a sharp break or a Commander hammer, I'll use one.

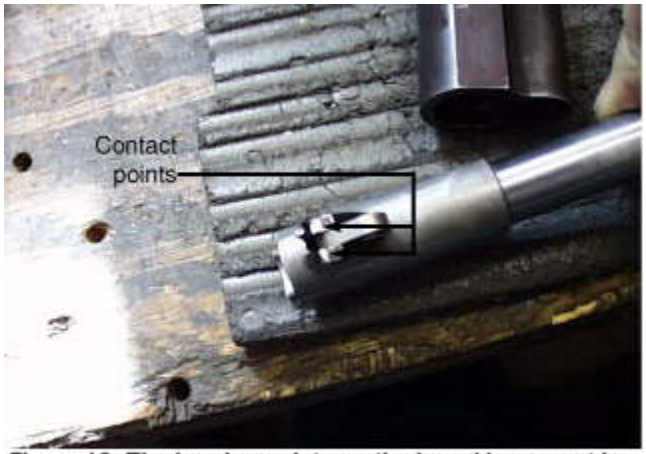


Figure 16: The bearing points on the barrel lugs must be perfectly even for long-term, consistent accuracy.



Figure 17: All of the components that affect trigger pull and some of the tools and stones required for fitting and adjustment, not including vises, hammers ...



Figure 18: A microscope allows me to see the sear and hammer contact in great detail. Technology works.

The sear-to-hammer contact is crucial. Perfect contact and angles take hand-stoning on special jigs of my own design, installation in the gun, testing, then disassembling and examining the parts under high magnification. The testing tells me, through my trigger finger, what's going on. The exam tells me why. I've found that a microscope is very helpful in getting perfect, bring-a-tear-to-your-eye triggers (see figure 18). I go through this whole process over and over, until the trigger is right.

Which brings us back again to "Why a Bullseye pistolsmith?" Because you want an educated Bullseye trigger finger tuning your Bullseye trigger. For example, in IPSC, your trigger pull might take a fraction of a second and you'd use a two-handed hold 90% of the time. You'd

have to be very good to notice, say, that the disconnector is binding ever-so-slightly in its tunnel as you pull the trigger, and the IPSC pistolsmith who built that gun might not ever notice it. It's just not that important in IPSC. But in Bullseye, that same trigger bind is as big as a house and even if a Marksman doesn't know there's anything wrong with the trigger, it's affecting his/her trigger pull and, therefore, his/her score. And I guarantee you, a trigger finger with just a few matches to its credit will know there's something wrong in there.



Figure 19: The customers personal tastes dictate the details such as stock or high-rise grip safety.

Then I do the details -- what the customer has specified in fittings and accessories: beavertail grip safety or not, front grip strap stippling or checkering, scope mount, finish, whatever (see figure 19 and 20). Finally, I function-fire the gun, a minimum of 50 Bullseye rounds. If it so much as hiccups, it gets torn completely down and the cause is found. I function-fire by hand, so the gun has the same resistance and movement as it would in a match. Finally, I accuracy-test it in a Ransom Rest, with several kinds of ammo, including factory match (using lots I know are accurate) and my best handloads. If it doesn't shoot between one and two inches at 50 yards, it gets torn down completely and gone though again. I send out all guns with a test targets and the load or Lot # of the ammo that shot the target (see figure 21).



Figure 20: Frame-mounted or slide-mounted scope rail and a myriad of other details.



Figure 21: Finished gun and test target (with load specs) ready for shipment. Some shooter is going to be awfully excited when Santa in the brown suit (the UPS man) shows up!

You might have noticed I've said "hand-fitted" a lot. I think that's the key to an exceptionally accurate and comfortable bullseye gun. Sure, CNC technology can make slides and frames that are precisely-mated. But it takes the eyes and hands and commitment of a craftsman to make a masterpiece -- a gun where everything fits and works together perfectly for the shooter. And that's what a Bullseye shooter wants and needs. The perfect tool matched to their needs, with which to hone their skills and win matches.

When you get a gun from any nationally recognized Bullseye pistolsmith, it'll shoot. Why? Because our reputation is on the line. That's true for me, and it's true for any other dedicated bullseye gunsmith.

Resume

It always helps a reader to know the history of the writer, so that he/she may weigh the words knowing the experience of the speaker. So here's a brief history of myself, and why I think I can say these things.



Figure 22: Look at that handsome AMU shooter! Well, yes, now that you mention it, that is me in younger days. Notice the Sgt. behind me giving instruction.

I began my shooting career in 1972 in Fort Hood, Texas and I began my gunsmithing career in 1973 in Killeen, Texas, at Scotty's Gunshop. I got out of the Army in 1973 and joined the Wisconsin National Guard and became part of their pistol team. I traveled to the Army Match with them in 1974. After the matches were over in 1974 I went back on active duty, and back to Fort Hood, Texas. That year I shot very well and won most matches that I shot. The next year, 1975, was my first year to go to Camp Perry. After shooting with the team from Fort Benning, they decided this is where I should be. I was stationed at Fort Benning in 1977 with the pistol team (see figure 22). I stayed with the pistol team till 1979. Things that I did while with the pistol team:

I was a member of the 1978 and 1979 Interservice team champions. (the '79 score is still a record), and the winning 1979 National Trophy pistol team at Camp Perry (see figure 23). I made Distinguished in '77, the 2600 Club in '79, the President's 100 in '80 and '90 High Service Shooter at the Coral Gables match in '78. and part of the U.S. International Pistol Team.

In 1979 after Camp Perry, I was having some problems with my shooting elbow, so I asked if I could work as a pistolsmith in the AMU Custom Gun Shop. Bill Pullum, the shop officer, gave me that chance. For the first year, I was the Test Man. That means you test hundreds of lots of ammo to see if they meet AMU standards. You also test all the pistols being built with many lots of ammo and a Ransom Rest, to determine if the guns shoot and which lots of ammo each pistol likes (shoots well). Then particular lots of ammo are reserved for particular pistols.

With XXX match pistols and XXX lots of ammo, you did a lot of testing. But you also learn a lot about what makes 45s shoot and function. In 1980, I got my own bench as a full-time pistolsmith. Sometimes it was hard for me to watch the bullets go down range and not be the one shooting them, so I tried to shoot two to three matches a year just for fun.



Figure 23: Me on the winning 1979 Interservice Team. Our record still stands.

From 1981 to 1984, I was stationed in Hawaii with the 25th Inf. Div. Marksmanship Training Unit where I was a shooter, gunsmith and marksmanship instructor.

In 1984, I returned to Fort Benning and back to the USAMU as a pistolsmith.

In 1988, a new Army team was formed for Action Pistol. I asked if I could try out for it. I was told I could participate as long as it didn't interfere with my pistolsmithing duties. I was allowed to shoot, but I had to practice on my own time (lunch hours and week ends). It was a little difficult because I was also appointed Noncommissioned Officer In Charge and Senior Pistolsmith of the AMU. But the honor of that position took the sting out of having to practice on my own time (like civilians have to).

Things that I accomplished in Action Pistol:

In 1988 at the Bianchi Cup, the USAMU Team took a third place finish. I was the winner of the Tin Cup Shoot in 1990 at Robertsville, Mo. In the 1990 Bianchi Cup, I took the High Service Trophy and made the top shoot-off.

One of my proudest achievements is being chosen as the Military gunsmith to represent the U.S. Military at the Conseil International Du Sport Militaire (CISM) in Lahti, Finland in '88, Santiago, Chile in '89, Fort Benning, Ga., in '90 and Jaji Nigeria, in '91. CISM is the equivalent of the military Olympics for all NATO countries.



Figure 24: Just a handful of my 25-years worth of shooting medals and awards from around the world.

I've attended virtually every manufacturer training school offered, except maybe the Russian ones (but their team gunsmith trained me on the sly). From the Rock Island Arsenal for .45s and M14s, to Ruger, Colt, Smith & Wesson, Glock, Walther, Fienwekbau, FAS and Hammerli. I also attended the FBI Special Tactical Firearms school and the U.S. Army Sniper Marksmanship School.

I've also been a instructor of Pistol Marksmanship for the Army Marksmanship Unit from '77 to '79, and from 1984 to 1992, a Marksmanship Instructor for the 25th Infantry Division from '81 to '84. I also attended the NRA Class C Coaches School in 1990 (see figure 24).

After retired from the Army, I tried a couple of other jobs, Machinist running CNC lathes and mills, and a barrel maker (making rifle and pistol barrels). I went back to school and received a Machine Tool Operator Diploma. In 1998, I became a co-holder of a patent for the Patternmaster Choke Tube.

But Bullseye is still my first love. Making guns that put bullet after bullet through the same hole. I've built a lot of guns that have done well. The Past National Champions that are or were customers of mine include John Smith, Charlie McGowen, James Laguana, Roger Willis, Eric Buljung, Tom Woods, Max Barrington, Bonnie Harmon and Kim Dyer, just to name a few. Check out the historical listings in your Perry book. You'll see those names. Past ISU National Champions that are my customers include a lot of those same folks, plus Ken and John McNally, Terry Anderson (Eric Buljung is also an Olympic Silver Medalist).

My most recent Customer-In-The-News is Jim Henderson. He was in the hunt for a while at Perry in 1998, but ended up second overall (also taking Army Reserve Champion). He's been up there the last few years, shooting over 2670 twice each year. I build and maintain all of his guns. (I also end up cleaning them most of the time, but that's another story for another time.)

TOOLS OF THE TRADE

Troubleshooting and Preventing Malfunctions

By by John A. Dreyer

Source: https://www.bullseyepistol.com/malfunction.htm

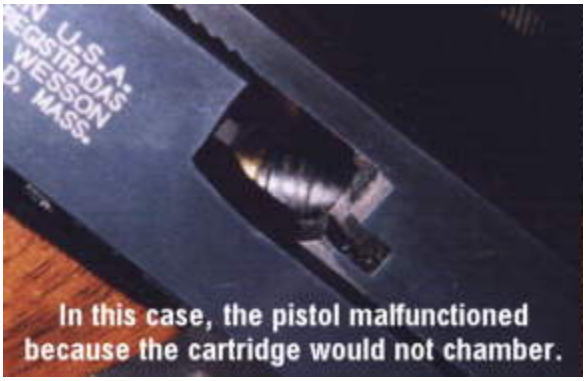
INTRODUCTION

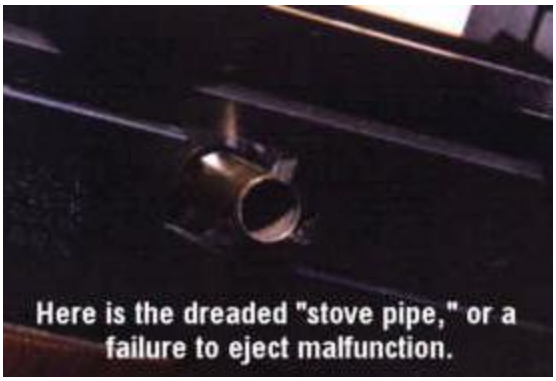
Over ten years ago, I began writing a troubleshooting article dedicated to solving a very common riddle: "Why has my .22 caliber pistol suddenly developed an urge to occasionally malfunction?" The inspiration for the piece was my own experience of developing an occasional alibi with my Model 41 after two years of literally *flawless* performance. Untimely perhaps, my problems first appeared during the NMC at the 1999 Illinois State Championship and just seemed to recur more and more often over the next several months.

This is undoubtedly one of the most frustrating experiences a shooter might have, especially if your pistol has run like a clock for years. In my opinion, it is imperative that you have complete confidence in your pistol to function 100% of the time during strings of sustained fire. After weeks and weeks of cleaning, measuring, testing and replacing parts, I finally solved the mystery with my pistol. Better late than never, I want to share my experiences with you, as well as other information on this subject that I have collected over the past decade.

TYPES OF MALFUNCTIONS

You can generally classify all alibis into one of three types: a misfeed, failure to eject, or a misfire.





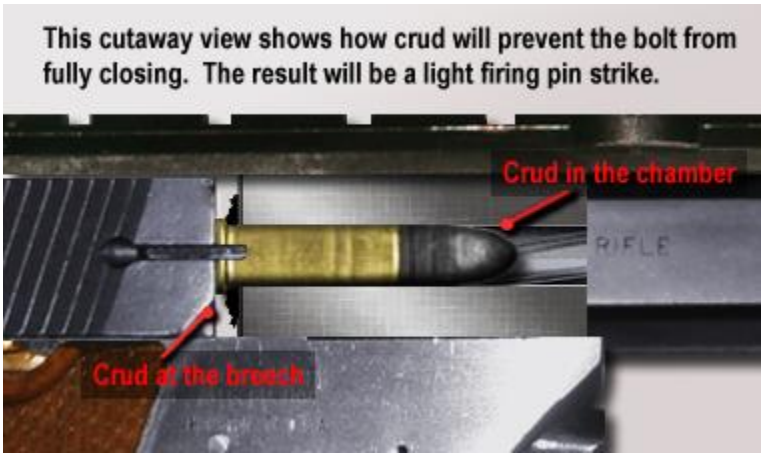


RULE OUT THE OBVIOUS

Before we go any further, if you are experiencing function issues, you must be absolutely sure that the pistol is clean. A dirty gun is the cause of possibly 90% of mechanical function problems. And just for the record, I had heard many of the bullseye *old-timers* claim that, "cleaning your .22 pistol does more harm than good." Most likely because they had always used improper methods in cleaning their pistols! To the contrary, a regular cleaning routine is paramount to accuracy and reliability.







Recognize that crud (in the form of primer, powder and bullet wax residue) will naturally build up within the active parts of the pistol. After even a short time of accumulation, it will impede the ability of the pistol to function correctly and efficiently. Furthermore, low temperatures have a way of magnifying the effects of accumulated crud by making it less pliable.

Crud at the breech or bolt face or in the chamber will prevent the the cartridge to load fully and will leave the its rim unsupported. This will cause a light firing pin strike and possibly a misfire.

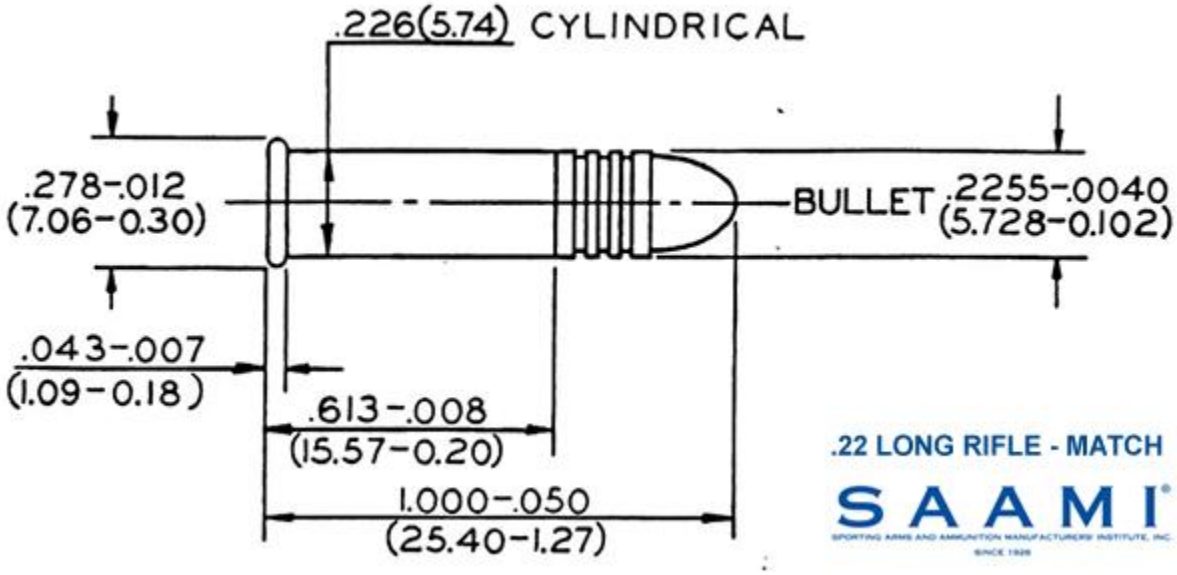
Crud on the inside, the follower or the lips of magazines may cause a misfeed. Crud on the feed ramp or in the bolt face recess or extractor groove can also cause a misfeed.

Crud on the top of the slide or recoil rod will possibly slow down the blowback action and cause either feeding or ejection problems.

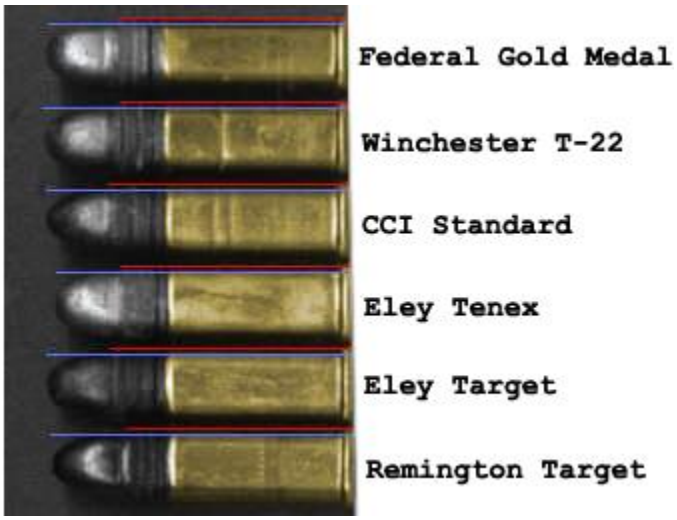
To read an excellent article by Larry Carter on proper cleaning and maintanance of a .22 caliber pistol, click HERE.

THE AMMUNITION

SAAMI (The Sporting Arms and Ammunition Manufacturers Institute) founded in 1926, provides and maintains a database of cartridge and chamber specifications for the many firearms calibers. As shown below, here are the SAAMI specifications for the .22 Long Rifle Match cartridge.



And while these dimensions are *not* vague, it is clear that they do allow for tolerance variations in application.



In the image on the right are six popular cartridges side-by-side. At a glance, you can easily see that each cartridge has a specific yet differing bullet profile and set of dimensions. Above each cartridge is a set of colored dimension markers: the blue line indicating overall cartridge

length and red indicating shoulder to rim length. As their red lines show, CCI Standard and Eley Target have a longer shoulder length than the other cartridges.

Similarly, pistol manufacturers oftentimes chamber their pistols slightly short to enhance accuracy. For example, Ruger target pistols and Clark replacement barrels have typically tight chamber dimensions.

This combination of long bullet shoulder and short chamber can prevent a cartridge from chambering completely, keeping the bolt from closing fully. In cases like this, all appears OK until the hammer falls. The inertia of the hammer closes the bolt and forces the cartridge the rest of the way into the chamber, but lacks the energy to cause the firing pin to dent the unsupported case rim. The result is a misfire.

While the cartridge dimensions will not change from day-to-day, there is an element of cartridge design that does change with the ambient temperature - the lubricant bullet wax. Each brand of cartridge is coated with a different type and quantity of lubricant wax. Some brands like Remington use a thin coat of a firm wax. Others like Eley use a heavy coat of a softer type of wax. In the accuracy department, the heavy soft wax seems to be the winner. However, be advised that cold weather will cause that soft wax to firm up and effectively increase the dimensions of the cartridge - thus, creating feeding problems. But worry not, the solution is easy. In cold shooting conditions, keep your ammunition in your pocket to allow your body heat to keep the cartridges relatively warm. Blowing once or twice on a loaded magazine before inserting it into the pistol is also a popular solution that works well.



And now the last, but most notorious contribution of a particular cartridge to reliability: primer compound. If you are experiencing occasional misfires and the struck rim of these have a good firing pin dent such as that in the image on the right, they can be ruled as *true* misfires or "duds." It is entirely the fault of the cartridge and not the pistol.

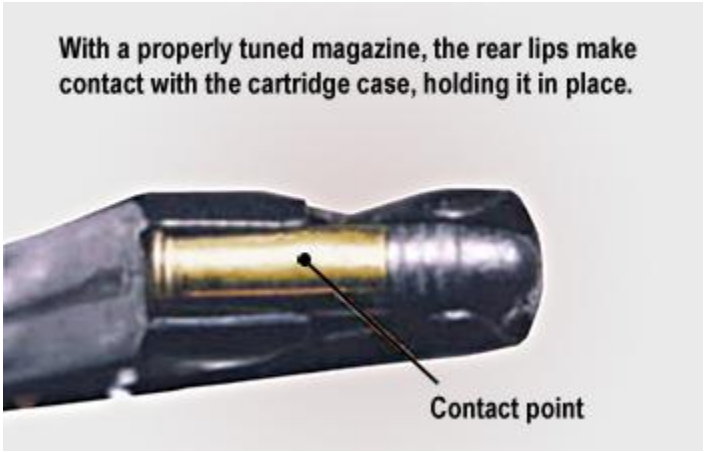
In more than 15 years of using Eley ammunition I have honestly never experienced a dud, a credit to their industry leading priming method. Other brands can boast this same reputation. However, to the contrary, some domestic brands such as Remington and Winchester are almost *expected* to have inconsistent priming and at *least* a few duds in every 100 rounds.

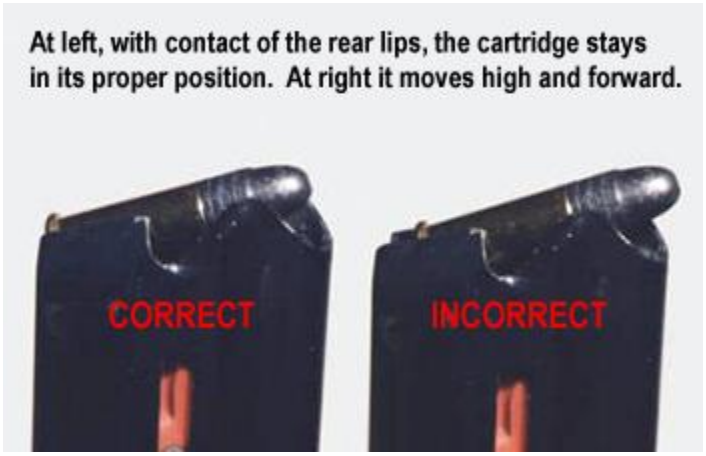
THE MAGAZINES

The health of your magazines has a huge impact on the pistol's functionality. The magazine can cause a variety of manfunctions including feeding problems and even failures to eject. Because of the critical nature of the magazine, I recommend that you take genuine precautions to keep them clean and healthy. For starters, I suggest using the Gun Box Magnet to keep your magazines secure instead of just tossing them into your pistol box.

A properly *tuned* magazine is truly important for 100% reliability in function. There really should not be much surprise in this when you consider that they are merely made of stamped and bent sheet metal. To qualify this theory, I purchased six new magazines directly from Smith and Wesson and, while all were excellent in quality, each had slightly different dimensions at the lips. And, each functioned differently in the pistol as well.

Tuning involves tweaking the top-front corner of the rear magazine lips in order that it makes contact with the brass of the top cartridge. The naked eye is the best gauge for this. Check for lip to cartridge contact and empty the magazine to make corrections if needed, pinching gently with pliars. Then repeat the process as needed. When properly tuned, the top cartridge will stay in its proper place, not too high and not too low.





Because tolerances in the specifications allow for variations in cartridge dimensions, different brands of cartridges will fit against the same magazine lips differently. In other words, if recently changing ammunition brands has caused immediate feeding malfunctions, the solution might be a simple a tuning of the magazines - and truly no fault of the ammunition.



Because magazine construction is done by merely stamping them from sheet metal, tiny burrs, slight defects on the edges of stamped cutouts, sometimes form on the inside of the magazine during the manufacturing process. These burrs can make contact with the internal spring or follower and cause it to stutter. This stuttering will cause a feeding problem.

Test your magazines for smooth movement throughout its five-round range. If you can feel hesitation as the follower moves up or down, you may need to disassemble the magazine and inspect the inside. A fine sandpaper of the 400-600 grit range, wrapped around a pencil, will easily clean up any burrs inside the magazine that may be catching the spring. Be sure that the magazine is *completely* clean of any filings before reassembling it.

EXTRACTOR AND EJECTOR

Often overlooked for importance in functionality, the extractor has a critical role in both the *extraction* and *feeding* processes. The extractor is commisioned with a very important duty: hold the blowing-back spent case in the correct horizontal position against the bolt-face until it strikes the ejector and flips to the right at the *correct* angle, exiting through the ejection port. Without a sharp and snug extractor on the case rim, the case will spin off-axis and have difficulty getting out of the chamber and/or hitting the ejector correctly. The spent case might also spin downward and impede the top cartridge in the magazine from entering the chamber.



As shown above on the left, this worn and misshapen extractor will clearly not hold the newly spent case in its proper position at ignition. Replacement with a new extractor claw would be the solution - but the new extractor will need to be prepared by checking and adjusting its dimensions against the old extractor and honing the claw a sharp point with a file or other abrasive.

The ejector on most pistols is a pretty maintenance-free part. However, some pistols (especially the Ruger) are plagued with ejectors that come loose and no longer function correctly. A loose ejector may cause excessive drag on the backward and forward motion of the bolt causing either ejection or feeding problems. A damaged or bent ejector might also cause an extraction problem. A quick inspection of the ejector in the frame is well worth it.

CHAMBER PROFILE AND CONDITION

Dry firing is an undisputed essential in the training process for competitive shooting. However, in some cases, it might be detrimental to the health of your .22 caliber pistol. The travel on many firing pins is longer than necessary and when sent flying, it will actually make contact with the metal of the chamber edge if a cartridge is not present.

This condition of excess firing pin travel may also slowly develop over time. A pistol whose firing pin had never touched the breech of the barrel when new may slowly make it there as the firing pin or the firing pin stop within the bolt slowly wears.

The ding that may have formed on the edge of the chamber will force metal inwards, thus shrinking the dimensions of the chamber opening at that point. This can cause both feeding and extraction problems.





But all is not lost. Available from Brownells is the Menck Chamber Iron, a tool that allows you to swage out the ding, and restore the original round profile of the chamber. It is inexpensive and easy-to-use.

To prevent problems from ever occurring, a best practice is to use a *dummy-plug* in the chamber as you dry fire. They are commercially available, but you can easily make your own. I suggest collecting some spent cases that you have fired (from this pistol or they won't fit) and grinding off one side of the rim to make a superb dummy-plug at no cost. Insert the case into the chamber with the grinded-off side facing the extractor and the case will not be extracted when you rack the slide to cock the hammer. I have used the same case for as many as 50 clicks.

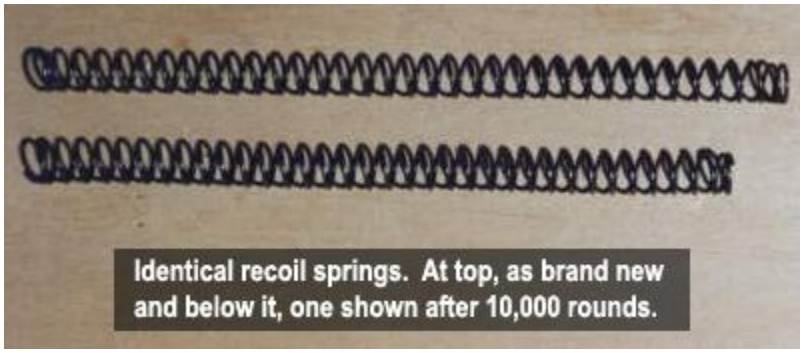
RECOIL GUIDE ROD AND SPRING



The condition of the recoil guide rod is quite important. If it is rusty, dinged up or has an otherwise rough surface, the recoil spring will drag upon it and slow the cycling slide in both its backward and forward motions. A real example is a 1971 pristine Model 41 that I had purchased from a frustrated owner for less than half its street value. In over 30 years, he just could not get it to function, so it sat mostly unfired in his gunsafe. After taking a chance and buying it, I discovered that the recoil guide rod had circular machining marks from one end to the other, the single cause of the extraction problems. Placing the rod in a drill press and sanding it smooth with 400 grit sandpaper solved all problems.

And finally, last but not least, consideration of the recoil spring. While the recoil spring is oftentimes the first thing that gets examined in the case of malfunctions, in this primer, it is the last. Many a shooter has a collection of various lengths and weights of types of recoil springs for his prized pistol. But they were likely purchased (at high expense) to remedy a problem caused by some other reason.

However, recoil springs do have a real life-span. As shown below, a recoil spring will compress and shorten over time.



It is quite likely that the older spring, in its shortened and compressed state, may be responsible for feeding problems. In this situation, a replacement with a new factory spring is probably the best solution. Pouring out extra cash on specialty springs such as the Wolff variable-weight springs is not necessary with a .22 caliber pistol.

FINALE

Once again, complete confidence in the function performance of your pistol is essential for confidence on the firing line. I hope that this document has provided you with the knowledge to solve any problem you might have encountered, and give you that needed confidence!

References

- HERE

TOOLS OF THE TRADE

Facts and Figures About Dot Sights

By John Dreyer

Source: http://bullseyepistol.com/dotsight.htm

The red dot sight is possibly the greatest innovation in competitive pistol shooting in over one-hundred years! This gadget offers the sighting process a level of precision and a simplicity of operation that cannot exist with open iron sights. It also has allowed shooters with failing eyesight to either return to the sport or to stay competitive for much longer.

This article is NOT a product review. It is intended to educate competitive shooters about the principles and characteristics of differing makes and models of red dot sights.

A History of the Dot Sight

The tradition of utilizing open iron sights in bullseye was first challenged with the introduction of the "Bullseye Pistol Scope" by Burris in the 1960s. It featured either no magnification or 1.7x for *"hard holders"* and a innovative reticle that exchanged traditional crosshairs for a big black dot.



The Original
1975 Aimpoint Electronic

Sold for decades by Gil Hebard, it was quite popular on the firing lines, despite being not only quite heavy, but also problematic in low-light situations. When the Swedish firm Aimpoint broke into the scene in 1975 with their original model the "Electronic," it would not be long before the bullseye game would be changed forever.

With any innovation comes imitation. The original UltraDots were mediocre copies of the high-priced Aimpoint sights and quickly gained popularity among shooters in the late 1980s because they were affordable. But it was probably the offering by the Florida-based Tasco company of the very affordable and well-made Japanese sight called the ProPoint that made the biggest impact on the dot sight market. A decade later, there would be dot sights of all shapes and sizes being manufactured by different companies all over the world.

A Myriad of Models and Claims

While dot sights come in both *"tube"* and *"open"* designs, the tube type is superior for the bullseye game. Durability, superior zero-holding, fewer stray reflections and better mounting options are just a few of the reasons.

While the objective lens (tube diameter) of sights range from 25mm (or 1 inch) all the way up to 54mm, anything larger that 25mm may be nothing more than an unnecessary addition of weight. Also, the larger the tube diameter, there is a greater potential for off-axis parallax error as well. Highever, some shooters feel that they can acquire the target after recoil in sustained fire more easily with the slightly larger 30mm sight.

Here are some images of a few of the popular dot sights currently available on the market.



Aimpoint Comp C3



UltraDot MatchDot



Nikon Monarch



Tasco ProPoint 2



Millet RedDot



BSA Red Dots

The Mechanics of a Dot Sight

Understanding how a dot sight works is important in determining why dot sights behave the way that they do.



A cutaway of a typical red dot sight showing the concave reflective lens and LED.

There is nothing magical about them; the system is actually quite simple. The sight includes a concave lens with an extremely thin metallic coating that reflects red light but transmits other colors freely. The "dot" reticle itself is simply a reflection of a light emitting diode mounted inside the sight tube. The result is a red dot that appears as if it is "projected" upon the target as you look through the tube. To make sight (point-of impact) adjustments, two screws that work opposite of coil springs adjust elevation and windage by moving the inner tube with the LED either horizontally or vertically.

Dot Pitch or Size

It is unfortunate, but expected, that a manufacturer's purported specifications on the "size" of their dot is seldom correct. The dot size, usually reported in minutes of angle, is often pure hogwash. While most dot sights are advertised as having 3 or 4 minute dots, only your eyes can tell you what size they really are. For example, the Aimpoint 5000 and the Tasco Pro Point 2 have smaller dots than basic UltraDots, and yet they all claim to have the same size dot!

The new flavor of the month in dot sights is adjustable dot sizes and even reticle shapes. I would suggest dismissing the idea that there is any better reticle shape than the simple dot. However, an adjustable dot size may possibly be an advantage. A historical example is that the black reticle of the ancient Burris scopes were larger than typical red dots. Many shooters reported that a larger round reticle made sight picture and target acquisition easier and wobble more manageable. Today, with adjustable dot sizes, some shooters are changing the dot size larger as they move from slow to sustained fire. Others, who suffer from astigmatism, find that a larger dot appears more round and exact than a smaller one.

The Dot is Not Truly Focused at Infinity

One popular fallacy that is commonly believed is that the dot itself is focused, or projected, at the same apparent distance as the target. By doing tests on four different models of sights, it has become quite apparent that this is NOT the case. With the aid of a SLR camera with a rangefinder focusing screen, by viewing the dot with 24 inches of eye relief, it becomes obvious that the dots on all four models are "focused" anywhere between 30 and 75 feet, regardless of the distance of the target. This distance is the sight's *"sweet spot"* and likely where it will perform best against *off-center* parallax, discussed next.

Parallax is an Issue with All Dot Sights

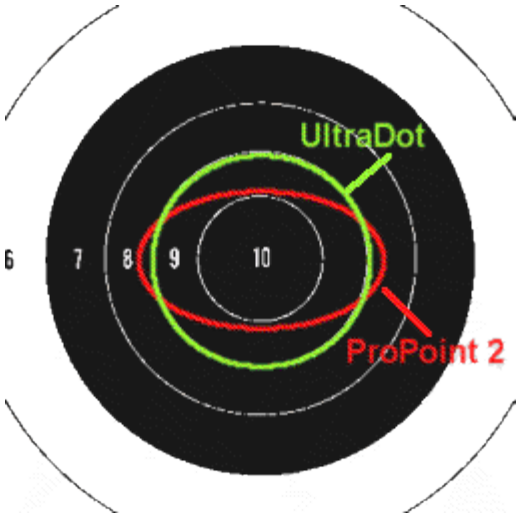
The most popular misconception about dot sights is that some models are blessed with complete freedom from parallax. Nonsense! Parallax exists in all dot sights because of the nature of the sight design itself. While it is true that all sights do adjust for *off-center* parallax at particular distances, they become more vulnerable to the problem at other distances.

What exactly is parallax? Parallax is the "error" that occurs when one of two vectors that are parallel is used as a reference for the other. This is a factor with a dot sight because its reflecting lens is optimized for reflecting the image of the LED right down the center of the tube into your eye. If the pistol is held incorrectly and the shooter views the dot near the edge of the tube, it will not actually be pointed at the actual correct point of aim. In an attempt to correct for this, the reflecting lens is concave. Unfortunately this design is only somewhat effective.

The following experiment analyzes this phenomenon of parallax at varying distances with our two test sights. The sights were mounted to a stable bench and pointed toward a special calibrated target placed at three different distances. While maintaining a consistent 24" eye relief, and keeping one eye closed, I moved my viewing eye to the left/right and top/bottom extremes that the dot was still visible, and charted my observations. The data was recorded as the measured point on the calibrated target at each viewing "extreme" where the dot appeared to actually be. All target coordinates are based on the location of the CENTER of the dot reticle since the two test sights have different sized dots.

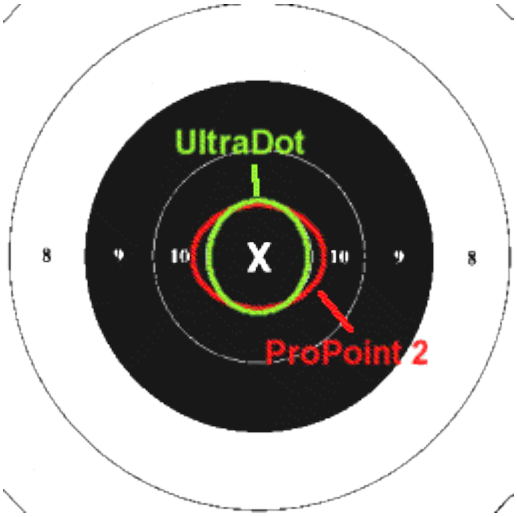
Before continuing, let me assure you that slight off-center viewing with either model resulted in imperceptible errors. If a shooter keeps a consistent rock-solid hold for every shot, parallax will not be as much of a concern. But nonetheless, knowledge of the great POTENTIAL for error is useful and reinforces the values of good technique.

The next three figures demonstrate the maximum potential for parallax with our two test sights as mapped upon the standard target for those three common shooting distances. The circles represent the greatest possible error resulting from parallax alone for each model. As you will see, each model is designed and acts much differently at various distances to the target.

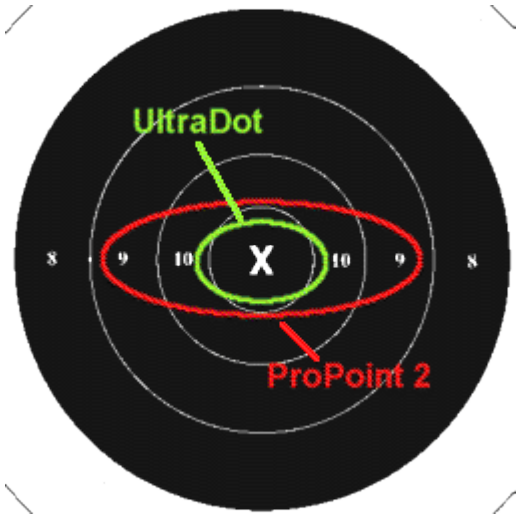


At fifty feet, the UltraDot can vary as much as 0.75" off-center. The ProPoint2 can vary as much as 0.875" horizontally and 0.50" vertically. Both models perform effectively at their worst at this distance.

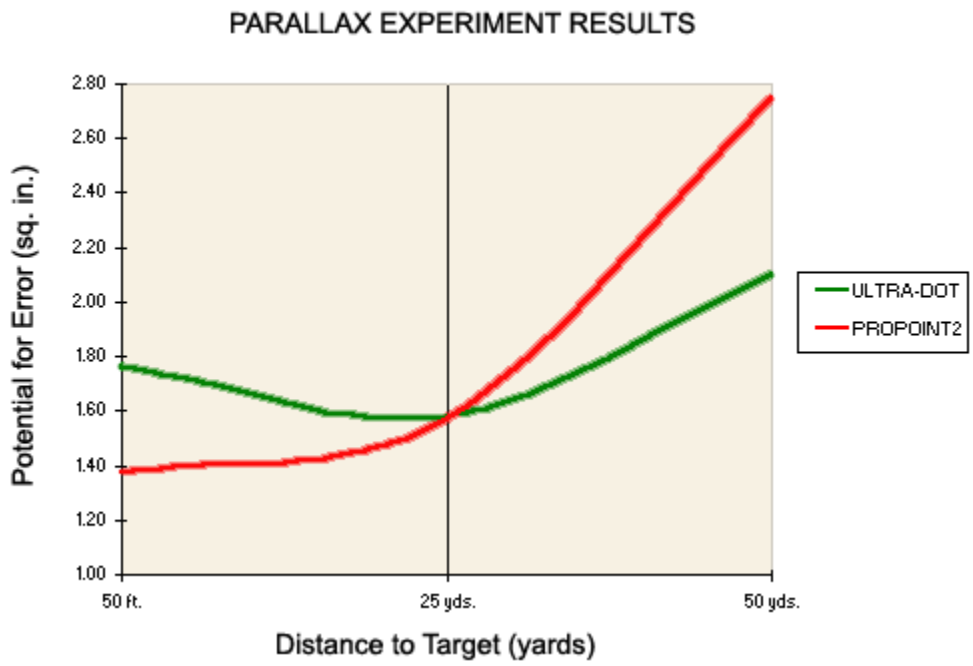
At 25 yards, both models perform effectively very well. Neither would distort the image so much that the actual point of impact would score outside of the X-ring of the timed and rapid fire target.



At fifty yards, the two models again demonstrate their big design differences. The UltraDot at maximum error will still pretty much hold the X-ring. The ProPoint 2 is plagued with errors up to 2.50" horizontally and 0.875" vertically.



To better compare the two sights, I charted these findings below as the "potential of error" for the sights at each distance. This figure was computed by calculating the area of the "circles" above. The results are perhaps deceiving, but demonstrate how different the two sights are designed.

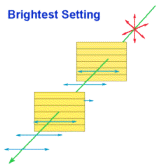


The Function and Mechanics of the Polarizing Filter

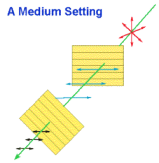
The most useful accessory supplied with a dot sight is the polarizing filter. Attached to the far end of the sight, it functions as a variable light filter to adjust the brightness of the target independent of the dot intensity. It is most valuable on a very bright sunny day.

The accessory is actually a set of two polarizing filters that rotate independently of one another. Each of the two filters is a sheet of transparent plastic with microscopic parallel lines etched upon it. These lines act to reduce the amplitude of light waves that arrive to the filter vibrating in directions other than that of the lines. The best image I could dream up to illustrate this is to imagine having to get a sheet of plywood through a wrought iron fence. Any attempt to get the sheet through other than with the plywood PARALLEL to the fence bars is fruitless, and the plywood bounces back. But the plywood still will get through that fence IN ITS ENTIRETY when it is turned the correct way. This is the same principle with light rays passing through a single polarizing filter.

Now let's add the second filter. Since it rotates independently, its lines can be parallel, perpendicular, or any angle between in relation to the first filter. Now back to the image of the iron fence, let's place another fence section behind the first. If the bars on each are parallel, that entire sheet of plywood will still pass through. If you tilt the rear fence section a little bit, you'll have to cut off some of the plywood to get it to go. As you tilt the rear fence section closer to perpendicular of the first, the plywood will have to be trimmed more and more. This is how the second polarizing filter works in conjunction with the first to reduce the amplitude of light waves.



Here you can see how light waves vibrating with any amplitude can pass through two parallel polarizing filters. This is the most transparent that the filter unit will become.



When the second filter is turned 45 degrees, light waves of greater amplitude than the reduced "openings" are blocked. Waves of lesser amplitude are allowed to pass freely.



Now that the polarizing lines are perpendicular to each other, only light waves with extremely small amplitude are allowed to pass. The filter unit will appear almost opaque.

In Conclusion

I hope that any doubts or misconceptions about red dot sights have been resolved with this article.

Before rushing out to purchase any dot sight, I suggest that you gaze through it to see if it meets your needs. In regards to quality control and longevity, one standard is fairly clear - avoid any brand or model that is manufactured in China.

On a positive note, in this game of bullseye shooting, I have seen more military shooters over the past 15 years at all of the big matches using the basic UltraDot model than any other sight - and without question, these guys are truly the pros of the sport.

Further Reading

- UltraDot (http://www.ultradotusa.com/)

TOOLS OF THE TRADE

The Benefits of Custom Grips

By John Dreyer

Source: https://www.bullseyepistol.com/goodgrip.htm

SPECIAL THANKS TO RANDALL FUNG FOR HIS RESEARCH DATA AND ILLUSTRATIONS!

Introduction:

In much the same fashion that a champion long distance runner values well-designed shoes, the competitive pistol shooter must consider the grips on his pistol. Recognize that the grips are the direct link between a shooter and his weapon, and grip design is critical if maximum control is desired. No shooter can be successful unless he can take a firm, yet relaxed grip on his pistol that is identical each time. Likewise, he must be able to hold directly on target without any movement while pulling the trigger through its break and then be able to quickly recover after recoil. Well designed and fitted custom grips encourage this process to become automatic. Veteran champion shooter Gil Hebard has praised custom anatomical grips as being "worth every cent," and he is right.

The Psychological Advantage of Custom Grips:

While an accurate pistol is an important part of the equation of becoming a great shooter, overcoming the common fundamental errors is far more significant. Most shooters do not shoot to the capabilities of their target pistol, regardless of its make or model. When one reaches a certain level of competence, pistol shooting becomes more of a mental challenge than a physical one. Your mental state greatly affects your performance at the competing levels and is certain to be affected by your subjective feelings about your equipment. If you can possibly blame your equipment and its functionality, there will be an adverse effect on your confidence and performance. However, if you have the utmost confidence in your equipment, you will perform to your best ability. If your pistol's grips fit you like a glove, then of course you will have confidence, thus eliminating worry over a vast many fundamentals. Even the aesthetics of your equipment can contribute to your performance.

The Mechanics of Anatomical Grips:

A key attribute of anatomical grips is the ease in taking an identical hold on the pistol every time. This is a huge factor in consistency and proper technique in every stage of the game. This feature, control of the placement of the web of the hand, is easily achieved by two separate elements of the grips, the palmrest and the thumb cutout. If the grips are fitted properly, one gets the impression that his pistol is merely an extension of his hand.



Proof why you need anatomical grips!

Another primary function of anatomical grips is to allow you to "clamp" your hand to the gun, and yet allow your trigger finger to operate independently. This is particularly important with pistols that are muzzle or top heavy. Two elements, the palmrest and the middle finger support, work together in this task. Hand contact with the rear of the palmrest and upward contact by the middle finger help offset the forward balance of the muzzle providing increased stability with less strain.
After a shot has been fired, the effects of recoil come into play. Various angles of force effect the direction of the recoil and these must remain very consistent in order to achieve small groups. Likewise, the recoil affects your grip and the slightest roll or movement can change the angles of force effecting the recoil for the following shot. Unless you compensate consciously, or subconsciously for this difference, you might get a shot placement other than you expected, or a "flyer." As the shot breaks, the upward climb of the barrel in recoil, or muzzle jump, is limited by the overhang on the rear of a pistol. But that is not sufficient because the overhang is quite close to the "pivot point" of recoil. This is another task for the palmrest, whose function now is not only to keep the hand secure between the overhang and itself, but by making contact with the little finger, limits the upward pivot of the muzzle.

Manufacturers of Anatomical Grips:

In the case of European pistols, models intended for competition are usually equipped with anatomical grips. Provided that they fit your hand comfortably and are adjustable, they will suffice. For those not equipped with anatomical grips, there are plenty of manufacturers at your disposal. Here are the most popular:

1. Randall Fung Match Grips. San Francisco, CA. *Available directly from Randall.*
2. Thomas Rink Formgriffe. Germany. *Available through Larry's Guns.*
3. Morini Grips. Switzerland.
4. Nill Griffe. Germany.

.

Further Reading

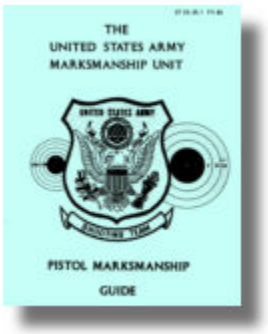
- RANDALL FUNG (http://jnb.com/~funggrip/)
- Randall Fung Match Grips (http://jnb.com/~funggrip/)
- Thomas Rink Formgriffe (http://www.formgriffe.de/)
- Morini Grips (http://www.morini.ch/)
- Nill Griffe (http://www.nill-grips.com/)

PERFECTING TECHNIQUE

Pistol Marksmanship Training Guide

By United States Army Marksmanship Unit

Source: http://www.bullseyepistol.com/amucover.htm



This section of the website is dedicated to the Army Marksmanship Unit's Pistol Training Guide. Quite simply, this is the "Bible" of Bullseye Pistol Shooting. No one serious about the game of shooting should miss the information in this booklet.

Use the green buttons to the right to navigate through this massive treasure of information.

PERFECTING TECHNIQUE

Foreword to the Training Guide

By Army Marksmanship Training Guide

Source: http://bullseyepistol.com/chapterf.htm

The United States Army Marksmanship Unit has compiled this guide with the intent of providing assistance to pistol marksmen throughout the United States. Much of the information has been contributed by championship caliber U.S. Army pistol shooters and coaches. The text is designed for ready reference and contains the fundamentals of pistol marksmanship and the most advanced techniques known to these experts. The data presented represents a comprehensive coverage of the many facets of pistol marksmanship and related subjects. Portions of the information contained herein were obtained from the research of authoritative articles, personal interviews and observation of foreign competitions. Noteworthy contributions were received from many experienced persons closely associated with pistol activities not connected with military training.
This guide is dedicated to the individual marksman who is seriously interested in improving his performance in the art of pistol competition. This publication was prepared in compliance with that portion of the mission of the United States Army Marksmanship Unit which requires the continuing improvement of training methods and techniques so as to raise the standards of pistol marksmanship skill in the United States Army. The information published in this book is not for sale and/or resale for commercial purposes but is intended for furthering the aims and objectives of the civilian/military marksmanship programs being conducted in the United States. Suggestions, criticism, or comments that will contribute to attaining these objectives are solicited and may be directed to the Commanding Officer, USAMU, Fort Benning, Georgia 31905. STANLEY PARMENTIER Colonel, Infantry Commanding * U.S. GOVERNMENT PRINTING OFFICE: 1979 - 640 - 000 - 32 ... REGION NO. 4 Special thanks to Richard Harbordt for taking the time and doing an excellent job of scanning the text and pictures from an original manual and to Chuck Falls for making it available to Bullseye-L subscribers. This document is a U.S. Government publication and therefore is public domain to be downloaded, reprinted and distributed as you desire.

Further Reading

- Richard Harbordt (mailto:richardh@texas.net)
- Chuck Falls (mailto:candjfls@eagnet.com)

PERFECTING TECHNIQUE

Elements of Pistol Shooting

By Introduction: Army Marksmanship Training Guide

Source: http://bullseyepistol.com/chapter0.htm

The fundamentals of pistol marksmanship embrace all of those physical factors essential to the firing of an accurate shot. Accuracy, in this sense, assumes that the weapon is zeroed, that a high degree of inherent accuracy exists in both the weapon and the ammunition, and that the firing is taking place under ideal conditions.
Essentially, accurate shooting with a pistol requires no elements other than those described in the following sentence: ALIGN THE SIGHTS PROPERLY ON THAT PART OF THE TARGET REQUIRED FOR YOUR GROUP TO CENTER IN THE TARGET AREA AND CAUSE THE HAMMER TO FALL WITHOUT DISTURBING THAT ALIGNMENT. All elements of pistol shooting such as position, grip, sight alignment, breath control, trigger control, physical condition, and psychology of shooting, when perfected, simply enables the shooter to perform the action described in the above key sentence. In order for a shot to be accurate, it is first necessary to make sure that the pistol will be held as motionless as it is possible to do so during the time that the shot is being fired. The stance assumed by the shooter must provide the greatest stability possible for both the shooter's body and the weapon. To obtain a minimum arc of movement the shooter must give the pistol a definite stability of direction. Proper body position points the pistol directly toward the target with no tendency to drift or move to either side. Likewise, the vertical movement of the pistol is confined to the aiming area. Breathing is accompanied by the rhythmic movement of the chest, and in order to keep the pistol as immobile as possible the shooter must hold his breath for the length of time required to deliver an accurate shot. To obtain correct sight alignment, it is necessary for the shooter to grip the pistol in a manner which guarantees that he is holding the pistol firmly and that trigger pressure is applied straight to the rear. The delicate balance of sight alignment and minimum arc of movement can be easily disturbed if the trigger is activated in a manner which causes excess movement. However, since the shooter cannot achieve complete immobility when assuming the stance and position, the trigger has to be pressed during some movement of the pistol. In order to deliver an accurate shot within his ability to hold, the shooter must not only press the trigger evenly, but he must to so with correct sight alignment. The size of the shot group will, therefore, not exceed the dimensions of the arc of movement, provided the shot breaks as a surprise and no reflex action of muscles disturbs the delivery of the shot. To help the shooter acquire the necessary knowledge to master all the factors that control his shooting we shall analyze In detail each separate element of accurate shooting - stance, position, grip, holding the breath, sight alignment, and control of the trigger. Also included will be certain methods of training that will accelerate the shooter's

development into a champion pistol shot. This status is achieved only after the shooter has mastered the technique of executing the fundamentals.

PERFECTING TECHNIQUE

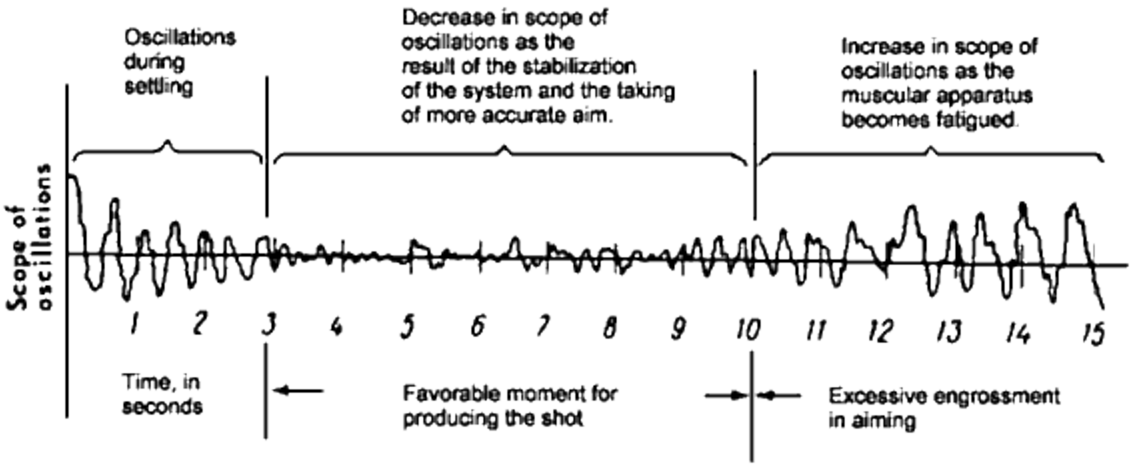
Attaining a Minimum Arc of Movement

By Chapter 1: Army Marksmanship Training Guide

Source: http://www.bullseyepistol.com:80/chapter1.htm

It is necessary during firing to press the trigger under varying conditions of pistol movement in conjunction with correct sight alignment. In order to apply coordinated pressure on the trigger, the shooter must wait for those very definite times when all control factors are optimum and firing conditions become favorable. The rule that must be observed as the first step in attaining control of your shooting is: "You must never attempt to fire until you have completely settled into a minimum arc of movement."

In order to learn how to fire a shot at the proper time, the shooter must make analysis of the time needed to settle and the duration of the minimum arc of movement. The entire system, consisting of the shooter's body and the pistol, always undergoes a degree of movement. This is sometimes a pulsating, swaying or erratic arc of movement during aiming and firing a shot. The cause of this movement aside from conditions such as weather, is the action of the muscles maintaining the shooter's body in a definite position. Other action such as blood pulsation, causes movement of individual parts of the shooter's body and the pistol. The nature and extent of the arc of movement changes within the time being devoted to delivering a shot. For example, when the shooter is first getting his sight alignment and has not yet had time to settle his body and pistol, the extent of the movement is relatively great. As the body becomes balanced and the aiming is more precise, the arc of movement minimizes. After a certain length of time, the minimum arc of movement begins to increase, because the muscles begin to fatigue, and the shooter does not have enough air in his lungs to continue holding his breath. If we record the arc of movement, we will see a wavelike line with varying amplitude of oscillation (Figure 1-1).



Arc of Movement

Figure 1-1. Basic Scheme of Minimum Arc of Movement.

It is obvious that under such circumstances the shooter must begin his smooth pressure on the trigger while not devoting too much attention to the arc of movement as long as it remains at the minimum. Continue to apply pressure on the trigger and intensely concentrate on keeping the sights in alignment. The resulting five to seven second period is the most favorable time for firing an accurate shot. Taking into consideration the direct relationship between accuracy of shooting and the degree of immobility of the pistol when the shot is being delivered, the marksman must give greatest consideration to the selection of a stance, a position, a grip, and a means of breath control which will guarantee the greatest stability to both the pistol and the body. The relatively small degree of movement thus obtained provides a stable foundation, permitting use of the other fundamentals. A. THE STANCE. The excellence of the stance is a major factor in creating conditions for maximum control. A high degree of control is necessary for the delivery of an accurate shot. Every individual possesses a combination of individual characteristics that are peculiar to him alone. Among these are height, weight, proportion of body, development of muscle system, etc. It follows, then, that there cannot be any definite, all-purpose stance which applies equally to all shooters. Therefore, the shooter himself, on the basis of his own particular configurations, must find the variation of stance which provides the greatest degree of stability for his body. 1. The Main Requirements of the Stance: The assumed stance is the position of the human body to support a pistol aimed at a target. Despite the great number of physical differences encountered in any cross-section of shooters, the stance must provide for: a. The greatest possible degree of equilibrium and stability in the body-weapon system with the least possible strain on the shooter's muscles. b. A head position which will allow for the most efficient use of the shooter's eyes throughout the sighting and aiming process. Throughout the process of training it is necessary, therefore, for the shooter to exercise special care in the selection of a

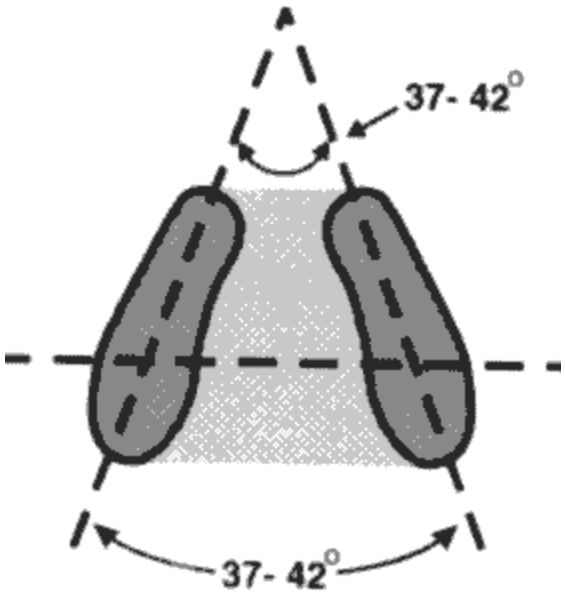
stance. The development of a poor stance should be detected and corrected early in the training program. Otherwise, it may require the breaking of deeply ingrained habits later. Considering the role played by the muscles, bones and ligaments in the creation of stability in the shooter's stance, it is necessary for the shooter to understand the makeup of the human body. See Section Five, Annex I for supplemental information entitled, "Characteristics of the Human Body Relevant to Stance, Position and Grip ". 2. Assuming the stance: a. When assuming the firing stance, the head must be held as level as possible, so that the shooter can see the target directly in line with the arm and sights. It is necessary to take all steps to eliminate the tilting of the head to the right or left or an excessive tilting forward. It is not necessary to look sideways or to look at the sights from beneath the eyebrows. The head should not be pushed forward closer to the rear sight; neither should the head be tilted back excessively. This causes undue tension upon the neck muscles and, as a consequence, a slight movement of the head develops from fatigue. This may hinder the maintenance of perfect sight alignment. b. When assuming a firing stance, the shooter must support the extended arm holding a weapon. As a result, the muscular system undergoes considerable strain. It must not only maintain the shooter's body in a definite position but must also exert a counteraction to the rather large weight of the suspended gun. c. A shooter supporting a weapon constitutes a single system with a common center of gravity (Figure 1-2). Since the entire system is in equilibrium only when its line of gravity runs through the support area, (Figure 1-3), the holding of the weapon causes a change in the relative position of the individual parts of the body. A compensating displacement is brought about by the necessity to create a counteraction to the weight of the pistol and supporting arm. This compensating displacement of the parts of the body changes the shooter's posture. As a result, when he assumes a firing stance, his body takes on an asymmetric position which is unnatural. The preservation of the body's equilibrium in this unnatural posture requires that a greater load be placed upon the muscles and ligaments reinforcing the movable portions of the body. d. The shooter has the task of finding for himself a suitable stance which will achieve immobility of the body without an excessive strain on muscles. Let us assume that the shooter takes a stance for firing that will preserve the natural, erect posture of the body. He will strive to keep it erect with small compensating deviations of the muscle system. Thus, the extended arm holding a pistol places great tension on the muscles in the back and shoulders. In addition, if the shooter's figure is examined from the side, it will become obvious that when the shooter's body is kept stiffly erect, the body will be slightly unstable. The keeping of the body rigid will result in early fatigue and cause undesirable movement. e. What posture should be given to the body in order to best support the weapon with the least expenditure of muscular effort?



Center of Gravity

Figure 1-2. Situation of the Over-all Center of Gravity of the Entire System

The shooter must hold his body in something less than an erect posture, with a slight rearward bend in the back and the pelvis brought slightly forward. In this posture, the body has the vertical line of the center of gravity shifted back of the axis of the hip joints. In such a pose, the body is kept stable in the hip joints not so much by the work of the muscles, but by strong ligaments. The relaxed immobility of the body is attained by counterbalancing the weight of the upper body against the extended firing arm and pistol, and transfer much of the weight to the spinal column. f. The selection of the most stable stance will include giving the body a certain degree of bend. As shown by practice, the shooter has nothing to fear in giving his body an asymmetric pose. g. The stability of the firing arm and weapon depends to an extent upon the correct placement of the feet. This determines the support area for the shooter's body. The most stable and most comfortable stance will be when the feet create a support area in the shape of a trapezoid with the feet placed apart, approximately shoulder width. The toes should be spread apart slightly (Figure 1-3). This placement of the feet creates not only a comparatively large support area, but is also the most favorable positioning of the feet for avoiding muscular strain in the legs.



Foot Placement

Figure 1-3. Placement of the Body's Support Surfaces - Both Feet - in Relation to One Another, Creating the Support Area for Firing.

h. When assuming the firing stance, the shooter should not attempt to bring the legs too close to one another. Narrow placement of the legs decreases the support area and will result in a lose of stability, causing movement of the firing arm along the horizontal. Do not place the legs too far apart, as this creates undesirable strain on the inner arches of the feet, straining the leg muscles and holding the hip joints rigid, which leads to fatigue and an increase in the arc of movement. I. In order for the stance to be a stable one, the shooter must, first, distribute the weight of his body, the arm and pistol evenly on both legs; second, the load placed on each leg must pass through the middle of the foot or close to the balls of the feet. When the weight of the body is distributed in this way the body's line of gravity will run through the middle of the support area. The stance will be the most stable when the muscles of both legs carry the same load. The coordinated work of these leg muscles results in the body's weight being alternately shifted in slight corrective moves in order to maintain balance. j. The degree of strain upon the muscles and ligaments of the knee joints is of importance in the stability of the stance. The insufficiently rigid position of the knee joints will lead to an increase in the body's movement as a whole. By holding one leg straight and keeping the other one partially bent, varying tensions will exert excessive tension on the leg muscles. Inflexible straightening of the legs also causes tension of the leg muscles which leads to a loss of stability. k. The group of muscles which do not directly participate in maintaining the shooter's body in the vertical position or holding the pistol aimed at the target is the muscles of the left arm and hand, the left-hand portion of the chest, the neck muscles, the

abdomen, and the buttocks, must be relaxed as much as possible. It is necessary to properly position the left or free arm and hand (for right-handed shooter). The free hand should be inserted into the left aide pocket in a relaxed manner, or you may hook the left thumb over the waist belt. In relaxation of the left arm and shoulder, the free arm must not be allowed to hang loose, as any wind or recoiling of the body during firing will cause the free arm to swing, transferring to the body any movement. l. The pistol arm should be extended with the wrist stiff and the elbow locked without strain. The arm must be straight, firmly extended and with no unnecessary tension of the muscles. This establishes solid arm control. m. The body weight center of gravity should be brought forward slightly from the center of the support area, with a very slight shift toward the tips of the foot to reduce the action of the balance correcting mechanism. This is apparent in the alternate tensing and relaxing of the muscles of the legs, abdomen and lower back. This action to regain equilibrium is continuous. The body cannot remain motionless because the equilibrium does not remain constant. The constant corrective process causes an almost imperceptible weaving or sway. 3. The stance factor is so essential that a step-by-step summary of all of the points important to a proper stance is in order. a. Stance must provide for: (1) The greatest possible degree of equilibrium and stability of the shooter's body and weapon with the least possible strain on the shooter's muscular system, and the smallest movement possible of any part of the body, the shooting arm and the pistol. (2) A head position which will allow for the most favorable conditions for the operation of the eyes during aiming. b. During training, the shooter must take special care that he is not developing an incorrect stance or body posture which will require a breaking of habits later. c. The shooter should become familiar with assuming the proper stance and practice getting the same stance each time it is assumed. The requirements are: (1) The feet are separated about the width of the shoulders or slightly less, toes pointed out slightly. (2) Stand up erect and relaxed. (3) The legs should be straight, but not stiff, knees firmly straight but not rigidly locked. (4) The hips should be level and in a natural position. (5) The abdomen should be relaxed.



Stance

.



Figure 1-4. Three Views of an Effective Stance.

(6) The shoulders and head should be level. No humping over or slouching with an unnatural tilt to the head. (7) The non-shooting arm should be relaxed, the free hand in the side pocket or thumb hooked over the belt, not hanging loose. (8) The pistol arm should be extended with the wrist stiff and the elbow locked without strain. (9) The body weight center of gravity should be brought forward slightly from the center of the support area, with a very slight shift toward the tips of the feet to reduce the action of the balance correcting mechanism. B. POSITION. When preparing for accurate shooting, it is insufficient merely to assume a comfortable and stable stance. You must be able to aim or point at your target in a natural,

consistent manner. Improper position will affect your ability to establish or maintain the hold in the center of the aiming area. Before each shot or string of shots, it is necessary to check the correctness of the assumed position with respect to the target. Avoid unnecessary muscular tension in the effort to hold in the aiming area. An adverse effect upon the movement of the shooting arm and weapon is caused by extra muscular effort. Match competition requires the shooter to fire a large number of shots in one day. It is necessary to find the most effective position, which will allow the shooter's body to assume an identical position over a long period of time without causing undue strain upon the muscular system. Any feeling of discomfort, fatigue, constraint or a continuing necessity to correct the orientation of the body to the target distracts the shooter's attention from the principal goal; the uniform, absolute control of an accurate shot. The finding of the most effective position when firing will provide for a consistent pointing of the shooting arm and weapon and provide a free and unforced feeling of natural alignment with the target during the entire period of shooting. 1. The shooter must position himself so as to naturally align or point himself and his weapon with the target so the hold will remain in the desired area without a tendency of the shooting arm to drift away from the aiming area. 2. To orient or align yourself properly with the target, use the following method: a. First face approximately 40 to 50 degrees from the target using the methods of assuming the stance previously mentioned. b. Look at the target by turning only the head. Keep the head level and turn it far enough toward the target to allow the eyes to look straight out of the head. c. Raise the arm to align with the target. Close your eyes, raise your pistol arm a foot or two above the horizontal and then allow it to settle back relaxed and naturally to the horizontal. Completely relax the arm and shoulder not being used. Repeat this procedure once or twice and settle into a natural point. A true, natural point is not obtained with the eyes open. d. After settling into a natural point, open your eyes to check if your arm and pistol are aligned with the target. If the pistol has settled in the center of the target, you have your natural position. e. If the arm settles to one side of the target center, move your rear foot in the direction of error. Maintain without change the stance of the body as a unit from the feet to the shoulders and head. Swing the whole body by shifting the position of the feet until the arm and pistol are naturally aligned on the center of the target. Tests such as this will readily indicate your natural position. In no instance must the shooter correct errors in hold by moving the arm independent of the body. This type of correction is purely artificial and the arm will revert to the original error after recovery from the recoil of a shot. f. Recheck after each error is found until no error exists. 3. The shooter must always remember that the improvement of his marksmanship skill requires an unceasing search for an even better stance and position. The position and stance assumed must not be considered as something constant. As marksmanship skill develops, changes in the stance and position are necessary in order to improve performance. 4. Many expert marksmen, as a result of long and persistent training have completely developed their stance and position to the point of automation, sometimes not even noting its individual shortcomings. It sometimes happens that some

experts, even though they know about their shortcomings, do not attach the proper importance to them. Only when the individual shortcomings in position or stance become a serious hindrance to their progress do they begin to change. The overwhelming majority of the leading shooters actually work seriously and creatively to improve themselves by evaluation of their position and stance. 5. Young shooters must not blindly copy, and instructors and coaches must not mechanically, without any analysis, instill in their pupils a particular variation of position or stance. It is necessary to make an intelligent approach to the problem of selecting the particular stance and position that is acceptable to oneself, taking from the experts desirable aspects and rejecting undesirable ones.



The Grip - Taking

Figure 1-5. Getting the Grip.



The Grip - Results

Figure 1-6. The Grip.

C. GRIP. The proper grip is one which provides the shooter with the maximum control of the weapon. To maintain a natural sight alignment, he must hold the weapon firmly and be able to apply positive, straight to the rear pressure on the trigger that will not disturb sight alignment. 1. Uniformity: For maximum control, all of the requirements for a proper grip must be uniformly applied at all times. 2. Requirements: The proper grip on a pistol is one that meets the following requirements: a. The grip should be such that the front and rear sights will stay in natural alignment without any extra effort to maintain the relationship. Without this feature, there will be a tendency for the front sight to move over to one side of the rear sight notch, or be moved above or below the horizontal surface of the rear sight. Sight alignment, quickly regained after recoil without the need for correction, speeds up recovery and improves timed and rapid fire control. Maintaining sight alignment should be an effortless action before the next shot. Positive trigger pressure can be applied if the sight alignment is being maintained without effort. Sight alignment is easier to maintain if no adjustments are necessary such as moving the wrist or head. b. Grip the pistol firmly enough while firing a shot so that shifting or slipping of the grip will not cause loss of control of the pistol. Recovery from recoil for the next shot in sustained fire is seriously hampered by the loss of sight alignment. The trigger pressure under these conditions is usually reluctant and timid. Unless the proper grip can be renewed quickly, (next to impossible in the middle of a timed or rapid fire string) maintaining sight alignment during the application of positive trigger pressure is a difficult operation. The tighter the grip, short of setting up a tremble, the better the control. The degree of pressure that should be exerted in gripping the pistol is determined by the condition of the muscles that do the gripping. Frequent practice, experience and certain exercises promote a strong grip and have a bearing on when a tremble will begin. c. There must be no change in the tightness of the grip because a variation of gripping pressure will adversely effect sight alignment. Any degree of tightening or loosening of the grip from an established grasp will cause the sights to move out of alignment. The pressure of the grip must remain constant. It cannot be increased or decreased as trigger pressure is being applied because sight alignment will be altered. d. The trigger finger should apply positive pressure on the trigger as an independent action, completely free of the other muscles of the shooting hand. The trigger finger should not touch the stock or the frame of the pistol because of the added friction and drag on applying trigger pressure. Dry fire a few shots watching the front sight carefully. If the front sight moves at the instant the hammer falls, reposition the trigger finger to the left or right, up or down, on the face of the trigger. Repeat the dry firing and adjusting the position of trigger finger until the release of the hammer causes no movement of the front sight in the rear sight notch. e. There can be no variation in the grip from one shot to the next, from one series of shots to the next, from one day's shooting to the next, ad infinitum. In the final analysis, there is only one correct grip for each shooter. Each type of pistol, caliber . 22, caliber . 38, caliber .45 has its peculiarities and the shooter must adapt to each. The proper grip can be discovered through trial and error,

practice and analysis. It must become, by extensive use, a familiar operation that eventually can be assumed without much difficulty. When the experienced shooter checks his grip out before shooting, it seldom needs adjustment. One of the frequent variations of grip that plagues new shooters is the grasping of the pistol grip with the hand slightly displaced to the right or left from the normal. As a result the placement of the trigger finger on the trigger will be different, thereby jeopardizing the requirement that the trigger be pressed straight to the rear. f. The grip must be as comfortable as possible. The muscles of the hand and lower arm, after sufficient time has passed for the hand to become accustomed to the added stress, should experience little discomfort from the way the pistol is placed in the hand. If the grip is awkward and possible cramping and the hand muscles continue to tire easily, look for another solution or use an exercise device to strengthen the hand. To avoid the formation of painful, blisters, calluses and cracked tissue, reduce the tendency of the skin to stretch. Tautly stretched skin may also pull or exert force on the pistol frame in such a way as to cause eight alignment deviation. An equalization of the stretching of the skin and muscles of the gripping hand is paramount. Straight-in contact should exist between the skin of the fingers and palm and the surfaces of the frame and grips when the gripping pressures are brought to bear; not a sideward, sliding or grazing pressure. g. The force of recoil must be controlled by being transmitted straight to the rear into the shooting arm. Recoil against the base of the thumb, which causes the weapon to twist in the hand, will allow a shift or grip and/or a bending of the wrist. Either event jeopardizes quick recovery from recoil in timed and rapid fire. The pistol should be held by being gripped normally, not by a choking grasp that endeavors to press on the stock in an all enveloping grab. The best points of pressure to hold the sight in alignment are the semi-flat grips on each side of the frame. However, the gripping hand cannot exert equal pressure on each of these surfaces simultaneously and such pressure would not overcome the effect of recoil. Therefore, the obvious pressure points of the shooting hand that will channel the effect of recoil straight to the rear and allow relative ease in maintaining sight alignment are: the middle bones of the three lower fingers, the base of the thumb high on the stock, the depression on the center of the heel of the hand, and last, the base joints of the four fingers along the upper palm. The primary pressure points on the .45 caliber pistol are the front surface of the grip and the mainspring housing-grip safety surfaces. The secondary points are: high on the left side of the stock near the slide lock and the forward curve of the right grip, each of which have to have gripping pressure applied equally to prevent loosening of the over all grip, and to maintain sight alignment. h. Holding the grip too long without an occasional relaxation will result in early fatigue. Fatigue destroys control. Excessive force of gripping for control of the pistol assures that fatigue will exist if the gripping power of the hand is weak. Undue fatigue in the muscles of the hand and forearm will also cause erratic application of trigger pressure. The tremble level is lowered to a point where the shooter cannot hold the pistol still, even for a few seconds, while trigger pressure is being applied. 3. Method of getting the proper grip: The proper grip must conform to all of the foregoing

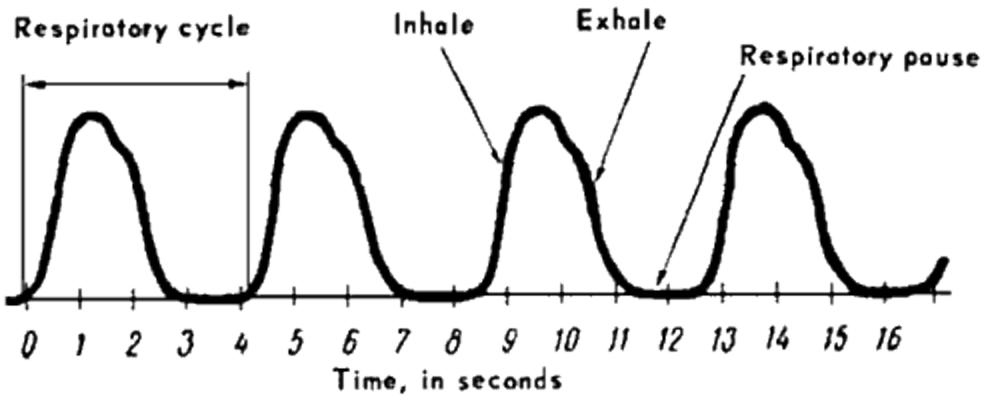
requirements plus it must be a hard grip and it must be adapted to the hand of the individual shooter. NOTE: FOR THIS INSTRUCTION IN OBTAINING THE PROPER GRIP, THE WEAPON IS THE .45 CALIBER SERVICE PISTOL. THE FOLLOWING STEP-BY-STEP SEQUENCE WILL PROVIDE THE PROPER GRIP: a. With the non-shooting hand, pick up the pistol by the barrel and of the slide, being careful not to mar the blackened sight and keep the muzzle pointed down range. b. Spread the index finger and thumb of the shooting hand apart to form a "V", with the thumb held slightly lower than the index finger. c. Bend the wrist slightly downward to obtain proper angle of contact. d. Fit the pistol into the "V" of the thumb and index fingers by seating the grip safety straight and firmly into the loose "web" of akin in the "V". e. Press downward on the barrel to pivot and push the mainspring housing firmly against the inside of the bulge of flesh at the base of the thumb and into the depression in the approximate center of the heel of the palm. f. Stretch the fingers forward, letting the trigger finger come to rest flat against the pistol frame just above the trigger guard. Safety dictates the trigger not be contacted at this time. g. The lower three fingers should come to rest closely touching each other, with the center bone of each finger resting on the curved front surface or "front strap" of the receiver. Little or no pressure should be exerted on the finger tips extending around the front strap to the surface of the left handgrip. Pressure exerted on the front strap by the little finger should be lighter than that brought to bear by the middle and ring fingers. Too much pressure with the little finger may cause the muzzle to depress slightly, resulting in the front sight aligning low in the rear sight notch. h. The thumb should be raised to a level higher than the index or trigger finger. Only the joint at the middle of the thumb is high against the stock in the vicinity of the slide safety. The end of the thumb is turned up and away from the stock as it has no function. Pressure exerted on the aide of the pistol by the end of the thumb has a tendency to disturb sight alignment. The thumb should not exert great pressure on the aide of the pistol as early fatigue will result. Only required substantial supporting force should be exerted to hold the weapon firmly in place in the shooting hand. i. A controlling grip can be affected by the three lower fingers directing primary pressure on the front strap straight to the rear, pressing the mainspring housing and grip safety firmly against the side of the center depression and the heel of the palm at the base of the thumb, and the loose flesh in the "V" of the thumb and index finger, respectively. This can be compared to a vise with the inner surfaces of the palm as the stationary jaw of the vise and the three lower fingers pressing on the front strap of the pistol as the moving jaw. j. The non-shooting hand should be used to adjust the "fit" of the pistol into the shooting hand. A slight rotation of the weapon in the gripping hand as it is alternately gripping and releasing will allow the equalizing of a forceful grasp. The gripping hand must reach around to the right far enough to allow the trigger finger to reach into the trigger guard and also to position itself on the trigger at the exact point at which the trigger pressure can be applied straight to the rear. According to the size of the hand, the trigger finger will apply pressure with the tip, ball of the first section or the crook of the first joint or elsewhere. The primary concern is not what portion or

spot along the trigger finger is the standard point of contact, but at what spot on the finger you can bisect the trigger, press straight to the rear without disturbing sight alignment. k. When the "fit" is correct, remove the trigger finger from the trigger, free the pistol from the non-shooting hand and tighten the grip with great force until a tremor is noticed. Release a small percentage of this gripping pressure immediately, enough so that the tremor disappears and leaves the shooter with a hard, solid grasp that will result in absolute control. The tighter the grip, the better the control. The shooter is now exerting correct pressure for maximum recoil control. 4. Checking For Proper Grip: The proper grip is a natural grip that will meet all the requirements in paragraph 2, above. To assure a proper grip, it should be checked against the requirements. A deciding factor in knowing whether your grip is proper is one of familiarity. By use of the proper grip innumerable times, a flaw is immediately sensed. a. To assure the sights will stay in alignment, the following test is made: extend the shooting arm and observe the sight alignment. If the front and rear sights are out of alignment, grasp the barrel with the non-shooting hand, loosen the grip sufficiently to slide the pistol in the hand, rotating it slightly away from the direction of error in sight alignment. Re-grasp the pistol firmly and extend the arm. Check the alignment without an effort being made to align them by wrist or head movement. If the alignment is natural, you may check for maintenance of sight alignment. With the arm extended, close the eyes, raise and lower the arm and settle. Open the eyes and observe. If the alignment has deviated, reposition the pistol in the shooting hand and repeat the closed eye test until natural alignment of the front and rear sights is achieved and maintained. During shooting, a constant check should be conducted of the tendency of the sights to continue to align themselves. The grip obtained at the beginning of shooting will not necessarily remain correct because the jolting recoil and build-up of fatigue will require correction to the grip to maintain sight alignment. b. To check for a grip firm enough to prevent shifting after making sure the pistol is unloaded, have the coach bump the pistol rather forcefully, up or to the aide with the heel of his hand. Also, have the coach grasp the pistol by the barrel and make an effort to tear it from your grasp. c. To check for variations in tightness or correctness of grip, it is best to dry fire a few shots before live shooting starts and watch for slight variations in sight alignment. d. Checking for independent trigger action should be accomplished before shooting by a visual check of the trigger finger clearance from the grip. Check by dry firing to detect any drag or undue friction noticed in the trigger. Also, check for a sympathetic tightening of the muscles of the hand as trigger pressure is applied. This can cause as much disturbance of sight alignment as the failure to press the trigger straight to the rear. e. The rapid onset of fatigue and soreness of the shooting hand is usually the result of an incorrect grip. f. Checking for straight to the rear recoil directly into the shooting arm and shoulder can best be done in practice with an unloaded pistol by having a coach or team mate stand in front of you and forcefully and abruptly push against the muzzle of your tightly gripped pistol driving it straight back toward your shoulder in simulation of recoil action. 5. Aids to Developing a Good Grip: The great pistol shooters have: strong hands

and a hard grip; a method of gripping without change unless analysis dictates a change that will improve it; a different grip mastered for each shape of stock or different type of pistol; molded, shaped or custom grips, that fit perfectly; and if they use powdered rosin or a like substance, they use it every time the hand becomes moist before they grip the pistol. a. The "top guns" have a grip like a vise. Exercise devices such as rubber balls, spring grip builders, etc. will develop a strong grip. Exercise devices require constant use. Another approach, to reduce reliance on artificial exercisers, is to engage in work or a sport that places demands on your manual strength and dexterity, for example, chopping wood, digging in the garden, using hand clippers on the hedge; playing tennis, baseball, ping-pong, etc. Use of the hands in meticulous work also develops an exacting touch and coordination that is valuable to the pistol shooter. b. Never thoughtlessly change your grip. A correct grip is a precious commodity. It evolves from much hard work, thinking, and planning, plus painstaking analysis. Each satisfactory grip found among the better shooters comes from trial and error. The good grip that is the end product of much effort should not be changed except when sharply critical analysis dictates a change that will improve it. The shooter who is desperately changing his grip hoping that he will chance upon the right solution will generally lower his scores. In the event that a better score is fired under these conditions it comes on an occasional basis with no tangible reason for the improvement. Analysis and trial, in a never ending quest to improve your marksmanship, is the answer. c. A modification of the shooter's proper grip is necessary on different types of pistols. The firmness of the grip remains the same for all calibers and types of pistols and revolvers, but nature of the grasp must correspond to the shape and size of the grips in meeting all the requirements of the proper grip. For example, the caliber .22 grip is sometimes found to be smaller in circumference than a caliber .45 pistol. In this instance, the reach of the lower three fingers may extend further around the stock, resulting in one of the primary pressure points (the middle bones of each of the three lower fingers) coming to rest beyond and partially around to the left side of the front strap. Pressure exerted would not be straight to the rear. As it is fully applied in the normal grip, it would no doubt effect the natural alignment of the sights. Also, shooters with small hands have trouble with stocks of varying sizes. One example is having to compromise, due to a short trigger finger which can reach the trigger only with the finger tip, between a straight to the rear trigger pressure and the best position of the pistol in the shooting hand that tends to give natural sight alignment. d. Shaped, molded or tailored custom grips are required to fit perfectly. Fitted grips are primarily used to help the shooter who can't consistently duplicate the proper grip when using standard factory grips. The individual shooter must first decide what features and characteristics of a shaped grip suit his hand. Stocks can be made to fit exactly, but it is a difficult job. Only an experienced shooter is capable of knowing what he actually needs in a custom grip, because only he knows what his proper grip looks and feels like. e. Powdered rosin dusted on the hand can help to maintain a solid, controlling grip but it is not absolutely necessary. Normally, a strong hand and the checkering and stippling on the

stocks and metal surfaces is sufficient. In hot weather when the hand may perspire or a hand that becomes wet in the rain may cause slippage, powdered rosin or a like substance, that will temporarily dry the skin of the palm and fingers, is then justified. In the final analysis, there is only one correct grip for you. It is one that is firm; affords the individual shooter the maximum degree of control over maintaining sight alignment and allows positive, straight to the rear pressure on the trigger without disturbing sight alignment.

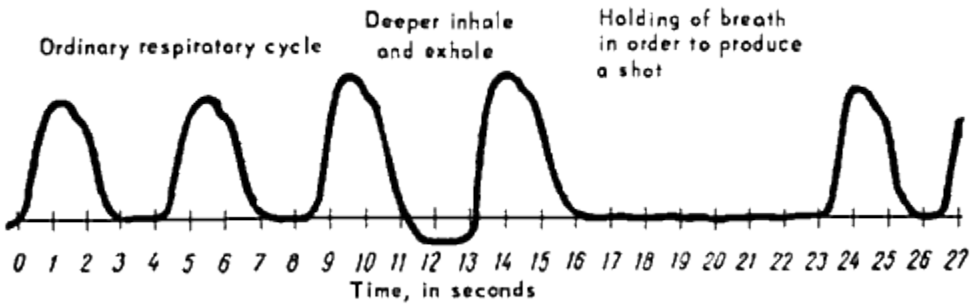
D. BREATH CONTROL. The correct method of breathing is an essential part of the shooter's system of control. Most pistol shooters know less about the proper method of breath control than of any of the other fundamentals. The object of proper breath control is to enable the pistol shooter to hold his breath with a comfortable feeling long enough to fire one shot slow fire; five shots in twenty seconds timed fire; or five shots in ten seconds rapid fire without loss of the ability to hold still or concentrate on sight alignment. 1. To be Effective, Breath Control Must Be Employed Systematically and Uniformly: The ability to concentrate and maintain rhythm is aided. a. Promote a steady hold: It is generally known that one must not breathe during aiming. Breathing is accompanied by the rhythmical movement of the chest, abdomen, and the shoulders. This causes the pistol to move about excessively, making it almost impossible to produce an accurate shot. Therefore, one must not simultaneously breathe and try to fire a shot, but must endeavor to hold the breath for a short period of time. b. The physiological processes involved in breathing: The shooter however, must not view the breathing process solely from the movement of the chest and the gun. He must not forget that the process of breathing, which consists of a combination of processes which occur constantly in the human body, determine in general the condition of the human being. Therefore, proper breathing is of great importance during shooting exercises which last several hours. Incorrect breathing technique has an adverse effect upon shooting, especially if the concentration is disturbed by sensing of the need to breathe. (1) During the process of breathing, there is an alternating increase and decrease in the volume of the chest, as a result the person inhales and exhales. A person inhales when the dimensions of the chest increase. Once inside the lungs, the air provides oxygen to the blood and in turn it absorbs carbon dioxide and aqueous vapors. Exhalation occurs when all the muscles relax, the diaphragm presses upward, and, under the action of the weight of the chest and the elasticity of the lungs, air is forced out of the body. Exhaling does not require muscular effort; it occurs as the result of the resiliency of the ribs and the muscular tissues and the elasticity of the lungs. (2) When breathing calmly a person produces an average of 12 - 13 respiratory cycles a minute. Consequently, one respiratory cycle lasts 4 - 5 seconds. If one traces the respiratory cycle, it is not difficult to note that the strained position of inhalation is replaced very quickly by exhalation. The very next inhalation begins after a respiratory pause of 2 to 3 seconds, (figure 1-7) during which time the carbon dioxide accumulates in the lungs. The duration off the respiratory pause is determined by the ratio of oxygen and carbon dioxide in the air remaining in the lungs.



Breathing

Figure 1-7. Scheme of a Person's Breathing.

(3) The respiratory pause and the problems of the ventilation of the lungs are of great importance to the shooter. It is obvious that during aiming and applying pressure on the trigger, the breath must be held only after the shooter has exhaled, timing it so that the breath is held at the moment of the natural respiratory pause. During that time the muscles are not strained and are in a relaxed state. c. A person can prolong by several seconds this respiratory pause, that is, hold his breath comfortably for 15 - 20 seconds, without any special labor and without experiencing unpleasant sensations. This time is more than adequate to produce a shot or shots. Experienced shooters usually take a deep breath before firing and then, exhaling slowly, hold their breath gradually, relax and concentrate their entire attention upon sight alignment and the smooth application of pressure on the trigger (Figure 1-8).



Breathing

Figure 1-8. Scheme of the Manner in Which A Person Holds His Breath in Order to Produce a Shot.

2. Recommended method a. Prior to fire commands: (1) When expelling the air from the lungs before aiming, no effort whatever must be exerted. The exhaling must be natural and

free, as in ordinary breathing. The air must not be held in the lungs; incomplete exhaling before aiming leads to straining and to stimulation of the nerve centers regulating the breathing, and the shooters concentration on aiming is distracted. (2) In order to make sure that during prolonged firing the interruption of the rhythm of breathing does not have an influence upon the shooter, the breath must not be held for an excessive period when trying to fire a slow fire shot. If the shooter does not produce a shot in 8 - 10 seconds, he must stop aiming and take another breath. (3) Before holding his breath for the next shot he must empty his lungs well, taking several deep breaths. The same should be done between shots and strings of shots throughout the firing. This facilitates the lengthening of the respiratory pause before aiming and provides for regular rest between shots and strings. The oxygen level in the blood is slightly increased. As a result the shooter is relaxed and comfortable during all shooting without excessive and premature fatigue. b. During the fire commands: Take a deeper than normal breath at the command, "READY ON THE RIGHT", take another at "READY ON THE LEFT", extend your pistol and take the final breath and exhale to the point of comfort at "READY ON THE FIRING LINE". As the shooter gains experience in proper breath control, he will find that he will hold his breath, or extend his normal respiratory pause, without being too conscious of the action and allow intense concentration on sight alignment and trigger pressure. c. During actual firing: The shooter should not be conscious of the need to breathe. If during practice a shooter finds that he cannot hold his breath the twenty seconds necessary to fire a timed fire string, he should make a practice of firing his timed fire strings in less than twenty seconds. However, if during a timed or rapid fire string, the shooter feels compelled to breathe, he should take a short breath quickly and continue to fire. This causes a lapse of concentration on sight alignment and should not be the normal technique used.

PERFECTING TECHNIQUE

Sight Alignment

By Chapter 2: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter2.htm

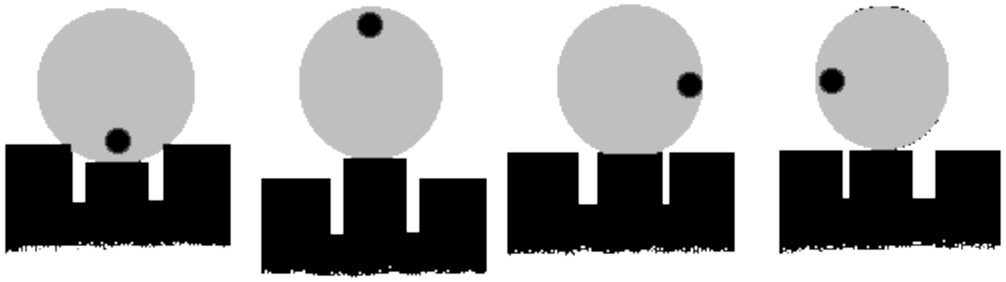
Sight alignment is the most important contribution to firing an accurate shot.
In order for the bullet to hit the center of the target, the shooter must aim the pistol and give the barrel a definite direction relative to the target. In theory, accurate aiming is achieved when the shooter places In exact alignment the rear sight with the top and sides of the front sight and holds them In alignment In the aiming area. A requisite for correct aiming is the ability to maintain the relationship between the front and rear sights. When aiming the front sight is positioned in the middle of the rear sight notch with an equal light space on each side. The horizontal top surface of the front sight is on the same level as the top horizontal surface of the rear sight notch (Figure 2-1) A. RELATIONSHIP OF SIGHTS. It is necessary to be acutely aware of the relationship of the rear sight to the clearly defined front sight. Normal vision is such that the rear sight of the pistol will be as nearly In focus as the front sight. Some shooters may be able to see only the notch of the rear sight In sharp focus; the outer extremities may become slightly blurred.



Sights

Figure 2-1. The Relationship of the Sights.

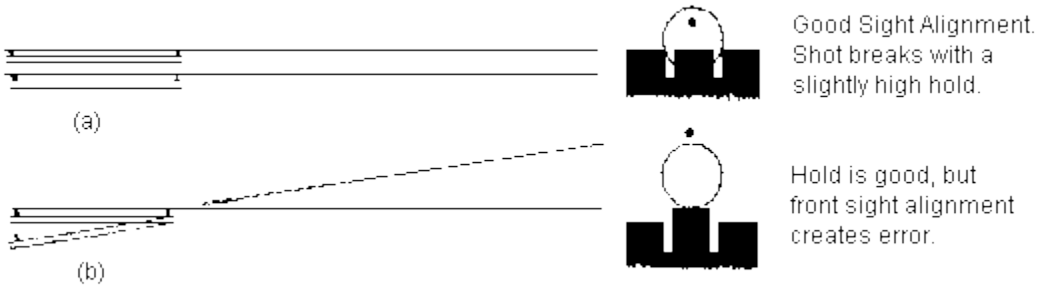
1. Angular Shift Error: If the shooter does not observe correct aiming (maintaining the top surface of the centered front sight on a level with the top of the rear sight and equal light space on each side of the front sight) there will be few accurate shots. Most often, he locates the front sight in a different position In the rear notch. This accounts for a greater dispersion of shots on the target, since the bullets will deviate In the direction In which the front sight is positioned In the notch. (Figure 2-2). This aiming error is known as angular shift error.



Angular Shift

Figure 2-2. Displacement of the Bullet When There Is Angular Shift Error in the Alignment of the Front Sight.

2. Parallel Shift Error: If the hold (arc of movement) is deviating In near parallel error from the center of the aiming area, the shooter should know that these deflections will not lower the score to the extent of angular shift error. Therefore, sight alignment is the most critical of the two. Thus, the accuracy of a shot depends mainly upon the shooter's ability to consistently maintain correct sight alignment. The main effort should be toward keeping your sights aligned, Holding the pistol perfectly still is desirable but It is not mandatory.



Parallel Shift

Figure 2-3, Displacement of the Bullet When The Pistol is Shifted: (a) Parallel, (b) Angular



Impossible

Figure 2-4a, Impossible! The Human Eye Cannot Focus On A Close-UP Object and A Distant Object Simultaneously.

B. POINT OF FOCUS. Correct sight alignment must be thoroughly understood and practiced. It appears on the surface as a simple thing - this lining up of two objects, front and rear sights. The problem lies in the difficulty in maintaining these two sights in precise alignment while the shooter is maintaining a minimum arc of movement and pressing the trigger to cause the hammer to fall without disturbing sight alignment. The solution is partly in focusing the eye on the front sight during the delivery of the shot.



Wrong

Figure 2-4b. Improper. Control of sight alignment is not precise. Distinct focus on target renders sight indistinct. Error incorporated here is the same as Figs 34a and is not as readily apparent.



Correct

Figure 2-4c. Proper. Control alignment is precise. Focus limited to front sight only, renders the sights distinct and target indistinct and sight relationship can be controlled constantly.
1. It is imperative to maintain 'front slight point of focus" throughout the sighting and aiming of the pistol. The shooter must concentrate on maintaining the correct relationship between front and rear sight, and the point of focus must be on the front sight during the short period required to deliver the shot. If the focus is displaced forward, and the target is momentarily in clear focus, the ability of shooter to achieve correct sight alignment is jeopardized for that moment. Frequently, this is the moment that the pistol fires. A controlled, accurate shot is impossible under these conditions. 2. When the eye is focused on the target the relatively small movement of the arm appears magnified. However, when the eye is correctly focuses on the front sight this movement appears to have been reduced. C. CONCENTRATION. 1. If the sights are incorrectly aligned, the net result is an inaccurate shot. Carelessness in obtaining correct sight alignment can usually be traced to the shooter's failure to realize its importance. Many shooters will, in the initial phase of holding, line up the sights in a perfect manner. However, as the firing progresses, and the shooter is concentrating on delivering the shot, he often loses correct sight alignment which he attained in the initial phase of his hold. Usually, when the shooter is unable to maintain a pln-point hold, his concentration on sight alignment wavers. An accurate shot is lost because the shooter is thinking of his arc of movement and not the perfection of sight alignment. 2. Another factor which contributes to the deterioration

of sight alignment, is the feeling of anxiety which arises over the apparently stationery pressure on the trigger when attempting to fire. An impulse is generated to got more pressure on the trigger, so that the shot will be delivered. When the shooter thinks about increasing the trigger pressure, a degree of the intense concentration required to maintain correct sight alignment is lost. Even if trigger control and the hold are good, the net result will be a poor shot. Sight alignment must remain uppermost in the shooter's mind throughout the firing of the shot. Positive trigger pressure must be applied involuntarily. Consistently accurate shots are produced when the shooter maintains intense concentration on sight alignment during the application of trigger pressure, while experiencing a minimum arc of movement. Control of the shot is lessened in direct proportion to the loss of concentration on sight alignment. 3. The average, advanced shooter is probably limited in sustained concentration to a period of 3 to 6 seconds. This short space of time is the optimum period in which a controlled shot can be delivered. This concentration interval should be attained simultaneously with acquiring a minimum arc of movement, a point of focus, satisfactory sight alignment, and the involuntary starting of positive trigger pressure. If exact sight alignment is maintained, and the trigger pressure remains positive, the shot will break during the limited time the shooter is able to control his uninterrupted concentration. Result! A dead center hit on the target. D. THE EYE. The principal difficulties which confront the shooter during aiming are determined to a great extent by the inherent characteristics of the eye and its work as an optical apparatus. All shooters should familiarize themselves with the optical properties of the human eye as explained in Annex II, entitled "Optical Properties of the Human Eye Relevant to Sight Alignment".

References

- Annex II
- "Optical Properties of the Human Eye Relevant to Sight Alignment"

PERFECTING TECHNIQUE

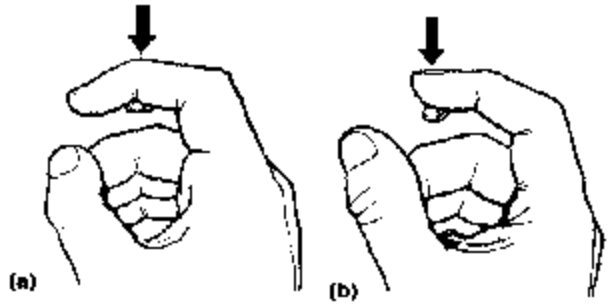
Trigger Control

By Chapter 3: Army Marksmanship Training Guide

Source: http://www.bullseyepistol.com:80/chapter3.htm

A. GENERAL.
Correct trigger control must be employed in conjunction with all other fundamentals of shooting. The physical act of applying pressure on the trigger to deliver an accurate shot may vary from individual to individual. Proper trigger control for each individual gradually assumes uniformity when the techniques of proper application are mastered. Many shooters, for example, maintain a degree of trigger control with a relatively light grip, while another shooter may use a very tight grip. Some shooters prefer to apply consistent trigger pressure at a rapid rate, while maintaining correct sight alignment. For another shooter, a slower, deliberate application may achieve the same results. An ever increasing number of shooters use the positive approach to trigger control, that is, once it is initiated, it becomes an uninterrupted, constantly increasing pressure until the weapon fires. Trigger control is of very great importance in producing an accurate shot. When the shooter exerts pressure on the trigger, he must do so in a manner that does not alter the sight alignment, or position of the pistol. Consequently, the shooter must be able to exert smooth, even pressure to the trigger. Furthermore, the trigger must be pressed in conjunction with maximum concentration, peak visual perception of sight alignment and minimum arc of movement. In order to produce an accurate shot, the shooter must carry out many diverse, but related, actions. Fulfilling this action is compounded by the fact that the pistol is in some degree of motion throughout the period of sighting and aiming. The movement varies according to the stability of the shooter's stance. Consequently, the sight alignment deviates from the aiming area. Often it will move through the aiming area, pausing only for a short period of time in perfect alignment with the target. It is impossible to determine when, and for how long the properly aligned sights will stay in the center of the aiming area. This difficulty is aggravated further by the fact that the shooter is trying to execute coordinated actions when reflex action seeks to contradict them. Such a situation requires the development of conditioned reflexes, and improvement of coordination. The coordinated action of correct aiming, timely pressure on the trigger, and the correct delivery of the shot is difficult and can be accomplished only by overcoming former uncoordinated reflexes or by acquiring new ones. Only through constant training and attention to accepted techniques can these new reflexes be acquired. The peculiar nature and characteristics of the human nervous system are covered in detail in Annex III entitled, "Processes of the Human Nervous System Relevant to Equilibrium, Trigger Control and Hearing". B. FACTORS PROVIDING FOR THE CORRECT CONTROL OF THE TRIGGER.
The pressure put on the trigger must come from independent movement of the trigger finger

only. The gripping fingers and the thumb do not move or tighten. Keep the grip pressure constant. Align the sight, settle into your normal aiming area and exert positive, uninterrupted, increasing pressure, straight to the rear, until the hammer falls. You must not look for a perfect sight picture combination of rear sight-front sight-bull's eye. Instead, focus your eye on the front sight, keeping it perfectly aligned in the rear sight notch. The blur of the out-of-focus target may move about slightly, but this movement is relatively unimportant. Any time the weapon is fired with good sight alignment within the normal arc of movement and it is a surprise shot, the shot will be a good one, and will hit the target within your ability to hold. Trigger control has a series of actions that take place if a smooth release of the firing mechanism is accomplished. 1. Slack and Initial Pressure: Any free movement of the trigger, known as slack, has to be taken up prior to a light initial pressure. This action assures that the tolerances in the firing mechanism linkage are taken up and are in firm contact before positive trigger pressure is applied. Initial pressure is an automatic, lightly applied pressure, approximately one-fourth or less of the total required to fire the weapon. This careful action is an aid in the positive pressure that will release the hammer quickly and smoothly. In order to fire a controlled shot the shooter must learn to increase the pressure on the trigger positively, smoothly, gradually, and evenly. This does not mean, however, that the trigger must be pressed slowly. It must be pressed smoothly, without interruption, but the release of the trigger must take no more than 2 to 5 seconds. Numerous accurate rapid fire strings of five shots in ten seconds are fired in a cycle that allows only one second or less to employ the principals of correct trigger control. Smooth trigger action makes special demands on the trigger finger when pressing upon the trigger; its correct functioning determines to a great extent the quality of the shot. The most carefully attained sight alignment will be spoiled by the slightest error in the movement of the trigger finger. 2. Function of Proper Grip: In order for the index finger to be able to perform its function without spoiling the aim, it is first necessary to have the hand grasp the pistol correctly and create the proper support; permitting the trigger finger to overcome the trigger tension. The pistol grips must be grasped tightly but without any tremor. It is also necessary that the index finger clears the side of the stock. The movement of the index finger must be independent as it presses on the trigger, and also not cause any lateral change to the sight alignment. 3. Proper Placement of the Trigger Finger: It is necessary to apply pressure on the trigger with either the first bone section of the index finger, or with the first joint. The trigger must be pressed straight to the rear. If the finger presses the trigger to the side, undesirable things will happen. The weight of trigger pull will increase; because of additional friction on certain parts of the trigger mechanism an otherwise flawless trigger action will take on the characteristics of a poor trigger when side pressure is exerted on the trigger. Another consideration is the effect that side pressure has on sight alignment. Only slight pressure to the side is required to bring about an error in sight alignment. The prime cause of exerting pressure to the side is improper placement of the trigger finger.

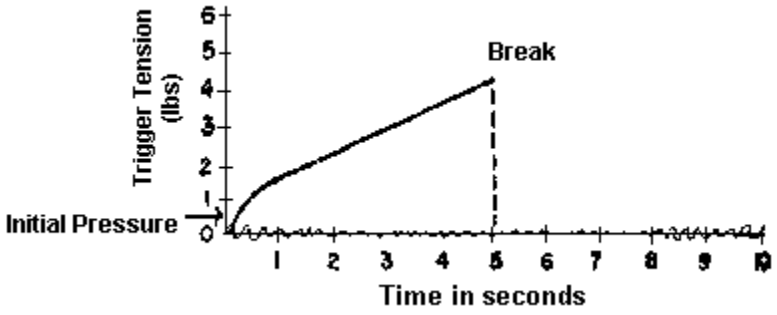


Placement

Figure 3-1. Correct Placement of the Index Finger on the Trigger. (a) With Joint of Index Finger. (b) With First Bone Section of Index Finger .

Ideal trigger finger placement may be modified to a degree by the requirement that the grip provide a natural alignment of the front and rear sights. The shooter frequently must make a compromise to overcome the undesirable effects of not being able to utilize each factor to full advantage. 4. Coordination: It must be emphasized that match shooting is successful only when all the control factors are consistently in coordination. Ability to control the trigger smoothly is not sufficient in itself to produce an accurate shot. The trigger must be activated in conjunction with correct sight alignment, minimum arc of movement, and maximum undisturbed concentration. This might be called cadence, rhythm or timing. Under any name, it comes only to those who practice frequently. Occasional ability is not the answer to championship shooting. A three-gun aggregate requires 270 successful results. Consistent, exacting performance is enhanced by an ability to compensate automatically for errors. It is necessary during firing to press the trigger under varying conditions of pistol movement in conjunction with correct sight alignment. In order to apply coordinated pressure on the trigger, the shooter must wait for definite times when all factors and conditions are favorable. Frequently, it will be impossible to exercise maximum control. However, the shooter must never attempt to fire until he has completely settled into a minimum arc of movement. C. APPLICATION OF TRIGGER PRESSURE. 1. Positive Uninterrupted Trigger Pressure - Surprise shot method - is primarily the act of completing the firing of the shot once starting the application of trigger pressure. The shooter is committed to an unchanging rate of pressure, no speed up, no slowdown or stopping. The trigger pressure is of an uninterrupted nature because it is not applied initially unless conditions are settled and near perfect. If the perfect conditions deteriorate, the shooter should not fire, but bench the weapon, relax, re-plan, and start again. In instances when the pistol is stable and steady, and the periods of minimum arc of movement are of longer duration, it is immaterial whether the release of the trigger is completed a second sooner or a second later. Anytime that the shot is fired with

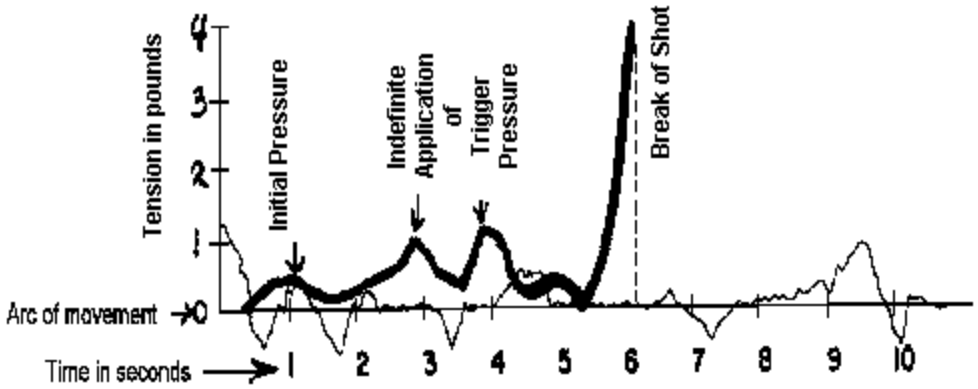
minimum arc of movement and the sights are in alignment, it will be a good shot. Therefore, when the shooter has established stable minimum arc of movement and sight alignment, he must immediately begin to press on the trigger, smoothly but positively, and straight to the rear without stopping, until a shot is produced. This method of controlling the trigger action will give the shooter a surprise break of the shot before any muscular reflex can disturb sight alignment.



Surprise

Figure 3-2. Surprise Shot with Positive Uninterrupted Trigger Pressure.

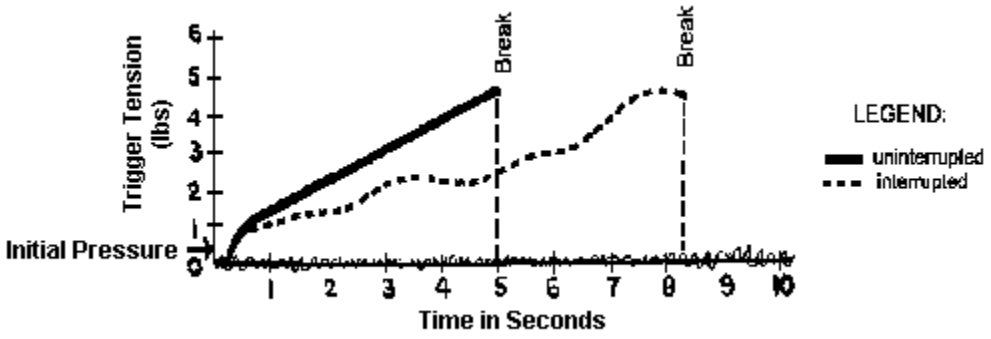
2. Interrupted Application of Trigger Pressure or the "Point" shooting Method: This is a method of trigger control not recommended, although used by some shooters. Some shooters think they can pick the trigger release time even after years of experience. a. The shooter will align the sights and exert initial pressure on the trigger. He will then make every effort to hold the weapon motionless. During extremely brief moments of motionlessness, pressure is applied on the trigger. If the sight alignment changes and is not perfect, or the arc of movement of the weapon increases, the pressure on the trigger is halted and trigger tension maintained. When sight alignment is again perfect and movement diminishes, pressure on the trigger is resumed until the shot breaks, or after the slack in the trigger is taken up, initial pressure is applied and the shot released by a single swift movement of the trigger finger when there is a decrease in the minimum arc of movement. In this case the presence of perfect sight alignment is not considered essential in initiating trigger action. Abrupt action in applying trigger pressure will disturb the existing sight alignment and other fundamental control factors are subordinated to a minimum arc of movement. The application of all other fundamentals is required regardless of whether or not they are optimum.



Release

Figure 3-3. Application of Trigger Pressure When Based on Perfect Sight Picture.

b, While applying positive trigger pressure straight to the rear, if any thought enters the shooters mind to speed up or slow down this trigger pressure, it will result in the concentration on sight alignment being broken down. e. The decision to increase the trigger pressure may result in a reflex action commonly known as anticipation and usually results in heeling the shot (The bullet strikes the target at approximately one o'clock). The recoil becomes more imminent and the brain will send a signal for the arm and hand muscles to react prematurely a split second before the shot is fired; resulting in frequent bad shots and low scores.



Trigger

Figure 3-4. Shot fired With interrupted Trigger Pressure Compared to Shot Fired With Uninterrupted Positive Trigger Control.

D. ERRORS MADE IN TRIGGER CONTROL AND MEANS OF COMBATING THEM. 1. The most serious and disrupting error made by the shooter is jerking - that is, the abrupt application of pressure on the trigger accompanied with muscular action of the hand and arm

muscles. If Jerking was limited to abrupt pressure on the trigger, and the rapid displacement of the axis of the bore, it would cause only part of the results. a. Jerking is usually accompanied by: (1) The sharp straining of all the muscles in the arm arid shoulder. (2) The abrupt tightening of the hand on the grip. (3) Failure to press the trigger directly to the rear. All of these factors taken together, lead to a great shifting of the pistol to the side and down and only a very poor shot can result. b. Most frequently, jerking is observed in new shooters. Usually because of a large arc of movement, favorable moments for producing a good shot are of very short duration. c. The cause of trigger jerking is the practice of "snatching a ten-pointer", as the expression goes. The shooter tries to fire at the moment when the centered front sight, as it moves back and forth, passes under the lower edge of the bull's eye, or comes to a stop, for a brief time, near the center of the aiming area. Since these moments are fleeting the inexperienced shooter strives to exert all the necessary pressure on the trigger at that time. This rapid and abrupt trigger pressure is accompanied not only by the work of the muscles in the index finger, but also by the sympathetic action of a number of other muscles. The involuntary action of these muscles produces the "jerk", and the inaccurate shot that results. The young shooter, in anticipation of the recoil of the pistol and the loud noise, strains his muscles by flinching, to counteract the anticipated recoil. This is also known as heeling the shot. d. Practice has shown that a young shooter must be warned sufficiently early in his training about the dangers of jerking the trigger and effective steps taken to instruct him in the correct technique of accurate shooting. e. Difficulty In detecting errors In trigger control is frequently because the pistol shifted during recoil and errors are not recognized. The shooter has a more difficult time in evaluating than a coach, and often does not realize that he is jerking the trigger, blinking his eyes, or straining his arm and shoulder muscles. f. The easiest way to correct jerking in the young shooter is by the coaching of an experienced coach. A coach can more readily detect errors and correct habits that will produce poor trigger control. Frequently a shooter does not consider it necessary to prove conclusively whether or not they are jerking on the trigger. It is necessary, though, to know that if he does not get rid of the detrimental habit of jerking on the trigger, he will never succeed in achieving good results. g. Signs of jerking are an increased in the size of the area of the shot group or shots off to the side which are not called there; chiefly to the left and down (for righthanders). To correct the condition, the shooter must make a change In his training exercise, but in no Instance must he stop them. (1) Dry-fire practice will enable the nervous system to rest from the recoil of the shot. By this practice some of the reflexes which are detrimental to firing (tensing of the arm In order to counteract the recoil, the straining of the muscles in anticipation of the shot, blinking from the noise of the shot), are not being developed. They will, In fact, begin to decrease and may completely disappear. (2) Secondly, the shooter may continue regular training, but occasionally he may practice "dry". This way, he will not lose the stability of this position, as well as the useful reflexes which the shooter has developed during the process of previous firings. {3) By aiming carefully and noting attentively everything that happens to the

pistol when he presses on the trigger, the shooter will discover his errors and eliminate them. Training by means of ball and dummy and dry firing is of great benefit. It makes it possible to develop correctly and carefully the technique of pressing the trigger, and contributes to acquiring proper habits In controlling the trigger. {4) When beginning to use dry firing the shooter must first overcome the desire to "grab" for a shot when the centered front site is under the bull's eye. Despite the arc of movement the shooter must teach himself only to press smoothly on the trigger and to use the uninterrupted positive control method of trigger action. When the smooth control of the trigger again becomes habitual and he no longer has to devote special attention to it, he can again shoot live cartridges. After starting again to shoot live cartridges, the first training exercises should involve firing at a square of blank white paper, rather than at a target with a black aiming area. Simultaneously, the shooter must devote special attention to analyzing his performance, counteract the desire to jerk on the trigger, and be conscious of reacting incorrectly to the firing of a shot. 2. Another error committed by a shooter when controlling the trigger is "holding too long", that is, the drawn out action of pressing the trigger. a. A consequence of holding too long is that the shooter does not have enough air to hold his breath, his eye becomes fatigued, and his visual acuity decreases. In addition, his stance loses part of its stability. Consequently, when he holds too long, the shooter presses on the trigger under unfavorable conditions. b. Holding too long is a consequence of excessively slow and cautious pressure on the trigger. This is caused by the shooter's fear of producing a bad shot. Such indecisiveness and over caution may be regarded as the opposite of jerking. Moreover, holding too long stems from the lack of coordination of movement which frequently occurs during those stages of training when the process of inhibition outweighs the process of stimulation. Simply stated, the shooter cannot force himself to exert positive pressure on the trigger at the proper time. One favorable moment after another goes past, and soon the chances for an accurate shot are gone. Naturally, the trigger control phase has been extended far beyond its effective duration. This situation frequently occurs after a period of dry-fire training exercises. The shooter loses the sense of the trigger's true weight when he fires for extended periods of time with a round in the chamber. When the trigger is released in a dry shot, the trigger seems to be rather light, but when the shooter switches to live rounds, the trigger weight seems to be considerably greater. He feels he must exert greater effort to overcome this seemingly greater weight. Frequently, the shooter will blame his troubles on faulty adjustment of the trigger mechanism. Nothing is gained from such assumptions. More times than not, the shooter returns to normal trigger control since the root of the evil is lack of coordinated control and not trigger adjustment. e. The restoration of coordination of movement, and the return to the correct balance between stimulation and inhibition is brought about primarily through systematic practice, match training and dry-fire exercises. It is precisely this method of training which develops the necessary coordination of the shooter's actions. When the shooter's movements become automatic, the trigger finger will operate in an unstrained manner, and the shot will break at

the proper moment. It is important that each training session begin with a few dry-fire exercises. It has been demonstrated that such exercises are necessary for the development of accurate shooting. Such exercises may also be repeated after record shooting to restore equilibrium in the nervous processes. d. Frequently, a shooter, when firing for record, is unable to fire a shot. After several unsuccessful tries, a loss of confidence will arise. Rather than risk a wild shot the shooter should unload the pistol, time permitting, and dry-fire a few shots. After restoring coordination of movement, and regaining his confidence, the shooter is far better prepared, both physically and mentally, for the delivery of an accurate shot. Firing the shot during the first few seconds after settling into a good hold, will guarantee confidence. 3. We have considered the fundamental errors that arise in trigger control. Let us now consider a problem that is also closely related to trigger control - trigger adjustment. a. The firing of an accurate shot depends to a great extent on the quality of the trigger adjustment. An incorrectly adjusted trigger aggravates the errors committed by the shooter as he exerts pressure on the trigger. Incorrect adjustments include: (1) Excessive trigger weight. (2) Excessive long creep (movement of trigger). (3) Too light trigger weight. (4) Variable trigger weight. b. The shooter should not try to overcome these difficulties with modification in his trigger control but take the problem and pistol to the armorer (gunsmith) for solution.

PERFECTING TECHNIQUE

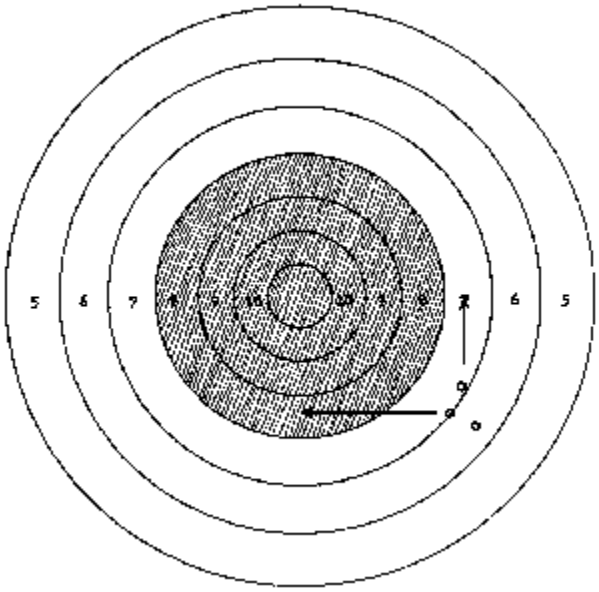
Establishing a System

By Chapter 4: Army Marksmanship Training Guide

Source: http://www.bullseyepistol.com:80/chapter4.htm

The requirement that 120-270 record shots be fired in one day of match shooting demands great expenditure of energy. For this reason the shooter must plan his actions with special care in order to use his energy intelligently and conserve his physical and mental strength for the duration of the shoot.
The best method in which the shooter can impart all of his shooting skill to his shooting is by careful organization of maximum control. Most of the points lost in an aggregate score are slow fire points. It is therefore imperative that this stage of fire become the first goal of exact control. One shot in one minute is sufficient time to organize the delivery of each shot. Successful delivery is assured when control of the cycle of action and thought is uniformly established - prepare, plan, relax, deliver, analyze and correct. A. PREPARATION Complete preparation and prior planning is essential. The firer must be both mentally and physically ready. So that he can concentrate on performance he must have checked the range conditions, his equipment, zero of weapons, range commands, ammunition, etc. prior to actual participation on the line.
1. Zeroing: As a competitive shooter you must know how to zero your weapon in order to place the strike of the bullet in the center of the target. There is no excuse for losing points in competition due to an improperly zeroed pistol. Inexact zeroing is a demonstration of lack of preparation. a. There are two types of sights, fixed and adjustable. (1) You may have fired a pistol with fixed sights. It could be that you are using a pistol with fixed sights now. The fixed sights found on the . 45 caliber service pistol are somewhat difficult to adjust and therefore not primarily used for competitive shooting. For elevation correction, you must use the trial and error method of changing the sight height with a file, or, if a good armorer is available, he will be able to cut off the exact amount from the sights. Windage must be corrected by moving the rear sight with either a sight mover or a hammer and punch. You cannot have both a 25 yard and 50 yard zero in elevation with fixed sights. Your point of aim at 50 yards will be higher than at 25 yards. Correcting for windage with a fixed sight is very difficult due to the method used. Except for initial determination of normal zero for ideal conditions, it is recommended that fixed sights not be moved during competition. (2) Most pistols that are used in competition today are equipped with adjustable sights which are easily moved by a screwdriver or a small coin. (a) There are many makes of adjustable sights available on the market today. The sight adjustment screws do not all move in the same direction for a given adjustment. Clockwise will move one sight to the right and move another sight to the left. This also applies to elevation adjustment. Another difference is that each sight does not move the

strike of the bullet the same distance per click. Some of these sights are more durable than others. (b) The primary consideration when you are using adjustable sights is that you know the capabilities of your sights. How far one click in elevation moves the strike of the bullet at 50 yards, etc., on your pistol, is important to correct zeroing and adjusting. (1) We recommend that you start your zero at twenty-five yards. Fire at least three rounds slow fire before moving the sights.

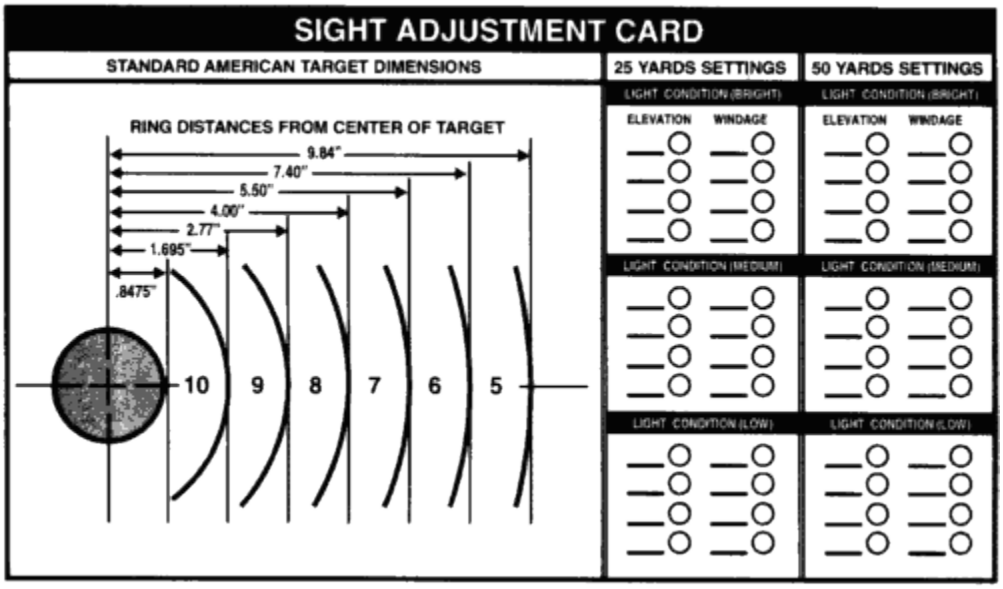


Sights

Figure 4-1. Sight Change Example.

(2) Check your group in relation to the center of the target and if necessary, make a bold sight change. Do not creep to the center of the target one click at a time. Also, repeat the shot group method to establish a 50 yard zero. Do not try to zero by moving your sights after you have fired only one round. One round can easily be off because of errors committed by the shooter. The shot group is equivalent to an 8 ring or a four inch error at 6 o'clock in elevation. If the sight moves the strike of the bullet one-half inch per click at 50 yards - approximately eight clicks increase in elevation are needed. The group is equivalent to 7 ring or five and one-half inch error in windage at 3 o'clock, approximately eleven clicks of left windage are needed. (Refer to scoring ring dimensions in Sight Adjustment Card) (3) Exception to the shot group method may be made when a shooter with extensive experience is firing in a match and has called his shot good, but the shot was not located on the target in relation to his call. Analyze your shooting performance. You may determine that an immediate sight change is necessary, but do not be in too great a hurry to move your sights, as the chances for error caused by faulty technique are many. (4) When the group is centered you may wish to fire 10 and 20

second strings to confirm the zero at 25 yards. There will possibly be a change required when shooting timed and rapid fire. (5) Never "Hold Off" or use "Kentucky Windage" with adjustable sights. The shooter will force the shot to break at a specific spot instead of allowing the normal arc of movement. If you are grouping away from the center of the target, adjust the sights to compensate for the error. Mark and record your zero sight settings (a) It is a good idea to mark the sight adjustment screw with a small drop of nail polish or airplane dope to indicate the position of the 25 yard and 50 yard zero. (b) You may also set the elevation screw down to its lowest point, counting the clicks as you do and record the number of clicks up from the bottom for both settings, This way you record the number of clicks from the base of the sight up to your 25 yard and 50 yard zero. (This method is used only for elevation.) (c) You may also make a sight adjustment card on which you can record the zero position of the windage and elevation screw for all weapons. At certain times on different ranges, and under various weather and light conditions, you may find your zero changes slightly. (Figure 4-2).



Card

Figure 4-2. Example of Sight Adjustment Card.

(6) A scorebook which has provisions for you to write in your sight settings for specific ranges, light and weather condition, and hard to remember data is a necessary item of equipment (Figure 4-4). Unusual zero changes frequently appear without reason. Some things to check are as follows: (a) Check the spotting telescope: You may be scoping the adjacent target. Adjusting your zero to compensate for another shooter's grouping frequently happens. (b) Check front and rear sights. The sights may have become loosened from the recoil or bent or damaged. The rear sight may have worn notches in the adjustments and would allow recoil to disturb the sight setting or when an adjustment is attempted, the sight would not move in the

desired direction. (c) Errors in position and grip may cause the hold and sight alignment to drift even though the shooter may have made a correction. An artificial correction is made to compensate for errors. At the shoulder it affects the hold, and at the wrist it affects the grip. An integral shift in body position should be made by moving the rear foot in the direction of the error in the hold and the grip should be shifted by sliding the grasp of the hand on the grips of the weapon toward the error. If a natural position and grip are attained, the tendency to point at other than the aiming point will be removed. (d) Positive action is urgently needed when any of the above situations are present. Make a quick check of the scope, position, and grip and then continue until it is apparent that the difficulty is not being remedied. A bold change in sight setting should now be made. (e) In the case of a sight damaged beyond use, ask the range officer to permit you to declare the weapon disabled and change guns or have it repaired by a qualified gunsmith on the range. (f) The scorebook may indicate a large change in sight adjustment for the pistol range on which you are firing. Changes in light angle or light intensity may affect the zero that you use on the home range. (7) Upon completion of firing, a consultation with your gunsmith might uncover certain changes in the fitting of components, wear, burrs or foreign matter that can alter the mating of finely fitted parts. Any of these may cause a change in zero. 2. Preparation in the Assembly Area: Before you can employ a systematic technique of shooting, a systematic preparation phase must be established. Actually the shooter's over-all preparation starts the first time he attempts to improve his shooting ability. However, important steps can be taken during the period immediately before shooting that will favorably affect control and coordination. a. Physical: The shooter must first complete his preparation in order to fully concentrate on shot delivery. The important thing to remember is to perform all operations and checks before record firing. Otherwise, the shooter will have to alter his system of control to compensate for the time lost in searching for an emergency solution. (1) A shooter should eat light but nourishing food so that there will be no feeling of hunger before shooting, or a sensation of being stuffed. Avoid drinking a lot of water before and especially during shooting. Heavy food and too much liquid raises the pulse beat and induces excessive perspiration, thus weakening the body during shooting. (2) Before shooting the shooter should do mild limbering up exercises: static tension exercises, walking, dry firing, etc., will revive blood circulation after sleep and help balance the nervous system. If a shooter starts shooting shortly after awakening, without limbering up, he will not be steady. His reactions are too slow, and his movements will lack coordination. (3) The shooter should report to the range at least 15-30 minutes prior to firing time. This is necessary for equipment preparation and the shooter will also have a chance to relax and become settled during this period. Carrying a heavy gun box puts a strain on the shooter and no effort should be made to shoot for a period after such exertion. (4) Checking the firing point from which a competitor will shoot must be given due attention. (a) The shooter must choose the best place to shoot from. On a bright day, for example, the position should be slightly back, to allow shooting from shadow if possible, or he must place himself to the rear of his gun kit so as to be in its

shadow. Establish your position in such a way that it will not be necessary to change once the shooting begins. Shooting glasses to prevent the sun from shining into the eyes should always be used. (b) Frequently, the surface of the firing line has bumps, slopes, and other irregularities in it. In order that there be no unevenness on the firing line to interfere with a proper position, a shooter should level his area, scraping smooth the bumps or filling in the depressions with loose dirt or gravel before shooting starts. (5) Before arriving at the firing line a shooter must have previously inspected his clothing and shoes carefully for comfortable fit. (6) Before your relay is called you should move your equipment to a location directly behind your firing point. Make a final check to be sure you have the proper weapon, magazines and ammunition. Blacken your sights with the carbide lamp and make sure the sight setting is correct. Clean your shooting glasses, check for a pencil, screw driver, ear plugs and stop watch. Listen to the range commands and observe the range operation. Be aware of conditions and adapt your performance to take advantage of knowing beforehand the conditions under which you will have to fire. For example: Check for rapidly spoken fire commands, evaluate the wind and light by observing the effect on previous relays, etc. As soon as you are satisfied that you are familiar with the range and range operation, you should mentally review each step you must go through in delivering a good shot. b. Mental: At the beginning of a shooting day it is a good idea to concentrate on planning each shot as a prelude to actual firing. Experienced shooters as a rule, take into account their feelings, energy and fitness and plan a definite method for themselves. (1) Stimulate your confidence by developing a conviction that a controlled, uniform and exacting performance will produce good scores. Accept scores that are within your ability to hold. Confidence is a deciding factor. You will achieve a flawless performance if you are convinced you are capable of winning the match. (2) Prior planning of your actions in the delivery of each shot will minimize the destructive effect of tension and pressure. (3) Delay and irregularities in range operation upset some shooters. Remain relaxed and exercise patience. (4) When you are shooting, you must think shooting, and only shooting. Be mentally alert and remove all stray thoughts from your mind. Condition your mind to concentrate on the match. (5) Mentally review the entire shot sequence, with emphasis on how you are going to perform each act. 3. Preparation on the firing line: Having made sure that we are prepared to shoot the match, the shooter sets up his gun box and scope and makes a final check of his weapon, equipment, and of himself. (Refer to "Three Minute Preparation On The Firing Line Checklist" Figure 4-3).
a. Check your squadding ticket and place your shooting box on the correct firing point. This eliminates the unnecessary shuffle when another competitor informs you that you are on his firing point. Make sure you are correct and let him do the walking. b. Set your scope on your target. c. Scope your target and inspect for holes. Notify range operating personnel if the target is not ready for firing. d. Adjust ear protectors. e. Load magazines with proper caliber ammunition. f. Make a final check of your weapon to see that the sights are still blackened. If the blacking has been rubbed off, then re-blacken at this time. g. Place all of your accessory

shooting equipment that is located in the gun box on the shooting bench where it will be immediately available. h. Most top shooters use approximately three minutes to carefully check out stance, position, and grip. Simultaneously, with this checking process, the shooter should be very careful not to shorten the depth of his breathing and at the end of the preparation sequence, should breathe deeply two or three times. 4. Preparations after the command "LOAD" has been given. a. Upon the command to "LOAD", assume the stance, position and grip that you previously checked out to be correct. Verify this by extending the arm to check for a natural center hold and sight alignment with the target. b. Load your weapon: Pay particular attention to see that the magazine catch has engaged the magazine. Failure of the weapon to load, caused by the magazine being partially placed in the weapon is not a valid reason for an alibi. c. Keep the weapon pointed downrange. d. With the non-shooting hand, grasp the weapon by the barrel or slide. Recheck for a good comfortable grip on the weapon. Check the position of your trigger finger. Extend the shooting arm briefly again to check if the hold and sight alignment are naturally obtained. e. Check again that you are lined up on proper target. f. Relax with the pistol muzzle touching the bench. g. Continue the mental process, knowing that you are prepared to do the job. Mentally review the sequence of events necessary to deliver a correct shot or string. Concentrate on shot sequence. Visualize perfect sight alignment. A checklist that can be stapled inside the lid of your gun box is recommended for use during all preparation.

ASSEMBLY AREA PREPARATION	FIRING LINE 3 MINUTE PREPARATION
(a) Check Squadding for Relay and Target (b) Have the Proper Gun & Ammunition (c) Check the Sight Settings (d) Carbide Light WITH CARBIDE (e) Blacken the Sights (f) Position Ear Plugs or Protectors (g) Oil Can - Check Lubrication (h) Screw Driver for Sight Change (i) Stop Watch for Timing (j) Scorebook BE READY for the Relay to be Called Look to the Weather Time Range Commands Concentrate on Fundamentals Rehearse Mentally	DO NOT HANDLE Weapons Until the Range Officer Gives Clearance to do so. (a) Set Up Scope ON TARGET (b) Check Target For Holes (c) Adjust Ear Plugs (d) Load Magazines (e) Recheck Magazines (f) Check Grip & Position (g) Locate All Shooting Equipment (h) Breath Control RELAX PHYSICALLY Continue Mental Processing THINK POSITIVELY Act Aggressively YOU - ARE - READY!!!

Checklist

Figure 4- 3. Preparation Checklist.

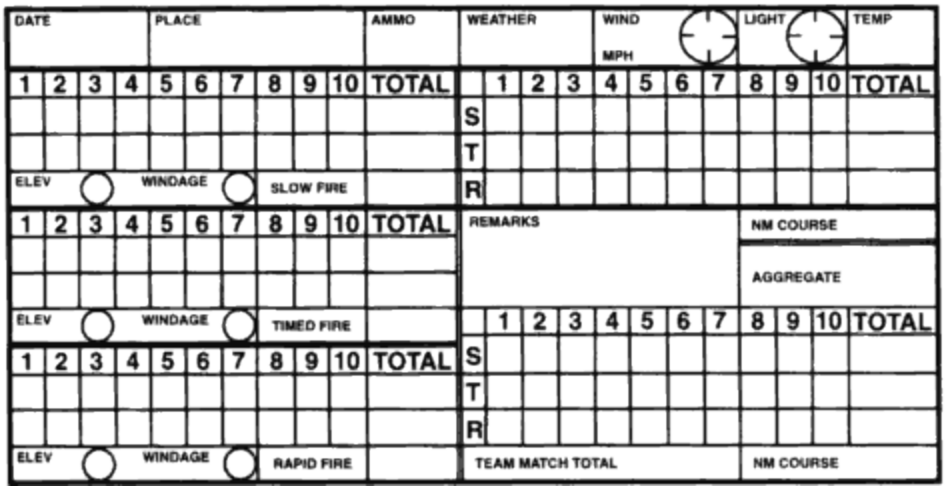
5. The items that are important aids to the shooter should be present and operative prior to shooting. a. Carbide Lamp with Carbide: This small item may be termed one of the most important accessories in the shooter's kit. But it does not work without carbide. Be sure the lamp Is ready and spare flints and carbide are on hand. b. Magazine: Extra magazines are necessary to insure continuous operation. Be sure they are clean, operative and on hand in the gun box. c. Ammunition: Have the correct amount and caliber of ammunition for the match being fired. Include enough for re-fires due to malfunctions, range alibis, etc. d. Pencil and Score Book: Have a pencil or pen on hand. In some matches you will be called upon to score. Your scorebook should always be kept up to date whether in practice or in a match. e. Ear Plugs: Protect your ears at all times from possible damage and eliminate breaks in concentration by closing out gun blast and extraneous sounds by use of ear protection. f. Glasses with Cleaning Tissue: Colored lenses afford glare protection and are designed to let in only glare-free light. Corrective, plain or colored glasses also protect the eyes from possible damage from ejected shells, etc., and should be worn at all times when on the line. Use them to best advantage by keeping them clean. Have lens tissues In your kit. g. Screwdriver and Tools: Sights were made to be moved. Tools that will prove useful are: cleaning rod, barrel bushing wrench, alien wrench set and screwdrivers. Be sure they are all on hand. h. Weapons and Magazines: A properly cleaned and lubricated weapon will have much less chance of malfunctioning. Look and see If you have clean magazines for the correct weapons. i. Squadding Tickets: It is best to rely on your squadding ticket to Inform you of your firing point and correct relay. Do not rely on memory. Keep them on your person or in your gun box at all times. j. Stop Watch: A stop watch is an efficient means to pace your shooting during slow fire. k. Sight Setting: Check to see if you have the sights set for the range at which this match will be fired. l. Lubrication: Have a can of light machine oil available. As a student of advanced pistol marksmanship, there should never arise any circumstances under which the results of your efforts are jeopardized by your failure to make complete and painstaking preparation. B. PLAN SHOT SEQUENCE There must be a systematic approach to obtaining shooting control. To successfully employ the fundamentals the shooter must develop a plan of action and fix it so firmly in mind that distractions do not interfere with his ability to follow a planned sequence. Simply giving yourself the order to watch the sights hold, and squeeze is not sufficient. A shooter with natural talent may find it possible to occasionally fire good strings without having a plan of action. But regardless of his talent his performance is going to be erratic until he uses a comprehensive plan. The shooter must realize that his ability to consistently perform well under pressure is related to the uniformity of his preparation. A planned sequence of thinking that will guide his physical actions through the complete string of fire is absolutely necessary. You must control your mind and stop disconcerting thoughts of the possibility of failure. Picture yourself as you felt and thought while firing good strings and then ask yourself what technique you were using that enabled you to employ the fundamentals so successfully. The difference between champions and the also-rans lies in the

ability to control thinking and plan actions from this point on. Prior planning of the delivery of the shot is the shooter's only insurance that the delivery will be consistently controlled. Knowledge of a successful shot sequence is the basis of the plan. The best assurance that a good performance can be duplicated is that the action follow a uniform sequence. 1. Remind yourself that when you consistently controlled your shooting you were using a shot sequence. You proved that there is a shot sequence that will work successfully for you. You must recreate precisely those same conditions to get the same results. The course of fire may have a successful conclusion only if the shooter in setting-up each shot, goes through all the stages - getting completely ready, planning, relaxing, and delivering the shot, analyzing and correcting in the same manner each time. 2. The following sequence is recommended for slow fire: a. Extend arm and breathe. b. Settle into a minimum arc of movement. c. Pick up sight alignment in the aiming area. d. Take up trigger slack - apply initial pressure. e. Hold breath. f. Maintain sight alignment and minimum arc of movement. g. Start positive trigger pressure. h. Concentrate point focus on front sight. i. Follow through. (Occurs with surprise shot only) (No reflex action) 3. The following sequence is recommended for timed and rapid: a. Extend shooting arm and breathe. b. Find sight alignment. c. Find aiming area on edge of target frame (final deep breath). d. Settle into minimum arc of movement. e. Point focus of front sight (Partly release breath). f. Take up slack - apply initial trigger pressure. g. Maintain sight alignment (target faces). h. Start positive trigger pressure. i. Concentrate on sight alignment (first shot is fired). 4. When a shooter has a system to follow, he can concentrate on performance and not be worried about results. Care should be taken during early stages of instructional practice to comply with each of the items on the shooter's worksheet. As the shooter becomes more capable, sequence, analysis and corrective action becomes more important. Repetition of these steps will instill in the shooter good habits that will enable him to repeat good performance. Further, the worksheet will help the shooter form the habit of not overlooking any factor that will help his shooting. Winning scores are produced by being ready, confident, performing uniformly and being in complete control. C. RELAXATION 1. Relaxation is best achieved by methodically bringing about a loosening of the muscular masses of the body. Think of the neck muscles, the shoulder, back, abdomen, buttocks and upper legs. Systematically reduce the tension of these members to one of support of an upright stance only. 2. A relaxed muscle does not become as tired as quickly as a tense one. It is also important to rest and relax after two or three shots during slow fire. D. DELIVER SHOT OR STRING OF SHOTS ON TARGET The successful delivery of an accurate shot on the target, embraces the proper employment of all the fundamentals. Do not compromise. Follow through and continue to apply all control factors. If the shot is fired as a surprise, there will be no reflex action. When you are absolutely sure you have set up conditions for a controlled shot, put your plan into action. Confidently and aggressively follow each mental step with physical action until the sequence is complete and the shot is delivered on the target. Remember that in slow fire you do not have to shoot before bringing your gun down to

rest. When you fatigue, run short of breath, experience difficulty in maintaining concentration on sight alignment or cannot maintain a suitable arc of movement, lower the weapon to the bench and relax. Re-plan the delivery of the shot, breathe deeply and try again. Some excellent slow fire shooters try two or three times before being able to deliver a controlled shot. Full control of the application of the fundamentals insures the correct control of a shot. 1. Example of a system delivering rapid fire strings with fire commands: As the Range Officer starts his commands, he will announce: a. "ON THE FIRING LINE, FOR YOUR FIRST STRING OF RAPID FIRE,WITH FIVE ROUNDS LOAD." (1) You should load at this time and assume your grip. b. "IS THE HE LINE READY ? " (1) Continue your rhythmic breathing. (2) Check that you are going to shoot on the proper target. c. "READY ON THE RIGHT. " (1) Extend the arm with a stiff wrist and a locked elbow. (2) Align the sights. (3) Breathe deeply and exhale. d. "READY ON THE LEFT. " (1) Find the aiming area on the edge of the target frame. (2) Take a final deep breath. (3) Settle into a minimum arc of movement. e. "READY ON THE FIRING LINE. " (1) Partly release the breath and hold the remainder. (2) Point of focus is on the front sight. (3) Take up the slack - apply initial trigger pressure. (4) Maintain sight alignment. f. Target faces toward shooter - commence firing. (1) Start positive trigger pressure. (2) Shift concentration to perfecting sight alignment. g. First shot is fired (Surprise Shot). (1) Maintain eye focus (follow through). (2) Quick recovery with the sights approximately in alignment, and hold approximately in center of aiming area. (3) Renew positive trigger pressure. (4) Strive to correct errors in sight alignment, but do not delay positive trigger pressure. h. Follow through with four more surprise shots. i. Good rhythm indicates coordinated application of the fundamentals. E. MAKE AN ANALYSIS Complete and instantaneous shot analysis is a prerequisite for improvement in performance or score. It is a complete waste of time and ammunition to fire haphazardly without any comprehensive attempt to improve. A mental impression of the sight alignment should come at the instant the shot breaks. It is advantageous to analyze why you are shooting well on a particular day. Some shooters have a tendency to ignore their good scores. It is important to analyze the good shot string so the combination of factors that produce these good strings can be remembered and repeated. In making the analysis, an important point to remember is to be honest with yourself and your coach. By admitting mistakes the shooter can correct them. The coach can help find a solution to mistakes which he was not aware of if you reveal everything you saw, heard or thought of during the firing of the shot or string. If any shooter critically re-examines and analyzes his performance and his technique of shooting, he will find "minor points", which to a certain extent hinder him from improving his results. The elimination of individual shortcomings and poor methods are the method the shooter has at his disposal to increase his competitive stature. 1. Slow Fire a. Steps: (1) Call each shot. Base your call primarily on the relationship of the front and rear sight. Also consider any unusual occurrences in the arc of movement and whether or not concentration on sight alignment was maintained. (2) When you have decided where your shot should be located on the target, verify your call by

observation with the spotting scope. (3) If the shot or call is good or bad, determine the cause. Generally one of the following situations will occur: (a) Shot call and shot location coincide and you have a good shot. (b) Shot call and shot location coincide, but shot is bad. (c) Shot call and shot location do not coincide. NOTE: ANY TIME YOU FAIL TO RECOGNIZE THE ERROR, YOU MUST EXAMINE YOUR PLAN TO MAKE SURE YOU HAVE NOT NEGLECTED A FUNDAMENTAL. (4) Evaluation: Now consider the question - Did you or did you not follow the planned sequence? If your answer is yes and you had an acceptable shot, this should stimulate your confidence. Review the technique you used to deliver the successful shot. Make every effort to reestablish the same conditions that existed for the first controlled shot and repeat the sequence for each succeeding shot. If your answer was no, you must identify the specific point in your shot sequence where control was lost. The following examples cover only a few of the errors that may have occurred: (a) Failure to establish a minimum arc of movement. (b) Inability to maintain point focus on sight alignment. (c) Concentration drifts from sight alignment to trigger control. (d) Trigger pressure intermittent and uncertain with considerable effort required to fire the pistol. (e) Lack of aggressive, determined attitude, and confidence in the technique. b. The shooter's slow fire work sheet lists the following steps as a guide to complete shot analysis: (1) Follow through check. (2) Call shot (describe sight alignment). (3) Compare target hit location with shot call. (4) If shot or call is bad, determine cause. (5) Watch for error pattern to form. (Same error on more than one shot) (6) Did shot break in normal arc of movement? (7) Did you hold too long? (8) Did you apply positive trigger pressure? (9) If you benched weapon on a shot effort, why? (10) Did you lose concentration? (What did you think about other than sight alignment?) (11) Did you get a surprise shot break? (12) Were you worried about results? 2. Strings of Five Shots : After each five shot string, the shooter should remember each shot as one of five individual sight alignments that enables him to accurately call the shot group. If the call and the group are not together it is necessary to determine the cause and apply positive correction. If the group is not centered then the weapon isn't zeroed, the position was bad or the grip incorrect. If the shooter is sure of the zero of his weapon, then quickly check out the position and grip before firing the next five shot string. Look for one of the five following situations to occur. (It is possible however, for two or more of the situations to occur in one string.) a. Shots off call and grouped within ability to hold, but not centered on target. b. Shots on call but group larger than normal holding ability and may or may not be centered on target. c. Shots off call, group larger than holding ability and may or may not be centered on target. d. Shots on call grouped within ability to hold, but off center on target. e. The shooter's rapid fire worksheet lists the following steps as a guide to complete shot group analysis: (1) Follow through the proper recovery checkout. (2) Shot group call (describe five individual sight alignments). (3) Compare group location with call. (4) If shot group or call is bad, determine cause. (5) Did you get a surprise break on each of five shots? (6) Was the first shot fired on time? (7) Was rhythm maintained throughout string including last shot? (8) Did all shots break in normal arc of

movement? (9) Did you apply positive trigger pressure on each of five shots? (10) Did you lose concentration during string? (What were you thinking of)? (11) Did you ignore minor errors in hold? (12) Were you worried about results? F. POSITIVE CORRECTION (If Necessary) 1. After a shot analysis, corrective measures have to be incorporated into the shooter's performance if the error is to be avoided on the next shot. Slipshod analysis, if any, is usually compounded by the absence of corrective action. Without analysis and corrective measure, improvement is at an end. The shooter who has not learned what he is doing wrong, or what to do about it if the trouble is found is lost and will never excel. 2. Corrective measures prevent the recurrence of poor performance and must be immediately applied. Much has been written about why we shoot poorly; however, it is just as advantageous to analyze why you are shooting well on a particular day. It is more helpful to know the right way to perform than to have your mind cluttered with a multitude of "don't". Coaches in particular should concentrate on and emphasize the positive factors. 3. Look for, analyze and correct mistakes every time a shot is fired. 4. There must be agreement on corrective measures between coach and shooter. The coach must explain the reasons for specific corrective measures recommended. G. USING ADDITIONAL AIDS TO ORGANIZE A SYSTEM OF SHOOTING 1. The sight adjustment card furnishes information on sight capabilities, aids in obtaining a zero and provides a convenient record of sight settings for all caliber weapons. 2. The preparation check sheet provides a convenient list of actions necessary to shoot successfully. 3. The pistol scorebook: The pistol scorebook is a valuable aid to the competitive pistol shooter. It is an individual shooter's record of all firing by stage, such as slow, timed, and rapid fire, plus National Match Course and the value of all shots fired should be recorded. Practice scores should be kept separate from Match Scores. The score book is valuable only if it is kept accurately and up to date. Record the bad scores as well as the good ones. By keeping the scorebook accurately it will be an aid in finding weak points in the shooter's performance. The scorebook will show over a period of time a trend in the shooter's progress. It will reflect progress in the different stages of fire. Averages may also be kept in the scorebook for each stage of fire. Record of ammunition used, sight adjustment, windage, and elevation, weather, light, wind, temperature, may be an aid to the shooter at a later day. There are many type scorebooks that will do the job well, but choose the one that is simple and will furnish the data you need quickly. An example of pistol scorebook page appears on the following page. 4. Due to the great number of functions necessary for the shooter to apply in exercising control of his shooting, a worksheet for both slow and rapid fire is furnished. a. The slow fire worksheet provides the shooter with a guide to organization and application of a system for the control of a slow fire shot. b. The timed and rapid fire worksheet provides the shooter with a guide to the organization and application of a system for the control of a five shot string of timed or rapid fire.



Scorebook

Figure 4-4. Example of the Pistol Scorebook Page.

H. ORGANIZED TRAINING The champion shooter must be capable of intense concentration and have the sheer guts and determination to adhere to the application of fundamentals in spite of match pressure, adverse weather conditions, discouragement, gun trouble and possible sickness. The best method of sustaining good performance is good organization. To shoot well, do more of the little things right. If you are well organized, overall superiority will result. The knowledge of a systematic approach to controlled shooting must be thoroughly ingrained in the minds of all shooters. The "edge" lies with the shooter with a system. It should be added that any system will be worthless to a shooter who has indifferent attitudes toward training and who turns the training sessions into a fixed routine, repeating in robot-like fashion the prescribed shooting aspects of a competitive shooter's activity. It requires not only systematic and painstaking work, but a thoughtful analysis of his work, continuous striving for the new and better, a bold search for the best, and an unwavering strengthening of his will. Only such work on his own shooting methods will turn a shooter's training into a reliable means of assuring his further development and make it possible for him to win. SLOW FIRE WORKSHEET PREPARATORY (In assembly area) 1 2 3 4 5 6 7 8 9 10 1. Squadding - Proper Relay & Target 2. Using clean weapons - Proper Caliber 3. Proper sight setting (Zeroed) 4. Blacken sights 5. Ammunition, Ear Plugs & Screwdriver PREPARATORY (On firing line) 6. Focus scope on proper target 7. Dry Fire for natural Position & Grip 8. Adjust ear plugs 9. Ammunition loaded into magazine 10. Mental Stimulation of Confidence PLAN DELIVERY OF SHOT (Review Shot Sequence) 11. Extend arm & breathe deeply 12. Settle into minimum arc of movement 13. Pick up sight alignment in aiming area 14. Take up trigger slack - apply initial pressure (take final breath and hold part of it) 15. Maintain sight alignment and minimum arc of movement 16. Start positive trigger pressure 17. Concentrate point focus on front sight RELAXATION 18. No unnecessary muscular tension (Relax each

major portion of body) DELIVER SHOT (Follow Plan - No Compromise) 19. Follow Through (Continue all control factors) 20. Shot fired as a surprise - No reflex action SHOT ANALYSIS 21. Call Shot - describe sight alignment 22. Compare hit location with call 23. If shot or call is bad, determine cause 24. Watch for formation of error pattern 25. Did shot break in minimum arc of movement? 26. Hold too long? 27. Positive trigger pressure? 28. Benched weapon why? 29. Lost concentration? 30. Surprise shot break? 31. Worried about results? POSITIVE CORRECTION (Prompt Application) 32. Include in plan of delivery of next shot. RAPID FIRE WORKSHEET PREPARATORY (In assembly area) 1 2 3 4 1. Squadding - Proper Relay and Target 2. Using clean weapons - Proper Caliber 3. Proper sight setting (Zeroes) 4. Blacken Sights 5. Ammunition, Ear Plugs & Screwdriver PREPARATORY (On Firing Line) 6. Focus scope on proper target 7. Dry fire for natural position & grip 8. Adjust ear plugs 9. Ammunition loaded into magazine 10. Mental Stimulation of Confidence PLAN DELIVERY OF STRING (Review Sequence) 11. Extend arm & breathe deeply 12. Find sight alignment 13. Find aiming area on edge of target frame (Final deep breath) 14. Settle into minimum arc of movement 15. Point focus on front sight (partly release breath) 16. Take up slack - Initial pressure 17. Maintain sight alignment & minimum arc of movement (Target faces) 18. Start positive trigger pressure 19. Concentrate on sight alignment (First shot is fired) RELAXATION 20. No Unnecessary Muscular Tension (Relax each major portion of body) DELIVER STRING OF FIVE SHOTS 21. Follow through and recover with good sight alignment and central hold on each shot of string SHOT GROUP ANALYSIS 22. Group call. Describe five individual sight alignments 23. Compare group location with calls 24. If group or call is bad, determine cause 25. Surprise break on each of five shots? 26. First shot on time & rhythm maintained? 27. Did shots break in minimum arc of movement 28. Positive trigger pressure? 29. Lost concentration? 30. Ignored minor errors? 31. Worried about results? POSITIVE CORRECTION (Prompt Application) 32. Include in plan of delivery of next five shots

AN OUTLINE SUMMARY OF THE COMPETE SEQUENCE OF FIRING ONE ACCURATE SHOT

1. Preparation: In assembly area and on firing line

a . Physical (1) Personal preparation. (2) Limbering up. Arrive early. (3) Check out firing line. (a) Protection from wind and bright or changing sun light. (b) Smooth and even horizontal surface. (4) Fit of clothing and shoes (5) Make final check in assembly area of all necessary equipment and become aware of firing conditions both weather wise and in range operation. (6) Move to firing line and place equipment on your firing point. (7) Set telescope up on your target. (8) Make final check of weapon. Check sight black. Should be Zeroed. (9) Check for proper caliber and amount of ammunition and load magazines. (10) Check out stance and position for natural pointing at target center. (11) Check out grip for natural alignment of sight when incorporating all requirements of a

good grip. (12) Breathe Deeply for increased oxygen retention. b. Mental (1) Stimulate confidence. (2) Think only of shooting and expel all stray thoughts. (3) Let the coach worry about any distractions or interruptions on the firing line. (4) Mentally review shot sequence: (a) Think of the act of extending arm and breathing deeply. (b) Think of how your minimum arc of movement looks and feels. (c) Picture properly aligned sights with point focus on front sight. (d) Decide at what point in sequence you are going to take up slack in trigger and apply initial pressure. (e) Here you take final deep breath, exhale and hold only part of it. (f) Remind yourself to maintain sight alignment and a minimum arc of movement while; (g) Positive, uninterrupted trigger pressure is being applied. (h) Concentration must shift to and remain unbroken on sight alignment while positive trigger pressure is being applied. (i) The pistol will seem to almost fire itself because positive trigger pressure is almost involuntary. (j) The surprise shot is now a possibility because with the employment of positive trigger pressure, the normal reaction time suffices to delay anticipatory reflexes that could disturb sight alignment, until all the continuously applied control factors can bring about the delivery of the shot on the target. This is, in effect, follow through. 2. Plan the Shot a. Stance (1) Stable balance. (2) Immobility. (3) Head position. (4) Uniformity. (5) Position of feet. (6) Body erect. (7) Shoulders level. (8) Legs firmly straight. (9) Hips level. (10) Head level. (11) Rest non-shooting arm and hand. (12) Solid firm shooting arm and hand. (13) Center of gravity slightly forward. b. Natural Position Orientation. (1) Start at 45 degree angle. (2) Turn only head to target. (3) Extend arm. (4) Close eyes. (5) Raise arm and settle. (6) Open eyes. (7) Shift trail foot in direction of error if necessary. (8) Recheck. c. Grip. (1) Natural sight alignment. (2) Firm to prevent shift. (3) Unchanging tightness. (4) Independent trigger finger. (5) Uniformity. (6) Comfortable. (7) Recoil straight to rear. (8) Avoid fatigue of hand. d. Breath Control. (1) Systematic. (2) Oxygen retention. (3) Minimize movement. (4) Respiratory pause. (5) Comfortable. (6) Concentration aided. (7) Prior to and during fire commands. e. Sight Alignment (Relationship of Front and Rear Sights, not Sights to Target). (1) Front sight point focus. (2) Rear sight awareness. (3) Exclusive concentration. (4) 6 to 8 Seconds Duration. (5) Coordination with other control factors. f. Trigger Control. (1) Positive uninterrupted trigger pressure. (2) 2 to 5 second duration. (3) Based on perfect sight alignment. (4) Undisturbed sight alignment. (5) Coordinate with optimum perception and minimum arc of movement. g. Shot Sequence. (1) Extend arm and breathe. (2) Settle into minimum arc of movement. (3) Pick up sight alignment in the aiming area. (4) Take up trigger slack and apply initial pressure. (5) Take final breath and hold part of it. (6) Maintain sight alignment and minimum arc of movement. (7) Start positive uninterrupted trigger pressure. (8) Concentrate point of focus on front sight. 3. Relaxation a. No unnecessary muscular tension. b. Relax each major portion of body: (1) Neck (2) Shoulders (3) Non-shooting arm (4) Abdomen (5) Back (6) Buttocks (7) Upper legs. 4. Deliver Shot a. As planned - Do not compromise. Apply all control factors. b. Follow through - continue to apply all control factors. c. Shot will fire as a surprise - no reflex action. 5. Shot Analysis (Use a target center to plot shot calls) a. Follow through check. b. Call shot - Describe sight alignment. c. Compare hit location with call. d. If shot or call is bad, determine cause. e.

Watch for error pattern to form. f. If analysis is vague or unsure, ask yourself some or all of the following questions: (1) Did shot break in minimum arc of movement? (2) Hold too long? (3) Positive trigger pressure? (4) If shot could not be fired & shooter benched weapon, what was wrong? (5) Lost concentration (6) Surprise shot break? (7) Worried about results? 6. Positive Correction (If Necessary) a. Promptly applied. b. Agreement between coach and shooter. c. Incorporate into plan for next shot.

Figure 4-7. The Complete Sequence of Firing an Accurate Shot

PERFECTING TECHNIQUE

Technique of Slow Fire

By Chapter 5: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter5.htm

A. GENERAL

1. A technique is the method of performing the procedures of an operation. 2. A technique of slow fire that will permit a pistol shooter to compete successfully will include: a. Painstaking preparation. b. Thorough planning. c. Systematic relaxation of muscular system. d. Intense concentration. e. Coordination of all the fundamental factors in delivery of an accurate shot. f. Analysis of shot delivery. g. Correction of errors. h. Strict uniformity of execution of the above is required to insure duplication of a good performance time after time. 3. The factors regarding the technique of slow fire shooting should not be regarded as hard and fast which demand strict execution of each point. In competitive shooting, as in any other sport, there are no established rules on technique that apply to all persons. 4. The shooter should accept the material in this chapter in a critical manner, using it as a guide in finding his own style of shooting - a style that fits his temperament, individual ability and physical construction. B. TECHNIQUES IN SLOW FIRE CONTROL. The best technique of control is slow fire, for each individual shooter, is the proper employment of a combination of the fundamental factors that contribute toward attaining satisfactory performance. 1. Dry Firing Before Shooting: A shooter should rarely shoot the first time he settles to aim. Form the habit of firing a "dry" shot or two before the first shot. This will aid in attaining the sharp, clear focus necessary for a smooth, coordinated performance. Avoid dragging out a shot which results in increasing the arc of movement and dulling of the vision. A shooter, after trying unsuccessfully for five or six seconds, should stop dry-firing, take his finger from the trigger and rest, and then make another attempt to dry fire a satisfactory shot. Do not do this too many times, however. A long day's shooting will make heavy demands on the muscles and eyes. 2. Carefulness: Great care is one of the mainstays of the control of slow fire shooting. Every shooter will get a certain percentage of good shots, the rest will be mediocre to poor. The poor shots result from carelessness. Accepting conditions as almost good enough is a form of carelessness. There are few top shooters who are absolutely without some compromise. The shooter who is not willing to devote himself fully and fails to pick up the tie breaking point is the one that loses the match. The principal damage that carelessness inflicts on controlled performance, strikes at the point where it can be least afforded; uniformity. For example, the failure to properly re-grip the pistol for each new attempt to shoot, violates the stipulation that if a good performance is to be repeated, you must uniformly duplicate the employment of all the fundamentals. 3. Patience: Patience is of great importance to the conduct of slow fire, in that without it, the shooter may disrupt an otherwise good performance an instant from successful

completion. When the conditions for a controlled, accurate shot are met, the problem facing the shooter is reduced to one of patiently allowing proposed events to follow their normal course. The synchronization of the factors in delivering an accurate shot require a certain time for completion. Any impatience that would cause one control factor to be disrupted will reduce the whole to a shambles. Improvised, desperate measures to reconstitute a lapse cannot possibly succeed. As an example, a shooter is on the firing line, making a concerted effort to maintain conditions that have been set up to control an accurate shot. His sights are aligned, the arc of movement is settling and he is positively pressing straight to the rear on the trigger, and then must momentarily wait. At this point in his technique of employing the fundamentals he will succeed or fail, depending upon his ability to allow the necessary time to pass for the smooth, undisturbed release of the hammer. 4. Over Sighting: A shooter must remember that his scores cannot be improved by sighting for a long time. On the contrary, "over-sighting" sharply reduces accuracy. After about ten or twelve seconds of sighting, vision becomes gradually blunted and the shooter's eye ceases to observe some minute errors. The result is a deceptive aim, and the shooter without noticing his error in sighting makes an inaccurate shot. In order to achieve the highest accuracy, sighting should be accomplished within six or eight seconds. This system of sighting is good because a shooter who does not over-sight and strain his eyes, permits them to recover rapidly and to keep their sharpness for the duration of the whole match. 5. Establishing a System: A system of operation must be devised by each shooter individually without which he is not capable of attaining a sustained high performance standard. Consult Chapter IV, "Establishing A System". The shooter's guide to comprehensive organization of control of a slow fire shot, "The Slow Fire Worksheet" is included in Chapter IV. Completing the day's schedule of shooting successfully requires the shooter to go through all the stages of shooting for every shot in exactly the same way. This is possible only if the shooter can conserve sufficient physical energy, sharpness of vision, quickness of reaction, etc. while he is shooting. The ability to repeat everything the same way requires an intelligent and economic use of ones energies. 6. The Tempo of Shooting: Experience has shown that the best way to shoot is to shoot each shot rapidly, within six seconds after settling. The time spent between each shot, preparing and planning, is limited only by the total time allowed for the stage of fire. All shooting should be done at a definite tempo and with a definite rhythm. a. In order that the shooter not only expend his energies with care, but actually build them up during the course of shooting, adequate breaks should be taken both between shots and between series of shots. It is beat to shoot at regular intervals, using all the time allotted for a particular match. In slow fire, circumstances rarely permit unbroken rhythmical shooting. Therefore, the shooter must take a particular situation and his own daily capabilities into account and shoot, sometimes at an accelerated or reduced rate, thereby operating at a tempo that will permit maximum performance. b. Before beginning to shoot for record, it is recommended the shooter fire "dry shots". If he feels that after the first shots that his shooting is going easily, he should proceed to shoot live ammo,

making no attempt to slow down, so that the established coordination of his movements will not be upset. c. When a shooter finds it difficult to shoot, he should not speed up his shooting. He must re-plan, relax, wait a short time until he has the "feel" of his pistol, shoot some "dry shots", patiently reestablish control of the trigger, aim with care before each shot, and then begin shooting with assurance at a quickened tempo in order to make up for lost time. d. No matter how well the shooter is doing, a shooter should always be careful to control his movements at all times so as not to cause a performance failure because of negligence. There is no such thing as shooting for a long period of time without some lose of control. However, one must be constantly on guard against becoming nervous and rushing because of a bad shot, as if it is necessary to compensate in a short a time with good shooting for the poor shot. It usually happens that rushing leads to repetition of the same mistakes. Under such circumstances, analyze, correct errors, shoot without haste, and develop the next shot with more care. e. The regular tempo of shooting applies when a shooter is shooting under favorable conditions in good shooting weather. If, on the other hand, the weather is not favorable, the shooter should use a different tempo and approach, depending on the conditions prevailing. 7. Resting During the Breaks Between Shots: During the rest period between shots, a shooter should see that uniformity in assuming body stance and positions is not destroyed. The angle at which the body is turned from the target and the relative position of the feet should not change. In order that the rest interval be most effective, it is a good idea to assume a posture which will make it possible for the muscles to relax and rebuild their strength. The best way is to put the pistol down on the table and sit down in a chair without moving the feet. You can mark the position of the feet with chalk if you are afraid the exact position of the feet cannot be duplicated. It is necessary, when shooting in gusty winds, to rest while standing, with the pistol in a slightly relaxed grip. The shooter must be prepared to assume the firing position quickly, to shoot between gusts, and during the best periods for firing. D. COMMON POINTS OF DEFICIENCY IN CONTROL. There are a multitude of causes for bad shots. Listed below are those most frequently found. It is not intended to be a complete list nor is it intended to provide the shooter with a convenient list of bad habits. It is, however, intended to assist the shooter in finding the source of his trouble. 1. Jerk or Heel: The abrupt application of pressure either with the trigger finger alone or in the case of heeling, pushing with the heel of the hand at the same time. 2. Vacillation: Lack of know-how and skill causes constant changing of technique. The end result is usually that you hope to get a good shot. 3. Anxiety: You work and work on a shot, meanwhile building up in your mind a doubt about being able to fire the shot while your control is optimum. Impatience sets in. Finally you shoot just to get rid of that particular round so you may work on others. 4. Not Looking At the Sights: This is listed frequently as "looking at the target". A shooter may be focusing his eye on neither the sights nor the target, but since he does not see the target in clear focus he assumes he is looking at the sights. Concentrate on sight alignment. 5. Loss of Concentration: The concentration will shift between sight alignment and the relative position

of the sights to the target if good sight picture is the objective. Concentrate only on attaining perfect sight alignment with minimum movement and the application of positive trigger pressure comes much easier and is almost involuntary. 6. Holding Too Long: Adverse conditions that disturb a shooter's ability to "hold" will cause him to delay his positive application of trigger pressure while waiting for conditions to become better. The disturbing factor about this is that you will do it sometimes when you have your normal minimum arc of movement, therefore, you must try to freeze all arc of movement momentarily to get off a perfect shot quickly before any movement is resumed. 7. Control: Maintaining control of your shooting is a continuous battle. The battle builds tension. Tension tightens the muscles and finally the abrupt motions made in compensation for errors cause the shooter to go beyond the desired area and deliver shots in exactly the opposite place from where the error was causing him to shoot originally. Smooth coordinated actions are best assured by the relaxed, confident and carefully planned approach. 8. Lack of Follow Through: Follow through is the conscious attempt to keep all control factors applied through the break of the shot. For example, you are continuing to maintain concentration on sight alignment even after the shot is on the way. This is accomplished by having a surprise shot break and no reflexes of anticipation to disturb sight alignment. Follow through is not to be confused with recovery. Merely recovering from recoil and reestablishing the hold and sight alignment after the shot is fired is no indication that you are following through. 9. Match Pressure: (See Chapter VII - Mental Discipline) If there are 200 competitors in a match, rest assured that there are 200 shooters suffering from match pressure. You should exert all your mental energy toward planning and executing the fundamentals correctly. E. WIND SHOOTING AND ADVERSE CONDITIONS. In a shooting day the weather is apt to change drastically but not necessarily very often. This means that a shooter must be able to react to all changes taking place around him and to change his method of shooting accordingly. A shooter should also be guided by the rule that shooting should not be rushed when the weather conditions are changing frequently. He should be particularly attentive and carefully analyze any weather condition. After making a decision to shoot under a certain condition, adopt the appropriate method, and shoot only under that condition, time permitting. 1. The Wind not only causes a bullet to drift to one side while shooting, but it reduces accuracy by increasing the sway of both the shooter's body and his pistol. The shooter should try to determine a proper compensating sight change if there is a constant side wind. If there is a head wind, take care that the wind is deflected away from the eyes by shooting glasses. Powder fragments and acrid fumes blown back into the face cause smarting and watering of the eyes. Body sway can be minimized by a concerted effort to resist the wind pressure. A slight increase in general muscular tension is necessary. a. Wind shooting is conducive to jerking the trigger because as the arc of movement increases, the shooter develops a tendency to relax his positive trigger pressure. Usually the shooter is waiting for a more stable sight picture. His concentration on sight alignment will diminish and he will make an effort to set the shot off on the move as the sights pass the vicinity of the

aiming point. b. The obvious answer is to first wait for a lull in the wind; next, concentrate as one normally does on sight alignment. When the smallest arc of movement that is possible to obtain under existing conditions is achieved, positive pressure is applied to the trigger. c. Do not continue the hold during extreme gusts. Always take advantage of a chance to rest. Each subsequent attempt to fire a shot should be made with a firm resolve to align the sights and to apply constantly increasing trigger pressure until the shot is fired. d. The surprise shot continues to be the indicator, even under these conditions, of whether you are applying the fundamentals correctly. Your shot group will be somewhat larger as a result of the wind disturbance increasing the arc of movement, but the wild shots resulting from faulty sight alignment, flinching, jerking and over correction will be minimized. e. Extensive practice under wind conditions is not recommended, but enough firing should be conducted under those conditions to familiarize the shooter with the technique and method best for him. f. Changing of Wind: When shooting at 50 yards and the whole air mass is moving approximately the same velocity in one direction, fairly accurate sight corrections can be made for wind. When doing this, however, it is not wise to think an all-purpose correction has been made. The changing nature of the wind must be taken into account. The grass and weeds, etc., should be watched attentively while shooting to detect a change in the force and direction. A shot should be made only when wind conditions have been accurately determined. g. Sometimes the necessity to shoot when the wind is gusty requires a shooter to shoot accurately in a very short time, say within two seconds. The successful firing of an accurate shot under such conditions will be achieved only if a shooter has remained in the firing position in the intervals between lulls. When shooting under such conditions, he should figure out the most advantageous posture for himself in which he can wait out the gusts of wind. As soon as there is a lull he will be able to take aim quickly and fire his shot. h. To aid accurate shooting when a gusty wind is blowing and when the wind is changing, a shooter must alter his tempo of shooting. Sometimes he must shoot rapidly, sometimes shooting twice when the wind is quiet or when the lighting is right, sometimes taking fairly long breaks. Wait out the unfavorable conditions for making a shot; but watch your time! i. In order to manage the difficulties arising during windy conditions a shooter should be prepared beforehand to change both the tempo of his shooting and alter his system of control as the situation in which he finds himself changes. 2. Adverse weather conditions such as cold, hot or rainy weather or extreme light conditions pose problems that can be solved in much the same manner as in wind shooting. Be determined to adhere to the fundamentals and ignore distractions to the competition. Compensate for disagreeable conditions. a. It is advisable to carry a raincoat with you at all times and possibly a plastic cover for your gun box to keep your equipment dry. Most ranges, except for those at the National Matches, have covered firing points that help to keep the competitor dry during rainy weather. A rain suit or short heavy coat are the best garments for shooting in rain or cold windy weather. The folds and loose ends of a rain coat or overcoat flapping in the wind will cause body movement. b. During cold weather the shooter

must obviously wear warm clothing. Use insulated underwear to avoid wearing many layers of heavy clothing. When the shooter becomes shivering cold it is difficult to hold the sights in perfect alignment or retain sensitive trigger control. Hand warmers are very good and are small enough to keep in the gun box or pocket. Light weight lubricating oil must be used in cold weather to prevent malfunction of weapons. c. During hot weather, perspiration becomes a problem. A sweat band on the forehead keeps the sweat out of the eyes and it is recommended that powdered rosin be used to dry the hand. When not on the firing line the shooter should relax in the shade. Covered firing points provide protection from the sun while shooting. Salt tablets prevent heat prostration. Eat lightly. 3. Changes in light intensity and direction have a great influence on the accuracy of aiming. Under changing conditions, the eye does not perceive the relationship of the front and rear sights consistently. There may be a change in the point of impact of the shot group on the target. Experienced shooters usually settle on one lighting condition when the cloud cover is changing, aiming only when the sun is shining brightly or shooting only when the targets are in the shade. The choice of a lighting condition must be made depending on the relative length of time that the targets are well lighted or shaded. a. Light condition varies from extremely bright to very dim and the shooter must keep a record of the light conditions on every range he fires on in his scorebook. Some competitors are affected more by changes in light than others. A note should also be made as to how much his zero changes in these different light conditions. Sight should be blackened with care on bright days. As part of your shooting accessories, you should have both amber and green shooting glasses not only for light conditions but for protection against oil, powder fragments, fumes, wind and empty brass. Firing from an uncovered firing line usually requires different sight settings than the firing from under a shed. Ammunition should be kept out of the sun as its accuracy is affected if it is exposed to the direct rays of the sun. b. A shooter must also be able to complete the match rapidly in order to avoid firing under marginal conditions. The necessity for rapid shooting arises when twilight sets in. There are times on any range when a shooter must either accept an interruption during a match or shoot at a stepped up pace. For example, when the light on the target is shifting he must finish shooting before the sun's rays come through the target from the rear. This causes the target to appear blotched and makes accurate sighting and shooting difficult. 4. Our accomplishments on the firing line stem from our mental capacity to face up to the out of ordinary, and parlay these conditions into winning. Poor conditions must never become an excuse to quit or compromise. Good scores are produced by hard work in the application of the fundamentals regardless of the conditions. Proper control of the application of the fundamentals is the most important factor in shooting winning scores under adverse conditions. F. TRAINING METHODS: 1. Competition: Any top shooter will agree with this word of advice: Shoot every match you can afford. The special conditions created by shoulder-to-shoulder competition can best be controlled by lessons learned in match experience. To learn how to apply this control to your slow fire technique is the result of continuous match competition. If numerous

matches are not available, try to make your practice sessions duplicate match conditions as nearly as possible. 2. Dry Firing: Developing the ability to apply the fundamentals to your shooting Is a tedious, painstaking process. If all the effort is confined to range practice and competitive matches, years of hard work and great expense for ammunition are involved. The use of dry fire practice can reduce the cost in both respects, Dry firing develops the ability to control your shooting in all the primary factors - coordination, eyesight, arc of movement, uniformity of applying fundamentals, analysis and correction, etc. Achieving the ability to control your body in its job of delivering a good shot is one of repetition of good shooting habits. Dry firing is a definite aid in this stage of development. To got the most out of your dry firing, use the appropriate work sheet to guise your work. Prepare and plan each shot as if it were a live round. Relax when you are ready and give yourself fire commands. Deliver the shot with the same amount of effort as in live practice. Use a target center on the bench to record your shot calls. If the shot calls embraced a detectable error, analyze and determine why the errors were in your performance. You may notice errors in your performance that have been hidden in the recoil and sound of the weapon as it is fires. Precise identification of these errors can be made only during dry fire sessions. A positive correction is necessary before proceeding to the next shot. A bulls-eye, a blank wall or the open sky may be used to conduct dry firing. Dry fire practice can be overdone. Initially, the new shooter should limit himself to ten minutes of effort. Later, as performance Improves, maximum time should be about thirty minutes. 3. Ball and Dummy: Ball and dummy exercise is another important aid in accelerating your improvement. It is most effective when another person loads the weapon and observes the shooter as he attempts to fire. The coach loads either a live or dummy round at random. The shooter must never know whether he has a live or dummy round in the chamber. If the shooter is disturbing the weapon with muscle reactions In any way, the coach will be able to identify the error immediately. After identifying the error pattern, the coach and the shooter must agree on a positive correction. Ideally, the correction will prevent recurrence of the error. In most cases however, the coach and shooter must be satisfied with the error occurring less frequently. 4. Instruction in Fundamentals and Techniques Must not be Overlooked: Every shooter must be thoroughly schooled in pistol marksmanship fundamentals. His future success will be based on how well he masters the fundamentals and the techniques of employing them. 5. A Review of Fundamentals and Techniques must be conducted periodically. The shooter must be reminded of the fundamental points of pistol marksmanship to assure that his shooting skill will constantly improve. Each shooter must develop a personal technique of employing the fundamentals. Improvement is at a standstill when analysis of the technique no longer identifies and removes the flaws from poor performance.

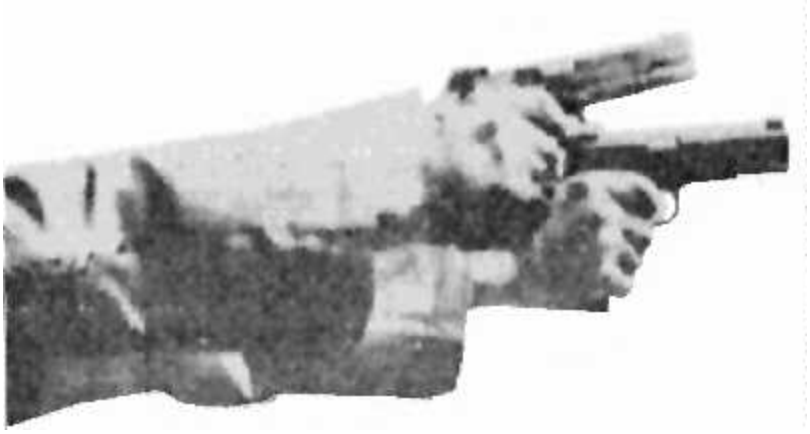
PERFECTING TECHNIQUE

Technique of Sustained Fire

By Chapter 6: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter6.htm

Timed and rapid fire stages can be stumbling blocks, especially if attempted in a haphazard manner. However, through the development of proper techniques and careful planning, you can improve your scores and become more consistent in your performance. A recent development in the thinking of shooters is the concept of a sustained fire technique instead of the more generally accepted idea of a separate method for each of the timed fire and rapid fire stages. Many shooters that have attained national and world recognition find a distinct problem in the build-up of tension caused by the time limitation of rapid fire. It has been found that the best method is to practice the technique developed for rapid fire by employing it during the timed fire phase as well. A further advantage is gained in that should an error pattern become apparent during timed fire, the shooter has ample opportunity to take corrective action. He can then test the effectiveness of the correction before being forced to employ it under the more strenuous conditions of the ten second time limitation of rapid fire. A. EMPLOYMENT OF THE FUNDAMENTALS. When a shooter makes his plan on the firing line to shoot timed and rapid fire, usually a number of things have already taken place. Normally, his slow fire has already been fired and the same fundamentals used there apply to the shooting of timed and rapid fire. The planning for a string of shots is an extension of the basic factors involved in firing slow fire shots with the addition of recovery and rhythm. 1. Recovery is the return of the weapon to the original holding position in the center of the aiming area, accompanied by a natural alignment of the sights. If the shooter has a good solid stance, correct natural position, a firm grip, wrist stiff and elbow locked, the recovery is more natural and uniform. In the preliminary check out, if the weapon recovers to the right or to the left of the target center, it may be corrected by simply moving the rear foot in the direction of the error. If the sight alignment deviates, a compensating shift in grip must be made. Recovery must be accomplished as quickly as possible to allow more time for precise alignment of the sights and applying positive trigger pressure. The instant that the weapon was fired, the shooter must immediately resume the sequence of applying fundamentals for the next shot. A distinct rhythm will develop that enables him to deliver his string on the target under control and within the time allowed.



Position

Figure 6-1. The Stance, Position, and Grip Must Be Firm Enough to Absorb the Shock of the Recoil without the Wrist or Elbow Bending.



Recovery

Figure 6-2. And Correct Enough So That Your Recovery Will Return the Weapon to Your Aiming Area Quickly and Precisely With a Center Hold and a Natural Alignment of Sights.

2. Developing a good rhythm is very difficult but is absolutely necessary for good, consistent time and rapid fire. By using a uniform technique, executing a planned sequence of actions correctly and applying careful timing for each shot, we achieve good rhythm. A regular cadence indicates smooth employment of the fundamentals provided the five-shot group is centered and tightly clustered. It is particularly true during rapid fire that you do not have time to correct minor errors in hold. Any attempt to correct minor errors in hold may result in loss of rhythm. This attempted correction cause a hesitation or pause in the sequence of firing a shot while the correction is being made and results in a speed-up of trigger pressure for the

remaining shots of the string. The lack of rhythm causes more bad rapid fire strings than any other factor. The first shot must be fired within one second after the target turns in rapid fire. A common error is to try to attain a perfect sight picture in an effort to make the first shot an X thereby losing valuable time in getting the string started. When this happens, usually the shooter becomes worried about the time, loses his concentration, speeds up his deliver rate for the remaining shots of the string and as a result has poor rhythm and a bad string. Another common error is to shoot the first four rounds with good rhythm then knowing there is a lot of time left, hesitate and try to set up a perfect sight picture so as to shoot an X on the last shot. Usually this last shot will be bad because the shooter does not apply trigger pressure properly. He invariably becomes worried about the time, loses his concentration and forces the shot to fire. In doing so he disturbs the sight alignment by either jerking the trigger or heeling the shot. In timed and rapid fire, a rhythm or cadence of firing must be acquired. This rhythm is needed for coordination and also for assuring the shooter, in a subconscious manner, that an equal amount of time is being allotted for each shot, and that he is abreast of the time schedule. Any mechanical operation has a certain rhythm. Timed and rapid fire is definitely a mechanical operation. B. TECHNIQUE OF SUSTAINED FIRE. The shooter must employ a technique tailored to give him the ability to control the employment of the fundamentals under all conditions of competitive stress. The shooter must consider the following known factors which have a bearing on the control of a five-shot string of timed or rapid fire. 1. Find your aiming area on the edge of the target frame. Look directly at the faced target with your head in shooting position. Determine precisely where your aiming area is going to be when the target turns away. Relate this area to a spot on the edge of the frame that will be nearest you when the targets are edged. The time limitation precludes the luxury of looking at the target as it turns toward you so you can adjust your hold before applying positive trigger pressure for the first shot. 2. Stiffen your shooting arm as it extends the weapon toward the target and settles into the aiming area. Remember the degree of muscle tension required to give you solid arm control and a minimum arc of movement. 3. You should look at the aiming area and relate it to a spot on the edge of the target frame as you settle into a minimum arc of movement. Then shift the point of focus back to the rear sight before making a final point of focus on the front sight. This system is used to make absolutely sure the eyes are not focused somewhere between the front sight and the target. 4. The shooter must never forget that once he attains final focus on the front sight he never again allows a focus shift until all five shots of the string have been fired. To look at the target at any time during the string is inviting disaster. Trust your stance, position and grip to give your precise recovery and to maintain a minimum arc of movement within your aiming area. 5. In rapid fire, the first shot should break soon after the target turns. It is not necessary to try to get the first shot to break while the target is turning but it should break within one second. It is advisable to use the first motion of the target as the signal to apply positive, steadily increasing pressure on the trigger. The target's turning may sometimes produce a feeling of

surprise and is accompanied by a momentary hesitation. This can cause a break in the shooter's composure and the firing of the first shot is delayed. By assuming a more determined attitude and stimulating your competitive aggressiveness you can overcome this problem. We suggest this approach: "When that target moves I am going to punch the 10 ring full of holes." You will be surprised at the effect this action has of eliminating any remaining doubts and at the resulting surge of confidence that it incurs. 6. During recovery, reestablish sight alignment without a focus shift. This action is important because a focus shift during recovery will delay the reestablishment of sight alignment. Approximately one-half second is needed between shots for this ill-advised operation and could total two full seconds of lost time. Successful rapid fire requires the use of ten full seconds for proper coordination and full control. 7. If you allow your eyes to follow the pistol during recoil, you may inadvertently move your head out of its original position. Any head movement during firing will disrupt the relationship between the aiming eye and the front and rear sight alignment. Correction will require a wrist movement which only artificially corrects the error. Upon recovery from recoil of the succeeding shot, the same error is once again apparent and likewise needs correction. 8. Reaffirm your determination to concentrate upon sight alignment the instant positive trigger pressure is resumed. Maintain your concentration on sight alignment until the pistol fires again. 9. After the pistol fires it will be moved out of the normal hold area by the recoil of firing, and it must be recovered instantly to the position it occupied prior to the dislocating effects of recoil. Recovery must be natural, uniform and quick. 10. You must immediately reestablish a positive, steadily increasing pressure upon the trigger. This should occur shortly before recovery is complete and the minimum arc of movement reestablished. The increasing pressure should neither be stopped nor varied in rate until the weapon has again been discharged. As soon as the positive, constantly increasing trigger pressure has been reapplied, shift your attention from thoughts of trigger control to the problem of sight alignment, just as you did on the first shot of the string. 11. The sights will be in near perfect alignment at the end of recovery, if the grip, control and head position are maintained. However, this ideal situation occurs only intermittently. 12. Remind yourself that this technique, repeated for each shot, insures that continuity is established from one shot to the next. Assure yourself that you can deliver a successful string on the target with an absolute minimum of wasted thought and time by following this system. C. COMMON DEFICIENCIES IN CONTROL: A number of deficiencies peculiar to timed and rapid fire are: 1. Follow through, applies to slow, timed and rapid fire and should not be confused with recovery. Follow through is the attempt by the shooter to keep everything exactly as it was set up until after the round is on its way to the target. Lack of follow through is a breakdown of one or more of the factors set up by the shooter to control a good shot. For example, lack of follow through might be caused by a speed up of trigger pressure resulting in anticipation of recoil and a heeled shot at one o'clock. 2. Recovery must be made quickly to allow time for aligning sights and positive trigger pressure. Recovering too slowly takes up excess time, alters the shooter's rhythm, and when

he realizes that he has very little time left causes him to speed up his delivery rate. Each shot of a five shot string must be fired individually and uniformly, each one treated as a single shot. The shooter must see five distinct sight alignments. 3. Grip: An incorrect grip will cause misalignment of the sights on recovery after each round is fired. This is corrected by carefully shifting the grip before the next string. Any tilting or turning movement of the head from its normal level position will cause the weapon to appear to recover either to the right or left of the bull's eye. Both of these errors may cause a delay in firing on the shooter's part in an effort to correct them, or break his concentration on sight alignment, losing valuable time and causing a loss of rhythm. All these factors add up to a poor string of five shots. Check out and dry fire the position and grip during the three minute preparation period just prior to the range officer's command "LOAD". 4. Calling The Shot Group: Many shooters fail to remember each shot on the basis of five individual sight alignments and cannot call the shot group accurately. If the shot group call is made and the call and the group are not together, it is necessary to determine the case and apply positive correction. The weapon probably is not zeroed, the position was bad or the grip incorrect. If the shooter is sure of the zero of his weapon, then dry fire the position and grip before firing the next five shot string. 5. Rhythm is absolutely essential. A common error in sustained fire is trying to make the first shot an X and thereby losing valuable time in getting the string started. When this happens, the shooter usually has poor rhythm and a bad string. When a determined shooter causes the first shot to be fired on time, this same determination brings about a continuous application of the fundamentals that assures a rate of fire that will complete the string on time. 6. In shooting rapid fire the shooter does not have time to correct minor errors in hold. Trigger pressure is applied on the basis of sight alignment and not sight picture. The shooter should make every effort to keep his arc of movement at a minimum, continue positive trigger pressure, maintaining sight alignment, and shoot with a definite rhythm. 7. Lack of A System: When a shooter has a system to follow it relieves his mind so that he can concentrate on performance. Care should be taken during the early stages of instructional practice to comply with each of the items on the worksheets. As the shooter becomes more capable, only the key items of preparation, shot sequence, shot analysis and positive correction are relevant. Methodical repetition of these essential steps will instill in the shooter good shooting habits that will enable him to repeat a good shooting performance. Further, the rapid fire worksheet will help the shooter form the habit of not overlooking any factor that will help his shooting. 8. Complete and instantaneous shot analysis is a mandatory prerequisite for any improvement in your performance or scores. A mental impression of each sight alignment should come at the instant the shot breaks. Corrective measures to prevent the recurrence of a poor performance must be immediately applied. Much has been written about why we shoot poorly; however, be reminded that it is just as advantageous to analyze why you are shooting well on a particular day. It is more helpful to know the right way to perform than to have your mind cluttered with a multitude of "don'ts ". Coaches in particular should concentrate on and

emphasize the positive factors. 9. Overeating during the shooting day has lowered may aggregates. The delicate edge that a shooter attains before a match can be completely shattered by one hearty repast. The minimum arc of movement is greatly increased by the pulsations of a heartbeat imprisoned between an overloaded stomach and a suet incased ribcage. 10. Inability to control mental processes indicates a fear of failure or lack of motivation to do your best. The shooter must develop more effective method of stimulating confidence. Review the reasons why you are here as a shooter. You came to win the match. Encourage your competitive instincts by setting a goal as high as you can possibly reach. 11. The shooter's concentration breaks as target turns. More attention is required in developing a determined attitude and mental alertness. Review the system you use in starting positive trigger pressure and maintaining point of focus on front sight. Apply any correction needed. Remove all doubts as to the location of the center of the aiming area of the target in relation to the edge of the target frame during the preparation stage. D. TRAINING METHODS. 1. Frequent shoulder-to-shoulder competition and regularly scheduled record practice on the firing range is the most effective method of accelerating your development as a top competitive shooter. 2. To be most effective, each practice session must have a goal. You should approach the training period with the idea that you are going to distinctly improve one aspect of your shooting technique and at the same time continue the general improvement of your ability to employ the fundamentals more effectively. 3. To improve your ability to deliver your first shot quickly and accurately, we advise a practice session of about ten rounds delivered in the following manner. Adjust the target turning mechanism to face the target and turn it away after one and one-half seconds. Use your normal preliminary preparation with maximum attention on delivering the first shot without hesitation as the target turns. Fire one shot only. Repeat the exercise ten times with sufficient time between shots to allow for mental reorganization and preparation. Fire two, fire-shot strings with the proper 10 second interval to establish your rhythm and then shoot a rapid fire, 20 shot match for record practice. 4. To improve your ability to achieve rhythm and maintain a point focus on the front sight, place a target on the frame backwards so that no bull's-eye or aiming point is visible. Assume your stance, position and grip with meticulous attention to detail. Without a point to aim at, you will find that you must trust your stance and position to maintain an acceptable arc of movement in the center of the aiming area on the blank target. You will find it easier to apply the fundamentals and discover that you can deliver the string with amazing accuracy. Rhythm and sight alignment can be maintained with a startling degree of control. This is because the distracting effects of having an exact point of aim has been eliminated. You have no way of knowing when a perfect hold occurs. A perfect sight picture is not necessary. You simply accept minor errors in hold caused by your minimum arc of movement and go ahead and follow your plan of delivery of each shot. After firing on the blank center you should immediately go into a rapid fire stage of fire with a normal target for record practice. 5. Avoid training and shooting alone. Use a training program that duplicates as near as possible the

competitive atmosphere of a match. Develop and use a comprehensive plan that improves your ability to employ the fundamentals reliably under pressure and continuously strive for improvement. 6. Dry firing practice should be conducted with the same careful attention to detail as live ammunition practice. The shooter's rapid fire worksheet (para C, "Establish a System"), this Chapter, is a guide to perfecting your system of shooting control. 7. Improvement of recovery must be approached from two angles: Reestablish a hold in the center of the aiming area, and realignment of the front and rear sights in perfect relationship. Practice and re-practice assuming a proper position that furnishes the shooter with a natural hold that points the shooting arm and weapon at the center of the aiming area. Get a proper grip and head position that gives the shooter a natural sight alignment. Quick recovery is essential. For example, with a 2 second interval to deliver a rapid fire shot, there should not be more than 1/2 to 1 full second devoted to recoil and recovery. At least 1 second of the interval must be used to dress up sight alignment while settling into a minimum arc of movement. The simultaneous application of positive trigger pressure may be delayed if either factor, hold or sight alignment, is imperfect. E. WIND SHOOTING AND ADVERSE CONDITIONS: During timed and rapid fire, the shooter has to fire when the commands are given, wind or no wind. The means of overcoming this disturbing handicap are found in strenuous application of the fundamentals. Usually these efforts attain less positive results because the shooter cannot maintain his normal, minimum arc of movement in the wind. Concentration on sight alignment regardless of movement caused by wind will result in groups only slightly larger than those fired under ideal conditions. 1. During Wind Shooting: As the arc of movement increases during wind shooting, the shooter develops a tendency to relax his trigger pressure. He is waiting for a more stable sight picture. His concentration on sight alignment will diminish and he will make an effort to fire as the sights pass the vicinity of the target center. The obvious answer is to concentrate on sight alignment and maintain as small an arc of movement as possible and to start a constantly increasing pressure on the trigger until all shots are fired. Each attempt to fire a string of shots should be made with a firm resolve to align the sights for each shot and to apply increasing trigger pressure in spite of the increased arc of movement due to the wind. Your shot group will be larger, as a result of the increased arc of movement, but the wild shots resulting from faulty sight alignment, flinching, jerking and over-correction will be minimized. a. Rhythm must be maintained, with the uncompromising determination not to hesitate in applying positive trigger pressure despite the abnormal movement of the shooting arm. b. Extensive practice under windy conditions is not recommended but enough firing should be conducted under windy conditions to prevent a stampede to the nearest wind shelter when a wisp of air movement stirs the pine tops. c. The shooter should not place too much reliance on indications of flags high above the line of targets and the firing line. In addition, do not accept the indications of flags flying at the edge of a forest, steep precipice, ravine, or depressions, since the wind speed, at various levels of the atmosphere and terrain are different. It is necessary to be

guided by the indications of high grass, tall weeds, strips of paper, etc. in the vicinity, which are nearer the level of the weapon-target line. d. It should also be kept in mind that wind can blow around terrain irregularities and create all kinds of turbulence. If flags were set up along the entire length of the range, they often would indicate a different, even opposite, wind direction. For this reason, the shooter should not always rely on one indication at the line of targets. Determine wind direction and intensity for the entire length of the range, by carefully observing the motion of grass and bushes located between the firing line and the target. e. With time, the shooter will develop a subconscious feeling and acquire experience that enables him to become rapidly oriented to wind conditions and to make the necessary corrections for carrying out accurate fire under adverse conditions. 2. Adverse weather conditions such as cold, hot or rainy weather or extreme light conditions pose problems that can be solved in the same manner as in wind shooting. Be determined to adhere the fundamentals and ignore the distraction of adverse weather. a. It is advisable to carry a raincoat with you at all times and a plastic cover for your gun box to keep your equipment dry. Most ranges have covered firing points that help to keep the competitors dry during rainy weather. b. During cold weather the shooter must obviously wear warm clothing to include insulated underwear. When the shooter becomes shivering cold it is difficult to hold the sights in perfect alignment, or retain sensitive trigger control. Hand warmers are very good and are small enough to keep in the gun box or pocket. Light weight lubricating oil must be used in cold weather to prevent malfunction of weapons. c. During hot weather perspiration becomes a problem. A sweat band on the forehead keeps sweat out of the eyes and it is recommended that powdered rosin be used to dry the hands. When not on the firing line the shooter should relax in the shade. Here again covered firing points provide protection from the sun while shooting. Salt tablets prevent heat prostration. Eat lightly. d. Effect of temperature on the shot dispersion: (1) The lower the air temperature, the greater the air density. A bullet traveling in denser air encounters a larger number of air particles, with the result that it loses its initial velocity rapidly. Therefore, when shooting in cold weather, the bullet velocity decreases somewhat and the center of impact moves downward slightly. (2) In firing a large number of rounds for an extensive period of time, when the pistol barrel becomes hot, the shooter should not permit a round to remain in the chamber too long. The relatively high temperature of the barrel is transferred to the propellant by means of the cartridge case, and can lead to a change in center of impact and to high shots, depending upon the length of time the round remains in the hot chamber. 3. Light varies from extremely bright to very dim and the shooter must keep a record of light conditions on every range fired on in his score book. Some competitors are affected more by changes in light than others. A note should be made as to how much his zero changes in the different light conditions. Sights should be blackened with care on bright days. As a part of the shooting accessories, you should have both amber and green shooting glasses not only for light conditions but for protection against oil, wind and empty brass. Firing from an uncovered firing line usually requires different sight settings

than the firing from under a shed. Ammunition should be kept out of the sun as its accuracy is affected if it is exposed to the direct rays of the sun. 4. The major portion of our accomplishments on the firing line stems from our mental capacity to face up to the out of the ordinary and parlay these conditions into a winning margin. Poor conditions must never become an excuse for expending less effort and consequently a poor performance. Good scores are produced by hard work in the application of the fundamentals regardless of the conditions. Proper application of the fundamentals is the most important factor in shooting winning scores under adverse conditions.

PERFECTING TECHNIQUE

Mental Discipline

By Chapter 7: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter7.htm

A. PURPOSE: The purpose of this chapter is to acquaint you with the need for and the method of controlling your mental and emotional processes and extending your span of mental concentration while under conditions of competitive stress.
B. GENERAL: Mental discipline is the broad term used in describing the shooter's actions and reactions when facing competitive pressure. A distinguishing feature of successful competitive shooting is that it is associated with overcoming obstacles and difficulties which require the utmost exertion of a person's mental capacity. The ability to keep control of oneself to force oneself to overcome difficulties, and to maintain presence of mind in any difficult situation is a necessary human quality. Without this quality, you will not shoot high scores in a match. To sustain mental discipline, you must have high moral qualities, a sense of duty and responsibility to the team and a sense of honor. These traits are the source of the will to win. In moments of crisis, they help you to mobilize all your resources for victory. No person is born with these qualities. They are partly developed in the course of the shooter's life and the activities of daily living. Good marksmanship training will solidify these traits and develop the minds ability to control mental processes. C. ESSENTIAL TO MARKSMANSHIP 1. Mental control is essential to marksmanship. Mastery of the physical skills alone does not provide the control of performance necessary to compete at the highest level. Emphasis must be placed on how and what to think. The capacity for intense concentration will provide for exacting control. Coordination of the essential factors is necessary for the delivery of an accurate shot on the target. 2. Mental discipline provides the control you must have xx of your mental faculties to maintain confidence, positive thinking, and sustain the ability to duplicate a successful performance. Mental discipline will help to avoid overconfidence, pessimism and withstand conditions that disrupt mental tranquility. It also provides the emotional stability necessary for the development of a champion shooter and confidence in his ability to successfully employ the basic skills of marksmanship for a dependable performance under all types of stress. D. DEVELOPING MENTAL DISCIPLINE AND CONFIDENCE: The continuously repeated, successful execution of a completely planned shot results in the gradual development of mental discipline. If your mental discipline has developed sufficient force you will be able to control your thoughts and exercise unhampered mental concentration. Also, your preparations and shooting routines will always be the same. 1. Response to a problem: Psychologists have determined that there are four basic methods of responding to a problem. Two methods are positive and classified as direct or indirect. Two methods are negative, classified as retreat and evasion. a. Positive Response. (l) The direct,

positive approach. This is the self-confident, self-sufficient, direct, positive attack. You realistically face the facts, analyze them, identify and evaluate the obstacles to a successful solution. You know what you want to accomplish and you take direct steps to attain it. (2) The indirect, substitute or compromise approach. This is characterized by small diffident, tentative, indirect action. Sidestepping leads to seeking shortcuts. When the probable solution is tried, there is much fervent hoping that the fates are on your side. You are only hinting and probing instead of establishing definitely what you need to do. There is only a minimum of positive effort here. b. Negative response. (1) The negative retreat: The failure to give the honest try to see what you are capable of accomplishing. Surrendering without a sincere effort. The flight habit can become chronic. This is the man that cannot accept the responsibility for a mistake or failure. A bad shot produces excuses. (2) Evading the issue: Evasion is the lack of incentive. Why? Is the approach. Why do I have to do better then anyone else? If the desire to excel is not there, you will never aimlessly or otherwise achieve the degree of accomplishment that crowns the champion. 2. Analyze the problem. a. Psychologists have discovered that one of the chief reasons for difficulty in the solution of problems is inability to soundly analyze. Pose a clear-cut plan of action in full array. Face the specific difficulty and make a determined effort to break it down. If it can be identified there is a solution for it. There are shooters on your team or some other team that are operating without this specific problem putting a brake on their performance. Talk it out. A communal pondering session will break it wide open, b. There is a four-point system of analyzing and solving specific problems. It reduces the whole big problem to four small ones: "STEPS IN THE PLANNING"; " SPECIFIC DIFFICULTIES"; "SUCCESSFUL SOLUTIONS"; "DOUBTFUL OR NO WORKABLE SOLUTION". Weigh your "specific difficulties" and "doubtful solutions" and start an improvement campaign to resolve each area of deficiency. 3. Confidence. Confidence results from repeatedly bringing under control all the factors that create conditions for firing an accurate shot. An accurate shot is one that hits the center of the target. You must have confidence to shoot well. Confidence in what? How do you get it? How do we keep it once it is obtained? a. You must have confidence in the fundamentals. You must be convinced that if you control their employment correctly, you will achieve excellent results. b. You must also have confidence in your ability to execute the proven fundamentals correctly. You will have proven your ability to do this in your practice sessions. c. Think big! Think positive! "I will do it", and you will succeed. However as soon as you admit the slightest possibility of failure, your chance of success is questionable, d. It has been said that a shooter must have an open mind, implying that we must have the ability to accept new ideas. What we should also strive for is a mind that is open to positive thoughts and completely closed to negative thoughts. You have heard so many times "Don't jerk that trigger". True as this axiom may be, it is of no advantage to have this thought enter your mind when you are trying to get off a shot. It is negative, it implies failure. Such thinking continually occupies your mind with something you don't want to do, rather than something you should do. Would it not be more

advantageous to think, "I must follow through, for when I do this, I will get an "X"". This is the positive side of the picture, it implies success. It gives you something that you should do rather than something you should not do. What the shooter needs is a mind full of positive "do's" and "wills". There is no room or necessity for those distracting "don'ts" and can'ts". However, just thinking positively is not enough; we still must have definite ideas of how we are going to employ positive thoughts. There is no room for vagueness or vacillation in our technique of shooting. e. A confident attitude adversely affects your competitors. A match is generally conceded to a small number of confident individuals who expect to win. Confidence is contagious and favorably affects your teammates. Smile. Give no comfort to your competition by revealing by word or by act that anything is wrong that might affect your chances of winning the match. 4. Channeled mental effort resists the tendency of the mind to drift during the period when intense concentration on the relationship of the front and rear sights is essential. a. Channel mental effort relentlessly toward the final act. b. Complete exclusion of extraneous thoughts for a brief period (three to six seconds) is necessary for controlled delivery of the shot. c. Prior planning of the sequence of action gradually enables the shooters to sustain concentration for a longer period. d. Coordination of thought and action is the result of experience obtained through extensive practice and match shooting where the same satisfactory plan of action is followed repeatedly. Precise coordination is absolutely necessary in controlling the delivery of each shot during the entire match. Split second coordination and timing are maintained by frequent practice. When the practice time is insufficient, do not be overconfident and expect to be able to sustain coordination through prolonged match shooting conditions. E. WHY CAN'T YOU BE A WINNER? (THE DANGER OF NEGATIVE THINKING.) 1. Who won the last match in which you participated? If you did not win, what was the reason? Why is it so difficult to shoot championship scores? It is not that most of us have not been taught the fundamentals of shooting, the fault usually lies in that we open our minds up to thousands of negative reasons why we cannot shoot good scores. The following is a discussion of each of the reasons that bring about a poor performance, and what can be done about them: a. When the weather is bad, it is simple to say "It is raining, snowing, the wind is blowing. All my scores are going to be bad. ". This may be a true assumption. You can follow this vein of thought throughout the match and you probably will continue to shoot just average scores as compared to your competitors. Why not think and convince your self that good winning scores have and will be fired under the same bad conditions. Positive application of the fundamentals has produced good results in spite of the numerous difficulties. If your thoughts are directed strongly enough towards planning and executing a controlled performance, you will not have time to worry about the weather. b. Don't "Sunday - morning - quarter back" the operation of the range. Convince yourself that, "As long as there is a target to shoot at and I have the proper amount of time to shoot, I will shoot good scores. ". c. Have you asked yourself, "why must I shoot exceptional scores ? ". The answer to this question will vary with each shooter. You must be motivated to constantly

improve your performance. One of the most common excuses for not trying your best is because there is no challenging competition. A tendency to accept a passable score in a match becomes a habit. You tolerate an average performance without becoming alarmed. Regardless of the competitive ability present, you must employ the fundamentals to the utmost of your ability. You must retain not only the desire to win, but strive to set new records at all times. Failure to accept the challenge will cause a decline into the habit of treating your shooting as a daily task instead of a challenging adventure. d. The main components necessary to shoot championship scores are an accurate gun, good ammunition, an individual with the ability (physical and mental) and the desire to be a champion. Therefore every time you let the thought of inferior equipment enter your mind, STOP! Think: "This gun and ammunition will shoot possibles if I control it. " e. The potential winner is always thinking about applying his plan of action and not about how he is going to beat you. He knows that most of the other competitors are beating themselves with their own uncontrolled thoughts. You can be one step ahead of all your competitors by directing your mental effort toward your plan of controlling each shot. f. There is a first time for winning in shooting as in everything else. A first time for a national champion to be beaten, and a first time for you to become a national champion. If you want to win all the marbles, you can. The best way is to believe you are as qualified to win as anyone else. Make up your mind that you are going to shoot your next tournament as one big match. Let the individual stages and gun aggregates take care of themselves. A good performance on each individual shot is now your aim. g. Carelessness is a state of mind that overwhelms an individual who is aimless and hap hazard in his approach to a challenging task. Organization of all the factors having a bearing on the task will in most instances assure that the action will be successfully executed. h. Overconfidence dulls your normal responses. You ignore or are unconscious of the development of unfavorable conditions. False assurance can upset the sensitive balance on which your performance depends. Do not relax your determination to work hard even if competition is not keen. Strive to reach a happy medium between overconfidence and pessimism. i. Pessimism detracts from your ability to concentrate. Anxiety over possible failure undermines the ability to control the shot. Impatience and uncontrolled actions are the results. A negative approach hampers the repetition of a uniform, satisfactory performance. j. Avoid distracting conditions which you know will upset you. Avoid emotional upset such as anger, worry, giving up under adverse conditions or after unsatisfactory shots, ignore boasts, rumors, misinformation, and snide remarks. Avoid adding up individual shots as the buildup to the final scores. F. MATCH PRESSURE: If you think that you and you alone have the problem of match pressure, look around - we all have it. The man who has never experienced match pressure has never been in a position to win a match. What is the difference? What is the dividing line between champion and plinker? Both may shoot comparable scores in practice, yet one is invariably at the top of the list and the other at the bottom. The dividing line is clear and obvious; the ability or lack of ability to control their thinking. Mental discipline. Some have learned to control their

emotions and anxieties and go right ahead and perform within their capabilities. Others, even with years of experience, pressure themselves out of the competition every time they step up to the firing line. 1. First, in the treatment of match pressure, we must find what causes it. Without knowing the reasons, we can never combat it. Match pressure is simply a condition created by suspense, and the uncertainty and anxiety which generally accompanies suspense. For example, it is easy for the relatively inexperienced competitor to feel suspense building up as he finds himself amassing a superior score; or for even the experienced competitor to feel, as he nears the finish of a match, knowing he can win. This is when worry and fear creeps in and, unless controlled, the resultant tension will undermine efforts for maximum performance. 2. The main thing that will help a shooter under these conditions is experience. Long hours of practice in working on his shortcomings and tournament participation against the best competition will serve to gradually calm our emotions and anxieties when under stress. The champions, in spite of their nervousness in match competition, mobilize all their energies and resources and on occasion, do even better in a match than in practice. The emotional and physical upsets of competitive stress are experienced differently by different persons. The condition varies for every shooter both in its character and in its intensity. However, regardless of experience or ability to exercise self-discipline, shooters are to some degree nervous in competition. The better you are trained, the more confidence you will have. If you have trained under conditions approximating match conditions and have participated in many tournaments in the past, you will be less nervous. At the beginning of a shooting season, even with experience, you may be somewhat nervous. It is important that you must not remain passive to these disturbances. Do not let yourself become a victim of your emotions. Resist stubbornly and force yourself to shoot to win. If you feel that nervousness in competition is unknown to you, you may be indifferent to the best interests of the group. You may lack an elementary understanding of pride in doing a job well. You are showing indifference to one of the strongest, natural excitements which present a challenge to the human animal. When anxious, you add to your distress when you feel that everyone is watching you. Yet with all this, our counterpart, the Champion, appears to be calm and enjoying himself. Let's face it, he is! 3. How do you control match pressure? First, realize that it can be controlled and actually used to your advantage. Individuals have learned to control their shooting to the extent that their match and practice scores don't vary appreciably. a. Prior mental determination. This is the most helpful factor that is available to you. By thinking through the correct procedure for firing each shot, just before you shoot, you can virtually eliminate distraction. If you fail to do this and approach the shot without a preconceived plan of attack, your results at best will be erratic. b. Channel your thinking to the more important fundamentals. You must continually think fundamentals and review them in your mind. Train yourself so that as many of these fundamentals as possible are executed automatically without tedious effort on your part. When you do this, you have only the most difficult fundamentals to contend with in the actual firing. This will enable you to direct all of

your mental and physical efforts toward keeping your eyes focused on the front sight and following through. c. Establish a Routine: Keep from becoming excited. In establishing a routine, you eliminate the possibility of forgetting some trivial item of preparation or technique that may throw you off balance. d. Work on each shot individually. Each shot must be treated as an individual task. There is no reason to believe that because your first shot was bad, your next one will be the same. Nor is it logical that if your first three shots were good, you have a guarantee that those to follow will also be good. Each one is merely a representation of your ability to apply the fundamentals. Your performance will vary if you let it. e. Relax your mind. Right from the time you get up in the morning. Nothing will put you in a greater state of mental agitation than to have to rush through breakfast and rush to make your relay. If this happens, your score is ruined at about the third red light you hit. Take it easy. Shooting is fun, enjoy it. f. Practice Tranquility. Are you the guy that loses his temper every time he has a bad shot? With whom are you mad? You are doing nothing more than admonishing yourself for your vacillation in the execution of a shot. If you had worked a little harder on applying the control factors, the shot would have been better. On the other hand if you do everything within your power to make the shot good and for some reason or other it is not good, you should have no cause for undue irritation. Although you must exert all of your mental and physical ability toward shooting a good score, infrequently you will fail to do this. Needless to say that when this happens, if you chastise yourself severely, or fall into a fit of depression because of poor score, you will hurt your performance for the rest of the match. It is not intended that you laugh off or treat lightly a poor performance; however, you must possess the presence of mind to accept the bitter with the sweet. Preparing, planning, relaxing and care in delivering the shot with careful analysis and positive corrective measures, is the cycle of action you must force yourself to conform to. You can then be assured that the next shot will be delivered under the most precise control you are capable of exerting. g. Match Experience. Without question, competitive experience is one of the ingredients necessary to an accomplished competitor. However, experience alone is of limited value. You must flavor experience with an accurate and honest evaluation of performance. You must strive for increasing mental control. It is often left out of training until the physical ability to shoot far exceeds the ability to exercise mental control. h. Argue with your Subconscious. Not only argue with it but win the argument. Even as you are reading this you are hearing that little voice in the back of your mind that keeps saying "Yes, this sort of thing may work for Joe, but I know damn well I am going to goof the next time I get close to a winning score.". Whose voice is this? Where did all these ideas come from in the first place? Where did this little guy get all his knowledge? Let us be realistic. Your conscious mind puts these ideas into your subconscious, so don't ever believe that you can not over power it. It is not easy. He has been saying what he pleased for years and now he isn't to be routed easily. But don't give in to him and eventually you will find that the subconscious mind is not in conflict with your conscious efforts "don'ts. " k. With all of this emphasis on the positive approach you are now going to get

two big "don'ts". (1) Don't expect spectacular results the first time you try mental discipline. There is coordination of employment of the fundamentals to be mastered. If you find that you exercise satisfactory control only for a short period of time, work on extending this period by practicing and perfecting your system. Remember that your returns are in proportion to your investment. (2) Don't use alcohol and drugs. One or both of these may control some of the symptoms brought about by match pressure. However, in doing so they incapacitate you in other ways that will prevent good performance. G. REDUCING TENSION AND ATTAINING RELAXATION 1. Types of Tension: a. Normal tension is the prevailing condition of any organism when it is mustering its strength to cope with a difficult situation. All animals, including man, tense in situations which involve the security of themselves and their loved ones. b. Pathological tension is an exaggeration of normal tension and fairly rare. This type of tension usually requires that the subject be put under the care of a physician. c. The vast majority of people and shooters who are concerned with tension have nothing more than normal tension. All they need is a technique for relaxing. You should know what tension is and a few hints on how to minimize its effects. 2. In normal tension, your body undergoes certain definite changes. Adrenalin pours into your bloodstream and your liver releases sugar, giving a supply of energy to your muscles. Your entire nervous system shifts into high gear. It causes your sense of smell, hearing and sight to become sharpened and all your mental faculties to become razor keen. Your stepped-up nervous system also causes the large voluntary muscles of your legs, arms and torso to contract, ready for action. The muscles of your digestive tract cause your digestion to slow down for a while. Your chest and arterial muscles contract slightly so that your breathing becomes a bit shallower and your blood pressure increases. When all these things are happening, you are experiencing normal tension. Most of us experience this kind of tension one or more times a day. When the problem which caused you to be tense has been solved, your tension will subside and you will return to a normal state of relaxation. It may leave slowly but it will leave. Normal tension is self-limiting, it does not continue unabated after you need it. 3. Pathological tension is when your whole body over-reacts, as if the difficulty confronting it were a life or death matter. It is the kind of reaction a normal person would have only in an extremely dangerous situation. In pathological tension, blood pressure, heartbeat and pulse go way up and stay up. Excessive adrenalin may result in jitteriness, flushing and trembling. The digestive actions of the stomach usually stop entirely and will not resume, causing loss of appetite or indigestion. Muscles tense for action but may end by cramping. There is rapid, shallow breathing to the point of dizziness. The inevitable, and often swift result is a sense of deadening fatigue. Normal tension may make you feel exhausted too, but not to this degree. 4. Tension Reducing Techniques: Take a breather. Breathe deeply, three times, very slowly; at the end of each exhalation, hold your breath as long as possible. When you have finished, you should feel noticeable relaxed and much-calmer. By forcing yourself to breathe deeply, you break the tension of your voluntary breathing muscles. This causes the involuntary muscles of the lungs,

gastrointestinal tract and heart to relax too. This is the simplest method for relaxing. For some, it can be used to end tension completely. It can be used by others for temporary relief when they do not wish to "let down" completely. b. Let go. Sit down and let your head droop forward. Try to actually concentrate on relaxing the muscles. Make one arm relax completely; then the other. Now let your legs go completely limp; now your torso muscles. Stay in this posture for several minutes. Momentarily divorce the competition completely from your mind. This technique is aimed at relaxing the voluntary muscles. It is especially effective when you have had to maintain normal tension for several hours on end. c. Stop and Think. When the tension-making job allows a respite, sit down and calmly review the things in your life that you value highly. Think of the long range purpose of your life, of the people you love, the things you really want. In a few minutes you may notice that you have involuntarily taken a deep breath. This is a sign that tension is dropping away rapidly. When you tense to face a difficult situation, you tend to exaggerate its importance. Judgment and reason can quickly change this mental state when it is time to relax again. d. Take a Break. This is a "Remote Control" technique for dealing with normal tension. Simply take a break for ten full minutes every hour. You may find that this allows you to ease out of your working tension more quickly and easily. e. Shift Into Low. Taper off at the end of the day by becoming involved in a diverting activity. If you like handiwork, pick a kind which is interesting but not too creative. Soap sculpture, finger painting, woodworking, and gardening all are excellent low-gear activities that will help you to "simmer down". This kind of tension-remover is aimed at changing your mental "set". It is helpful for those who have to operate at top capacity. After stimulation, a part of your mental capacity will continue to be aroused. To slow you down when you are in this state of mind, you require something which is engrossing but which demands nothing of you intellectually. Television entertainment and simple handicrafts are ideal. f. These techniques are based on the fact that tension can be ended in two distinct ways: through the relaxation of your voluntary and involuntary muscles; and by changing your mental "set". If you achieve either, you modify the other and hasten the process of normal relaxation. H. YOU CAN WIN! 1. Confidence furnishes the alloy to stiffen the will to win and not give up or compromise. Confidence is based on a full grasp of the complete technique of controlling employment of the fundamentals. Confidence combined with knowledge, good physical condition and a determination to win, will allow you to perform at your best. A chance at greatness lies in each man's grasp. You must have confidence that you are capable of a performance exceeding any previous level of personal accomplishment. Know that you can win if that is what you set out to do. 2. Be a hungry shooter. The slashing onslaught of a voracious appetite for victory destroys the resolve of the lesser competitor.

PERFECTING TECHNIQUE

Physical Conditioning

By Chapter 8: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter8.htm

A. GENERAL

1. The objective of physical training in a pistol marksmanship training program is to condition the shooter physically, to better withstand the rigors of match conditions. An individual in good physical condition has better developed reactions, better control of his muscles and better endurance; all of which promote consistency in performance. 2. It is important that physical training not be of haphazard nature, nor should it be timed immediately before a pistol match. A shooter should perform physical exercises regularly, both during the preparation period between shooting seasons and during the period when the shooter is training for tournament participation. Morning limbering up exercises are important in this connection, and they should become a part of a shooter's daily routine. It has been found that a physical training program should be discontinued approximately 3 days prior to a match and resumed immediately thereafter. 3. Physical Conditioning must consist of exercises of a general nature directed toward strengthening the muscles, proper breathing, developing body flexibility and precision of movement. The requirements of marksmanship are such that drills must consist of exercises which develop the muscles and flexor of the arms and fingers, and the muscles of the shoulders and waist. A certain amount of static tension (dynamic) type exercise is valuable if it is not overdone. 4. Whenever the shooter exercises, he must put the maximum effort into the exercise. Merely going through the motions of an exercise is of no advantage. Physical conditioning is a gradual process and results will not be apparent immediately. As the shooter's physical condition improves, the number of repetitions may be gradually increased. Heavy exercise such as serious weight lifting should be discouraged. 5. Any sport that encourages regular physical activity is beneficial to a shooter. It is recommended that each shooter cultivate an interest in a sport that will insure sufficient exercises for all around physical fitness. 6. A muscle builds more rapidly under tension applied vigorously. 7. The stronger the muscle structure is developed, the surer movement can be coordinated and positions held. Besides general conditioning practices, durable muscle tension exercises of the trunk, shoulder and arm muscles make the most sense. Resistance exercises and grip exercises are in order. Physical training should take place at least three times a week for 30 minutes to an hour. 8. Sleep: During the training period the shooter needs plenty of sleep to give all the organs a chance for sufficient rest. Eight hours should be the rest interval. Before matches, insomnia sometimes occurs due to excitement. Under such conditions no sleeping pills should be taken the night before a match for they work out unfavorably the next day. Short walks in the evening, warm showers or a small snack will

sometimes induce sleep. 9. Detrimental habits: Nicotine, caffeine and alcohol reduce the performance ability of the body and effect the ability to concentrate. (See Chapter X, "Effects of Alcohol, Coffee, Tobacco and Drugs.") 10. Overall behavior: Before a match the shooter should avoid all types of excitement. For example, he should not drive fast and, if possible, arrive at the range one-half hour before the beginning of the match. That will give him plenty of time for his last preparations and he can prepare himself inwardly for the test ahead. The individual feeling of well being is the best measure of whether or not your living habits and daily routing is in the best interest of your shooting. B. BASIS FOR A GOOD PHYSICAL CONDITION, A physical training program should be progressive. It is not necessary or is it generally considered wise to strive for the peak condition sought by a track athlete or a professional football player. Violent and strenuous athletics which may result in injuries should be avoided. The competition shooter must possess the following basic physical and physiological characteristics: 1. An adequately developed muscular system (this is especially true for the muscles of the abdomen, arms, and legs) and the endurance to fire many shots without perceptible worsening of results. 2. Lungs must have a high oxygen assimilation factor so that long pauses between inhalation will not cause oxygen starvation. NOTE: A smoker's lungs do not have a desirable oxygen assimilation factor. 3. Precision and coordination of bodily actions and thoughts. The physical training of a pistol shooter must be directed to the development of these qualities. C. TYPE OF EXERCISE. There are many different general types of exercises and activities that a shooter can use to his advantage. 1. Walking is a very good exercise. When walking, don't just take a slow window shopping walk. To get any good out of it, you must make the walk very brisk. 2. Running: Results already obtained indicate a running program improves overall physical fitness, especially in endurance capacity and overall improvement of the heart, lungs, and the entire circulatory system. Running and walking for a period of 12 minutes at least three times a week is, in itself, a splendid conditioner. Here again, the individual must apply himself diligently for the duration of the exercise period. 3. A series of mild, non-strenuous exercises of the type that require body bending, stretching, deep breathing and moderate muscular tension are best suited to obtaining a condition defined as good body tone and a feeling of well-being. Sore, aching muscles tend to fatigue quickly, and nervous tremor usually results. 4. In Swimming almost all of the muscles get a workout. Here again, the exercise should be pursued with moderation. 5. A good exercise to build the wrist and arm muscles in the wrist and forearm developer, roll-up exercise. A mop handle, a short length of rope and a weight, Roll it up and let it down slowly. 6. A method of developing the grip is by using a sponge rubber ball about 3" in diameter, cut in half. Squeeze the ball with the shooting hand. You can take this aid with you almost any place you go, and exercise any time that you are not using your shooting hand. D. THE PISTOL TEAM DAILY DOZEN EXERCISES. The Pistol Team Daily Dozen was especially developed to affect those muscles used in pistol shooting. Figures 8-1 thru 8-12 will assist the instructor in understanding how these exercises are accomplished. 1. Warm Up: A

four count exercise done in moderate cadence. This exercise is designed to get you ready for the forthcoming exercises. Starting position is standing with feet spread approximately 12 inches apart, hands extended overhead. At the count of one, bend at the waist ant knees, reach down between the legs, and place hands on ground. On the count of two, straighten body up, extending the hands over the head; at the count of three, perform same as number one; at the count of four, repeat number two. 2. Cat Stretch: The starting position is a modified leaning rest, the buttocks being higher. This is a four count exercise. Count of 1 is upward, pushing the buttocks higher. Count of 2, back to starting position. The count of 3 upward and on the count of 4, back to the starting position. This exercise uses the back and shoulder muscles. 3. Body Twister: The starting position is standing with arms extended parallel with ground, feet spread approximately twelve inches apart. This is a four count exercise. At the count of one, swing the arms to the right, keep the shoulders and arms rigid so the twisting movement is from the waist. On the count of two, swing to the right and front. At count of three, repeat count one to the left. At count of four, face to front. This uses the muscles along the sides and back of the trunk of the body. 4. Push-Up: The starting position is standing. This is a four count exercise. At the count of one, squat with hands on ground; count two, extend legs. Count three, cause the body to move downward; keep the body straight at all times. At the count of four recover to the raised position. Continue counts three and four as desired. End exercise by recovering to standing position. This uses the arms and shoulder muscles. 5. Back Bender: The starting position is standing with the feet spread twelve inches apart, hands on back of neck. On the count of one, bend back at the waist. On the count of two, recover. Count of three, bend backward at the waist; count of four, recover. This uses the back and stomach muscles. 6. Hip and Leg Spreader: The starting position is with the hands and toes on the ground. This is a four count exercise. At the count of one bend the elbows and touch the chin to the ground at the same time extending the left leg back and up. Count two, recover. The counts of three and four are repetitions using the right leg. This uses the arm and leg muscles. 7. Shoulder Exerciser: Starting position is standing with the feet spread and hands and arms at the sides. This is a four count exercise. At the count of one, extend the arms sideward, parallel to the ground. Count of two, rotate arms. Count of three, touch shoulder with hands. Count of four, recover. This exercise is to be done with dumbbells if available. This uses the muscles of the arm and the shoulder muscles. 8. Abdominal Kick: The starting position is lying flat with the arms extended to the side; at the count of one, raise your body with the weight on the buttocks and hands, legs straight together, about 15 degrees off the ground. At the count of two raise the legs toward the chest. Count of three, extend the legs outward, keeping the feet off the ground. Count of four, recover to starting position. This uses the stomach and leg muscles. 9. Side Bender: The starting position is standing with the feet together, arms at sides. At count of one, extend arms overhead, shift weight to bended left leg. On the count of two, bend at the waist to the side, count of three, recover to count one. Count of four, recover to starting position. Count 5, 6, 7, and 8 are a repetition, only bend to right.

This uses the muscles along the sides of the body. 10. Body Kick: The starting position is flat on the back, arms behind head. At the count of one raise the body at the waist. Count two, resume flat position. At the count of three raise body and lift the right leg upward. Count of four, drop the right leg and lift the left leg. Count of five, drop left leg and lift right leg. Count of six, resume flat position. This uses the stomach and leg muscles. 11. Triceps Exerciser: The starting position is standing, arms at sides. At count of one, bend forward at the waist, arms hanging down. Count of two, raise arms parallel with the body. At the count of three, extend the arms to the left foot. Count of four, recover to count two. Count of five, extend the arms to the right foot. Count of six, recover to starting position. This exercise is to be done with dumbbells if available. This uses the triceps muscles. 12. Leg Spreader: The starting position is lying flat for count one. At the count of two, throw the legs back over the head, legs spread; extend the arms parallel with the ground. At the count of three, recover and reach for ankles. At count of four, resume lying flat. Count of five, spread legs and arms. Count of six, recover to count one. This uses the leg and stomach muscles. To be effective, physical training for the pistol shooter must be realistic and continuous. The objective is to continue to condition the body so that the general health is excellent and that the muscular and nervous systems are fully capable of withstanding the grind of match conditions and enable the shooter to continue to assert his utmost skill.

PERFECTING TECHNIQUE

Diet and Health of the Pistol Shooter

By Chapter 9: Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/chapter9.htm

A. GENERAL.
What is meant by good nutrition? Gayelord Hauser, America's famed Diet and Health specialist, answers as follows: "First, it is adequate nutrition, giving the individual cells of the body not only the quantity but also the quality of nourishment they require. Second, balanced nutrition, supplying the body cells with vital nutrients in the proper proportion. As a simplified illustration, think of your body as a motor car. It is made of protein, inside and out. Arteries, glands, colon, connective tissue; muscles, skin, bones, hair, teeth, eyes: all contain protein and are maintained and rebuilt with protein. Fats and carbohydrates are your body's oil and gasoline, they are burned together to produce energy. Vitamins and minerals are its spark plugs, essential to the utilization of food and its assimilation into the blood stream. It is a marvelously sturdy motor car, this body of yours - marvelous in its ability to maintain and rebuild itself. Given care, consideration, and respect, it will function smoothly. Neglected or abused and it will break down. Scientists are unanimous in agreeing that over-nutrition, through excess calories stored as fat, can contribute materially to physical deterioration. " B. THE IMPORTANCE OF PROPER DIET AND NUTRITION TO COMPETITIVE SHOOTERS. The army trains men to be winners. Winners in combat and winners in any competitive activity. If the pistol competitor falters, his energy exhausted, it must not be because of a diet deficient in essential nutrients. Proper nutrition will assure strength and endurance to perform your job well. Your job is to shoot championship scores in every match fired all day long, not just the first match or two of the day. The expert marksman must feel well and be energetic to shoot well. No stone can be left unturned in today's level of competition that will provide that important "edge" that may be the narrow margin of victory. Good nutrition is based on a diet that includes all the essential nutrients. These nutrients are found in a wide range of foods. C. THE ESSENTIAL NUTRIENTS. The essential nutrients are Protein, Fat, Carbohydrates, and protective ingredients or Vitamins. 1. Proteins are the body's building blocks. As the house builder uses various building materials to construct a house, proteins build and replace the body's tissue of muscle, nerves, tendons and all of the vital organs, etc. This is done in the form of amino acids. For instance, the lining of the gastrointestinal tract is renewed every three days. 2. The protein built body must have fuel to consume as it moves about and also take care of the energy demands of the vital body functions. The body maintains a reserve of energy by converting fats to glucose and storing it in the liver. Fats and fatty acids are the long range source of energy. The fat in the diet provides the full gas tank reserve for endurance. Part of the fatty acids are converted to easily digested sugar forms that

are stored in the liver. The fatty acids that are not held in reserve in the liver provide for part of the present energy needs of the body or they are deposited in and about the muscular system. The complete absence of fat in the diet means slow starvation even if the other nutrients are ample. 3. There is another source of energy but a smaller portion of the amount eaten goes into the ready reserve. Carbohydrates are the high octane energy source. Compare fats and carbohydrates to the relative combustion qualities of oil and gasoline respectively. Both will burn but gasoline is ignited easier and burns quicker. Starches and basic sugars are converted into a more readily useable glucose at a relatively faster rate than fats. In this form, they are absorbed by the blood system and supplied to the muscular tissue to furnish needed energy. For example: Alcohol, a starch food, can pass through the entire gastrointestinal tract without being altered in any way and yet be absorbed readily. There is no question about the energetic effect it has on some people. 4. Vitamins aid the breakdown of nutrients into usable forms. Protective ingredients or vitamins are not considered to be a food but they are essential to proper nutrition. Vitamins are a means of triggering the chemical reactions that convert the essential nutrients into useable forms in the digestive process. For example, the automobile carburetor converts gasoline from a liquid into a more usable form of a vapor of gasoline molecules mixed with a large proportion of air. D. TYPES OF FOOD THAT SUPPLY NEEDED NUTRITION. The pistol shooter can be assured of receiving the important elements of nutrition in his diet if certain representative foods of each type are known to him. 1. There are three foods that are the main source of protein in the diet. Proteins are mainly supplied by eggs, meat and milk. A shooter of 140 pounds should consume 100-120 grams of protein in his daily diet. Approximately one half should come from the animal source foods; eggs, meat and milk. The remaining amount can be obtained from the usual sources of bread and vegetables. 2. The same basic foods provide a source for another essential nutrient, fat. Fats or fatty acids are obtained from animal fats and vegetable oils. The average shooter consumes too much fat. While in training, a 140 pound man should consume 70 to 90 grams of fat tally. Half of this amount should come from animal sources such as eggs, meat and milk. The remainder should come from vegetable oil sources such as peanuts, corn, soya and olives. Moderation in the use of fats is especially beneficial to the digestive process. 3. There are two main sources of carbohydrates. Carbohydrates in the form of starches and sugar are contained in nearly all foods. It is important to give preference to such starch and sugar sources such as dark bread, fresh fruit, unpolished rice, milk, oatmeal, fresh vegetables and potatoes with the jackets on, because of the high content of protective ingredients or vitamins. Stay away from that enormous birthday cake with its gleaming white icing! E, VITAMINS. 1. Protective ingredients or vitamins are found in most foods. In addition the human body manufactures some vitamins but usually in amounts too small to meet its needs. Some vitamins cannot be stored in the body and must be replenished daily. Vitamin pills which contain a minimum adult requirement of most identifiable vitamins should be taken only on the advice of a doctor. Your diet may be furnishing all the vitamins you need. 2. The best

source of vitamins are foods in which they are naturally present. Of special importance to the shooter are the following vitamins and sources: a. Vitamin A; is found in carrots, spinach, sweet potatoes, milk, liver, egg yolk and green and yellow vegetables. This vitamin builds resistance to the infection and helps the eyes to function normally in light of varying intensity. This feature is an aid in night vision. Vitamin A prevents and cures pellagra, a disease of the eyes and skin. b. Vitamin B1; found in yeast, most meats, especially beef and pork, whole grain cereals, beans, peas, nuts and green vegetables. A deficiency of this vitamin causes great and persistent fatigue, aching leg muscles and bones and illness of the nervous system. Vitamin B1, prevents and cures beriberi, primarily a disease of the nervous system. Chronic alcoholics sometimes develop symptoms of the disease because alcohol diminishes the appetite and they fail to receive proper nourishment. c. Vitamin C; the body does not store this vitamin and it must be replenished daily. It is found in citrus fruits, tomatoes, raw cabbage, strawberries and cantaloupe. Vitamin C promotes a healthy circulatory system, which is important to body energy; develops good sound bones, and teeth. This vitamin prevents and cures scurvy, a disease known by general listlessness and fatigue; sore, inflamed gums and various other dental disorders. d. Vitamin D; is a fat soluble vitamin that prevents rickets. It becomes active when exposed to ultraviolet rays. Two variations of this vitamin are important in nutrition. One is vitamin D supplied by plants. The other, vitamin D3, is found in fish-liver oils, irradiated milk, and in all irradiated animals proteins. Scientists believe that vitamin D3 forms in the skin when the body is exposed to sunlight. Because of this it has been called the "sunshine vitamin. " Doctors have found that lack of vitamin D leads to serious bone changes. However, they have found that excessive amounts also cause serious bone changes. e. Vitamin E; is a fat-soluble substance that scientists believe may be necessary for reproduction. All the functions of this vitamin are not known. The best sources of Vitamin E are wheat-germ oil and lettuce. Whole-grain cereals, meat, milk, eggs, liver, and most Vegetables also contain it. f. Vitamin K includes a group of fat-soluble vitamins that are essential for making the blood clot. These vitamins are rather abundant in food. They are found particularly in the green leafy vegetables such as spinach, cabbage, kale, and cauliflower. Pork liver is also an excellent source. Intestinal bacteria manufacture vitamin K in the body. Therefore, deficiencies of this vitamin are rarely the result of poor diet. But deficiencies can result when something interferes with the normal function of the intestinal tract. g. Certain minerals, including iron, copper, sulfur, calcium, phosphorus, iodine, and sodium, are needed for growth and to maintain tissues and regulate body functions. E. YOU MUST HAVE STAMINA AND READY ENERGY TO BE A CHAMPION. The lack of forceful energy to carry on under a prolonged condition of stress can be corrected to a degree by proper, balanced nutrition. You must select the various types of food that provide this needed energy. A strong, well nourished body will allow sustained physical and mental effort under the stress of competitive shooting. Last years beginner can become the champion this year if

he learns to eat the foods that will supply him with the stamina to keep up the fight, regardless of the odds against his winning the match.

PERFECTING TECHNIQUE

Alcohol, Coffee, Tobacco and Drugs

By Chapter 10: Army Marksmanship Training Guide

Source: http://www.bullseyepistol.com:80/chapter10.htm

A. GENERAL

The habitual use of alcohol, coffee, tobacco and various drugs is harmful to the average person and in no way promotes better body function. We can be easily fooled by misleading advertisements into believing that such things are helpful. For example, an advertisement may tell us that cigarettes are an aid to digestion. Cigarette smoking after meals does cause the saliva to flow more freely and the heart to beat faster, aiding digestion. But, this may also result in overwork for the salivary glands and the heart. In like manner, many people may believe that a highball or cocktail at the beginning of a meal promotes digestion because of the greater flow of the digestive juices that alcohol causes. And what about that change of pace drink, tea? It is no different in caffeine content than coffee but possesses increased amounts of tannic acid. 1. Inform yourself. Any drug which causes the body organs to perform their work at a greater rate than normal, fatigues them sooner and causes them to age more rapidly. Stimulants and depressants overwork many vital organs, often when their best performance is needed for normal body activity. The effects of the use of such substances depend upon how much is used and whether or not the body is strong enough to repair the damage done. 2. In order to understand the discussion that follows there are certain terms whose specific meaning you should know. A stimulant is a chemical which, when taken into the body, excites the organs to greater effort. Depressants are chemicals which slow down body action but may also speed up body functions by reducing the influence of the nerve centers which slow down body action. For example, nicotine increases heart action by depressing the nerves that slow the heart beat, thus causing a faster pulse. Depressants deaden pain and lessen discomfort and thus make us feel better without removing the cause. 3. Much has been written and said both pro and con concerning the habitual use of alcohol, coffee, tobacco, and drugs, their temporary and permanent effects on the human body, both mental and physical. Material covering these subjects is available at any well stocked library. Part of the information that follows was derived from this source but much of the evidence against alcohol, coffee, tobacco and drugs that we are concerned with has been contributed by the shooters themselves. a. Although all shooters are not in agreement that complete abstinence by habitual users is the solution, all will agree that these agents will in no way help to improve shooting performance or scores. b. To learn the fundamentals of pistol shooting is no great achievement in itself. Anyone interested in becoming a pistol shooter can with persistency and training learn to shoot with some degree of proficiency. What then, is necessary to become a skilled shooter ? The top shooters in the nation today unanimously

agree that control is the most important factor in becoming a top competitor. Control can best be explained as the coordination of mental and physical effort, born in thought and culminating in a concentrated, precise action. This effort must be natural, unstrained and smooth flowing. Any habit or action that results in departure from perfect coordination will lesson the degree of control and reduce the effectiveness of the action. In shooting, lessening of control shows itself in lower scores and poor performance. c. What can you the shooter do about sustaining control? The same thing you would do when training for a match. When you find yourself having difficulty in maintaining your shot groups in the center of the target, you analyze and make corrections, be it position, grip or sight adjustment, etc.. Sometimes when control is declining, analysis may pinpoint some cause other than faulty technique in employment of the fundamentals. What did you have at breakfast? Coffee, two cups and two cigarettes. Enough to ruin anyone's control. Perhaps a few too many last night and a loss of several hours of sleep. Whatever the reasons, they should be noted in your score book just as you would enter unusual conditions at a match. In a short period of time, if you are honest with yourself you will be able to piece together enough information upon which to take remedial action. The most difficult person to convince is yourself. No one who habitually smokes or drinks coffee wants to admit that such habits have the effect of destroying control. So they remain slaves to habits which, in affect, they attempt to overpower by mental and physical exertion, often ending in frustration and exhaustion. The following paragraphs cover the effects that alcohol, coffee, tobacco and drugs have on control of pistol shooting. If you have been plagued with a built-in error, it may be that the answer to your problems lies herein. B. ALCOHOL (ETHER) 1. Effects of alcohol on the human body: The name alcohol is used for a number of organic substances some of which, like glycerin, are necessary to good health. The scientific name for the alcohol sold for drinking purposes is ethyl alcohol. Ethyl alcohol is generally considered to be a habit forming narcotic. However, in the strictest scientific sense it is an anesthetic or pain killer like ether, which is made from it. a. Alcohol taken into the body passes through the walls of the stomach and the small intestine and thence into the blood stream. It is rapidly distributed through the body and promptly affects the brain by decreasing its ability to take up oxygen. Even a small percentage of alcohol in the blood may sometimes cause remarkable effects. Inhibitions and the corresponding cautions are removed, reactions are slowed, coordination is impaired. The senses become less acute, particularly that of sight. The field of vision is reduced - ordinary objects become darker and indistinct - poorly lighted objects are lost entirely. Reactions are slowed down and concentration becomes difficult. b. A peculiar property of ethyl alcohol is its ability to take up water. It is a valuable dehydrating and preserving agent. When used as a drink, alcohol produces a burning sensation as it takes up water from the delicate mucous membranes of the throat, stomach, and intestines, thus causing the drinker to become thirsty. Once alcohol becomes a part of the blood, its dehydrating properties are much reduced. c. Although alcohol is a source of heat energy, its depressing effect upon the nerve centers that control the size of

blood vessels causes the blood vessels of the skin to enlarge. So long as alcohol remains in the blood to affect the brain, extra heat loss by radiation will take place through the skin and prevent any benefit that might be derived from its oxidation and the resulting warmth. For this reason, in severely cold weather, the man who drinks whisky to keep warm is in much greater danger of freezing than the person who does not. 2. Effects of Alcohol on Shooting: a. Contrary to popular belief, alcohol acts as a depressant rather than a stimulant. It dulls the senses, lessens the desire to win, destroys coordination and lessens the shooter's ability to concentrate. Alcohol taken at the proper time in the proper amount might possibly lessen the shooter's anxiety but by doing so other effects are released that are far more harmful to the body and detrimental to the shooter's score. No one can say what the right amount is or when it should be taken. Some shooters may shoot a good score with a hangover. But, the second day is when the after affects become acutely noticeable and the shooter's control may disintegrate on the firing line. b. Experimental research scientists using delicate tests and sensitive instruments, have been able to demonstrate the adverse effect of even small amounts of alcohol on various isolated bodily functions such as sensory perception and discrimination, reaction time, fine coordination, judgment, alertness and efficiency of dexterity. The changes observed have no apparent difference in quality, magnitude or expression from those due to fatigue, hunger, distraction and a host of other environmental factors. These facts establish that one small drink of intoxicating beverage places the shooter under an enormous handicap. The false feeling of well-being is deceptive. Alcohol, and gun powder do not mix. C. COFFEE (Caffeine) What's wrong with drinking coffee? That is easy - caffeine. Each cup contains an amount equal to about two pinches of salt. That doesn't sound like much, until you realize that it is one-third of the amount given by doctors as a heart stimulant. With three cups of coffee you are getting a dose of caffeine calculated by scientists to be medically effective for making a weakened heart work as hard and fast as a normal heart. When a heart is ready to quit, and won't pump another beat without the help of caffeine, maybe such a measure is justified. Are you sure your heart is ready for a synthetic jolt three to six times a day? 1. Effects of Coffee on the Human Body: Many coffee drinkers say they can't do without it as a pick-me-up during the day. But let us see what really happens after that coffee break. Dr. Rolf Ulrich, in his book, "Coffee and Caffeine", reports that after coffee consumption, mental tempo rises first, and speed of association increases, but there is a noticeable decrease in the quality of work being done. In test examinations it was seen that the subjects finished quicker, but that false conclusions were more frequent. Reliability and accuracy definitely took a beating as a result of a coffee pep-up. The physical result is the same. Caffeine raises muscular output temporarily, but in severe physical demands of longer duration, the muscular output decreases. As a famous scientist has said, "Coffee acts like a spur, which drives a horse to do its best, but cannot replace oats. ". That is the whole problem in humans - many of them do expect coffee to take the place of "oats". They pass up a solid breakfast because they can get by with coffee. The stimulating and exhilarating affects coffee

produces is usually followed by a loss of energy and a feeling of unsteadiness. No matter how we look at it, coffee takes more from the body than it gives. All coffees contain caffeine but in varying amounts. Fresh ground coffee is the most potent in caffeine. Instant coffees contain half as much and decaffeinated coffees contain about one third as much. It is imperative that a shooter refrain from drinking coffee before and during the shooting session and be moderate in coffee consumption when not firing. 2. Are you considering a change of pace drink, like tea? Before you do, read the following: It is not generally known that tea has larger amounts of caffeine and tannic acid (the two most detrimental ingredients) per weight, than coffee. Caffeine in tea leaves is about three percent in ratio of one to two percent in coffee. The general effects of caffeine are cerebral, cardiac and diuretic (copious urination) stimulation. As to tannic acid, tea leaves have about ten percent while coffee berries have only about five percent content. Tannic acid, when brought into contact with mucous membrane, acts as an astringent and diminishes its secretions. It coagulates albuminous substances and thus hardens animal source food matter in the stomach with which it comes in contact. It also leads to more rapid clotting of the blood when absorbed into blood circulation. There is evidence of liver damage from extensive use. In solution, it is unstable and should not come in contact with metals. Since coffee is made about twice as strong as tea in liquid form, a strong cupful of either will contain about two grains of caffeine and over three grains of tannic acid. 3. A shooter should stay away from the colas. Cola drinks, in addition to other soft drinks, contain that well known perk-up ingredient, caffeine. The bottles of some brands contain a listing of cola contents which should serve as a reminder. Know them and avoid them while shooting. D. TOBACCO (Nicotine) For a period after January 1964 when the U. S. Surgeon General (see paragraph D-3, this chapter) revealed to the American public the results of an investigation into cigarette smoking and health, many smokers quit the habit. At the end of one year a poll taken revealed that one out of every four hundred smokers had quit. This small percentage points up the fact that most smokers will continue the habit no matter what the future consequences might be. The smoking habit is easily acquired and even after a short duration becomes a difficult habit to break. For this reason one who has not yet acquired the habit should be encouraged to abstain. The objective of this section is to provide you the shooter with information that will enable you to establish control of smoking in order to improve your shooting. Who knows, once you gain control of the smoking habit and can turn it on or off at will you might be inspired to quit all together. 1. The Effects Tobacco has on the Human Body. Nicotine is a powerful alkaloid poison. Its chemical formula in $C_{10}H_{14}N_2$, which means that it contains carbon, hydrogen, and nitrogen in the proportions indicated by the numbers. a. Being a volatile substance, it is carried along with the burning smoke of the tobacco. In cigarettes about 61 percent of the nicotine is burned and destroyed, 27 percent is ordinarily exhaled, and about 12 percent is absorbed by the smoker. The absorbed nicotine specifically affects the nerves that regulate the heart rate and the size of the blood vessels, and, therefore, alters the pulse rate and the blood pressure. For about ten minutes after

smoking is begun, the pulse rate is slowed about five beats per minute because of an increased stimulation of the nerves that slow the heart beat. After this temporary slowing effect, nicotine depresses these same nerves. This results in an increased pulse rate that lasts for two or three hours. The increase, for the average person, is from five to ten extra beats per minute. One cigarette after breakfast will step up heart beat for half the shooting day. With the damage already done, abstaining for the rest of the day's shooting is to small avail. The work of the heart is affected not only by the increased pulse rate but also by the decrease in size of the arteries. Both of these factors raise blood pressure and increase the work of the heart. b. The carbon monoxide which is also present in tobacco smoke will, if inhaled, reduce the capacity of the hemoglobin of the red corpuscles to carry oxygen. This is due to the fact that hemoglobin absorbs carbon monoxide about 300 times faster than it does oxygen with which it ordinarily combines. Therefore, to the extent that the blood takes on carbon monoxide it cannot in that same proportion, carry oxygen. This results in "cutting the wind", or breathlessness, whenever there is exertion. c. In 1959, the American Cancer Society began a study to prove that there exists an association between cigarette smoking and many physical complaints. The study involved 1,079,000 men and women (smokers and nonsmokers). For comparison purposes we have listed five of the more important complaints:

COMPLAINT	CIG. SMOKERS	NON-SMOKERS
Cough	33.2%	5.6%
Loss of appetite	3.3%	0.6%
Shortness of breath	16.3%	4.7%
Easily fatigued	26.1%	14.9%
Loss of weight	7.3%	4.5%

One can readily see that the complaints were more prevalent among the cigarette smokers than non-smokers. The study further revealed that lung functioning is affected if one inhales cigarette smoke regardless of age. For example: A young man who smokes one pack of cigarettes per day has the same efficiency of lung functioning of a man 20 years senior to him who does not smoke. 2. Effects of Tobacco on Shooting: a. The combined effects of nicotine and carbon monoxide explain why the pistol shooter must avoid smoking if he is to shoot with the greatest possible skill. This conclusion does not mean that an individual or a team whose members smoke may not win, if it is competing against inferior opponents, but it does mean that any individual shooter on a team cannot perform at his best if he uses tobacco. The top competition today does not allow a margin of indulgence if you expect to win. Denying yourself a quick drag on the weed is not a sacrifice, it is a necessity for victory. b. Simply explained, cigarette smoking affects the smoker by: (1) Initially, slowing the pulse rate. (2) Increasing the pulse rate. (3) Increasing blood pressure and overworking the heart. (4) Reducing the oxygen capacity of the blood, causing shortness of breath. (5) Bringing on

fatigue more quickly. c. From the standpoint of shooting, smoking does affect performance, and more importantly, control. It is possible to become more proficient in shooting and still continue smoking, but the road is long and the progress slow. Many of our top shooters can attest to that fact. Today most of the top shooters are in the non-smoking class. It is not that they have never had the habit, but that the determination to reach the top was stronger than the addiction to tobacco. Occasionally, you may see one sport a cigar, but seldom, if ever, will you see one inhale tobacco smoke. Consequently, the crux of the problem of smoking is the inhalation of tobacco smoke. Herein lies the answer. Stop inhaling and you have solved the problem. It is the inhaled nicotine and carbon monoxide that are responsible for upsetting the normal body balance. For this reason we find many shooters making the switch from cigarettes to pipes and cigars, since the smoke of each is generally too toxic for the normal person to inhale. Performance as well as health improves accordingly. For the shooter who will consider quitting, we offer a plan, entitled "You Can Quite Smoking", paragraph D-4, this chapter. Mark Twain once said that he had no difficulty giving up smoking, and added "I have done it a thousand times". If you decide though, make it permanent. 3. The report of the Surgeon General of the US Public Health Service, released on 11 January 1964. "SMOKERS DIE EARLY" it says, "Smoking cigarettes is a health hazard that calls for corrective action - and is a major cause of lung cancer and other death-dealing diseases, especially in men", a blue-ribbon federal panel reported. In short, the panel indicated, the more you smoke, the greater your risk of an early death. Deeply inhaled cigarette smoke sends a threat of premature death spreading through the lungs, arteries and the heart itself. a. Lung tissue was obtained from more than 1000 postmortems, put on microscope slides and carefully examined by pathologists. The slides were identified only with coded numbers, and pathologists did not know their origin. Later, statisticians matches the pathological findings with the histories of the dead patients. The results added up to an elaborate description of progressive smoke damage. b. Deeply inhaled smoke, the researchers found, irritates the cells that line the tiniest chambers of the lung (alveoli). The walls of the alveoli thicken, lose their elasticity and much of their ability to do their vital job of exchanging carbon dioxide for oxygen. Subjected to sudden stress -- such as a cough or sneeze -- the alveolar walls rupture; a minute part of the lung becomes useless. c. Even while it is attacking the alveoli, smoke also damages the small arteries that carry blood to the interior lung surfaces for oxygenation. The artery walls become fibrous and thickened. Soon, internal deposits on the thickened walls make the arteries so narrow that little blood can get through. Eventually many tiny arteries are blocked completely. These two sets of events alone would be enough to explain why thousands of Americans are "lung cripples", suffering from what most U. S. Doctors call pulmonary fibrosis and chronic emphysema. But the damaging chain of events runs on. d. The destruction of smaller blood vessels in the lung and the thickening of slightly larger ones increases the blood pressure in the pulmonary arteries and puts a strain on the right side of the heart. It also prompts the left side of the heart to work harder to pump blood against

increased resistance. A healthy heart could probably stand the extra work; a heart already weakened by other difficulties might fail. e. Even while the heart is being asked to overexert, carbon monoxide from cigarette smoke combines with red blood cells decreases their capacity to carry oxygen. As a result, the hardworking heart muscle is given less fuel to do its job. At the same time, tobacco's nicotine causes a constriction of small arteries in the extremities and speeds up the heart, increasing its need for oxygen and complicating the coronary problem. f. Smoking dulls the sense of taste causing loss of appetite, thus creating a loss of weight. For this reason people who give up smoking tend to stop losing weight. They begin to taste food again and enjoy eating. 4. You Can Quit Smoking! a. A vicious, velvety trap; Light a cigarette, smoke it, taste its bitterness, put it out. Even as you do, you know that you will want another. Not that you enjoy it. You simply want it, and why ? (1) When you smoke a cigarette, for instance, nicotine, carbon monoxide, small amount of hydrocyanic acid, pyridine and various phenols and aldehydes are absorbed into your lungs and mouth. Your nervous system is momentarily stimulated. Your blood pressure goes up. Your pulse rate increases. (2) Most important of all to the satisfaction of the habitual smoker, your blood vessels undergo a constriction. This "slows you down". That is, after the momentary stimulation, smoking depresses, for a far longer period. (3) When you smoke, you are artificially slowing down most of your body's normal activities. If you are suddenly confronted with an emotional psychological emergency: adrenalin is pumped into your blood stream, your muscles tense, you breathe faster and get edgy, jittery -- "nervous". Tobacco smoke retards these natural processes by slowing the blood circulation and thus "calming you down". You find a smoke is "good for your nerves". (4) If you smoke a pack and a half of cigarettes a day, you smoke an average of one cigarette every 32 minutes of your waking hours. That many crises don't arise every day. You need cigarettes simply because your body has come to expect this depressant effect every so often. You begin consciously to want to cigarette. (5) There is little true pleasure in smoking. The harsh taste, the hot dryness is tolerated, for the sake of tobacco's mild narcotic effect. If it were possible for you to go without cigarettes for the next 24 hours, and then light one, you would find out how distasteful and noxious tobacco smoke really is. If you think this is an exaggeration, try it. (6) Think back to the time many years ago when you smoked your first cigarette. How did it taste? Gaseous, strong, biting, wasn't it? This is the experience that you may give your system 30 to 60 times a day. You are able to do it because the human mechanism is a marvelously adjustable piece of machinery which can get used to almost anything. b. What can you do about it? You have already taken one big step toward giving up smoking: you have been thinking about the detrimental effects of smoking and about giving it up. If you want to stop smoking, think about giving it up at one fell swoop. Think of it cooly and calmly, without fear or hopelessness. Think of what it would be like to never have to smoke. Giving up smoking isn't all self-denial; there are compensations. There are so many good things to enjoy more when you give yourself a chance to fully appreciate them. You will never want to go back to smoking. (1) When you give up smoking, your food

will taste much better. Your nose and throat and lungs will not be continuously permeated with smoke. You will begin to smell the world around you. When you walk into a garden you will smell as well as see the flowers. When you get up in the morning, you won't find your throat clogged with phlegm, and you won't cough or clear your throat so often. (2) You will actually feel far less nervous. That is hard to believe -- for during the first days of non-smoking you will be nervous. The depressant effect smoking has exerted on your body for years suddenly ends, and the unfamiliar effect is almost overwhelming. You will possibly be more emotional; you may laugh at trivial things and, for a while, be tense, jumpy. But gradually the nervousness diminishes. You will be calmer, more poised. For when you stop slowing down your body and cutting your energy with tobacco you will find that you have much more energy. There will be more time to get things done. (3) A word of caution here. It is generally believed that a reformed smoker gains weight. If you have trouble with your waistline, remember this: when you stop smoking, you will not gain more than a few pounds. When you stop smoking, you will have a great increase in energy. In using up that energy, you will burn away a lot of the weight that you would otherwise put on. (4) If you have read this far, you probably think you are about ready to swear off. Don't do it yet. c. To stop smoking, follow these rules: (1) Watch and wait until some time when your life is on a fairly even keel. Don't try it when you are leaving on an important trip, or preparing to give a big party, or when you are facing some personal emergency. Don't postpone it too long, either, or you will lose the momentum you are gradually building up. (2) But some sunny morning -- maybe on a weekend -- you will wake up feeling especially good. You will have had a good night's sleep; you will feel fit for anything. The idea of stopping smoking will pop into your head. Why foul up a wonderful day with the noxious fumes of burning tar and nicotine? Decide, then and there quietly and firmly, that you are through with smoking! This is the moment, intelligently selected and properly prepared for, when you can get off with a running start. (3) After you have started yourself off with as much momentum as you can, tell your friends that you have given up smoking. Don't be smug or complacent or boastful, but let people know what you are doing. Then, at some point when you are seriously tempted to smoke, the thought of all the derisive laughter you will get for giving in may well carry you over the crisis. (4) Most smokers have fixed ideas about the occasions when a smoke tastes best. The first cigarette after breakfast, or the one with a cocktail before dinner. If such associations are likely to tempt you to smoke, brace yourself in advance for such temptations; tell yourself that such an occasion is coming, and that you must be prepared to want to smoke badly. If you hold out only for a moment, that sudden strong temptation will die almost as quickly as it arose. (5) Don't permit yourself to make a single exception. Until the non-smoking habit is firmly implanted, "don't". If a habit is not fed, it dies relatively quickly, but it can subsist for a long time on the slightest food. If you occasionally let yourself have one cigarette or pipe on the ground that "just one won't hurt", you will keep alive the desire to smoke. Just one drink is too many for an alcoholic, as one cigarette is too many for the heavy smoker who is trying to reform. Win the

battle of the moment -- every time you say no to the temptation to smoke, you are making the next "no" easier. (6) Baby yourself to an extent. Most of us are inclined to launch sudden, ambitious programs of self-improvement. We try to do more than we can reasonably expect of ourselves. On the contrary, indulge yourself a little. Eat what you want and enjoy it. Make it a habit to carry mints, gum, or salted nuts. During the first few weeks keep substitutes on hand -- and pop one into your mouth whenever you feel like smoking. (7) Let your sleep work for you. On the night of the first day that you give up smoking, think for a moment when you go to bed of how today you did not smoke. Then tell yourself, "Tomorrow I am not going to smoke". Repeat it to yourself as you get drowsy. This will be the last thing in your conscious mind as you drop off to sleep. When you wake in the morning, remind yourself that you are going to get through this day, too, without smoking. Don't make a big issue of it; just briefly say: "This day I don't smoke". Even if you don't follow the other rules set down here, this exercise in "controlled sleep" could get you over the hump. You will find a sense of freedom and independence and self-assurance results from simply going half a day without tobacco. This is a sharp, continuing pleasure, and every minute helps to strengthen you against the next minute's temptation. Above and beyond this pleasant, heartening knowledge is the awareness that you are doing something of which you will be proud -- not to mention healthier and happier -- for the rest of your life. Six months or six years from now, when someone offers you a cigarette, you will refuse it, but not weakly or defensively. You will say "Thanks -- I use to smoke, but I gave it up.". 5. Expense: If a man smokes two packs of cigarettes a day for 365 days it will cost him $290.00 a year! Quit smoking and automatically you save money, remain healthy, and start winning pistol matches. This is a bargain you can't afford to overlook. E. DRUGS At one time or another some shooters have probably tried a sedative drug or tranquilizer to see what effects it would have on their shooting. Drugs affect different people in different ways, so dosage would be a problem even if they did any good toward reducing anxiety, nervousness, etc. Any time medication or drugs are used that affect the body functions, there is a chance that the side effects will do more harm than good to the shooter's performance. Some shooters no doubt prescribe certain remedies for themselves when they have a cold, a stopped-up nose or a headache. Here are some of the effects of the drugs found in these and other preparations. Most of the effects are not conductive to good shooting. Most drugs are habit forming and all are a deterrent to good health if used frequently without proper medical advice. There is not substitute for good clean living, a healthy body and just plain GUTS! 1. A depressant slows reflexes, lessens the desire to win, promotes carelessness, causes loss of concentration and coordination. 2. A stimulant causes nervousness, hypertension, increases heartbeat, excessive movement of the hands, trembling, etc. 3. Drugs in daily use. a. Barbiturates. (To induce rest and sleep) Phenobarbital has special effects against insomnia. Continued use increases tolerance and leads to dependence. Acute anxiety may result if the drug is abruptly discontinued after long use. Alcoholics substitute barbiturates for alcohol and become just as devoted to it. Even after moderate doses,

lassitude, dizziness, headache, nausea and diarrhea may occur. Other toxic effects are respiratory depressions peripheral vascular collapse, feeble heart beat, low body temperature and continued stupor with depressed reflexes. b. Analgesic (Pain relief and reduction of symptomatic discomfort) Aspirin - acetylsalicylic acid (relief of headache, fever and other symptomatic discomfort). Gastrointestinal distress due to irritation is common. Continued dosage symptoms same as quinine (Cinchonism): Dizziness, ringing in ears, impaired hearing, acidosis and depressed blood clotting mechanism. c. Stimulants or adrenergics (Relief from drowsiness, depression, curbing the appetite and relief from nasal congestion.) Benzedrine, amphetamine and ephedrine elevate blood pressure, accelerate the heart beat, causes headaches, nervousness, insomnia and spasms of the urinary bladder's sphincter. (Muscular control that permits urination) d. Antihistamines (Relief of colds and fever and relief or prevention of allergy symptoms) Exerts a potent sedative effect. There is a danger of toxic action, especially drowsiness. A form of antihistamine namely, methapyrilene, is used for sedative purposes. Used in conjunction with alcohol, this sedative action is especially dangerous as alcohol heightens the depressant effect. e. APC pills: (Relief of headache and other symptomatic discomfort) Basic ingredients usually are acetanilid or acetophenetidin and caffeine. Continued use developes a blood condition known as methemoglobinemia or simply a union of oxygen and iron in the blood instead of oxygen and hemoglobin, the natural state. The oxygen in this instance is retained in the blood and not exchanged normally. Another combination used is aspirin (acetylsalicylic acid), phenacetin and caffeine. The phenacetin adds the property of antipyretic (reduction of fever). Other effects similar to above. f. Decongestant tablets. (Relief of colds, fever and prevention of allergy symptoms) Basic ingredients quite similar, usually as follows: (1) Phenylephrine hydrodoride. A stimulant of the sympathomimetic group. It is a local vaso-constrictor, elevates blood pressure, reduces swelling of nasal membranes. It is usually mixed with a local anesthetic to retard rate of absorption. Used in treatment of vasomotor collapse which is a condition where the nervous system cannot control the dilation and contraction of the blood vessels. (2) Phenindamine tartrate. An antihistamine. The tartaric acid may be detrimental to the kidneys. (3) Acetylsalicylic acid (aspirin). See aspirin described above. (4) An antipyretic (reduce fever) and analgesic. (5) Caffeine. A stimulant, produces wakefulness and respiratory stimulation. When combined with an analgesic it is used to relieve headache. Continued use may produce nervousness and insomnia. (6) Vitamin "C", (ascorbic acid): large dosage leads to gastrointestinal upset.

PERFECTING TECHNIQUE

Optical Properties of the Eye

By Annex: Army Marksmanship Training Guide

Source: http://www.bullseyepistol.com:80/annex2.htm

A. GENERAL

The principal difficulties which confront the shooter during aiming are determined by the inherent characteristics of the organ of vision -- the eye -- and its work as an optical apparatus during the aiming process. It is well known that the aiming process makes very exacting demands upon the vision, since consistency and degree of accuracy are directly dependent upon the sharpness of vision and the conditions determining them. Therefore, it is necessary for the shooter to have a knowledge of certain of the optical properties of the eye. He must know the degree and the conditions under which the optical imperfections of the eye can affect the accuracy of aiming.

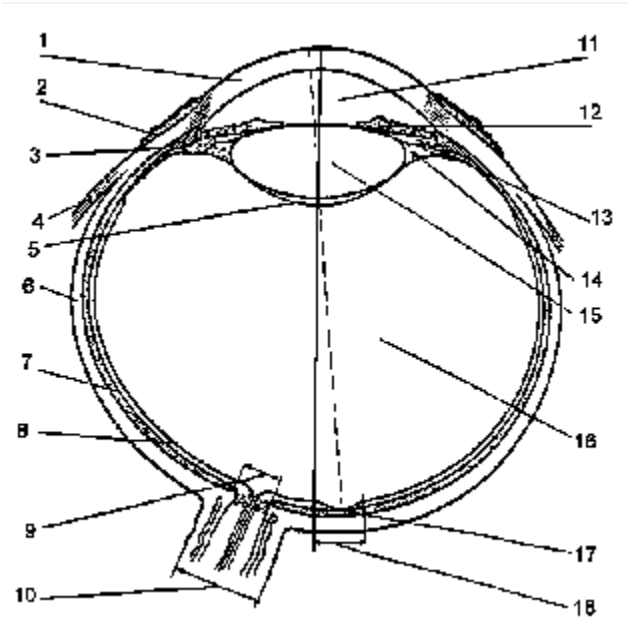


Figure A2-1. Horizontal Cross Section of the Human Eye (Right Eye)

1. Cornea (transparent, grasslike portion of coat of eyeball). 2. Conjunctiva (mucous membrane which lines eyelids and is reflected onto eyeball). 3. Ciliary muscle (smooth muscles, controlling alteration of crystalline lens). 4. Muscle of eyeball. 5. Space posterior to crystalline lens. 6. Sclera (toughest of the three membranes, forming the outer protective and supporting layer of the eye ball). 7. Choroid (vascular coat of the eye). 8. Retina (innermost

tunic of the eye, containing receptors, rods and cones). 9. Optic disk (blind spot of retina). 10. Optic nerve (transfers images from retina to visual nerve centers located in the brain). 11. Anterior chamber (filled with aqueous humor). 12. Iris (opening in center is called the pupil, contains groups of smooth muscles that dilate and contract pupil). 13. Posterior chamber (filled with aqueous humor). 14. Suspensory ligaments (zonule of Zinn). 15. Crystalline lens (transparent biconvex body enclosed in transparent sheath, suspended from ciliary body by suspensory ligaments). 16. Vitreous (transparent jelly-like substance contained within transparent membrane close to retina). 17. Fovea central is (a pit in the middle of the macula lutea). 18. Macula Lutea (point of clearest vision, contains greatest number of cones). B. OPTICAL PROPERTIES OF THE HUMAN EYE. The human eye as a visual analyzer makes it possible to distinguish rather accurately between colors, form, dimensions, degree of illumination, and the location of objects in the immediate environment (Figure A2-1). 1. The forward portion of the eye, which is turned toward the light, contains a light-refracting apparatus which transmits the image to a light-sensitive membrane -- the retina; this apparatus consists of a system of refracting media and surfaces -- the cornea, the crystalline lens, the aqueous humor, and the vitreous humor filling the optical cavity. (The light-refracting apparatus also includes the ciliar humor and the iris, which has an opening, the pupil, in the center of it.) 2. The degree of illumination, the form and location of the objects surrounding us are perceived by the internal light-sensitive membrane -- the retina -- which is linked by means of the optic nerve to the cerebral cortex. In order to obtain the correct visual perception of any object, the image of it upon the retina must be sharp. This is achieved as a result of the eye's ability to adapt its light-refracting system and thus to obtain on the retina a sharp image of objects located at varying distances from it. 3. In our eye, the role of photographic lens is played by the crystalline lens which is a transparent, biconvex body similar in form to an ordinary lens. When the eye observes objects located at varying distances, the curvature of the crystalline lens changes, as a result of which the eye's optical system adapts by reflex and very rapidly to the perceiving of objects located at varying distances from us. As a result, the image produced on the retina is a sharp one and this makes it possible to perceive correctly and sharply the form and outlines of the objects surrounding us. This ability of the eye to adapt to the viewing of various objects located at varying distances from the eye by means of changing the curvature of the crystalline lens (changing the refraction) is called accommodation. 4. Consequently, the human eye is constructed in such a way that it is not able to see sharply, simultaneously, objects located at varying distances from it. Therefore, it is not possible when aiming, to see with identical sharpness the sight alignment and the target which is located at varying distances from the shooter's eye. Understanding this, one must not strain the vision excessively in vain attempts to see everything sharply at the same time. 5. The normal eye in the state of rest is set up to perceive distant objects, that is, it is set to infinity. In order to switch the vision to perceive objects located close by, it is necessary to exert a definite muscular effort. The mechanism of

accommodation lies in the fact that the ciliary muscle contracts, and the crystalline lens takes on a convex form, thus increasing its refraction. As a result, when aiming, one must not abuse the eye by shifting the glance with excessive frequency, from one point of clear vision to another. For example, a shift of focus from the rear notch and the front sight to the target, and back again. The protracted muscular effort expended under such conditions leads to the rapid and considerable fatiguing of the oculomotor muscles. For the same reason, the shooter must not aim for long sustained periods. He must make sure that in the intervals between sequences of aiming, he does not concentrate his glance on some object, but looks momentarily into the distance "with an absent-minded stare" in order to rest his eye muscles. 6. When there is a change in the force of natural illumination, the level of the eye's light-sensitivity changes and the eye adapts to the different amount of light entering it. A role similar to that played by the diaphragm in a camera is played by the pupil, the opening or aperture in the middle of the iris. Under the action of various eye muscles, the diameter of the pupil can be made narrower or wider. It is this action which regulates the amount of light entering the eye and which improves the depth of focusing of the image upon the retina when the pupil becomes narrower. The question of the speed at which the pupil reacts to a change in illumination is also deserving of attention. When the light changes to greater brilliance, the pupil contracts much more rapidly than it expands after finding itself again in conditions of lesser brilliance. For example, the contraction of the pupil to the stable level of average light intensity takes about 5 seconds, but the process of its reverse dilation after the stimulus created by low intensity light requires about 3 minutes. From this the shooter must also make the corresponding conclusions: in order to preserve the eye's working efficiency without reducing the accuracy of aiming, before or during aiming, one must not look at brightly illuminated objects or, moreover, subject the eye to the action of sharp transitions from light to shadow. In the intervals between shots, one must not rest the eyes by closing them. Between shots, it is necessary to rest the eyes, but the best way is to look at distant dull surfaces having even tones of gray, green, or blue. C. FUNCTION OF THE HABITS OF THE NORMAL EYE. All shooters should make a conscious effort to improve the condition of their eyes in the intervals when they are not actually aiming by allowing the habits of normal sight of function. The following will give an idea of how this should be done. There are three things that every healthy eye does: Blink, center its attention (called Central Fixation) and shift. 1. Blinking, the first habit of normal sight an involuntary action. The blink is the quick, light, easy closing and opening of the eye, and it is done intermittently by every normal eye. The rate of blinking varies with people and also varies with the use an eye is put to. You blink more, for instance, when you look at something brilliant than you do when you look at something soft in tone. a. Frequently the dividing point between a normal and abnormal pair of eyes is its impulse to blink under a given situation. If the eyes are perfectly normal, they will blink; suppression of the act of blinking shows a tendency to become abnormal. b. The action of the eyelids in blinking is most essential to normal eyes and sight. The fluid that

keeps the eyes moist is produced by a small gland called the lacrimal gland under the outer portion of the upper lid. When one blinks, this fluid is washed down and over the eyeball and keeps the eye moist. This moisture has several functions: (1) There is a definite antiseptic and cleansing action of the fluid. (2) The brilliance of the eyes and their ability to reflect light are largely due to the fluid on their surface. (3) The fluid is essential to the cornea, which is the small translucent front part of the eye. Since the cornea has no blood vessels, it needs this fluid to keep it moist or it may develop corneal ulcers. (4) When particles of foreign matter get into the eye, the lacrimal fluid tends to float them off, while on an eye that is dry, the particles may stick and imbed themselves. (5) In cold weather, frequent blinking tends to keep the eye warm. An eyeball can be very uncomfortable when cold. (6) In strong wind or when the weather is very dry, blinking comforts and protects the eye. Under these conditions, one should blink frequently, almost continuously, because the fluid is lost so rapidly. (7) In the short interval of blinking, the muscles of the pupil have a chance momentarily to relax their tension. (8) Blinking also enables the eye to move slightly and thus allows the recti muscles to make the small amount of movement essential to their well-being, since motion is necessary to the health of any muscle. (9) The circulation of the lymphatic fluid around the eye is aided by blinking, and the eye is strengthened by this good circulation, just as any body is benefitted by keeping the circulation of the blood active around it. c. Blinking is not an interruption of continuous vision. Continuous vision is the illusion that a normal eye produces, authentic in effect but nevertheless an illusion. When an image falls on the retina, there is another image or an after-image produced. In other words, the image remains on the retina for a short period longer than the image is kept before the eye. It is as if your image in the mirror, stayed there a moment after you had gone away. Thus, it is not necessary for the eye to be seeing actively all the time in order to produce the illusion of seeing constantly. In fact, nothing in the body works more than half time or so much as half time. More than half of the time of every organ is consumed in the repair and replacement of its own tissue and the elimination of its waste products. d. The frequency of the visual impressions made by the eye is between thirty and forty images per second in the average person. Therefore the blink does not interfere with constant vision. It is possible for the eye to blink so frequently that the eye is closed half of the time and yet it will see as much as if it were open all the time. e. In fact, blinking increases the actual amount of time you may actively see, since failing to blink constitutes strain and this may reduce the number of images from thirty or forty to twenty or fewer images per second. There is no single instance where blinking interferes with sight. It is a natural, constructive performance and improves the eye. If for some reason the eye has not been blinking normally, the resumption of normal blinking improves its vision. f. Do not confuse a wink or a spasm of the eyelid with blinking. A spasm of the eye lid is a forceful, involuntary constriction of the lid and usually involves the muscles around the eye as well as the muscles of the eyelid and is frequently associated with some nervous disease. A blink is a light, easy, smooth, barely noticeable movement of the eyelid. g. If you have formed a habit of

looking too fixedly at things, remind yourself to blink. Blink consciously and often. Condition your reactions until you again have the unconscious blink. 2. Central fixation: The second habit of normal sight is to have the eye and the mind so coordinated that they fix on a small area at one and the same time. In other words, when you look at an object you should localize your attention, fasten it on the one small area, not scatter it. a. For example, when you look at a page of print, you cannot see the whole page clearly. If you fix your eyes on the upper right-hand corner of the page, you can see that clearly, but the remainder of the page, although it is within your field of vision, is much less clear. To see the last word on the page clearly, you will have to shift your eyes so that they are directed straight at that word. b. The same is true if you take words quite close to each other. To see the first word of a line clearly you must look directly at it, and to see the last word on that line it is necessary to shift the eye. The same is true if you want to see the second word on the line clearly. You can see it well enough to read it, but you do not see it perfectly clear when you are looking at the first word. A definite strain is involved if you try to see it that way. This is true down to the very smallest degree of space. c. There is a basic, structural reason for this. The Macula Lutea, the only part of the eye that sees perfectly clear, is in the center of the retina and is no larger than the head of an ordinary steel pin. This dot of perfect sight is placed in the eye like a point at the bottom center of a bowl whose sides slope gently like an arena. This one tiny point has clear, strong vision. When your vision departs from that point, there is a tremendous reduction in clarity of sight. There is, instead, blurred, collateral vision. And this is increasingly blurred as you continue out from the center until near the outside edge there is only perception of general shape, color and motion. d. Since only this point, the Macula Lutea, has perfectly clear vision, only a very small area can be seen clearly at one time. But the movement of shifting is so swift that the illusion of seeing a large area is given. The images falling on the Macula Lutea are carried swiftly into the visual brain centers, one succeeding another with such rapidity that there are thirty or forty and sometimes more images a second, thus creating the illusion of a whole picture in the brain. e. This ability of the brain to carry successive images and so produce the illusion of clearly seeing the whole object or a considerable area is an impressive and beautiful fact, but it is also a cause of trouble to the shooter. One comes to believe that the eye itself can see a large area clearly, and so misuse slips in because any attempt to do this is to use the eye without focusing! f. "Large Area" means trying to see for example, two words or more at a time. The healthy, normal eye habitually sees only a very small area at given moment. The mind and the eye normally coordinate perfectly on each word or point of observation with no effort or impulse to see more, just as it does when one is writing. If the practice of seeing a large area at one time persists over a sufficient length of time, the ability to focus perfectly is lost and the blurred vision naturally to the collateral area is the only vision possible. It is then necessary to retrain the eye and mind to look at only a small area in order to again have central fixation without which no vision can be clear and normal. g. One can read indefinitely without undue tiring or harming the eyes in any way if the eyes are relaxed and the vision is

localized. But, if the seeing power of the collateral field of vision is used, the eye is straining and there is a resulting fatigue and loss of efficiency. The fact that the eye sees clearly only a very small area at any one time cannot be over stressed. In the awareness of this fact rests the coordinating of the mind with the structural limitations of the eye, without which there cannot be normal vision. If you grasp this fact of focused vision and mentally close your sight to a large area, you will attain this valuable habit of central fixation and find increased efficiency in the use of your eyes in shooting. 3. Shifting. The third beneficial habit of normal eyes is to shift. This seems to quarrel with the second habit which is to localize your gaze but in reality it does not. You must point your gaze, but you must, too, constantly shift your point of vision. If you do not shift it, you will stare, and staring is one of the worst and commonest forms of eye strain. a. Shifting is a normal function and is normally done unconsciously. The frequency with which your eyes shift varies with the type of demand upon the eyes; for instance, looking at a book or watching a tennis match. The book is stationary and the eyes do not tend to move, while the tennis balls and players are constantly in motion so the eyes must move continually in order to follow them. b. But, in any event, shifting should be as frequent as possible. The time required for an image to register on the retina, about 1/50 of a second, allows for a great frequency of shifting with no loss or interruption of vision. c. People who are inclined to look at one area too long, and every abnormal eye does this, would benefit both in vision and in eye comfort if frequent shifting from the point being looked at is consciously practiced. If your vision is abnormal; without wearing your glasses look at a word, then look at a word three word spaces beyond it; then back up to the original word. Do this until both words become clear. Be relaxed while you practice. d. Or, if your vision is good; look at the moon and while blinking frequently, shift your vision from one point to another on the moon. Do this a number of times and the moon will stand out much more clearly and appear in its true form as a solid spherical body with depth and shape instead of a flat disc. e. Shifting is both voluntary and involuntary in character. The involuntary shift is continuous, automatic and very slight. This movement is not visible and is believed to correspond in frequency with the rate of image production in the retina. f. There is in every muscle a faint tremor, since muscle tone is not a constant factor but is a rapid succession of contractions producing a relatively steady muscle pull. And, since the eyes are held in position by muscles and all focusing is produced by these muscles, the eyes are naturally subject to all conditions that muscles produce incidental to their normal functioning. g. When the eye is relaxed, the voluntary shifting is frequent and the movement is short in scope. The tense eye can make a large movement, but it requires relaxation and normality for an eye to keep shifting in relaxed condition with a very small movement. This is true of all muscles -- the finer the movement, the better trained and the more relaxed must be the muscle. When an eye is strained and the vision is abnormal, practice in shifting frequently will give relief from the strain and produce improvement in the vision. h. An exercise that accomplishes this is to focus definitely on each word and consciously shift to the next one. A few minute's practice each day will make this an

unconscious habit. i. Normal shifting is absolutely essential to normal sight. Loss of vision is frequently in direct proportion to the loss of motion. 4. In addition to acquiring the three habits described above, a shooter may find it desirable to strengthen his tolerance for light. This may be done as described in the following paragraphs: a. Sunlight is very beneficial to the eyes. It both relaxes and stimulates. But it is necessary to know how to use the sunshine to get the most out of it. Sunlight directly on the eyes may cause great damage. The eye can be strengthened in its light tolerance by judicious exposure to light. One of the most effective and simple ways of strengthening the eyes is to expose them to the sun's rays in the following manner: (1) Close the eyes lightly as the face is turned directly toward the sun. Keeping the eyes closed, slowly turn the head from side to side. Keep this up for four or five minutes. Then, when the eyes are relaxed from the heat of the sun and the motion of the head, they may be opened, but only momentarily, and when the head is turned to the side. The eyes must not look directly at the sun but may look near it. Make no effort to see, and open the eyes only in flashes. As this exercise is continued, and the eyes become accustomed to the increased light, the glance may be directed closer and closer to the sun. (2) By doing this with regularity on successive days and for a gradually increasing length of time, any eye will be strengthened and its vision improved. b. The eye is admirably equipped to protect itself and function under widely varying light conditions. When the natural protective mechanism is used, as just outlined, strong light will be handled easily by the eye. D. OPTICAL IMPERFECTIONS OF THE HUMAN EYE. As a result of various optical imperfections of the eye, the images of objects on the retina have edges which are not completely sharp, or are to a degree totally fuzzy. As a consequence, there exists a certain limit of varying sensitivity of our eye which determines the sharpness of vision. It must be said that sharpness of vision, in and of itself, is inconstant. It has a certain variable value which depends upon the degree and the conditions under which, the optical imperfections of the eye have a noticeable effect. Therefore, the shooter must know, at least in overall features, the conditions which influence the sharpness of vision and thus the degree of accuracy of aiming. As an optical instrument, the eye has inherent in it, the phenomena of aberration and diffraction of light. 1. Spherical aberration is a function of the eye in which rays of light falling upon the crystalline lens are refracted differently and are not focused at a single point. The extreme outer rays are refracted more strongly than the central ones (Figure A2-2). As a result of spherical aberration, a beam of parallel rays entering the eye is focused on the retina not in the form of a sharp image, but in the form of a circle of light diffusion. The size of the circle of light diffusion resulting from spherical aberration is in direct proportion to the size of the pupillary opening. It is obvious that the sharpness of the image is increased if one eliminates extreme rays. Consequently, as the pupillary opening contracts, the sharpness of the image of the object upon the retina increases. The degree to which spherical aberration can hinder the seeing of objects sharply, and to which the sharpness of the image depends upon the size of the pupillary opening, can be convincingly shown to the shooter by means of a simple example. Small orienting marks

and objects which can be distinguished only with difficulty during overcast weather become incomparably more discernible if one looks at them through a small peep hole which, in this instance, fulfills the role of an artificial pupil.

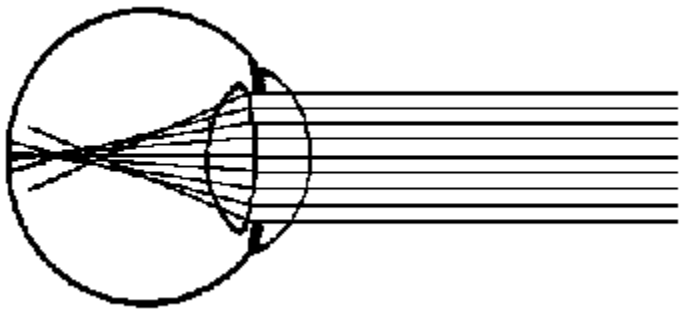


Figure A2-2. Phenomenon of Spherical Aberration.

2. The phenomenon of light diffraction lies in the fact that light rays passing through small openings, particularly through the crystalline lens; seem to bend (figure A2-3) and produce on the retina an image not in the form of a single sharp point, but in the form of a circle surrounded by a number of concentric light rings of decreasing sharpness. This occurs as a result of the wave nature of light. a. As the pupillary opening decreases, the diameter of the diffraction ring of light diffusion increases. The diffraction rings around the images have a noticeably telling effect only when there are extremely small dimensions of the pupil, and this, as we can see, is a certain opposite of the phenomenon of spherical aberration. The phenomenon of diffraction makes itself felt when there is solar illumination from the front and the sun in shining into the eyes; when there are bright patches of sunlight on the horizontal surfaces of the front and rear sights, etc.

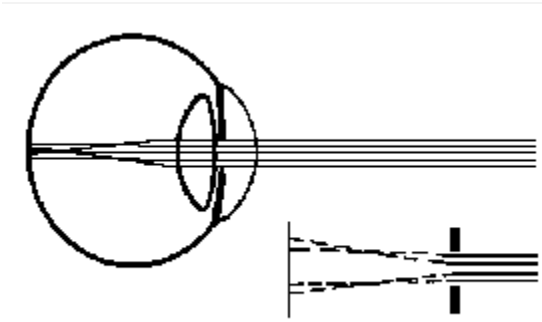
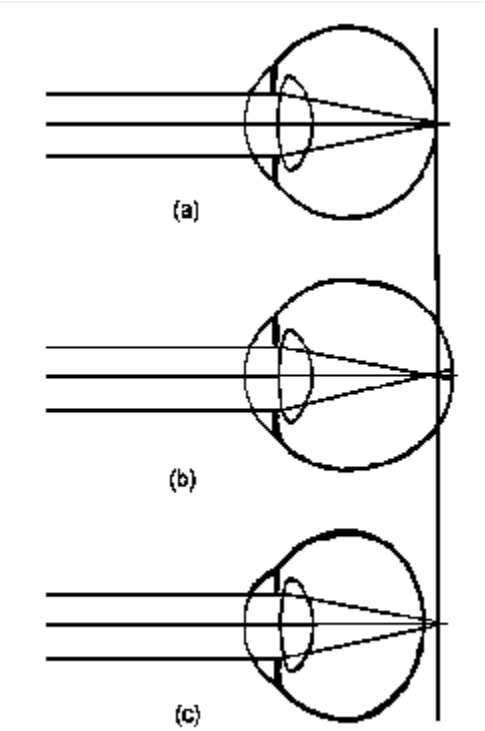


Figure A2-3. Phenomenon of Diffraction of Light on the Pupil.

b. The operation of the eye as an optical apparatus is also harmed to a certain degree by the light diffusion occurring with in it. It is especially discernible when one views brightly illuminated objects located against a dark background. The effect of light diffusion in the form of a more or less noticeable radiation, covering the field of vision, is caused by media which do not possess absolute transparency -- the crystalline lens and the vitreous humor. The light diffusion in the optical media is responsible for the halos of light. They are especially noticeable where the targets are strongly illuminated by sunlight. In such an instance, the white background of the target casts a sharp reflection and causes a considerable light diffusion in the optical media. This causes a blinding effect. Both the bull's-eye, perceived by the eye in the form of a gray spot with indistinct edges, and the front and rear sights are perceived with unclear outlines. c. It is obvious from what has been said that the amount of light diffusion from spherical aberration is in direct proportion to the size of the opening of the pupil, and the amount of light diffusion from diffraction is in inverse proportion to the size of the opening of the pupil, and thus it is not possible to eliminate these types of diffusion completely. As a result of this inverse dependence of the effects of aberration and diffraction upon the size of the pupil, the best conditions of sharp vision correspond to a certain average size of the pupillary opening -- a diameter of approximately 3 mm. Taking this into consideration, depending upon the conditions of illumination which influence the size of the pupillary opening, the shooter must strive, insofar as he can, to create the most favorable conditions for the operation of the eye. He must protect his eyes from the action of light by using visors, filter type shooting glasses, or by possibly using an artificial pupil. An eye disc device is attached to the shooting glasses with adjustment for varying diameters. He must also make sure that the sights do not shine and thus produce a blinding effect upon the eye: They must be carefully and evenly blackened. d. Brilliant sources of light harm the eye chiefly by means of the violet sector of the visible and invisible portions of the spectrum. The complete elimination of the violet sector of the spectrum is achieved by yellow, yellow-green, and yellow-orange light filters. Such light filters not only do not reduce the acuity of visibility, but, on the contrary, increase it. Type of darker glass protecting the eyes from brilliant sources of light somewhat reduce the acuity of vision. However, by having an assortment of shooting glasses of varying shades, it is possible to select and use them in such a way that the shooter's eye perceives the correct sight alignment under the brightest illumination in almost the same way that he does during overcast weather. 3. Near sightedness, farsightedness and astigmatism: The optical imperfections of the eye also include nearsightedness, farsightedness, and astigmatism, the existence of which also hinders the correct focusing of the optical system of the eye and the obtaining of sharp images of objects upon the retina. a. If the eye is constructed in such a way that rays of light entering it in a parallel beam are focused exactly on the retina without any effort at accommodation, we say that the eye is a normal one. (Figure A2-4(a)). b. The eye is nearsighted if the rays entering it in a parallel beam are focused in front of the retina (Figure A2-4(b)). Nearsightedness is caused either by the fact

that the eyeball is excessively long from front to back or by the fact that the eye has great refracting force, or by a combination of both factors. Nearsightedness can be corrected comparatively easily by means of eye glasses. Many pistol shooters suffer from nearsightedness, but this defect of vision, after being corrected by the proper choice of eyeglasses; does not prevent them from achieving record-making competitive results.



Figures A2-4. Schemes Showing the Refraction Rays in the Eyes: {a) Normal, (b) Nearsighted (c) Farsighted.

c. The eye is farsighted if the rays entering it are focused in back of the retina (Figure A2-4(c)). This can occur either as a result of the eye having a weak refracting force, or that the eye is too short from front to back or by a combination of both factors. In such an instance, in order for the rays to focus upon the retina, they must enter the eye in a converging beam, and therefore a farsighted eye sees near objects worst of all. This type of eye is harder to correct but eyeglasses help it overcome the difficulty. Shooter's suffering from farsightedness will see the sights very poorly. The characteristic complaint of farsighted persons is that the rear notch seems to fuse with the front sight. d. An indistinct, hazy image of objects on the retina can also result from astigmatism of the eye. (1) An eye in which the refracting surfaces of the cornea and crystalline lens do not have a perfectly spherical form is called astigmatic. When an eye is astigmatic, parallel rays entering the eye cannot produce an exact focused image on the retina. The refraction of the light rays at various points in the eyeball occur at different

angles. As a result, the eye possesses not a single principal focus, but several foci, which are located at various distances from the retina; that is why the image on it is indistinct and uneven (Figure A2-5). It must be said that certain strange phenomena frequently observed under practical conditions of marksmanship (for example, when two shooters are firing the same pistol, with an identical sight setting, and the centers of impact differ sharply from one another) are, in all probability, connected with astigmatism of the eye.

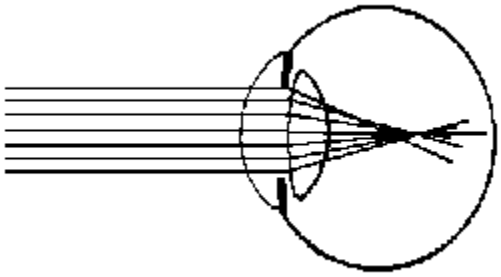


Figure A2-5. Scheme showing the Refraction of Rays in an Astigmatic Eye.

(2) One can easily be convinced of the existence or absence of astigmatism by making use of a circle chart (Figure A2-6). For this purpose it is necessary to look with one eye from the distance of best vision (approximately 10 feet) at a disk on which concentric circles are drawn. If the person has astigmatism, only certain areas of the disk will be clearly visible and the remaining areas will seem hazy.

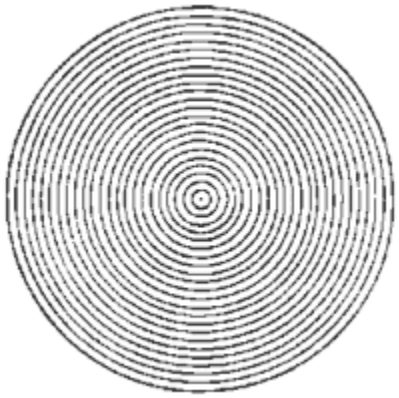


Figure A2-6. Chart for Discovering Astigmatism.

4. Correction of Defects: If even insignificant defects in vision are discovered, it is necessary to wear corrective eyeglasses when firing, since the excessive accommodation of the eye

resulting from aiming will greatly fatigue vision and this can lead to a still greater decrease in its accuracy. It must also be kept in mind that when eyeglasses for firing are chosen by the ordinary method, that is, in an optometrist's office, the choice is not completely satisfactory for the shooter. It is very desirable when selecting the lenses to check them immediately on the pistol range, to make sure that one can see well the sight alignment located at arm's length distance away from the eye. This selection is linked not so much with the determination of the lens dioptrics, as with the determination of the quality of their grinding, since all defects will make themselves known quickly during such a check. When wearing shooting glasses (including those with filter lenses) it is necessary to make sure that the line of sight runs perpendicular to the surface of the lens and through its center, since the central portion of the lens is usually ground considerably better and therefore has less distortion of the vision. To hold the lenses perpendicular to the line of sight requires no change in the ordinary level placement of the head when assuming the firing position. E. MONOCULAR AND BINOCULAR VISION. It is necessary to dwell on one more peculiarity of our eyes which has tremendous importance in aiming -- the existence of monocular and binocular vision. 1. Vision with one eye is called monocular and vision with two is called binocular. The fact that a person has two eyes does not always mean that he also has binocular vision. There are instances when the eye which has the poorest vision is not included in the act of vision and the person actually uses just one eye, the better one. The dominance of one eye over the other also occurs, even when both eyes possess identical sharpness of vision. The eye that the person prefers to use is called the dominant, or directing eye. There exists a very simple method by which one can determine which of the eyes is the stronger. 2. In order to determine which is the dominant eye the shooter must hold his hand out slightly making a ring out of the fingers and thumb, and look through it in such a way that some small object can be seen by both eyes (Figure A2-7). Then, by alternately closing one eye then the other, it is necessary to see whether the object stays within the ring or leaves it. The dominant eye is the one with which the shooter sees the object as un-shifted, remaining in the ring. In most people the dominant eye is the right one. 3. The protracted work of one eye (for example, by laboratory assistants, microscopists) contributes to the fact that the eye used becomes the dominant one. This naturally pertains also to shooters, who, when aiming, use one eye. The right eye is, in the overwhelming majority of cases, their dominant one.

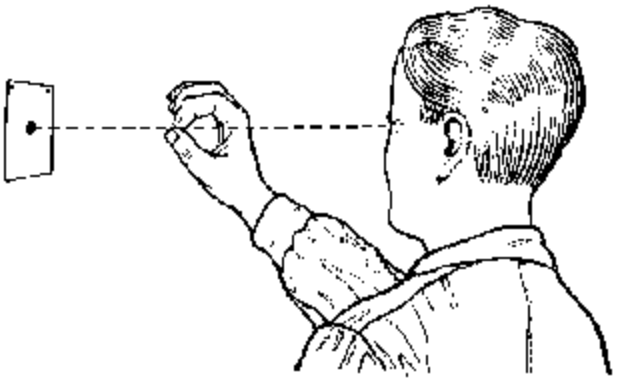


Figure A2-7. Determining which eye is dominant.

4. In the past the shooter has usually been instructed to squint his left eye and aim the pistol at the target with his right eye. During subsequent instruction it is no longer necessary to continue this device of closing the eye, since it has its major shortcomings, which are attested to by numerous instance of medical research. a. The first shortcoming is the strain which is involved in squinting the eye and is hard for most people to endure. Another undesirable aspect is the fact that the squinting of one eye is almost always accompanied by the tension and pressure of the closed lid on the eyeball. This pressure of the eyelid upon the eyeball affects the refraction of the crystalline lens and thus has an adverse influence upon the sharpness of vision of the open eye. The third factor having an adverse effect upon the accuracy of fire is the involuntary sympathetic dilation of the pupil of the open eye in response to the closing or squinting of the other eye. Thus, when excluding the second eye from work, it is best not to squint but suppress the visual impressions of the open left eye with an opaque disc which will allow equal light intensity to affect the eye. b. With binocular aiming, that is, with both eyes open, the line of sight still is achieved with one eye. Consequently, this method does not involve anything new in principle and therefore the shooter is not required either to learn something new or to relearn something, but must simply stop closing one eye when aiming. 5. Binocular aiming has a number of major advantages: The shooter does not have to expend the additional effort involved In squinting the eye, and this is very important when he is engaged in prolonged firings. The binocular acuity of vision is usually better than the monocular. The visual perception of one eye intensifies the total stimulus sent to the central nervous system from the visual perception of the other eye. In such aiming, the stimuli sent by two eyes are more natural than those sent by a single one. 6. All the movements of the eyeball, as well as the holding of its fixed position at moments when the glance is fixed on some object, are effected by the work of three pairs of eye muscles. During the time when the eyes are at work, including the times when the eye is aiming, these muscles are in a state of indiscernible, slight vibration or quivering. For

example, when aiming a pistol and the shooter turns his head down and to the right, the eyeball turns respectively upward and inward, it is held in the least desirable position; one that requires the combined, intensified work of all three groups of muscles (Figure A2-8). When the eye muscles become fatigued, the involuntary quivering of the eyeball increases considerably and this lessens the accuracy of aiming. Therefore, the shooter must devote major attention to the position of his head when firing. He must select that firing stance in which the head position is the most natural one, with the least amount of tilt, so that the shooter does not look at the target from under his eyebrows or sideways. This results in the rapid fatigue of the eye muscles and, hence, the lessening in the accuracy of aiming.

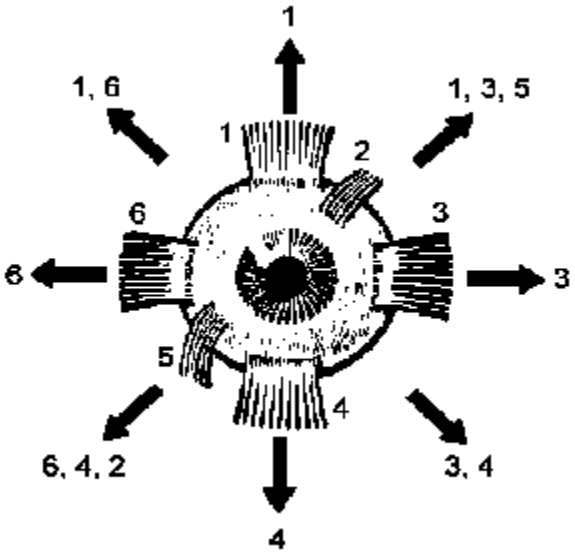


Figure A2-8. Muscles of the Right Eye. Arrows Show the Direction in which the Eyeball turns when the Muscles designated

by the numbers are contracted. F. THE SHARPNESS OF VISION. The shooter is interested chiefly in the degree of the eye's differentiating sensitivity and its resultant sharpness of vision, as well as the degree of accuracy of aiming which the eye can guarantee. 1. Basically, sharpness of vision depends upon the physical properties of the anatomy of the eye. The physical condition of the eye determines primarily the greater or lesser accuracy of the image of the object upon the retina. Those anatomical and physiological conditions determine how well we can see the object. The sharpness of the image upon the retina is the impression received by the brain. 2. Sharpness of vision is usually determined by the minimum space that we are able to see between two objects. In order for this space to be visible, it is necessary for at least one retinal element lying between the images of those two points to be stimulated. Thus, the normal sharpness of vision is generally considered to be that at which the eye can

distinguish between two visible points at an angle of one minute. 3. However, the anatomical dimensions of the retinal elements (the rods and cones) do not completely determine the limit of visual acuity. Therefore, the visual acuity of the normal eye actually can be considerably greater than the medical norm. Research works have shown that the average visual acuity of the normal human eye, at one hundred yards under normal illumination, can distinguish distance between objects separated from one another within the limits of 40 angular minutes. This means that the normal eye can distinguish sufficiently clear, for example, a space of .1 inch between the side of the front highs and vertical inside surface of the rear sight notch on the pistol sight at a distance of one yard (the approximate distance to the muzzle and front sight). But the eye of an experienced shooter can distinguish a considerably smaller space between two objects. A number of experiments carried out by specialists attest to the greater accuracy of a trained shooter's sharpness of vision. For example, the vertical space between front and rear sight against a white background can be discerned down to the minute width of .01 inch. 4. Many experiments confirm that the sharpness of vision can be considerably increased by means of exercises. This increase in the sharpness of vision is achieved by searching for new signs, new criteria for recognizing the form of objects. Such a sign for shooters is undoubtedly their highly developed sense of symmetry and visual memory. 5. Visual Memory: Therefore, in order to achieve symmetry, a visual memory of correct sight alignment, with its symmetrical interrelationship of the front and rear sights, mainly, the equal amount of space between each side of the front and rear sights and the levelness of their horizontal surface must be ingrained into the mind and never violated. No shot should be fired with less than perfection. All these factors, together with an existing sharpness of vision, will provide for accurate and consistent aiming and the accurate calling of the shot. 6. Calling of the shot: Accurate calling of a shot is dependent upon exact recall of the mental image of the sight alignment at the instant of firing. As demonstrated, the presence of 1/100th of an inch error in sight alignment will result in approximately a 3 inch error from target center at 50 yards. Acute awareness of the slightest degree of error in alignment is an absolute requirement of accurate shot calls from a clear visual memory. The overwhelming majority of shooters try to see the front sight sharply, and thus accept the blurring of the bull's eye. With the passage of time and with regular practice, the eye develops the ability to obtain the identical space relationship between the front sight and the rear sight and with increasing frequency can position the sights uniformly in the center of the aiming area which is blurred and out-of-focus. Eventually, the shooter develops his visual powers to such an extent that the eye will consistently perform the act of aiming automatically and he can call his shots without error. 7. Changing degrees of accuracy: When the eye performs intensified work, not only the motor apparatus of the eye, but also its light-sensory apparatus has reduced efficiency. When the glance is fixed steadily upon some object, the eye possesses its greatest sharpness of vision for the first several seconds, after which the sharpness of the image on the retina, that is, the clear seeing of it, gradually decreases. Consequently, the shooter must not be captivated by

excessively prolonged aiming, since, after the elapse of 12-16 seconds, his eye ceases to notice certain inaccuracies in aiming. By relying on the false assumption that the rear sight and front sight are in correct relationship to one another, the shooter makes grievous errors without noticing and therefore, does not know why they were committed. a. If one calculates the time between the moment when the visual attention is concentrated on obtaining the precise alignment of the front and rear sights and the moment when the shooter makes the decision to positively press the trigger, the aiming process must not exceed 6-8 seconds. b. When aiming, some shooters run their eye from the front sight to the rear sight notch and then to the third object, the target, doing this quickly several times, until these three points are all located on the same line. It must be said that his method of aiming causes rapid fatiguing of the muscular apparatus of the eye and fails to provide a constant objective for the formation of a visual memory. With this method of aiming it is very difficult for the shooter, for example, to conduct rapid fire, which is very limited in time. He simply does not have time to run his eye back and forth between objects located at varying distances. The shots may be fired when he is focused on any of the three objects. Consistently accurate shot calls are impossible. Therefore, when aiming, the shooter must strive to see only the front sight sharply and distinctly.

PERFECTING TECHNIQUE

Glossary of Terms

By Army Marksmanship Training Guide

Source: https://www.bullseyepistol.com/glossary.htm

ASSEMBLY AREA is a designated zone in the rear of the firing line, approximately 25 yards, where the next relay of competitors can complete their preparations for the match and receive instruction and advice from their coach or team captain.
AIMING AREA is the center area of the target as against an aiming point on the target which is extremely difficult to attain due to the universal presence of movement in the shooter's hold. BREATH CONTROL is the ability to temporarily suspend breathing while firing a shot without being conscious of the need to take another breath. This awareness disturbs the shooter's concentration on maintaining perfect sight alignment. BALL AND DUMMY is a training exercise wherein the shooter makes an effort to employ all the control factors for firing a good shot. The ammunition inserted into the pistol by the coach is a random selection of live or dummy ammunition. This training aids the shooter in overcoming reaction to the recoil and loud noise of firing, preventing uncontrolled reflexes from disturbing the hold, and perfecting sight alignment. THE CENTER FIRE PISTOL FOR INTERNATIONAL COMPETITION is similar to the pistol used in NRA domestic competition. The trigger pull is three pounds as compared to the NRA requirement of 2 1/2 pounds. Caliber .38 is the normal size used. DRY FIRING is a training exercise wherein the shooter employs all the factors of controlling the firing of an accurate shot without using live ammunition. FOLLOW THROUGH is the effort on the part of the shooter to continue the employment of the fundamentals throughout the deliver of the shot exactly as they were planned and set-up. THE FREE PISTOL is a special design of competitive weapon for achieving the ultimate in accuracy from a handgun. It is 22 caliber long rifle. The majority of free pistols have a longer barrel than the normal pistol and an extended sight radius. The trigger weight is extremely light and is adjustable from 1/2 oz to 1 1/2 ozs. There is a form fitting grip for the hand. THE FUNDAMENTALS OF PISTOL MARKSMANSHIP are those essential factors that the shooter must know and learn to control in order to fire an accurate shot. THE GRIP of the shooting hand on the pistol provides the shooter with a firm hold on the weapon that prevents shifting during recoil and a natural alignment of the sights without moving the head or wrist from their normal attitudes. HARD BALL ammunition is the term used to describe the .45 caliber service type of pistol ammunition. INTERNATIONAL SHOOTING UNION is the organization that sponsors and controls all international firing competitions between nations. The headquarters is in Weisbaden, West Germany. MENTAL DISCIPLINE in pistol marksmanship is the shooter's ability to maintain his concentration on sight alignment while the other fundamentals of minimum arc of movement and trigger control are being employed

at their optimum. MINIMUM ARC OF MOVEMENT is the smallest degree of movement that the shooter can attain in the body shooting arm and weapon during the time of firing a shot. THE NATIONAL MATCH COURSE OF FIRE is a three stage pistol match. The first stage is ten shots in ten minutes, fired at 50 yards. The second stage is two strings of five shots each timed fire, 20 seconds each string at 25 yards. The third stage is two strings of five shots each rapid fire, ten seconds each string at 25 yards for a total of 30 shots. POSITIVE PRESSURE on the trigger is an uninterrupted, constantly increasing pressure applied by the trigger finger in an effort to fire the shot. This pressure is initiated by the presence of a satisfactory minimum arc of movement in conjunction with perfect sight alignment, not perfect sight picture. A perfect sight picture is the absence of movement combined with perfect sight alignment. POSITION of a pistol shooter is the relationship of the shooter's body to the target. Proper or natural positioning of the shooter's body points the shooting arm directly at the target center without deviation. RAPID FIRE is a type of pistol match or a stage of the National Match Course of fire where a string of five shots are fired consecutively during a period of ten seconds. A Rapid Fire Match is a series of four such strings and the National Match Course includes two strings. THE RAPID FIRE PISTOL is a special design of competitive 22 caliber short weapon for firing in international rapid fire competition. It's outstanding characteristics are minimum recoil, dependability and fast semi-automatic operation. SQUADDING TICKET is a card issued to each shooter in a pistol match that indicates the caliber of weapon, the relay of firing, and the target number for each of a succession of matches scheduled to be fired during the tournament. SLOW FIRE is a type of pistol match or a stage of the National Match Course of fire where a period of one minute is allowed for each of a maximum of 10 or 20 shots. SIGHT ALIGNMENT is the relationship of the front sight to the notch of the rear sight as seen by the shooter's eye. The top of the front sight must be level with the top of the rear sight and the light space must be equal on each side for the front sight. STANCE is the posture assumed by a pistol shooter while firing a shot. THE INTERNATIONAL STANDARD PISTOL is similar to the caliber .22 weapons used in domestic NRA competition. The barrel length is limited to 15 centimeters and the trigger pull is limited to 1000 grams. TIMED FIRE is a type of pistol match or a stage of the national match course of fire where a string of five shots are fired consecutively during a period of 20 seconds. A timed fire match is a series of four such strings and the National Match Course includes two strings. TRIGGER CONTROL is the ability of the pistol shooter to apply pressure on the trigger to fire the weapon without disturbing sight alignment. WAD - CUTTER pistol ammunition has a type of bullet that is shaped to cut a clean hole in the target face. This sharp edge enables the scorer to readily and accurately evaluate the hit. ZEROING is the technique of setting your sights so that a shot called good, (undisturbed perfect sight alignment and minimum arc of movement) will hit the center of the target on an ideal day with no wind. * U.S. GOVERNMENT PRINTING OFFICE: 1979 - 640 - 000 - 32 ... REGION NO. 4

PERFECTING TECHNIQUE

A Clinic on the Fundamentals

By Brian Zins

Source: http://www.bullseyepistol.com/zins.htm

Brian Zins is clearly the greatest pistol shooter of all-time. His unprecedented ELEVEN national championship wins is a milestone that will never be broken. His thoughts are especially valuable because he doesn't always follow tradition on technique.

Let me start by stating that some of what you're about to read may sound contradictory to a lot of articles that have been written and lessons that have been taught in the past. Some of this may say the same thing in the end, just a different route to get there. By no way am I suggesting they are incorrect. I am merely giving my approach to learning bulls-eye and what I do as a shooter.

Trigger Control

Started out this segment as "Fundamentals" but, honestly if you know me there is only one that really, really counts when the shot breaks. Don't get me wrong stance and grip and aiming are important. Keep in mind that you can have a perfect grip and hold perfect sight alignment all day long. It only counts when the gun goes bang and only one action causes the gun to go bang. Pulling the trigger. You can hold the gun upside down, squeeze the trigger with your pinky, and align the sights to the target in a mirror and shoot tens as long as you don't jerk the trigger.

Please DO NOT try that at home.

Again this is what I was taught and this is what I do. It is a way, not THE way, that's all. But I have had pretty good results with it. Also this is for a trigger with at least a little roll. My preference that's all.

First let's look at trigger finger placement.

Where should the trigger make contact on the finger? The trigger should be centered in the first crease of the trigger finger. Remember this is an article on Bulls-eye shooting. If this were an article on free pistol or air pistol it would be different. So International shooters please don't unleash the hounds on me.

Why you ask? We have always been taught to place the pad on the trigger. If you a trigger that weighs in between 2 and 4 lbs. That is enough weight to move the fleshy part of the pad of your finger. Try it! Get something that weighs at least 2 lbs that has a tip or point on it the size

of a pen or a trigger would be ideal. Put your hand on a table or desk, palm up and slowly lower the weight onto the pad of the finger. It moves. A lot. Now lower the weight onto the first crease of the finger, right on the joint. Sure it moves, a whole lot less than the fatty, fleshy part of the pad of the finger. Now having conducted this little experiment think about the pulling the trigger with the pad of the finger. The first part of movement you feel is flash and fat moving out of the way. This is not part of the movement involved in trigger control.

Ever been shooting well and in slow fire you start to get “chicken finger?” The trigger starts moving and then it stops and feels like it weighs 30 lbs, or did it move at all? Maybe what you felt was the flesh moving out of the way cause you are shooting well and don’t want to screw up the match, so you are really in tune with what you are feeling in the trigger.

Now think if the trigger were placed at the crease or first joint of the trigger finger. When the trigger moves, what you are feeling is really the trigger moving.

Let’s go even deeper.

Take a pen or a pencil and start tapping the fleshy part of the web between your trigger finger and the thumb. Keeping the same intensity move the tapping up the large knuckle on your trigger finger. Feel the difference? The flesh acts as a shock absorber to the tapping where the tendon in the joint is more of a conductor. I know what you’re thinking. Why do I want to feel that? Because it is that important to be able to feel every little movement of your trigger so you know that you are squeezing the trigger and not jerking the trigger.

Types of trigger squeeze

I remember when I was in Boot Camp and they taught us about trigger control before we went to the rifle range. I now look back realize they were, well not wrong, but mistaken. Just in case I run into one of my old Drill Instructors and by some strange twist of fate he read this article. We were taught two different types of trigger control: Interrupted and Uninterrupted. I still believe that there are two types: Uninterrupted and Wrong.

If trigger control is ever interrupted in slow fire the shot needs to be aborted and the shot started over. If trigger control is interrupted in a sustained fire stage then we revert to our “key word” that triggers us to kick-start our shot process. Shot process and “key word” if not already covered will be covered later.

The relationship between sight alignment and trigger control

Often when the fundamentals are explained these two are explained as two different acts. Well truth be told, it’s really kind of hard to accomplish one without the other. They have a

symbiotic relationship. In order to truly settle the movement in the dot or sights you need a smooth steady trigger squeeze. Trigger control is merely a reaction to what the eye sees.

What? One can hold perfect sight alignment / sight picture for a long time. Now apply pressure to the trigger, what happens? Something moves right? So if this happens in a slow fire shot what typically happens? We stop squeezing until everything settles down again and we start the trigger. It moves. We stop the trigger. It settles. We start the trigger. It moves and so on and so on and so on. So from this we can see how the finger already acts to what the eye sees. We do not have to train the eye to accomplish this it already does it. We need to make it work for us and get the finger to react at a more opportune time. BEFORE we have obtained perfect sight alignment / sight picture. Yes I just said to start squeezing the trigger before you have obtained a perfect sight alignment / sight picture. Your sight alignment should be pretty close to perfect since we have mastered a perfect grip and the sights are aligned, not just to each other but also to the our eye, as soon as we raise the gun. Or we will master that soon. So before the sights or dot have settled into the center of the target we should start our trigger squeeze taking advantage of the pressure that is being applied to the trigger to help stabilize the sight. Continue to squeeze the trigger uninterrupted using that pressure to help move toward and stay center and allow the shot to break.

Said nothing about a surprise break. Not a big believer in the surprise break. Just being honest. I know when my gun is going to shoot. I know what it feels like right up until it shoots. I really know my trigger. I have done lots of dry firing without looking at the sights just to know what it feels like.

It has been called steering the sights with the trigger, but I don't like the connotation that carries. More of a sight alignment/sight picture through trigger control concept.

Proper Grip

Not saying that this is the only grip to use. This is merely an option, it may work for you it may not, depending on your hand size and structure. It works for me and it has worked for many shooters that Andy and I have taught it to. It is not UNSAFE for those who have questioned it, trust me the gun will move less with this grip. If you do use a grip that aligns the sights and gun up your arm to your shoulder you are putting yourself at a huge disadvantage.

Here we go.

A proper grip is a grip that will NATURALLY align the guns sights to the eye of the shooter without having to tilt your head or move your wrists around in order to that. Also a proper grip, and most importantly, is a grip that allows the gun to return to the same position that

is one in that allowed the sights to be aligned without having to search for the sights after each and every shot.

To many times I have seen, and I am sure most of you have also, may do it yourself. Someone shoots and you can see the front of the gun waving around as they try to get sight alignment back. This is not a problem with recoil management, grip strength or position. This is a grip issue.

If you line the gun up you arm and too your shoulder in order to align the sights one of 2 things must happen. Your head leaves a natural position in order to see the sights since they are lined up with your shoulder or you have turn your wrists until the sight are slignes to your eye and then the gun is no longer aligned up your arm. This was called “grip alignment.”

Yes back in the day they taught “grip alignment.” Well grip alignment as they described it is BS, just like Natural Point of Aim with a pistol is BS. But that’s a different conversation. Two handed shooters don’t even shoot with the gun aligned up their strong arm to their shoulder. Unless you have some deformity and you head is growing out of your shooting side shoulder why would you want the gun aligned up your arm to your shoulder?

This is for a 1911 slab style grip or a .22 with slabs. If you are using orthopedic grips you can’t do it with them.

My grip is simple, it’s not hard to get into and once you get it, it will feel better and recoil will be better managed and the sights will always come back to alignment. If you rotate the meaty portion of your hand below the little finger behind the back-strap of the gun, everytime you shoot, it will want to move off of it. It is just a squishy, fatty portion of your hand that cannot control the gun or offer any resistence to recoil whatsoever. The fatty portion of the hand there cannot be moved, it is just fat. Try to move it. The movement caused by making fist does not count because the rest of the hand causes that movement. The place the gun tends to move to is the valley formed between that meaty portion below the little finger and the meaty part of the thumb. So I ask why not start with the gun in the place it wants to be.

Even with a picture this hard to describe and even when done in person we still tend to have to work individually with shooters to show them how to do it.

The best and easiest way to get the proper grip, at least a good starting position as you may need to tweak it around a bit until it feels good, is with a holster.

Put your 1911 in a holster on the side of your body not in fron or behind, but on the side of your hip. Put your hands in the surrender position, like the action shooters from. Keep your eyes and head straight. Allow your shooting hand to come down naturally to the gun, don’t move it around just let it come down and grab the gun. The fatty part of the little finger should all be on the right panel of the grip. Now keeping the gun in your hand with the grip assume your one handed shooting position. The sights should be pretty close to being aligned.

If they are not the you will need to tweak the grip a bit. Like I said this is best if in person and someone who uses the grip can actually show you.

So if you ever run into me at a match or wherever please ask and I will show you or talk to someone who has attended a clinic or uses the same grip and ask them.

I hope the photos below are of some help.





Aiming the Dot Sight

So ya understand what it is to aim with iron sights, or so I guess, as I have not had any questions in regard to the Aiming with Iron Sights post. Then again maybe there are only a few people reading my posts. Spread the word guys, free stuff here. I know we all like free stuff. LOL

So what seems to be the difference between aiming with irons and dots that shooters can either make a smooth transition or they cannot? Excluding the obvious that as we get older our vision may or may not allow us to switch to a dot or back and forth from irons to a dot.

So with a dot, I was taught to look at the dot as if it were the front sight of irons. As a young Lance Corporal on the Marine Corps Pistol Summer Team in 1990 I was approached by then Gunny Moody who said try this. "Next time you shoot slow fire, turn your dot down and instead of looking at the dot try looking at the target." Being a young Marine and not one to question a Gunny, I did it. I shot my first 100 slow fire at 50 yards with my .22. Part of the Team training was to share information with the rest of the team. The OIC (Officer in Charge) Capt. Belkes said, " Zins here shot his first 100 slow fire, tell everyone what was different and how you did it."

Oh boy! He didn't like the answer when I told him I turned the dot down and looked at the target instead of the dot. As matter of fact his response was, "what the hell gives you the right to look at the target just because we put a $100 sight on your gun?!?!?" My answer was "The Gunny told me to do it sir."

This was the beginning of a new thought process for the Team when it came to shooting dots. I know that we were taught to look at the front sight, I teach it, but the dot is a different creature all together. It is in essence a one point aiming device. Where as irons you have 2, a front sight and a rear sight. I know, "what about the target?" I am dealing strictly with the aiming process of the gun not the in relationship to the target at this point. So, we can aim the gun and then we can aim the gun "at" something. Does that make any sense at all? The difference between sight alignment and sight picture.

So it was pointed out in the iron sight post that a big problem is wanting to look at the target all the time. Well now you can and shoot well while looking a the target! It is only natural for us to wanna look at the target, after all that is why we struggle with iron sights. One of the first things as kids we were taught is how to throw something, well that and " Don't eat that!" When we were taught to throw a ball we were told to keep our eye on where we were throwing the ball. Shooting is merely a method of throwing an object, albeit faster than Nolan Ryan ever could have imagined, at a target.

I know it sounds easy right? Just go grab your guns and gear and head to the range and look at the target and shoot. Keep in mind like looking at the iron sight, we have to make sure that the focus stays on the target or on the dot. You have to pick one. If you have a hard time seeing the target at 50 yards look at the dot and stick to it. If you can see the target clearly at 50 yards give it try, but like I said you have to keep the focus on the target. The big problem with this is still the same, too often shots are breaking as the shooters eye is going back and forth from the target ot the dot and the shot breaks while the vision is in transition between 20 inches away, the dot, and 50 yards away, the target. That's a big area to be looking at and yet focused on nothing at all.

Remember the article on trigger control? Well that once again comes back into play here. Trigger control needs to be started before the dot to target relationship reaches that perfect

sight picture. If you have obtained perfect sight picture and there is no pressureon the trigger as soon as you start that pressure to the rear **THE GUN IS GOING TO MOVE**! So when the gun is presented to the target and the dot is in a place NEAR your AIMING AREA that is acceptable for you, continue the pressure to the rear. I say continue because there is already some pressure on the trigger already from when we established our proper grip. Right? As the pressure on the trigger builds the dot will begin to stabilize and move less until ultimately the shot breaks at the precise moment that perfect sight picture has been obtained. If only it were really that easy. It will take time. I wish I could find a pharmacist to make a pill to make it happen as esy as it sounds. But you will get there.

Questions?

Does the dot need to be in the center of the tube? Yes and no. Really depends on the sight and if there is any parallax. If there is you need to stay in the middle. It is aesthetically pleasing to the human to see circles in circles so it easier to accept the dot in the middle of the tube than not.

Aren't the dot and the target on the same "plane?" I really don't know the answer to that. I have heard it and truly don't understand the statement cause I cannot see the dot and the target clearly at the same time. Maybe it's just me. I'm not really smart enough to shoot bad, so I just may not be able to grasp the concept.

Anticipation

Yes, everybody. In some form or another practically every bad shot that was not a grip or position issue was a form of anticipation. Jerking the trigger is in essence anticipation, bucking, heeling, flinching, all of them are a form of anticipation.

The problem is I don't feel anyone has ever really addressed the issue of how to correct it. Like a lot of things when it comes to shooting pistol, coaches could tell ya what you where doing wrong. They could even show you what you were doing wrong. Remember the "Ball and Dummy drill?" Put a dummy round mixed in with the live rounds and show the shooter that they are jerking the trigger or anticipating the recoil. Remember that one? I absolutely hate it and think there is no place on the range for it. The shooter would see the sights go away and the coach would say, "See, I told ya were jerking the trigger." This usually followed with, "don't do that!" Well thanks genius!

It doesn't take you to read this to know that if you are jerking the trigger, heeling, flinching, bucking or closing your eyes and you are aware of it that obviously you don't want to do that. So how do we stop doing that? That's where we seem to lose a lot of coaches. They can identify problems but cannot fix the problem. Imagine going to your mechanic and he tells you that the Flux Capacitor on your car is broken because, when you drive at 8:23 PM on a Saturday

night in Myrtle Beach for 20 minutes looking for beer because you think they don't sell beer on Sundays in Myrtle Beach the Flux Capacitor will break, but doesn't know where it is or even how to fix it if he could find it. Would ya go back to that mechanic?

There is a solution to fix or identify a lot of these problems prior to them becoming shots in places we don't them to be. Say hello to the Shot Process. Welcome the Shot Process into your life. Get to know the Shot Process. Make it your own little Shot Process and raise it however you see fit, after all, it is YOUR Shot Process to cultivate and modify as you wish. If only children were that easy, well than again what fun would that be?

The Shot Process is your mental plan and checklist that ensures that you are doing everything the same every time in order to shoot tens. Everyone has shot a ten at some point. A ten that you shoot is no different then a ten that I, Jimmy Henderson, Steve Reiter or Phil Hemphill shoots. The difference is we shoot more tens. We have a routine that we follow everytime the gun leaves the bench. A lot of shooters even good shooters that took a long time to get good may not believe this. They may say they never developed Shot Process and they are good. Well guess what they did develop a routine and process over time. They developed a Shot Process or plan without really knowing it. What you need to do is actively develop that plan. Why would you wait and allow it to happen over thousands and thousands of rounds when you can be in control of the development and tweak the plan as you need.

I have heard shooters, good shooters, describe their Shot Process as, "Pick gun up, shoot gun, put gun down." While it is frustrating to hear a good shooter say that, that is almost my Shot Process now. This is a process now that I have been developing for 20 years. Yes I am still developing my Shot Process. In the beginning my Shot Process was long. I was told to write down everything I did in order to shoot one ten. Everything from the time I put the gun on the bench until the gun went into recoil. Just the part from loading the gun til the gun went bang filled one side of a piece of paper.

Everything that I had to do in order to shoot a ten needed to be confirmed that it was correct. I would dry fire and ensure that everything felt good from feet, hips, shoulders, grip, trigger felt good, confirmed the stage of fire I was about to shoot so I knew whether I was gonna shoot slow fire, timed or rapid. You have got to put to paper everything you need to do to shoot a ten. How many shots you dry fire in slow fire, timed and rapid fire? How many breaths do you take before raising the gun? When do you breathe before raising the gun? Do you inhale the gun up and exhale to settle into the black or vice versa? (note you really should inhale on the way up and exhale to settle into the aiming area) What are you looking at? Irons, the dot, the target if shooting a dot maybe. Do you talk to yourself as the guns comes up? What do you say? Is it a phrase or key word. is it the same thing for all 3 stages of fire? (it really should be or at least really close) What is your visual focus on as the gun is raised? Are you looking at the front sight are you looking at the firing pin. Did you straighten you arm

prior to raising the gun or did you just hap-hazzardly throw it up in the direction-of the target? When the gun settled in the aiming area was it acceptable or did you just figure that you could move your arm at the should or adjust the gun in your grip to get the sights where you want them to be? Was the grip proper when you dry fires and did you do something to screw it up when you chambered a round?

Those and **so many more** questions that need to be addressed in your Shot Process. The best way to learn this process is to have a partner there to write down everything you do and are thinking so that you do not take any short cuts. That is the easy part if you are honest with yourself and actually write down everything you do to shoot one well aimed shot. The hard part is following that process and being able to identify when you are not, or identify problems in the process and abort the shot in slow fire or fix it, expediently, in timed and rapid fire. You have got to be aware of all these steps and be honest with yourself and stop and fix them. If you think in slow fire that you muscle through an indicator and still make a good shot. I am here to tell you that if it does end up on occasion a good shot, you got lucky. I can't tell ya how many times I thought,"oh I can still shoot this and make it a ten." Almost every time I didn't shoot a 100 slow fire, there was that shot. Remember your training partner that helped write down your process? Now use him to ensure that you are in fact following that process that you developed Eventually, down the road, they will become subconscious acts. You will not have to ask yourself all these questions before each slow fire shot and each string of timed and rapid fire, your process will take over and you will automatically be able to identify the imperfections in all the steps and nuances of your process without asking yourself the questions. Maybe you to will reach the, "pick gun up, shoot gun, put gun down" Shot Process.

With proper trigger control and aiming, by following the Shot Process that you have developed for yourself, you can eliminate a lot of issues to include anticipation.

What makes a real difference is the ability to identify indicators earlier in the process. Most of us at some point have been able to call a shot and say where the shot went before scoping because you saw something happen as the shot broke. What you need to realize is that most of the time there was an indicator earlier in the process that was trying to tell you it was gonna be a bad shot. What we do is not pay attention to them and keep squeezing the trigger thinking that it will fix its self. Most of my shots that I abort are aborted before the gun settles into the aiming area. Yes that early in my process there is an indicator that tells me DO NOT let this shot go without fixing something.

Brian Zins

PERFECTING TECHNIQUE

Brian Zins on Trigger Control

By Brian Zins

Source: http://bullseyepistol.com/zins1.htm

Let me start by stating that some of what you're about to read may sound contradictory to a lot of articles that have been written and lessons that have been taught in the past. Some of this may say the same thing in the end, just a different route to get there. By no way am I suggesting they are incorrect. I am merely giving my approach to learning bulls-eye and what I do as a shooter.

Trigger Control

Started out this segment as "Fundamentals" but, honestly if you know me there is only one that really, really counts when the shot breaks. Don't get me wrong stance and grip and aiming are important. Keep in mind that you can have a perfect grip and hold perfect sight alignment all day long. It only counts when the gun goes bang and only one action causes the gun to go bang. Pulling the trigger. You can hold the gun upside down, squeeze the trigger with your pinky, and align the sights to the target in a mirror and shoot tens as long as you don't jerk the trigger. Please DO NOT try that at home. Again this is what I was taught and this is what I do. It is a way, not THE way, that's all. But I have had pretty good results with it. Also this is for a trigger with at least a little roll. My preference that's all.

First let's look at trigger finger placement.

Where should the trigger make contact on the finger? The trigger should be centered in the first crease of the trigger finger. Remember this is an article on Bulls-eye shooting. If this were an article on free pistol or air pistol it would be different. So International shooters please don't unleash the hounds on me. Why you ask? We have always been taught to place the pad on the trigger. If you a trigger that weighs in between 2 and 4 lbs. That is enough weight to move the fleshy part of the pad of your finger. Try it! Get something that weighs at least 2 lbs that has a tip or point on it the size of a pen or a trigger would be ideal. Put your hand on a table or desk, palm up and slowly lower the weight onto the pad of the finger. It moves. A lot. Now lower the weight onto the first crease of the finger, right on the joint. Sure it moves, a whole lot less than the fatty, fleshy part of the pad of the finger. Now having conducted this little experiment think about the pulling the trigger with the pad of the finger. The first part of movement you feel is flash and fat moving out of the way. This is not part of the movement involved in trigger control. Ever been shooting well and in slow fire you start to get "chicken finger?" The trigger starts moving and then it stops and feels like it weighs 30 lbs, or did it

move at all? Maybe what you felt was the flesh moving out of the way cause you are shooting well and don't want to screw up the match, so you are really in tune with what you are feeling in the trigger. Now think if the trigger were placed at the crease or first joint of the trigger finger. When the trigger moves, what you are feeling is really the trigger moving.

Let's go even deeper.

Take a pen or a pencil and start tapping the fleshy part of the web between your trigger finger and the thumb. Keeping the same intensity move the tapping up the large knuckle on your trigger finger. Feel the difference? The flesh acts as a shock absorber to the tapping where the tendon in the joint is more of a conductor. I know what you're thinking. Why do I want to feel that? Because it is that important to be able to feel every little movement of your trigger so you know that you are squeezing the trigger and not jerking the trigger.

Types of trigger squeeze

I remember when I was in Boot Camp and they taught us about trigger control before we went to the rifle range. I now look back realize they were, well not wrong, but mistaken. Just in case I run into one of my old Drill Instructors and by some strange twist of fait he read this article. We were taught two different types of trigger control: Interrupted and Uninterrupted. I still believe that there are two types: Uninterrupted and Wrong. If trigger control is ever interrupted in slow fire the shot needs to be aborted and the shot started over. If trigger control is interrupted in a sustained fire stage then we revert to our "key word" that triggers us to kick-start our shot process. Shot process and "key word" if not already covered will be covered later.

The relationship between sight alignment and trigger control

Often when the fundamentals are explained these two are explained as two different acts. Well truth be told, it's really kind of hard to accomplish one without the other. They have a symbiotic relationship. In order to truly settle the movement in the dot or sights you need a smooth steady trigger squeeze. Trigger control is merely a reaction to what the eye sees. What? One can hold perfect sight alignment / sight picture for a long time. Now apply pressure to the trigger, what happens? Something moves right? So if this happens in a slow fire shot what typically happens? We stop squeezing until everything settles down again and we start the trigger. It moves. We stop the trigger. It settles. We start the trigger. It moves and so on and so on and so on. So from this we can see how the finger already acts to what the eye sees. We do not have to train the eye to accomplish this it already does it. We need to make it work for us and get the finger to react at a more opportune time. BEFORE we have obtained perfect sight alignment / sight picture. Yes I just said to start squeezing the trigger before you

have obtained a perfect sight alignment / sight picture. Your sight alignment should be pretty close to perfect since we have mastered a perfect grip and the sights are aligned, not just to each other but also to the our eye, as soon as we raise the gun. Or we will master that soon. So before the sights or dot have settled into the center of the target we should start our trigger squeeze taking advantage of the pressure that is being applied to the trigger to help stabilize the sight. Continue to squeeze the trigger uninterrupted using that pressure to help move toward and stay center and allow the shot to break. Said nothing about a surprise break. Not a big believer in the surprise break. Just being honest. I know when my gun is going to shoot. I know what it feels like right up until it shoots. I really know my trigger. I have done lots of dry firing without looking at the sights just to know what it feels like. It has been called steering the sights with the trigger, but I don't like the connotation that carries. More of a sight alignment/sight picture through trigger control concept. Brian Zins

PERFECTING TECHNIQUE

A Focus on the Fundamentals

By Ron Steinbrecher

Source: https://www.bullseyepistol.com/steinbrecher.htm

STANCE

The main goal of a proper stance is to provide stable and consistent support for the entire body so that it can hold the gun as steady as possible as well as recover from recoil quickly with little movement. The foot positions and body twist should be duplicated naturally, with little effort. It should be comfortable and give a repeatable natural point of aim.

Finding the stance that works for you involves experimentation on your part. Everyone is biomechanically different, so one stance does not fit all. But there are some basics to adhere to as mentioned previously: comfortable, sense of equilibrium, no muscle tension in the legs or torso, arm raises naturally toward the target and 45 recoil is easily handled with no loss of balance.

To find the position that works for you, start by turning somewhere between 45-90 from the target. Keep feet about shoulder width apart. Knees should not be stiffly locked. Back and neck should be straight, shoulders relaxed. Head upright. Your non-shooting arm must be anchored to prevent body sway in recoil...most place their hand in their pocket, but hooking your thumb on you belt loop in front works even better for balance. Their are three twisting points to align you with the target. Use all three instead of just one or two. The three twist zones are hips-torso, shoulder and neck.

The shooting arm must be extended fully with elbow and wrist locked in position (but not with too much tension). To counterbalance the weight of the gun in your extended hand, some backward sway may be necessary so as to keep your center of gravity in line with the center of your body mass. Keep your head level with the sights; your face should be turned as far as possible toward the target without undue strain on the neck muscles.

To find your natural point of aim, close your eyes and assume the positions described above....open them and see where your sights are aimed. Shift your feet to align the sights on the target bull. Now lower your arm, close your eyes and repeat the process until your are properly aligned. Simulate recoil with your arm and see where it falls back naturally. Adjust again if necessary. Remember, as tension free as possible. You should be able to stand in this position for a long time without any noticeable fatigue or movement.

After a period of time shooting, you may notice that your foot positions may have altered somewhat; this may be your tired body adjusting to find a more comfortable position. It's

okay to accept this comfort adjustment. Check your alignment again and if it is off a bit, just use your rear foot to fine tune your point of impact by leaving the heel of the foot stationary and only making very slight (like and inch or two) toe movements left or right.

The stance forms the foundation for the shooter to hold steady and to build on the other fundamentals. All else remaining equal, an inconsistent stance will change your point of impact on the target. Like some other fundamentals, finding the stance that works for you is not given enough attention. But once found and habitualized, a proper stance becomes natural and automatic and leaves you free to concentrate on other things.

GRIP

Gripping the pistol correctly and consistently is essential for shooting groups. Orthopaedic grips fitted to your hand will aid in the consistency necessary every time you grasp the pistol. That is your main objective: to grasp the same way every time. The gun should feel comfortable in your hand. The trigger finger should be able to position itself on the trigger so that it can be able to press it firmly straight back without rubbing against the side of the grip.

Here is a step by step method to grasp the pistol:

1. Pick up the pistol with your non-shooting hand and place it firmly in your shooting hand.

2. The pistol is seated high and deep into the web of the hand (that fleshy area between the base of the thumb and index finger). You are trying to make the pistol barrel a natural straight line extension of your arm. If a string traveled from the muzzle to your eye, it would follow a line along the inner portion of your arm. The mainspring housing (or that area of the grips) should catch the meaty portion of your palm below the base of the thumb.

3. The thumb itself is loose and relaxed along the thumb rest (if you have one).

4. The primary gripping pressure is produced by the middle finger.

5. Secondary pressure is provided by the third finger. Also, this finger can produce tiny changes in front sight elevation by increasing/decreasing pressure.

6. The little finger is loose and not exerting any noticeable pressure. Essentially, it is just along for the ride. Be careful...this finger has a lot of leverage, pressure while shooting will bring the muzzle down.

7. The fingertips do not exert pressure on the pistol grip. They are part of the total "package", along with the thumb and little finger, that form the entire grip and help to control the gun in recoil, but are not consciously applying pressure as do the middle and third fingers. If you "let go" with the thumb, little finger, trigger finger and finger tips..the gun should still be held firmly by the middle sections of your middle and third fingers pressing straight back into the lower palm of your hand.

8. The trigger finger must not be in contact with the grip at all.

The trigger finger must be able to flex at the middle knuckle without moving the pistol or varying the pressure of any of the other fingers. This is important and demands a lot of practice. You must be able to pull the trigger without moving the other fingers. The hand is not meant to work this way, so you must train to overcome this tendency.

Total grip pressure is firm but not to the point where your hand starts to shake. Keep the wrist locked. A firm grip and locked wrist have much less play from left to right. If your finger nails are totally white, you are using too much pressure. If you can see a deep impression of the grips in the flesh of your hand, you are overgripping, back off a little (except maybe in hardball rapid fire). Try to keep a consistency in the amount of pressure you use to grip the gun throughout the entire shot process. Inconsistency will definitely change your point of impact.

Spend time working on your application of a good grip. Learn how to easily and consistently acquire it...like putting on a well worn glove. If the grip doesn't quite feel right, start over. Some shooters put memory markers on the grips to let them know that they are holding correctly.

If you feel you lack grip strength, do some exercises with a hard rubber ball. Dumbbell wrist curl exercises working both the inner and top of the forearm will strengthen the finger flexor tendons. Take a single sheet of a newspaper and using just your fingers, wad it up into a tight ball. Make sure you stretch your "tennis elbow" ligament.

Between shots in slow fire, relax your grip (without letting go) to let blood rejuvenate the hand and fingers.

Let me end by repeating:

1. A firm grip gives you a sense of controlling recoil.

2. The repeatability of the way you grasp the gun will give you tighter groups.

3. The trigger finger and nothing else tightens to squeeze the trigger.

BREATH CONTROL

To shoot well one must become motionless. Breathing causes motion, so it must stop....but in doing so you are robbing your body of the oxygen it needs to perform well. Lack of oxygen causes muscle fatigue and loss of vision (in fact, the eyes are the first thing in your body to go). Fortunately, these effects are not felt immediately because we have a reserve of oxygen in our system to sustain us for brief periods, and the more aerobically fit you are, the longer you can last.

To insure that you have enough oxygen in your system to last for the 10 to 20 seconds needed for a slow fire shot or timed fire string you need breath control.

Taking two deep breaths (inhale slowly nose, exhale quickly mouth) before a shot or string does two things....lowers your carbon dioxide blood level (which tells the brain when to breathe), thus reducing our urge to breathe and also gives you sufficient oxygen to avoid eye and muscle fatigue.

Then take a third normal breath as you bring your gun on target and exhale half the air to leave your lungs in a "neutral", comfortable state (too much or too little air in the lungs will cause an uneasiness to inhale or exhale to relieve the pressure). Some listers have proposed having a little more air than you need and slowly letting it out as you shoot. Try it and see if you like that method. But remember, your chest must not move.

Since your body is accustomed to pauses between breaths you can stay motionless in this state until you begin to run out of oxygen. Hopefully, you will have finished shooting before that happens. If you run out of air in slowfire, put the gun down and start all over again since you are definitely holding too long and your eyesight is losing focus and your wobble area is increasing. In sustained fire, you can take a quick inhalation of air during recoil to revive yourself (the effect is almost instantaneous for your eyes and muscles).

If you cannot hold your breath for timed fire, even after taking two deep breaths prior, you are not aerobically fit and need to improve your aerobic capacity by walking or jogging exercises. If you are a smoker...well, duh!

This is the easiest fundamental to master since your body "knows" involuntarily how to do it. Just help the process along by remembering to oxygenate first and then leave the lungs and diaphram in a relaxed state while shooting.

I know it seems as if I went into a too long explanation for what amounts to just "take a couple of quick breaths and hold it", but understanding the mechanics of "why" is, in my opinion, a reinforcement in understanding how important subtle things are when you are trying to something as difficult as precision shooting. Plus, I like to "hear myself talk".

AIMING AND SIGHT ALIGNMENT

There are different techniques to aiming and sight alignment depending on whether you are using iron sights or red dot scopes.

AIMING involves correct positioning of the sights in relation to the aiming mark on the target. Since the eye very easily centers concentric circles...with dots the aiming mark is the center of the bullseye. It is as simple as that. With iron sights, however, aiming presents three common options. Center and six o'clock holds plus sub-six area aiming. All three are valid

aiming marks and the type of shooting and targets used determines which is best. For the less skilled shooters, center hold may be the best option for sustained fire since it allows the brain to accept more readily what it sees. Six o'clock and sub-six aiming are more suitable for slow fire. Sub-six is sometimes referred to as "area aiming" As in center hold it allows the brain to have a more acceptable area in which to pull the trigger. Six o'clock hold is a very precise point and trying to hold it before squeezing the trigger causes hesitation and can lead to poor trigger manipulation by novice shooters.

Make sure you are using your dominate eye for aiming. To find out which eye is dominate, extend your arms, palms forward, then bring your hands together making an opening with your thumbs at the bottom. Center an object in the opening, then close one eye, then the other. The eye that kept the object centered in the opening is your dominate eye. There has been a lot of discussion on the list about cross dominate shooters...refer to the archives for that discussion.

SIGHT ALIGNMENT is one of the most critical of the fundamentals. Minor alignment mistakes make for major variations in you point of bullet impact. For dots, alignment consists of keeping the dot in the center of the scope lens and the bullseye. Again, it is as simple as that. Unfortunately, no one can keep the dot perfectly still and centered, especially while squeezing a trigger.

Iron sights are in perfect alignment when the front sight blade is centered in the rear sight notch and the tops of both are on level plane with each other. Since the eye can only focus clearly on one object at a time, your focus must be on the front sight blade...for it is only here that you can see misalignment....the brain determines if there is equal light on either side of the front blade and the top of the blade is level with the rear notch opening. You CANNOT do this if your eye focus is on the target or the rear notch. Both of them will be seen in relation to the front sight blade, but slightly fuzzy, that's okay.

The combination of sight alignment and aiming point on the target forms what is known as the "sight picture". Maintaining the sight picture while squeezing the trigger is the secret of successful shooting. Let me write that again...for it is the holy grail of successful pistol shooting. Squeezing the trigger straight back so as to not cause any misalignment or excessive movement will result in a good shot. Everytime. Guaranteed. Unless, of course, your gun and ammo suck.

A couple of clarifications. Don't confuse movement within your aiming area with movement which causes misalignment. If you maintain perfect alignment (no wobble of the dot from center, the front blade centered and level) you can still have side to side and up and down movement of your aiming area as much as the size of the ten ring and still shoot a ten. Jim Lenardson describes it thusly: imagine a pipe with a 3" inside diameter (the size of the ten ring at 50 yds) running perfectly straight from your muzzle to the target. AS LONG AS YOU

MAINTAIN PERFECT ALIGNMENT, you can have a 3" movement by the combination of your arm and body sway and still hit the ten ring. But with a very slight misalignment, the path of the bullet is altered and will it hit the side of the pipe before it travels 50 yds.

So you can have three kinds of movement: wobble (misalignment), arm sway and body sway. All three contribute to the point of impact of the bullet, but the wobble is the one that does the most damage toward a good shot.

TRIGGER CONTROL

The controlled release of the trigger lies at the heart of performing a good shot on target. All of the other fundamentals can be perfectly performed but they mean nothing if the trigger release is faulty, because it will spoil the aim and sight alignment at the crucial moment when the shot is fired. Thus, trigger control is an equally critical fundamental with sight alignment.

Your trigger finger must apply pressure to the trigger in a direct line back toward the aiming eye. For this, the finger must be correctly positioned, with the center of the pad of the first joint (or whatever works for you) touching the trigger. The trigger finger must be clear of the frame and grips so as not to move the pistol as the pad is pulled straight back. ONLY THE TRIGGER FINGER MOVES. Grip pressure of the other fingers must remain static (see the post on "Grip").

Trigger finger pressure must be constant, smooth, and progressive. Once you start applying pressure, continue until the gun fires... do not "stop-start" the trigger for usually only a very skilled shooter can do that successfully (because each stop-start requires the breaking of the trigger's inertia and that causes movement). The "keep the trigger moving" mantra for rapid fire applies to all trigger pulls.

In slow fire, the timing of the hammer fall should practiced so that the release of the shot coincides with the steadiest part of your hold, which is approximately 6-10 seconds after you start the aiming process. In sustained fire, the hammer fall coincides with returning to an acceptable sight picture.

Keep your finger pad centered on the trigger. Too high decreases leverage and the pull feels harder, too low and the shot may fire before you are ready. You must be consistent in your trigger finger placement. Don't be afraid to experiment in training with different placements till you find the one that gives you the steadiest straight back pull.

Trust in your ability to hold and minimize wobble and the eye's ability to center the sights to form a sight picture and pull the trigger smoothly and continuously without trying to "snatch" a shot or jerk the trigger. You can start out with a slow release and with experience and lots of dry fire trigger time, you can increase your trigger speed without additional movement.

With two stage triggers, take up the first stage completely and add slight prepressure to the second stage (the amount of which depends on your "trigger sense" acquired with experience) just before settling in on your aim and alignment. This works well with slow fire. For sustained fire, take up the first stage in recoil recovery and keep the trigger moving. Again, do not "stop-start" as it will destroy your rhythm and cadence (the only exception is if you recover badly and do not have a sight picture which will give you a shot in the bull).

Work toward unconsciously activating the trigger, such that the shot becomes a surprise break to you. This will help in avoiding flinching or anticipated recoil. This is difficult to master, but with practice and patience, it will be achieved. When trigger pull becomes automatic, your brain is free to focus on the other critical fundamental....sight alignment. Just as your eye cannot focus on two objects simultaneously, you brain cannot focus on two separate fundamentals at the same time. All the fundamentals should be done on auto-pilot, but if you have to focus on one, then let it be sight alignment when shooting with iron sights and when you are using a scope, you can let your focus be on trigger pull (since a sight picture is so easily done with a scope).

One of the best ways to develop trigger control, speed and a feel for your particular gun's trigger is to dry fire. I know it is boring compared to live fire, but if you have long periods between matches, it can substitute as practice.

FOLLOW THROUGH

Follow through is an often neglected fundamental; yet, without it, all your good efforts in applying the other fundamentals may be wasted.

Releasing the shot should not be your final effort. You need to do one more thing. It is argued (especially on this list) that at the moment of ignition, the bullet has left the barrel before you could possibly do anything to affect its flight. I believe that also for the simple reason that the nervous system of your body reacts in tenths of a second and the gun fires in thousands of a second. It's no contest. But the problem lies in doing something BEFORE the moment of ignition. Without follow through, a bad habit may develop which is initiated just at or before ignition in which case, the flight of the bullet will be affected by your action or movement.

In all sports the achievement of a quality performance mandates follow through. The pitched baseball, the sprinter keeping form when crossing the finish line, the martial artist breaking a board, the golf shot.....the list of examples is long. What we do is no exception. There can be no let up of form and technique until it is absolutely certain that the act is completed. The danger is in relaxing your grip too early, losing your concentration on the sights, or beginning recoil before it has actually started; all of which will produce a bad shot. To practice follow through, imagine that the bullet is like a rocket leaving the barrel a few seconds after ignition. Try to maintain your form as long as possible to allow the rocket to leave the barrel, keep your

focus on the sights or dot and try to call the shot. If you have trouble calling your shot(s)....you are not following through.

This technique of using follow through as a SEAMLESS, CONTINUING FLOW OF ALL THE FUNDAMENTALS will reduce the chance of giving up on the shot before it is fired.

Well, that's it. I hope you will or have been able to improve your score by the application and practice of these fundamentals. I've tried to explain them as best I could. I've already gotten some feedback from one individual who has improved his shooting. That's great.

RED DOT SCOPES

Most participants in matches use the red dot scope. Not that there is anything wrong with using iron sights, it is just that the dot scopes are easier. As our eyes weaken with age, the dot scope allows us to continue to compete at a level comparable with those of sharp vision.

With iron sights, three elements must be aligned to form a sight picture...the target, front sight blade and the rear sight notch. Since our eyes can only focus sharply on one thing at a time, our focus is thus on the front sight blade to ensure perfect alignment.

Red dots, however, are on the same focal plane as the target (infinity) and thus the "eye" has only to look at two things...the target and the dot and they are both in focus when you aim. That is quite an advantage! It allows for faster target acquisition and alignment and because the brain easily aligns concentric circles (the scope lens, the dot and the bullseye), you can concentrate more on trigger squeeze and let the aiming process become more automatic.

Low light levels (relative to the outdoors) at indoor ranges can be a problem with iron sights, but not with dots. Most dots have several brightness intensities to take care of the dimmest indoor ranges to the brightest outdoor sun.

Some things to remember when using scopes: Parallax is a factor even with the best dots (my choice is the Ultradot) so it matters that the dot is centered in the scope when aiming especially when shooting at 50 yds. You can check the amount of parallax your scope has by keeping it rested on the bench and held stationary with the dot centered on the bull. Now moving only your head slightly side to side and up and down, you can see how much the dot moves on the target when it is in different positions in the lens. The smaller the movement= the lesser the parallax= the better the scope. Keeping the dot centered in the lens avoids misaiming.

Always carry a spare battery. Dots will weaken gradually with silver oxide batteries and will fail very quickly with lithium batteries. The batteries last a long time if you can remember to turn off the scope when finished.

Ultradots (and some other brands) offers different dot and lens sizes. The choice is just a matter of personal preference and the style of shooting that you are doing.

The scope and mount will add 6oz. or more to the gun and if you are slide mounting, you may need to increase your powder charge slightly or change your recoil spring to give you reliable functioning.

I'll go out on a limb and say that dot scopes will probably add 10% to your iron sight scores especially if you have older eyes. They take some getting use it, as all changes do, and your scores may initially drop somewhat, but eventually you will be glad you switched. There are those who believe that all shooters should start out with iron sights when learning the fundamentals before transitioning to dots, I never really understood this, except of course if becoming Distinguished is important to you.

PERFECTING TECHNIQUE

Bullseye Pistol Basics

By Allen Fulford

Source: http://www.bullseyepistol.com/fulford.htm

Introduction

In my job as a County Agricultural Agent, I found that I was working many hours and weekends, and the pressure of the job was really getting to me. For many years I was a plinker and I always liked shooting. I wanted to become involved in competitive shooting, so I really picked up the sport as a kind of therapy through a stressful period. When I was out on the range concentrating on my shooting, I put all of my worries out of my mind, and I found that it was a wonderful source of inspiration for the rest of the week. I was able to go back to work and accomplish a whole lot more. Competitive shooting has added years to my life because it has allowed me to relax, enjoy life more, and allow me to get into something that I could set goals, reach those goals, set more goals, and so on. This process of shooting, and the fellowship that goes with it, I'm sure, has added years to my life. If I had to put one thing above all facets of shooting, it would be the fellowship with other shooters. I think that by and far, shooters are some of the best folks in the world.

Fundamentals:

I would like to discuss several fundamentals of precision, or bullseye, shooting..

Stance

The first thing we need to consider is stance. I suggest that you face the target and turn your non-shooting side away about 45 degrees to start to try to establish the stance that is best for you. What we like to do is relax EVERYTHING in the body except the shooting arm, elbow and wrist. We like to relax the non-shooting hand, and do something with it, either hook it on your belt, put it in your pocket or hook your thumb on your pocket. Extend the gun and your arm above the target, close your eyes, and let the gun settle into the normal aiming area. Once settled, open your eyes and see where the gun is pointing. If your sights are either right or left of the target, you need to move your trailing foot around so that your natural point of aim is on the target. This is important so that we are not using muscles to move the gun (horizontally) onto the target. We should be using muscles only to support the gun vertically. We need to find a comfortable stance. We want to almost lock our knees, but not quite. We

want to relax our stomach and all other parts of the body except our wrist, elbow and entire shooting arm; we want to remain as RIGID as possible without putting those muscles in a strain. If we strain, we will experience muscle fatigue, and our performance will be compromised quite a bit.

The Grip

The way we recommend to get a grip is to hold the pistol in the non-shooting hand, by the barrel or the slide, and take the shooting hand and assume the grip. Most experienced pistol shooters agree upon the amount of grip, or how hard, you should hold a gun. However, not everyone agrees about the position of the hand upon the gun. My personal preference is to put the trigger finger farther through the trigger guard than you would normally find in printed literature. Because my fingers are fat and short, I do that so I will have more leverage with my trigger finger. This is different from the traditional method of the wrist and arm being straight in line with the barrel. Regardless, we must grip as high as possible on the backstrap so that we will have more control. Most of the good shooters I know do not hold their thumb down toward their other fingers. They keep the thumb relaxed and high and this is very important. Once you establish your best grip, consistency becomes very important. We can really change our point of aim because of our grip. Now in as far as how much pressure to use, I like to imagine using the using the same amount of pressure as holding a hammer or a very firm handshake. Most of the pistol shooters I know use a fairly strong grip. One way to determine how much pressure to grip the gun is to extend the gun and take your grip as tight as you can get it until you start to tremor -- then back off. This is probably the grip pressure that is right for you. It is necessary that you maintain pressure on the forestrap that is straight to the rear. It is also important that we have constant pressure during the shot in order that we are not milking the grip -- that is squeezing all the fingers while pulling the trigger.

Sight Alignment

The human eye is not capable of focusing on two separate planes at the same time. Therefore, with iron sights, we cannot clearly see the sights and the target at the same time. What we have to do is place our concentration and our focus on the front sight and accept blurring the target. With the optical sights, we eliminate having to line up two separate front and rear sights. We are either using a dot or a cross-hair recticle, but we all agree that it is still important to focus ON THE RETICLE and not on the target.

Pistol Movement and Trigger Control

One of the hardest things for a beginning shooter to accept is the movement of the pistol while they are trying to fire the shot. I cannot stress this too much as none of us can hold a

gun absolutely motionless. Good shooters sometimes have the sensation of holding the gun perfectly motionless for some three seconds or so, but no one can hold perfectly steady for the entire time required to fire the shot. The principal of accepting this movement, and applying trigger pressure straight to the rear while at the same time keeping correct sight alignment, is the key to pistol shooting. I cannot say it enough, accepting this movement is very important. Your "arc of movement" is the entire area that your sight alignment encompasses while you are holding. Let's say your arc of movement is within the nine-ring on the slow fire target at fifty yards. And note that almost anyone can train themselves to do this. Remember then, if you can hold the nine ring, and you can mash the trigger straight to the rear causing it to break without any additional pistol movement, your shots will go within your arc of movement, and you will score a nine or better on each shot. If there is a secret to pistol shooting ... that's it.

Training and Practice:

Strength Builders

Since I first started shooting I have used a five pound weight to develop the muscles in my arm that I use to hold the gun. I lift the weight upward with an extended arm and like to imagine that it is being suspended from my neck and shoulder area instead of being pushed up from the bottom. That is the same mental picture that I get when I am actually holding the pistol out there. A series of repetitions will do wonders for your ability to hold the gun. I strongly recommend using a weight to develop the muscles used to hold a gun because in everyday life we do not use these muscles. This is a big part of reducing your arc of movement and learning to hold still.

Sight Picture Exercises

Here's a trick that my old buddy Bill Blankenship taught me years ago. He would draw a vertical line on the wall and cross it with a horizontal line. Holding up the pistol toward the lines, he would study the movement of the pistol on each. He would concentrate on vertical movement for a while, and then on horizontal movement for a while. I have found that through deep concentration and study of my movements, I have been able to mentally picture my arc of movement, kind of like a circle, and am getting it smaller and smaller. I recommend this technique for your training program. Exercises like this will carry over and help you during a match. If I am experiencing movement as I hold the pistol, and it doesn't matter if it is a little or lot, and the sights move off the center off the target, I've found that if I GENTLY try to bring it back in rather than a quick jerky movement, that is gently try to FLOW with the

movement instead of being tense and making jerky correction, my wobble is able to settle down. And even if it does not, I am still able to break a better shot this way.

Trigger Control

There are two methods of trigger control. In both methods, when you settle in your aiming area, or even before, you take up the slack in your trigger. The first method I am going to describe is the one we all recommend to beginners. Once slack is taken, you actually begin the put pressure on the trigger. You can pull that early pressure, taking part of the poundage off, and continue in a gradual consistent pull until the shot actually breaks, accepting your movement all the time. The second method I will describe is pulling on the trigger only while the sights are aligned in an almost perfect picture. As they move off center, HOLD the pressure that you have. When the sights move back on center, with a movement you can accept, then you CONTINUE the pressure. This is "staging" the trigger. Press when it is on, hold when it is off. When everything is going well, I can shoot better scores when using this second method. On other days when I am not as coordinated, I have to use the first, or the straight-through pull.

Dry-Firing

We can practice all the above elements, but we still need to put them together in a dry-firing exercise. We do that by practicing all the elements we are going to be doing in firing a shot. We will put a target on the wall in corresponding size to that of one at 25 or 50 yds. and practice our breathing, our grip, our stance, and practice breaking the shot. All the time, we are trying to minimize our arc, analyzing the movement of the gun, studying the direction of the sights, and trying to break that shot. This is a good time to learn to break that shot with steady, constant pressure. I recommend dry-firing highly.

Shooting Practice Sessions

During practice sessions at the range, I usually shoot an entire (2700 or reduced) course. Afterward, I will practice things again that bothered me a little bit more than others. For example, after shooting the whole course, if I messed up a few strings of rapid fire, I would shoot a little more rapid fire at the end. I don't recommend practice shooting at the range just to expend the rounds. If an individual is actually learning something then a lot of range practice could be justified. In converse, dry firing is never a waste, I don't care how experienced you are. My old buddy John Farley declares that he shoots better with a minimum amount of practice. With me, it is exactly the opposite. I need much more practice than he does, and I try to learn something from every shot that I fire. Just to say that you fired

1000 rounds this week might be worthless. However, if you are learning from each shot you fire, it might be hard to ever say that you are practicing too much!

Match Experience

For me, there is no substitute for match experience. There is nothing like going to matches, seeing good scores starting to build, dealing with the mental aspects of a good score, and also dealing with the mental aspects of that buddy you want to beat. Taking all of this into account is no substitute for the experience of shooting real matches. After you get over that initial "stage fright," you will concentrate more on trigger control, etc. when it is in "real" competition. For me, concentrating on my "routine" keeps me from thinking about anything else. Thinking about the fundamentals of grip, stance, trigger control etc. pushes any negative thoughts out of my mind. If I can maintain this kind of positive thinking, then I am able to overcome match nerves.

PERFECTING TECHNIQUE

Three Lessons of Pistol Shooting

By Bill Blankenship

Source: http://www.bullseyepistol.com/billblnk.htm

The following condensed article on pistol shooting comes from one of the greatest shooters of all time, Bill Blankenship. Bill's experience as a shooter on the U.S. Army Pistol Team and a winner of the National Championship five years in a row, plus a sixth time two years later, speaks for itself.

When reminiscing over the years I've been in competitive shooting, many things come to mind. The feeling of doing something well has indeed been very satisfying and well worth the effort. Winning the first medal, the first match, and to have had the good fortune and the friendships established will rank high among my finest memories. I've also been impressed by the sportsmanship displayed by shooters from the time I became involved.

To learn to shoot a pistol, there are certain basic fundamentals every interested person should be familiar with. In the beginning, most shooters make the same basic mistakes. Some of these are made because of the lack of knowledge, while other mistakes are made because it is simply the natural thing to do, such as looking at what you are shooting at. This is the first lesson a new shooter should learn: The focus of the eye must be one the front sight (with iron sights) or on the recticle (with red dot or scope sights) to get the most accuracy out of the pistol.

There are 3 major lessons that should be taught to new shooters. The first is proper sight picture. The second is trigger control. The third is learning to hold still. When working on these LESSONS, one is learning the majority of the FUNDAMENTALS; therefore stance, grip, breathing, and other fundamentals will be discussed within these 3 lessons. The ability to CONCENTRATE has a direct bearing on anyone's scores, but the fundamentals must be learned first.

Lesson One: PROPER SIGHT ALIGNMENT

Sight alignment has to do with only the alignment of the front and rear sights and has nothing to do with the target. Where you hold on the target is not too important as long as you hold as still as you can and in the same place for each shot. The greatest lesson I learned and one that made the greatest difference in score was to bring the focus of the eye from the target to the front sight. I found that during my experimenting that the focus of the eye was shifting back and forth from the front sight to the target and sometimes the focus was on neither, but

somewhere inbetween! Perfect sight alignment is necessary to get good scores, and having a natural alignment of the arm, hand and gun to the eye will keep the sights perfectly aligned.

There are several things that cause the sights to become mis-aligned. The wrist not being held in a stationary position is one. To eliminate this, concentrate on getting the arm straight with the elbow locked and by gripping the gun as hard as you can grip with out it starting to tremble. After you've taken the grip on the pistol and the sights do not align themselves naturally, DO NOT compensate by turning the wrist. This is especially a problem for those who shoot with optical (dot) sights. To do this will cause you to lose control of the solid arm, wrist, and grip. Practice holding the wrist in a stationary position at all times with a locked elbow and a hard grip. The will strenthen forearm muscles which control the grip.

The position of the head is very important in keeping the sights aligned. The head should be kept in the same exact position at all times. A drooping head will cause may difficulties.

Lesson Two: TRIGGER CONTROL

To new and old shooters alike, controlling the trigger seems to be a major problem. The new shooter has trouble because he has no conception of how delicate this problem is. The older shooters, myself included, are always trying to perfect their trigger control.

All the new shooters I've had the chance to coach have had just about the same problems. For example, the first time a person picks up a .45 caliber automatic, their first shot is usually somewhere between a point one yard in front of their feet to anywhere in the vicinity of the target! Most everyone knows that when the sights get aligned you're supposed to do something that will make the gun fire. This they do, but most of the time all at once. This sudden movement of the trigger finger disturbs the gun and is referred to a "jerking the trigger." Let me explain my idea of the problems and progress of new shooters. I like to use the example of a picture puzzle. When a new shooter starts to understand the fundamentals its like picking out the pieces that have border lines and putting them together. There is a lot accomplished but a lot ahead. Each new piece of the puzzle is like learning another lesson in shooting. Sometimes a puzzle takes a long time to figure out; likewise, there are many problems in shooting, and each time one is solved, the results produce better scores.

Trigger control can become a habit. I believe that any muscular movement of the body can become a habit through repetition. The reflex action of pulling (squeezing or mashing) the trigger, is the subconscious mind evaluating the situation, making the decision whether or not to shoot that shot, and without any conscious thought on your part starting the trigger finger to move and continuing pressure until the hammer falls. Just as if an object is about to hit you in the face you automatically close your eyes and usually bring your hand(s) up to protect your face, it is a reflex action controlled by the subconscious mind.

Without hesitation, the pulling of the trigger is a positive pressure straight to the rear in such a way as not to disturb the perfect alignment of the sights or the stillness of the gun. Taking a good position and stance is important. Also, you must have a solid arm with ELBOW LOCKED, the wrist straight, and a good hard grip on the pistol. Then try to hold the gun with perfect sight alignment as still as you can with the focus of the eye on the front sight (or recticle). If the gun settles in the PROPER AIMING AREA on the target and the movement is such that you can accept, the commit yourself to deliver that shot and squeeze with a positive pressure without any hesitation. The saying "he who hesitates is lost," applies very well to trigger control.

Trigger pressure must come from the trigger finger only. You must place the trigger finger on the trigger in such a way that you can squeeze straight to the rear and thereby have no disturbance to the sight alignment in any way. Having the arm so solid, the wrist straight, and the grip hard will aid you in trigger control. By gripping hard you can keep the same grip on the pistol throughout the shot or series. The trigger finger must move indepently from the rest of the hand. Consistency is the secret of doing anything well. This is especially true of trigger control.

Lesson 3: LEARNING TO HOLD STILL

To the new shooter holding a handgun relatively motionless is most difficult. He must learn this lesson early in his training. Even the most experienced shooters never hold the gun absolutely still, but the attempt to hold it still is always present. The experienced shooter has many shots break when the gun is almost motionless, and this is what the new shooter must be striving for in his performance. But when the gun still moves slightly he must accept this movement, always attempting to hold the weapon as still as it is possible for him to do so. As the new shooter progresses in his training, he will learn what movements he can expect and those that will cause a bad shot. The important thing to remember is that the shooter must always be trying to hold the weapon as still as he possibly can, with perfect sight alignment.

How to Stand: The feet should be at least shoulder width apart. This is a comfortable way to stand and also helps to keep the body balanced. The legs should be straight but it is not necessary that the knees be locked in the joint. The hips should be near level, the back should be straight. The head should be erect and turned toward the shooting arm. If you lower the head gradually toward the arm, your front sight will dip, causing low shots. The reverse of this is true when the head is gradually moved to the rear. In this line of thought a number of shooters droop the head to the left while attempting to fire the shot or a string of shots. This causes the front sight to block toward the left side of the rear sight. This brings out the importance of holding the head erect and in the same position during delivery of every shot or string of shots. To get the arm still you must first make sure that the arm is straight, the muscles firm, the elbow locked. The wrist must be set so that there is no movement of the

hand. This will help minimize your movements because the arm, hand, and weapon are then supported by the muscles in the shoulder. As the shoulder muscles get stronger the ability to hold more still becomes more apparent. While studying movement in general, it was helpful to aim at a cross on the wall. I found that by aiming perfect sight alignment on a horizontal line that the movements up and down were exaggerated and while aiming at a vertical line the movement from side to side was exaggerated. Working on these two separate lines enabled me to get an idea of what was necessary to stop movements that I did not desire. I did not stop all the movements of the gun or arm but there was less movement than I had ever had before. There were times in the next year after spending the time studying movement, that I had the sensation of the weapon being perfectly still for a short period of time just after settling the arm. I was not aware of this short period of stillness of the arm and gun just after settling, until I spent the time studying movement. To find this fact out made the considerable amount of time spent a very good investment.

How to Grip the Weapon: The first point to be emphasized is that the shooter must learn to grip the gun hard to get consistent results. He must also learn to grip the same way for each shot or string of shots. To do this, he must use both his senses of seeing and touch to see if it is right as well as to feel if it is right. The point here is that just getting a hold of the weapon is not enough. There must be a decisive effort each time to get the same firm grip on the gun for each shot or string. The young shooter, especially, has trouble here because he hasn't yet found out exactly how to grip the weapon or how hard he should grip it to get the best results. This can only come through time and practice. To get a good grip, first watch closely how you do it each time. This educates the mind to see as well as feel and it comes easier to duplicate one's efforts each successive time. I use the following method to get a grip on the gun:

1. Pick up the gun by the barrel or slide.
2. Place the stock between the thumb and forefinger of the right hand and push the gun firmly to the heel of the hand, being careful to watch how the gun seats.
3. While pushing the gun into the hand, wrap the fingers firmly around the grip
4. All the fingers are used to grip the gun and equal pressure by all the fingers is necessary to control the weapon. No excessive pressure should be exerted on the side of the gun by the thumb.
5. The greatest pressure should be between the heel of the hand and that portion of the fingers on the front of the stock.
6. While gripping very firm, the trigger finger must be able to work back and forth enough to give it an independent action. This can be accomplished through dry firing and practicing taking the grip on the weapon.
7. This firm grip should be maintained at the same pressure while squeezing the trigger to fire one shot or an entire string of shots. A common mistake made is to not

> get the grip in the same way, causing the gun to feel differently in the hand and thereby causing the individual to hesitate to squeeze the trigger. A frequent mistake made is to gradually release the grip while squeezing the trigger especially during a string of sustained fire and will result in a definite loss of control. To grip the weapon with a very firm grip and with the same pressure throughout the delivery of a shot or string of shots will reduce the chances of making this mistake.

Breathing: It is not often that the average individual would have to think about breathing. The body functions are such that the lungs take care of the body without conscious thought. However, where the breath must be held, an individual must plan ahead for the period that the breath will be held. The brain must have sufficient oxygen to function properly. To illustrate: a deep sea diver who does not have sufficient oxygen loses his coordination and then begins to have difficulty seeing. The shooter is frequently holding his breath for periods of 20 seconds and more. If there is not sufficient oxygen in the blood stream, the eyes are not clear and the lungs want to take a breath causing movement in the body. To illustrate this point, without taking a deep breath try to hold your breath for 20 seconds. You will find that you become very anxious for the 20 seconds to be over with. You will also gasp for breath as soon as the time is up, if not sooner. The purpose of this is to bring to your mind the importance of taking a few deep breaths in order to store up the greatest amount of oxygen possible in your body. You will also find that deep breathing tends to relax the body and has a calming effect on the nervous system. This is reason enough to point out the importance of the shooter practicing deep breathing just before a shot or string of sustained fire. I would like to point out that to deep breathe is not natural to all individuals, so it becomes necessary to practice it until it is a habit. The time for deep breathing is immediately before each slow fire shot and immediately before every string of sustained fire.

Learning to Settle the Arm: Although it is possible to get good results in shooting by expecting movement in the arm, it is possible to get better results if the arm and gun are almost motionless It is possible for most individuals to hold the arm and gun almost motionless for a short period of time just after the arm has settled. By settling I mean the arm and gun stopping at a certain area on the target and becoming as still as it is possible for the individual to hold. To get the stillness of arm and gun that I refer to takes a lot of practice, concentrated effort, and a great deal of thought on the subject.

PERFECTING TECHNIQUE

Application of the Fundamentals

By Army Marksmanship Unit

Source: http://bullseyepistol.com/armymark.htm

A. Slow Fire Technique:

Because most of the points lost in any aggregate are lost at slow fire it is imperative that this become your strongest stage. Accept shot values in your aiming area. Focus on the front sight and squeeze all the way through until the shot breaks.

Shot Sequence:

1. Settle Into Your Aiming Area. The normal area of movement is readily apparent. If it happens to be larger or have abnormal characteristics and you are unable to reduce it to normal, accept it and proceed to fire. However, make every effort to hold the weapon motionless.
2. Find Sight Alignment. Your sight alignment must be exact and in such distinct focus that the bullseye becomes a blurred gray mass somewhere beyond the front sight.
3. Start Positive Squeeze. Be determined that once started, a positive, constant rate of squeeze is to be completed without interruption. Any hesitation, change of rate, doubt about results or loss of concentration must be avoided.
4. Concentrate on Sight Alignment. Any distraction warrants benching the weapon and starting over. Do not try to fire the shot if any controllable irregularity disturbs the existence of ideal conditions. Do not think of impending results at the target Any surprise shot within the aiming area with good sight alignment will be a good shot.
5. Try for a Surprise Break of the Shot. Your reflexes cannot act quickly enough to disturb sight alignment or a smooth, positive squeeze if the shot breaks as a surprise. Shots breaking in the aiming area with good sight alignment will form a group that represents the equivalent of your holding ability.
6. Additional Suggestions. It may be advantageous to rest or relax after three or four shots. Remember that you do not have to shoot before bringing your gun down to rest. When a shooter fatigues, runs short of breath or experiences difficulty in concentration, by all means he should lower his arm, relax, breathe deeply and try again. Some excellent slow fire shooters try two or three times before getting a shot fired. Do not expect ever to have a perfect sight picture. You can shoot groups only within your ability to hold. If you can hold within the ten ring, they should go there. With experience and practice your ability to hold will increase and your groups will

consequently grow smaller. Sometime during the 6- 15 seconds required to fire a shot in slow fire, your arc of movement will be sustained at a minimum. The shot sequence should progress at such a rate as to attain a surprise break during this period. Example: If a shooter's minimum arc of movement is reached at about nine seconds he should train himself to break his shots in 8-10 seconds.

B. Timed and Rapid Fire Techniques:

Prior to firing any string of timed or rapid fire it is imperative that you mentally run through the sequence of the string itself. You must be psychologically as well as physically ready to fire at 25 yards. You will find that this prior determination will assist in your rhythm, squeeze, and recovery and ease any match pressure that you are subjected to at the time.

1. Timed Fire. Prepare your lungs by breathing deeply prior to firing and holding the breath with lungs approximately half full just as you align your sights. Make rhythm the prime object. Never vary your rhythm. Adjust your recovery so that you'll have your sights aligned in time for the shot to go, but do not wait for the perfect sight picture. If you maintain your rhythm and fail to get the sight alignment just right, you may get nines, but if you make the weapon fire just as the sight picture is perfect, you will get the axe.
2. Rapid Fire. Rapid fire is essentially the same as timed fire. You can improve your rapid fire by learning to fire the first shot within the first second after the target faces you. Immediate recovery of sight alignment and hold after each shot depends on perfection of uniform position and grip. Special attention to an uninterrupted, unchanging rate of squeeze will help develop for you, the coordinated start your squeeze before the sights become perfectly aligned. Make every effort to prevent extraneous thoughts, which may disturb rhythm and concentration.
3. Shot Sequence for Timed and Rapid Fire.
    1. Find aiming area on edge of the target frame in line with your aiming area.
    2. Settle into aiming area.
    3. Find sight alignment. The front sight should settle naturally into alignment with the rear sight, vertically and horizontally. Concentrate on looking at the front sight.
    4. Start positive squeeze on turn of target. If your position and grip are correct and you are concentrating on alignment of the sights a squeeze started at the turn of the target will give you a 10 every time.
    5. Squeeze continuously. In rapid fire you must start your squeeze before you have perfect sight alignment. This does not mean that you subordinate

sight alignment to squeeze. Start your squeeze as quickly as possible while continuously concentrating on and perfecting sight alignment. Sight alignment not a sight picture. Your natural aversion to not firing without correct sight alignment will delay your squeeze until the sights are aligned.

6. Recovery. Correct recovery with the sights approximately aligned in the aiming area is obtained only if your position and grip are correct.
7. Concentrate on sight alignment. Any time the shot breaks with good sight alignment, it will strike the target within your ability to hold.

C. Shooting on Windy Days:

Wind shooting is conducive to jerking the trigger. This is true because as the arc of movement increases the shooter develops a tendency to relax his trigger pressure. He is waiting for a more stable sight picture. His concentration of sight alignment will diminish and he will make an effort to set the shot off on the move as the sights pass the vicinity of the target center. The obvious answer is to, first: wait for a lull in the wind, next: concentrate as one normally does on sight alignment and as a minimum of movement is sensed, start an ever increasing pressure on the trigger until the shot is fired. Do not continue the hold during extreme gusts. Always take advantage of a chance to rest. Each attempt to fire a shot should be made with a firm resolve to align the sights.

PERFECTING TECHNIQUE

Goal Setting: Training for that Big Match

By GySgt Mario A. Lozoya

Source: https://www.bullseyepistol.com/zoyola.htm

The shooting season is upon us. Shooters from all over the country who are planning to compete in a major local event, state event, or even at Camp Perry this year are starting to, or have already started, to train for the big event. Most of us have either just thought about it (started to train mentally), or have actually developed a plan and began to implement it in some form or fashion. Goals have been set, refined, established, or just pondered. Goal setting is a difficult process and seldom implemented correctly in the training process.

We have all heard the different goals set by either ourselves or our friends: "I want to break 2600 this year." "I am going to get a gold medal in the Leg Match at Camp Perry this year." "I am going to win my category/class this year." "I am going to shoot a better score this year than I did last year." "I am going to make the President's 100 list." "I am going to shoot my personal high score." Here is a key thought to remember and stick with: If your goal is a number, or it can be affected by another person, then it is not a good goal. Back in 1991, GySgt Andy Moody, the Center Fire National Match Course record holder with a 300-19x, told me a story. He told me that all year, he had been working on his goal for Camp Perry that year. He told me that he was going to shoot a 290 or better during the National Trophy Individual (NTI) match. You see, normally, a 290 would almost guarantee you a win during the NTI. At least that was the case for the last few years. His hidden agenda in all of this was that if he accomplished his goal, then he would be the NTI match winner. He trained all year with intermediate goals within the strings of fire every time he got up to the line to shoot. He had MINIMUM scores to shoot at every yard line. 95 or better during the slow fire stage, 98 or better during the timed fire stage, and 97 or better during the rapid fire stage. In his mind, he had it licked! Guess what he shot at Camp Perry that year? Correct!....he shot a 290. He met his goal. He did not, however, accomplish what he really wanted to do. He was not the match winner. Someone shot a 291. I often wonder what goals the other shooter had. In this case, we had a National record holder actually limiting himself to a particular score. Someone with his abilities should have just relied on his capabilities and focused on the immediate task at hand: to shoot each shot or stage of fire to the best of his mental and physical ability. That would be, "almost flawless." At that time, he was one of the best shooters on the line at any given event! Why did he limit himself and allow his self-image not to exceed those numbers? His lesson to me was: "Do not make a certain score or number as your goal if you want to be the match winner. Why? Because that might not be the score required to win that day. The best performance that day wins." Performance: that was my lesson learned. How do I alter my goals and training in a manner which will allow my best performance, at whatever time I want

to produce it? A good or flawless performance is obviously one with the least amount of errors or none at all. To do this, I needed to reinforce an almost flawless routine. What is a flawless routine? Can I identify that? Can I assess that? Can I implement that? Here is where Goal Setting begins! Everytime a shooter comes to the line to shoot, he or she should have some sort of goal. If it is at training or practice, what are the goals for that particular session? These would be your daily goals. Perhaps to refine your mental plan, to fix a common error, to simply shoot 10 good shots, or even to keep all the shots in the repair center. No matter what, never shoot without a purpose. I have heard people say that to get better, one has to "bury themselves knee-deep in brass." You can shoot all day without a purpose, and all you have accomplished was to send lead down range. A medium or intermediate goal can be to master to 25-yard line, to have established a good routine, or perhaps to shoot well enough to make your local or state team. All of this would assist you in acquiring your ultimate or long-termgoal: to get that High Master card from the NRA (just an example). Always assess every training or practice session at its conclusion. What did I learn today? What did I do wrong? What am I going to do with what I learned? What am I going to do to fix what I did wrong today? Why am I doing this wrong? If there were good points about your training sessions, write them down and decide how you are going to incorporate them into your routine. Ask yourself how you will continue to do those good things. Examples of good goals are: 1. I will eliminate anticipating any shots today. 2. I am going to pay close attention to trigger control today. Pressing the trigger to the rear while maintaining the best sight alignment possible. (eliminate jerking the trigger) 3. I am going to stick to my routine (mental plan/program) for the entire day. 4. I will put on the best performance I ever have. 5. I will identify, assess, and correct my errors. 6. I will successfully implement a solid routine in my shot process. 7. I will learn to be aware enough during a shot and only shoot shots that I am prepared to shoot. 8. While in the shot process, I will stop if I veer from my routine and start again. Try to set goals that are positive in nature. Make a serious attempt to eliminate any negatives in your goals such as; "I will NOT shoot any late shots." This will only improve your chances of doing just that. Also, set goals that cannot be affected by anyone else, and see if they are realistic or not. I know I could not set a goal to win the next Olympics, or set National Records, if I just started shooting yesterday. Through practice, I will learn to assess and readjust my goals. Does practice make perfect? What if I don't know if I am doing something right? What if I can't really identify, assess, and correct my errors? If I continue to train in this "lost" state, is my flawed practice making me or my routine "perfect"? The answer is obvious: NO. Perfect practice makes perfect. That is a series of goals in itself. Identifying errors, assessing errors, correcting errors, and finally implementing and refining a new routine. That is crucial! This applies to both the mental and physical aspects of your shot process. One has to work on making his or her sessions "perfect." What you do to accomplish this should be your short-term goals. Intermediate goals are designed when you have identified an error and you work to fix it. An Ultimate goal should be one that truly measures your overall performance.

Perhaps an errorless performance would be a good Ultimate goal for you. To achieve that certain performance at the time that you need it most is a goal not easily achieved. If I don't have a clear understanding and mastery of the fundamentals, I have no business setting goals other than those types of goals that will allow my self-image to believe that I have mastered the fundamentals. Then we can proceed with a higher level. Many shooters go straight into studying Mental Management and some even the art of Zen. To me, this is crazy since they have not harnessed the application of the fundamentals in their shot process. It is like expecting a driver that just got his or her driver's license to outperform a NASCAR Professional at the Daytona 500. These are two complete ends of the spectrum. A systematic approach to success has to be developed. Success is relative. To me, to have an almost errorless day at the range is success. Can you imagine your score at the end of the match if you completed the match without errors? By that, I mean that you did not jerk the trigger once, you didn't hang up on any first shot during rapid fire, you didn't shoot a shot that you didn't want to shoot, or you never veered off of your mental plan or mental program (your routine). Establishing goals such as these and meeting them, will equate to a good performance and eventually achieving levels that perhaps even you didn't think you could achieve. In my opinion, Lanny Bassham has written the best material to understand this process. In a book he wrote, titled "With Winning in Mind," he describes the "Triad State" as the state in which your sub-conscience mind, conscience mind, and self-image are balanced. He goes on to describing how one can acquire this balance through perfect practice and mental program (routine) development. Recently, I talked to some competitors at the Marine Corps Western Division matches about this very subject. This particular event is a three-week event. Classes are given during the first few days, then practice at the pistol and rifle ranges, and 4 days in match conditions. I noticed that a number of shooters kept making notes of the scores they shot every day. When I asked some of them as to why they were doing it, I got different answers back: "I want to see my progress from day to day." "I want to find what my average score is." "I want to see in what stage of fire I'm weakest in." "I want to see if I'm shooting the scores required to get a medal here." I respond to each of these with: "Progress is not measured with scores but through performance." "Have you identified the errors that you make each day and averaged that out in an effort to correct them and ultimately improve your average score that way?" "Are errors more prominent in a specific stage of fire? Do your errors differ from one stage to the next?" "How do you know what is going to take to get a medal? What if there will be adverse weather conditions that day? What if the weather was really bad the years prior and recorded bad scores. What you need to identify, is what sort of performance is going to get you in the medal bracket." The discussion continued in form of explanations to my answers. What if the wind was blowing really hard, and some rain added to the variables around me? What if I had an Olympic level performance as far as my mental state, state of awareness, effort, etc.? If the conditions were different, would my score be the same? I don't think that would be likely. I can put on the same type of effort from one day to

the next, but that does not mean that the scores are going to be equal. There are so many variables and conditions to take into account to worry about a score. The key, again, is consistency in performance. Make a serious attempt to eliminate the errors. Stick to your task at hand. That should be to meet your immediate goals. Stick to your routine! A performance such as that will take care of the score for you. Good luck and keep them in the middle. Gunnery Sergeant, United States Marine Corps GySgt Lozoya first started his competitive shooting in the Marine Corps Western Division Matches in 1989. He was shooting for the Marine Aircraft Group 13 shooting team out of Yuma, Arizona. He did not place with either the rifle or pistol during that event. The following year (1990), at the same event, he was the Western Division Pistol Champion and earned his way to the Marine Corps Championships in Camp Lejeune, North Carolina. He earned a bronze medal at that event and was selected to augment the Marine Corps Pistol Team as a member of the Summer Team for that year. The following month, he earned another bronze medal at the Interservice Championships held in Camp Robinson, Little Rock, Arkansas. At the 1990 National Championships in Camp Perry, Ohio, he was third overall during the National Trophy Individual Match, earning a gold medal and a Pistol Distinguished badge. He returned to Yuma, Arizona to assume his regular duties as a Computer Operator. Later that same year, he received orders to Weapons Training Battalion, Quantico, Virginia and assigned to the Marine Corps Pistol Team. During Desert Storm/Desert Shield, he was a combat weapons instructor in Stone Bay, Camp Lejeune, North Carolina. In 1991, then Sgt Lozoya, was back in Quantico with the shooting team. This would be his first year as a permanent member of the Marine Corps Pistol Team. The Highest achievement that year was winning the Regular Service National Championship in Camp Perry, Ohio. In 1992, Sgt Lozoya was the Virginia State Champion and match winner at the West Virginia State Championship. At Camp Perry, Ohio that year, he set a match record during the .45 caliber pistol match with a 891-57Xs, was the Regular Service Champion, and National Champion. He was also the match winner during the Presidents Match for which he received a congratulatory letter from President George Bush. He was meritoriously promoted to Staff Sergeant for winning the National Championship. In 1993, SSgt Lozoya won the National Trophy Individual Match in Camp Perry, Ohio. In 1994, SSgt Lozoya was again the Virginia State Champion. During the Marine Corps Eastern Division Matches, he set a Division Match record with the M-9 shooting a 581 aggregate (that record still stands today). He then won the Marine Corps Championships earning the coveted Walsh Trophy badge. In 1995, SSgt Lozoya was the only Marine in the U.S. military team, which participated in the First Military World Games in Rome, Italy. Currently, GySgt Lozoya, is stationed in Marine Corps Base Camp Pendleton, California and assigned as a Legal Officer. He is distinguished with both the rifle and the pistol.

PERFECTING TECHNIQUE

Method of Aiming the Red Dot Sight

By Brian Zins

Source: https://www.bullseyepistol.com/zinsdot.htm

Before I get into MY theories on looking at the target or the DOT (not iron sights) I have a couple of things that I need to throw out there. First, I am a very jovial kinda guy and take very little personal or too serious. Second, The comments submitted by those who quoted The Great Bill Blankenship and the Late Don Nygord - They are legends in the game and at no point would I argue to say that they are wrong. I submit to you another way of accomplishing the same goal of shooting better in another way. I learned to shoot under the guidelines that Nygord and Blankenship wrote about. I have had the opportunity to shoot with and learn from some of the greatest marksmanship instructors the Marine Corps has had to offer. We were allowed to think out of the box and try new ideas. I only stake a small bit of claim to the ideas that I propose as they were a collaboration of Andy Moody, Mitch Reed, Mario Lozoya, Chris Hill and myself.

Basically we were trying to get Marines to 2600 as fast as possible because unlike the Army we Marines do not get the luxury of staying on the Team for life. We came up with ideas to get Marines better, faster. Last, before I get into the Dot conversation, I have found that many shooters think entirely too much. This is not rocket science, as if it I were I would not be a shooter. Hence, I am not smart enough to shoot bad. Erich Bujung, when he was the Olympic Pistol Team coach always said, "No stinking thinking". The mind is a terrible thing to waste and in Bullseye on the firing line, the mind is just a terrible thing.

First to address a couple of questions. I use a 1" Ultra Dot, not the match four dot, though it is an awesome scope and recommend it for anyone with failing vision. On a bright day outdoors my dot level is on about 8 in order to have small dot. With a dot I am not a big advocate of the Iris thingy on my shooting glasses. With iron sights, thats okay.

At some point or another we have thrown a ball to someone right? Think about, when you threw the ball what were you looking at? The Ball or the person or object you were throwing at. Iron sights involved 3 items. Front sight, rear sight and the target or aiming area. With the dot you have but 2 items in play. The DOT and the target. (Before I forget, I do understand the difference between sight alignment and sight picture, but thank you for the reminder.) When I first started shooting Andy Moody and I were talking about looking at the target vs. looking at the sight. I was young and knew no better so I tried it. Keep in mind this was my first year as a bullseye shooter. I proceeded to shoot my first ever 100 long-line with my .22. Heck it was my first 100 long-line period. The Team Captain called everyone into circle and

asked me tell them what I did that string and if I learned something. I told him I turned my dot down and looked at the target as opposed to the front sight. Oh My God. He about had a fit, but learning had occurred and not just for me. From that moment on it became not so uncommon for Marines to look at the target.

Here lies the problem. Many shooters that try this and do not get the desired results are not truly looking at the target. Keep in mind, to shoot a dot in Bullseye and do it well, you have to either focus on the target or on the DOT. Too many people are focusing somewhere between the DOT and the target. This is equivalent to looking at the target with iron sights. Not gonna work. Be true to yourself when determining where your focus is.

FOR ME, I find that when looking at the target I accept the wobble of the dot more and do not get hung up on the trigger. NOTE: if you have a big wobble area this is harder to acheive. There are days that my dot may not sit as still as I like and on said days I will bring my focus back to the dot. So if you have a decent hold you should be able to do this with a good deal of success. If you a big wobble, the true method of looking at the dot is great. Remember that is how I learned to shoot.

Does the DOT need to stay centered in the scope. The answer to that is a great big MAYBE. First of all we humans like to see things geometrically correct (I don't even know if that makes sense) What I am trying to say is We like to see a circle in a circle in a circle. If we had change sitting on our desk at some point or another we are going to stack it in the following order quarter, nickel, penny and then dime. If you stack another way that just means you are little more creative than the average bear and good on ya. So it makes sense to us to put the dot in center of the scope. Some dots you have to put in the center or you will get some parallax and the dot is really not where you think it is. Andy Moody shot with the dot in the bottom of his scope. Because that is where it rested when he brought the gun to the target. Why fight it, why MUSCLE the gun to place you want it to be as opposed to allowing the gun to rest where it wants to be? Make sense?

Please keep in mind that when it comes to shooting pistols, that I have had the opportunity to shoot with and learn from the likes of the earlier mentioned Marines and Steve Reiter, Doc Young, Jim Henderson, Jimmy Dorsey and have learned from all of them. I would like to think that I have an original idea to make people shoot better, but I am pretty sure that all I can do is relate what I have learned and maybe shed a different light on it. However I would like to think that if I wrote a book, someone would still buy it. If not for the informational value, at least from the entertainment.

I hope that this helps.

Brian

PERFECTING TECHNIQUE

Advanced Beginner Questions & Answers

By Jim Henderson

Source: http://bullseyepistol.com/henderson.htm

Jim Henderson is one of the very best shooters to come along in the past 20 years. Setting records at the National Matches and the Interservice Matches, he is a two-time National Pistol Champion (2006 and 2009) and will surely win more times in the future. I have had the pleasure of shooting next to Jim on several occasions at small matches and have enjoyed his sense of humor and generosity with sharing his experience.

Here are some of the questions Ive fielded over the years from aspiring shooters:

- *Slow fire: visualize the perfect shot EACH time, before firing each shot?*
  Yes, I think a good visualization program is key. But remember, perfection is a hard thing to see. So keep your thoughts obtainable, but ever striving forward.
- *Engage in conversation between targets or stay intently focused?*
  That to me is a match to match thing. Sometimes I can talk and cut up, sometimes I need to step back and keep my train of thought. This is something you will have to experiment with.
- *Scope a shot you called "out" (bad) just to confirm?*
  Nope. Why let that negative thought in to your game plan? If it didn't feel right why look. Concentrate on the next shot. You can call them anything but back. When it's gone it's gone.
- *Scope at all on the short line?*
  I use to, just for the first string. To reaffirm my zero. After I knew it was good I put the scope away. Now since I don't have a scope mount on my pelican box I hardly use my scope at all.
- *Intently think about breaking the shot or relegate to subconscious? (thinking about trigger pull)*
  This I think is an experience thing also. If you are just starting out your subconscious doesn't know what to do. If you are farther down the road, a little less attention to the finer things may be OK. Your focus is just on the big picture or the outcome of the action. Kind of like being in upper middle management. I generally just concentrate on relaxing now. Just because I have complete faith in my plan and my ability to stick to it.
- *Is it really worth visualizing your entire match beforehand, or is mental training like other training: getting "pieces" mastered to make up the whole?*

Well, I don't know about you but I have a hard time remembering where I left my car keys, never mind 270 individual items at one time. Try this; one slow fire shot, fired to the best of your ability. You have to be comfortable with yourself before you can make yourself comfortable. Then strive for the next level. One point at a time, one shot at a time. It's like learning as a child all over again. Crawl, walk, run. Then run like the wind.

- *Why is it such a battle to beat my average?*
  If you find that when you don't stick to your plan that you shoot right where you always do, I have a thought for you. Self-sabotage. Lets look at it for a minute before you start calling me names. Look at it this way. "Oh man I'm shooting great, I'm right on track to shoot my best score ever. Hey wait a minute, am I capable of this? (scratch 9) Can I shoot as good as I'm doing? (solid 8) I've never shot this good before, (slider 7) can I shoot as good as those Big Guns? (slammed in on paper miss)." And you know the funniest thing about all these doubts running through your mind. They flash by in the blink of an eye. They happen even before you know your defeating yourself. And I'll tell you something else. Sometimes you won't even know you did it until youre talking to someone after the match. Listen to yourself and you will hear your own best critic.

Jim Henderson

PERFECTING TECHNIQUE

Strength and Stamina of the Pistol Shooter

By John Dreyer

Source: http://www.bullseyepistol.com/stamina.htm

"Fatigue makes cowards of us all." - Vince Lombardi

The Importance of Physical Strength

I attribute the greatest rise in my personal scores to the increased physical strength from a weight training program. If you are weak, you will soon find that the exersion of holding the pistol toward the target will cause fatigue and uncontrollable shaking. In other words, greater strength reduces your "arc of movement."

The objective of physical training for a pistol shooter is to condition you physically to better withstand the rigors of match conditions. An individual in good physical condition has better developed reactions, better control of their muscles and better endurance; all of which promote consistency in performance.

Developing a Physical Training Program

Physical Conditioning must consist of exercises of a general nature directed toward strengthening the muscles, proper breathing, developing body flexibility and precision of movement. The requirements of marksmanship are such that drills must consist of exercises which develop the muscles and flexor of the arms and fingers, and the muscles of the shoulders and even the waist. A certain amount of static tension (dynamic) type exercise is valuable if it is not overdone. Whenever a shooter exercises, he or she must put the maximum effort into the exercise. Merely going through the motions of an exercise is of no advantage.

Recognize that physical conditioning is a gradual process and results will not be apparent immediately. As your physical condition improves, the number of repetitions may be gradually increased. Be advised that very heavy exercise like serious barbell weight lifting is discouraged as it can create injuries that will hinder your performance more than it will help. I would suggest a program that consists of daily walking and calesthenics accompanied by every-other-day light dumbell exercises as most ideal for our purposes.

The Muscle Groups that Matter

While you might benefit from a physical training program that is general in nature and targets all the muscles of the body, a more limited program may suffice. While the legs and back

muscles play a role in steadying the stance, you can see from the diagram below where the greatest muscular demand on a shooter lies:



Try to visualize turning your entire arm into an "iron rod" that begins at the shoulder socket and ends at the middle finger. The "rod" is securely held in place by isometrics or contracted opposing muscles. The opposing muscles of the shoulder, tricep, bicep and forearm are at work here; it is not accomplished by "locking the joint" of the elbow. You must recognize that unless your triceps, biceps and forearms are quite strong, you can NEVER visualize or experience this feeling. That is why a weight training program is in order for any pistol shooter.

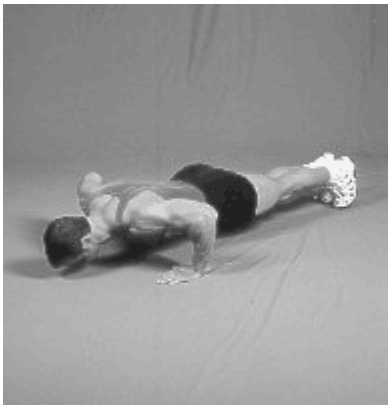
Targeted Exercise Program

A very effective program may only involve a remarkably small investment of time on your part. For example, the training program that I utilize includes two general-fitness exercises that are performed every day, and four strength-building exercises that are performed every other day. Regardless of what program you put together, you must remember four things:

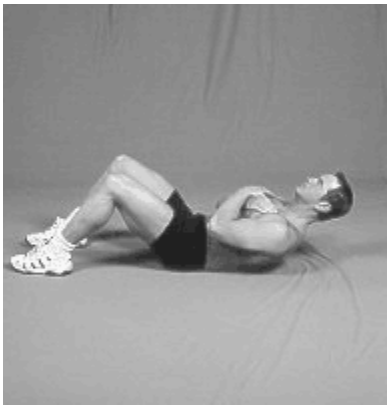
1. Skip any strength-building exercises the day before and the actual day of any match, resuming the program immediately after the match.
2. Avoid injuring tendons or ligaments by starting moderately and progressively increasing repetitions first, and then increasing weight.
3. Increasing the strength of the triceps should be the highest priority of any exercise program.
4. Balance the strength building exercises on both the left and right sides of the body, not just the shooting arm side!

The Daily General Fitness Calesthenics

Targeted at general fitness and stamina, these familar calesthenics should be performed every day. Two sets of as many repetitions as you can handle should suffice.



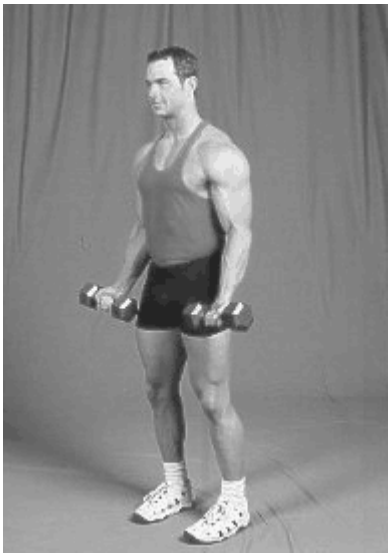
Push-ups



Crunches (Sit-ups)

The Every-Other-Day Strength-Builders

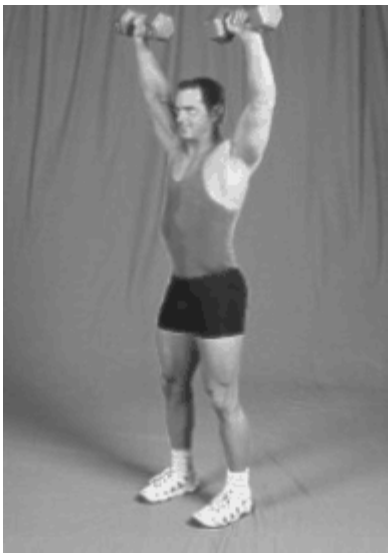
Performed every-other-day to allow for adequate recovery of the stressed muscles that you are developing, you are cautioned not to "overdo it" with these. Two or three sets with "moderate" weight of as many repetitions that you can handle should suffice..



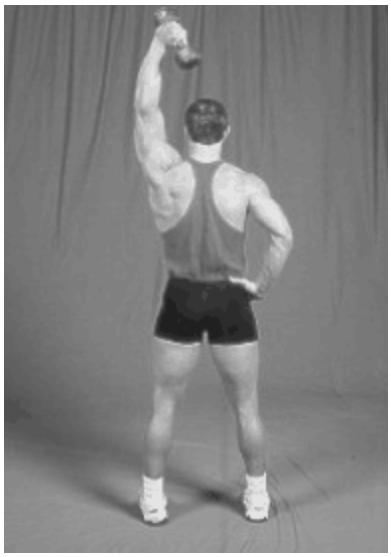
Biceps Curls



Forearm Curls (Both Sides)



Shoulder Presses



Triceps (French) Curls

Establishing Muscle Memory

Much of the goal of training for pistol marksmanship is to program your "auto-pilot." It is a fact that the muscles themselves have "memory" - a familiarity with the "feeling" of the processes that occur automatically with a learned procedure. While the isotonic exercises described above are excellent for building muscle fiber, it is not entirely enough! You must continue to practice often and utilize the muscles in "real-life" scenarios so that the newly developed muscle fiber also acquires this necessary "memory" needed for effective and consistent results.

Ideally, you would be able to get to the range and fire a practice session every single day. However, if that is not a possibility, in conjunction with short dry-fire sessions, I suggest that you take a proper shooting stance and hold out a five-pound weight for as long as you can. This will condition and discipline the developing muscles in preparation for match conditions.

PERFECTING TECHNIQUE

Error Analysis and Correction

By Army Marksmanship Unit

Source: http://bullseyepistol.com/training.htm

Insanity: doing the same thing over and over again and expecting different results. - Albert Einstein

The Wheel of Misfortune

Every day of the shooter's life brings a new lesson. Identifying errors are crucial in order that these lessons be learned. The following chart can help pinpoint such basic flaws in a shooter's technique by analyzing group locations. As printed, it is for a right-handed shooter. (A left-hander's chart would be mirrored horizontally.)



Top Eleven Bad Habits of Shooters

1. Not Looking at the Sights. This quite frequently is listed as "looking at the target." A shooter may be focusing his eye on neither the sights nor the target, but since he does not see the target in clear focus he assumes he is looking at the sights. You must concentrate on sight alignment.
2. Holding Too Long. Any adverse conditions that interrupt a shooter's ability to "hold" will cause him to delay his squeeze, waiting for conditions to better. The disturbing factor about this is that you will do it unconsciously; therefore, you must continuously ask yourself, am I being too particular?
3. Improper Grip or Position. Suffice to say that you cannot fire a decent score with any gun at any range if you continually change your grip or position.
4. Jerk or Heel. The application of pressure either with the trigger finger alone or in case of the heel, pushing with the heel of the hand at the same time. Apply pressure to the trigger straight to the rear and wait for the shot to break.
5. Anticipation. Anticipation can cause muscular reflexes of an instant nature that so closely coincide with recoil that extreme difficulty is experienced in making an accurate call. Anticipation is also the sire to flinching.
6. Loss of Concentration. If the shooter fails in his determination to apply positive pressure on the trigger while concentrating on the front sight his prior determination needs renewal and he should rest and start over.
7. Anxiety. You work and work on a shot, meanwhile building up in your mind doubt about the shot being good. Finally you shoot just to get rid of that particular round so you may work on the others.
8. Vacillation (Plain Laziness). This is a mental fault more than a physical one, which results in your accepting minor imperfections in your performance which you could correct if you worked a little harder. The end result being you hope you get a good shot. Just like you hope you will get a gratis tax refund, and you will get one just about as frequently as you get the other.
9. Lack of Follow Through. Follow through is the subconscious attempt to keep everything just as it was at the time the shot broke. In other words you are continuing to fire the shot even after it is gone. Follow through is not to be confused with recovery. Merely recovering and holding on the target after the shot is no indication that you are following through.
10. Lack of Rhythm. Hesitancy on the first shot or any subsequent shot in timed or rapid fire. Develop a good rhythm and then have the fortitude to employ it every case. Frequently many shooters will have fine rhythm until the last shot of a string and then hesitate, doctoring up that last shot.

11. Match Pressure. If there are 200 competitors in a match, rest assured that there are 200 shooters suffering from match pressure. So what makes you think you are so different? If you are exerting all your mental energy toward executing the correct fundamentals rather than the arithmetic evaluation, your shooting match pressure will be what you feel when people congratulate you on a fine performance.

PERFECTING TECHNIQUE

Fundamentals of Pistol Marksmanship

By William Joyner

Source: http://www.bullseyepistol.com/joyner.htm

In preview, I would like to say that shooting excellent scores with a pistol requires no other elements than those described in the following sentence. **ALIGN THE SIGHTS PROPERLY ON THAT PART OF THE TARGET REQUIRED FOR YOUR GROUP TO CENTER IN THE BLACK AND CAUSE THE HAMMER TO FALL WITHOUT DISTURBING THAT ALIGNMENT.** All elements of pistol shooting such as position, grip, sight alignment, breath control, trigger control, physical condition, and psychology of shooting, when perfected, simply enable the shooter to perform the action described in the above key sentence.

Body Position or Stance

We are all constructed differently and have different natural positions. To find your natural position, face away from the target 45 degrees. Look at the target by turning your head and eyes only and raise the pistol to the eye, target, line. Close your eyes, raise your pistol and arm several feet and allow it to fall relaxed, and naturally to the horizontal. If it falls right down the center of the target, you have your natural position. If it falls to one side, shuffle on your feet, keeping the body axis from the feet to the shoulder the same, until the pistol is aligned on the target again. Several tries such as this one will readily show you how far to face away from the target. This test need only be made during one shooting session. At all following sessions start out with the position that you have decided is natural for you and stay with it. The feet should be spread apart about the width of your shoulders or a little more. I have noticed that I spread my feet farther apart than when I first began shooting. Others have told me that they do the same. However, if you spread your feet unnaturally at first, you will have to exert undue muscular effort to maintain balance. The object is to be well balanced and comfortable. The legs should be straight, but not stiff. Allow the knee joints to fall into a locked position, but still be relaxed. The thigh muscles should be relaxed. If you are tense anywhere, it is a sign of strain and will show up in your trigger control. The hips should be level and in an easy, natural position. Let your abdomen relax. We have a lot of fun admiring each other's "pots" during pistol matches, but no one ever attempts to hold it in. Allow the shoulders to hang naturally and relaxed. I prefer to place my free hand in my side pocket. Some shooters, especially those with long arms, can perform best by just letting their free arms and hand hang naturally at the side. The object is to entirely forget about it. It must be relaxed and

forgotten. An instructor can easily spat a student who is not relaxed by the attitude of the free arm. The head and neck should be in an easy natural position. The shooter must look at the target by turning his head and eyes slightly without moving from the neck down. The simplest way to do this is to face your entire body away from the target at the angle you have selected and then turn your head and eyes only to the target before raising your pistol to the firing position. While looking at the target from this natural position, raise your pistol until you can align the sights on the target. The important thing is to make your pistol arm fit the body position instead of ruining a good body position by craning the neck and shoulders trying to get behind the pistol. The body position must be selected first, then use the pistol arm only to bring the sights in line with the eye and target. The pistol arm should be extended directly toward the target. The wrist is locked without strain, (this requires practice), the elbow is locked also but with no sense of strain or tenseness. The gull and arm supported by the muscles on top of the shoulder, (the trapezius group). Try holding a ten or fifteen pound weight out in the firing position and feel the top of your shoulder where the arm joins and you will find the small hard muscles that support your gun arm. You should feel that the pistol is hanging from above, and not that you are pushing it up from below.

Breath Control

The object of breath control is to enable the shooter to hold his breath with a comfortable feeling long enough to fire one shot slow fire; 5 shots in 20 seconds timed fire; and 5 shots in 10 seconds rapid fire. I recommend taking several deep relaxing breaths immediately prior to extending the pistol, and as you extend it, take another breath and exhale until your lungs feel normal. Hold until you fire the required shots. If you have too much air in the lungs, you will feel the pressure and it will interfere with your ability to hold. If you completely empty the lungs your arm will begin to shake in about 5 seconds. You are likely to have more trouble in the timed fire stage than the others. In order to be comfortable for 20 seconds, you must time your breathing just right and prepare for the string beforehand by taking several deep breaths. Take a deeper than normal breath at the command "Ready on the right"; take another at "Ready on the left"; at the command "Ready on the Firing Line" extend your pistol and take another breath and exhale to the point of comfort just as the targets turn.

Physical Conditioning

Many shooters discount the element of proper physical conditioning. They think that so little effort is required to extend a two pound pistol and fire it that they need no exercise. I have spent many days at hard labor such as cross-tie loading, woodcutting, ditch-digging, football, etc., but I have never felt as much fatigue from those labors as I have from a full day at match

shooting. I realize that some of my fatigue is due to a certain amount of nervous tension, however, I have learned that when I am in top condition, I feel good even after two or three days of match shooting. The real payoff for good condition lies in the score. I know several shooters who have added fifty points or more to their Grand Aggregates by conditioning themselves with systematic weight lifting pro- grams prior to the matches. I recommend a mild weight-lifting program and some road work to put the shooter in a good general condition, then some special exercises for the shooting arm. These special exercises consist of dry firing with a weight weighing several times more than the pistol. A quart milk bottle full of water, or a six pound dumb-bell are some of the things I have used. Extend the weight just as you would a pistol and line it up on an object and try to hold it steady until your arm starts throbbing. Rest for a few minutes and repeat the exercise. 10 minutes of this each day that you do not shoot on the range will enable you to hold steadier and longer than before.

Trigger Control

I do not like to use the word "squeeze" in connection with trigger control. When we think of the action of squeezing, we usually close all four fingers and thumb together at the same time. This is definitely not proper trigger control. The pressure put on the trigger must come from the trigger finger only The gripping fingers and base of thumb do not move. Review the chapter on grip. Get the proper grip on your pistol and keep the pressure constant, align the sights on the target properly, then with the trigger finger only, exert a steady, constantly increasing pressure, straight to the rear, until the hammer falls. There is a slightly different method of trigger control that I recommend for master shooters only and even then with extreme caution. The difference is that while the sight picture is not perfect, the trigger pressure is maintained, but not increased. When the picture becomes good again, the pressure is continued. This method when used correctly, insures that all shots go off with a perfect sight picture. The danger in this method is the tendency to flinch. I have been successful in the timed and slow fire stages, but I revert to the constantly increased pressure method in rapid fire. I just don't have time to interrupt my pressure in the rapid fire stage. There is one very important element common to both trigger control methods: the shooter does not pick out a definite moment to fire the gun. He knows by the amount of pressure on the trigger about when the hammer will fall, but not the exact instant. If he does pick out one exact instant to make the hammer fall, he will invariably flinch.

Flinching

Flinching is the convulsive movement made just as the hammer falls that causes shots to miss the target, or strike anywhere from the 5 ring to the 8 ring. All shooters suffer from this

malady at one time or another. When Joe Benner gets an eight he has flinched because he would never put pressure on his trigger with his sights aligned in the eight ring, (windy shooting excepted). Your progress in the competitive field of target shooting depends largely on your ability to overcome flinching. I include all such movements as "Bucking," "Jerking" in the general term "Flinching." Here is exactly what happens: If you know the exact moment your pistol is going to fire, your subconscious mind orders you to brace your body against the recoil, and you do so, resulting in a flinch. The remedy is to never know the exact instant the hammer will fall. Even then your subconscious mind will make brace, but the reaction time between the explosion and your bracing will allow the bullet to leave the barrel without being misdirected by your flinch.

Psychology of Shooting

This is a serious problem to many shooters and to some degree a problem to all shooters. I'm referring to the building up of pressure inside the shooter that makes him shoot like a novice when he is capable of shooting 2600. It is sometimes called "Buck Fever" or "Monkey on my Back." It prevents the shooter from shooting in matches, the scores that he shoots in practice. The best cure for this feeling is self confidence. If you shoot 870 with your .22 in practice, walk up on the line with the feeling that you can shoot 870 and will. 870 probably won't put you in the first 5 places, but it is your normal score and you can always shoot it. Sometimes you get hot and shoot 880. Don't keep such an accurate count of your scores that you end up in the National Match Course knowing that if you shoot 295, you will set a new record. Just shoot your matches as they come, record your score, and forget about them. Absolutely don't count your competitor's score to the point that you know exactly how much you need to beat them. Sometimes a shooter shoots 5 or 6 consecutive tens in the slow fire string. It is awful hard to stay with it. My advice is to spot your shots until you are sure that your sights are set right and then finish your string without spotting any more. The match shooter has a complicated problem. He wants to win and when he sees a chance to win because of some good strings, his breath quickens, and his heart beats so fast that he can feel it in his trigger finger. As a result he usually blows a five shot string and then for the rest of the match shoots normally. If we could just go to a match and be satisfied with our practice score; refrain from counting up our aggregates as we go; refuse to speculate on how much it will take to win; refrain from comparing competitor's scores, we would probably shoot much better. Here again experience strengthens our ability. The match shooter who has been to match after match and been disappointed time after time soon finds that it just doesn't seem so important to win. Then he begins to shoot his best scores in matches.

Suggestions to the Beginner

We will begin with equipment. I will not discuss equipment any further than to say that you must have complete confidence in your pistols and your ammunition. If you doubt either, you will blame equipment for your errors, and not correct them. Dry firing will develop and improve every element of shooting except recovery from recoil. It develops that machine-like precision in the timings of your timed and rapid fire stages. I suggest a fifteen minute session of dry-firing every day that you do not shoot on the range. Simulate your range conditions as much as possible. When you are troubled with flinching, use the roulette system, until you conquer the fault. By the roulette system I mean that you load all cylinders and spin the cylinder between each shot. This insures that you will soon be putting pressure on the trigger without knowing whether or not a live one is under the hammer. When the hammer falls and snaps, you will be able to see your flinch and soon eliminate it. You must do more than just shoot during your practice sessions. Call your shots slow fire and analyze your weaknesses. No amount of shooting will improve your score unless some thought and planning go along with the shooting. I shoot a complete aggregate, (900), with one caliber during each practice session. If you possibly can, practice on the range 3 times a week and dry fire at home all other days. Don't try to shoot too much during one practice session. One 900 aggregate is just about enough, especially with the .45. Keep an accurate record of your progress. If you fail to write down your scores, you will soon remember only the good ones. Always time yourself by some method or have someone time you during practice sessions. It is second nature to shoot your rapid in 12 seconds if you are not timed.

Slow Fire

Remember that you do not have to shoot before bringing your gun arm down to rest. When a shooter feels any fatigue or feels that he is running short of breath, by all means he should lower his arm, breathe deeply and try again, after relaxing. Some excellent slow fire shooters try two or three times before getting a shot off. Don't insist on having the perfect sight picture before applying pressure to the trigger. You can shoot groups only within pour ability to hold. If you can hold within the ten ring, then should go there, but if you are like most of us, even after years of shooting; you are satisfied to hold within the nine ring and get your tens from the law of averages, and cuss your eight's.

Timed Fire

Prepare your lungs by breathing deeply prior to firing and holding it just as you align your

sights. Make rhythm, (interval between shots), the prime object. Never vary your rhythm. Adjust your recovery so that you have your sight picture in time for the next shot to go, but do not wait for perfect sight picture. If you maintain your rhythm and fail to get perfect sight picture, you'll get nines. If you make the gun fire just as the sight picture is perfect, you will get misses.

Rapid Fire

Rhythm is of prime importance. Rhythm is important because you develop rhythm only by putting a uniform pressure on the trigger after each recovery. Your can improve you rapid fire by learning to fire the first shot within one second after the target turns to you.

Conclusion

The theory of shooting is simple: You create a machine rest with your stance, grip and breath control. Then with the gun in the machine rest, you apply pressure directly to the rear until the hammer falls. In practice we sometimes find our machine rest wobbly because it has a brain and can count scores and anticipate wins. Through experience and practice you must make the brain machine-like also.

PERFECTING TECHNIQUE

Mastering the Rapid Fire Stage

By David Cartes

Source: http://bullseyepistol.com/rapdfire.htm

When I asked people in the past for a personal account of rapid fire, they have usually said, "Just shoot timed fire twice as fast." I think that this is what hinders most new and old shooters alike. True, you do shoot it twice as fast, but unless there are some alterations in stance, squeeze, recovery, etc., there tends to be a great deal less effectiveness. In other words, a definite drop off in score. When I first started to shoot, I often heard the older shooters say, "It's rapid fire that divides the men from the boys." It is not the rapid fire itself that makes the division, but rather a lack of belief in oneself that he is capable. I believe that all shooters are capable of rapid fire scores comparable with their timed fire and slow if they will concentrate on the fine techniques during practice and matches.
To get a closer look on just what I have been doing during rapid fire strings, I went to the range and fired a few strings with each gun. The things that I noted are many but the main ones are as follows:

- Number One: There was a deliberate thought process prior to each string. While I was loading my magazine, I concentrated on just what I was going to do as soon as the commands started. I will call this "organization of the graymatter."
- Number Two: I was always certain to make sure that my grip was exact. Equal pressure throughout strings.
- Number Three: I took quite a few deep breaths prior to each string to ensure my lungs of needed oxygen.
- Number Four: During firing I always placed importance on sight alignment.
- Number Five: I fought off panic. With these things well fixed in my mind, I took the positive approach with myself and said, "Combine and control these, Cartes, and you can really set the world afire." I must say that this is true. Follow these rules and anyone probably could, but the key words are combine and control. Both easier said than done.

Now what do I do to assist me in an attempt to combine and control. I think that knowing my equipment and ammunition are in tip-top shape relieves me of some worry. I If you have ever had to fire an alibi string of rapid fire, you know how costly this sort of thing tends to be. Also, I know that in the past I have been able to fire rapid fire with a certain amount of respectability. Knowing this, I worry over it as little as possible. That doesn't mean that I have little to fear about not doing it again. It simply means that I am capable. After this comes the

guts and determination that you find only if you look hard enough and then say to yourself, "I will not fall down because of stupidity. I will not allow myself to give way simply because I lost control for a split second. I must recover immediately from all distractions either mental or physical. I must be the dominant one for ten seconds and not my pistol."
It is a fairly simple matter to "throw a match" to take off pressure but you have accomplished very little when all is said and done and you know why you lost. I say this now because most matches are booted out the window during the rapid fire stage. It is not a common rule for humans to push themselves to distraction but with pistol shooters, it becomes a second nature. Again it is organization of the gray matter. The mental picture is an important one if not the most important. It is through complete control over yourself that you can perform not only as a respectable shooter, but even occasionally as a winner. How often as a winner, depends upon the individual. Physical aspects of shooting rapid fire vary with all shooters and with me, they vary with the wind. I can't say how many matches I have won in the wind, but I do know that my ability to shoot rapid fire during the wind at the National Mid-Winter pistol matches in 1958 was the deciding factor in wrapping up the championship. Though I did not win the 45 cal. rapid fire match (I was out'X'ed) I did manage to gain up to fifteen points on my two closest competitors. The important change I made that day in my shooting habits was during rapid fire. I turned almost face in to the targets, spread my legs like I was straddling a mud puddle, turned in my toes and leaned forward. This nay sound uncomfortable and it is. Not only uncomfortable, but completely off balance; but I can guarantee that it is the most wind resistant stance that I have found. What happens is that the body being off balance, there is less tendency to sway. Turning the toes slightly inward braces you from falling on your face. Also, leaning forward tends to cut down on the recoil. Since then, I have adopted this stance, with certain variations for weather, with all three weapons. Another habit I think a must to good rapid fire scores is last second concentration prior to the turning of the targets. This concentration must be placed on the front sight. We all know that the front sight must be aligned with the rear notch during the squeeze to insure a good shot. What most of us don't realize is that it takes about three-tenths of a second for a person with normal vision to accommodate their eye to focus on any given point. This time element is costly if when the targets turn, you are watching the target line instead of your gun. There is a tendency at times to let your eyes drift to the targets but fight it off and just keep watching your alignment. When the targets turn,, don't look down; just lay down on that trigger and break the shot. No jerk, but a firm movement to the rear of the trigger with the finger. This movement must be of an ever increasing pressure or else you will "freeze." Okay, the first shot has been broken and probably a good one. Regardless, don't start searching for it. Just keep your eyes focused where they should be and recover. Now on that recovery. Don't dip your front sight or else you will find yourself wasting time trying to get it back in the notch. Keep it high and recover quickly. Don't become unloosened. Keep your wits about yourself and fight all urges to jerk the next one. Also, don't start the squeeze till those sights are back in alignment. Keep your

eyes on the front sight and be as calm as possible. They're aligned and now in the area of the black. Squeeze hard. The shot breaks and an explosion. Don't look down range, just follow the same procedure as on the previous shot. Now just three more shots to go and you will have finished your first string. Don't lose faith in yourself. Hang on and follow the rules. You know you have fired a couple of really good shots and there is a strong tendency to look down and admire them. Don't be foolish, for if you do your next shots will look like a couple of satellites in orbit. This is where the determination comes in. This is where you must master the pistol. A little pressure is off now. You have finished the first string and you have a few minutes to check things over in your mind prior to the next five shots. Don't spend this time discussing history with your neighbor on the next point, but slowly load your magazine or cylinder and methodically remember what went right and what wrong. If all was right, wonderful. If not, carefully account to yourself how your rhythm continued or discontinued. What disturbed it? How will you correct it in the following string? Were the sights aligned and did you have the intestinal fortitude to squeeze instead of jerk? Did you lose control and can you remember where and why? Don't expect possibles but, in turn, don't get excited if one does show up on the target. You have to get used to good scores if you are to become a winner. If it turns out to be the opposite, a really bad one, ask yourself why. You must be able to find out why you are making mistakes before you can correct them. Remember this. Everyone in the match will make costly mistakes. Don't let a mistake get you down. Rapid fire never came easily to even the best of shooters and to some it will probably never come. One thing we have to face though, in a pistol aggregate, thirty-three and one-third percent of your score is comprised of rapid fire. This must not be treated lightly. Experience is only a good teacher if you are a good pupil!. There are too many shooters today who have been shooting for ten years but only have one year's experience ten times. All shooters commit ridiculous mistakes and all feel like the world has fallen down upon them at one time or another. It is the man who can snap back into form by immediately realizing what he was doing wrong and can make the necessary corrections, that improves and in time becomes a winner. It is not easy for a man in any profession or sport to become a champion overnight. Nor is it necessary for anyone who is physically qualified to stay at the same level year after year and not improve. Hard work, an open mind, and a desire to try new things are the points which put most people on top in any game.

PERFECTING TECHNIQUE

Attributes of a Champion

By Army Marksmanship Unit

Source: http://bullseyepistol.com/champion.htm

Qualities of a Great Team Shooter

The one popular fallacy that good pistol scores are engendered by stupidity has been proven completely erroneous. A quick look at the nation's outstanding pistol marksmen will show you individuals of higher than average level of intelligence. The necessity for intense concentration and strict adherence to a multitude of sometimes unnatural but correct shooting fundamentals, quickly eliminates those of slower intellect.
A good team shooter need have many attributes; however, if he lacks the ability for complete concentration and the intestinal fortitude to make up his mind to adhere to fundamentals regardless of match pressure, adverse weather conditions or any other conceivable distractions he is useless to you and the team.

1. Compatibility. A team member must so conduct himself so that his presence is enjoyed by his team mates, win or lose. Hot tempers, temperamental prima donna activities and arbitrariness will do little to improve shooting ability and will do a lot to alienate shooting companions and quite possibly have a detrimental effect upon the team score.
2. Sure-Footed and Careful. This is no game for the indecisive. Each step must be planned and deliberated. Decide what has to be done, then carefully and methodically do it.
3. Confidence. A shooter must have no doubt whatsoever about his ability or the accuracy of his guns.
4. Consistency. A team shooter must be consistent enough that his performance can be predicted within reasonable limits.
5. Tranquillity. Although a good shot must place all of his mental and physical ability toward shooting a good score, infrequently he will fail in this. Suffice to say that when this happens if he admonished himself severely, or falls into a fit of complete depression because of a poor score, he will hurt greatly his chances for the rest of the match. It is not intended that you laugh off or treat lightly a poor performance; however, you must possess the presence of mind to accept the bitter with the sweet.
6. Good Health. The eyes are important so they must be perfect or corrected to this condition. Good physical condition is imperative to give resiliency to the muscles and better nerve control. The day of the drinking, smoking, have-a-good-time pistol

champion is long past. To win today a shooter must refrain from any habits of either eating or living that will impede his ability to perform at his best.

7. Open Mind. The shooter who has placed himself in the mental state that he can accept no help, no coaching-, nor a frequent re-evaluation of his technique, can never In addition to those proven ill effects on general health and longevity, the following effects and reactions will definitely prevent the shooter from reaching his maximum potential. Do not attempt to fool yourself, or to justify your own inability to refrain from either (or both) of these bad habits by pointing out the "exceptions" who are apparently able to fire good scores in spite of their habits. Most good shooters do not smoke at all or drink before or during their shooting.
8. Sportsmanship. A poor sport has no place on any Army team. He is a representative of the US Army and a member of the team that produces more winners than any other. Shoot to win, but if you are not the winner you should be proud of a team mate who is. There is no honor in winner over a team mate who did not shoot his best scores, so don't play underhanded and try to create unfavorable conditions for him. Win your matches by shooting.

Physical Conditioning

To be effective, physical training for the pistol shooter must be realistic and continuous. The objective is to so condition the body that the general health is excellent and that the muscular and nervous systems are fully capable of withstanding the grind of match conditions and enable the shooter to fire his maximum scores.
Physical training should be progressive, either in repetitions performed or in the resistance used. Conditioning must remain short of the fine drawn conditions sought by track athletes, as this is generally considered detrimental to good pistol shooting. Violent and strenuous athletics which may result in injuries should be avoided. A series of non-strenuous exercise of the type that require body bending and stretching, deep breathing and moderate muscular tension are best suited toward obtaining a condition defined as body tone and a feeling of well being. When you exercise, go at it with enthusiasm; if you are going to spend the time, get the results your time deserves. Simply going through the motions of an exercise is of no advantage whatsoever. You must put some effort into your exercising. Don't expect that just because you started exercising on Monday you are going to realize an appreciable raise in score by Friday. Physical conditioning is a relatively slow process.

Detrimental Habits for Shooters

In addition to those proven ill effects on general health and longevity, the following effects and reactions will definitely prevent the shooter from reaching his maximum potential. do not

attempt to fool yourself, or to justify your own inability to refrain from these bad habits by pointing out the "exceptions" who are apparently able to fire good scores in spite of their habits. Most good shooters do not smoke at all or drink before or during their shooting.

1. Smoking (these are proven facts).
    1. Shrinks capillaries causing increased heart action resulting in blood circulation difficulties, rapid, shallow respiration, increased pulse activity and frequent nerve pulsating in the extremities of the body.
    2. Dry and "cures" large areas of the lungs, preventing proper utilization of oxygen breathed, and causing thereby the need for much faster and deeper breathing when under pressure.
    3. Dulls certain sensory nerve endings in mouth and throat causing increased nervousness when under pressure.
2. Alcohol.
    1. Causes temporary (eventually permanent) loss of sense of values and judgment.
    2. A depressant, alcohol and its tail end effects will lessen the desire to win.
    3. It will dehydrate the body causing permanent ill effects to blood, nerves and certain brain areas.
3. Drugs.
   The debilitating after effects of even the mildest drugs are well known. Once used, requirements increase rapidly, causing eventual habitual need.
4. Coffee.
   Coffee overcomes depression. Three cups of coffee are equal to a five-alarm fire in your nervous system during a match.
5. The following suggestions are offered concerning the above.
    1. The use of tobacco is to be discouraged. The smoker is slowly tightening an unbreakable linkage around his nervous control that will eventually destroy that control.
    2. Alcoholic beverages when used should be in moderation and of the light variety, i.e. beer, and only after completion of the day's shooting.
    3. Mild drugs of a nature intended to calm the nerves and give a false feeling of stability in hope of attaining high scores under pressure are worthless.
    4. Overeating and late hours when you are trying to accomplish the difficult task of developing into a better shooter or attempting to produce scores in a match that will enable your team to win, will definitely remove that

brisk, enthusiastic alertness so necessary when you need every point you can get.

PERFECTING TECHNIQUE

The True Fundamentals

By Greg Derr

Source: http://bullseyepistol.com/gregderr.htm

I have been fortunate to meet and shoot with many of the "legends" of shooting over the past number of years. Being young with a thirst for knowledge and the "secret" of shooting, I was always tugging at their pant legs.

It didn't take too long to realize that all these great shooters had one thing in common. They all mastered the fundamentals of shooting. And they all had no secret.

I grew to know many shooters who have the ability to shoot any type of pistol in any type of match and- WIN. On some days I thought they could shoot a capgun on one leg in a gale and still win.

I shot for many years with Don Hamilton who I grew to love and respect like a dad. It was through this relationship that I had access to the "gods of shooting" It always amazed me how anybody could shoot a 560 with a free pistol on Saturday and a 2640 on Sunday. However, this is common for these shooters, for they have mastered the fundamentals.

If you really think about it, shooting is shooting regardless of what you shoot -- this is the concept these shooters have grasped which most others have not.

You can ask any great shooter from anywhere on the globe and I am sure that they will tell you "sight alignment" and "trigger control" are the most important aspect in their successful shooting.

Now I know many have expounded volumes on topics like grip, stance, breathing, etc etc, etc. However, these are not fundamentals.

A fundamental is a process by which "all" shooters must perform in the same way to achieve a result. In this case a center shot. Regardless of age, body size and strength, right hand, left hand, male or female. If you cannot align the sight and control the trigger. You may win at you local sand pit or whatever but you will never be a "master" of shooting a pistol.

I teach precision shooting to many clubs and individuals to support my shooting habit. It never ceases to amaze how many shooters will spend hundreds of dollars on some do-dad to try and gain five point in a NMC when they could spend two hours of their time and gain ten points. Don't get me wrong there is nothing wrong about having nice stuff. I just think like a house must have a solid foundation before it can have a fancy kitchen so should shooters have a firm understanding of how to shoot a proper shot.

Getting back to the Gods of shooting. If you only knew how different some styles of shooting were like Free Pistol and Center Fire and Hardball you would have to wonder how any one person could be a champion in all three, there are many shooters who do this, there are more than you think. For them it is easy, It's just shooting. They approach each gun, match and championship, weather it be a state, regional, national, international or Olympic championship. Just shooting to them.

The other things like grip, stance, etc are all important, but only in a supporting role, they help sight alignment and trigger control but they are so individualized that for ten shooters there are ten ways to do each one, not like a fundamental. If you watch a group of masters shoot you will see what I mean.

These other supporting aspect of shooting are techniques. Lets take sustained fire for instance since it was a recent topic on the list. Some shooters say shoot at the buzzer, turn of the target, some say shoot slow, make the first one count. Some try to use all the time and so on. Well all that is great, but what happens to your plan if the sights are out of line, a poor shot for sure.

In techniques there is no right or wrong way to teach since we are all different in some ways, have different bodies, strengths etc. But we all must align the sights and control the trigger. I offer many ideas about techniques in class but never tell anyone one is better than the other. A good shooter will try them all and find the one which gives the best result, results after all are what this game is all about. Remember the scoreboard says nothing about how you stood and how hard you held the gun. It does show if you had good trigger control and sight alignment.

Chances are if you are a High Master in one gun you have the ability to shoot any gun you pick up very well. I guess I have told all my secrets. Hope this helps someone shoot a little better.

Greg Derr

SHOOTING PSYCHOLOGY

Mental Aspects of Match Shooting

By Joe White

Source: http://bullseyepistol.com/joewhite.htm

During the last ten years I have been beaten by some of the best pistol shots in the world--also by some of the worst. By now I think that I know as much as anybody does about how to lose a pistol match. My personal experience in this field has been supplemented by countless hours of post mortem discussion. I have listened to a thousand stories about how some luckless wretch blew a match which he already had in his pocket. In about 90% of these cases "Buck Fever" was the culprit.

The "Buck" is the bane of the pistol shooter's existence. In its most severe form it can drive an intelligent, phlegmatic man into a state of idiotic convulsion. You may have heard of the deer hunter who ironically worked his slide until he pumped all his ammunition out on the ground, and then snapped his empty gun at a big buck 30 yards away. Or maybe you heard about the pistol shooter who loaded his gun with empty .38 brass for his last string of rapid after shooting 15 straight tens. These are extreme cases of course. Sometimes the pressure will show itself only in a breakdown of concentration--a slight unfocusing of the mind. The shooter won't realize that he is coming unglued until his group starts spreading and even then he may blame his ammunition, his gun, the wind or the light conditions, before he finally realizes that he is a little bit shook. I would like to give you the solution to this problem--a plan of action which would enable you to whip the pressure once and for all. That is what I would like to give you. But the only thing I can give you is a few solutions and suggestions which may or may not work for you. I'm sorry I can't do any better, but in my opinion, neither you nor I nor anybody else will ever be able to whip the pressure completely and for good. The best we can do is to learn to minimize its effect and to anticipate its attacks. Sometimes we can sidestep these attacks, and other times we can contain them to a degree, but we will never be immune. A man's own nervous temperament has a great deal to do with his ability to handle pressure. If he is so jittery that he leaps eight feet into the air when a cat steps on a rug behind him, then he shouldn't try to be a pistol shooter. He would have a better chance at fame and glory if he would sell his guns, buy a rug and a cat, and go after the Olympic high jump record. You might ask at this point, "But what about old Tommy Tenring!" Tommy doesn't have a nerve in his body, he uses ice water for blood, he has tunnel vision, he can concentrate like a Hindu mystic, he measures eight inches between the eyes and he holds a Ph.D. in both physical education and psychology. "Tell me," you ask, "How can this guy get shook!" The truth is that Tommy can shoot well enough when he is a little bit unglued to beat most of us on our best day. Also his attacks of pressure will come much less often and will be milder than yours and mine. But wait until he shoots 100 slow and 100 timed in the National

Match course. He knows full well that nobody has ever fired an individual score of 300 X 300 in a registered match, and as he gets ready for the rapid fire stage the thought keeps gnawing away at his mind that he is only ten easy 25 yard tens away from being the first man in history to do it. If he is not very careful to keep his imagination under control he will see a big neon sign above his target flashing out the headline-"Tenring goes clean in the National Match course." Tommy calls up all his powers of concentration and gets ready to fire the string. He does not drop his clip in the sand, he does not load up with five empties and he does not fire on the wrong target. But he does leak out two nines at six o'clock. His mental meanderings have taken their toll. The Buck has just cost him a new National record. He finds little consolation in the knowledge that he already holds the NMC record with 299-29X. He is furious with himself for being so spineless under pressure. But you and I look at his 298 on the scoreboard and say, "Wow, look at that. Not a nerve in his body, etc." If you are another Tommy Tenring, or have been close to it, all I can say to you is, "I wish I had your problems and you were writing this." But if you have the same troubles that I do, you may be interested in some of the things that have helped me. If you try them a few times, and they don't work for you, then by all means forget them. First of all, the most human thing to do is to look around for something which requires no training or mental effort on our part. I am talking about such things as alcohol, tranquilizers, or dope. I do not recommend any of these. You can hear plenty of stories around a pistol range about how old so-and-so shot thus-and-such when he was full of beer. If you are tempted to try it yourself, first find out how the pistol shooter makes out in the grand aggregate. You can then make your own decision. I have seen a man get so steady from alcohol that he couldn't find his shooting box which was twenty feet away in plain view. I know a man who got so tranquil from tranquilizers that he considered it a bit funny every time he shot a six. I heard about a man who smoked a marihuana cigarette before rapid fire because he had heard that it would make the time seem to pass slowly. He said that the time passed slowly enough, but that he couldn't lift his gun. There are several things that I try to remember to do that have helped me a lot. One of these is to be completely ready. I like to have my sights blacked and adjusted and a couple of clips loaded before I walk up to the firing line. Being ready is important in slow fire, more important in timed fire and absolutely critical in rapid fire. Very few things will unglue me quicker than to start a string of rapid when I am not quite ready. Like most everyone else I have a little routine I so through before the targets come around. I watch the bullseye when the targets are turned away to find some reference point to aim in on. I get my feet placed. get a good grip on the gun, aim in on the reference point. control my breathing just the way I want it, and watch the sights very carefully for alignment. Then when the targets start around I am completely ready to start mashing the trigger. If I am busy loading a clip or blacking my sights and forget to find an aiming point, it bothers me. If I let my routine get behind the commands it bothers me more and if I don't have my mind geared to rapid fire, I am in real trouble even though I went through the routine exactly right. If you also have trouble getting your mind geared to rapid

fire, I recommend that you try something that has helped me a lot. I prefer to do this on the line just before firing the match. Hold a stop watch in your left hand and go through a string of rapid in your mind. Try to picture everything exactly as you want it to happen, with the preliminary commands from the range officer, with your own target, and with the gun that you are using that day. When the imaginary target comes around. start your stop watch and go through the entire string in your mind including the recoil and recovery from each shot and the sight alignment and trigger squeeze for the next one. Pay particular attention to your rhythm as you go along. After the last shot stop your watch and see how many seconds you used. Repeat the string if necessary until you hit a satisfactory time which you can set to your liking. I like between 9 and 10 seconds. By doing this you have the right rhythm in your mind and you will start the match with confidence that you know how long it takes ten seconds to tick off. Confidence and rhythm won't necessarily give you 20 tens, but they will sure save you from a lot of wild shots. Probably the most common use of the stop watch is to time slow fire strings. If you have ever been worried about time on a windy day and hastily cranked off two bad shots because you expected the cease fire any second. and if the cease fire command came a full two minutes after two beautiful 30 second lulls, then nobody has to tell you that a stop watch is important. You don't need a watch to get off ten shots in ten minutes, but with a watch you won't worry about time. This is important because worrying will unfocus your mind and open up your group. With a watch you can space your shots better. If your hold is a little shaky, you know you have time to take it down and start over. It gives you more confidence, and confidence is conducive to better scores. On my better days when they are going pretty good I use the stop watch to time my rest periods between shots. This can help you fight off the buck when you have a real good string going. Make up your mind before you start how much time you will rest between shots; you can revise it later if you need to. When you fire one, check the watch. Try to stick to your self-assigned rest periods to the second. I sometimes time my breathing the same way while I am resting, trying to make it come out right for the next shot. This keeps your mind busy, and you are less likely to start sweating your score. It also introduces another element of precision into your routine which may improve your ability to think positive thoughts about what you are doing. Our mind has a tendency to relax before the shot breaks. We stand there working hard on a shot and the old subconscious is screeching "shoot it, so you can relax." The best way to overcome this is to follow through. This is very important. I would write follow through on the blackboard a thousand times if that would make me remember to do it on every shot. I am talking about a mental follow through which keeps all the attention on the business at hand until the bullet is safely in the ten ring. Bad shots are very often caused by a slight relaxing of the attention lust as the shot goes off. This is usually accompanied by a spasmodic jerk of the trigger finger when the subconscious gives us its mental elbow in the ribs to "shoot it and get it over with." One of the most effective procedures for me to follow when I am having this trouble is to tell myself that I am dry firing and that, on this particular shot, I will be very careful to keep the

sights aligned before, during and after the fall of the hammer. Then during the trigger squeeze I work up a mental picture of the hammer falling and the sights remaining in perfect alignment after the hammer has fallen. This is what I mean by mental follow through. The idea is to continue working at keeping the sights aligned even while the bullet is traveling toward the target. It is the best insurance you can get against relaxing your attention too soon. They say that intense concentration is just the old story of mind over matter. The brain must have complete control of the body and its actions. I think in my case it would be easier if I had a larger brain and a smaller body. Unless someone kept score at a match we would never know who won. My advice is to let the statistical officer do the worrying about who is winning. When you finish a match, check your score card and make sure that every individual shot has been recorded correctly and that the totals are correct. Then sign your card, turn it in and forget it. One of the surest ways to put the monkey on your back is to walk up to the line knowing that you need a certain score in order to win the aggregate, to set a new record, to beat old Tommy Tenring for the first time in your life, or to do anything else that you want very much to do. If the doctors could feed us a pill which would make us forget how to count or add, our aggregate scores would surely improve. But then we would still have the problem of the well meaning friend who says, "Boy, you have got it made. All you need on this last ten shots of rapid is 98 and you will have a new aggregate record. The way you are going today you can't possibly miss." I have promised myself that if this ever happens to me again, I will take out my pocket knife, open the dullest blade, and slowly whittle off his head. This should clear the air and ease the tension so that shooting the 98 will be easy. It is possible, however, that by the time I clean up the gore, dispose of the body, and have a long talk with the police, I may wish that I had gone ahead and shot the 10-10-9-8-7 on the last string, like I did the last time it happened to me. There are many mental gymnastics which you might employ to keep the neon sign from starting to flash out the glad tidings of your sensational victory when you are only halfway through the match. You might try naming all the New England states with their capitals, or some similar stunt. I sometimes conjugate Spanish verbs. I have to be careful with this one though. It's pretty easy for me to get confused trying to figure out the preterite form of some obscure radical changing verb, and the stop watch can easily get ahead of my shot string. If you will work hard on ideas such as these, concentrating on the ones that seem to help, you will find that you will gradually begin to control the pressure better. It goes without saying, however, that no amount of mental power can make you shoot any better than you know how to shoot. Lots of regular practice under match conditions will not only improve your ability, but as your practice scores improve, you will have more confidence in yourself. The ability to withstand the pressure of competition will then enable you to shoot your good scores in matches. When you get into a tight situation, you will lose your points, a few at a time, and not by the handful Then one happy day, the other shooters will start pointing to your scores and saying, "Wow, look at that, not a nerve in his body!"

SHOOTING PSYCHOLOGY

Competitive Shooting - Practical Psychology

By International Sport Shooting Federation

Source: http://bullseyepistol.com/issfguide.htm



This is the very informative publication of the ISSF covering the many aspects of psychology in competitive shooting.
The 84-page book is in PDF format and may be accessed by clicking on the cover of the booklet to the right.

SHOOTING PSYCHOLOGY

The Psychology of Routine

By Don Nygord

Source: https://www.bullseyepistol.com/d-nygord.htm

Don Nygord has been shooting for over 30 years and for the last 21 straight years has been a member of the US Shooting Team competing in Olympic style pistol shooting all over the world. He has been National Champion 16 times, has been on the 1984 and 1988 Olympic Teams, and was World Champion in Air Pistol with a score only two points off the World Record at that time. He is the current holder of the US Free Pistol record at 574 - a record that has stood for 14 years now! - and has held over 40 other National Records. The following essay is a result of this experience and is offered to help both the new and the experienced shooter improve their performance.

THE PSYCHOLOGY OF SHOOTING

There are some serious contradictions in the messages we get whenever the subjects of "How to acquire shooting skills", "How to perform at matches", and "What were the results of your last match performance" are discussed. These contradictions occur at all levels of the game, from the guys at the club to the National Team. Almost everybody that has been shooting for any length of time and not living on a desert island has heard "don't think about your score ..." when shooting in a match. Right, sure. Unless you are brain dead, *of course* you think about your score! We used to joke that the Russians (who were whuppin' us) all had received pre frontal lobotomies because they seemed so stoic in victory or defeat - no expressions of anxiety or joy. This was not true, of course, but we speculated that this procedure would probably improve your match performance, but you wouldn't care! The principal expressed in "don't think about your score while shooting ..." is sound enough. Actually, the whole idea is "To acquire great skills, one MUST think about every aspect of execution, and do so all of the time. But to achieve great PERFORMANCE, one must *not* "think" at all!!" What this means is that you must be extremely analytical and cognitive in the effort of learning and perfecting the skill of shooting in order to come up with a good technique. But, when actually performing, you must let the mind/body combination operate on the subconscious level, and so do what it was trained to do **without** the interference of the conscious mind. And, all effort and focus is to be on execution of the act of firing a shot or a series in perfect conformance with the model

you derived from all that cognition and rehearsal - not what the results of this execution might yield.

O.K. So you do this, or try to at least. What is the first thing the guys (or the National Coach) asks when you come off of the line? DO they ask, "How many times did you correctly execute your model technique?" Most likely not. Surely they say, "What score did you get?" And, they haven't yet put on a match where the medals are awarded on a percentage of perfect executions, nor do they publish the names of the shooters and the number of excellent executions in the match bulletin. No-sir-ee, they publish the *scores*. So live with it. Forget the silliness of "Don't think about your scores" but DO realize how one accomplishes skill versus how one achieves high performance (which of course equals high scores). Also recognize that *everyone* at every level experiences increased "arousal" (a word much preferred over fear, anxiety, dread, etc.) when in a serious competition. What most of the top performers have is the confidence that the work they did in learning the skill of shooting will carry them through in spite of (or even *because* of) this extra arousal. And, because the same adrenalin rush that makes your mouth dry and your palms sweaty also increases your visual acuity, tactile sense, and cognition speed - darned if you might not just end up with a personal best!

Having a **plan**, i.e. "I am going to execute correctly as much as possible" and *knowing* what that *is* will put you ahead of 90% of the pack before the first shot is fired!

MAKING CHANGES TO YOUR ROUTINE

Many times I am asked, "What should I change so I can do better?" We know that change is necessary to go from one state or level to another, but how and when and what do you change if you want to "improve"? One man asked me how to hold his gun "still". He was disapoointed when I told him this was not where he should be putting his effort! What he really wanted, of course, was better performance (scores). His analysis of what change would bring him the most improvement was faulty in thinking that his "stillness" or hold was the biggest problem he had. (Of course, we all know what *really* is the biggest problem, don't we?)

I am an inveterate changer myself. I believe this has helped me considerably in my career (although some of the coaches the US team has had were made crazy by this!) All my changes, though, were **SYSTEMATIC**. This means I analyzed the current situation, decided on which element would yield the most performance gain if changed, made the change I thought would be for the best, set up a test of efficacity, set a timetable for implementation and then **RECORDED** everything I could think of: All of the above (the change, the why, the how, the goal for the change, the schedule, etc.) and finally I documented **ALL** the results (scores). Warning!! All changes have side effects: In our eagerness to change something that seems to be an obstacle, sometimes we get negative, unintended results. One of my students decided

that he would stop drinking his 8 cups of coffee a day for the duration of the National Championships. This surely would make him less "nervous" and he would do better. The only problem was the intense withdrawal headaches from the sudden cessation of caffeine! Moral: If the change will affect your physical state, make it early enough to allow for acclimation. Another example of this was the student who started weight training in the middle of the shooting season. His body was getting stronger, but his fine muscle control, needed for good performance, was shot! (Pun intended.) This also applies to changes to your equipment: A different gun, adding weights to the gun, a different grip, etc. Make these kinds of changes early enough to allow for adaption by the time the "big match" must be shot. More subtle changes like trigger weight, trigger position, shooting glass lenses, rhythm of firing, etc. can be made without so much concern about adaptive time, but should be planned, adhered to for a predetermined period and **DOCUMENTED** - then correlated with the results obtained. Only this way can you make an objective evaluation of the value of the change and decide whether to keep the new conditions or go back to "zero" and start again. Of course, sometimes the change is either so good or so bad it is obvious immediately that should either keep it or drop it! Either way, you've learned something you didn't know before and that is progress. And, the smarter we become, the better we perform. So change away - wisely!

Don

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SHOOTING PSYCHOLOGY

The Art of Shooting Under Pressure

By Dr. ke Lundeberg, Psychotherapist and Violinist

Source: https://www.bullseyepistol.com/musician.htm

> *Dr. ke Lundeberg is a German physician specializing in psychotherapy and also trained as a violin teacher. He works full time giving courses in the art of performing under pressure. A paper he delivered in York England in March of 1997 during a conference on "Health and the Artist" superbly detailed and analyzed the condition of stage fright in orchestral musicians. By taking the liberty of doing a little editing of his paper, I have painstakingly translated his thoughts to apply to the experiences of the competitive pistol shooter. That translation is what follows...*

Negative Pressure

Anxiety is likely to occur whenever you must perform well in some endeavor of great importance to you in the presence other people, whose judgement you fear. Murphys law is at work: Anything that can go wrong will do so, but much more and much worse than you ever thought possible.

> *Thinking about the possible disastrous outcome of your performance and its consequences is the core of match pressure.*

It has been known for a long time that the best way to deal with match pressure is to concentrate completely on the plan of shot execution. True, but alas, easier said than done. About 300 BC Chuang Tzu said that "an archer competing for a precious golden prize shoots as if he were blind." His concentration is on the reward and the consequences of missing it, instead of the shooting itself.
I have developed a specific method of concentrating on shot execution and routine. This kind of concentration does in fact happen intuitively whenever a shooter is in the groove, i.e. when the shot execution simply flows effortlessly with intense concentration. But intuition all too often fails to function while you are under negative pressure. I teach a way to use consciousness, so that you can always know that you are in fact concentrating on shot execution rather than the possible disaster.

Common Misconceptions about Match Pressure

A great shooter has an inborn confidence and needs not feel any match pressure.

This certainly appears to be true. Great shooters rarely show negative feelings, which is not the same thing as saying that they never have such feelings. The higher your classification, the greater the fall can be. However, great shooters are generally very good at managing match pressure.

All it takes is confidence

Of course, confidence helps. But there is no such thing as 100% genuine confidence. Most great shooters report that they at times are troubled with self-doubts.

If you have a reliable technique, you will not suffer from match pressure

There are many reports of the opposite, shooters of great technical ability, who have suffered severely. But, of course you can get anxious knowing that your technique is not in shape.

The Art of Shooting Under Pressure

Most shooters feel negative pressure at times. The art of performing under pressure is a technique that you can learn and develop. The goal is to cope with pressure in a way that allows your true shooting potential to be unfolded. You can learn effective techniques for this purpose. Nevertheless, there are no short cuts in the shooting sports. You need to work with the area of nerve control continuously. Placing the plan of shot execution on center stage is a great foundation for coping with pressure.

I used to believe that match pressure symptoms are caused by all kinds of frustrating experience in the past. While there might be some truth in that, it is evident that some shooters while performing superbly on the firing line, they can have deep personal troubles including lots of bad experiences in the past. In contrast many sympathetic and mature persons can experience very negative pressure on the line. The conclusion is that the art of shooting under pressure is largely a technique. And such a technique can be taught and learned. The crucial questions are thus: What do you actually do before and during a match? How is your consciousness organized?

There are basically three ways of reacting to match pressure:

1. Everything goes better, the shooter feels elated and his/her concentration and execution is enhanced.
2. Performance suffers from various symptoms, such as fears, panic, body parts shaking, loss of concentration, etc.
3. The shooter becomes apathetic and may still shoot competently but not at his/her best.

The first reaction is, of course, the ideal one. The other reactions lead to achievements below the true potential of the shooter, because mind and body get occupied with other things than shot execution itself. When you do experience negative pressure, Murphys law is at rule: Anything that can go wrong will do so, but much more and much worse than you thought possible. Say you are to walk across a plank located between two cathedral towers. The task as such, i.e. the technique of walking is very easy indeed. But Murphys law will take over. Your brain will be flooded with pictures of coffins, undertakers, broken limbs, etc, and these pictures will govern your behavior, making it likely that you will indeed fall. Murphys law is very persistent. Say you have experienced a match pressure symptom such as a shaking forearm or a lack of concentration and you shot far below your potential. The experience was frightening and you will try to avoid it at all cost in the future. Then what happens? In the case of a shaking forearm, you hope intensely that your arm will not get that tremor again. But the primitive part of the brain regulating these fear reactions does not understand the words no or not. It thinks in pictures. The picture communicated will be that of a tremoring arm! The brain thus receives a command to make the arm shake again. It works like ***"Do not think of the color red!"*** Red is precisely what you think about, possibly trying to paint it over with other colors. The art of shooting under pressure is all about reversing Murphys law. For a great range performance, two things are needed, an excited state of alertness (fed by your desire to excel) and a relaxed state inside of that. You might call it the ***"eye of the storm"*** or *"**peace of mind in an eager body**."*

Reversing Murphys Law

Whether you feel pressure or are downright bored, you give energy to things other than shot execution. The remedy is thus to make sure that you devote yourself entirely to the plan of shot execution. Body and soul are different aspects of one unity, mental fears and physical tensions coexist, alertness and relaxation of mind and body are needed, getting into the shot plan is the best remedy of match pressure.

Relaxation and Attitude Training

Our attitudes deep down will largely influence our performances. Therefore it is effective to work with negative attitudes to performance (we all have them more or less). Shooters can utilize relaxation procedures including the use of a relaxation trigger, that is a mental image that displaces negative thought. This trigger can be used before shooting and also on the firing line. The excited state of mind and body will then mix with the relaxation that the trigger represents, helping to achieve the desired mixed state of both tension and relaxation called the eye of the storm. Furthermore the shooter is taught to control thought processes whenever needed before performance and on the line. Thinking constructively will not exactly erase all negative feelings, but rather limit their effects. It is more a question of living

creatively with an appetitite for success, because if there is no hunger at all, performance is likely to suffer and become over-relaxed and indolent.

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SHOOTING PSYCHOLOGY

Staying Focused Despite Match Pressure

By John Dreyer

Source: http://bullseyepistol.com/focus.htm

> *"The truly great shooters pay little or no attention at all to their competition, or anything else for that matter. For them, the contest takes place inside their head. The real struggle is to get in the zone. When they find it, the rest just seems to happen. It's as if the world around them melts, the distractions disappear and the universe is reduced to the few simple elements of eyes, hands, gun and target."*
> - Gabby Hulgan (1996 NSSA World Skeet Champion)

You have confidence in your equipment. Your practice sessions yield satisfactory scores. Now the challenge is to simply execute what you know that you are fully capable of doing when it really counts. It sounds easy enough, but unfortunately, our minds don't always cooperate. Feeling nervous about having a great performance is certainly natural. However, when that feeling turns to panic because you think that you are the only shooter on the line feeling this pressure, your mind has defeated you. You will not be able to focus on the task at hand. Rest assured, every competitor feels this pressure to execute to the best of his ability. The challenge for you is to channel this same pressure constructively and to not let it overwhelm you.
Your mind is the control tower for all your physiological reactions. If your hands start shaking, your palms get sweaty, you can't think straight because you "feel nervous," it is probably because you're thinking of the situation negatively instead of as an opportunity for success. Dr. Bob Rotella, a specialist in sports psychology, offers six steps to help deal with pressure, anxiety and nervousness:

1. Think good, pleasant, soothing thoughts rather than worrisome or negative thoughts.
2. Keep your mind on the present, on the shot you're going to execute right now. Think about what you want to happen. Remember, anxieties are always about what just happened or what might happen, so stay in the present.
3. Assume the best is going to happen, rather than anticipating the worst. You wouldn't go to work every day thinking you were about to be fired, so why try to shoot with that type of mental approach?
4. Use the power of perception to dwell on your strengths.
5. Feel as if you were destined to have good things happen to you rather than as if you were born to have bad things happen.

6. When you start to feel tension, stop and take deep, slow breaths.

Frankly, I think performing through nervousness is what sport is all about. Sport is supposed to teach you how to deal with your mind and emotions. Ultimately, when you're in a situation that makes you nervous, you need to remind yourself that this is right where you want to be - this is YOUR DREAM COME TRUE.

Dr. Debbie Crews, a sports psychologist from Arizona State, has done a lot of work on mental training and testing in sports, and gave a presentation at the 1997 Shooting Coaches College at the Olympic Training Center. Her topic: athletes who choke under pressure. She feels that her results are applicable to shooters, as well as golfers, with whom she specializes. Her research shows that when an athlete needs to perform a highly-skilled action, it must be the subconscious that does it, not the conscious, and our minds must be relaxed and in the subconscious mode to do it. The left and right sides of the brain must be in harmony (balanced activity) and there must be no conscious self-talk or activity 1-3 seconds before the action takes place. Establishing a routine is possibly the greatest combat against a lack of *initial* mental focus. Going through the motions of a pre-established and familiar plan can get your "mental wheels" moving in the proper direction. If necessary, a written checklist can serve as a tool to assist in "getting your head screwed on right." Now that all of your equipment is in place and you are at ease, the stage is set for a great performance. The first string of slow fire begins. You raise the pistol and flawlessly execute the fundamentals, confidently firing a ten. Now the challenge is to stay focused. You *can* and *will* succeed, each and every time. Remember the *little red engine* that could?

SHOOTING PSYCHOLOGY

Achieving 'The Zone' in Pistol Shooting

By Dave Salyer

Source: http://bullseyepistol.com/salyer1.htm

Many top bullseye competitors speak about something strange that goes on mentally as they compete. They use terms like "Zen" and "the zone." To the rest of us, what are they talking about?

I will start by saying that I am not yet qualified to explain what "Zen" really is. The American Heritage Dictionary defines Zen or Zen Buddhism as, "a Chinese and Japanese school of Buddhism that asserts that enlightenment can be attained through meditation in which dualistic thinking is overcome". (This dictionary didn't define dualistic.)
I believe they mean total singular focus on the task or thought at hand with no distractions bearing on the conscious or subconscious mind. I have experienced something spiritual (not in a religious sense) like this a few times in my shooting career. Unfortunately, I have not been able to hold this focused feeling except for very short periods. I have not yet learned how to bring this relaxed mental state on when needed most. The best personal example came once in .45 practice at 50 yards. I started by shooting an off-hand target in the mid nineties more or less by following the "mechanical" fundamentals and procedures. This is a little better than average for me. Somehow, this bolstered my confidence for the next 10-shot string. This string resulted in a 100-7X! The target is framed on my den wall. Normally, after shooting 5 or 6 slow fire tens in a row, doubts and fears come into my mind. I seem to fear failure, and even success! Mental pressure increases until I shoot a 9 or worse. Then I can relax a bit and usually finish with the rest as mostly tens. When I shot the clean 7X target, the feeling and awareness was totally different! After about the third shot within the 10-ring, I was sure the rest of the string would be good. Absolutely no doubts entered my mind during the string. I felt that I was holding so well that no shot could miss the 10-ring! I was in a "zone" that I cannot describe in words except it seemed that the red dot was moving very slowly and staying in the black. The dot moved without effort to the middle and the pistol fired, without conscious effort. The slide seemed to move in slow motion. There was never any hesitation or mental reservation. It is probably more important to try and recall what I was not thinking about. I can assure you that I was not thinking about the fundamentals. I was not conscious of position, stance, grip, nor trigger control. I didn't even consciously hear the pistol fire; nor did I feel the recoil. I was not thinking about past successes and failures. I was not worried about the future. I was in my own little world for about ten minutes! My mind was relaxed and apparently both sides of the brain were contributing to the process of firing well aimed shots. Contrary to my normal habit of coaching myself through the correct process

of firing a well aimed slow fire shot, I seemed to be just observing the process while feeling the right amount of confidence. It is important to be confident, but detrimental to be overly so. I think each experienced shooter has a "comfort zone" that he or she normally performs within. If performing below this level frustration sets in, further inhibiting abilities. Occasionally, we will perform above what we are accustomed to. This can create anxiety in the form of "fear of success". Either way we are likely to get in a judging mode and clutter our minds with thoughts that keep us out of that illusive relaxed "zone". I asked present .45 caliber National Champion and 2670 shooter, Al Dorman to give me his thoughts on the Zen concept. Here are his words: "First Zen is not a philosophy, it is a place that causes action. A chapter that when explored temporarily and permanently changes perception, both at the same time at different degrees. I cannot give directions as I don't remember getting there or even picking up the book to read the chapter. Maybe it was too long ago and time has eroded the path or maybe it has always been there in varying degrees and there was never a path." He explains further. "When I shoot with competitors who can beat me, it is the same as when I compete with no High Masters. I have a job to do and I go about getting it done. Wind or other elements do not change my perception; just my sight adjustment is changed. Maybe Zen is a type of focus as perceived by others and giving it a name within one's self causes it to evaporate with the morning mist. Fragile place if so." Note that Mr. Dorman doesn't worry or even think about what the competition is doing. His attitude and effort are at the same levels. I'm sure most of you experienced shooters have noticed how easy it was when you shot your best string. Once we are on the "stage" competing, extra effort does not pay off. Trying harder keeps the mind out of its relaxed state and best working mode. I am, however, convinced that extra effort during training does pay off by ingraining good fundamental habits that must be followed with little or no conscious effort once the competition starts. I certainly do not want to downplay the importance of the well established fundamentals. They must be applied by all shooters at any level. The goal is to apply them without having to double check for them once you are in competition. If you have a coach, that is his or her role. I can recommend two excellent books on the subject: Peak Performance by Charles A. Garfield and The Inner Game of Tennis by Timothy Gallwey May your comfort zone and scores continue to rise.

SHOOTING PSYCHOLOGY

Sleeping Well the Night Before a Match

By Elizabeth Somer

Source: http://bullseyepistol.com/sleep.htm

One of the biggest factors in my ability to shoot well is how much and well I am able to sleep the night before. I have included this article because it may be of value to those, like myself, who sometimes have difficulty getting a good night's rest when there is a big match the next morning. -JD

Do you toss and turn during the night instead of sleeping soundly? If so, your battle with insomnia might start at the dining table, not in the bedroom. A cup of coffee or tea or a glass of cola are quick pick-me-ups that might undermine your sleep. Even small amounts of caffeine (like the amount in a chocolate doughnut) can affect your sleep, especially if you are sensitive to caffeine. Try eliminating all caffeine-containing beverages. If you feel and sleep better after two weeks of being caffeine-free, then avoid caffeine permanently. You can try adding back one or two cups after the two-week trial, but cut back if sleep problems reappear. As for alcohol, a nightcap might make you sleepy at first, but in the end you'll sleep less soundly and wake up more tired. Alcohol and other depressants suppress a phase of sleeping called REM (rapid eye movement) during which most of your dreaming occurs. Less REM is associated with more night awakenings and restless sleep. One glass of wine with dinner probably won't hurt, but avoid drinking any alcohol within two hours of bedtime. And never mix alcohol with sleeping pills!

Sleep-Friendly Table Tactics

Big dinners make you temporarily drowsy but prolong digestion, which interferes with a good night's sleep. It's best to eat your biggest meal before midafternoon and have a light evening meal of 500 calories or less. Include some chicken, extra-lean meat or fish at dinner to help curb middle-of-the-night snack attacks. Spicy foods can contribute to sleep problems: Dishes seasoned with garlic, chilies, cayenne or other hot spices can cause nagging heartburn or indigestion. Avoid spicy foods at dinner. Gas-forming foods and hurried eating also cause abdominal discomfort, which in turn interferes with sound sleep. Limit your intake of gas-forming foods to the morning hours, and thoroughly chew food to avoid gulping air.

Bedtime Snacks: Great Alternative to Sleeping Pills

A high-carbohydrate snack, such as crackers and fruit or toast and jam, triggers the release of a brain chemical called serotonin, which aids sleep. And although the traditional glass of

warm milk, a protein-rich beverage, probably doesn't affect serotonin levels, the warm liquid soothes and relaxes you and makes you feel full, which might help facilitate sleep. A new product on the market called 5-Hydroxy-L-tryptophan, or 5-HTP, is touted as a building block for serotonin, which is a mood elevator, brain stimulant and sleep enhancer. However, since its safety is questionable and no optimal dose has been established, you're better off raising serotonin levels naturally with high-carbohydrate snacks.

Curbing the Midnight Snack Attack

Do you awaken in the middle of the night, unable to fall back to sleep unless you eat something? These midnight snack cravings may be triggered by hunger, or they may just be habit. In either case, your best bet is to break the cycle. Try eating more during the day, and stop rewarding your stomach by feeding it every time it wakes you up. Instead, read a book, drink a glass of water or ignore the craving. It takes up to two weeks to break a midnight snack habit.

Exercising to Relieve Stress

Stress is a common cause of insomnia. Often, relieving tensions and anxieties eliminates sleep problems. One tension reliever is exercise. In a study from Stanford University in Palo Alto, CA, healthy adults with mild sleep problems who exercised twice a week for at least 40 minutes per session fell asleep faster and slept about 45 minutes longer than people who didn't exercise. Physical activity also helps you cope with daily stress and tires the body so it is ready to sleep at night. Vigorous exercise should be done no closer to bedtime than six hours; mild exercise should be done no closer to bedtime than four hours. In short, sleeping pills are a temporary fix, but a few simple dietary and lifestyle changes could do wonders for your long-term snooze control.

Excerpted from *Doze Control: Eat Right and You'll Sleep Like a Baby*
From: Focus on Health By: Elizabeth Somer, M.A., R.D. Source: Unknown via Internet

ZEN IN THE ART OF SHOOTING

Zen in the Art of Pistol Shooting

By Foreword by John Dreyer

Source: http://bullseyepistol.com/zeninfo.htm

Introduction

Volumes of material have been written about acquiring proper physical technique and mental discipline in precision pistol shooting. I have found that most are tremendously valuable to any serious competitor and do a good job of describing the mechanics and philosophy of shooting. However, very few truly explored the ongoing tremendous internal battle of the competitor or the many other spiritual aspects of the sport.

In my journey from Marksman to Master, I learned a great deal about not only punching holes in paper, but about myself and my perception of life itself. Because I was unable to put together the words to relate my experiences, I sought for credible written works that properly illustrated my discoveries both on and off the range. As a result, I collected and dissected over fifty books plus almost that many articles from periodicals and other media on a variety of aspects of life and shooting. After almost a year of research and reflection, I was finally able to assemble that massive and mysterious puzzle of intellect and emotion that encompassed life both on and off the range. Surprisingly, the final piece of my puzzle turned out to be an incredibly simple way of looking at life, an ancient Eastern philosophy called Zen, a Japanese subset of Buddhist tradition.

Shooting as a Zen Art

Deceptively simple in appearance, yet vastly complex is the art of pistol shooting. Without question, its mechanics are simple. As Bill Joyner explains, "Create a machine rest with your stance, grip and breath control. Then with the gun in the machine rest, apply [trigger] pressure directly to the rear until the hammer falls." Attaining the physical prowess to accomplish this task is one thing. However, the mind's influence makes the process a bit more difficult. As Frank Higginson has said, "In shooting, you learn more about yourself than any other sport." This self-discovery that exists in shooting is nothing more than Zen itself.



Kenjuudo, literally translated as the "Way of the Pistol," is the appropriate Japanese name for our sport when practiced as a traditional Zen martial art.

"Kenjuudo"

While you may have thought of Zen as distant and mysterious, bear with me while I present it here in a series of diverse articles for you to explore. I encourage you to set aside any preconceived notions you have about Zen and check out these works. At minimum, you will gain insight into an aspect of competitive shooting that you may never have considered before.

As far as I know, this is the first extensive collection of writings published in *any* medium that explore the practice of precision pistol shooting as a Zen art.

Use the buttons on the navigation bar to the right to journey through the many writings on Zen in Shooting.

ZEN IN THE ART OF SHOOTING

What is Zen? Shooting Meets Real Life

By John Dreyer

Source: http://bullseyepistol.com/spirit.htm

The word *Zen* is the English derivative of the Japanese *dhyana* which means "to think." In China, dhyana was rendered as *shii-shu* or "thinking practice," possibly more applicable to the Zen arts.

The Japanese have been masterful in taking apparently mundane daily activities and elevating them to art forms that create paths to spiritual awakening. In Zen, what makes an activity an art or "Way" is when its practice is not done merely for the immediate result but also with a view to purifying, calming and focusing the psycho-physical apparatus. Indeed, the sport of precision pistol shooting provides a perfectly suitable discipline to practice it in such a fashion. Understanding a little more about Zen will not only benefit your shooting tremendously, but it will allow you to extract more value from this sport.

"When you are hungry, eat. When you are tired, sleep. This is Zen." - Hiakajo Roshi

If one would try to describe what Zen really is, it is important to understand that Zen, in its essence, is not a doctrine. There's nothing that you are supposed to believe in. It's not a philosophy in our sense, that is to say a set of ideas. It is not some intellectual net in which one tries to catch the fish of reality. Actually, the fish of reality is more like water--it always slips through the net. And in water, as you know from swimming in a body of it, there's nothing to hang on to. Well, all of the universe is like water; it is fluid, it is transient, it is changing. And when you're thrown into the water after being accustomed to living on the dry land, you're not used to the idea of swimming. You try to stand on the water, you try to catch hold of it, and as a result you would drown. The only way to survive in the water, particularly the waters of modern philosophical confusion, is to learn how to swim. And to swim, you relax, you let go, you give yourself to the water, and you have to know how to breathe in the right way. And then suddenly you find that the water now holds you up; indeed, in a certain respect, you have become the water.

"The perfect man employs his mind as a mirror. It grasps nothing, it refuses nothing. It receives but does not keep." - Chung-Tzu

There is a Zen poem that says of wild geese flying over a lake, "The wild geese do not intend to cast their reflection, and the water has no mind to retain their image." The thoughts here are to live without "hang-ups", that is to say, to be able to drift like a cloud and flow like water, seeing that all life is a magnificent illusion, a plane of energy, and that there is absolutely nothing to be afraid of. That is, fundamentally. You will always be afraid on the surface. You

will be afraid of putting your hand in the fire. You will be afraid of getting sick, etc. But you will not be afraid of FEAR. Fear will pass over your mind like a black cloud will be reflected in the mirror.

"If you see a weed, pick it." - Suhara Kohn Osho

"Eliminate the space between" is an old Japanese saying that describes an important element of Zen. While anyone should always achieve to do the "right thing," in Zen, this is not the ultimate goal. Rather, to do those right things INSTANTLY, without any conscious deliberation is a much higher goal. Jackson Morisawa stated, "When the operation of the mind and the body coincide with one point in time and when the space between thought and conduct is eliminated in such a way that they are in perfect unison, we may regard such a moment as the present." Just as seen in the familiar image in Japanese art of the moon reflecting on water, so united should be our thoughts and actions. If we live our lives in this fashion, we will be one with our surroundings. In this light, the Suhara Osho saying, "if you see a weed, pick it" is a masterful image. If you see a weed, you will pick it without calculation. The weed becomes an integral part of you, you do not have to give picking weeds any thought, you just do it. And in that manner, we should live our lives.

"Thousands of repetitions and out of one's true self perfection emerges" - Zen saying

An element of Zen teaching is to use various "arts" as a vehicle to attain enlightenment - that is, to better understand yourself and ultimately "Universal truth." While the Zen arts are many and quite diverse, they all share that same common goal - to confront our true nature. Because of that, the Zen arts have been called "fractional expressions of Zen in limited fields." Mere technical mastery of any of these arts is not TRUE mastery. Mastering a Zen art is not based on cognitive or intellectual understanding - it is based on an intuitive awareness of the underlying principles of the Universe as they apply to that particular art. The philosophy of learning a Zen art is to learn its underlying principles through the repetitive practice of techniques. In that respect, in becomes appropriate for more modern activities to be practiced in the method of the Zen art, provided they are studied and mastered with this philosophy in mind. In our case, it will be to study precision pistol shooting in that light, a discipline that bears much resemblance to the ancient Zen art of Kyudo (archery).

"A man who has attained mastery of an art reveals it in his every action." - Gogen Yamaguchi

In Zen, there is "no inside and no outside." There should be no distinction between life on and off the firing range. One should approach all activities and situations in the same sincerity, intensity and awareness that they have while shooting. In other words, while shooting, you are learning to attain an empty mind - acting without calculation - with the underlying objective to be able to maintain it in your everyday life. When a pistol shooter is able to sustain this "empty mind" in daily life, he is able to see the underlying principles of everyday

life and become aware of the life in all things. It is in this manner that the simple practices, such as Kyudo or pistol shooting become elevated to a Zen art.

"The challenge of Zen is to be happy when you are sad." - Tanouye Tenshin Roshi

By learning to maintain equipoise, or continuity of emotion, whether you fire a good shot or bad shot, you learn to maintain equipoise in daily life. As a result, you will face hardships and disappointments with the same calm that you must maintain after firing a poor shot or have a dreaded gun malfunction. You will accept responsibility for your mistakes and simply try again. This behavior is called achieving an "ordinary mind," one that is calm, well-balanced and disciplined at all times even when confronted with unexpected or unpleasant circumstances. Alas, this is the true Zen lesson of shooting. He who strives to live his life in that way understands that the "target" is not a piece of paper downrange; the true target is within.

"The secret of a successful learner is to become like a child again." - Hideharu Onuma Sensei

Learning ANYTHING new is difficult. This is especially true for adults who must endure embarrassing situations and fumble with awkward techniques as they struggle to acquire new information and skills. It is far easier to find reasons why not to pursue the study of complex thoughts or techniques than to suffer through the learning process. Much like anticipating the chilling shock of the water in a particularly cold swimming pool, it sometimes takes someone to figuratively "push you in" to get the learning process rolling. Quite possibly, the few articles on this site exploring "Zen in Shooting" will be that push. The first task is to set aside any preconceived notions you might have about Zen and have a willingness to abandon much of your present shooting mentality. I encourage you to get a fresh start - restore "a beginner's mind" to your shooting.

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ZEN IN THE ART OF SHOOTING

Beginning Zen in Shooting

By Translation by John Dreyer

Source: http://www.bullseyepistol.com/polmind.htm

Excerpts from an interview with Kanjuro Shibata XX, Sensei. While initially written for Japanese Archery (Kyudo) it has been translated and heavily edited to explore our modern precision pistol sport.

The Developing New Shooter

For the beginner, the fundamentals provide the basis of shooting. New shooters should carefully reflect on each and every shot as if it was their only shot. In all sports there are competitors, winners and losers. However, this is not the primary focus in shooting. If understood through a Zen perspective, shooting is based on the idea "one shot" (issha). The focus and reflection on each shot is the most important thing. In the beginning, learning technique is the first priority of the new shooter. However, most beginners forget about this and think of the target and their score too much. It is not important where your shots go. That is only a reflection of the accuracy of your technique and the purity of your mind. Working on the fundamentals refines your technique. Scores are secondary - improved scores will come in time. As the new shooter continues to work on proper technique, he begins to encounter the Zen "Seven Defilements" of mind. The way to set aside these defilements is through the Zen Art or "Way" aspect of shooting. This Zen "Way" is endless. Practice never ends, and begins again with each shot. You don't shoot the target. Keeping to the fundamentals, the shot will go to the bullseye, just as if the shot had its own mind. It is not "you" shooting at the target. The right mind and the right heart, not just the right form, shoots the target. This right mind will be gained by the fundamentals, reflecting on both accuracy of the technique as well as on the Seven Defilements. The Seven Defilements: 1. Happiness (Yorokobu) 2. Anger (Okoru) 3. Greed-Uneasiness (Urei) 4. Expectation (Omou) 5. Sadness (Kanashimu) 6. Terror (Osoreru) 7. Surprise (Odoroku) The ultimate goal of shooting is to cleanse or polish your mind, precisely the same as sitting meditation, or "zazen." By perfecting technique, you are not polishing your style of shooting, but rather the mind. The dignity of shooting is the important point. Without the right mind, no matter how long you shoot, this dignity won't be gained.

More Than Just a Sport

Today, shooting is being practiced by thousands of people all over the world for the

development of mental discipline as well as for spiritual development. The simple elegance of the mechanics, the beauty of the equipment and the atmosphere of solitary dignity predominant at the range, have a great attraction for those who wish to walk upon the path of self-knowledge. If you look at it from the outside, shooting seems to be just a game. Raising the pistol and shooting at the target resembles a test of skill, but shooting is much more than just a sport. To discover the true nature of shooting one has to look inside, to cut through and go beyond any kind of preoccupation, whether it be worry, hope, doubt or fear. Although the actual forms of competitive shooting have changed over and again and equipment has become more sophisticated over the last two centuries, the essence of true shooting practice always remains the same. It is the epitome of traditional Zen "standing" meditation.

A Spiritual Path

Although shooting is certainly not a religious practice, we may still apply the philosophies of Zen Buddhism and Shinto to understand its true value. In this light, the heart of shooting is truly linked to Buddhist philosophy, with the interpretation of Tao merged into it. The teachings of Zen Buddhism tell us that our true selves are hidden within deep layers of habitual thought patterns, self-delusion and ego. We live in a dream-world of our own making. The aim of Zen practice is to wear away these layers of illusion and ego so as to be free from the dualistic outlook that keeps us from understanding our true nature and living harmoniously with ourselves, others and the universe at large. In sitting meditation, one strives to unify body and mind, the medium of the breath, and maintaining a strict sitting posture. Shooting as "Zen in action" incorporates the same concepts of mind and body working in unison. To the sincere practitioner, shooting is a way of life, and there should be no separation between shooting training and everyday activities. Each shot is shot as if it were the only one, just as each moment of one's life is the ultimate. The shooting practitioner does not look at the target for the result of his or her practice, but inward, for the target is not a target - it is a mirror. And if the heart is right, each shot clears away some more of the obstacles clouding the vision of one's true nature.

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ZEN IN THE ART OF SHOOTING

Zen Elements of True Shooting

By Earl Hartman

Source: http://bullseyepistol.com/zelement.htm

Makoto, Tekichu, Sae, Shahin, and Heijoshin - The Elements Of True Shooting

The Shaho Kun touches upon various qualities that are fundamental to true shooting. While there are many elements that combine to create true shooting, and no list can be all inclusive, there are certain qualities which must be present in the shooting for it to be true. These are:

- Makoto - sincerity and whole-heartedness
- Tekichu - accuracy of shooting
- Sae - clarity of shooting
- Shahin - quality of shooting
- Heijoshin - everyday mind

Makoto - Sincerity and Whole-Heartedness

To reach the highest level in shooting, you must have what is called "Makoto," or sincerity and whole-heartedness. *Makoto* is an attitude toward the act of shooting which is absolutely pure, untainted by any selfish desires, thoughts of success or failure, fear of loss, or desire for gain. The ultimate expression of this sincerity is expressed in the phrase "Issha Zetsumei," or *"one shot, one life."* Issha Zetsumei refers to a spiritual state where you pour every faculty of body, mind, and soul into the shooting, treating every shot as though it were the last shot of your life. As the introduction to the Shaho Kun says, you must "cleanse your mind, correct your body, and, with singleness of purpose, nurture the proper spirit, train yourself in proper technique, and throw yourself into your training with all of the sincerity you can muster. This is the only way." If you can do this and shoot with all your heart, soul, and might, you will rise above the petty attachments and delusions that cloud your mind and your shooting will become pure. This is Makoto.

Tekichu - Accuracy Of Shooting

As the Shaho Kun says, "spirit settled, this becomes harmonious unity". If you can achieve *makoto*, your mind, body, and pistol will finally become harmoniously united. This is called Sanmi Ittai, or "the three essentials as one body". When you achieve sanmi ittai, the "Muhatsu

no hatsu," (release of no release) is born, and striking the target becomes an inevitability, a natural consequence of true shooting. This is the pinnacle of accomplishment in shooting, referred to as Seisha Hitchu (a true shot never misses), or Seisha Seichu, (true shooting, true hitting).

It is from this point of view that accuracy in shooting must be evaluated. In the preface to the Shaho Kun, Yoshimi Junsei states clearly that the discipline of shooting means to shoot tens; thus, hitting the target is not without value, as some people may mistakenly assume. In shooting, it is axiomatic that a person who has reached a high level will possess impeccable technique and therefore will shoot accurately. Indeed, you must master technique before anything else can be accomplished. However, hitting the target does not, by itself, prove anything other than that you have learned a set of mechanical skills. The quality of the shooting and how the accuracy is achieved are more important than the accuracy itself. Indeed, what might be called "hitting-the-target-ism", for lack of a better term, is despised as the lowest form of shooting, and such skill is always limited and will eventually fade away. shooting values the accuracy that flows naturally from sanmi ittai. Therefore, you strive to achieve the state of sanmi ittai so that the spirit will bring the technique to life. True accuracy will surely follow. You must shoot with absolute faith in this principle, or harmony will never be achieved.

Sae - Clarity Of Shooting

When the mind, body, and pistol are in harmony, shooting comes to display the quality called "Sae." *Sae* is a very multi-faceted term and refers in shooting to a completely unclouded clarity, sharpness, vibrancy, and power where the purity of the shooting shines through. When the shooting has *Sae*, your spirit, enlivening the pistol through the natural functioning of the body, allows the true nature of the pistol to be clearly displayed without any obstruction. When this happens, the pistol seems to be truly alive and animated by a tremendous, free-flowing power that transcends the simply physical.

In a very real sense, the liberation of your spirit liberates the pistol and brings forth *Sae*. Like a musical instrument that will not reveal its true sound except under the hand of a master musician, the pistol will not display *Sae* unless you can shoot it as its inner nature demands. Since it cannot be imitated, *Sae*, or the lack of it, shows whether or not you have grasped the truth of shooting. This is why it is said that "the true pistol does not lie" and why shooting is seen as a way to understand truth.

Shahin - Quality Of Shooting

With *Sae* must inevitably come "Shahin," or the "quality of the shooting." This concept can also be expressed as "Shakaku," the "Dignity of the shooting." Both these terms refer to the

natural grace, refinement, and nobility displayed by an shooter who has grasped the truth of shooting. *Sae* presupposes the mastery of your own mind, which is expressed through the mastery of technique. In achieving this mastery, you must honestly face your own true self, and, in the process of training, lose all the false bravado and swagger that are the marks of the immature and shallow person. This refined character naturally appears as *shahin* and *shakaku*, which Yoshimi Junsei describes as resembling the beauty of the dawn sky, "the golden body, shining white, and the half moon positioned in the West." Like *sae, shahin* and *shakaku* cannot be imitated or created. Someone who lacks shahin yet tries to imitate it will only seem pretentious and flamboyant. The degree of shahin proves beyond all doubt the level of your progress.

Heijoshin - Everyday Mind

The Shaho Kun says to place the "Kokoro (spirit) in the center of the whole body." This is none other than Heijoshin, or "Everyday mind". *Heijoshin* refers to a state of mind where everything can be viewed with equanimity. This is expressed in the old shooting saying "Treat everyday practice like an important event (a competition or a test) and an important event like everyday practice". Heijoshin means that regardless of whether you are shooting in your backyard or at Camp Perry, your mind will always be naturally focused yet composed, not viewing either situation as being less or more important than the other. Without *heijoshin*, your mind will always be selfishly attached to the results of the shooting, and you will be unable to cultivate the true shooting mind.

The shooting approach to achieving heijoshin can be expressed in the following verse from the Fudochi Shinmyoroku (Divine Record Of Immovable Wisdom) taught by the Zen priest Takuan: "Keeping the mind tranquil as it moves in the myriad directions in the midst of uproar and commotion is true tranquillity. Tranquillity in tranquillity is not true tranquillity; it is tranquillity in action that is the true tranquility" You cannot achieve *heijoshin* by thinking about it or pretending that you have it. You achieve true heijoshin through overcoming your limitations by facing situations that test your intestinal fortitude. Your shooting then becomes clear, steady, and flawless, the reflection of a clear, steady, and flawless mind.

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ZEN IN THE ART OF SHOOTING

Issha Zetsumei: One Shot and Expire

By John Dreyer

Source: http://bullseyepistol.com/issha.htm

Introduction

"Issha Zetsumei" is a Japanese phrase whose image provides a masterful foundation for the art of pistol shooting. When translated to English it means literally "one shot and expire." The saying conjures the image of the last act of a dying man - a shooter summarizing his entire life with a single shot. In other words, the goal is to establish in your mind that you have no other purpose or concern in life than to properly execute the current shot.

"When you do something, you should burn yourself completely, like a good bonfire, leaving no trace of yourself." - Shunryu Suzuki Roshi

In Zen, it is recognized that there are no second chances in life. One must throw himself fully into any activity he is participating, regardless of how mundane it might be. Each activity should be done as if it were their only activity on Earth. In shooting, this means to concentrate on each shot as if it were the only shot you will ever shoot. Jackson Morisawa Sensei explains this concept as "each [shot] is final and decisive as each moment is the ultimate."

It is both the challenge and valuable purpose of the art of shooting to help us attain the freedom of our being, both physically and mentally. Shunryu Suzuki Roshi has said, "When you do something, if you fix your mind on the activity with some confidence, the quality of your state of mind is the activity itself. When you are concentrated on the quality of your being, you are prepared for the activity."

Mushin

When we practice the art of shooting, our minds must become calm and quite simple. But this does not happen by accident. Because by nature our minds are complicated and very busy, all of our activity in life is usually shadowed by traces of other thoughts. These traces and notions make it hard to concentrate fully on what we are doing. In fact, most people have a double- or triple- notion in one activity. The saying, "to catch two birds with one stone" states what is happening in their minds. Because they are trying to catch too many birds, it becomes difficult to concentrate on one activity, and as a result, they may not catch any birds at all! In other words, if you leave a trace of thinking on your activity, you will become attached to the trace and not the activity.

"To take [your stance] is to have the right state of mind. There is no need to obtain some special state of mind."- Shunryu Suzuki Roshi

Our first priority in preparing to fire a single shot or string of shots is to bring all of our awareness to the present. The routine processes of taking our stance and focusing on our breathing should act as a trigger to automatically bring our thoughts fully to the moment at hand. To understand this mindset, try to picture the image of a digital clock - this instrument has no sense of past or future; all it has is the moment. Surely you've watched one to see how it changes - take this as a challenge to "catch the ceaseless flow of quality in the passing moment." How much quality can you compress into any moment by simply being there and nowhere else? This is the magic and life-lesson of our art.

The state of mind that we are trying to attain in Zen and the arts is called "mushin" in Japanese, roughly translated to mean "empty mind." While in English there is no exact equivalent to the word, Professor Daisetz Suzuki describes it as a state of "conscious unconsciousness." Because of the inseparability of mind and body, you cannot view mushin as purely a mental phenomenon. You cannot simply reason yourself into mushin, you must utilize proper breathing and stance to create it. When you have assumed your proper shooting stance and are preparing to raise the pistol, recall Thich Nhat Hanh's saying, "Breathing in, I calm my body. Breathing out, I smile." Awareness of our breathing is the biggest key in uniting our mind and body, the first step in creating mushin. Focusing on our breathing will enable us to lose any thoughts that might be clouding our minds and will allow us to "become" the present moment. As a result, you will lose all sense of yourself and other and will be able to complely concentrate.

Heijoshin

Attaining mushin while shooting is certainly a challenging task. Any number of delusive thoughts can distract us and cloud our minds. It can be thoughts of trying aspects of our daily lives such as work, finances or relationships that can intrude our concentration. But the thoughts that are most difficult to keep from following are those relating to our shooting performance itself. Each shot must be made without any regard for past performance or to one's performance in the future - concern about how the aggregate score is going to be affected cannot be a consideration. However, the truth is that most of the internal dialogue that people have in their minds center upon thinking about and planning what they will do. These thoughts are delusive - this planning for the future simply keeps them from concentrating fully on the present and impairs their performance. "Have you ever gone on a well-deserved vacation and in the middle of all of the fun and relaxation found yourself thinking about when the trip will end?" asks Sensei Dan DeProspero. Everyone has done so - and this is precisely the delusional type of thinking we must avoid in shooting.

"The Way is ... not picking things up along the road." - Tung-shan Shou-ch'u

It can be said that the real value of the art of shooting is not what we gain from it but what we lose. Lao Tzu has said, "If we learn, we accumulate day by day. If we study [a Zen art], we reduce day by day." In the practice of a Zen art, unnatural fixations ranging from muscular tension and immature attitude to viscious circles of interpersonal interactions are to be lost. It is this manner that our true self emerges. Calmness of spirit in all circumstances, acting spontaneously without restraint, acting without concern for public recognition, seeing things as they truly are, and realizing that every day is a fine day are positive qualities that are already within us. Our practice of shooting will allow us to unlock those qualities by shedding the delusive thoughts we have collected that weigh us down that mask our true self.

One objective in our practice of shooting is to achieve "heijoshin," which translates from Japanese to "ordinary mind." The Zen definition of an "ordinary mind" is one that is calm, well-balanced and disciplined at all times. The secret in life to attaining this mindset is to treat ordinary moments as special and as a result special moments will seem ordinary. In shooting, this means not grieving over bad shots, but also not to rejoice over good shots either! In either case, analyze the execution of the just fired shot and leave it - do not carry its memory any further. Preparations for the next shot, the only shot that matters, must begin immediately. This is particularly difficult because of the cumulative nature of an aggregate - good shots or bad - there is a natural tendency to foolishly reflect on how the aggregate score is going to be affected even BEFORE we fire a shot.

Zanshin

"Zanshin" is an important term in any activity we practice as a Zen art. It translates literally to "remaining heart" and partly refers to an element of the fundamentals known as "follow-through." However, follow-through is somewhat of a trivialization of Zanshin. On a deeper level, Zanshin means that the state of mind and body used in executing an action is not dissipated by it, but is carried over into the next activity. On a physical level, breathing and stance remain correct; the body remains in balance, arm and shoulder muscles remain flexed, and we are ready to fire another shot. On a mental level, one's awareness and concentration remain after the trigger break and "calling of the shot." It is at this point where we must come to terms with the quality of the shot just fired, but we cannot dwell on it. If so, our concentration will lapse. Zanshin, therefore, is the critical phase during which is is determined whether the shot just fired will haunt the shooter or whether he will be able to mantain mushin.

Understand that the ego is the source of any grief or anger we might feel after a poor shot, not the moment of the shot itself. The problem of grief arises when we subordinate the moment to our self-centered thoughts: not just this moment, but what WE want. Most emotions do not

arise out of the immediate moment, such as witnessing a child getting struck by a car, but are generated by our self-centered demands that life be the way WE want it to be. While it is not bad to have such emotions, they have no importance in themselves. When such emotions emerge, our feelings should be identified and looked at with interest, but nothing more. Our attention should remain in the present moment. If we become upset, we have lost our attention to the present moment. Conversely, if we can pay complete attention to the present moment, we will never become upset.

Fudoshin

The ego is a form which has been developed over countless generations of life. It is a technique used by everyone for living in a civilized manner. The practice of pistol shooting allows us to transcend the ego by mastering it. While we progress in our art, we will gain increasing insight into our own psychological makeup. This can be a painful process, for we must acknowledge flaws in our character that make us shoot poorly at times. Greed, competitiveness, vanity, self-criticism, shyness, fear, need for approval - are but a few of the personality traits that can lead to delusive thoughts that cloud our awareness. We must recognize these flaws before we can transcend them. Recognize that shooting is an ongoing struggle with the ego, a battle with oneself. In this battle, one must face physical and psychological pain in order to experience the fruits of victory. From our practice, one learns to endure pain and austerity, delay gratification, concentrate fully on the task of the moment, each moment, and refine form to the nth degree. The ultimate objective of our continued practice of shooting is losing the ego and realizing the freedom of the true Self.

"Don't be afraid of failure. See it as a learning experience that provides an opportunity for new growth." - Hideharu Onuma Sensei

Quite possibly, a person's most powerful weapon for survival and success is the ability to believe in oneself. It is true that one of the most difficult points in the practice of any Zen art is coming to terms with the fact that we alone are responsible for everything that happens. As a result, this reality creates a natural stress arising from doubts of our ability to perform well. However, it can be said that these self-doubts are nothing more than delusional thoughts. Just as you do not drive to work every day fearing that you are going to be fired, you cannot allow thoughts of the fear of failure cloud your mind while shooting. It is imperative that any stress we experience be identified in order that we can set our delusional thoughts aside. Once identified, stress can be seen as a positive thing. Armed with self-confidence, stress can be rechanneled into positive energy for the intense mental concentration that is necessary to succeed.

"Fudoshin" is a Zen term that translates to "immovable mind," a mind that is so clear and unclouded that nothing can move it. Since it sees everything as it really is, the immovable

mind harbors no hesitation, confusion, doubt or dishonesty, and therefore fears nothing. Like an irresistible natural force that sweeps away everything in its path, this mind is invincible. Nothing can stand before it and nothing is beyond its grasp. Takuan Roshi explains, "Keeping the mind tranquil as it moves in the myriad directions in the midst of uproar and commotion is true tranquillity. Tranquility in tranquility is not true tranquility; it is tranquillity in action that is the true tranquility." While some people may view this state of mind in an overly romantic way, divorced from mundane considerations, it has, so far as shooting is concerned, a very practical application. This mind gives birth to flawless shooting and so the results of your shooting will show plainly whether you have achieved it or not. Therefore, you can only grasp it through shooting, and when you grasp it, your shooting will be perfect.

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ZEN IN THE ART OF SHOOTING

Zen Practice, Action, and Life

By John Dreyer

Source: http://bullseyepistol.com/zenpract.htm

Introduction

Minutes after firing an outstanding 2672 at a sanctioned match one day, James Henderson was approached by a timid sharpshooter-class competitor who asked, "What's the key to shooting those kind of scores?" The questioner was hoping for a magical answer emphasizing some particular aspect of the fundamentals or mental focus that would quickly loft him into personal success. To his disappointment, Jim's response was a masterful, "Lots and lots of practice." To expand on that thought, it is your *results* that will show to what extent you understand the art of shooting -- and the only way to understand the art is to *practice it*. That is accomplished as Yoshimi Junsei Sensei explains, "Searching only in yourself, you must cleanse your mind and correct your body, and, with singleness of purpose, nurture the proper spirit, train yourself in proper technique, and throw yourself into your training with all of the sincerity you can muster."

"Since shooting is honest and is a mirror, if you have any dishonesty or problem, this will surely reveal itself in the results." - Suzuki Hiroyuki Hanshi

The physical act of shooting cannot be distanced from the mind and spirit and there is no separation between the mind and the body or between technique and spirit. They are interdependent, so by properly training in technique you train your mind and spirit. To the average Western mind this may not be as obvious as it sounds; but it is one of the most fundamental things to understand in the practice of pistol shooting. What it means in practice is that every physical action is viewed as a mirror of the state of your mind and spirit.

Seishin Tanren

It is important that you base your shooting practice on an understanding that will help you focus your energies on shooting correctly. You can then visualize and internalize the spirit of the art and express it through action. If you keep the following points in mind, you will be able to practice with the right mind and spirit.

- The ultimate truth of shooting is understood through the union of mind, body, and pistol, or "Sanmi Ittai" (the three essentials as one body).

- Incessant practice, done correctly with a pure mind and spirit, is the only way to understand shooting. Nothing can take the place of this.
- Growth in shooting has a natural progression. Instruction and practice must be in accord with this natural progression.
- Shooting tradition is the living legacy of past masters and the guide for future development. Therefore, dedicated practitioners must always follow tradition and the structure it imposes.
- It is impossible to ever be perfect in shooting; however, we must never stop training. It is the devotion to constant training that gives meaning to shooting as a Way of life.

True shooting can only come from within the depths of your own being, from your guts and your blood. For this process to work, there is one vital trait that you absolutely cannot be without: the capacity for honest self-evaluation. This does not mean evaluating little details of technique. It means to really look into your own soul for the true causes of things. If you are not honest with yourself, you will always make excuses for poor shooting and will never be able to accept whatever shortcomings you may have. Everyone wants to think highly of themselves, but if this prevents you from seeing your shooting as it really is, you will never make any lasting progress. Each shot gives you the opportunity to see your true self. When you do, you must honestly accept whatever deficiencies you may find, reflect clearly on them, resolve to correct them, and then act on that resolve, come what may.

People do things for many reasons. Without realizing it, most people have ulterior or impure motives for their actions and these motivations affect their shooting. Some people want to be liked and admired for their accomplishments; some aspire to position and power; some love competition and the adulation achieved through victory; some fear losing and so shun anything having to do with competition; some hold themselves in high regard and therefore believe that their way is right while others are wrong; others think little of themselves and value everyone's opinion but their own. Peoples' deepest character traits, even those unknown to themselves, affect everything they do, including their shooting. A physically strong person will shoot with confidence and power, but with too much aggression and brute force. A spiritually weak person's shooting will vacillate and lack decisiveness. A person whose mind is erratic will shoot erratically. A sloppy person will shoot sloppily. A tense person's shooting will be tense and an angry person's shooting will be angry. As the person is, so is their shooting.

"It is necessary, through the exercise of technique in your shooting practice, to sufficiently master such mental and spiritual aspects as mental calmness, the development of spiritual power, mental and spiritual concentration, and decisiveness." - Murakami Hanshi

This is the ability of the art of shooting to act as a vehicle for "Seishin Tanren," which translates to the "forging of the spirit." It is held by most masters of any art that it is not possible to ever be completely perfect in one's art or one's life, yet you must never give up the effort, since it is this effort itself which trains you. Therefore, it is not the result of training but rather the act of training itself that is important. Pistol shooting is a Way in that it seeks to use the art of the pistol primarily as a medium for training your mind and spirit. This training will lead to great proficiency in the art of shooting if you are sincere and diligent, but this is not the sole objective. The development of character through training is the real goal. However, shooting is like anything else in life: it will give back to you exactly what you put into it. If you treat it as a sport, it will be a sport. If you treat it as a diversion, it will be a diversion. If you treat it as a Way, it will be a Way. Therefore, it is up to you yourself to imbue shooting with meaning through your own efforts.

Makiwara Hanshi

It is a common occurance among the developing shooter to be able to shoot high scores in practice sessions, but then cannot display the same ability when shooting in competition. This is because the mind, being swayed by negative attitudes such as attachment and desire, becomes agitated, making it impossible to put forth all of one's skill and technique. In short, they are defeated by the working of your mind *(shin)* and spirit *(ki)*.

This phenomenon, described simply as "Makiwara Hanshi", or *"master of the practice target"* refers to someone who can shoot high scores in practice but falls to pieces the minute he step to the firing line in real competition.

All shooters sometimes lose their composure. At important events, you can be affected by the atmosphere of the event. As the strong internal and external stimulation causes increasing excitement, both your mind and body become abnormally tense; technique which you can normally perform smoothly and without thinking loses its cohesiveness, continuity, and sustainability; your body and mind become rigid; and you are seized by uncertainty and become unable to control your own mind. In short, you lose your self-control. No matter how strong and skillful you are, you cannot fully exercise your ability if this strength and skill is not accompanied by the proper working of the mind and spirit.

The target and the mind have a relationship similar to that of a catalyst and that which it catalyzes. When a catalyst is introduced into a seemingly stable chemical solution, for instance, a violent reaction will occur if there is anything in the solution that is receptive to the catalyst. In the same way, the target, taken by itself, is meaningless, and your mind, not affected by thoughts of the target, can remain calm. When the target and your mind are joined through the act of shooting, however, the target suddenly assumes overpowering importance. You see the 10-ring as an object that must be penetrated with every shot. It is the goal, the

objective, the physical representation of all your hopes and fears. Hitting it means success, and missing it failure. Everyone reacts to it in a different way, based upon what it represents to them.

Therefore, it is of the utmost importance for you to face the target and, through shooting, to learn to deal with the emotions it engenders. You cannot lie to yourself about how you feel about the target, for the shooting does not lie. It will be obvious for everyone to see. An honest shooter, therefore, does not shy away from this confrontation, for this confrontation, more than anything else, holds the key to understanding the essence of shooting as a Way.

Murakami Hanshi says that the "target on its wooden frame is just a physical object, a medium for your own mind. This physical target is unmoving. However, the target of the mind is always restless and tumultuous and is never still." When, through constant practice, you have rid yourself of the *"Seven Barriers"* brought into sharp focus by the target, your mind will become still and pure and the shooting will be natural and flowing, like a stream of pure water gushing from an unpolluted spring. As Ise Tadatake says, "training the mind is the most vital thing. If the mind is agitated the spirit is agitated; if the spirit is agitated the heartbeat is agitated; and if the heartbeat is agitated the whole body is agitated, so the target will not be struck". Therefore, in order to train your mind, your selfish attachment to the target, which is the source of your mental agitation, must be faced squarely and overcome.

For this process to work, there is one vital trait that you absolutely cannot be without: the capacity for honest self-evaluation. This does not mean evaluating little details of technique. It means to really look into your own soul for the true causes of things. If you are not honest with yourself, you will always make excuses for poor shooting and will never be able to accept whatever shortcomings you may have. Everyone wants to think highly of themselves, but if this prevents you from seeing your shooting as it really is, you will never make any lasting progress. Each shot gives you the opportunity to see your true self. When you do, you must honestly accept whatever deficiencies you may find, reflect clearly on them, resolve to correct them, and then act on that resolve, come what may.

Heion

If you can truly shoot with all your heart, soul, and might, you will transcend your attachments and delusions and achieve "Makoto," the "stainless mind". At that instant this stainless mind will, like a smooth and perfect mirror, intuitively perceive the undistorted essence of the shooting. All doubt and fear vanish and are replaced with an unshakable confidence and certainty. This spirit of truly knowing, deep down in the marrow of your bones, that everything is all right, that there is no need to hurry, to be agitated, to be afraid, or to hesitate, is called "Heijoshin", or "everyday mind," the mind that can face everything with calmness and equanimity. It is also called "Fudoshin," or "immovable mind;" that the mind

that is so clear and unclouded that nothing can move it. When the mind is freed in this way, tremendous power, energy, and vitality are unleashed. The Zen priest Takuan has taught, "Keeping the mind tranquil as it moves in the myriad directions in the midst of uproar and commotion is true tranquillity. Tranquillity in tranquillity is not true tranquillity; it is tranquillity in action that is the true tranquillity". This "uproar and commotion" definitely occurs in shooting - but not from the outside, but from the inside, from the target of your mind, "restless and tumultuous and never still." When you shoot, you are always assailed by doubts and fears that sap your strength and vitality. "Am I on pace for the new slow fire record? Will I clean this rapid fire target? Am I going to choke and throw a shot? Will I break 870 to win the .45 aggregate? What will my competitors think of my score? Am I better or worse than so-and-so? " Fudoshin cuts through all of these phantoms of the mind, reducing them to nothing, making it as though they never existed. All that is left is the brimming power of the immovable mind and the incandescence of pure shooting, the "shower of sparks born of the collision of iron and stone". The result is true "Heion" or "tranquility."

"Not to be attached to external forms, not to be unsettled within, not to think this and that, not to be cluttered with extraneous things, not to think about gain and loss and whether we are happy or sad. This can be called Zen." - Shodo Harada Roshi

Here is an magnified image that will help demonstrate how thoughts of fear can be can cloud your focus... Imagine that after a period of imprisonment, you have been selected to be executed. The guards have now come for you and are marching you from your cell to a guillotine in the courtyard for the execution. Would you be able to remain calm and composed during the walk? Would you be able to notice all of the details along the way? Count the number of steps up to the guillotine? Breathe normally? Would you be able to willingly stick your head into the guillotine?

Or possibly this more familiar image... Imagine taking your vehicle on a road trip to arrive at a distant destination by a specific time. On the road, the average driver's thoughts would be fixated upon the destination itself, possibly even planning what will happen after they arrive. Because of their preoccupation with their destination, they fail to truly experience all of the many events of the trip itself. Furthermore, if there are any difficulties or delays during the drive, they will begin to think about the ramifications of being late to their destination. They will spend the remainder of their trip consumed in thought of the various potential ramifications of their possible late arrival. Dwelling on these delusional thoughts will likely make your driving less efficient and safe. But it will *definitely* make the long journey unpleasant! Now compare this road trip to the firing of a pistol match. You will quickly see how destructive it is to dwell on our destination *(aggregate score)* as we encounter difficulties *(poor shots)* along the way. It will do nothing but cause us undo misery and increase the possibility of further lessened performance.

Mui Shinnin

While delusive thoughts of potential poor performance may haunt us and erode our performance, the elation from our *successes* during a match may easily distract us from executing the fundamentals. For example, knowing that you just fired an outstanding NMC may cause your mind to drift, resulting in a disasterous timed and rapid fire.

"Be satisfied with your accomplishments only at death." - Yamada Jirokichi Hanshi

There is an old saying, "don't get caught up reading your own press." In the short-term, it is a warning to not admire that last good shot or string and stay on focus to execute that at the moment. You must postpone your celebrations until after the grand aggregate is posted. Likewise, in the long-term, it is just as easy to become distracted as victories and classification promotions come our way. Yamada Jirokichi's, "be satisfied with your accomplishements only at death" is an admonition to never stop training and seeking to improve yourself. But we must remember that the truth of this statement is not limited to shooting; it truly applies to everything in life.

"Make yourself master of your situation; wherever you stand is truth." - Rinzai Gigen

From the Rinzai doctrine of Zen comes the masterful image of the person as "Mui Shinnin" or the "True person of no status." Rinzai Gigen asserts that it is one's "original nature" to be sought through Zen instead of no-thinking. Rinzai has told that "there must be a true person before there can be true knowledge." The *true person* is one of Rinzai's common descriptions for the sage who acts spontaneously, responsively, and without a plot. Thus, the Rinzai image of the *true person of no status* is one who functions in his *original nature* and does not sense the demands to meet expectations or social standing. In this perspective, the competitive shooter's satisfaction must come from his focus on breathing and perfectly executing the fundamentals. He manifests his total self in the activity at hand, executing the current perfect shot, and has no concern for the accolades that may come.

"The individual we think of as body, vision, and consciousness, of what constitutes existence and function, is what Rinzai calls the true person of no status." - D. T. Suzuki

Every Day is a Good Day

If shooting is to be of any use to our lives we must be able to bring what is discovered to our daily life. Shooting does not remove us from life and the world but gives us the insight necessary to help better it. As a practitioner, one has the freedom to take the teachings to whatever purpose one desires. Practicing shooting as a Zen art lays its foundations on this very world we call everyday life. Zen principles help us notice how to view this world, and the life we are living. This is like a mother teaching her child how to gain its first steps on their

own. The potential for walking is already there and the mother just encourages that potential to arise. Once the student of Zen has gained the first steps, it is like a water spring that suddenly rises from the ground, for what that stream encounters and where it meanders, is the Dharma life of the Awakened one.

The ability to apply the teachings of Zen to daily life is very important. Zen uses our ordinary life as a vehicle to enlightenment. One must always keep mindful of what one does. One must understand the importance of ordinary activities, that they are all methods of Zen.

Although knowledge is generally viewed as a key virtue, it is not so in Zen. Action is more important, and knowledge that cannot be used to enhance life is of little value. It is interesting that many people read all about the benefits of Zen and try to intellectualize the experience with out ever really having it. Although this can be beneficial in following the path it can never substitute for the actual experience.

Again, Zen practice can not only be limited to our time while shooting, it extends to the times we are just doing our normal activities. It is only in *this* way we will find that every day is a good day. The reality is that everyday is just a day there are no good days or bad days. It is up to our attitudes to decide. Life is always in a state of change and there is nothing wrong with this because it is teaching us something each day. If we can only detach ourselves from the past and enjoy what we have right now, how much better will we feel? We must learn to flow from one moment to the next. If we can do this, every day will be a good day.

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ZEN IN THE ART OF SHOOTING

A Brief History of Zen

By A Look at the Origins and Evolution of Zen

Source: http://bullseyepistol.com/zenhist.htm

Zen is a subset of Buddhist tradition founded first in China and then later in Japan. The name of the sect derives from the Sanskrit dhyana [meditation]. In China the school early became known for making its central tenet the practice of meditation, rather than adherence to a particular scripture or doctrine.

The founder of Zen in China was the legendary Bodhidharma, who came to China from India in the late 5th cent. A.D. He taught the practice of "wall-gazing" and espoused the teachings of the Lanka-Vatara Sutra (whose chief doctrine is that of "consciousness-only"; see Yogacara), which he passed on to his successor Hui-k'o (487-593). According to tradition, Hui-neng (638-713) became the sixth patriarch of Chinese Zen by superseding his rival in the intuitive grasp of the truth of enlightenment, even though he was illiterate. The Platform Sutra, attributed to Hui-neng, defines enlightenment as the direct seeing of one's "Original Mind" or "Original Nature," which is Buddha, and this teaching has remained characteristic of Zen. A number of teaching lineages arose after Hui-neng, all claiming descent from him, and teaching the method of "sudden enlightenment best known in the West by the term satori. In its formative period Zen was influenced by both Taoism and elements of Prajna-Paramita Buddhism. The 8th and 9th cent. were the "golden age of Zen," producing such great masters as Ma-tsu, Nan-chuan, Huang-po, Lin-chi, and Chao-chou. The unique Zen teaching style developed, stressing oral instruction and using nonrational forms of dialogue, from which the later koan was derived. In some cases physical violence was used to jolt the student out of dependence on ordinary forms of thought and into the enlightened consciousness. Scholarly knowledge, ritual, and performing good deeds were considered of comparatively little spiritual value. After the great persecution of Buddhism in 845, Zen emerged as the dominant Chinese sect, due partly to its innate vitality and partly to its isolation in mountain monasteries away from centers of political power. Two main schools of Zen, the Rinzai (Lin-chi) and the Soto (Ts'ao-tung), flourished and were transmitted to Japan in the 14th cent. The Rinzai sect placed greater emphasis on the use of the koan and effort to attain sudden enlightenment, while the Soto patriarch Dogen (1200-1253) emphasized sitting in meditation (zazen) without expectation and with faith in one's own intrinsic state of enlightenment or Buddha-nature. The austere discipline and practical approach of Zen made it the Buddhism of the medieval Japanese military class. Zen monks occupied positions of political influence and became active in literary and artistic life. Zen monasteries, especially the main temples of Kyoto and Kamakura, were educational as well as religious centers. The Zen influence on Japanese aesthetics ranges from poetry, calligraphy, and painting to tea ceremony, flower

arrangement, and landscape gardening, particularly the distinctive rock-and-sand temple gardens. Japanese Zen declined in the 16th and 17th cent., but its traditional forms were revived by the great Hakuin (1686-1769), from whom all present-day Rinzai masters trace their descent. Zen thought was introduced to the West by the writings of D. T. Suzuki, and interest in the practice of Zen meditation blossomed after World War II, resulting in the establishment of Zen centers in many parts of the United States.

ZEN IN THE ART OF SHOOTING

Zen: Recommended Reading

By Superb Zen Books of Interest to Pistol Shooters

Source: https://www.bullseyepistol.com/zenread.htm

These are my favorite books on Zen, all inexpensive and readily available.



Zen in the Art of Archery
by Eugen Herrigel The first Western book on Zen, it was written by a German philosopher who studied Zen through the art of Kyudo. Almost all recent Western writings on Zen refer to it.



One Arrow, One Life
Zen, Archery, Enlightenment by Kenneth Kushner A tremendous recent book that explores the discoveries of the author as he progresses in Zen through his practice of Kyudo.



Zen Mind, Beginner's Mind
By Shunryu Suzuki An incredible collection of images and explanations of Zen teachings by a gifted and accomplished Zen master who established Zen in America in the 1970's.



Zen and the Art of Anything
By Hal French An easy-reading book that uses imagery and quotations to explain Zen in everyday life.



Illuminated Spirit
Conversations with a Kyudo Master By Dan and Jackie DeProspero The authors discuss all aspects of their studies under a Kyudo master in Japan. This is an outstanding practical book on Zen.



Zen Action, Zen Person
By T. P. Kasulis A very thorough examination of Zen practice and teaching. It is the first book on Zen to be written by an American philosopher and deeply dissects Zen for a Western audience.



Samadhi
Self Development in Zen, Swordmanship and Psychotherapy By Mike Sayama A superb work on Zen in practice written by a PhD in psychology.



Nothing Special - Living Zen
By Charlotte Joko Beck and Steve Smith A quality work by an American Zen teacher that discusses solutions in Zen to the many mysteries and difficulties of the human experience.

THE REFERENCE ROOM

Match Bulletins from the National Championships

By Maintained by the NRA Competitions Division

Source: https://www.bullseyepistol.com/natbulls.htm

National Championship Results Bulletins

Listed by the various Aggregate and Fired Matches. Click the year to view the match bulletin.

Preliminary Pistol Championship (Match 111)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber Slow Fire (Match 135)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber National Match (Match 136)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber Timed Fire (Match 137)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber Rapid Fire (Match 138)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber Championship (Match 101)	2005	2006	2007	2008	2009	2010	2011
CF Caliber Slow Fire (Match 139)	2005	2006	2007	2008	2009	2010	2011
CF Caliber National Match (Match 140)	2005	2006	2007	2008	2009	2010	2011
CF Caliber Timed Fire (Match 141)	2005	2006	2007	2008	2009	2010	2011
CF Caliber Rapid Fire (Match 142)	2005	2006	2007	2008	2009	2010	2011
CF Caliber Championship (Match 102)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber Slow Fire (Match 143)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber National Match (Match 144)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber Timed Fire (Match 145)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber Rapid Fire (Match 146)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber Championship (Match 103)	2005	2006	2007	2008	2009	2010	2011
National Pistol Championship (Match 100)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber Team Championship (Match 121)	2005	2006	2007	2008	2009	2010	2011
CF Caliber Team Championship (Match 122)	2005	2006	2007	2008	2009	2010	2011
.45 Caliber Team Championship (Match 123)	----	2006	2007	2008	2009	2010	2011
Pistol Team Aggregate (Match 112)	2005	2006	2007	2008	2009	2010	2011
.22 Caliber 2-Member Team (Match 124)	----	----	----	2008	2009	2010	2011
CF Caliber 2-Member Team (Match 125)	----	----	----	2008	2009	2010	2011
.45 Caliber 2-Member Team (Match 126)	----	----	----	2008	2009	2010	2011
2-Member Pistol Team Aggregate (Match 114)	----	----	----	----	2009	2010	2011

Indoor National Pistol Championship	2003	2004	2005	2006	2007	2008	2009	2010	2011

Further Reading

- 2005 (http://www.nrahq.org/compete/natpdf/cp111-05.pdf)
- 2006 (http://www.nrahq.org/compete/natpdf/cp111-06.pdf)
- 2007 (http://www.nrahq.org/compete/natpdf/cp111-07.pdf)
- 2008 (http://www.nrahq.org/compete/natpdf/cp111-08.pdf)
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- 2011 (http://www.nrahq.org/compete/natpdf/cp121-11.pdf)
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- 2011 (http://www.nrahq.org/compete/natpdf/cp112-11.pdf)
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- 2011 (http://www.nrahq.org/compete/natpdf/cp125-11.pdf)
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- 2009 (http://www.nrahq.org/compete/natpdf/cp114-09.pdf)
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- 2011 (http://www.nrahq.org/compete/natpdf/cp114-11.pdf)
- 2003 (http://www.nrahq.org/compete/secpdf/open-pistol-03.pdf)
- 2004 (http://www.nrahq.org/compete/secpdf/open-pistol-04.pdf)
- 2005 (http://www.nrahq.org/compete/secpdf/open-pistol-05.pdf)
- 2006 (http://www.nrahq.org/compete/secpdf/open-pistol-06.pdf)
- 2007 (http://www.nrahq.org/compete/secpdf/open-pistol-07.pdf)
- 2008 (http://www.nrahq.org/compete/secpdf/open-pistol-08.pdf)
- 2009 (http://www.nrahq.org/compete/secpdf/open-pistol-09.pdf)
- 2010 (http://www.nrahq.org/compete/secpdf/open-pistol-10.pdf)
- 2011 (http://www.nrahq.org/compete/secpdf/open-pistol-11.pdf)

THE REFERENCE ROOM

Studio-Recorded Range Commands as .MP3s

By Bring a MP3, CD or Tape Player to the Range for Practice

Source: https://www.bullseyepistol.com/rangecmd.htm

I think everyone would agree that the most ideal practice environment would be actual match conditions. Unfortunately, most of us don't have a buddy willing to call range commands and turn targets for us whenever we feel like going to the range to practice! The second best thing is a dedicated timer, an effective "range-officer-in-a-box," like the Personal Linesman. And although you might not have the cash for one of these, you probably do have a cassette tape player, a portable CD player, or even cooler, a portable .MP3 file player! The .MP3 audio files of the Range Commands will allow you to use your existing gadget as that coveted dedicated timer.
The below "download" links are to the two audio (.MP3) files, each containing the range commands for the particular stage of sustained fire. Once "downloaded," you will have two files, one a set of commands for a string of the *timed fire* stage, and the other for *rapid fire*. Once on your computer, you have several options. You may:

1. Burn them onto a CD that you can play on your portable player.
2. Transfer them electronically into a portable .MP3 player.
3. Play the two .MP3 files and record them with an analog cassette tape recorder.

Download the

Timed Fire MP3

,

or the

Rapid Fire MP3

THE REFERENCE ROOM

Administrative Tools

By For Match Directors and Stat Officers

Source: http://bullseyepistol.com/matchdir.htm

Tournament

- Competition Administration Program

The culmination of seven years of research and development, this program will handle all administrative aspects of a Pistol or Rifle competition! You may explore its amazing versatility, time-saving features and letter-perfect report generation at the Official Tournament Website. You gotta see it to believe it! Take me to the Official Tournament Website!

Score Cards for a 2700 Pistol Tournament

All you need is legal-sized paper and you can print out these score cards, 4 cards to the page. Bring the three printouts to your local copy or print house and have them printed on colored paper or cardstock and cut and you will be running one classy match. Download the Tournament Score Cards

Range Officer's Outline & Checklist for a 2700

Having a simple, yet complete outline of the program for a 2700 makes calling a match a pleasure, and helps prevent embarrassing mistakes that frustrate the competitors. This MS Word Document serves as both an outline and a checklist for your range officer. Download the Range Officer's Outline/Checklist

Mechanical Range Officer - MP3 Files

While this is not an option for Sanctioned 2700 Competitions, less formal practice sessions and gallery league competitions may be controlled with mechanical range commands. These two .MP3 Audio files contain the commands for operating either a string of timed or rapid fire, respectively. Download the Timed Fire MP3,or the Rapid Fire MP3

Sample By-Laws for an NRA-Sanctioned League

NRA Sanctioned Leagues make up a large percentage of the participation of competitive shooters. They are tremendous for introducing new shooters to the sport and offering competitive practice during the off-season. However, setting up teams and administration for a Sanctioned League can be quite a chore. This MS Word Document contains a recent edition of the by-laws of a 55 year-old League that can well serve as a foundation for your own League's operating guidelines. Download Sample NRA-Sanctioned League By-Laws

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Further Reading

- Take me to the Official Tournament Website (http://tournament.bullseyepistol.com)

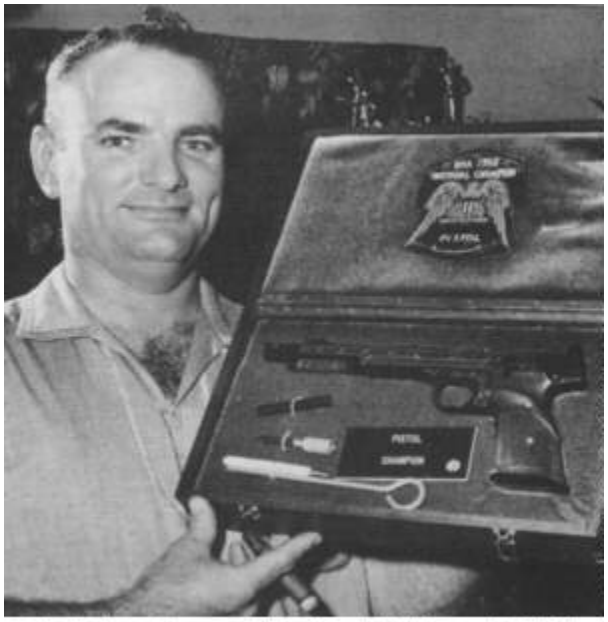
BIOGRAPHY AND HISTORY

James E. Clark - Bullseye Legend

By February 15, 1923 - January 7, 2000

Source: http://bullseyepistol.com/clark.htm

- excerpted from a 1989 article by Dean Grennell



Jim Clark receiving a gold inlaid Smith & Wesson Model 41 after winning the National Pistol Championship in 1958.

JIM CLARK shot NRA Bullseye Pistol competition for twenty-eight years, finally retiring from competition in 1975. Clark won the National Pistol Championship in 1958 and is the only civilian ever to accomplish this feat. He won the National civilian title five times. He was the fifth man in the U.S. to break 2600 and the fourth to break 2650. But his accomplishments don't end there. He has been fully active in the pistolsmith business since 1950, well known as one of the true innovators of pistol accurizing.

In 1985, the American Pistolsmiths Guild awarded Clark a trophy proclaiming him American Pistolsmith of the Year. Bagging double brass rings as the best pistol shooter in the country and its top pistolsmith is a unique coup: accomplished by James Edwin Clark and by no other known human being. I would say it serves as more than adequate excuse to feel a bit chesty about things. Mrs. Clark works in the office in the shipping-receiving department and they have a son, Jim Clark, Jr., who' s also a pistolsmith and shooter. The younger Clark racked up his first listed win in 1971, his twenty-eighth in 1988. Give him time and he may catch up with his sire, but that's a tough row for anyone to hoe, though his genes should be helpful. The

story of Clark's early years constitutes one of those triumphs over bleak adversity through sheer guts and perseverence. He was born on the fifteenth of February in 1923, down in Fort Worth, Texas. When he was 3 years old, his father deserted his mother and her brood of three. She struggled to keep the family intact as best she could, working as a seamstress. After a time, she remarried and, from Jim's viewpoint, everything didn't start coming up roses, as of that moment. His stepfather wouldn't allow Jim to own or fire guns during the lad's early years. The children had no toys other than what Jim was able to construct and share with them. Put another way, when the basic alloy is just right, adversity can serve as the tempering agent that, somehow, produces an uncommonly remarkable human being. Browning certainly was one example and Clark is another. Does this mean we should set up homes for over-privileged children and systematically mistreat the wee nippers so they' 11 grow up to be the leaders and innovators of generations to come? I hasten to note that I'm being facetious... probably. Despite being 8-1/2 months older than myself, Clark made a more routine passage through the school system and graduated from high school in 1942, rather than 1940. We both went into uniform in '42, however. Clark wanted to be a Navy pilot, but the swabbie docs thought they spotted a trifling defect in the cartilage of his nose and declined his offer. So he went around to palaver with recruiters for the U. S. Marine Corps, all too innocently unmindful that they "take" from the USN. A glib-tongued Marine recruiter solemnly assured Clark that, if he'd only sign the paper and hold up his right hand, he'd go off to pilot training straight as a shot. So Clark signed up and they zipped him off to training...as a rifleman. Clark's family had moved to Shreveport, Louisiana, in 1939, about the time he got into high school and his favorite course had been mechanical drawing getting nothing but straight A's for that on his report card. He'd also joined the ROTC and became the captain of its rifle team. Thus, the USMC's arbitrary assignment wasn't all that grotesque. All they really did was nail the feet of a would-be birdman to terra ever-so firma. Clark got through USMC boot camp and was assigned to Camp Elliot, California, for scout/sniper training with the newly organized 4th Marine Division. Graduating with honors from that program, Clark saw service at Roi-Namur and Saipan with the division. Scout/ snipers serve as sort of the eyes and flashing fangs of combat USMC groups; a spectacular but small band of specialists. Most of Clark's duties were carried out well in advance of the American lines. He was issued a Springfield with a 10x Lyman target scope sight and came to like the rifle, meanwhile feeling somewhat less warmth toward the unwieldy and painfully vulnerable scope. During the Saipan landing, he stashed the scope in his pack and later, on unpacking it, discovered it had been demolished by a shell fragment, even though he didn't have so much as a scratch. He discarded the sightless Springfield in favor of a scavenged M-l Garand and an M1911A1 pistol, dealing himself into the surrounding fray. After a time, he acquired another Lyman-scoped Springfield from a sniper who'd collected some punctures and was being retired from cornbat, thus putting Clark solidly back into business as a scout/sniper. As Clark' s unit came up on the small village of Garapan, Clark had lost three different spotters to enemy fire. All by himself

and unattended, he perched alone upon a cliff, a thousand yards or so above a road leading out of the village. The road below was all a-crawl with Jap troops engaged in a frantic bugout. Clark managed to get his new scoped Springfield sighted in for the generous distance and fired over three hundred rounds, scoring nothing but hits after he'd "found the range." It seems likely he accounted for more enemy soldiers than he would had he gotten his original wish and won his wings. Clark put in twenty-one days of combat before catching a rifle bullet in the shoulder as he attempted to rescue a wounded partner. He was sent to the hospital and his left arm was paralyzed for six months. He spent the next year in various stateside Naval hospitals before receiving a discharge in 1945. Returning to Shreveport, Clark enrolled in college, majoring in architecture and took a job after classes, working part time in a local gun shop.



Jim Clark circa 1969.

It was in 1947 that a friend invited Clark to shoot in a pistol match. Using an as-issued GI Remington Rand .45 auto and hardball military ammo, Clark chalked up a seventy-eight percent at fifty yards on his first try. Considerably intrigued, he borrowed the needed three guns -- a.22, a .38 Special and an accurized .45 auto -- to enter a registered NRA match. He qualified as sharp shooter and made expert in his second match, master in his third. Many pistol shooters work for grim and dogged years to achieve master rank and a lot of them never make it.

When Clark shot his first score over 2600 -- out of a possible 2700 - he did it with borrowed guns, except for the Colt Match Target .22 he'd bought second-hand for $50. It was 1950 when Clark broke 2600 and 1960 when he became the fourth man to top 2650. With the outbreak of the Korean War, Clark was called back into service in 1951 and spent considerable time at Camp Pendleton, California, doing accuracy jobs on Marine competitors' .45 pistols. He encountered a .38 Super Colt that had been converted to handle the .38 Special wadcutter and was thoroughly fascinated by the prospects he saw in it. Upon his discharge in late 1951,

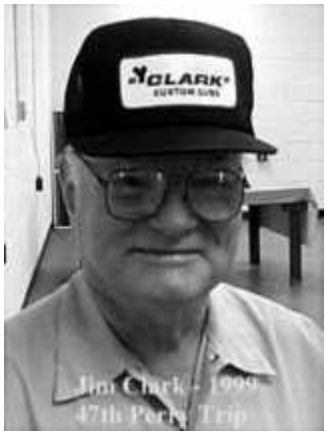
Clark went back to Shreveport with a Sheldon lathe and milling attachment he'd purchased for $600 from a Marine gunner. His friend Bill Gooch -- the same one who had introduced him to pistol shooting in the first place -- Loaned him money and helped him set up a building. Clark arranged a bank loan and bought twenty .38 Super Colts, converting them to handle the .38 Special midrange wadcutter cartridge. He ran an advertisement in the American Rifleman and it drew a heavy response from interested would-be customers. The Clark name quickly became known among pistol competitors and there has been heavy demand for his products and services ever since. Despite the time and effort of maintaining a successful business, Clark continued as an active competitor. Goaded by the heavy traveling entailed in match shooting, Clark learned to fly and bought an airplane, thus finally realizing a long-term ambition.


Clark's son "Jimbo" is a master pistolsmith and a prominent combat shooter.

During the years as a match competitor, Clark spent upward of $10,000 a year on match fees, travel expenses and practice ammunition. There are some who do not need a great deal of practice, but Clark admitted he is not one of them.

His top place in the U.S. Open Pistol Championship in 1958 was a singular achievement, as Clark was the first shooter to win that event without being sponsored by a military or police group. He footed all of the considerable expenses out of his personal pocket and, yes, he feels it was worth it. It was 1975 when Clark finally decided the game was no longer worth the candle and retired from the competition circuits so he could devote full time to running his shop. One of the house specialties was a target conversion of the .22 LR Ruger auto pistol to which he fitted a steel trigger of his own design, as well as stocks and barrels. He has turned out close to 30,000 such guns, along with nearly a quartermillion .45 accuracy jobs and .38 Special Colt conversions. As time permitted, Clark made a number of experiments aimed at

improving performance of the .45 ACP Colt. Among the things he discovered was that six inches probably represents the ideal length for the barrel. He began using Douglas barrels that incorporated an integral feed ramp of his own design. The novel concept served two vital purposes: First, it cured a lot of the feeding problems that stem from the juncture between the feed ramp at the upper front of the magazine well and the relieved area at the lower edge-of the chamber. Second, it supported the case head and thus prevented the occasional problem from blown case heads. Clark eventually manufactured such barrels entirely in his own shop.



Another Clark innovation was the long-slide conversion and that came about rather fortuitously. Visiting a war surplus store in Arkansas, Clark came upon an entire oil drum full of GI pistol slides that had been cut in two by the government in the process of deactivating service pistols. Clark bought the entire lot of destroyed slides for a dime apiece and spent the next several years turning them into long slides. With the supply finally exhausted, he had them made from scratch by an outside vendor.

With the advent of bowling pin competition in the mid-Seventies, Clark came upon another area of endeavor. He produced what probably was the first auto pistol expressly designed for that challenging event, As refined and finalized, the Clark pin gun has a coned sleeve at the muzzle extending in front of the slide and the same width. At the buyer's option, it can have recoil-control vents or it can be plain, with a choice of 5-1/4, 5-3/4, or 6 inch barrel lengths. Several other helpful features can be had such as a bevel around the base of the magazine well, stippled front strap, low-mounted sights by Millett, Bo-Mar or others, extended ejector and a lowered ejection port, Jim Clark was also an enthusiastic hunter and has taken dozens of deer, using an iron-sighted Model 29 Smith & Wesson with an 8-3/4 inch barrel, along with a great many smaller game species bagged with assorted other handguns. To read more about Jim Clark and the company he founded, Clark Custom Guns, visit their website at www.clarkcustomguns.com. Up until the day he died, Jim and company faithfully made the trip to Camp Perry every year.

Further Reading

- www.clarkcustomguns.com (http://www.clarkcustomguns.com/)

BIOGRAPHY AND HISTORY

The Legendary Pistol of Charles Askins

By In his own words

Source: http://bullseyepistol.com/askins.htm

This is the infamous story of the .22 caliber pistol that was converted to centerfire and helped Charles Askins win the National Matches at Camp Perry. The pistol was a Colt Woodsman with the gunsmithing done by no other than the legendary Frank Pachmayr. This story has been told so many times that the details are rarely straight. Here is the *real* story of the pistol, adorned with real photographs, in the words of the man who fired it, Col. Charles Askins:

"In 1937, while preparing for the National Matches, I developed the .221 Askins. This was a .22 caliber automatic which fired a .22 centerfire ctg. The NRA rules in '37 stipulated that "any centerfire caliber" could be fired in the Nationals. At that time the only useable centerfire cal was limited to the revolvers. You could fire the .32 S&W long which was the smallest caliber and one yet accurate enough to compete with. I concluded if I could develop a .22 centerfire I'd have the boys over a barrel."



"The .221 Askins was the 5.5 Velo-Dog cartridge shortened in length to exactly the .22 long rifle casing; the bullet was the standard .22 long rifle leaden slug at 40 grains. The Velo-Dog was designed to be fired in a French revolver; it had a rimmed casing. My gunsmith and I ran every case through the lathe and reduced the diameter of the rim to the same specs as the .22 long rifle rimfire shell."

"I had Frank Pachmayr alter the .22 Colt Woodsman to fire the centerfire ctg. This was not too difficult. He simply countersunk the face of the bolt to take the slightly thicker rim of the 5.5 ctg. He also redesigned the extractor to handle the new ctg. Changing the firing pin to strike center and not on the rim was no great chore."

"The pistol performed very reliably. There were no more jams than with the .22 rimfire and the accuracy was very good. I have forgotten what powder charge I used now but it was about 1-1/2 grains Bullseye. The biggest problem was saving empties. I had to erect a canvas sheeting to act as a baffle."

"I won the All-Around championship of the United States in 1937. In this match you had to fire once over the NM Course with the .22 rimfire, once over with the centerfire (everyone but me shot a .38 revolver) and once over the course with the .45 auto. The .221 Askins accounted for my winning."





"It also caused a storm of controversy! Word got all over Camp Perry about the pistol and those jazbos who were most likely to beat me howled the loudest. I got word before the Orton Match that my gun would be protested not because it was a .22 caliber (remember the NRA rules said, in those days, "any centerfire caliber") but because the rear sight was behind the hammer. I got wind of this and ran into Port Clinton and had an auto mechanic move the rear sight forward on the slide and simply solder it in place. If you get this gun you will see the somewhat crude job he did."

"The pistol really upset C.B. Lister and old Gen Reckord. They got the Board of Directors together, those who were at Perry, and they discussed barring my pistol somehow. But they could not quite decide how the rule book could be changed on such short notice. So I fired the highly controversial .221 Askins and won the All-Around Championship."

"The year following, let me assure you, the rules had been changed to read ".32 caliber and larger." It has read that way ever since!"

The above quotations are from a 1980 letter from Col. Askins to Col. Rex Applegate.

BIOGRAPHY AND HISTORY

Making Master in One Year

By Chris Johnson's Story, in his own words

Source: https://www.bullseyepistol.com/cj.htm

> *Special thanks to Chris Johnson for taking the time to chronicle his incredible story! You can quickly tell that he is a great one by how humble his is about his accomplishments. New shooters, you can look to this story for inspiration in pursuing your goals. You can do it! -JD*

I was asked to write an article entitled "On Making Master in a Year" but feel a better title might be "One Bullseye Shooter's Experience". Although I am currently classified Master both indoor and outdoor and a member of the 2600 club, I am far from a Zen-master or gun-guru or anything like that. But, I know first hand how much a new shooter can learn from the experienced ones and in that spirit, I will gladly share what I've done.

Background

I grew up on a dairy farm in northern Minnesota where there was never a shortage of work. If something didn't work, you had to figure out what was wrong, ask for help if you needed it, but you got it done. We hunted deer, grouse, and squirrel but never with handguns. College, a desk job (I work with computers) for the past 15 years, and three kids have changed my lifestyle but I still swim at least once a week. I'm about 5'8", 170 pounds and medium build.

Early Bullseye

I believe the first time I fired a handgun was when a friend invited me to an informal action pistol event. That was fun but you'd shoot for about a minute then stand around a wait for an hour. I heard about Bullseye and decided to give it a try. I fired in indoor league then an informal outdoor league two summers ago with a used Smith & Wesson Model 41. First few weeks my scores were about 60-65% average but by the end of the summer, I was up around 70%. I now know that is pretty typical for a beginning shooter. A beginner will have about five good shots on a slow-fire and five bad ones. Timed-fire will be OK but in rapid-fire a beginner will miss the target about once in every five rounds. Been there, done that.

Competition

That fall, I borrowed a friend's 1911 (iron sights) wad gun and fired my first 2700. I didn't break 1900 but I learned a lot. The most important thing I learned was that my guns most certainly shot better than I did. (As a matter of fact, they still do by a long ways) I enjoyed it so much that I talked to a few shooting friends, did my research, and ordered a 1911 .45 ACP custom build. I still shoot that gun and it still shoots way better than I do. Early competition also taught me to never change anything just before a match. (New reloading press, didn't get the crimp die set right, double alibi) Over the winter I fired off and on in a league with some good shooters and watched what they did and asked some questions. Most shooters are nice folks and will almost always share information. I fired a couple indoor matches and, after six months of shooting, shot a target with ten holes in the black. Later that season, I fired a 100 with the .22 and I almost fainted.

Perry

Camp Perry is like going to Chicago O'Hare for the first time; it's wonderful, so many people and so much to see and do, but you can't let the place overwhelm you. I had just read the AMU Handbook and been to Small Arms School and decided what I needed to do was FOCUS. I focussed my way to an 860 in the Preliminary Match on Thursday as a Marksman. I didn't sleep the entire night. I was not used to shooting those kinds of scores and the whole thing overwhelmed me. I was fortunate enough to place on the NRA/USA Civilian Marksman Team and came home from Perry thinking I had it all figured out.

Mistakes

After Perry, I fired four or five other outdoor matches and quickly received a sharpshooter and expert card in the mail. I began to realize that my scores were inversely proportional to the number of mistakes I made during a match. I would be clipping along with good slow-fires and then loose concentration or not set the gun down on a shot and fire a six. Then I'd follow it with another six. Rapid-fire would give me problems because I wouldn't stay focussed. I would have a 98 rapid-fire followed by a 84 with an entire five shot string out of the black. That was last fall. During the winter indoor season, I found my scores improving when I worked fundamentals on each shot or string. With the fundamentals (stance, grip, sight alignment, trigger control, follow-thru) under control, all I did was look for things I could do better (mistakes teach us this) and improve them. Now, I learn much more from a slow-fire

90-1x then a 97-5x because the 90-1x has four or five "mistake" shots and I figured out what I was doing wrong.

Running with the Big Dogs

This past spring things seemed to settle in nicely for me. Each time I would shoot, I would focus on fundamentals and look for mistakes. I saw my scores increasing on average and was very pleased to have shot 875 to win an Indoor Sectional and place first Expert in the Indoor Championship. Later in the spring, I went to the Dixie Match in Jacksonville with four goals. I focussed on fundaments, had some bad targets (because of loss of focus) but still met three of my goals; a personal best 2587, first expert civilian (those service guys/gals can really shoot), and I legged with a 270-10 in the ball match. Even though I had some bad targets, I figured out what went wrong and kept track of those things to make sure I focussed on them in the future. My big problem in Jacksonville turned out to be trigger finger placement.

The Wailing Wall

In order to stay mentally focussed, I learned that I must focus my conscious attention on the current shot or string. Worrying about my score, someone else's score, or the last bad shot will not help. Just before Perry this year, I went to Valporaso, IN thinking I was hot stuff. I came out blazing and fired a 881 rimfire score at my first match as Master. Then I made the mistake of looking at the Wailing Wall only to see Dave Glenn (a high master from Illinois and a great guy) had shot a 890. I didn't let it bother me until .45 rapid-fire match. I couldn't get it out of my mind and fired 92 and 93 at the end. I missed 2600 by four points because I didn't focus. The lesson I learned that day was to focus on me and my shooting, the round in the chamber now, and nothing else. Experienced shooters will chat and relax before a match but most will put on a game-face during the match. Mental focus; it can and will make a big difference.

Perry II

The huts aren't nearly as bad as some folks say. I went to Perry this year and had an even more enjoyable time than last year. One thing that contributed to my enjoyment was the success of some of my friends. For some strange reason, that seemed to help me relax and just shoot. I did well the first two days and knew that I had a good chance to break 2600. But every time I started to think about it, especially during slow-fire .45, I would stop myself from doing mental arithmetic. I would tell myself to just shoot one at a time and figure the score

when it's all over. My last rapid-fire target was 100-3x and I finished with 2603-91x. A couple of high masters overheard me when I did my final addition on the ready line and came over to congratulate me. You could have tipped me over with a feather. They signed my last target and I brought that one home.

Equipment

I currently shoot a Hammerli 208S with a 25mm UltraDot on a BME mount. I have anatomical grips made by Horton of Illinois. In practice I shoot CCI Standard Velocity and in matches I shoot either Eley Yellow or Federal UM-1. The CCI groups 1 " at 50 yards from a Ransom Rest, Eley Yellow about " and the Federal UM-1 just over a ". For centerfire and .45, I shoot a Masaki wad gun: Caspian frame and slide, Kart barrel, Wilson rib mount with 25mm UltraDot on the slide. I also have anatomical grips from Horton for this gun. For long-line I reload Hornady 200gr JSWC with 4.05 VV N310 and they've grouped x-ring with a flyer at 50 yards from a Ransom. For short-line and all indoor, I reload Behnke 185gr cast LSWC with 3.9 Bullseye and they group just outside x-ring with a flyer or two. Overall length of these is 1.255 with a crimp of .468. With the UltraDot on the slide, I use an 11lb recoil spring. When I get a stovepipe, it's because the gun/chamber is dirty. I've experimented with a lot of bullets (Nosler, Star, Bull-X, Meister, Speer) and powder (Clays, AA #5, 231) and this combination seems to work well for me. I use a Dillon 550, Federal primers, and WCC many-fired cases. I inspect my lead bullets and only use the visually clean and perfectly shaped ones (especially bases) for matches. That means about one in ten goes into the practice box. Before each match, I will quickly visually inspect each round and set aside any that look different, just in case. For a ball gun, I purchased a basic mil-spec 1911A1 from Springfield Armory and had Masaki do a accuracy job on it. I believe that included a hand fitted Kart barrel, slide to frame fit, Bomar adjustable rear sight, and a trigger job. I've never really benched it but I've fired in six leg matches and am now fortunate enough to be distinguished. Using the CMP issue WCC Match ball, I fired 277-5x in the NTI match this year at Perry and won a Gold Custer Medal. The key to shooting the ball gun for me is one thing and one thing only; front sight. I remove everything else from my mind but front sight. I actually say "front sight, front sight" over and over to myself all during a ball match.

Getting it Done

As I get ready for each string, I will set my feet, look at the target, close my eyes and take a few breaths. Then I raise my shooting arm and mentally play a shot in my mind (the Army calls this visualization). As my imaginary shot breaks, I open my eyes and if my hand isn't pointing directly at my target, I adjust my back foot. Early in a match, I will take as many as three

times mounting the gun in my hand to get the right feel of the grip. I find my hand changes slightly because of temperature, stress, and dampness during a match. Once I feel the proper grip, it starts to become second nature later in the match. Then I do the same with trigger finger placement. It might take three or four times, but I don't shoot unless it feels just right. I try to dry-fire 8-10 times before a slow-fire string; it helps me to identify subtle problems with my stance, grip, and trigger control. Relaxing and clearing my breathing helps me to mentally focus between shots. After the dot settles (about 2-3 seconds), I apply pressure to the trigger. If the shot doesn't break after about 6-8 seconds, my mind starts telling me to set the gun down. If I don't, it will always be a mistake. I shoot slow-fire in about 7 minutes but find that time decrease with later slow-fire targets because I get in the zone. If I am really in the zone, the dot stops moving for a few seconds but most of the time, it just slows down and the arc of movement gets smaller. From time to time I'll have a timed or rapid-fire target with a group that is tighter than any of my slow-fire targets. That makes me want to shoot my slow-fire in a timed-fire cadence. That might work for some folks, but not me. What I have found is bad slows and good sustained's happen because my sub-conscious is pulling the trigger for sustained but my conscious mind is pulling the trigger in slow. If I consciously pull the trigger it will usually be a bad shot. I need to apply pressure to the trigger and simply have the gun go off subconsciously. In sustained fire, I reset the trigger and recover the gun at the same time. By then, the sights are coming back on target and as soon as the dot (or front sight) is entering the black, I begin to apply pressure. The breaking of the shot needs to surprise me or it's probably a bad shot. I'm getting better at this, but I have a long ways to go. For me, timed-fire is the same as rapid-fire except the first shot doesn't have to happen so quickly. I focus my vision through the dot to the target. As the shot breaks, I try to freeze that image for just a moment and call my shot. On indoor targets, I can pretty consistently call my shots to within a half-inch, outdoors about one to two inches. If two or three shots are consistently away from my call location, I'll consider changing my sights. A friend of mine once told me, "You have a screwdriver in your gunbox for a reason; USE IT!" My Hammerli seems to require no sight adjustment between 50 and 25 yards. However, the .45 requires about 7 down and 3 left when moving to the short line with the different load.

Maintenance

I swim once a week for exercise. While working on the computer, I'll pick up exercise putty and squeeze it for 10-20 minutes twice a day. I believe this really helps my grip and forearm strength. I am not a coffee drinker but still avoid any caffeine or sugar 24 hours before shooting. I am moderately near-sighted but shoot with contacts that correct my vision to 20/20. I completely clean my .45 (field strip then remove firing pin, extractor, sear, disconnector, trigger, etc) about every 500 rounds. In between time, I'll field strip and clean

the barrel and slide/frame about every 200 rounds. In addition, I'll brush out the barrel after shooting lead before shooting jacketed. The Hammerli gets the barrel cleaned with a string and wet patch about every 500-700 rounds. At that time, I clean the extractor, bolt face, and chamber/feed ramp.

Keys to the Game

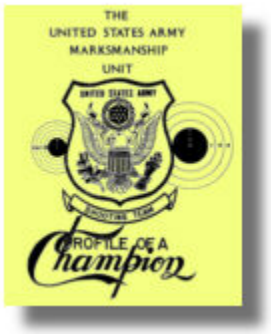
I have good days and I have bad days. When things don't go well, especially slow-fire, I try not to let it affect my mental concentration. I just run through my fundamentals and try to diagnose the problem. At an outdoor match recently, I fired two slow-fire targets with scores in the low 80's before I figured out what I was doing wrong. I'd like to practice and train with 100-200 rounds per week but my time just doesn't allow that. Because of that, I try to make my best effort to improve whenever I have an opportunity to shoot; be it practice, an informal league, or a match. Buy as many points as you can and then earn the rest. I find this an accurate statement when it comes to Bullseye shooting but there is a point of diminishing returns. Spending the extra $1000 on a Hammerli or similar gun over a Smith & Wesson Model 41 is not going to make a Marksman a Master. When a shooter has the fundamentals down (stance, grip, sight alignment, trigger control, and follow-thru), this sport becomes simply a test of mental concentration. Learn from your mistakes and concentrate only on the current shot or string. Expect X's and 10's; be satisfied with 10's; improve all the other shots until they are X's and 10's.

BIOGRAPHY AND HISTORY

Army Marksmanship Unit Champion Profiles

By United States Army Marksmanship Unit

Source: http://bullseyepistol.com/amuprfil.htm



This section of the website is dedicated to the Army Marksmanship Unit's Profiles of Champions Booklet. Well worth the reading, It contains autobiographical information and tips from the top pistol shooters and coaches of the Army Pistol Team during the 1960's and 70's.

- FOREWORD
- Frederick J. Kiefer, International Coordinator USAMU
- Oscar K. Weinmeister, CONUS Coordinator USAMU
- SFC Bonnie D. Harmon, USAMU, 1972 National Champion
- SFC Harland V. Rennolds, USAMU, Top Competitor
- CPT Robert L. Merritt, USAMU, OIC Pistol Division
- MSG Aubrey E. Smith, USAMU, NCOIC Pistol Division

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BIOGRAPHY AND HISTORY

Foreword

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile1.htm

FOREWORD

These articles were written by members of the Army Marksmanship Unit who have demonstrated an individual standard of excellence within either the National and/or the International levels of competition. SO YOU WANT TO BE A CHAMPION ... In reading these articles certain personal qualities are mentioned by many as important or essential in climbing the ladder of success.

Dedication, establishing intermediate goals, personal sacrifice, desire and mental/physical discipline are deemed necessary to achieve the ulitimate which is National Champion or Gold Medalist in World Competition. As pointed out by many, "Winning isn't everything, it's the only thing" and a true champion must develop this attitude or he will become satisfied with less.

Competitive shooting is a participant type sport and personal acclaim is rarely commensurate with achievement outside the "shooting community." Therefore, the competitive shooter must be even more dedicated to "pay the price" for success. You are not born a champion and only you can decide if you should become one.

It is hoped that you as an aspiring shooter and hopeful future champion can personally benefit by the inner-thoughts and driving forces of some of those who have successfully negotiated the road to "championship." As you ponder the tips these contributors are sharing with you -- consider, don't they really apply to personal achievement in any competition in life???

STANLEY J. PARMENTIER
Colonel, Infantry Commanding

BIOGRAPHY AND HISTORY

Frederick J. Kiefer, International Coordinator USAMU

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile2.htm

SHOOTING ACCOMPLISHMENTS:

- Distinguished Pistol Shot
- USMA Pistol Team '40-41-42-43
- U. S. Constabulary Pistol Team '47-48-49
- U. S. Army Pistol Team '50-52-53-54-55
- CISM Pistol Team '57-59
- Champion First Fiesta of Five Flags '52
- Fort Benning Pistol Champion '55
- Adjutant '64 Olympic Team
- Adjutant '67 Pan American Team
- CISM Pistol Team Captain '68-69
- NRA & ISU Referee

WHAT MAKES A CHAMPION SHOOTER - MR. KEIFER

On the verge of victory, many falter and fail to demonstrate the courage and greatness that separates the participants, really merely a matter of degree...from the winner, for there is only one winner and he is called the champion.

In spite of the thousands of words on the subject of champions, like an iceberg with only 10% visible, probably more remains to be learned about their development than is presently known.

The list of traits must remain forever incomplete since we must consider each as an individual. What may make one person a winner does not necessarily apply to another champion.

Two traits, ATTITUDE and CONTROL play a most important part in the makeup and personality of any champion. Without these two basic ingredients, he will never be one.

Being a champion is an attitude in itself...a winning frame of mind. In competition, his will to win is greater than that of all others competing. A passionate overwhelming desire forced him to pay the price, make the sacrifices. endure the monotony, pain and drudgery along the road to the pinnacle. He knows loneliness as well as exultation for he has experienced them to a greater degree than you or

Many fine competitors drop out because they did not have this singleness of purpose that was greater than all else in life. To win IS LIFE for the champion. It is not unrealistic to call a champion a fanatic. To him winning is not the most important thing. Winning is the ONLY thing.

The will to win, since it is an attitude, is more important than ability. Ability can improve with work, practice and coaching, but the attitude must be there as part of the body, soul, and brain since it is the motivating drive that forces him onward and upward. Attitude makes the work and practice worthwhile and meaningful. He is WIN oriented and blind to distractions.

Included in this attitude of the champion is confidence gained by careful, meticulous preparation of himself and equipment. He is a true professional. He makes fewer mistakes. He remembers. He knows what he has to do and does it because it is important to make not even one mistake.

The champion has control, physical and even more so, mental. He can stand the heat of pressure. He is emotionally conditioned to win. He will not let a moment of carelessness or loss of concentration occur.

Most shooters know how to shoot a good shot. The higher the level of competition, the more it becomes a battle of nerves between the top contenders. The champion will be the one who maintained control of himself to the highest degree.

The perfect score fired in practice may be important for it instills confidence, but it does not count for record. Practice for the champion is a compass to point to areas of improvement and to give him an equipment status check. The champion strives to control his training so as to be at his strongest during competition. Many do not become champions because they fire their best scores in practice. They cannot stand the strain of competition.

Many traits are found in champions. The list is unlimited. All are important but in varying degrees according to their personalities. The man with the resolute ATTITUDE and CONTROL, actually WILLS himself to be the winner and CHAMPION. He refuses to settle for less.

...and there he will remain until someone with greater will, greater control, more resolute attitude, wills otherwise.

BIOGRAPHY AND HISTORY

Oscar K. Weinmeister, CONUS Coordinator USAMU

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile3.htm

SHOOTING ACCOMPLISHMENTS

Oscar K. Weinmeister, Sr., a Department of the Army civilian employee with the U. S. Army Marksmanship Unit has been shooting and coaching competitive handguns for the past three decades.

Weinmeister entered the Army at Atlanta, Georgia, in November 1934 and served on combat duty during World War II. For his military service he has been decorated with the Legion of Merit, Bronze Star Medal and Army Commendation Medal. He was awarded the Distinguished Pistol Badge in 1951 for his Outstanding pistol marksmanship.

While serving 30 years active duty with the Army, Weinmeister fired competitively in National type .22 caliber, Center Fire and .45 caliber. He fired in his first match in company level competition in 1937 and became a member of the U. S. Army Infantry Pistol Team in 1940. He won the Georgia State Pistol Championship in 195i and 1953 in three-gun competition. In the same years he fired on the .22 caliber team which won the championships of the National Matches at Jacksonville. Florida, and the Kentucky State Championships where they established a .22 Caliber Team National Record. He also held the NMC National Record with Center Fire in 1953 with a score of 298, fired in Coral Gables, Florida.

Weinmeister was a member of the 1962 International Center Fire team in the Counseil du Sport Militaire (CISM) Matches held in Buenos Aires, Argentina and was a member of the Army "Blue" team in three-gun competition during the years 1951 through 1956.

Prior to his retirement from the Army as a Chief Warrant Officer W4 in 1965, Weinmeister served as Head Coach for the U. S. Army Pistol Team from 1958 through 1965. The U. S. Army Pistol Team made its first clean sweep of the National Matches in 1962. Since his retirement he returned to the Marksmanship Unit where he authored and compiled several of the Marksmanship Training Manuals. He served as Officer in Charge of the U. S. Army Pistol Team in 1969-1970. He is now serving in the USAMU as the Coordinator of Marksmanship Training for the Continental United States. His duties now include administering the Active Army, Army Reserve and National Guard Distinguished Marksmanship Award Program.

TO BECOME A CHAMPION SHOOTER

There is no substitute for knowledge and experience in becoming a shooting champion. This fact establishes the need for a marksmanship training program that will improve the shooting

skill, provide increased knowledge of shooting techniques and motivate the individual and team competitor to continuously strive toward producing the winning score. Marksmanship skill of championship caliber has never been acquired quickly. The champion shooter is the result of astute coaching, possession of a commanding knowledge of how to employ the fundamentals of marksmanship, manual and mental dexterity, a consuming desire to excel and a willingness to work hard during the long hours of practice and preparation for match competition.

There are four phases in the development of a shooter to be a champion; exploring, questioning, organizing, and accomplishing his goals.

First, the new shooter finds himself becoming interested in the skill of marksmanship by association with competitive marksmen or a shooting coach. The tyro senses that he is involved in a challenging, skillful endeavor. In his limited experience, he has learned that this is a dynamic sport and embraces certain vital factors that arouse his curiosity. His continued participation is sustained by the encouragement and help he receives from his contemporaries. He has lasting interest only if he is able to improve or feels that he has the potential to become a contender.

Secondly, initial exploration is followed by inquisitive expansion of his knowledge of the fundamentals and techniques of shooting. A new shooter's learning possibilities are limited if he is left alone to ponder the multitude of factors that make up the framework of shooting skill. Beyond a certain point, if he is to gain an understanding, he must ask questions, look for the answers to his problems and devote extensive discussion to the methods of employing the fundamentals. Real conversation begins when the developing shooter does not fully grasp or accept the answers he is getting. A meeting of minds results and there ensues an exchange of ideas and information between coach and shooter. During this period the shooter may veer toward excessive self doubt or lean in the direction of overconfidence. The skillful coach however can bring him back to the middle ground of solid progress. Less and less the shooter is merely probing, he repeatedly convinces himself that he is approaching a new plateau of performance from which he will not retreat. He has now materially expanded his knowledge of marksmanship.

Thirdly, intelligent coaching then helps the shooter to hurdle the next important challenge, organization of his shooting knowledge into a technique of operation. Organization of a full grasp of the fundamentals, factors and principles of marksmanship contributes to the shooter's ability to attain control of his performance. His system of operation is peculiar to him and now he can progress on his own at a rata suitable to his capacity. There has come into being a degree of personal accomplishment that generates confidence that the goals will be achieved; winning in competition.

Fourth and last, the consummation. Personal coaching becomes a sustaining factor when the inducement to accomplishment is reached. Instruction is replaced by counseling and guidance. If the established method of operation is momentarily disrupted, the resourceful coach and the enterprising shooter will devise a corrective action to overcome the error in performance. The relentless adherence to a system that produces unswerving control will result in consistently high scores. By exercising skillful judgment, the shooter can overcome variations of the will, of the weather and of the competition. He has now developed a determination of how he will accomplish his end. The degree of accomplishment depends upon the peculiar nature of this entity he has created. The limitations, if any, lie in the ambition, force of character, completeness of knowledge, confidence, enthusiasm and his response to group incentive. Whether he will fulfill his desire to become a champion is ultimately only his business.

BIOGRAPHY AND HISTORY

SFC Bonnie D. Harmon, USAMU, 1972 National Champion

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile4.htm

SHOOTING ACCOMPLISHMENTS

- Center Fire National Champion, August 1965.
- Instructor at various clinics, 1966-1969-1971-1 972.
- 2650 Club, 1966.
- Pan American Games, CF Team Member, 1967.
- International Shooting Badge, July 1967.
- .22 Caliber National Champion, August 2964.
- Mid Winter Pistol Champion, 1968.
- Second Place CF, Bucharest, Romania, 1969;
- Second Place CF, U. S. International Championships, 1969.
- Mayleigh Cup, High Firing Team Member, 295 - 1969.
- Sniper School, Nov-Dec 1969.
- 1971 CISM Rapid Fire holder new record.
- 1972 All Army Champion:
- 1972 Georgia State Champion.
- 1972 Florida State Champion.
- 1972 Interservice Champion.
- 1972 22 Caliber National Champion.
- 1972 National Champion.
- 1972 Individual Service Pistol Champion, 293-14X.
- 1973 CISM Champion.
- 1973 Tennessee State Champion.

PROFILE OF A CHAMPION

Why do I shoot the way I do? When I am asked this question, I find it not easy to answer in one sentence of 25 words or less because many physical and mental factors contribute to my ability to shoot rather well.

In 1962, I was introduced to competitive pistol shooting, green, no experience, and didn't know what to expect. In 1963, I was a new shooter all the way and that year I was awarded the

gold Distinguished Pistol Shot Badge. In retrospect, more emphasis was on service pistol than any other pistol, but now I realize it takes strong emphasis with each pistol in today's type of competition.

In my tyro years, a pistol match didn't excite me like it does now. In my younger years, a pistol match was somewhat considered just another military duty with no significance. This was a result of inexperience, youth, and attitude. But with seniority comes experience, and with experience comes wisdom (hopefully).

Climbing the ladder to reach the elite 2650 group, periodically I would shoot a single gun aggregate that was considered excellent. I found it difficult to mass all three gun aggregates that represented any degree of winning. I would either shoot a winning score with the .22 Caliber or .38 Caliber and fall flat with the .45 Caliber or I would shoot a mediocre score with the .22 and .38 and shoot a hot score with the .45 Caliber.

I am convinced that the problem was that if I shot a winning score with the .22 or .38 I was over- joyed, overwhelmed (not having won too often), and oversimplified the remainder of the shooting match, settling for just one gun aggregate. However, if I won only the .45 Caliber, I believe it was because my attitude changed to a more aggressive form of determination.

The longer I shoot the more I realize I must set my goal high and be determined to reach that goal. That's how records are broken. My technique of reaching this goal is to shoot just one point higher than the best score I ever fired with that particular gun.

Attaining these standards for myself I find it necessary to know each and every one of the basic fundamentals, haw each works, and why. I must have confidence in my weapons. I feel more comfortable if I can handle my weapons the night before a match to reaffirm that certain feel, because each weapon fits differently in my hand. I like to feel and dry fire my guns the night before a match. I feel more close and intimate; I call it shooter and weapon relationship. The two are inseparable.

When I train, I need or would like to train as close as possible to actual match conditions. This method eliminates major changes in physical or thought processes. I also need to train sufficiently with a gun until I'm having no problems with the trigger or grip, to a point where I am confident. On the other side of the coin, I can reach a saturation point because of too much practice beyond that point of confidence. When time permits. I prefer to practice three days before a pistol match, having a day off prior to the match for weapons cleaning, reaffirming my zero, and working on any shooting problem that might have developed during practice.

When I know the exact date of a match, my preparation starts then and slowly builds up as the match date advances. I also like to know the level of competition, so I can adjust my

psychic preparation. Knowing who I must beat helps me to develop a proper attitude toward winning.

I try not to change my regular daily routine. I try to get 6-8 hours sleep prior to a match when possible. I always eat a hearty breakfast 1 1/2 to 2 hours before the match starts. I take natural vitamins each morning and at evening meals. I eat a light lunch and what I want at the evening meal. I never drink alcohol to the point of being intoxicated at least 3 days prior to a match.

Experience has taught me. The most damaging and detrimental state of being for me is the loss of sleep. I find the loss of sleep causes a feeling of fatigue, with reaction time slowed down and an un- stable feeling.

I would advise any new or young upcoming shooter that it isn't the act itself but rather it's the pursuit of fulfillment that will agitate and lessen your ability to win and think clearly.

I find experience in shooting NRA and international events helps a great deal. In general, 9 times out of 10 an individual firing for the first time in an important match will find himself rather nervous and tense.

I remember my own experience in the 1967 Pan American Games (first big international match). I was nervous and even scared because people were watching, pictures were being taken, and because of lack of experience, I paid more attention to my surroundings. I now know I should have envisioned the small details pertaining to shooting. Match pressure never really leaves me completely, but experience helps to ease anxieties.

I try to create a state of tranquility in my domestic affairs. Observation and personal experience proves to me a shooter can't put all his effort into competitive shooting if his wife is constantly nagging about his shooting, money, TDY, children sick, etc. My wife encourages me to win and is happy as I am when I do win. I never leave on a TDY trip with domestic possibilities dangling as loose ends. My wife never annoys me with petty phone calls.

After having learned how to use the basic tools for shooting and having won a victory, I find it similar to a child's first taste of ice cream; they like it. I to like to win because it gives me a feeling of accomplishment. One victory creates a desire in me to win again and/or be a part of a winning team. The prestige and recognition further stimulates my attitude toward achievement.

Travel to foreign countries is another incentive for me to shoot well. The thought of competing against another country always stimulates my psychic-circuit to the point where I want to try a little harder.

The many contributing factors that enable me to shoot the way I do are:

1. Understanding the basic fundamentals.

2. Good training habits.
3. Taste of victory developing the proper approach to attitude.
4. With a positive attitude a desire to win is developed.
5. The incentive of winning is a large motivating factor.

It is my conviction that, if I know-how to apply the fundamentals, have a positive attitude, and high spirits, I have more time to devote to the small, delicate, and perplexing details of shooting one shot or 5 shots.

My mental preparation consists primarily of a technique I call "visualization" combined with know how, experience, and confidence. The visualization technique consists of formulating a mental picture of what you want to accomplish, whether it be one single shot or a 5 shot string, it can be used to train yourself to stand still and hold still, to see your score in Increments of 100 or 300, or to visualize your entire aggregate. This is not a new technique, but it is not really used too often in shooting. However. I do believe it's used more often than the shooter realizes.

Some days this technique works better than other days with due respect to your physical and psychological well-being and attitude.

Mental discipline, I believe, is developed in the process of good sound training techniques. Also paying particular attention to small details, such as small errors and not accepting these errors when it's possible to correct them.

Recognizing an error can be corrected only if I am aware of the mistake, if I am aware of the error, I can either stop--correct--or shoot the shot. If I shoot a shot while being aware of an error, I have either lost concentration, patience. or mental control--generally the latter. Prior to shooting a shot or a series of shots I try to hold a vivid picture of the preconceived mental picture of the end result desired, which is, of course, a 10-50- or 100, after each shot or series. I must reaffirm this mental picture each time. Having faith in my own abilities is derived from having fired good scores in practice for the last match.

This technique of visualization is always accompanied with confidence, desire, mental discipline, and faith in my convictions. I suppose I could base my philosophy of thought on an age-old clich "The end result desired is equivalent to the degree of faith you have in your thoughts", and "that which cannot be conceived in the mind's eye first will have much difficulty in materializing"

In conclusion, the tools I find necessary for shooting good scores are: Sufficient practice with each weapon I will fire in the match, test-proved equipment, mentally preparing myself by keeping my goal in mind, developing positive attitude toward these standards and formulating a mental picture of what I want to shoot, which is one point higher than I've ever

fired. And keeping my mind, body and psychic spirit on the infinite details of shooting one shot or five shots.

BIOGRAPHY AND HISTORY

SFC Harland V. Rennolds, USAMU, Top Competitor

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile5.htm

PROFILE OF A CHAMPION

I started my pistol shooting career in January 1958. My battalion told me I would be in charge of training a pistol team. I drew 4,000 rounds hard ball and put out bulletins for volunteers. No one came out to shoot so I had 4,000 rounds of hard ball ammo to shoot each month. This all took place in Fort Sill, Oklahoma, where it was extremely cold. Using Arms Room Weapons I came out each day to shoot. Of course, I did not have any training with pistols before so I got off to a bad start. I did not have the training as a new shooter on the proper use of fundamentals. I sure shot a lot of ammo and in March we had the Pistol Matches to select the personnel for 4th Army Tryouts. We shot a 900 aggregate with hard ball and I won the first match (slow fire 20 rounds) with a pair of 82's which gave me 164 total. This 164 I fired beat the best shooter in 4th Army, which at that time was Major Dunn. This was not a high score to be winning a match, but the adverse conditions were: We had a 30 to 40 MPH wind, no cover, cold, raining, and I had all the shots on the target. As a result, I was 20 points ahead of my nearest competitor.

No one had told me about the proper use of the fundamentals, so I would squeeze the trigger when I was in the black and quit squeezing when the wind blew me out. As a result, I won the match as a new shooter and placed third in the aggregate (900).

At this point I was selected for the 4th Army Team and got the proper instruction on the fundamentals of pistol shooting.

At this time I had decided on some goals to shoot for and it took many years to accomplish these:

1. To become a Distinguished Pistol Shot (1959).
2. To become a 2600 Pistol Shot (1959).
3. To win a match at Camp Perry (1970).
4. To win a National Record (1966).
5. To later become a 2650 Shot (1973) 2662.

When you are a new shooter starting at the bottom, just becoming a Distinguished shooter is like climbing a high mountain. You learn a lot by listening to the old shooters and gaining small bits of information. They will help you add a little more to the fundamentals and enable

you to shoot better. When you stop at a certain level of shooting you have to find some means to pull yourself a little higher and progress further.

Every person who shoots does not become an outstanding shooter. Some do not have the time or the desire. Others do not have the power of concentration.

To become one of the top shooters in a particular field you have to develop a technique that fits you. What works for one does not always work for another.

Let's talk about my technique. I am short and heavy and I have small hands. My stance is about the same as stated in the book, about 45 degrees from target, feet should be wide. My grip is modified to fit my small hand. To get a good grip I place my three fingers around the front of the receiver (2d joint) and hold while I pull my hand and thumb around to get a firm grip which will not shift In my hand while firing. Try to keep the 2d joint directly in front and under the trigger guard. I do not jam the grip safety tang down into my hand, as you can not squeeze the trigger without moving the pistol. In slow fire I do not grip as hard or tight as when I shoot timed and rapid. I grip tighter in timed rapid because it aids me in recovery with my arm locked up. I also shoot very fast slow fire. I have a good fast even squeeze which I have perfected in my technique. Very often I will shoot two shots in slow fire before coming down. If I have trouble squeezing, I shoot the same as if I were shooting timed fire and concentrate on trigger control. This also helps me when I shoot in the wind. When the wind comes down to its lowest level it is several seconds before it picks up again. If you practice this, it is not a new technique when you have to use it.

In timed fire use a firm grip, locking arm, wrist, and hand so recovery is faster. You have to be very aggressive so as to start your shot, shoot and follow through, and repeat for four more shots. You have to have faith in your hold, depending on sight alignment and trigger squeeze.

In rapid you have to speed up the technique you developed for timed fire. You can squeeze a trigger fast and with even pressure in one to 1 1/2 seconds, allowing you to align sights in same period of time.

Let's talk a little about preparation time. The night before you should get together what you will need to shoot the next day. Get the proper amount of rest. Don't strain your eyes by watching TV or movies.

Some of the things I do to enable me to keep up with the top shooters are:

1. The night before a match, I get a good night's sleep.
2. No smoking during the matches.
3. No coffee except decaffeinated.
4. No carbonated drinks (due to gas on stomach and some have caffeine).
5. No strenuous exercise or work before or during matches.

6. No alcoholic beverages before or during match.
7. To prevent eye strain, no movies or late TV.
8. I do not eat until after each day's matches.
9. I read between strings of slow fire because I shoot so fast (approximately 2-3 minutes). During my shooting I concentrate on one shot or one string. This reading helps me keep my mind from building up so much match pressure.

Regardless of what method you use and technique you perfect you have to master the fundamentals of sight alignment and trigger squeeze that will not misalign the sight while waiting for the shot to break. recover, analyze and correct mistakes. By making less mistakes you are able to win over your competitors.

BIOGRAPHY AND HISTORY

CPT Robert L. Merritt, USAMU, OIC Pistol Division

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile6.htm

SHOOTING ACCOMPLISHMENTS

- 1964-68 Member, Pistol Team, U.S. Military Academy, West Point, New York.
- 1966 Member, 1st All American Pistol Team, Collegiate Pistol Champion, Camp Perry, Ohio.
- 1967 Member, 1st All American Pistol Team, Collegiate National Trophy Pistol Champion, Camp Perry, Ohio.
- 1968 Team Captain, Pistol Team, West Point. Member, 1st All American Pistol Team.
- 1969 8th Infantry Division Pistol Champion.
- 1970 Commanding Officer, U. S. Army Europe Marksmanship Unit.
- Distinguished Pistol Shot.
- President' s Hundred.
- 2600 Club.
- Member, U. S. Army Team, Camp Perry, Ohio.
- 1972 Officer in Charge, Pistol Division, U. S. Army Marksmanship Unit, Fort Benning, Georgia.
- 1973 Member, National Indoor Championship Team.

WHAT MAKES A CHAMPION SHOOTER - CPT MERRITT

I started shooting pistol competitively in 1964 at the U. S. Military Academy, West Point, New York. Until then I was totally inexperienced with firearms. After initially making the team at West Point, I was content to merely keep my place on the team. I subconsciously set a low standard--to keep all slow fire shots in the black on the gallery course. As the shooting season progressed, I noticed other teammates excelling; they were shooting better scores than I.

I assumed that they had more ability until one day late in that first season, it dawned on me that I was accepting any slow fire shot so long as it was in the black. So I shot many sevens and did not mind. With this realization came the determination to tighten my slow fire groups. The following season, with higher standards, I became a member of the varsity team, scoring in all team matches for the remaining three years of my cadet career. In retrospect, I am always amazed when I consider how much effect that single negative thought/low

standard had on my scores. During my nine years of shooting, I have often seen competitors who were satisfied with marginal performance--it is a common pitfall.

One of the unique aspects of the shooting sport is that the age of the competitor has very little effect on performance. Experience is far more significant--I have seen a fifty-year old man break 2550 for the first time and set open national records. The physical aspects of shooting are apparently only important insofar as they permit the shooter to hold the pistol steady. Looking at a number of champion shooters makes it obvious that unusual strength or size is not a prerequisite for this sport.

If age is unimportant and exceptional physical attributes are unnecessary, then, by elimination, the discriminator which separates champions from losers must be the mental element. A strong will to win, intense concentration, a thorough knowledge of the fundamentals, and self-knowledge these are some of the noteworthy aspects that comprise the champion's mental attitude.

It has been my privilege to be the Officer- in- Charge of several exceptional pistol teams. Last summer (1973) the U. S. Army Blue Team won all team matches at Camp Perry, Ohio. As a result of this association with some of the best pistol shooters in the country, I have developed several attitudes based upon observation that are perhaps worthy of mention.

Shooting is basically an individual sport, contrasted with a team sport such as football.. Therefore, emphasis must be placed upon the natural development of each individual separately. To stereotype shooters and prescribe the same training program for all is to tend toward mediocrity. That is to say that a good coach will give each of his shooters individual assistance and will be able to accommodate each individual's training needs so far as is practical. One man may not feel that he should practice during a given training session. He should generally be entrusted with this decision; his judgment should be respected. When a man shoots, he should want to shoot. One particularly valid generalization is that all pistol champions enjoy shooting.

Coaching a shooter, especially during an actual team match, can either help or hinder performance. I have seen good, bad, and indifferent coaching. The most important attribute that a good coach has, I believe, is a keen awareness of everything concerning the members of his team. He will observe and know many things about a shooter that are unknown to that shooter. For example, a team member may have better scores on the second relay than on the first; he may shoot much better with one man than with others. The coach must be aware of his personal problems on his team. Any external worries or problems must be excluded from his team as much as possible. Certain words said to one shooter might upset him, while those same words could have no effect on his teammate. That is what I mean by awareness.

An aware coach must then be able to tactfully communicate with his shooters. Animosities among the coach and his team should be minimized. An observant coach can spot potential problems in this area and tactfully correct them. The shooters on a consistently successful team will generally have a rapport among themselves. When one shooter does poorly, causing the team to lose, he should never be told that he dropped the team. If he is embarrassed by the coach, he will feel uncertain in the next match. Also, the other shooters will worry that they might receive the same treatment should they score low. Lf a coach detects something specific which may have caused the law score, he should tactfully inform the shooter. Ideally, I think, a shooter would be more interested in making his coach look good in the team match, than in his own particular accomplishments. In my shooting career, I have had only one coach that instilled this attitude in me.

One final consideration, which may be obvious, is that a shooter must enjoy shooting for its own sake. The satisfaction he derives from willing his mind and body to shoot competitive scores should be sufficient. If he has other purposes, such as policing everybody's brass, trading guns, winning enough awards to pay for his entry, etc., then his chances of becoming a champion approach zero.

Robert L. Merritt
CPT, US Army Pistol Division, USAMU

BIOGRAPHY AND HISTORY

MSG Aubrey E. Smith, USAMU, NCOIC Pistol Division

By Army Marksmanship Unit - Profile of a Champion

Source: http://bullseyepistol.com/profile7.htm

SHOOTING ACCOMPLISHMENTS

- 1955-70 Member U.S. Army Team, Camp Perry, Ohio.
- 1971 Head Coach, U.S. Army Team, Camp Perry, Ohio.
- 1973 NCOIC, Pistol Division, U.S. Army Team, Camp Perry, Ohio.

PROFILE OF A CHAMPION

There are many types of competitive shooters in the United States: The weekend shooters, the two hour a night shooters, shooters who are satisfied with being second or third place, and the champion, the shooter who will accept nothing short of winning. The weekend shooter is usually not dedicated or motivated. It's a hobby to him. He is usually satisfied with just shooting along with the champion. The two hour a night shooter is a little more dedicated but after his two hours he goes home and forgets shooting altogether. He will never be a champion. Shooters who are satisfied with a second or third place do not have much more going for them. They will never be champions. Champions are never satisfied. They are a hungry lot. They live shooting day and night. They make many sacrifices and their loved ones make many sacrifices. For example, he might enjoy two or three cups of coffee for breakfast but he knows coffee isn't good for shooting so he either cuts down or quits. The same goes for tobacco and his bad drinking habits. His wife and children will also make sacrifices. Their every day lives must be formed around his every day living.

My shooting career started in 1954. Ever since I was a child guns always fascinated me, so it was very easy for me to get into the competitive shooting program in the U. S. Army. I began shooting on a battalion team and I was a winner in my first match. Winning that match gave me the boost to drive myself toward becoming a top competitive shooter. I remember shooting much more than my friends. I was always the first to the range and the last to leave. I couldn't learn fast enough. I constantly sought out shooters with experience to learn more about shooting, and learned quickly to accept everyone's techniques of shooting. I always had an open mind and still do. I believe all shooters should keep eyes and ears open. Maybe an inexperienced shooter will help you in some phase of your performance.

During training a shooter should work on one phase of his technique. Too many little things can be wrong with your performance; it is very difficult to correct them all at once. In training I never shoot for score. I try to perfect my technique. If a shooter cap perform in all phases of

shooting, the good scores will come. At all times think performance; concentrate on delivering a well controlled shot each time the gun fires. If he can do this in training and it becomes a habit, it will carry over in competition. Never underestimate the aid of dry firing. During dry firing a shooter can see all errors he is making and easily correct them. All phases of shooting can be perfected by dry firing and this includes building confidence in yourself. If a shooter does not have confidence that he can fire well controlled shots at any time, he will never become a champion.

I believe in shooting in as many matches as possible. I think a shooter can learn more by shooting in one match than he can with ten training sessions on the range. If possible, shoot with as many top competitors as you can. Sure, you might get beat, but defeat will give you that drive to learn more to become a better shooter. Defeat makes you hungry; learn from it. This also helps your confidence. You might think "The winner didn't beat me by too many points, he didn't learn anything--l did. " You are motivated to work harder and train more to perfect your technique and you are building confidence in yourself. Confidence combined with lessons learned, good training habits, and determination will allow you to perform at your best.

In the shooting sport, I believe, physical conditioning is over emphasized. Physical exercise should be on an individual basis. A shooter should do any exercise he thinks will make him hold the gun still. For example, if a shooter feels that running helps him, let him run. The only exercise I ever did was a few pushups each day and hold the gun and arm extended for as long as 30 or 40 seconds. What I'm saying is I don't believe one particular exercise will help 3 or 4 different shooters. I believe in each shooter doing his own thing. It has been tried in the Army to train a shooting team like a football or baseball team. It just doesn't work. Again I say find an exercise that will help you hold a gun still and stick with it.

My last competition was the 1970 World Championships in Phoenix, Arizona. and I can't say that I've missed it. First of all it gave me an opportunity to devote full time to coaching. I've always enjoyed teaching and coaching so I was constantly observing other coaches; how they performed their duties and applied their techniques in certain situations. I felt, with all my years of shooting experience and the knowledge I've gained from observation, that I could become the best coach in the world. I quickly found out that experience and knowledge are only a small part of coaching. I learned that each shooter must be coached in a way that you can reach that individual. If you have a team of four shooters or forty you are dealing with four or forty individuals. That means a different approach to each shooter.

A good coach must be a salesman; a persuader. First a coach must sell himself. Convince the shooter that if we work together as a team, his scores will rise. Some shooters are set in their ways; therefore, they are hard to get to. But never give up on a person--there is always a way to reach each individual. If you let it be known that you have given up on an individual this

will be seen by other shooters on the team, with the result that some of them could lose confidence in you as a coach.

Although I believe you approach each individual in a different way, there are times when a team must be treated the same. For example, you might have a 2670 shooter and a 2550 shooter. These two shooters must be given the same treatment off the firing line. If not you will have a split team. A split team will never perform up to its capabilities. A coach must always be on the alert and prevent preferential treatment to the few top shooters. There are other ways to award the champions.

I have been a firing member of three United States World Championship Teams and three Pan American Teams and on not one of these has there ever been a coach. This I cannot understand. A pistol shooter needs a coach to help him if and when he's in trouble and to encourage him along when he is going great. Many times during the matches I mentioned I wished that I could have looked back and there was a coach available to give me encouragement and to know he would stick with me regardless of the results. All the U. S. Teams that I've been associated with were not teams, we never trained as a team, we were a group of individuals who gathered together to shoot a match, When the United States shooting officials realize that top coaches are needed for training the shooters as a team, the United States will be back on the top in world competition.

TECHNICAL INFORMATION

Vision Topics for Bullseye Pistol Shooters

By Norman H. Wong, O.D.

Source: http://www.bullseyepistol.com/wong.htm



Here is a series of articles by Master Bullseye Pistol shooter and highly successful Optometrist, Dr. Norman H. Wong, O.D. I have not met anyone who has not learned a lot from the writings of Dr. Wong, on a topic of particular importance to competitive shooting.

- For Eyecare Professionals
- ABBE and Shooting Glasses
- Inaccuracies in RX Glasses
- The 20 Foot Eye Exam
- Red Dot Focus Position
- Obsolete RX for Shooting
- Keeping Focused
- About Dr. Norman Wong

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TECHNICAL INFORMATION

Bullseye Shooting for the Eyecare Professional

By Norman H. Wong, OD

Source: http://www.bullseyepistol.com:80/wong1.htm

Dear Colleague:

My name is Norman H. Wong, O.D. and I am a member of the California Optometric Association and the American Optometric Association. I am also a veteran, a life-member of the NRA and a competitive shooter in the sport of Conventional Pistol, known to its adherents as bullseye shooting. In bullseye shooting, competitors fire one-handed at paper targets from distances of 25 and 50 yards (50 feet during the indoor season) under exacting time constraints: In the most extreme instance, five shots must be placed on the paper within 10 seconds, while the slow-fire phase allows shooters as much as 10 minutes in which to make 10 shots. At the upper levels of competition, a millimeter or two of accuracy can make the difference between winning and simply placing in the top twenty. Like golf, bullseye shooting is a sport in which incremental improvements in equipment, technique and consistency can result in vastly improved scores.

My reason for writing is that many bullseye shooters have told me that, while their day-to-day eyewear is often fine for conventional use, shooting glasses prescribed by the same eyewear professional often leave something to be desired. I believe this situation can be alleviated with a bit of guidance from someone familiar with the shooting discipline, so I hope you will not mind if I pass along some advice in the form of this document.

Bullseye shooters are a wonderful and convivial group of individuals with critical visual needs. I urge you to set aside any political views against firearms and treat the bullseye shooter as you would any other patient.

The patient will ask, or has asked, for permission to bring his/her pistol(s) and/or revolver(s) to the office for the eye examination and will demonstrate that the firearm is unloaded and safe to handle. If permission was denied, sighting parts of the equipment were brought instead. Please take a moment to review the following important steps before and during your eye examination. Besides your routine health tests of the eyes, I have emphasized a few areas of concern. Your kind attention to these details would be most appreciated.

EYE EXAMINATION

1. CASE HISTORY: A thorough case history revealing any medical problems relating to eye health and vision stability is essential. Note all medications taken, including over-the-counter medications, and advise of possible visual side-effects.

2. DOMINANCE: Review and confirm eye- and hand-dominance.

3. REFRACTION: Your best effort is needed to obtain the most accurate results. After you obtain your best distance Rx, see if the patient is sensitive to 0.12 diopter steps. Please double-check vertex distances especially for higher powers. Check to see if the patient's line of

sight is "continually" centered through the phoropter lenses. Do not reduce the full strength of the prescription even if there was a large change from the previous examination results. Small prescriptions such as +0.25, -0.25 diopter sphere and +0.25, -0.25 diopter cylinder, even if obliquely orientated, may be significant to the bullseye shooter.

4. RED DOT SCOPES: For decades, the sport of bullseye shooting was conducted almost exclusively with conventional, open sights (also known as iron sights). An issue with iron sights is the impossibility of keeping the front sight of the pistol and the bull of the target (the black center portion) in focus at the same time. Because of this, shooters in recent years have adopted the use of 1X red dot scopes, which superimpose an illuminated dot on the image of the target and allow simultaneous focus. Normally, the best lens for the red-dot scope viewing will be the best distance prescription. Demonstrate this lens while the patient looks at the red dot while holding out the scope. Because the red dot in the scope is not focused at "optical infinity" (it is closer), try a +0.12 or a +0.25 diopter lens over the best distance prescription to see if the dot becomes even clearer. If possible, judgment would be best if the patient can view at a distance greater than the standard 20 feet and with outdoor lighting. If the dot is distorted, use the phoropter once again to verify cylindrical power and axis as the patient holds the pistol (or scope only) in front of the phoropter. Final results should be demonstrated with trial lenses. If the red dot never becomes clear and round after all lens possibilities have been demonstrated, then a careful determination of ocular health involvement needs to be assessed.

5. IRON SIGHTS: Iron sights are still preferred by some competitors and are mandatory in certain forms of competition. With iron sights, we are concerned with three separate entities, the clarity of the front sight, the clarity of the rear sight and the relative blurriness of the bullseye. These three positions cannot be focused simultaneously with just a lens. The rear sight is separated from the front sight by about 6 3/4 inches in most standard model-1911, .45-caliber pistols. Measure the EXACT distance from the patient's shooting eye to the rear of the front sight while he/she is in the proper stance. Write this down for the patient's record. Set this distance for the reading card on the rod of the phoropter. Find the best lens for this position and then try 0.12 diopter higher and 0.12 diopter lower and note if the patient responds to this small change. Presbyopic patients will give good responses. For the pre-presbyopes, low power lenses may allow for a more stable focus. Younger patients (those under 35 years of age) may benefit from a minimal plus power for a steadier focus, or perhaps none at all.

The patient needs to know the best plus lens because there are specialty type shooting glasses available with interchangeable lenses. The two popular ones are Knobloch Optik and Neostyle Champion systems which include lenses, occluders, apertures and side blinders. There are a number of lenses available with these systems (+0.50, +0.75, +1.00, +1.25, +1.50). The base lens (patient's distance prescription) can be custom made at the optical lab to incorporate into this system. A cylindrical lens must have an edge marking to coincide with an edge marking on the round lens holder. This would alert the shooter if the lens rotates within the holder and may cause blurriness and distortion.

Interchangeability allows for a quick change when different powers are needed for various shooting distances and lighting conditions. Personally, my best lens for the front iron sight stays the same under day and night lighting situations. In my case and in many other patients', we prefer a dedicated pair of shooting glasses for the iron sights, which may also be helpful as a computer Rx and for other hobbies at a similar working distance.

Next, have the patient view this best lens with the pistol in hand at a distance greater than 20 feet, and if possible, outdoors at 25 and 50 yards. It is understood that the majority of offices may have limitations in this regard. Depending upon a few factors such as arm length and pupil size, typical lenses that work best are +0.50, +0.75 and +1.00 diopter over the best distance prescription, but also try +0.37, +0.62 and +0.87 diopter lenses, even though these are uncommon powers. You may be surprised how sensitive some shooter's eyes are. In my case, +0.75 diopter is my ideal lens. A +0.62 diopter lens blurs the rear sight and a +0.87 diopter lens blurs the bullseye too much.

Experienced shooters would know that the distance bullseye will be out-of-focus. As different lenses are tried, this will allow the patient to compare the relative blurriness to the bull. Stress that front iron sight clarity is more important than bull clarity. Most shooters know that an adjustable aperture will then help clear the bull. When too much emphasis is given to the bull clarity, then rear-sight clarity will be compromised. The patient will always shoot better if the front and rear sights are perfectly clear and aligned while the bull is blurred, as opposed to a clear bull with sights that cannot be seen well enough for proper alignment. Always use the lowest plus-power lens to achieve this goal.

As a quick reference guide, here are the focal lengths of the powers discussed:

> +0.37 diopter: 2.66 meters
> +0.50 diopter: 2 meters +0.62 diopter: 1.6 meter +0.75 diopter: 1.33 meter +0.87 diopter: 1.14 meter +1.00 diopter: 1 meter

Lens determination by focal lengths alone may cause erroneous results. Use these only as a starting point.

6. BINOCULARITY: Advise the patient if there are any binocular problems which may affect focusing stability. Most shooters occlude one eye but some shoot with both eyes open and suppress the non-dominant eye. Hyperopic patients who are esophoric may have more of a difficult time if eyeglasses are not worn. Again, low power lenses need to be prescribed if the patient desires clear and stable focus.

7. CORNEA: Carefully inspect for any corneal defects including beginning signs of keratoconus. Note and advise the patient of any old scars and dystrophies along the visual axis that may compromise the focus of the red dot. Check corneal curvatures with the keratometer, or use corneal topography for irregularities. Check the tear film and advise of any possibility of dry eyes, which may cause unstable focus. Recommend dry-eye therapy as needed; use of ocular lubricants may be beneficial before and during shooting.

8. LASIK and RADIAL KERATOTOMY: Foreign matter introduced in the interface and other complications during LASIK procedure may or may not affect vision. Advise accordingly. Irregular astigmatism may result from radial keratotomy and distort the red dot. Frequently, we will have an undercorrection or an overcorrection after surgery and the full lens prescription needs to be given.

9. CONTACT LENSES: Patients correctable to 20/20 or better frequently see only 20/25, 20/30, or worse with contact lenses. This may be due to small uncorrected astigmatism, contact lens surface deposits, or desiccation of the soft lens material. Contact-lens lubricants may help when used before and during a match. A shift in vision may be noted when toric contact lenses rotate. These minor problems may be acceptable to the patient.

10. PUPILS: Note if the patient's pupil is unusually small or large. A small pupil will allow for a longer depth-of-focus but may cause more symptoms with lenticular opacities. A wide pupil will cause a short depth-of-focus and will make it a little more challenging to find the best lens possible. As the amount of ambient light changes throughout the day, the pupil size will also change and may give different sighting appearances. Inspect the iris for colobomas and for trans-illumination defects that can cause diplopia and glare.

11. CRYSTALLINE LENS: Note and advise the patient of any lenticular opacities which may affect the viewing of the sights. Commonly seen opacities that may not affect non-shooters will affect the shooters' clarity of the red dot or iron sights. Senior patients who have had intra-ocular implants need to be closely inspected for signs of posterior capsular opacities.

12. MACULA: Closely inspect the macular area for any signs of defects including ARMD. Use of Amsler Grid may be helpful.

EYEGLASS SELECTION

1. LENSES: Review past lens types and materials used. Discuss what has worked and what has not. Review ABBE VALUE of various lens materials and possible distortions especially in higher powers. Polycarbonate is the only FDA approved safety material that is widely used today and is always recommended as the first choice. Other materials may be used ONLY with the patient's understanding that they are not approved safety materials. If needed, the patient may be required to sign a waiver of responsibility. Many patients choose to have a dedicated pair of shooting glasses, which should be polycarbonate, and employ other materials for their dress eyeglasses. Note that Knobloch Optik and Neostyle Champion lenses come only in CR-39 plastic. When writing up the lab order, request for "exact power required" and reject lenses that are not .

2. EYEGLASS FRAMES: Frames should not be too small and flimsy (such as rimless). Safety frames following ANSI standards would be best. Wrap-around style frames may require steeper base curve lenses, which may cause distortions. Some sports frames with a one-piece curved front shield may have inserts, which will allow for the patient's prescription lenses. These goggle-type frames have no adjustability, important with moderate to high prescriptions. Being that there are four surfaces in such eyewear, fogging and cleaning may be a problem.

3. MULTI-FOCAL HEIGHT MEASUREMENTS: Have the patient demonstrate their head posture while at their "shooting stance" as you take the proper height measurements. Extreme diligence needs to taken so that the line of sight is not impeded by the near or intermediate portion of the lens. When measured correctly, there should be no restrictions with bifocal, trifocal and progressive lenses. Verify that the measurement taken with the shooting stance is also compatible with the patient's normal daily use.

4. ANTI-REFLECTION COATINGS and ULTRAVIOLET COATINGS: Discuss the advantages of AR coatings which will reduce glare as well as reflections and UV coatings for blocking harmful UV rays of the sun.

5. TINTS: Kalichrome (yellow) enhances contrast but offers very little shading and may be helpful in low light settings. PLS 530 (orange) and PLS 540 (orange-brown) are tints that block all wavelengths above their stated levels. I have found that it is best for the patients to view tint samples and have them report what they find most comfortable. Transitions photochromic lenses lighten and darken with the amount of direct sunlight and their use is

very convenient. Avoid "fashion tints" with no protective qualities. If possible, loaning samples for the patient to view at the range would be most helpful.

I am constantly learning more of the bullseye shooter's needs as I continue to participate in this rewarding sport, and have been revising and adding onto this list. I will, from time to time, update this list as needed, so successive patients may come in with guidelines that differ slightly from these. A check of the version date (under my name, below) will tell you which are the most recent guidelines.

Bullseye shooters are intelligent, honest, law-abiding citizens with great depth of character. You will now have a very happy patient and should you care to see the product of your work in action, we invite you, your family and friends to join us and see what BULLSEYE SHOOTING is all about.

Regards,
Norman H. Wong, O.D. 10/05

TECHNICAL INFORMATION

ABBE Values of Shooting Glasses

By Norman H. Wong, OD

Source: http://www.bullseyepistol.com:80/wong2.htm

Greetings Shooters,

Have you ever experienced the following? You had a thorough eye examination by your Optometrist or Ophthalmologist and purchased a new pair of eyeglasses with all the bells and whistles. Of course, you had expectations of seeing like an eagle. You put on the new eyeglasses and did not focus very well and worse yet, you felt a little dizzy. Your immediate thoughts were, "The lab made my eyeglasses wrong!," " I messed up during the eye exam!," or "The doctor is a quack!" A fourth possibility may be the ABBE Value of your eyeglass lenses.

Different lenses have different properties. The density, known as the index of refraction, will vary from one material to another. The ABBE Value will also be different for the various materials. ABBE Value is a number given to quantify the amount of chromatic aberration of a specific ophthalmic lens material. Chromatic aberration occurs when white light is broken up into component colors. A color fringe may be noted around a light source or around an object. In reality, my patients never complain of color fringes, but rather, complain of blurriness and peripheral distortion.

Here is a list of ABBE Values. With a higher number, there is less aberration. With a lower number, there is more aberration.

MATERIAL	INDEX	ABBE VALUE	Crown Glass	1.523	59
High Index Glass	1.60	42	High Index Glass	1.70	39
Plastic CR-39	1.49	58	Mid Index Plastic	1.54	47
Mid Index Plastic	1.56	36	High Index Plastic	1.60	36
High Index Plastic	1.66	32	Trivex	1.53	43
Polycarbonate	1.58	30			

Ernst Abbe was a brilliant German mathematician and physicist who developed this rating system. Carl Zeiss was so impressed with Ernst Abbe that he was hired by Zeiss in 1866 and eventually became the owner of the company. Different manufacturers may have a different ABBE Value for the same exact lens material. Crown glass is most commonly used for eyeglass lenses with an index of refraction of 1.523. There are, however, 50 glass formulations used today for various optical applications, all with their unique properties. Perhaps this may be one of the explanations as to why there are different qualities to our scopes. Polycarbonate lenses are used extensively for safety eyewear such as in our shooting glasses and have the

lowest ABBE Value (most aberration). Note that crown glass has the highest ABBE number (least aberration). I've communicated with shooters who thought that their lenses made out of glass seemed to provide the best focus. High index glass has more aberration than standard crown glass. The higher the index of refraction, the denser the material and therefore a thinner lens. Cosmetically, thin lenses appear more attractive but have more aberration. You might want to print out this list and refer to it when you purchase your next pair of eyeglasses. Despite the ratings, I must emphasize that most of us may see just as well with low ABBE Value lenses as compared to high ABBE Value lenses. I would always recommend polycarbonate as the first choice of materials for our shooting glasses because of its safety characteristics. For your everyday lenses used under non-hazardous conditions, the chart may prove to be very useful. An informed shooter is a better shooter.

Good Vision and Good Shooting To All,

Norman H. Wong, O.D.

TECHNICAL INFORMATION

Eyeglass Prescription Innaccuracies

By Norman H. Wong, OD

Source: http://bullseyepistol.com/wong3.htm

Greetings Shooters,

After communicating with some shooters, I thought that "Inaccuracies with Eyeglass Prescriptions" would be an important topic for bullseye shooters. We've discussed supplements to improve cellular structure in the retina. We've discussed ABBE VALUE which may account for lens distortion in our eyeglasses. We've discussed PRESBYOPIA which affects our iron sight vision. We've discussed pupil size as well.

With all this wealth of information, some of you may still be unsatisfied with your vision, and rightly so. No matter how meticulous your eye examination, if your prescription eyeglass lenses were made poorly, your eyesight will not be optimum.

First, as a review, let us consider the eye exam. As long as you, as the patient, do your part correctly, we can decrease the variables. Do not close the opposite eye while being tested through the phoropter (the machine with all the lenses). Do not squint your eyes when viewing the different lenses. Be sure to look through the center of the phoropter lenses and not drift your line of sight towards the edge. Stop when your eyes are tearing. Let the eye care professional know your medical background and all medications, including over-the-counter medicines. Beware of the eye doctor, no matter how reputable, who uses technicians relying on automated instrumentation for verification of your eyeglasses and for your refraction. (Easier to use for the limited trained but could be inaccurate).

As bullseye shooters, I believe we have more acute needs than the average individual. In the vast majority of eye exams, the doctor checks your eyesight in 0.25 diopter increments. For some of us however, our exact prescription may fall in between these increments and our final prescription may be a rounding off value, and could be 0.125 diopter off. Most patients may not see this difference, but some may. To compound this error, the accepted standard of tolerance by the optical industry is + or - 0.125 diopter of your exact prescription. This is known as the ANSI standard. In other words, your eyeglass lenses may be wrong, but it may be considered "close enough." If the exam was 0.125 diopter off and the lenses were made 0.125 diopter off, then we may be 0.25 diopter in total error at this point, and yet, this scenario is not given a second thought or even contemplated by most people in the eyecare profession.

The optical centers of both lenses must align with our eyes otherwise there will be induced unwanted prism in our prescription. It is possible for the lenses to be misaligned vertically, horizontally or a combination of both. Some may tolerate this prism while others may experience eyestrain and occasional double vision. If you could never adjust to a pair of eyeglasses in the past, don't blame yourself. It just may be poor craftsmanship.

Let us now consider the location of service. "You get what you pay for!" holds true for everything we encounter in day to day life and eyeglasses are no different. With optical locations offering quick turn-around time such as one hour service or one day service, the emphasis is on quickness and not as much on accuracy. With locations offering low prices, the emphasis is on cost and less on accuracy. With lower profit margins, borderline Rx's will probably be given to the patients and not be redone to the exact requirements. In some locations, the lenses are not even verified for accuracy before given to the patient. This is not to say all work is done poorly, only a greater chance. I've seen lenses substituted when the exact power was not in stock. I've also frequently seen the lenses mounted reversed, that is, the right lens on the left side and left lens on the right side. I speak from past experience with the hopes of enlightening my fellow shooters to understand what may happen "behind the scenes" of the bright lights and mirrors of the showroom.

For those of you who think you are getting your proper lens Rx when filling your prescription, you might consider these points I've discussed. Your eyecare professional should be willing to discuss your concerns. My best advice,..... be as critical with your eyewear as you are with your gun's trigger action.

Good Vision and Good Shooting To All,

Norman H. Wong, O.D.

Since the writing of this article, a major optical chain has agreed to settle a multi-million dollar class action law suit because of substituting lens material with a lower quality (price) material and without the patients' knowledge.

TECHNICAL INFORMATION

The Eye Exam at 20 Feet

By Norman H. Wong, OD

Source: http://bullseyepistol.com/wong4.htm

Greetings Shooters,

"I need to shoot at 50 yards and fly a jetliner as my occupation so why does my eye doctor tests my eyes at 20 feet?" "My eyes were tested at 20 feet but wouldn't I see with better eyeglasses if I was tested at 1,000 yards." "I see great with my new eyeglasses even though I was tested in a small exam room, how can that be?" These are, of course, hypothetical questions which may have been lurking in the back of our minds but we didn't think of asking our eye doctors.

I briefly discussed this topic with shooters recently and I thought there may be some interest to expand it further. The discussion was centered upon that 20 feet was considered "Optical Infinity." In other words, our eyes focus the same at 20 feet as compared to the horizon. This is true and false depending on how critical we would like to evaluate this situation. We had a few comments that focus seemed different at 50 feet, 25 yards and 50 yards. There is, in fact, a dioptric value differential in the focus at 20 feet versus out to the horizon. This value is a fraction of a diopter and the typical human eye may not be able to detect any difference in focus and this small power is not available in prescription lenses. If a "stronger" lens is placed in front of the eyes, vision will not become clearer but rather worse, once the best prescription lenses have been determined.

Here are some other points to consider. Probably the most striking perception of different distances is from our use of depth perception with our binocular vision. Each eye sees a different angle of the object and our brain interprets how far the object is located. We tend to loose this ability to differentiate distances when one eye is covered. Pupil size will definitely have an affect on our perceived clarity at 20 feet and 1,000 yards due to depth of focus. Pupil size may be a function of age, "Night Myopia" and use of certain medications.

From a practical standpoint, exam room settings typically involves mirrors to obtain a projected eye chart at 20 feet. This method has been used by Optometrists and Ophthalmologists for years and will not likely change in the future. The best minds in Medicine and Optics would have considered a better option if it was considered necessary.

Good Vision and Good Shooting To All,

Norman H. Wong, O.D.

TECHNICAL INFORMATION

Red Dot Focusing Position

By Norman H. Wong, OD

Source: http://bullseyepistol.com/wong5.htm

Greetings Shooters,

I would like to suggest everyone who will be having their eyes examined in the future to request their eye doctors to show you the RED-GREEN test, also known as the duochrome test. This test may give you an understanding of how colors work within the human eye.

Some doctors may routinely use this test to confirm and finalize your lens prescription. The line of letters is shown half with a red background color and half with a green background color. When your final lens prescription is exact, you will see the line of letters with equal sharpness on both the red and green sides. If a slight plus lens is used, the letters on the RED side become clearer while the green side becomes blurred. Conversely, if a slight minus lens is used, the letters on the green side now become clearer while the red side becomes blurred. Different colors are focused differently within our eyes.

If we now apply this principle to our red dot scopes, I believe this explains why the red dot is focused at a different position or plane from the target. We see colors of different wavelengths at different positions in relationship to our retina. Red does not focus onto our retina the same as other colors. I have found that although many of my patients are sensitive to this red-green test and give good responses, many cannot see any difference at all. This may explain why some see the red dot at the same plane as the bull while others see the red dot closer, and it has nothing to do with being nearsighted or farsighted.

My intuition tells me that there was never any manufacturing designs for the dot to be at any specific distance, but because of color wavelength properties, the red dot may not be seen in the same plane as the target. A picture is worth a thousand words, so have your eye doctors show you this red-green test, which then may make you a believer rather than relying on my words.

Good Vision and Good Shooting To All,

Norman H. Wong, O.D.

TECHNICAL INFORMATION

Using Obsolete RX Glasses for Shooting

By Norman H. Wong, OD

Source: http://bullseyepistol.com/wong6.htm

Greetings Shooters,

I have just returned from Camp Perry, had a great time and made some good friends from North Carolina, Iowa and Illinois who had neighboring huts. The revolver matches were the most fun for me as well as the CMP EIC ball matches. After talking with a number of shooters, I've come to the conclusion that many shoot iron sights only when they come to Camp Perry. Because of the infrequent use of iron sights, the need for the exact prescription was not important for many shooters. This was not the case for those who routinely shoot iron sights.

Whichever the case, I would like to make a recommendation for shooters to keep all of their old eyeglasses even though the prescription may have changed. The logic behind this thought is that as we age, we will find that one pair of eyeglasses will no longer serve all our visual needs. Depending upon the changes, older prescription lenses may provide the exact focus for specific tasks such as with our iron sight guns.

As we age, those of us in our 50's, 60's and beyond will develop signs of crystalline lens opacification (various forms of cataracts). This may cause our eyes to become more nearsighted or less farsighted. The vitreous humor, the gelatinous material inside the globe of the eye, will become less dense and will cause our eyes to become less nearsighted or more farsighted. Along with these physiological changes, we may have prescription lenses that may have changed 0.50, 0.75 or 1.00 diopters. Guess what? The old eyeglasses you were about to donate may now be your perfect iron sight shooting glasses. You might have heard old grandpa say, "I see better with my 20 year old eyeglasses!" Contrary to misleading popular belief, you won't ruin your eyes by using older eyeglasses.

Here is an example to illustrate my point. A shooter at age 39 has a prescription of -3.00 diopter lens for the shooting eye. At age 49 and now presbyopic, the same shooter has become more nearsighted and needs a -3.50 diopter lens. By using the older Rx for the iron sights, the shooter is essentially using a +0.50 lens over the latest distance prescription. This may very well be the same lens I would have prescribed for this shooter using the iron sight pistol.

If changes in the prescription eyeglasses are small, you might try to see if the red dot becomes clearer with older eyeglasses. We had discussed that the red dot may not be focused at the same distance as the bull. As long as the older eyewear is in a safe condition and the changes

happen to be in the correct direction and magnitude, you may be in for a pleasant surprise. This won't work for everyone, but it will for some.

I would like to make a suggestion for all shooters to ask for a copy of their eyeglass prescription(s) after the exam. Also, bring your older eyeglasses for the doctor to inspect and to see if they would be compatible as shooting glasses. The vision guide, "Bullseye Shooters' Guide for the Eyecare Professional," may be helpful for your eye doctors because many eye care professionals are not aware of the shooters' visual needs. Keep in mind that routine eye examinations are still important.

Here's a summary:

For myopic (nearsighted) eyes: If you have become more nearsighted with aging, try your older eyeglasses which will have a lesser amount of minus power in the prescription.

For hyperopic (farsighted) eyes: If you have become less farsighted with aging, try your older eyeglasses which will have a greater amount of plus power in the prescription.

Good Vision and Good Shooting To All,

Norman H. Wong, O.D.

TECHNICAL INFORMATION

Keeping Focused

By Norman H. Wong, OD

Source: http://bullseyepistol.com/wong7.htm

Greetings Shooters,

Via the Bullseye-L Mailing List, Roddy Toyota had discussed great shooters of the past using iron sights in a thread entitled, "Blast from the Past……Scores." Rest assured, these elite shooters had perfected their focus. My friend Ross, who is not presbyopic, had problems seeing his iron sights in a recently attended "Two Day NRA Three Gun Pistol Shooting Clinic" at Richmond, Virginia in which Brian Zins was one of the coaches and Jerry Keefer provided technical information. Ross was informed that if the sights were misaligned by the thickness of a business card, this would then be equivalent to 2.4" at 50 yards. The coaches talked about being so focused, they could see little defects on the front sight. Jack had mentioned that although he can see the front iron sight with the aid of a lens, keeping focused for any length of time was difficult. Ed discussed with me that he noticed changes in his "aiming ability" with his eyes as he tries to focus onto the iron sight. Ed, of course, is an extremely astute observer and was able to convey to me one of the visual principles known as PHORIA, which I'll be discussing. Camp Perry is just around the corner and it's never too early for us to hone up on our skills for the upcoming ball matches for those of us who do not regularly shoot with iron sights. The topic of interest is, "Keeping Focused."

When we look at a close object, three things happen which is known as the "Triad." The triad consists of accommodation, convergence and decreased pupil size. Accommodation is the ability to change distance focus to near focus. We have a certain amount of focusing ability depending upon our age, easy when we are young, more difficult as we become older and almost impossible at ages 60's and above. As we accommodate, our two eyes converge inwards so that each eye is pointing to the close object. Simultaneously, our pupils diminish in size. These three things happen smoothly and effortlessly for most of us, but not all.

Accommodation and Convergence:

When we get new eyeglasses, the distant part of the lens will allow our eyes to have a "resting focus" at infinity. This is the upper portion of the lens if we have bifocals or progressive lenses. When we get lenses for the iron sight, our "resting focus" will now be at the front iron sight. What this means is that we do not have to struggle with refocusing our eyes when the lens does its job. Both eyes cannot see the front and rear sights simultaneously so we must use

only one eye, typically, the dominant eye. Essentially, we are disrupting the natural binocularity of our eyes.

When we see the front iron sight clearly with the use of the proper lens, we no longer need to converge our eyes. Some may need an occluder for the non-shooting eye while others have developed "suppression" in their non-shooting eye and not need an occluder as they keep both eyes open. Since we know that the front iron sight is in close proximity, there is a tendency to want to converge our eyes. If one converges the eyes, the eyes will then try to accommodate. If the front iron sight was already clear (with the use of a lens for us older presbyopic shooters), the extra accommodation caused by the convergence may now blur the front iron sight. This may be one of the reasons why the front sight goes in and out of focus.

The best way to avoid this extra accommodation is to use a plus lens of sufficient magnitude and not use such a low lens power that it does not clear the front sight properly. Granted, the bull will be blurred, however maintaining focus will be easier and aligning front and rear sights is the key for good scores. If the power of the plus lens is too low, then we will try to compensate the extra focus needed with our own accommodation and this can lead to fluctuating focus. All this relates to how well we maintain focus onto our iron sights. Remember, younger shooters who do not need prescription shooting glasses will see the bull equally blurred if they focus onto the front sight properly.

Pupil Size:

Since there is no way we can control our own pupil size effectively, we can use use various makes of apertures to help further refine our focus. Decreasing the size of an aperture will extend our depth of field, thereby allowing simultaneous clear focus for the front sight, rear sight, and the bull as well. On the downside, we need to carefully position our line of sight to view through the aperture, a little challenging as we're shooting sustained fire. Since the apertures are adjustable, you will be able to fine-tune the opening for the best compromise.

Apertures that attach to your eyeglasses with a clip may not fit because of shape and size of your eyeglass frame and the thickness of your lenses. Nose pad arms on metal frames may further hinder a good fit as well. A consistent fit may not be possible for some. Apertures with suction cups would eliminate these problems but care must be taken to avoid scratching your lenses. The suction can fail and the aperture may pop loose during an inopportune time such as during rapid fire in a match. My best advice here would be to borrow a friend's aperture before buying your own, the cost being around $50 to $80.

Phoria:

Just as the eyes have a "resting focus," our eyes also have a "resting aim" known as the PHORIA. Often, our eyes may not normally have a "resting aim" at the object we are observing. What I mean by this term is best described by an example. If we are looking at the

bull at 50 yards, our resting aim could be inwards or outwards, but we then must force our two eyes to point at the bull.. A crude example would be a car wheel alignment with a toe-in or toe-out problem. If the resting aim is way off, then we may see an individual with a "crossed-eye" (esotropia) or a "walled-eye" (exotropia). If the resting aim is slightly off, then we normally re-align our aim by forcing our external eye muscles to help point our eyes at the object we are seeing.

A simple way to test your own resting aim is to stare at a small object at 20 feet and cover your right eye with your right hand. Quickly uncover the right eye and now cover the left eye. Note if there was any motion to this object. If the object moved to the left, then the resting aim of the right eye is outwards. If the object moved to the right, then the resting aim is inwards. Repeat this alternating cover several times and you may note the movement of the object to widen, representing that you are reaching your maximum resting aim. Next, try this again at an object approximately 2 to 3 feet away. Again note the movement when you switch your hand from one eye to the other. Some may notice a vertical displacement as well. If there was no movement whatsoever, then your resting aim is perfect for the distance you are viewing. If there was movement, then your external eye muscles must pull to align the eyes to avoid double vision.

By occluding the non-shooting eye and using the best plus lens for "your" eyes, you will have the perfect iron sight picture as long as you allow your non-shooting eye to "drift" to its natural resting position and not try to converge your eyes. Those who shoot with both eyes opened must avoid converging also. Younger shooters who are not presbyopic will also benefit with the plus lens if keeping focus is difficult. The plus lens is a must for all older shooters.

Holding breath and oxygen deprivation would be a good future topic for its effects on keeping focused but getting the correct lens should be the first concern. Please refer to the front magazine cover of the January 2005 issue of Shooting Sports USA. This picture of the seven time national police shooting champion, captioned, "Shows his match-winning form," will help summarize some of my points. Note that the shooter in the featured front cover is right eye dominant so he aligns the sights with only his right eye, and allows his non-shooting left eye to drift to his "resting aim" which is outwards. In this case, he then suppresses vision from his left eye to avoid double vision rather than using an occluder. (Police shooting is an entirely different discipline than bullseye shooting but I wanted to use this picture for illustrative purposes)

Shooters of the ball matches at Camp Perry participate on an equal playing field in that we all must use comparable ammo and ball guns. It is my hope that we also "see" our iron sights well so that qualifying for the President's Hundred will have a new meaning because I suspect that a large percentage of shooters struggle with their focus judging from the number

of messages I receive. We can be sure that the top hundred shooters see their sights fairly well.

Good luck and see you at Camp Perry!

Norman H. Wong, O.D

Further Reading

- Bullseye-L Mailing List (http://www.lava.net/~perrone/bullseye/)

TECHNICAL INFORMATION

About Dr. Norman Wong

By A Brief Biography of Norman H. Wong, OD

Source: http://bullseyepistol.com/wongbio.htm

Dr. Wong's military service was coming to an end within eight months, back in November 1976. As an optometrist in the United States Navy, he examined thousands of eyes but did not once handle a firearm until one day after he examined the Master Chief Petty Officer in charge of the rifle range at Great Lakes, Illinois. After attending an introductory course for the 1911 .45 caliber pistol, he was invited to join the Recruit Training Center Pistol Team.

After this course, the very first target was kept as a souvenir, a grand total of 6 points. The second target improved to 20 points, and the third to 30 points. After two months, scores improved to consistent high 90's. Scopes were not used and only military hardball ammo was issued. He was later proud to have been a member of the winning Commanders Cup Championship Team and also became a two time qualified expert in the Navy, the highest ranking possible.

Dr. Wong's short shooting career ended in June of 1977, but memories of the fun, excitement, commitment and the camaraderie never subsided, even after a couple of decades. It was not until 2003 when Dr. Wong once again, picked up a gun. However, those glorious past memories of excellence were elusive as he found himself placing typically 10th place out of twelve shooters in a local bullseye league.

Upon learning of shooting websites such as Bullseyepistol.com and the Bullseye-L Internet Forum, he had the determination to continually improve his skills. His first formal NRA conventional match was in February 2004. One of the highlights of attending the National Matches at Camp Perry that year included having the distinct honor of making the President's Hundred list, and then again in 2005. This was soon followed by the arrival of his Master NRA classification card on Christmas Eve 2005.

Dr. Wong was impressed by the generosity and character of the bullseye shooters. As he was discovering pearls from these sites, he offered his knowledge as an eyecare professional to many shooters with his numerous posts and articles. He was able to understand the visual difficulties experienced by many of his fellow shooters by being an active participant himself. His commitment to the shooting world was undeniable as this sport is his passion.

Nothing would please Dr. Wong more than to see another shooter benefit because of his contributions, as many shooters have personally commented. Never, did he imagine that many of the optical principles learned while attending the University of California at Berkeley, School of Optometry, would have such an impact 30 years later.

TECHNICAL INFORMATION

Reloading Data for Pistol Competition

By Compiled by John Dreyer

Source: https://www.bullseyepistol.com/reloads.htm

My Favorite Load for the .45

After a considerable amount of experimenting, I have determined the components and measurements that produce the most accurate load for my .45 caliber pistol at both 50 and 25 yards. You must experiment to find a load that works best for your pistol, yet utilizes components that are affordable.

Bullet	Star 185 grain swaged HP SWC
Case	Starline or Federal Brass
Powder	3.9 grains Alliant Bullseye
Primer	Federal Large Pistol
OAL	1.213" regardless of case length
Crimp	.473" at case mouth
Accuracy	1.923" *average* 10 shot groups at 50 yd.
Reliability	Flawless Feeding and Cycling (11# recoil spring)
Weapon	Les Baer Bullseye Model w/ Scope on Slide

A True Hardball Load for .45 Service Pistol

The following represents recipies to duplicate a true .45 caliber hardball load. The NRA has published some load info that should fill your needs. *American Rifleman* June 1993, pgs.

44-45
Loads to simulate Hardball recoil and energy 850 +/- 30 FPS vel

> 200-220 WC lead (such as Lyman 452460) OAL 1.16"
> 4.6-5.0 gr BE, Red or Green Dot, 700X 6.0-6.5 gr Unique 5.5-6.0 gr WW231 The above loads can also be used with a 230 gr FMJ bullet to get 850 FPS +/- 30 with overall length of 1.25"

American Rifleman January 1987, pg. 66
Loads to Duplicate Military Ball Ammo

> 230 gr FMJ
> 4.6 gr BE 5.6 gr WW231 6.5 gr Unique

A Load for the 9mm Service Pistol

Here is a 9mm load that David Sams has been using for testing Beretta 92 barrels and his completed guns that has been proven sub 1.5" groups at 50 yds is as follows:

- Starline brass
- 115gr Hornady FMJ-RN bullet
- 6.0 - 6.2gr Alliant Power Pistol powder
- Rem #5 1/2 SP primers
- O.A.L. -- 1.120" no-crimp

Out of a barrel test fixture using Bar-Sto barrels groups have consistently run between .975" to 1.5" averaging about 1". Through David's completed 92F's, he claims that they have been running 1.3" - 1.5" as a reminder these are hand-loaded.

A Load for the .32 Caliber Pistol

Here is a .32 S&W Long load, courtesy of Dr. Darius Young, for the .32 caliber center-fire pistol. It is designed for pistols that have had replacement barrels of .308" diameter and yields remarkable results at 50 yds.

- Starline brass
- 100gr H&N Swaged hollow-based wadcutter @ .312 diameter
- 1.8gr of Accurate #2 powder

Manufacturers of Reloading Components

Dillon Precision	The maker of the finest progressive presses for the money!
Nosler Bullets	Manufacurer of popular and accurate jacketed bullets.
Hornady Bullets	Manufacturer of bullets and other components for reloading.
Starline Brass	The Sedalia, Mo. company makes fine handgun brass.
Winchester	Winchester reloading components and reloading guide.
Barnes Bullets	Another well-known U.S. bulletmaker.
Alliant Powders	Alliant (formerly Hercules) manufactures the finest gun powders.
Vihtavuori Oy	Finland's superb manufacturer of gun powders.
Hodgdon Powders	The Shawnee Mission, Kansas manufacturer of gun powders.
IMR Powders	IMR manufactures a full line of gun powders.
Lyman Products	Lyman reloading components and equipment.

Further Reading

- Starline (http://www.starlinebrass.com/)
- Alliant Bullseye (http://www.alliantpowder.com/)
- Les Baer (http://www.lesbaer.com/)
- Dillon Precision (http://www.dillonprecision.com/)
- Nosler Bullets (http://www.nosler.com/)
- Hornady Bullets (http://www.hornady.com/)
- Starline Brass (http://www.starlinebrass.com/)
- Winchester (http://www.winchester.com/)
- Barnes Bullets (http://www.itsnet.com/home/bbullets/)

- Alliant Powders (http://www.AlliantPowder.com/)
- Vihtavuori Oy (http://www.vihtavuori.fi/)
- Hodgdon Powders (http://www.hodgdon.com/)
- IMR Powders (http://www.imrpowder.com/)
- Lyman Products (http://www.lymanproducts.com/)

TECHNICAL INFORMATION

Care and Feeding of Your Hammerli 208s Pistol

By Larry Carter

Source: http://bullseyepistol.com/22care.htm

Besides being the continent's premier Hmmerli dealer, Larry Carter is also one of our best shooters, a man who has broken scores of records over the years. His tips for care of the Hammerli 208s are excellent and are recommended for any fine .22 target pistol. Many thanks to Larry for contributing this article.

CLEANING

Chamber

In my opinion, while over-cleaning can do more damage than good, proper cleaning and maintenance can give your 208 a long and trouble-free life. The most important area to keep clean is the chamber. These guns obtain phenomenal accuracy, in part, because of their tight chamber dimensions. If the chamber is allowed to build up an accumulation of carbon or lead, feeding, extraction, ejection and worse problems can occur. The method of cleaning that I prefer is to bend a bore brush into an "L" shape with the short leg being only as long as a spent case. This will allow you to clean the chamber without getting the brush into the rifling. Insert it into the chamber and rotate back and forth to loosen any carbon or lead that may be stuck to the walls. Follow this with a "Q-Tip" to get this debris out of the gun before putting a patch through the bore.

Bore

I never use a metal brush in the bore. The bore should come clean with, at most, a nylon brush and some cleaner. NEVER clean from the muzzle. The crown of the barrel is immensely important to accuracy. A small nick or scrape that cannot be seen with the naked eye can open groups threefold or more. Take the time to disassemble the slide from the frame and do it from the breech. You'll be glad you did. A pull-through device made with nylon "weedwacker" string works good for a quick patch through the bore, but a thorough cleaning requires a jag from the breech end.

Frame

A toothbrush and "Q-Tips" work well to clean the area of the feed ramp, ejector and slide face between matches. (Be careful in the area of the slide stop, not to snag the small hair spring with a brush.) DO NOT IMMERSE THE FRAME OR WASH IT WITH SOLVENT SPRAYS. The sear and hammer are specially lubed at the factory for extended life of these parts. If you wash the lube off, you may feel no difference, but these parts will wear very quickly and they are not cheap. The frame should only be immersed during a complete teardown of the pistol by a qualified gunsmith. Relubing of the sear and hammer is not possible without removing them from the frame. Regarding oiling, do not drop oil on the gun from the bottle. Instead put a drop on a "Q-Tip" and wipe onto the areas of contact between the frame and slide. Too much lube attracts more contaminants and holds them there to wear your pistol. The lube is not intended to augment function, but to prevent wear. MORE IS NOT BETTER!

Magazines

Often forgotten in the cleaning regimen, magazines play an important role in the functioning of your Hammerli. Some of the by-products of combustion in a blowback action wind up in the magazines. This debris, added to the bullet lube that comes off inside the mags, can make them sluggish or cause them to stick in the process of feeding the next cartridge. After disassembly, use a nylon bore brush in 44-45 cal. to free up the crud. Finish with a good rinse in solvent followed by some light oil (again using a "Q-Tip) finishes the job. If you have an older gun or many rounds fired, check the track where the follower button rides for chatter marks. These will give problems sooner or later and should be replaced. (The metal housing is available separately).

NOTES

If you drop a round on the ground, no matter how clean you think it is, throw it away. A microscopic piece of dirt passing through the bore at 1000 feet per second can do serious damage The spring for the trigger bar is supposed to have a slight twist. Don't straighten it. Be careful with magazine bases. They are designed to blow off in the event of a cartridge rupture and are more fragile than you might think. If you change ammo, clean the chamber. Ammo with a shorter bullet may leave lead in the front of the chamber, creating an obstruction to ammo with a longer bullet. In some guns this can cause the gun to fire out of battery resulting in a case separation. Watch for wear on the recoil spring. With time, the sides of the spring will wear flat, weakening the spring. This is usually visible to the naked eye. Don't try to

remove all slack when adjusting the trigger. Some slack is required for the trigger bar to reset between shots. Likewise, do not take all of the over-travel out of the trigger. This will allow the sear to drag the hammer as it falls, shortening it's life. Dry fire only with a plastic plug in the chamber. Be careful of the rear sight mounting screw. It has a large head, but the screw is very small and can be twisted off if tightened too aggressively.

LARRY'S GUNS
HAMMERLI SALES & SERVICE 49 HAWTHORNE ST. PORTLAND, ME 04103
PHONE: (207) 772-0998 FAX (207) 772-0628 WEBSITE: www.larrysguns.com

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Further Reading

- www.larrysguns.com (http://www.larrysguns.com)

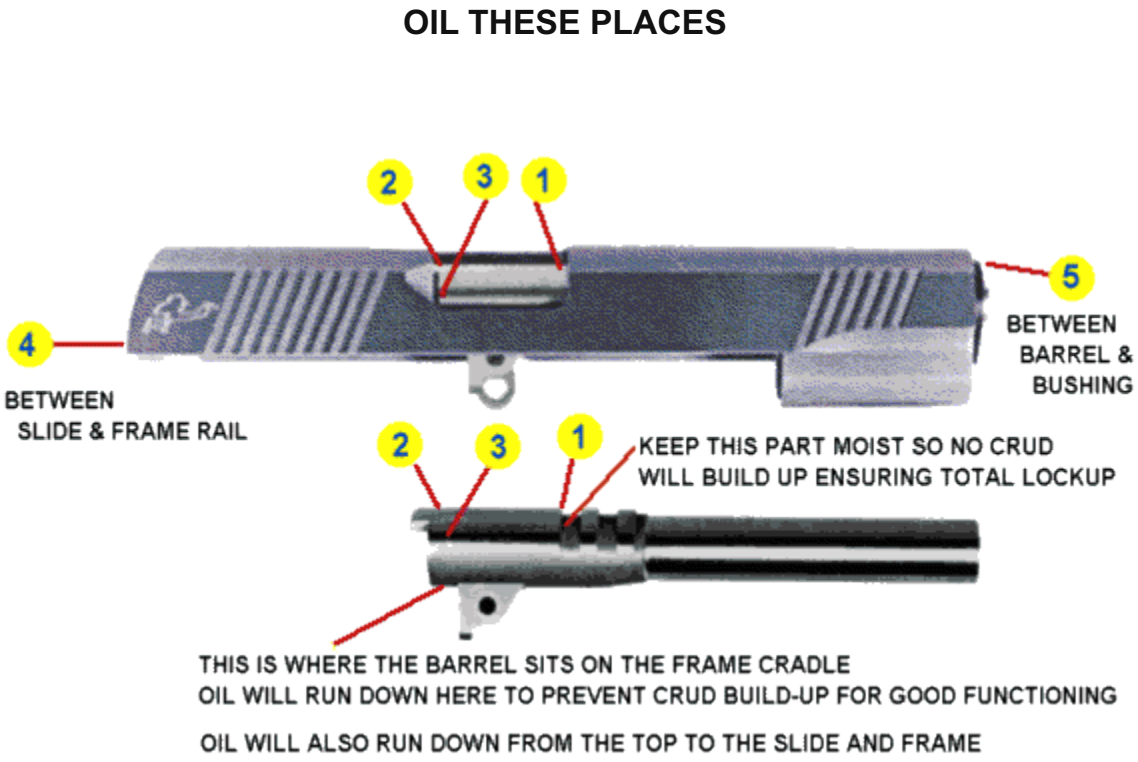
TECHNICAL INFORMATION

Oiling the M-1911 Pistol

By Edward Masaki

Source: https://www.bullseyepistol.com/oiling.htm

Ed Masaki is a master pistolsmith from Hawaii. If there was someone who would know the best methods for maintaining a fine accurized pistol, he is the man.



Ed Masaki on Oiling the 1911

When the numbered places look dry, oil them.

TECHNICAL INFORMATION

Restrictions on the .45 Caliber Service Pistol

By As Defined by the DCM

Source: http://bullseyepistol.com/dcm.htm

There is often confusion on the restrictions for the .45 caliber "service pistol" as designated in the NRA rules for CMP (or EIC) Matches. The following information answers commonly asked questions relating to hardball guns.
This information was distributed at Camp Perry a couple of years ago. It was prepared by Michael W. Rose of the DCM. Its intent is to clarify AR 920-30, the Army Regulation that applies to Leg matches.

GENERAL RESTRICTIONS:

- Trigger pull must be at least 4 lbs.
- Grips, right and left, must be "mirror image" no thumb rest.
- Pachmayr style wrap around grips are legal as long as they do not have finger grooves.

ADDITIONAL RESTRICTIONS:

LEGAL ITEMS	ILLEGAL ITEMS
Extended Triggers	Allen head or torx head grip screws
Adjustable Triggers	Ambidextrous Safety
Vented Triggers	Beavertail Grip Safety
Trigger Shoes	Combat style hammer
Bright Finish	Covered Mainspring Housing
Colt "Gold Cup" (no illegal items)	Extended Magazine Release
Front Strap Stippled	Extended Magazine Well Funnel
Gold Cup Dimple	Extended Slide Stop
Lowered Ejection Port	Extended Thumb Safety
Rear Sight "Melted" into Slide	Magazine Base Pads
	Ribs

TECHNICAL INFORMATION

Solutions for Pistol Malfunctions

By Dave Salyer

Source: http://bullseyepistol.com/salyer5.htm

If you have more than about one failure to eject in three hundred rounds with a particular pistol, you need to read on.

I will assume that you thoroughly clean the action and chamber of your pistol at least every 300 rounds. I will also assume that you are using commercial ammo or that you are a careful reloader.

Contrary to popular opinion few "stovepipes" are caused by the wrong recoil spring strength; but lets eliminate that possibility first. Fire the gun several times paying special attention to whether the slide cycles fully. It should come all the way to the rear with a light thud on each round. If it does you have eliminated the recoil spring as the culprit.

Next lets jump to the cause of more than 95% of stovepipes. This is improper extractor condition or tuning. To check this out, you can conduct a simple test in your home. Load an empty unswollen case into the chamber and slowly pull the slide backward while observing the case. The case must come out remaining horizontal and snug up against the face of the slide or bolt. If it falls before it contacts the ejector you will have many stovepipes! If it droops, even slightly you will have too many malfunctions for bullseye competition. This test should be done with and without an empty magazine in the pistol. The empty case must clear the magazine in its way back. (unless you have certain european guns that have the ejector built into the magazine near the rear.). If your gun doesnt pass this simple test go through the following corrective steps, or have a pistol mechanic do it for you.

- Check the extractor spring for dirt and tension. It doesnt have to be strong but it must provide enough force on the extractor to pivot it to the left to bite the empty case holding it against the opposite ledge without drooping. If you are working on a .22 you can usually correct the problem with a slightly longer spring or a short spacer at the back end of the spring. I have fixed several using a cut-to-fit bic® lighter spring. If its your .45 government model you can correct the tension by slightly bending the extractor in a gentile curve to the left. Be careful about too much spring pressure because the side effect of this is feeding problems caused by excessive drag on the rim as the loading round moves up under the extractor hook.
- If your empty case still droops or drops you may be able to correct the problem by filing a small amount of metal out of the throat of the extractor. This will let it pivot or move inward a little to bite the case.

- If your .22 still doesnt hold the case snugly in a horizontal position, you may then need to sharpen the point of the extractor to get the necessary bite.
- Your .45 acp doesnt work exactly the same... Dont sharpen anything on its extractor. The flat part just behind the hook holds the case rim correctly and the nose of the hook must not touch the case at all. (if you old timers like me have ever broken a .45 extractor it was undoubtedly caused by the nose riding up on the tapered part of the extractor groove in the round.) Many, if not most of the non-Colt® extractors have too much nose on them that should be filed off as part of the tuning process. Check your cases for small dings in the tapered part of the groove.
- One other test is required to prevent that occasional piece of brass from being trapped between the slide and barrel. That is, the brass must have a clear path away from the gun. It cant hit your scope or mounts on the way out. If you are a lefty, it cant hit your thumb either.

When you get your pistols in shape to pass the simple tests above, you will find that you wont have to select a special brand of commercial ammo to make your .22 have minimum stovepipes. You wont have any! Your .45 loads will not have to be too hot either.

GENERAL FIREARMS INFORMATION

Basic Firearms Safety

Source: http://www.bullseyepistol.com/safety.htm

WARNING: THIS IS NOT INTENDED TO BE INSTRUCTION IN THE USE OF FIREARMS. OBTAIN FIREARMS INSTRUCTION ONLY FROM A QUALIFIED INSTRUCTOR.

Since there is always concern outside the shooting community about gun accidents and safety, it's a good idea to know what the commonly accepted safety rules are. Also, while this article won't teach you marksmanship, you're better off knowing this material than not.

If all your knowledge of firearms comes from the popular media, everything you know is wrong. Wipe it out of your mind and we'll start from scratch.

An Important Word About Children and Guns

Even if you do not own a gun, there's a good chance that your children will encounter one someday. Your children's friends might live in homes with guns, or your children might find one that a criminal has discarded. There are about 200 million firearms in the U.S., so some sort of encounter is almost inevitable.

To prepare for this, the Eddie Eagle Gun Safety Program suggests you teach your children these steps to follow when encountering a gun:

1. Stop!
2. Don't touch.
3. Leave the area.
4. Tell an adult.

Eddie Eagle instructional materials are made available to schools, law enforcement agencies, and affiliated groups free of charge. For more information about the Eddie Eagle program, call (800) 231-0752.

See the last section for more information about children and gun safety.

Basic Rules of Safe Gun Handling

1. All guns are always loaded.
2. Never point a gun at anything you are unwilling to destroy.

3. Keep your finger off the trigger until you are ready to fire.
4. Know your target.
5. Keep your gun at the minimum level of readiness.

These rules are valid whether you are hunting deer, practicing your target shooting, or confronting an intruder.

(Every organization and person who teaches firearms safety has these rules organized a little bit differently. This is how I prefer to do it.)

Let's elaborate on those rules a bit...

(1) All guns are always loaded.

This means you should NEVER relax the amount of care with which you handle a gun just because it's not loaded. Always handle an empty gun with as much care as a loaded one.

There are three reasons for this rule. First, you could be wrong. The gun could be loaded. Maybe you just forgot to unload it, or maybe you forgot that you loaded it. Maybe you missed one of the cartridges when unloading. Maybe someone else messed with the gun when you weren't looking. Maybe you've picked up the wrong gun. There are lots of ways that the gun could be loaded without your knowing it.

Second, unsafe handling is a bad habit. Once you begin to do something unsafe with an empty gun, you will eventually screw up and do it with a loaded gun.

Third, people around you won't know the gun is unloaded. In fact, if they know to treat all guns as loaded, they will naturally assume yours is loaded. At the very least, you will frighten them. At worst, they will do something drastic in reaction to a perceived life-threatening situation. In particular, they might shoot you first.

Violations of this rule have lead to many of the most tragic and avoidable gun accidents.

(2) Never point a gun at anything you are unwilling to destroy.

Although there are lots of things that you probably don't want to destroy---walls, windows, furniture, appliances, pets---the most important point is that you shouldn't aim your gun at people. Don't point your gun in the general direction of people. Don't point your gun at objects with people in or behind them. Keep the gun pointed in a safe direction (usually straight up or straight down) at all times.

Aside from the obvious danger of an accident, pointing your gun at someone is rude and often a crime. Also, as already mentioned, the other person might shoot you first.

To elaborate on this, it is rude to even face a person while holding a gun by its grip. The difference between pointing a gun down and pointing it forward is 1/10 of a second, which no one's reflexes can beat. Thus you are forcing that person to make a choice: Either trust you with their life or draw and fire immediately. It is rude to force that choice on another person. It is also deadly dangerous. Above all, never ever face or approach a police officer with a gun in your hand. They will shoot you. [J.H.]

Obviously, in a self defense situation, you might end up pointing your gun at a person; but you should do so only if that person's destruction is a consequence you are willing to face.

Note: In some circumstances, particularly in urban settings, there are no safe directions. Don't handle guns in those situations.

(3) Keep your finger off the trigger until you are ready to fire.

The trigger is for firing the gun, not for resting your trigger finger between shots. You can rest your finger on the trigger guard or on the body of the gun just above the trigger---this is known as the "index position." Moving your finger to the trigger is such a small movement that it won't slow you down no matter how badly you need to fire.

If your finger is resting on the trigger, it's too easy to fire the gun accidentally. A sudden loud noise could startle you, or you could trip and fall, or run into something. [M.D.] Simple tension in a high stress situation can cause you to tighten the muscles in your hand without being aware of it. [J.H.]

(4) Know your target.

If you don't know what your bullet will be hitting, don't pull the trigger.

Keep in mind that your bullet may miss or pass through your intended target and hit another one. Also, watch out for ricochets off of hard, flat surfaces or even off of the flat surface of a pool of water. (Bullets will skip off of water just like stones.)

The first question that gets asked at a police shooting investigation is "What was your backstop?" The answer had better not be "A school full of children" if the officer wants to keep his job. [J.H.]

If you hunt, make sure you can see the animal in your sites when you shoot. A rustling in the bushes is not good enough, it could be anything from a fellow hunter to a couple of kids making out. This is especially tricky when your brain is telling you you're seeing more than you really are. If you see a deer dart into some bushes, your brain will match the rustling of the leaves with the deer's last visible motion and tell you where the deer is, almost as if you could see it. The problem is that you can't tell what else is in the bushes.

In a defense situation, the same rule applies. Shooting through doors or at shadows and sounds is a good way to hit an innocent person. You wouldn't want to shoot your child for sneaking into the house after being out too late, or your spouse for eating a midnight snack. Always identify your target.

(5) Keep your gun at the minimum level of readiness.

This usually means keeping your gun unloaded when you're not using it. Obviously, if you expect trouble at any moment, your gun should be loaded and ready. On the other hand, you shouldn't leave a loaded gun around the house if it isn't for self defense.

If you're at the shooting range, keep the action open until just before you start shooting so other shooters can clearly see that your gun is safe.

If you're out hunting, keep the gun unloaded until you get to the hunting area. Keep the firing chamber empty until you need to load it, and the action open until you're ready to shoot. (Obviously, for some types of game, the gun has to be ready at all times).

Additional Safety Rules

(6) Always check the load condition of any gun you handle.

A gun in your hand is your responsibility. Never take someone else's word that it's unloaded (or loaded). [J.H.]

On a revolver, simply pop out the cylinder. If it's loaded, just eject the cartridges. This is a little trickier on a magazine-fed gun. If the magazine is removable, take it out. If the magazine is fixed, check it and empty it. Then open the action to empty the firing chamber. (Some revolvers don't have a cylinder that pops out, so you have to unload the gun through a loading gate.)

If someone offers you a gun that you are unfamiliar with, don't be afraid to ask how it works.

This is a very good habit to get into. Get religious about it. Check every time you pick up a gun, every time you hand a gun to someone else, and every time someone hands you a gun. Check even if you see the other person check it first. Don't worry that the other person will feel insulted: Experienced shooters will appreciate that you are concerned with safety---both yours and theirs.

Most firing ranges have a rule against handing a loaded gun to another person. It's a good rule to follow at any time. The polite way to hand a gun to another person is unloaded, with the action open, barrel pointed down and trigger towards them.

In a combat situation, of course, you should still check the gun, but this time to make sure it's loaded. [J.H.]

(7) Know how the gun works.

It's hard to use gun safely if you don't know to use it at all. If you don't know how a particular gun works, read the manual, ask the person you got it from, write the manufacturer for a manual, or seek the advice of an expert.

Most gun manufacturers will be delighted to send you a free manual for your gun if they made it. It's a lot cheaper than fighting a law suit after you do something stupid. [J.H.]

(8) Make sure the gun is safe to operate.

Regular cleaning and proper storage are necessary for safe operation. If you are unsure about a gun, have it inspected by a gunsmith.

(9) Use only the correct ammunition.

The wrong type of ammunition can cause all kinds of problems, some of them fatal. Just because the ammunition fits in the gun doesn't mean it's right for the gun. Ammunition with extra powder (often labeled "+P" or "+P+") can blow up the gun. Ammunition that is too small can result in the cartridge sliding out of the firing chamber and into the barrel. If a correct round is then loaded and fired, the obstructed barrel could blow up the gun.

The correct type of ammunition is usually printed on the side of the gun. You can also get this information from the manual, from the manufacturer, or from an expert.

Always store ammunition in the original boxes, because it's hard to tell the cartridges apart once they're out of the box. Also, don't carry more than one type of ammunition, especially if they look enough alike that you could easily confuse them. (E.g. carrying .22 LR and .410 shells is okay, but don't carry both 12-gauge and 16-gauge shells.)

(10) Wear eye and ear protection.

Gunfire always involves fire and smoke, which can harm the eyes. In addition, autoloading guns are also auto-ejecting, and the ejected casings can bounce back into your eyes. Finally, a shot into a nearby object can produce splinters and other flying debris.

It's important to wear glasses that wrap around the sides or have protective side pieces. You should also wear a hat with a brim that comes down to the top of your glasses to prevent flying shell casings from lodging behind them. ("Deadeye" isn't a romantic nickname when it applies to you). [J.H.]

Hearing damage is cumulative. Firing a few shots without ear protection probably won't hurt you, but continual exposure to gunfire will eventually cause nerve damage and deafness. (Lots of old-time shooters are nearly deaf from shooting without protection.)

Hearing protection comes in two general flavors: exterior (muffs) and interior (plugs). Plugs may be adequate for outdoor ranges. Muffs are essential at indoor ranges. Indoors, the overpressure from firing the gun has nowhere to go and is reflected back at you from the walls. It can be conducted to your inner ear through the mastoid bone and do damage even though you don't feel anything. You should probably use both muffs and plugs simultaneously and always get the highest rated ones you can find (typically 29 db for muffs and 30+ for plugs). [J.H.]

(11) Know how to handle misfires, hangfires, and squib rounds.

If you pull the trigger and the gun doesn't fire, there could be several reasons. It may not be loaded, or it may not have fed the ammunition from the magazine after the last shot. If the cartridge is loaded okay, it might simply be a bad round that doesn't fire---a misfire. Finally, it could be a hangfire, a round that fires slow.

Since you can't tell without examining the round, and since a hangfire going off while you're looking at it would be dangerous, it's best to just hold the gun pointing at the target for at least 30 seconds. If it doesn't go off, you can begin to investigate.

A squib round is one that has a weak powder charge. The bullet could easily get stuck in the barrel. If this happens, the report will be quiet and the recoil will feel funny. You should immediately unload the gun and inspect the barrel and firing chamber(s) for a stuck bullet. You may want to have a gunsmith examine the gun before you begin firing again.

(12) Always store the gun and ammunition separately, and where unauthorized persons---especially children---cannot get at it.

See below for more information about children and guns.

(13) To unload, remove the magazine, then cycle the action.

A magazine-fed pistol, rifle, or shotgun loads the firing chamber from the magazine every time the bolt, slide or pump mechanism is operated. If you're not paying attention, you could accidentally load the chamber. An autoloader requires even more care because it will keep the chamber continuously loaded until the weapon runs out of ammunition---you don't have to do anything.

This can present a hazard when you try to unload the gun. Simply removing the magazine (or unloading a non-removable magazine) is not good enough because there could be a round in the firing chamber. You have to cycle the loading mechanism to get it out of there. However, if you cycle the mechanism before removing (or unloading) the magazine, you will simply load another round to replace the one you ejected.

The solution is to remove (or empty) the magazine first, then cycle the loading mechanism to eject the loaded round. With no cartridges in the magazine, the firing chamber will not be reloaded.

Accidentally loading the chamber is probably the single most common cause of accidental discharges: You want to show someone your gun, or you want to check the trigger pull to see if it's smooth enough, so you cycle the slide to eject the round from the chamber, then remove the magazine. Now you check the trigger with a careful tug and BLAM! Where'd that come from? Of course, if you were being careful where you were pointing the gun then there's no harm done except to your ego and your reputation.

(14) Drinking and shooting is as bad as drinking and driving.

The old adage "alcohol and gunpowder never mix" is as true today as ever.

(15) Use the safety, but don't rely on it.

A "safety" is a mechanism on a gun designed to prevent the gun from firing. Since keeping your finger off the trigger also prevents the gun from firing, safeties are mostly intended to prevent accidents if the gun is dropped or snagged on something.

(Note: Most long guns are not drop safe, regardless of safeties. In most cases long gun safeties only block the trigger. They do not block the firing pin or even the hammer.) [J.H.]

In gun training circles, the subject of safeties is a bit controversial. Instructors are concerned that students may attempt to use the safety as a substitute for safe handling practices. Consequently, many safety courses never even mention them.

The consensus seems to be that if your gun has safety mechanisms, you should learn how they work, and you should use them whenever possible, in keeping with the idea of minimal readiness.

However, a safety is a mechanical device and any mechanical device can fail. Use them, but never depend on them.

Keep in mind that even the best safety mechanism is not as safe as unloading the gun; and since you shouldn't relax your safe handling with an unloaded gun you certainly shouldn't be

any less careful with a loaded gun that has the safety on. In particular, don't assume that it's okay to pull the trigger or violate any of the other safety rules just because the safety is on.

(16) Non-firearm safety is important too.

Airguns, paint guns, and dart guns aren't firearms, but that doesn't mean you shouldn't follow the safety rules with them too.

First of all, they do fire projectiles which could hurt people under the wrong circumstances.

Second, some of these guns look enough like real firearms to frighten people into thinking you are dangerous. This is definitely rude, possibly illegal, and sometimes dangerous---people might take drastic actions such as jumping out of window to avoid you or trying to shoot you first.

Third, you can pick up bad habits this way.

Obviously, you have to make adjustments. For example, the whole point of having paint guns is that you and your friends can play with them and shoot each other. Still, it's good practice to treat these guns like real firearms by following all the safety rules (finger off the trigger, muzzle pointed in a safe direction, unload when not in use, etc.) except when it is necessary to shoot a participant in the game.

(17) You are responsible for any gun you own, no matter whose hand it's in.

Be cautious and thoughtful about who you loan your guns to and the circumstances under which you loan them. (State laws may also apply). [J.H.]

Even if you cannot be held liable in your state, you don't want the burden of guilt that comes with someone getting killed with a gun you were responsible for.

(18) There are often special rules for specific situations, such as at a gun range or out hunting.

It is your responsibility to learn these rules.

Children and Guns

If you're thinking of mixing children and guns in the same household, there are a few things you should know.

Some Ineffective Ideas to Avoid.

Hiding your gun is a good idea, but not good enough. Kids like to explore and find new things, so hiding your gun is of limited benefit. A child may very well go through all of the drawers and cabinets in the house. Putting the gun out of reach only works until your kid learns to climb or use a ladder or stepping-stool. (Don't think your kids don't climb on the furniture just because you've never seen 'em do it.)

Don't make the mistake of thinking your child is too weak to use the gun. Large autoloading pistols will have strong slide springs that even many adults find difficult to work, so you may be tempted to leave the chamber of a gun empty on the assumption that your child won't be able to load it. Also, there are gun locks that use a strong spring or rubber band to hold the gun, on the theory that only adults will be strong enough to release the gun. Both these methods are insecure against any child larger than an infant. Your child can probably find some way to put his entire weight on the slide or the lock, usually by carrying it to a stairstep and standing on it.

Basically, if your child-proofing mechanism depends on your child's lack of height, strength, or intellect and your child is beyond infancy, it's not good enough. Children spend many years being small and weak, and they learn lots of ways to compensate. They also have a lot of spare time to figure things out. Your children can probably defeat any mechanism you can. Thus, the only effective child-proofing methods are those that also work against unauthorized adults. You need real security.

Real Security

Gun stores and gun magazines have lots of useful security products for sale. A trigger lock will be good enough for most people (but never use one on a loaded gun). There are also locks which extend a long rod down the barrel or which go through the action. One of those plastic-coated bicycle locking cables can be threaded through the receiver and out the ejection port of many magazine-fed guns. You could even take the gun apart and put a padlock through a hole in one of the important pieces to prevent reassembly.

If you want protection from damage during transportation, you should purchase a lockable gun case. You could also invest in a gun safe. This would offer protection against theft, and it would allow you to store a self-defense gun loaded and ready.

Locks with multiple wheel combination mechanisms should have at least five wheels. More is better. Unfortunately many come with only three. This may fulfill the letter of the law in some states, but it is not secure. They can be opened in 10 minutes, on average, simply by systematically trying every possible combination (there are only 1000). [J.H.]

Keeping Your Children Safe From Accidents With Other People's Guns.

One weakness in this whole approach is that it only keeps your children from mishandling your guns. If your child finds a gun on the street, or discovers one at a friends house, all of your gun locks won't help. A different approach must be used. You need to "gun-proof" your child.

The idea is similar to the approach many parents use when they are concerned about their children's exposure to sex, AIDS, drugs, or any other issue with an important moral dimension. Children have a better chance of getting it right if their parents teach them than if they learn from television or from their peers. You need to talk to your children about what's important, about your values.

When your children are young, teach them what to do when they encounter a gun: Stop! Don't touch. Leave the area. Tell an adult. (The Eddie Eagle materials mentioned earlier may be useful here.)

If your children play with toy guns, begin teaching them the safety rules right from the start. If they learn to handle the toys safely, they'll be less likely to do something disastrous when they get their hands on the real thing. Help them form good gun safety habits.

When your children get a little older, they may become curious about real guns, especially if they've seen you with one. Answer your child's questions about guns. When you're watching television, point out unsafe gun handling and other mistakes.

It would be best if you satisfied their curiousity before they try to satisfy it themselves. Unload the gun and let them handle it. Remind them of the safety rules. Show them how it works. (If you think you can get away with it, teach them to clean it for you.) Just make it clear that they're not to touch the guns without your supervision.

Try taking them shooting. Even a very young child can shoot safely if you are always standing right behind him or her. It's not too unusual to see 7-year-old children shooting with their parents. Again, make sure they follow the rules and show them proper shooting behavior. You might want to find a place where you can shoot up a few melons or gallon water jugs. It's fun, and it's a graphic illustration of why guns are not toys.

There's always the chance that shooting will turn into a fun family hobby. Chances are, however, that your children---especially if young---will find shooting to be loud and scary, or more likely, loud and boring. Oh well, at least you've got a child who knows about gun safety, who isn't dangerously curious about guns, and who isn't likely to do something stupid.

If you don't feel comfortable training your own kids to shoot, there are plenty of instructors available. Contact the NRA for a list of instructors in your area, or call a local gun club or gun shop.

[M.D.]

References

- more information about children and gun safety
- more information about children and guns
- Eddie Eagle

GENERAL FIREARMS INFORMATION

Logical Commentary on Gun Control

By A Variety of Analysts and Media Personnel

Source: https://www.bullseyepistol.com/experts.htm

LOGICAL ASSESSMENTS:

"No criminal ... with the intent or predisposition to commit a violent crime is going to be deterred by a local ordinance under which he could be fined a few hundred dollars for possessing the dangerous weapon he intends to use to commit the violent crime ... If the prospect of spending much of his life in prison does not deter this particular individual, then why in the world would the threat of a minor monetary fine deter him?" -*Governor Allen, State of Virginia*

"One tenet of the National Rifle Association's faith has always been that handgun controls do little to stop criminals from obtaining handguns. For once, the NRA is right and America's leading handgun control organization [Sarah Brady's Handgun Control, Inc.] is wrong. Criminals don't buy handguns in gun stores. That's why they are criminals." -*Josh Sugarmann, Communications Director for the National Coalition to Ban Handguns*

"If one looks on the (gun-control) debate as being only about crime and the equipment with which it is committed, one misses the entire point; the "debate" over guns is fundamentally a cultural and political debate." - *James D. Wright, Department of Sociology at Tulane University*

"Cultural conditions, not gun laws, are the most important factors in a nation's crime rate. Young adults in Washington D.C. are subject to strict gun control, but no social control, and they commit a staggering amount of crime. Young adults in Zurich are subject to minimal gun control, but strict social control, and they commit almost no violent crime." - *David B. Kopel, NYC District Atty. & Policy Analyst at Cato Institute*

"A fear of weapons is a sign of retarded sexual and emotional maturity." - *Sigmund Freud*

"The weapons-violence hypothesis is far too simplistic a basis on which to base sound public policy." - *Daniel D. Polsby, Professor of Law at Northwestern University*

"Gun controls would not work in America ... legal controls ought to support, not replace, self-control." - *Thomas S. Szasz, Professor of Psychiatry at SUNY Science Center*

NATIONS ENACTING GUN CONTROL:

Countries having successful gun countrol laws in the 20th Century -- and the results of those laws: 1915-1917 Ottoman Turkey, 1.5 million Armenians murdered 1929-1953 Soviet Union, 20 million people that opposed Stalin were murdered. 1933-1945 Nazi occupied Europe, 13 million Jews, Gypsies and others that opposed Hitler, murdered 1948-1952 China, 20 million anti-communists or communist reformers, murdered 1960-1981 Guatemala, 100,000 Maya Indians, murdered 1971-1979 Uganda, 300,000 Christians and political rivals of Idi Amin, murdered 1975-1979 Cambodia, 1 million educated persons, murdered

EDITORIAL BY HUGH DOWNS:

I realize that what I am about to say may alienate people who think they know where I stand on issues. But I want to take a somewhat different look at the gun issue: Everyone seems to agree that we have a terrible problem with guns in this country. There are headlines as well as statistics to prove this. Few, however, agree on the causes of the problem, or what to do about it. Among the theories about the origin of this mess is one that says our frontier heritage cast into our culture a belief that because we got out from under the tyranny of England's George III with a shooting revolution, that implementing justice, redressing grievances, and correcting wrongs of any kind are all best handled with guns. The good guy--the guy on the proper side of the law--wins in a throw-down because he is faster on the draw. This is demonstrated by every B Western movie and a thousand pulp fiction novels. (And the genre of Western Fiction is not a passing fad. It is our Iliad and Odyssey and it's here to stay.) The theory is that deep in our collective psyche is the feeling that a gun is a social tool and an indispensable prop of American Civilization. Another theory cites a drug policy in this country that lacks sanity. Because we have put a monster drug traffic outside the law--we have outlawed drugs instead of regulating them--we have started with a medical problem and managed to turn it into a crime problem--and so we lost control of the situation, and the flow of guns into the streets and even the schools is a result of this. I subscribe to both these theories. But whatever other causes may be operative, the problem has little to do with the number of guns per capita. Israeli citizens are far more heavily armed than we are and there is less street crime there than in the US. In Switzerland, with its compulsory military service, every home has a gun in it--often a machine pistol--and they can boast the lowest gun-death rate of any industrialized country. Gun aficionados are fond of saying "Guns don't kill people--people kill people." Well, they are right. A firearm is a potent weapon that can provide action-at-a-distance--that can be lethal fairly far away from the person wielding it. There is a very real question whether humans have progressed socially and psychologically

enough to deal, not just with atomic bombs, but with small calibre firearms. But these things are here to stay. They are with us, and we can't un-ring that bell. It might seem that it would be nice if no such things existed--if there were no instruments to send pellets of metal tearing through flesh--but when there were no guns people impaled each other on spears and hacked each other to pieces with swords, so the difficulty is not guns, but human nature. And we aren't going to solve that problem with simple prohibitions. Abolishing guns in not in the cards. [Sensible regulation is.] I sympathize with people who want to ban guns, but I can't agree with them. We have to be careful that in our zeal to abolish guns we don't wind up pushing counter-productive legislation that will leave armed only those people most likely to do harm with the weapons. We have probably 200 million firearms in the United States, owned by 65 million Americans. About 25% of us are gun owners. (In Switzerland it's closer to 100%). More than 98% of U.S. guns are never involved in crimes. The headlines, of course go to those that are involved. When the magnitude of the gun problem dawned on the American Public, the reaction was two-fold: one faction sought to ban guns, or to put in place controls so stringent as to be unworkable; and the other decided to arm themselves individually. Both responses were in the main inappropriate. Outright bans are counter-productive; and untrained civilians carrying firearms are dangerous--to themselves and others. I have reservations about mass arming of everyone--the potential for increased incident of accident and crimes of passion looms as a great worry. But here is where well-thought-out regulation can play a part. Taking courses in the proper and safe use, transport and storage of firearms can go a long way to keeping accidents, etc., down. Waiting periods and qualifications for licensing for each type of weapon are in order. As for training in gun use and gun safety, the dreaded National Rifle Association has actually been in the lead on safety and proper use of firearms. Their image has been besmirched, probably unjustly, by zealotry in the ranks (and in management), which gave the appearance at times of wanting 9-year-olds to be able to walk into a gun shop and buy armor-piercing bullets. This was nonsense, and the organization is not as irresponsible as many think. (This statement is a little like George Bernard Shaw's when he was a music critic and said "Wagner's music is not as bad as it sounds."). But in fairness, the NRA has a point about the inadvisability of simply taking guns away from the populace. If that were possible, it would not disarm that small percentage of the populace willing to break the law, and the social environment. It would, in effect, disarm the honest bulk of the population. A narrow 1986 study, appearing in the New England Journal of Medicine stated that a gun in the home is 43 times more likely to kill a family member than an intruder. A breakdown of that number is instructive: First of all, 37 of those 43 deaths are suicides. And banning guns does not ban suicide. When Canada enacted strict gun controls in 1976 psychiatrists noticed that jumping off bridges replaced suicides previously committed with guns. In Japan, where they have tried to ban all guns, the suicide rate is consistently and dramatically higher than here in the U.S. And home defense by force of arms is not simply a matter of shooting intruders. There are cases in which burglars or intruders are wounded or

frightened away by the use or display of a firearm, and cases in which would-be intruders may have purposely avoided a house known to be armed. By and large, burglars, robbers, and muggers use guns because they assume their victims are unarmed and afraid of guns. They want control. They do not seek to have a shoot-out with their victims. Training in gun use deals with when to display a firearm and when to turn over your cash and jewelry. The total number of deaths attributed to firearms is between 25,000 and 30,000 a year. This includes suicides, homicides, the so-called justifiable homicides (mostly self-defense) and accidents. Even at the 30,000 figure it is far less than are killed by automobiles. Would we be justified in seeking to ban automobiles? American adults use many things that are dangerous, and we live with the risks. Most of America's 65 million gun owners collect and use their firearms responsibly. Punishing people who obey the law is backward thinking. It's true that there are many armed crimes in this country, but our collective delusion about firearms, obscures the real reason for almost all of that crime. It is the so-called War on Drugs, recently again called a failure by legislators who see the folly of our policy. Having abandoned our jurisdiction by putting drug traffic outside the law--outlawing substances that can be abused--we have created a situation where gang members and drug dealers buy guns on a black market where guns are cheap, untraceable and require no waiting period, and then overflow onto the streets and show up in the hands of school-children. We need to get that straightened out. In the meantime, let's not commit the folly of prohibition on yet another dangerous thing. We did it with liquor, and things got better when we woke up and brought liquor under the law and got rid of prohibition. We may yet display some common sense in the drug problem by replacing prohibition with regulation, and we have time to avoid the mistake with guns. - *Hugh Downs, ABC Reporter, in a 1995 Editorial*

GENERAL FIREARMS INFORMATION

Violent Crime and Fatherlessness

By David Kopel

Source: http://bullseyepistol.com/singlepr.htm



In 1996, Congress continued business as usual by enacting legislation which will do little or nothing to reduce violent crime, but which will infringe civil liberties. The Clinton/Dole terrorism bill is Exhibit A, while the anti-gun laws snuck through in the closing days of Congress are exhibit B.

But almost by accident, the last Congress ended up doing something which, in the long run, will likely lead to a major decrease in violent crime: welfare reform. For all the talk about the complexities of the "root causes" of crime, there is one root cause which overwhelms all the rest: fatherlessness. Almost 70 percent of juveniles incarcerated in state reform institutions come from homes with no father or without their natural parents. Most gang members, 60 percent of rapists, and 75 percent of teenage homicide perpetrators come from single-parent homes. Young black males from singleparent families are twice as likely to engage in crime as young black males from two-parent families. If the single-parent family is in a neighborhood with a large number of other single-parent families, the odds of the young man becoming involved in crime are tripled. These findings are based on a study conducted for the Department of Health and Human Services by M. Anne Hill and June O'Neill of Baruch College. The study carefully held constant all socioeconomic variables (such as income, parental education, or urban setting) other than single parenthood. Crime has often been thought of as a problem of race or poverty, since poor people and racial minorities comprise a larger portion of the violent criminal population than of the population as a whole. But in fact,

the causal link between fatherlessness and crime is so strong that controlling for family configuration erases the relationship between race and crime and between low income and crime," as Barbara Dafoe Whitehead noted in her famous "Dan Quayle was Right" article. There is also a second association between illegitimacy and crime: unwed fathers are more likely to commit crimes than are married fathers. So if illegitimacy leads to crime, then what leads to illegitimacy! Changing social mores are one factor that is beyond government's reach. But the current welfare system is another factor. You get what you pay for, and the current welfare system pays women to have illegitimate children. Doctor O'Neill found that a 50 percent increase in the value of food stamps and AFDC led to a 43 percent increase in the number of out-of-wedlock births. This finding is consistent with a Canadian study which showed that an increase of only $100 to $200 in annual welfare benefits increased by five percent the odds that a poor woman would become a single parent. Researchers from the University of Washington found that increasing welfare benefits by $200 per month per family led to a 150 percent rise in illegitimate births by teenagers. Much of the increase in illegitimacy since the mid 1960s can be attributed to increased welfare benefits and easier eligibility standards. Increased welfare benefits and easier eligibility have been shown to have led to higher illegitimacy rates by both black and white teenagers. Welfare almost always implies fatherlessness; ninety-two percent of welfare families have no father present. Thus, it should not be surprising that Hill and O'Neill discovered that a 50 percent increase in AFDC and food stamp benefits led, over time, to major increases in crimes perpetrated. The increase in crime was primarily a long-term result of the increase in illegitimacy. In other words, the government's promotion of illegitimacy through the misnamed "welfare system" creates concentric circles of victims. Illegitimate children are the first set of victims, as they grow up in homes without a father. The next set of victims are persons murdered, robbed, and raped by some of these children when they grow older. Finally, as Washington politicians slash at the Bill of Rights in a futile effort to be "tough" on crime, every other law-abiding American becomes a victim. Illegitimate children from welfare homes do not become criminals at birth; it takes about 15 years. So America is probably doomed to at least 15 more years of escalating crime, as the children whom the federal government paid teenagers to bear but not to raise properly become adolescents. But a massive change in the incentives of the welfare system will lead, in the short run, to a significant drop in illegitimacy. And that, in the long run, will make America much safer. David Kopel is the District Attorney of New York City and a Policy Analyst at the Cato Institute.

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Further Reading

- Cato Institute (http://www.cato.org/)
- Blue Press (http://www.Dillonprecision.com/)

GENERAL FIREARMS INFORMATION

The False Promise of Gun Control

By Daniel Polsby

Source: https://www.bullseyepistol.com/polsby.htm

Gun-control laws may save some lives, but they can never stem the flow of guns, and they divert attention from the roots of our crime problem.

During the 1960s and 1970s the robbery rate in the United States increased sixfold, and the murder rate doubled; the rate of handgun ownership nearly doubled in that period as well. handguns and criminal violence grew together apace, and national opinion leaders did not fail to remark on that coincidence.

It has become a bipartisan article of faith that more handguns cause more violence. Such was the unequivocal conclusion of the national Commission on the Causes and Prevention of Violence in 1969, and such is now the editorial opinion of virtually every influential newspaper and magazine, from *The Washington Post* to *The Economist* to the *Chicago Tribune*. Members of the House and Senate who have not dared to confront the gun lobby concede the connection privately. Even if the National Rifle Association can produce blizzards of angry calls and letters to the Capitol virtually overnight, House members one by one have been going public, often after some new firearms atrocity at a fast-food restaurant or the like. And last November they passed the Brady bill.

Alas, however well accepted, the conventional wisdom about guns and violence is mistaken. Guns don't increase national rates of crime and violence -- but the continued proliferation of gun-control laws almost certainly does. Current rates of crime and violence are a bit below the peaks of the late 1970s, but because of a slight oncoming bulge in the at-risk population of males aged fifteen to thirty-four, the crime rate will soon worsen. The rising generation of criminals will have no more difficulty than their elders did in obtaining the tools of their trade. Growing violence will lead to calls for laws still more severe. Each fresh round of legislation will be followed by renewed frustration.

Gun-control laws don't work. What is worse, they act perversely. While legitimate users of firearms encounter intense regulation, scrutiny, and bureaucratic control, illicit markets easily adapt to whatever difficulties a free society throws in their way. Also, efforts to curtail the supply of firearms inflict collateral damage on freedom and privacy interests that have long been considered central to American public life. Thanks to the seemingly never-ending war on drugs and long experience attempting to suppress prostitution and pornography, we know a great deal about how illicit markets function and how costly to the public attempts to

control them can be. It is essential that we make use of this experience in coming to grips with gun control.

The thousands of gun-control laws in the United States are of two general types. The older kind sought to regulate how, where, and by whom firearms could be carried. More recent laws have sought to make it more costly to buy, sell, or use firearms (or certain classes of firearms, such as assault rifles, Saturday-night specials, and so on) by imposing fees, special taxes, or surtaxes on them. The Brady bill is of both types: it has a background-check provision, and its five-day waiting period amounts to a "time tax" on acquiring handguns. All such laws can be called scarcity-inducing, because they seek to raise the cost of buying firearms, as figured in terms of money, time, nuisance, or stigmatization.

Despite the mounting number of scarcity-inducing laws, no one is very satisfied with them. Hobbyists want to get rid of them, and gun-control proponents don't think they go nearly far enough. Everyone seems to agree that gun-control laws have some effect on the distribution of firearms. But it has not been the dramatic and measurable effect their proponents desired.

Opponents of gun control have traditionally wrapped their arguments in the Second Amendment to the Constitution. Indeed, most modern scholarship affirms that so far as the drafters of the Bill of Rights were concerned, the right to bear arms was to be enjoyed by everyone, not just a militia, and that one of the principal justifications for an armed populace was to secure the tranquility and good order of the community. But most people are not dedicated antiquitarians, and would not be impressed by the argument "I admit that my behavior is very dangerous to public safety, but the Second Amendment says I have a right to do it anyway." That would be a case for repealing the Second Amendment, not respecting it.

Fighting the demand curve

Everyone knows that possessing a handgun makes it easier to intimidate, wound, or kill someone. But the implication of this point for social policy has not been so well understood. It is easy to count the bodies of those who have been killed or wounded with guns, but not easy to count the people who have avoided harm because they had access to weapons. Think about uniformed police officers, who carry handguns in plain view not in order to kill people but simply to daunt potential attackers. And it works. Criminals generally do not single out police officers for opportunistic attack. Though officers can expect to draw their guns from time to time, few even in big-city departments will actually fire a shot (except in target practice) in the course of a year. This observation points to an important truth: people who are armed make comparatively unattractive victims. A criminal might not know if any one civilian is armed, but if it becomes known that a larger number of civilians do carry weapons, criminals will become warier.

Which weapons laws are the right kinds can be decided only after considering two related questions. First, what is the connection between civilian possession of firearms and social violence? Second, how can we expect gun-control laws to alter people's behavior? Most recent scholarship raises serious questions about the "weapons increase violence" hypothesis. The second question is emphasized here, because it is routinely overlooked and often mocked when noticed; yet it is crucial. Rational gun control requires understanding not only the relationship between weapons and violence but also the relationship between laws and people's behavior. Some things are very hard to accomplish with laws. The purpose of a law and its likely effects are not always the same thing. many statutes are notorious for the way in which their unintended effects have swamped their intended ones.

In order to predict who will comply with gun-control laws, we should remember that guns are economic goods that are traded in markets. Consumers' interest in them varies. For religious, moral, aesthetic, or practical reasons, some people would refuse to buy firearms at any price. Other people willingly pay very high prices for them.

Handguns, so often the subject of gun-control laws, are desirable for one purpose -- to allow a person tactically to dominate a hostile transaction with another person. The value of a weapon to a given person is a function of two factors: how much he or she wants to dominate a confrontation if one occurs, and how likely it is that he or she will actually be in a situation calling for a gun.

Dominating a transaction simply means getting what one wants without being hurt. Where people differ is in how likely it is that they will be involved in a situation in which a gun will be valuable. Someone who *intends* to engage in a transaction involving a gun -- a criminal, for example -- is obviously in the best possible position to predict that likelihood. Criminals should therefore be willing to pay more for a weapon than most other people would. Professors, politicians, and newspaper editors are, as a group, at very low risk of being involved in such transactions, and they thus systematically underrate the value of defensive handguns. (Correlative, perhaps, is their uncritical readiness to accept studies that debunk the utility of firearms for self-defense.) The class of people we wish to deprive of guns, then, is the very class with the most inelastic demand for them -- criminals -- whereas the people most like to comply with gun-control laws don't value guns in the first place.

Do guns drive up crime rates?

Which premise is true -- that guns increase crime or that the fear of crime causes people to obtain guns? Most of the country's major newspapers apparently take this problem to have been solved by an article published by Arthur Kellermann and several associates in the October 7, 1993, *New England Journal of Medicine*. Kellermann is an emergency-room physician who has published a number of influential papers that he believes discredit the thesis that private ownership of firearms is a useful means of self-protection. (An indication of

his wide influence is that within two months the study received almost 100 mentions in publications and broadcast transcripts indexed in the Nexis database.) For this study Kellermann and his associates identified fifteen behavioral and fifteen environmental variables that applied to a 388-member set of homicide victims, found a "matching" control group of 388 non-homicide victims, and then ascertained how the two groups differed in gun ownership. In interviews Kellermann made clear his belief that owning a handgun markedly increases a person's risk of being murdered.

But the study does not prove that point at all. Indeed, as Kellermann explicitly conceded in the text of the article, the causal arrow may very well point in the other direction: the threat of being killed may make people more likely to arm themselves. Many people at risk of being killed, especially people involved in the drug trade or other illegal ventures, might well rationally buy a gun as a precaution, and be willing to pay a price driven up by gun-control laws. Crime, after all, is a dangerous business. Peter Reuter and Mark Kleiman, drug-policy researchers, calculated in 1987 that the average crack dealer's risk of being killed was far greater than his risk of being sent to prison. (Their data cannot, however, support the implication that ownership of a firearm causes or exacerbates the risk of being killed.)

Defending the validity of his work, Kellermann has emphasized that the link between lung cancer and smoking was initially established by studies methodologically no different from his. Gary Kleck, a criminology professor at Florida State University, has pointed out the flaw in this comparison. No one ever thought that lung cancer causes smoking, so when the association between the two was established the direction of the causal arrow was not in doubt. Kleck wrote that it is as though Kellermann, trying to discover how diabetics differ from other people, found that they are much more likely to possess insulin than nondiabetics, and concluded that insulin is a risk factor for diabetes.

The New York Times, the *Los Angeles Times*, *The Washington Post*, *The Boston Globe*, and the *Chicago Tribune* all gave prominent coverage to Kellermann's study as soon as it appeared, but none saw fit to discuss the study's limitations. A few, in order to introduce a hint of balance, mentioned that the NRA, or some member of its staff, disagreed with the study. But readers had no way of knowing that Kellermann himself had registered a disclaimer in his text. "It is possible," he conceded, "that reverse causation accounted for some of the association we observed between gun ownership and homicide." Indeed, the point is stronger than that: "reverse causation" may account for *most* of the association between gun ownership and homicide. Kellermann's data simply do not allow one to draw any conclusion.

If firearms increased violence and crime, then rates of spousal homicide would have skyrocketed, because the stock of privately owned handguns has increased rapidly since the mid-1960s. But according to an authoritative study of spousal homicide in the *American*

Journal of Public Health, by James Mercy and Linda Saltzman, rates of spousal homicide in the years 1976 to 1985 fell. If firearms increased violence and crime, the crime rate should have increased throughout the 1980s, while the national stock of privately owned handguns increased by more than a million units in every year of the decade. It did not. Nor should the rate of violence and crime in Switzerland, New Zealand, and Israel be as low as they are, since the number of firearms per civilian household is comparable to that in the United States. Conversely, gun-controlled Mexico and South Africa should be islands of peace instead of having murder rates more than twice as high as those here. The determinants of crime and law-abidingness are, of course, complex matters, which are not fully understood and certainly not explicable in terms of a country's laws. But gun-control enthusiasts, who have made capital out of the low murder rate in England, which is largely disarmed, simply ignore the counterexamples that don't fit their theory.

If firearms increased violence and crime, Florida's murder rate should not have been falling since the introduction, seven years ago, of a law that makes it easier for ordinary citizens to get permits to carry concealed handguns. Yet the murder rate has remained the same or fallen every year since the law was enacted, and it is now lower than the national murder rate (which has been rising). As of last November 183,561 permits had been issued, and only seventeen of the permits had been revoked because the holder was involved in a firearms offense. It would be precipitate to claim that the new law has "caused" the murder rate to subside. Yet here is a situation that doesn't fit the hypothesis that weapons increase violence.

If firearms increased violence and crime, programs of induced scarcity would suppress violence and crime. But -- another anomaly -- they don't. Why not? A theorem, which we could call the futility theorem, explains why gun-control laws must either be ineffectual or in the long term actually provoke more violence and crime. Any theorem depends on both observable fact and assumption. An assumption that can be made with confidence is that the higher the number of victims a criminals assumes to be armed, the higher will be the risk -- the price -- of assaulting them. By definition, gun-control laws should make weapons scarcer and thus more expensive. By our prior reasoning about demand among various types of consumers, after the laws are enacted criminals should be better armed, compared with non criminals, than they were before. Of course, plenty of noncriminals will remain armed. But even if many noncriminals will pay as high a price as criminals will to obtain firearms, a larger number will not.

Criminals will thus still take the same gamble they already take in assaulting a victim who might or might not be armed. But they may appreciate that the laws have given them a freer field, and that crime still pays -- pays even better, in fact, than before. What will happen to the rate of violence? Only a relatively few gun-mediated transactions -- currently, five percent of armed robberies committed with firearms -- result in someone's actually being shot (the statistics are not broken down into encounters between armed assailants and unarmed

victims, and encounters in which both parties are armed). It seems reasonable to fear that if the number of such transactions were to increase because criminals thought they faced fewer deterrents, there would be a corresponding increase in shootings. Conversely, if gun-mediated transactions declined -- if criminals initiated fewer of them because they feared encountering an armed victim or an armed good Samaritan -- the number of shootings would go down. The magnitude of these effects is, admittedly, uncertain. Yet it is hard to doubt the general tendency of a change in the law that imposes legal burdens on buying guns. The futility theorem suggests that gun-control laws, if effective at all, would unfavorably affect the rate of violent crime.

The futility theorem provides a lens through which to see much of the debate. It is undeniable that gun-control laws work -- to an extent. Consider, for example, California's background-check law, which in the past two years has prevented about 12,000 people with a criminal record or a history of mental illness or drug abuse from buying handguns. In the same period Illinois's background-check prevented the delivery of firearms to more than 2,000 people. Surely some of these people simply turned to an illegal market, but just as surely not all of them did. The laws of large numbers allow to say that among the foiled thousands, some potential killers were prevented from getting a gun. We do not know whether the number is large or small, but it is implausible to think it is zero. And, as gun-control proponents are inclined to say, "If only one life is saved..."

The hypothesis that firearms increase violence does predict that if we can slow down the diffusion of guns, there will be less violence; one life, or more, *will* be saved. But the futility theorem asks that we look not simply at the gross number of bad actors prevented from getting guns but at the effect the law has on *all* the people who want to buy a gun. Suppose we succeed in piling tax burdens on the acquisition of firearms. We can safely assume that a number of people who might use guns to kill will be sufficiently discouraged not to buy them. But we cannot assume this about people who feel that they must have guns in order to survive financially and physically. A few lives might indeed be saved. But the overall rate of violent crime might not go down at all. And if guns are owned predominantly by people who have good reason to think they will use them, the rate might even go up.

Are there empirical studies that can serve to help us choose between the futility theorem and the hypothesis that guns increase violence? Unfortunately, no: the best studies of the effects of gun-control laws are quite inconclusive. Our statistical tools are too weak to allow us to identify an effect clearly enough to persuade an open-minded skeptic. But it is precisely when we are dealing with undetectable statistical effects that we have to be certain we are using the best models of human behavior.

Sealing the border

Handguns are not legally for sale in the city of Chicago, and have not been since April of 1982. Rifles, shotguns, and ammunition are available, but only to people who possess an Illinois Firearm Owner's Identification card. It takes up to a month to get this card, which involves a background check. Even if one has a FOID card there is a waiting period for the delivery of a gun. In few places in America is it as difficult to get a firearm legally as in the city of Chicago.

Yet there are hundreds of thousands of unregistered guns in the city, and new ones arriving all the time. It is not difficult to get handguns -- even legally. Chicago residents with FOID cards merely go to gun shops in the suburbs. Trying to establish a city as an island of prohibition in a sea of legal firearms seems an impossible project.

Is a state large enough to be an effective island, then? Suppose Illinois adopted Chicago's handgun ban. Same problem again. Some people could just get guns elsewhere: Indiana actually borders the city, and Wisconsin is only forty miles away. Though federal law prohibits the sale of handguns in one state to residents of another, thousands of Chicagoans with summer homes in other states could buy handguns there. And, of course, a black market would serve the needs of other customers.

When would the island be large enough to sustain a weapons-free environment? In the United States people and cargoes move across state lines without supervision or hindrance. Local shortages of goods are always transient, no matter whether the shortage is induced by natural disasters, prohibitory laws, or something else.

Even if many states outlaws sales of handguns, then, they would continue to be available at a somewhat higher price, reflecting the increased legal risk of selling them. Mindful of the way markets work to undermine their efforts, gun-control proponents press for federal regulation of firearms, because they believe that only Congress wields the authority to frustrate the interstate movement of firearms.

Why, though, would one think that federal policing of illegal firearms would be better than local policing? The logic of that argument is far from clear. Cities, after all, are comparatively small places. Washington, DC, for example, has an area of less than 45,000 acres. Yet local officers have had little luck repressing the illegal firearms trade there. Why should federal officers do any better watching the United States' 12,000 miles of coastline and millions of square miles of interior? Criminals should be able to frustrate federal police forces just as well as they can local ones. Ten years of increasingly stringent federal efforts to abate cocaine trafficking, for example, have not succeeded in raising the street price of the drug.

Consider the most drastic proposal currently in play, that of Senator John Chafee, of Rhode Island, who would ban the manufacture, sale, and home possession of handguns within the United States. This proposal goes far beyond even the Chicago law, because existing weapons would have to be surrendered. Handguns would become contraband, and selling counterfeit,

stolen, and contraband goods is big business in the United States. The objective of law enforcement is to raise the costs of engaging in crime and so force criminals to take expensive precautions against becoming entangled with the legal system. Crimes of a given type will, in theory, decline as soon as the direct and indirect costs of engaging in them rise to the point at which criminals seek more profitable opportunities in other (not necessarily legal) lines of work.

In firearms regulation, translating theory into practice will continue to be difficult, at least if the objective is to lessen the practical availability of firearms to people who might abuse them. On the demand side, for defending oneself against predation there is no substitute for a firearm. Criminals, at least, can switch to varieties of law-breaking in which a gun confers little to no advantage (burglary, smash-and-grab), but people who are afraid of confrontations with criminals, whether rationally or (as an accountant might reckon it) irrationally, will be very highly motivated to acquire firearms. Long after the marijuana and cocaine wars of this century have been forgotten, people's demand for personal security and for the tools they believe provide it will remain strong.

On the supply side, firearms transactions can be consummated behind closed doors. Firearms buyers, unlike those who use drugs, pornography, or prostitution, need not recurrently expose themselves to legal jeopardy. One trip to the marketplace is enough to arm oneself for life. This could justify a consumer's taking even greater precautions to avoid apprehension, which would translate into even steeper enforcement costs for police.

Don Kates, Jr, a San Francisco lawyer and a much-published student of this problem, has pointed out that during the wars in Southeast and Southwest Asia local artisans were able to produce, from scratch, serviceable pot-metal counterfeits of AK-47 infantry rifles and similar weapons in makeshift backyard foundries. Although inferior weapons cannot discharge thousands of rounds without misfiring, they are more than deadly enough for light to medium service, especially by criminals and people defending themselves and their property, who ordinarily use firearms by threatening with them, not by firing them. And the skills necessary to make them are certainly as widespread in America as in the villages of Pakistan or Vietnam. Effective policing of such a cottage industry is unthinkable. Indeed, as Charles Chandler has pointed out, crude but effective firearms have been manufactured in prisons -- highly supervised environments, compared with the outside world.

Seeing that local firearms restrictions are easily defeated, gun-control proponents have latched onto national controls as a way of finally making gun control something more than a gesture. But the same forces that have defeated local regulation will defeat further national regulation. Imposing higher costs on weapons ownership will, of course, slow down the weapons trade to some extent. But planning to slow it down in such a way as to drive down crime and violence, or to prevent motivated purchasers from finding ample supplies of guns

and ammunition, is an escape from reality. And like many other such, it entails a morning after.

Administering prohibition

Assume for the sake of argument that to a reasonable degree of criminological certainty, guns are every bit the public-health hazard they are said to be. It follows, and many journalists and a few public officials have already said, that we ought to treat guns the same we do smallpox viruses or other critical vectors of morbidity and mortality -- namely, isolate them from potential hosts and destroy them as speedily as possible. Clearly, firearms have at least one characteristic that distinguishes them from smallpox viruses: nobody wants to keep smallpox viruses in the nightstand drawer. Amazingly enough, gun-control literature seems never to have explored the problem of getting weapons away from people who very much want to keep them in the nightstand drawer.

Our existing gun-control laws are not uniformly permissive, and, indeed, in certain places are tough even by international standards. Advocacy groups seldom stress the considerable differences among American jurisdictions, and media reports regularly assert that firearms are readily available to anybody anywhere in the country. This is not the case. For example, handgun restrictions in Chicago and the District of Columbia are much less flexible than the ones in the United Kingdom. Several hundred thousand British subjects may legally buy and possess sidearms, and anyone who joins a target-shooting club is eligible to do so. But in Chicago and the District of Columbia, excepting peace officers and the like, only grandfathered registrants may legally possess handguns. Of course, tens or hundreds of thousands of people in both those cities -- nobody can be sure how many -- do in fact possess them illegally.

Although though is, undoubtedly, illegal handgun ownership in the United Kingdom, especially in Northern Ireland (where considerations of personal security and public safety are decidedly unlike those elsewhere in the British Isles), it is probable that Americans and Britons differ in their disposition to obey gun-control laws: there is reputed to be a marked national disparity in compliance behavior. This difference, if it exists, may have something to do with the comparatively marginal value of firearms to British consumers. Even before it had strict firearms regulation, Britain had very low rates of crimes involving guns; British criminals, unlike their American counterparts, prefer burglary (a crime of stealth) to robbery (a crime of intimidation).

Unless people are prepared to surrender their guns voluntarily, how can the US government confiscate an appreciable fraction of our country's nearly 200 million privately owned firearms? We know that it is possible to set up weapons-free zones in certain locations -- commercial airports and many courthouses and, lately, some troubled big-city high schools and housing projects. The sacrifices of privacy and convenience, and the costs of paying

guards, have been though worth the (perceived) gain in security. No doubt it would be possible, though it would probably not be easy, to make weapons-free zones of shopping centers, department stores, movie theaters, ball parks. But it is not obvious how one would cordon off the whole of an open society.

Voluntary programs have been ineffectual. From time to time community-action groups or police departments have sponsored "turn in your gun" days, which are nearly always disappointing. Sometimes the government offers to buy guns at some price. This approach has been endorsed by Senator Chafee and the *Los Angeles Times*. Jonathan Alter, of *Newsweek*, has suggested a variation on this theme: youngsters could exchange their guns for a handshake with Michael Jordan or some other sports hero. If the price offered exceeds that at which a gun can be bought on the street, one can expect to see plans of this kind yield some sort of harvest -- as indeed they have. But it is implausible that these schemes will actually result in a less-dangerous population. Government programs to buy up surplus cheese cause more cheese to be produced without affecting the availability of cheese to people who want to buy it. So it is with guns.

One could extend the concept of intermittent roadblocks of the sort approved by the Supreme Court for discouraging drunk driving. Metal detectors could be positioned on every street corner, or ambulatory metal-detector squads could check people randomly, or hidden magnetometers could be installed around towns, to detect concealed weapons. As for firearms kept in homes (about half of American households), warrantless searches might be rationalized on the well-established theory that probable cause is not required when authorities are trying to correct dangers to public safety rather than searching for evidence of a crime.

In a recent "town hall" meeting in California, President Bill Clinton used the word "sweeps," which he did not define, to describe how he would confiscate firearms if it were up to him. During the past few years the Chicago Housing Authority chairman, Vincent Lane, has ordered "sweeps" of several gang-ridden public-housing projects, meaning warrantless searches of people's homes by uniformed police officers looking for contraband. Lane's ostensible premise was that possession of firearms by tenants constituted a lease violation that, as a conscientious landlord, he was obliged to do something about. The same logic could justify any administrative search. City health inspectors in Chicago were recently authorized to conduct warrantless searches for lead hazards in residential paint. Why not lead hazards in residential closets and nightstands? Someone has probably already thought of it.

Ignoring the ultimate sources of crime and violence

The American experience with prohibition has been that black marketeers -- often professional criminals -- move in to profit when legal markets are closed down or disturbed. In order to combat them, new laws and law-enforcement techniques are developed, which are

circumvented almost as soon as they are put in place. New and yet more stringent laws are enacted, and greater sacrifices of civil liberties and privacy demanded and submitted to. But in this case the problem, crime and violence, will not go away, because guns and ammunition (which, of course, won't go away either) do not cause it. One cannot expect people to quit seeking new weapons as long as the tactical advantages of weapons are seen to outweigh the costs imposed by the prohibition. Nor can one expect large numbers of people to surrender firearms they already own. The only way to make people give up their guns is to create a world in which guns are perceived as having little value. This world will come into being when criminals choose not to use guns because the penalties for being caught with them are too great, and when ordinary citizens don't think they need firearms because they aren't afraid of criminals anymore.

Neither of these eventualities seems very likely without substantial departures in law-enforcement policy. Politicians' nostrums -- increasing the punishment for crime, slapping a few more death-penalty provisions into the code -- are taken seriously by few students of the crime problem. The existing penalties for predatory crimes are quite severe enough. The problem is that they are rarely meted out in the real world. The penalties formally published by the code are in practice steeply discounted, and criminals recognize that the judicial and penal systems cannot function without bargaining in the vast majority of cases.

This problem is not obviously one that legislation could solve. Constitutional ideas about due process of law make the imposition of punishments extraordinarily expensive and difficult. Like the tax laws, the criminal laws are basically voluntary affairs. Our system isn't geared to a world of wholesale disobedience. Recalibrating the system simply by increasing its overall harshness would probably offend and then shock the public long before any of its benefits were felt.

To illustrate, consider the prospect of getting serious about carrying out the death penalty. In recent years executions have been running at one or two dozen a year. As the late Supreme Court Justice Potter Stewart observed, those selected to die constitute a "capriciously selected random handful" taken from a much larger number of men and women who, just as deserving of death, receive prison sentences. It is not easy to be exact about that much larger number. But as an educated guess, taking into account only the most serious murders -- the ones that were either premeditated or committed in the course of a dangerous felony -- there are perhaps 5,000 prisoners a year who could plausibly be executed in the United States: say, 100,000 executions in the next twenty years. It is hard to think that the death penalty, if imposed on this scale, would not noticeably change the behavior of potential criminals. But what else in national life or citizens' character would have to change in order to make that many executions acceptable? Since 1930 executions in the United States have never exceeded 200 a year. At any such modest rate of imposition, rational criminals should consider the prospect of receiving the death penalty effectively nil. On the best current evidence, indeed,

they do. Documentation of the deterrent effect of the death penalty, as compared with that of long prison sentences, has been notoriously hard to produce.

The problem is not simply that criminals pay little attention to the punishments in the books. Nor is it even that they also know that for the majority of crimes, their chances of being arrested are small. The most important reason for criminals behavior is this: the income that offenders can earn in the world of crime, as compared with the world of work, all too often makes crime appear to be the better choice.

Thus the crime bill that Bill Clinton introduced last year, which provides for more prisons and police officers, should be of only very limited help. More prisons means that fewer violent offenders will have to be released early in order to make space for new arrivals; perhaps fewer plea bargains will have to be struck -- all to the good. Yet a moment's reflection should make clear that one more criminal locked up does not necessarily mean one less criminal on the street. The situation is very like one that conservationists and hunters have always understood. Populations of game animals readily recover from hunting seasons but not from loss of habitat. Means streets, when there are few legitimate entry-level opportunities for young men, are a criminal habitat, so to speak, in the social ecology of modern American cities. Cull however much one will, the habitat will be preoccupied promptly after its previous occupant is sent away. So social science has found.

Similarly, whereas increasing the number of police officers cannot hurt, and may well increase people's subjective feelings of security, there is little evidence to suggest that doing so will diminish the rate of crime. Police forces are basically reactive institutions. At any realistically sustainable level of staffing they must remain so. Suppose 100,000 officers were added to police fosters nationwide, as proposed in the current crime bill. This would amount to an overall personnel increase of about 18 percent, which would be parceled out according to the iron laws of democratic politics -- distributed through states and congressional districts -- rather than being sent to the areas that most need relief. Such an increase, though unprecedented in magnitude, is far short of what would be needed to pacify some of our country's worst urban precincts.

There is a challenge here that is quiet beyond being met with tough talk. Most public officials can see the mismatch between their tax base and the social entropies they are being asked to repair. There simply isn't enough money; existing public resources, as they are now employed, cannot possibly solve the crime problem. But mayors and senators and police chiefs must not say so out loud: too-disquieting implications would follow. For if the authorities are incapable of restoring public safety and personal security under the existing ground rules, then obviously the ground rules must change, to give private initiative greater scope. Self-help is the last refuge of non scoundrels.

Communities must, in short, organize more effectively to protect themselves against predators. No doubt this means encouraging properly qualified private citizens to possess and carry firearms legally. It is not morally tenable -- nor, for that matter, is it even practical -- to insist that police officers, few of whom are at a risk remotely as great as are the residents of many city neighborhoods, retain a monopoly on legal firearms. It is needless to fear giving honest men and women the training and equipment to make it possible for them to take back their own streets.

Over the long run, however, there is no substitute for addressing the root causes of crime -- bad education and lack of job opportunities and the disintegration of families. Root causes are much out of fashion nowadays as explanations of criminal behavior, but fashionable or not, they are fundamental. *The root cause of crime is that for certain people, predation is a rational occupational choice*. Conventional crime-control measures, which by stiffening punishments or raising the probability of arrest aim to make crime pay less, cannot consistently affect the behavior of people who believe that their alternatives to crime will pay virtually nothing. Young men who did not learn basic literacy and numeracy skills before dropping out of their wretched public schools may not have been worth hiring at the minimum wage set by George Bush, let alone at the higher, indexed minimum wage that has recently been under discussion by the Clinton Administration. Most independent studies of the effects of raising minimum wages show a similar pattern of excluding the most vulnerable. This displacement, in turn, makes young men free, in the nihilistic, nothing-to-lose sense, to dedicate their lives to crime. Their legitimate opportunities, as always precarious in a society where race and class still matter, often diminish to the point of being for all intents and purposes absent.

Unfortunately, many progressive policies work out in the same way as increases in the minimum wage -- as taxes on employment. One example is the Administration's pending proposal to make employer-paid health insurance mandatory and universal. Whatever the undoubted benefits of the plan, a payroll tax is needed to make it work. Another example: in recent years the use of the "wrongful discharge" tort and other legal innovations has swept through the courts of more than half the states, bringing to an end the era of "employment at will," when employees (other than civil servants) without formal contracts -- more than three quarters of the workforce -- could be fired for good reason, bad reason, or no reason at all. Most commentators celebrated the loss of the at-will rule. How could one object to a new legal tenet that prohibited only arbitrary and oppressive behavior by employers?

But the costs of the rule are not negligible, only hidden. At-will employment meant that companies could get out of the relationship as easily as employees could. In a world where dismissals are expensive rather than cheap, and involve lawyers and the threat of lawsuits, rational employers must become more fastidious about whom they hire. By raising the costs of ending the relationship, one automatically raises the threshold of entry. The burdens of the

rule fall unequally. Worst hit are entry-level applicants who have little or no employment history to show that they would be worth their pay.

Many other tax or regulatory schemes, in the words of Professor Walter Williams, of George Mason University, amount to sawing off the bottom rungs of the ladder of economic opportunity. By suppressing job creation and further diminishing legal employment opportunities for young men on the margin of the work force, such schemes amount to an indirect but unequivocal subsidy to crime.

The solution to the problem of crime lies in improving the chances of young men. Easier said than done, to be sure. No one has yet proposed a convincing program for checking all the dislocating forces that government assistance can set in motion. One relatively straightforward change would be reform of the educational system. Nothing guarantees prudent behavior like a sense of the future, and with average skills in reading, writing, and math, young people can realistically look forward to constructive employment and the straight life that steady work makes possible.

But firearms are nowhere near the root of the problem of violence. As long as people come in unlike sizes, shapes, ages, and temperaments, as long as they diverge in their taste for risk and their willingness and capacity to prey on other people or to defend themselves from predation, and above all as long as some people have little or nothing to lose by spending their lives in crime, dispositions to violence will persist.

This is what makes the case for the right to bear arms, not the Second Amendment. It is foolish to let anything ride on hopes for effective gun control. As long as crime pays as well as it does, we will have plenty of it, and honest folk must choose between being victims and defending themselves.

NRA CONVENTIONAL PISTOL RULES

Conventional Pistol Rules

By The Abridged NRA Publication Effective January 1, 2000

Source: http://bullseyepistol.com/rulebook.htm

INTRODUCTION

These rules establish uniform standards for NRA sanctioned pistol competition. Where alternatives are shown, the least restrictive conditions apply unless the tournament program sets forth limitations. They do not apply to International Shooting Union type competition. They supersede the January 1, 2000 and all earlier editions and remain in effect until specifically superseded. Tournament sponsors may not alter these rules. If sponsors require additional rules for special conditions, the additions must be fully set forth in the program for the competition concerned. The arrangement and rule numbering systems are such that corresponding rules for other types of NRA competition are correspondingly located and numbered in the Rule Books for those competitions. Gaps in the sequence of rule numbers result from there being a rule in one or more of the other Rule Books which does not apply in this book. Anyone wishing to submit recommendations for rules changes may forward those recommendations to the Pistol Committee in care of the National Rifle Association.

PUBLIC AFFAIRS

All tournament sponsors are urged to give special consideration to the needs of news media personnel in order to achieve maximum publicity for the competition. Public news media personnel representing print and/or broadcast should be given every consideration and cooperation in keeping with the proper conduct of the competition.

NRA CONVENTIONAL PISTOL RULES

1. NRA Competition

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook1.htm

1. NRA COMPETITION

1.0 NRA Competitions - Competition which is authorized in advance of firing by the National Rifle Association. The program, range facilities and officials must comply with standards established by the NRA. The types of tournaments which are Sanctioned are listed in Rule 1.6.

1.1 Sanctioned Tournament - A series of matches covered by an Official Program. Such matches may be all individual matches, all team matches, or a combination of both which must be conducted by an NRA Affiliated Club or organization. They may be all fired matches or a combination of fired and aggregate matches. A tournament may be conducted on one day, on successive days, or may provide for intervening days between portions of the tournament, such as tournaments programmed to be conducted over more than one weekend.

1.2 Authorization - Before being publicized in programs or otherwise, the sponsoring organization of each type of competition mentioned in Rule 1.6 shall have agreed to comply with the current regulations for such competition and shall have received notice from the NRA that the competition applied for has been authorized.

1.3 Rules - The sponsor for each type of competition must agree to conduct the authorized competition according to NRA Rules, except as these rules have been modified by the NRA in the General Regulations, for that type of competition.

1.4 General Regulations - The sponsor of each type of competition must agree to comply with the General Regulations published by the NRA for the competition concerned. See Appendix in back of this Rule Book.

1.5 Refusal or Withdrawal of NRA Authorization - The NRA may refuse to authorize or may withdraw its authorization for any competition which cannot, or does not comply, with the requirements for that competition.

1.6 Types of Tournaments - The types of tournaments listed below are those which are Sanctioned by NRA in its Competitive Shooting Program.

(a) International Matches - Arranged by the NRA with the recognized national shooting organization(s) of the countries concerned. The officials thereof are appointed by the NRA.

(b) International Team Tryouts - All U.S. tournaments conducted under NRA Rules and authorized by the NRA as preliminary or final Tryouts for the selection of International Team members. The officials thereof to be appointed or approved by the NRA. (c) National Championships Organized by the NRA and, in some cases, in conjunction with the Corporation for the Promotion of Rifle Practice and Firearm Safety (CPRPFS), to form the

National Matches. The officials thereof are appointed by the NRA, in some circumstances in cooperation with the CPRPFS. The tournaments will be Registered. (d) Regional and Sectional Championships - Arranged between the NRA and a local sponsoring organization. These tournaments will be Registered. (e) State Championships - Annual Tournaments authorized and/or conducted by State Rifle and/or Pistol Associations affiliated with the NRA. Such State Associations may, if desired, authorize local organizations to sponsor and conduct State Championships. In states where there is no NRA affiliated State Association the NRA may authorize a local organization to sponsor and conduct the State Championship. State Championships will be Registered Tournaments. (f) Registered Tournaments - May be authorized by the NRA after application has been filed by the local NRA affiliated member organization that will act as the sponsor. Application forms are available from NRA on request. National records may only be established in Registered Tournaments.(See Rule 17.1). All competitors in Registered Tournaments must be individual members of NRA, except for juniors (Rules 2.3, 2.3.1, and 2.3.2), who may be either members of NRA affiliated Junior Clubs or individual members of NRA. (g) Approved Tournaments - May be authorized by the NRA after application has been filed by the local NRA affiliated member organization that will act as the sponsor. Application forms are available from NRA on request. Match sponsors may require all competitors to be NRA members if specified in the program. (h) Sanctioned Leagues-(shoulder-to-shoulder or postal) - May be authorized by the NRA after application has been filed by a local group or organization. Application forms are available from NRA on request. Sanctioned league scores are used for classification. A League need not be operated by an Affiliated Club or Organization. (i) Postal Matches - Organized by the NRA and publicized to groups concerned through the Shooting Sports U.S.A. announcements and special mailings. (j) Special Tournaments - may be sanctioned by NRA for types of shooting not otherwise a part of the NRA program. (k) State and Senior Games - tournaments which are part of the State and Senior Game program. Such tournaments are not charged fees, nor will any scores fired be used for classification. A simplified sanctioning procedure may be used, and they will be listed in Coming Events with the only contact being the State Coordinator. **1.7 Types of Matches**

(a) Match - A complete event as indicated in the program for the awarding of certain specific prizes. A match may consist of one or several stages. It may, in the case of aggregate matches. include the scores fired in several subsidiary matches. (b) Stage - A portion of a match which consists of one or more strings fired in one position, distance, time allowance (slow. timed or rapid fire), or target. (c) Open Match - A match open to anyone, that if so stated in the program an open match may be limited to one or any combination of the following: (a) United States citizens; (b) members of the National Rifle Association of America; and/or (c) non-U.S. citizens, persons who are members in good standing of their respective national shooting federations or associations. (See Rule 2.1.1).

(d) CMP ("Leg ") Matches - The Civilian Marksmanship Program (CMP)under the direction of

the Corporation for the Promotion of Rifle Practice and Firearm Safety, Inc. (CPRPFS) sponsors Excellence in Competition ("Leg") Matches which are organized and conducted under the direction and rules of the NBPRP in conjunction with NRA Regional and State Championships. In addition the CPRPFS authorizes the NRA to conduct the National Trophy Matches in conjunction with the NRA National Championships. The combined events are titled "The National Matches". All CMP matches are conducted in accordance with rules and regulations contained in the 2nd Edition of the Civilian Marksmanship Program Competition Rules and Regulations, effective date of January 1, 2000. They are not NRA Sanctioned Matches, and scores are NOT used for Classification. All NBPRP matches are conducted in accordance with rules and regulations contained in Army Regulation 920-30; OPNAVINST 3590. 7B; AFR 5017; and MCO P3590. 13, entitled "Rules and Regulations for National Matches" in its current form. They are not NRA Sanctioned Matches, and scores are not used for Classification. (e) Restricted Match - A match in which competition is limited to specific class, i.e., Masters, Experts, Sharpshooters, Marksmen, etc. (f) Classified Match - A match in which awards are given to the winners and the highest competitors in several specified classes, such as Masters, Experts, Sharpshooters, and Marksmen. The classification of competitors may be accomplished by the National Classification System (Sec. 19) or by other means. The program for classified matches must specify the groups or classes in which awards will be made. (g) Invitational Match - A match in which participation is limited to those who have invited to compete. (h) Squadded individual Match - A match in which each competitor is assigned a definite relay and target by the Statistical Office. Failure to report on the proper relay or firing point forfeits the right to fire. All entries must be made before firing commences in that match, except when otherwise stated in the tournament program. (i) Unsquadded Individual Match - A match in which the competitor is not assigned a definite relay or target by the Statistical Office. The competitor reports to the Range Officer within the time limits specified in the program and is then assigned to a target and a relay in which to fire. (j) Re-Entry Match - A match in which the competitor is permitted to fire more than one score for record; one or more of the highest scores being considered to determine the relative rank of competitors. The number of scores which may be fired. and the number of high scores to be considered in deciding the relative rank of competitors must be specified in the program. Scores fired in these matches shall not be used for classification purposes. (k) Squadded Team Match - A match in which the teams are assigned a definite time to fire. Teams may be assigned one or more adjacent targets. All entries must be made before firing commences in that match. The entire team must report and fire as a unit. (l) Unsquadded Team Match - A match in which the teams may report at the firing line at any time within the limits specified in the program, targets being assigned by the Range Officer. The entire team must report and fire as a unit, unless the program provides otherwise. (m) Aggregate Match - An aggregate of the scores from two or more matches. This may be an aggregate of match stages, individual matches, team matches, or any combination, provided the tournament

program clearly states the matches which will comprise the aggregate. Entries in aggregate matches must be made before the competitor commences firing in any of the matches making up the aggregate match.

NRA CONVENTIONAL PISTOL RULES

2. Eligibility of Competitors

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook2.htm

2. ELIGIBILITY OF COMPETITORS
The conditions of a match shall prescribe the eligibility of competitors, team or individuals, in accordance with Rule 1.6 and/or the definitions contained in section 2. Any limitation of eligibility to compete must be stated in the Match Program.

INDIVIDUALS
2.1 Members of the National Rifle Association - Any individual member, including Benefactors, Patrons, Endowment, Life, Annual, Associate, Non-Resident and Junior Members.
2.1.1 Non-U.S. Citizens - Non-U.S. Citizens who are also nonresidents, who are not members of the National Rifle Association of America, but who are members in good standing of their own National Association, and have adequate proof of such membership in hand, may compete in any NRA Sanctioned Tournament, unless further restrictions are imposed by conditions stated in the program.
2.2 Civilian - Any civilian including all members of the Reserve Officers Training Corps (ROTC, NROTC and AFROTC), personnel of the State Security Forces (e.g. State Guard organization having no federal recognition), retired members of each of the several services comprising the Armed Forces of the United States, and members and former members entitled to receive pay, retirement pay, retainer pay or equivalent pay, are classified as civilians except as noted in the example below. All competitors who are enrolled undergraduates of any of the service academies will be considered as civilians, and may compete in collegiate and ROTC categories.
Individuals of any Reserve or National Guard component who, during the present calendar year, have not competed as National Guard (2.5) or Regular Service (2.6) or Reserve component (2.7) and have not been provided Service support for competition (in the form of firearms, ammunition, payment of travel or other expenses, wholly or in part, may fire as civilians. The provision of firearms and ammunition for a specific competition (i.e., National Matches or NBPRP Regional Leg Matches), when such is available to both military and civilian competitors, it is not considered Service support under this rule. Unless specifically authorized to do so by the tournament program, members of the regular Army, Navy, Air Force, Marine Corps, Coast Guard; members of the reserve components on active duty; retired personnel of the several services comprising the Armed Forces of the United States on active duty; or police (2.4) are not permitted to compete as civilians. **2.2.1 Senior** - A person

may compete as a Senior beginning on January 1 of the calendar year in which the 60th birthday occurs.

2.3 Junior- A person may compete as a junior through December 31 of the calendar year in which the twentieth birthday occurs.

2.3.1 Intermediate Junior - A Junior may also compete as an Intermediate Junior from January I of the calendar year in which the 15th birthday occurs through December 31 of the calendar year in which the 17th birthday occurs.

2.3.2 Sub-Junior - A Junior may also compete as a Sub-Junior through December 31 of the calendar year in which the fourteenth birthday occurs.

2.4 Police

(a) Regular full time Law enforcement officers of a regularly constituted law enforcement agency of a municipal, county. state or federal government. "Full Time" Police Firearms Instructors in Law Enforcement or Police Academies are eligible to compete in the police category. Official current identification from the agency employing a civilian full time police firearms Instructor will be required. (b) Full time salaried Railroad Police; Penal Institution Guards; Industrial Police Including Bank Guards, and Armored & Express Company Guards. Persons employed as Industrial Police by private industry on a part-time basis, or where such employment is not the sole occupation of such person, may not compete in the police category in NRA pistol tournaments. (c) Officers of a regularly organized Reserve or Auxiliary named in (a) above provided that when on duty they are required to perform the same law enforcement functions and/or duties as the Agency to which they are in reserve and are authorized to be armed by the appointing authority. (d) Officers who qualify under sections (a) and (b) may after retirement, and if receiving retirement benefits. enter competition covered by these rules. Such retirees who are employed after retirement in any Agency which qualifies for entry in NRA sanctioned Pistol competition may not compete as a retired officer, but may compete as a member of the organization in which he is presently serving. (e) Military Police: Members of the Armed Forces of the United States, regular or reserve. private detectives. private bodyguards, honorary police, or sheriffs, consultants or any other persons who are members of a body organized for ceremonial purposes may not compete as Police in NRA sanctioned Pistol tournaments regardless of the titles given such members. **2.5 National Guard** - Federally recognized officers or enlisted of the Army National Guard, the Air National Guard, or the Naval Militia of the several states, territories, the District of Columbia, or the Commonwealth of Puerto Rico, who are not on extended active duty are eligible to compete as Juniors and/or Collegiate if otherwise qualified.

2.6 Regular Service - Officers or enlisted of the Regular United States Army, Navy, Air Force, Marine Corps, Coast Guard, and members of reserve components thereof, who are on extended active duty; provided the term "reserve components" shall include Army National Guard and Air National Guard called into federal service and while in such status are eligible to compete as Juniors and/or Collegiate if otherwise qualified.

2.7 Reserve Components - Officers and enlisted of any reserve components of the Armed Forces, exclusive of the Army National Guard and the Air National Guard of the United States, not on extended active duty are eligible to compete as Juniors and/or Collegiate if otherwise qualified.
2.8 College - Regularly enrolled undergraduate students who carry 12 or more semester hours or the equivalent in an accredited two or four year collegiate level institution and who have not received a Bachelor's Degree. Eligibility to compete as a collegian shall extend for a maximum of four years within a five year period beginning when a student first attends any class as a full-time (12 or more credit hours) student. The five year period may only be interrupted by extended military service. (60 consecutive days or more), service with a recognized foreign aid agency of the U.S. Government or certified missionary service.
2.9 School - Regularly enrolled undergraduate students of any primary or secondary school, who comply with the eligibility rules of their institution.
TEAMS
2.10 Team Representation - No individual may be a team captain, coach, firing member, or alternate firing member on more than one team in any one match.
Note: Entries will not be accepted from "Pickup" teams (teams whose members are selected without regard to club or other organization affiliation) unless the program specifically provides for such eligibility. Pickup teams are not eligible for national records. **2.11 Club Teams** - All members, including team captain and coach of such teams, must have been active fully-paid members of the club which the team represents for a period of at least 30 days immediately prior to the competition. The club must be affiliated with NRA and in good standing. If specially allowed by conditions of the program, a person who is not a club member may serve as a non-firing coach of such a team. There are two types of teams allowed under this rule:
(a) Local Club Teams - Local club teams must be composed of members who reside legally within the same state, territory or province. (b) Open Club Teams - Open club teams must be composed of members all of whom are not necessarily legal residents in the same state, territory or province. (A single club could have more than one team, one of which could be a Local Club Team; and another an Open Club Team.) **2.12 State Association Teams** - Members of State Association teams must be residents of the state represented. Members of such teams must be individual members of the State Rifle and /or Pistol Association represented, if such State Association provides for individual membership, or be members of a rifle and/or pistol club which is affiliated and in good standing with the State Association concerned at the time of the competition. State Association teams permitted to enter the competition concerned by the tournament program conditions, must be authorized and accredited by the State Association for that tournament. Authorization shall be signed by the State Association President, Vice President, or Secretary. Such State Association must be affiliated and in good standing with the NRA at the time of the competition. If specifically

allowed by conditions of the program, a person not a State Association member may serve as the coach of a State Association Team but may not be a firing member.
Note: Teams representing State Association, Leagues and other associations (composed of more than one club) are not club teams. Such teams may enter NRA sanctioned matches only when the program specifically authorizes such entry.
2.13 Regular Services, National Guard or Other Armed Forces Reserve Teams - Members of such teams must have been commissioned or enlisted members of their respective service for a continuous period of at least 30 days immediately preceding the day of competition. Army National Guard, Air National Guard. and Naval Militia personnel may be combined into a single team.
2.14 Police Teams - Firing Members of such teams must be from the same agency, regular or reserve, and in active service for a continuous period of at least 30 days immediately preceding the day of competition.
2.15 Civilian Club Teams - Members of such teams must comply with Rules 2.2 and 2.11.
2.16 College Teams - Members of such teams must comply with Rules 2.8 and 2.11.
2.17 School Teams - Members of such teams must comply with Rules 2.9 and 2. 11.
2.18 Junior Club Teams - members of such teams must comply with Rule 2.3 and 2.11.
2.20 Residence - In those matches which are limited to residents of any specified geographical area a "resident" is defined as:
(a) A person who lives within a specified area for at least 30 days immediately prior to the day of the match, whether or not employment is at a place requiring travel into some other area.
(b) A person who has been regularly employed within the specified area for at least 30 days immediately prior to the day of the match and who has maintained domicile in that area for the same period of time, although the permanent residence is located outside the specified area. (c) Military Personnel: The place of residence of members of the Military on active duty is defined as the place at which they are stationed by reason of official orders, provided they have been so stationed within the specified area for a period of at least 30 days immediately prior to the day of the match. In the case of retired, Reserve, or National Guard personnel not on active duty, the provisions of paragraphs (a) will apply. Naval personnel assigned on sea duty qualify for a residence in the area which is the usual base or home port of the unit to which attached. (d) Federal and State Law Enforcement Officers: The provisions of paragraphs (c) will apply.

NRA CONVENTIONAL PISTOL RULES

3. Equipment and Ammunition

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook3.htm

3. EQUIPMENT AND AMMUNITION

This section defines authorized equipment. Where alternative types of equipment are shown, the least restrictive conditions apply unless the tournament program sets forth limitations.

3.l (a) Service Pistol - U.S. Pistol, caliber .45 M1911 or M1911A1 or the same type and caliber of commercially manufactured pistol. The pistol must be equipped with the standard issue or similar commercially-procured stocks or other comparable design of grips which interferes with no functional or maintenance features of the pistol, is functionally identical for right or left hand use. is not more than 1 inches in thickness between right and left side extremities as installed, includes no thumb rest or finger channels, and would not prevent holstering in the standard service holster. Trigger pull must be not less than 4 pounds. The pistol must be equipped with open sights. The front sight must be non-adjustable. The pistol may be equipped with an adjustable rear sight with open U or rectangular notch, the distance between sights measuring not more than 7.25 inches from the apex of the front sight to the apex of the rear sight. The forestrap of the grip may be checkered. The mainspring housing may be either the flat or arched type. Trigger shoes may be used. Trigger stops. internal or external, are acceptable. Otherwise, external alterations or additions to the arm will not be allowed. The internal parts of the pistol may be specially fitted and include alterations which will improve the functioning and accuracy of the arm, provided such alterations in no way interfere with the proper functioning of the safety devices as manufactured. All standard safety features of the pistol must operate properly. It is the competitors responsibility to have his pistol checked prior to the firing of the match.

3.1 (b) U.S. Pistol, 9mm, M9 or a commercial pistol of the same design and caliber. Standard stock must be made of wood or synthetic material, a similar stock of commercial manufacture, or another comparable design that does not interfere: with the functional or maintenance features of the pistol. The stock must be functionally identical for right or left hand use. It must not be more than 1.5 inches thick between the right and left extremities. Trigger pull of at least 4 pounds. Open sights only with non-adjustable front sight. The rear sight may be adjustable and must have an open "U" or rectangular notch. The distance between the sights, measured from the apex of the front sight to the rear face of the rear sight, must be not more than 7.25 inches. Barrel length of no less than 4.9 inches or longer than 5.0 inches. Overall length of the pistol may not exceed 8.62 inches.

3.1 (c) Ammunition requirements for service pistols in Excellence in Competition (Leg) Matches are as follows:

(1) Pistol competitors with the caliber .45 pistol may use any safe ammunition with a 230 grain, full metal jacket, round nosed bullet. (See rule 4-16b CMP Rules and Regulations). (2) Pistol competitors with the 9mm pistol may use any safe ammunition with a 115 through 125 grain, full metal jacket, round nosed bullet. (See rule 4-16c CMP Rules and Regulations). **3.2 Any Center Fire Pistol or Revolver** - Center-fire pistols (single shot or semiautomatic) or revolvers of .32 caliber or larger (including 7.65 mm and .45 caliber pistols and revolvers); barrel length, including cylinder, not more than 10 inches; trigger pull not less than 2 1/2 pounds, except .45 caliber semiautomatic pistols not less than 3 1/2 pounds. Sights may be adjustable but not over 10 inches apart measured from the apex of the rear sight to the apex of the front sight. All standard safety features of the gun must operate properly. Programs may specify particular calibers or types of center-fire guns which will be permitted or not permitted in stated event. Any sights, including telescopic, are permitted with the exception of those sights which project an image on the target. Any sighting device programmed to activate the firing mechanism is prohibited.

3.3 Any .45 Caliber Semiautomatic Pistol or Revolver - Any .45 caliber semiautomatic pistol or revolver; barrel length, including cylinders, not more than 10 inches. trigger pull for revolvers not less than 2 1/2 pounds, .45 caliber semiautomatic pistol trigger pull not less than 3 1/2 pounds. Sights may be adjustable but not over 10 inches apart measured from the apex of the rear sight to the apex of the front sight. All standard safety features of guns must operate properly. Any sights, including telescopic, are permitted with the exception of those sights which project an image on the target. Any sighting device programmed to activate the firing mechanism is prohibited.

3.4 .22 Caliber Pistol or Revolver - Any pistol or revolver using a .22 caliber rimfire, cartridge having an over-all length of not more than 1.1 inches and with lead or alloy bullet not greater than .23 inches in diameter and weighing not more than 40 grains; barrel length, including cylinder, not more than 10 inches; sights may be adjustable but not over ten inches apart measured from the apex of the rear sight to the apex of the front sight. Trigger pull not less than 2 pounds. All standard safety features of the gun must operate properly. Any sights, including telescopic, are permitted with the exception of those sights which project an image on the target. Any sighting device programmed to activate the firing mechanism is prohibited.

3.5 Use of Authorized Equipment

(a) The .22 Caliber Pistol as described in Rule 3.4 must be used in the .22 caliber conventional pistol matches. (b) The Center Fire Pistol as described in Rule 3.2 including center fire pistols of .32 caliber or larger may be used in the center fire conventional pistol matches. (c) The .45 Caliber Semiautomatic Pistol, as described in Rule 3.3, or the Service Pistol described under Rule 3.1 may be used in conventional .45 caliber matches. (d) The Service Pistol, as described under Rule 3.1 must be used in all matches approved by the NBPRP in which credits toward Distinguished designation may be achieved. **3.6 Grips** - The grip, or any part of the firearm,

may not be extended or constructed in any way which would give any support beyond the hand. The wrist must remain absolutely free and visible when the pistol is held in the firing position.
No part of the grip or accessory may encircle the hand. Upward curvature of the heel of the grip may not exceed one inch ... Adjustable Grips are permitted as long as they conform to the dimensions as [described.] **3.7 (BLANK)**
3.8 Spotting Scopes - The use of the telescope to spot shots is permitted.
3.9 Shooting Kits - The shooting kit may be taken to the firing point when it is of such size and construction as to not interfere with shooters on adjacent firing points.
3.9.1 Deflecting Screens - Shooters may use a screen fastened to their shooting kit, or similarly placed, to deflect empty cases, provided the screen is made of material such as "Hardware Cloth' with wires not larger than 1/16 inch in diameter and spaced not closer than 1/4 inch. Screens of solid material may not be used.
3.10 - 3.15 Blank
3.16 Release Triggers - Triggers which function on release are prohibited.
3.17 Ammunition
(a) Service-Full charge cartridge ammunition manufactured for or by the Government and issued for use in service arms. (b) Any-Ammunition of any description that may be fired without danger to competitors, range personnel or equipment. Tracer, incendiary, armor piercing and similar ammunition is prohibited. **3.18 General** - All devices or equipment which may facilitate shooting and which are not mentioned in these rules, or which are contrary to the spirit of these rules and regulations, are forbidden. The Match Director, Official Referee, Jury Chairman, Supervisor, shall have the right to examine a shooter's equipment and apparel. The responsibility shall be upon the competitor to submit questionable equipment and apparel for official inspection and approval in sufficient time prior to the beginning of a match so that it will not inconvenience either the competitor or the official.
3.19 Eye Protection - All competitors and other personnel in the immediate vicinity of the range complex are urged to wear eye protection devices.

NRA CONVENTIONAL PISTOL RULES

4. Targets

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook4.htm

4. TARGETS

4.1 Official Targets - In NRA Sanctioned competition, only targets printed by NRA Licensed Manufacturers, bearing the Official Competition target seal, or military targets issued by the Armed Services, mithout modification except as authorized by the NRA, will be used. All NRA Official Competition Targets are printed by NRA licensed manufacturers only. They may not be modified by the user of the manufacturer, except with specific written permission from NRA Competitions Division. NRA Target masters are made using Computer Assisted Design and must be used in production of all NRA Targets.

4.2 to 4.8 (Summarized in Chart Form) The targets for official competition are:

Scoring ring dimesions for indoor targets (in inches)

Distance	Match	Target No.	X ring	10 ring	9 ring	8 ring	7 ring	6 ring	5 ring	4 ring
50 feet	Slow	B-2						4.16	5.56	7.33
	Timed/ Rapid	B-3				4.46	6.14	8.32		
20 yards	Slow	B-4						5.04	6.72	8.84
	Timed/ Rapid	B-5				5.44	7.46	10.08		
25 yards	Slow	B-16						7.22	9.66	
	Timed/ Rapid	B-8				8.00	11.00	14.80	19.68	

Scoring ring dimensions for outdoor targets (in inches)

Distance	Match	Target No.	X ring	10 ring	9 ring	8 ring	7 ring	6 ring	5 ring
50 yards	Slow	B-6					11.00	14.80	19.68
25 yards	Slow	B-16						7.22	9.66
	Timed/ Rapid	B-8				8.00	11.00	14.80	19.68

NRA CONVENTIONAL PISTOL RULES

5. Positions

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook5.htm

5. POSITIONS

Positions - The positions for use in a match shall be stated in the program under conditions of the match and shall be in accord with the definitions of positions prescribed in this section.

5.1 The Ground - All references to "the ground" in the following position rules are to be constructed as applying to surface of the firing point floor.

5.2 Artificial Support - Artificial support is defined as follows:

(a) Any supporting surface, except the ground, not specifically authorized for use in the rules for the position prescribed. (b) Any garment which can be interpreted as providing artificial support. (c) Any glove on the shooting hand. (d) Any band on the shooting arm (includes wrist watches) that are within two (2) inches of the wrist joint. The term artificial support shall not apply to the "tennis elbow" type strap or band if worn on the forearm no closer than two inches from the wrist or elbow joint. This rule recognizes that such a strap or band merely allows a competitor to compete up to their normal ability and offers no advantage to a shooter with no physical problems. (e) Other devices which can be interpreted as artificial support. (See Rule 3.18). Use of artificial support is prohibited except as individually authorized for a physically disabled shooter. (See Rule 5.5) **5.3 Ready Pistol Position** - The gun held in a safe forward position with the trigger finger outside the trigger guard.

5.4 to 5.11 (BLANK)

5.5 Firing Position - Standing, gun held in one hand only, the other hand being used in no way to support the gun; all portions of the shooter's clothing, body and gun clear of artificial support. Competitors will take their position at their numbered firing point in such manner as not to interfere with competitors on either side. No portion of the shooter's body may rest upon or touch the ground in advance of the firing line.

NRA CONVENTIONAL PISTOL RULES

6. Range Standards

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook6.htm

6. RANGE STANDARDS

6.1 Firing Line - The firing line is immediately in front of the several firing points. All ranges are measured from this firing line to the face of the targets when targets are hung in their proper firing position.

6.2 Firing Point - That part of the range provided for the competitor immediately in the rear of the firing line from which firing takes place. Each firing point is numbered to correspond with the target frames. Each firing point should have a minimum width of 4 feet.

6.3 Shelter - The firing point of outdoor ranges may be covered and enclosed on four sides, but there shall be an opening toward the targets for each competitor measuring no less than six square feet. There must be ample room for Range Officers, Block Officers, and witnesses to move freely to the rear of the shooters. This does not preclude the construction of ranges within areas surrounded or partially surrounded by safety walls.

6.4 Distance -

(a) Outdoor distances are 25 and 50 yards from firing line to target. Indoor ranges at these distances equipped with outdoor-type targets may also be used but record scores fired on such indoor ranges are recorded separately from scores fired on outdoor ranges. (h) Indoor pistol range distances are 50 feet, 50 yards, 20 or 25 yards from firing line to target. Official matches may be fired out-of-doors at these distances, **6.5 Illumination** - Artificial illumination of ranges is authorized.

6.6 Target Numbers - Target frames will be numbered on alternating background of contrasting color. The numbers will be large enough to be identified under ordinary light conditions. Numbers must correspond with the firing point numbers. Target numbers will be fixed in position so as to remain visible when the targets are exposed and when concealed.

6.7 (BLANK)

6.8 Types of Target Frames

(a) Outdoor Type (25 and 50 yards and 50 meters)-fixed frames may be used for slow fire only. Frames which rotate on a vertical or horizontal axis must be used for timed and rapid fire in registered competition. Not more than one record target for each competitor will be in firing position at one time. (b) Indoor Type (50 and 60 feet)-fixed frames, target trolleys, or rotating frames may be used. No more than one record target for each competitor will be in firing position at one time.

NRA CONVENTIONAL PISTOL RULES

7. Courses of Fire

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook7.htm

7. COURSES OF FIRE

The following courses and types of fire are most commonly found in NRA sanctioned pistol competition, fired on the targets described in Section 4. See Section 8 for time allowances. Rule 17.5 for courses of fire for which National Records are recognized, and Rule 19.4 for courses of fire used for classification. Other courses of fire, other time limits, or the use of other targets, may be scheduled by sponsors provided the conditions are clearly stated in the program. **National Match Course**

Stage	Range	Time	Strings	Target
Slow	50 yds.	10 shots/10 min.	1	B-6
Timed	25 yds.	5 shots/20 sec.	2	B-8
Rapid	25 yds.	5 shots/10 sec.	2	B-8

NRA Short Course

Stage	Range	Time	Strings	Target
Slow	25 yds.	10 shots/10 min.	1	B-16
Timed	25 yds.	5 shots/20 sec.	2	B-8
Rapid	25 yds.	5 shots/10 sec.	2	B-8

Gallery Course

Stage	Range	Time	Strings	Target
Slow	50 ft.	10 shots/10 min.	1	B-2
Timed	50 ft.	5 shots/20 sec.	2	B-3
Rapid	50 ft.	5 shots/10 sec.	2	B-3

Single Stage Course

Any number of strings (10 shot at 50 yards slow fire, or 5-shot at timed or rapid tire) fired at a single distance as specified in the conditions of the match and in the regular time allowed for that type of firing.

NRA CONVENTIONAL PISTOL RULES

8. Time Limits

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook8.htm

8. TIME LIMITS

8.1 Computing Time - When rotating targets are used time is taken for the period during which the target is fully exposed. Time is not checked on each individual shot but is computed on the indicated allowance for each five-shot string depending on the type of fire being conducted, except slow fire may be fired in 10-shot strings. The signal to commence firing may be given orally, by whistle or by having the targets turn. Shots fired when targets are turning will be scored in accordance with Rule 14.3.

8.2 Time Allowances - See Section 7.

8.3 Team Time - Same as for individual matches.

8.4 Passage of Time - Range Officers will not voluntarily warn competitors of the passage of time. Competitors and Team Captains in team matches, may inquire of Range Officers as to the time remaining before expiration of the time limit. The request for time and the response shall be given in a tone which will not disturb other competitors.

References

- Section 7

NRA CONVENTIONAL PISTOL RULES

9. Tournament Conditions

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook9.htm

9. TOURNAMENT CONDITIONS

9.1 Changing Pistols - No competitor will change pistols during the firing of any match (except aggregate matches), unless it has become disabled and has been so designated by the Chief Range Officer. Claim that a pistol is disabled must be made immediately. All shots fired up to the time the claim is made will stand as part of the official score. The exchange of barrels, portable weights, etc., shall not be restricted. (For timed and rapid fire see Rules 9.5, 10.9 and 10.10)

9.2 Sighting Shots - Sighting shots are not allowed as a part of the course of fire in pistol matches. This does not prohibit practice matches or practice targets provided all competitors have equal opportunity to enter such events or to use such targets. Practice must not be permitted to interfere with the safe, efficient and scheduled operation of the program

9.3 (BLANK)

9.4 Defective Cartridge - A defective cartridge is one:

(a) Which has such evident structural defect as to cause a misfire or to cause a pistol to fail to function (b) Which does not fire (c) From which the bullet has not left the barrel (No claim for a defective cartridge shall be allowed if bullet has left the barrel.) (For procedure in case of a defective cartridge see Rules 9.4, 10.9 and 10.10. For refiring privileges see Rule 9.14.) **9.5 Disabled Pistol** -A disabled pistol is one:

(a) That cannot be safely aimed or fired (b) That has suffered damage so it cannot be fired or will not function (c) That has suffered the loss of a sight or damage to the sights. The fact that sights are improperly adjusted does not constitute disablement A pistol once declared disabled by the Range Officer shall not again be used for competitive firing until the defect has been corrected and until the pistol has been ruled as safe by the Chief Range Officer. (For procedure in case of a disabled pistol see Rules 10.9 and 10.10. For refiring privileges see Rule 9.14.) **9.6 Malfunction** - Failure of the pistol to function properly due to mechanical defects or to defective ammunition. Functional failures due to improper manual operation are not to be considered as malfunctions. (For procedure in case of a malfunction see Rules 10.9 and 10.10. For refiring privileges see Rule 9.14.)

9.7 Continue to Fire - To break ties or establish National Records competitors who fire a perfect score (all X's when target has an X ring) in individual matches will continue to fire five-shot strings until a hit is made outside the scoring ring of highest value. In three stage matches containing slow, timed and rapid fire the competitor will continue to fire five-shot strings at rapid fire until there is a hit outside the scoring ring of the highest value. The

continuation of firing must in all cases commence during the day of the match in which the perfect score was fired. When using these scores to break ties, rules will apply as provided in Section 15. This rule does not apply in team matches.

9.8 Weighing Triggers - Triggers may be weighed with official NRA trigger test weights at the discretion of the Match Director, Official Referee, Jury or Supervisor. Triggers of pistols used in establishing National Records shall be weighed by a tournament official immediately after the firing of such record score and must meet the specifications of Section 3 for the pistol prescribed by the tournament program. Failure of the trigger to meet the trigger pull requirements shall disqualify the competitor in matches previously fired.

While trigger pull is being weighed, the pistol shall be held with the barrel perpendicular to the horizontal surface on which test weight is supported. The rod or hook of test weight shall rest on lowest point of the curve in curved triggers, or on a point approximately one quarter of an inch from lower end of straight triggers. (Figure E) To pass the weight test, a weight of the correct number of pounds shall be lifted by the pistol trigger while in the cocked position and while all safety devices are in firing position from the horizontal surface on which it is resting, until the weight hangs free and without releasing the trigger. Magazine must be removed and pistols unloaded while trigger is being weighed. Pistols equipped with a device to prevent firing while magazine is out must be closely inspected to see that no cartridges are in a magazine or chamber. Magazine will then be inserted and trigger pull weighed. Competitors will be permitted to adjust triggers which have failed to pass the weight test provided they do not occasion any delay. Failure of trigger to pass the weight test is the competitor's responsibility. **9.9 Competitor's Position** - Competitors will take their positions at their numbered firing points in such manner so as to not interfere with competitors on either side.

9.10 Coaching Prohibited in Individual Matches - Coaching is prohibited in all individual matches of an NRA Approved or Registered Tournament.

9.11 Matches Not Complete - When a match is not completed by all competitors in accordance with the tournament schedule, the match or stage may be rescheduled or canceled. Any match or stage which has been completed by all competitors will not be refired. Only scores of a match or stage which has been completed will be included in an aggregate event or for National Record purposes, and a match or stage is not completed unless all competitors have fired.

9.12 Interruption of Fire

(a) In timed or rapid fire when the firing of a string is interrupted by some occurrence which renders it impossible for one or more competitors to complete the string under the conditions of the match, the Chief Range Officer will proceed as follows. Without being permitted to examine their targets, competitors in the relay who have been so prevented from completing their strings will be asked if they wish to refire or to accept their score as fired. Targets will then be scored in the usual manner for all competitors except those who have elected to refire. Without being scored, the targets of such competitors who have elected to refire will be pasted

or new targets substituted and a complete string will be fired and scored. Reasons authorizing this procedure are: (1) Failure to allow full time. (2) Failure of targets to operate properly or uniformly. (3) Failure of target to remain in position on frame or carrier. (4) Damage to target rendering impossible proper aiming or scoring impossible. (5) The appearance of some object in line of Fire constituting a hazard. (6) Some incident involving a Range Officer or competitor on the firing line. (7) Unintended moving of the target during Fire. (b) In timed or rapid fire when, due to faulty target operation or error in timing, one or more competitors are allowed more time to complete the string than provided by conditions of the match, the Chief Range Officer will immediately order all such targets pasted or new targets installed. The fired target will not be scored. A complete new string will then be fired by the competitors who are allowed extra time. If in the same relay some targets operate properly in accordance with the legal time limit, such targets will be scored in the usual manner and competitors firing on those targets will not be required or permitted to refire. (c) In slow fire in case a target is unintentionally moved out of firing position just as a shot is fired the shot will be circled by the Range Officer if it can be identified, and disregarded when target is scored. If the shot cannot be identified, the competitor will be permitted to accept the score as fired or to Fire a completely new string. (d) In case of excessive hits see Rule 14.10. (e) If a shot hits the target frame or the target carrier causing the target to fall, the Range Officer will be notified. The target will be rehung and if the shot causing the target to fall strikes outside the scoring rings of the target, it will be scored as a miss. All other shots on the target will count as record shots. For slow fire the competitor will be permitted to continue to Fire any unfired shots. For timed and rapid fire the competitor will refire the suing on the same target as provided in Rule 9.14. **9.13 Interference** - With the exception of competitors actually firing and such range personnel as may be necessary, the firing points and line will be kept clear of other persons during the firing.

9.13.1 Interference with Targets - Competitors will not be permitted to interfere with the handling of targets by range personnel. Competitors shall not touch their targets after they have been fired until final score determination on the target has been made. Final score determination is not reached until all challenges have been settled.

9.14 Refiring

(a) No competitor will fire more than one score for the same award except as provided in the program or in accordance with Rule 9.18. (b) In single stage matches composed of several strings of fire only one refire per match will be allowed. (c) In multiple stage matches one refire will be allowed in the slow fire stage and one refire will be allowed in the combined timed and rapid fire stages. (d) Refires of slow fire refires are to be fired immediately after the relay in which the refire occurred. **19.15 to 9.22 - Blank**

9.23 Aliases - No competitor may fire under an assumed name nor substitute for another in a match; or register, enter, or fire in the name of another.

9.24 Score and Classification Falsification - No competitor will falsify score, or

classification, nor that of any other competitor, nor be an accessory thereto.

9.25 Cross Fires and Excessive Hits - No competitor will deliberately fire on the wrong target nor fire more than the required number of shots, including hits on some other competitor's target and misses.

9.26 Bribery - No person will offer a bribe of any kind to any of the range, statistical personnel or others, nor be an accessory thereto.

9.27 Disorderly Conduct - Disorderly conduct or intoxication is strictly prohibited on the range and anyone guilty of same will be expelled from the range. Expelled competitors will be disqualified from the competition with no return of entry fees.

9.27.1 Willful Destruction of Range Equipment - No competitor shall cause any range equipment to become damaged through a deliberate act, and anyone guilty of same, will be expelled without a warning from the range. Expelled competitors will be disqualified from the competition with no return of entry fee.

9.28 Refusal to Obey - No person will refuse to obey instructions of the Match Director, Official Referee, Jury, Supervisor, Range Officers or any other officer of the tournament, if instructions are given in the proper conduct of the office.

9.29 Evasion of Rules - No competitor will evade nor attempt to evade nor be an accessory to the evasion of any of the conditions of a match as prescribed in the tournament program or in these rules. Refusal of a competitor or

9.30 Penalties/Disqualification - The Official Referee, Jury, Supervisor or Match Director, upon proper presentation of evidence, may disqualify or order the expulsion of any competitor from the range for violations of rules or for other conduct considered discreditable or unsafe. In the event of a disagreement between Match Officials, the Official Referee or Jury shall prevail at the match with recourse only to the Protest Committee.

9.31 Suspension - For violations of these rules deemed so to justify, any member may be suspended or expelled from the National Rifle Association upon presentation of evidence and conduct of a hearing as prescribed in the Bylaws.

NRA CONVENTIONAL PISTOL RULES

10. Range Commands and Operations

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook10.htm

10. RANGE COMMANDS, CONTROL, AND OPERATIONS

10.1 Discipline - The safety of competitors, range personnel and spectators requires continuous attention by all to the careful handling of firearms and caution in moving about the range. Self-discipline is necessary on the part of all. Where such self-discipline is lacking it is the duty of range personnel to enforce discipline and it is the duty of competitors to assist in such enforcement

10.1.1 Action Open - Unless Pistols are holstered or cased, cylinders must be open or slides back and the magazine removed at all times until the competitor is in position at his firing point and the command, "THE PREPARATION PERIOD STARTS NOW", has been given.

10.1.2 Pistols Unloaded - Pistols will not be loaded until the competitor has taken position at the firing point the pistols pointed toward the targets and the command "LOAD" has been given.

10.1.3 Loaded Pistols - A pistol or revolver that has a cartridge in the cylinder or in a magazine which has been inserted shall be considered as being loaded. No pistol will be loaded until competitor has taken the assigned place at the firing point and the command "LOAD" has been given by the range officer. Loaded pistols shall be pointed in the direction of the targets at all times.

10.1.4 Cease Firing - Any person will immediately command "CEASE FIRING" if any incident occurs which indicates possible injury to some living thing should firing continue. In all other cases commands will originate with the Chief Range Officer, (See Rules 10.7 and 18.10.)

10.1.5 Not Ready - It is the duty of competitors to notify the Range Officer if not ready to fire at the time the Chief Range Officer asks, "IS THE LINE READY?" Should the Chief Range Officer cause firing to proceed, the competitor concerned will be given an opportunity to fire a score In the earliest possible relay or by time extension. Failure of competitor to notify the Range Officer "not ready" forfeits the right to fire. (See Rule 10.7).

10.1.6 to 10.1.8 - BLANK

10.1.9 Ranges with Target Carriers - When a range is equipped with a carrier system permitting the changing of targets from the firing line without suspending firing, competitors may be issued the targets necessary to complete the match. Each competitor will be responsible for hanging the target properly on the target carrier. Fired targets will be gathered by the Range Officer as soon as each stage is completed.

10.2 Range Courtesy

(a) Loud Language - loud or abusive language will not be permitted. Competitors, scorers, and Range Officers will limit their conversation directly behind the firing line to official business.
(b) Smoking - As a courtesy to competitors, smoking is prohibited forward of the "ready line."
10.3 Delaying a Match - No competitor may delay the start of a match through tardiness in reporting or undue delay in preparing to fire.
10.3.1 Preparation Period - In all cases competitors will be allowed 3 minutes to take their places at their firing points and prepare to fire after the relay is called to the firing line. Dry firing will be permitted during this period.
10.4 Policing Range - It is the duty of competitors to police the firing points after the completion of each match or stage. The Range Officer will supervise such policing and see that the firing points are kept clean.
10.5 Competitors Will Score - Competitors will act as scorers when requested to do so by the Executive Officer or Chief Range Officer, except that competitors will not score their own targets.
10.6 Repeating Commands - A Range Officer will repeat the Chief Range Officer's commands only when those commands cannot be clearly heard by competitors.
10.7 Firing Line Procedures and Commands - When ready to start the firing of a match the Range Officer commands "RELAY NO. 1, MATCH NO. (or naming the match), ON THE FIRING LINE." The competitors in that relay immediately take their assigned places at their firing points and prepare to fire but do not load. The Range Officer then states "THE PREPARATION PERIOD STARTS NOW." Range Officers check the location of each competitor as to correct firing point by comparing the number of each competitor's firing point with the relay and target number on the score card or on the range assignment card. At the end of the 3 minute preparation period the Range Officer states "THE PREPARATION PERIOD HAS ENDED."
The Range Officer having made sure that the range is clear (in timed and rapid fire the targets must be turned out of firing position) then commands "WITH 5 ROUNDS LOAD." Note: In slowfire events, which are scored after 10 shots, the command should be "10 SHOTS SLOW FIRE, 10 SHOTS IN 10 MINUTES, LOAD." (The number of rounds to be loaded by the competitor in slowfire stages will be determined by the competitor.) The Range Officer then asks, "IS THE LINE READY?" Any competitor who is not ready or whose target is not in order will immediately raise the nonshooting arm and call, "Not ready on target.... The Range Officer will immediately state, "THE LINE IS NOT READY:' and the Range Officer will immediately investigate the difficulty and assist in correcting it. When the difficulty has been corrected, the Range Officer calls. "THE LINE IS READY." When the Range Officer asks "IS THE LINE READY?" and the line is ready, he then calls, "THE LINE IS READY." The Range Officer then commands, "READY ON THE RIGHT. READY ON THE LEFT." Competitors may point their guns toward the target after the command, "Ready on the right." The Range Officer will then command, "READY ON THE FIRING LINE." In approximately 3 seconds the

targets will be exposed, or the signal to commence firing will be given. The Range Officer then commands "COMMENCE FIRING" which means to start without delay as timing of the string is started with this command. "COMMENCE FIRING" may be signaled verbally, or by a short sharp blast on a whistle or by moving the targets into view. "CEASE FIRING" is the command given by the Range Officer at the end of time limit for each string or at any other time all firing is to cease. FIRING MUST CEASE IMMEDIATELY, Even if a competitor is about to let off a carefully aimed shot the competitor must not fire and open the action of the gun. Failure to immediately obey this command is one of the worst infractions the next command of the Range Officer. "Cease Firing" may be signaled verbally, or by a short sharp blast on a whistle or by moving the targets out of view. When the command to cease firing is given at the end of a string the command is "CEASE FIRING- UNLOAD- CYLINDERS OPEN- MAGAZINES OUT- SLIDES BACK- GUNS ON THE TABLE. ARE THERE ANY REFIRES? The Range Officer than asks "IS THE LINE SAFE ON THE RIGHT? IS THE LINE SAFE ON THE LEFT?" On this command all Assistant Range Officers and scorers check their competitors to make sure each one obeys the command before signaling the Range Officer that their portion of the firing line is safe. When all Assistant Range Officers and scorers have given a signal that the range is safe the Range Officer commands "THE LINE IS SAFE, GO FORWARD, SCORE TARGETS AND PASTE" (or "CHANGE"). When the Range Officer has ascertained that all personnel have returned to the firing line after scoring or changing targets, the command "THE RANGE IS CLEAR, YOU MAY HANDLE YOUR GUNS" will be given. Other commands used less frequently are: "POLICE FIRING POINTS" means pick up fired cartridge cases, empty cartridge cartons and "tidy-up" the firing line. "AS YOU WERE" means disregard the command just given. For example, if the commands were given "READY ON THE RIGHT' followed by "AS YOU WERE" it would mean someone was not ready. "CARRY ON" means proceed with whatever was being done before some interruption occurred. Should a target blow out of the frame or otherwise become unserviceable the Range Officer will report the condition immediately to the Chief range Officer who will issue such commands and directions as seem advisable. Note: Commands controlling the start of the firing sequence commencing with "Ready on the Right" should have approximately a 3 second interval between firing line commands. **10.8 BLANK**

10.9 Procedure in Case of Defective Cartridge, Disabled Pistol In Slow Fire - If a cartridge fails to fire or a pistol fails to function in slow fire the competitor may replace the unfired cartridge or clear the jam and continue firing. Additional time may be allowed any competitor, equal to the time lost because of the defective cartridge, disabled pistol or malfunction, if the Range Officer has been notified at the time of the malfunction. (For refiring privileges see Rule 9.14(B). It is not required that the Range Officer attempt to fire a cartridge before it is declared defective. In case of malfunction where a pistol or revolver is disabled and must be replaced requiring a refire, the competitor will call the Range Officer.

10.10 Procedure in Case of Defective Cartridge, Disabled Pistol or Malfunction in

Timed and Rapid Fire - In the event of a defective cartridge (Rule 9.4), disabled pistol (Rule 9.5), or malfunction (Rule 9.6), before a suing is completed in timed or rapid fire, the competitor shall be privileged to fire another five shot string, provided he assumes the "Ready" position and calls the Range Officer by holding up the non-shooting hand at the end of the time period. The Range Officer will inspect the pistol, if satisfied that there the pistol or bullets that have failed to leave the barrel. The competitor will then fire another complete five shot string on the same target.
When scoring is after each 5 shot string, the competitor will be charged with firing 10 shots minus whatever number of cartridges were found unfired or bullets that failed to leave the barrel in the original string and will be scored the 5 shots of lowest value. When scoring after 10 shots, the competitor will be charged with firing 15 shots minus whatever number of cartridges were found unfired or bullets that failed to leave the barrel in the original string and will be scored the 10 shots of lowest value. Unfired shots in the refire string will be scored as misses. Failure of the competitor to notify the Range Officer of the malfunction, or the opening or clearing or attempting to clear the pistol by the competitor before the Range Officer has inspected the pistol forfeits the right of the competitor to refire. (For refiring privileges see Rule 9.14c). Example - In timed or rapid fire a competitor has a misfire on the fourth shot. The Ready Position will be assumed and at the end of the time limit calls the Range Officer. The Range Officer inspects the gun, finds conditions as claimed, and finds 2 unfired cartridges in the gun. The competitor is then given orders to "LOAD 5 ROUNDS" and fires another 5 shot string on the same target. When scoring is after 5 shot string - In the first string there are 2 10's and a 9, in the second string there are 5 10's. The score is the 5 shots of lowest value or 49. However, if in the first string there are 2 10's and a 9 and at the end of the second string the only other visible hits were 2 additional 10's and another 9 the competitor would be scored 2 misses in the second string and the score would be 28. When scoring is after 10 shots - In the first string there are 2 10's and a 9, in the second string there are 5 10's and in the third suing there are 5 10's, the score is the value of the lowest 10 shots of the 13 shots fired or a 99. However, if in the first string there are 2 10's and a 9 and at the end of the third string the only other visible hits were 7 additional 10's and another is 9, there would be scored 2 misses and the score would be 78. **10.11 to 10.17- Blank**

NRA CONVENTIONAL PISTOL RULES

14. Scoring and Marking

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook14.htm

14. SCORING AND MARKING
14.1 When to Score - Targets are scored after each 10 shot stage of each 5 shot string.
14.2 Where to Score - Targets may be scored on the target frames, at the firing line or in the Statistical Office. The scorer must be at the target when scoring.
14.2.1 Targets are Score Cards - Targets constitute the score cards when scored in the Statistical Office and therefore must be retained in good order until the time allowed for filing challenges and protests has expired.
14.3 How to Score - A shot hole, the leaded edge of which comes in contact with the outside of the bullseye or scoring rings of a target, is given the highest value (Fig. F). A scoring gauge will be used to determine the value of close shots. The higher value will be allowed in those cases where the flange on the gauge touches the scoring ring. No scoring gauge will be used unless the diameter of the scoring flange is within these limits:

> .22 caliber,.2225" - .2240"
> .32 caliber,.310" - .314" .38 caliber,.355" - .359" .40 caliber,.397" - .401" .44 caliber,.426" - .430" .45 caliber,.450" - .454"

Devices other than scoring gauges may be used to assist in establishing the correct value of hits. These devices are not to be inserted into the bullet hole and do not constitute a scoring gauge. (a) In case of keyholed or tipped shots, the higher value is awarded if the leaded edge of the bullet hole touches the scoring ring of higher value, even though the hole is elongated to the bullet's length rather than being a circle of the bullet's diameter. (b) In case of skid shots, the higher value is awarded if the leaded edge of the bullet hole touches the scoring ring of higher value, except the value of a skid shot may not be more than one ring higher than the original point of bullet contact with the target. The target shall be defined as the entire card or paper on which the scoring rings are printed but shall not include the backing. When the original point of bullet contact is on the target card, but outside the scoring rings, and the leaded edge of the bullet touches a scoring ring, it will be given the value of the lowest scoring ring. (c) When a bullet enters a target from the back side it will be scored as a miss. **14.3.1 Authorized Use of Plugs or Scoring Gauges** - The use of the plugtype gauge will be restricted to use by range operating personnel who may include the Range Officer or Block Officers, Pit Officers, Match Supervisors, Statistical Director, Match Director, Jury or Referee,

as appropriate to the type of tournament concerned. No competitor will use scoring plugs of any kind on a target at any time. Where targets are scored on the range only the above designated officials may use plugs to check values and determine the outcome of challenges. However, competitors acting as scorers may use official NRA overlays in the performance of their duties. The tournament program should state in the Match Conditions Section by whom and under what circumstances plug-type scoring gauges may be used.

14.3.2 Removal of Plug Type Scoring Gauge

(a) The plug type scoring gauge, if used, will remain in the shot hole until the shot value is agreed upon by the competitor and the scorer, or until removed by the Referee, Jury or Supervisor after a challenge. If the plug type scoring gauge is removed prior to an agreement of shot value, the shot hole may not be re-plugged and must be taken to the Referee, Jury or Supervisor for final decision. (b) Only the Referee, Jury or Supervisor may use a plug type scoring gauge to determine if a shot hole is a double. **14.4 Misses** - Hits outside the scoring rings are scored as misses. (Except as provided in 14.3(b).

14.5 Early or Late Shots - When a shot is fired early or late, that is, before or after the signal to commence or cease fire, when the required number of hits are visible on the face of the target, the value of the highest hit will be scored a miss. When fewer than the required number of hits are visible on the target, the competitor will be scored a miss for each shot not on the target, and those visible on the target will be scored in the normal manner.

14.6 All Shots count - All shots fired by the competitor after position has been taken at the firing point will be counted, even if the pistol is accidentally discharged.

14.7 Hits on Wrong Target or Bullseye - Hits on the wrong target or bullseye are scored as misses. A wrong target is defined as a target other than that:

(a) Assigned to the firing point upon which the competitor is squadded (assigned). (b) Intended to be used for the match, stage and distance in that event concerned. **14.8 Ricochets** - A hole made by a ricochet bullet does not count as a hit and will be scored as a miss. It must be noted that a bullet which keyholes is not necessarily a ricochet.

14.9 Visible Hits and Close Groups - As a general rule, only those hits which are visible will be scored. An exception will be made in the case where the grouping of 3 or more shots is so close that it is possible for a required shot or shots to have gone through the enlarged hole without leaving a mark and there has been no evidence that a shot or shots have gone elsewhere than through the assigned target. In such case, the shooters will be given the benefit of the doubt and scored hits for the nonvisible shots, on the assumption they passed through the enlarged hole. If such an assumption could place a nonvisible hit in either of 2 scoring rings, it shall be scored in the higher-valued ring.

14.10 Excessive Hits - If more than the required number of hits appear on the target, any shot which can be identified by the appearance of the bullet hole as having been fired by some competitor, other than the competitor assigned to that target, or as having been fired in a previous string, will be pasted and will not be scored. If more than the required number of

hits then remain on the target a complete new score will be fired and the original score will be disregarded, except:
(a) If all hits are of equal value, the score will be recorded as the required number of hits of that value. (b) The competitor shall be allowed to accept a score equal to the required number of hits of lowest value. (c) If a competitor fires fewer than the prescribed number of shots through the competitors own fault, and there should be more hits on the target than the shots fired, the competitor will be scored the number of shots of highest value equal to the number fired, and will be given a miss for each unfired cartridge. (d) If a competitor, by mistake, fires more than the required number of shots, the required number of hits of lowest value will be scored. This shall not be considered a refire as outlined in Rule 9.14. (e) If the competitor refires, the original target with excessive hits shall be retained by the scorer, and on refiring, the competitor may not receive a score higher than the required number of hits of highest value on the original target. If the score on the refired target is higher than the required number of hits of highest value on the original target, then the original target shall be scored using the appropriate hits of highest value; but if the score on the refired target is not higher, then such refired score will be recorded. **14.11 Scoring Altered Targets** - Targets (Targets in this instance include the target frame) intentionally altered or marked to benefit a shooter over other competitors will not be scored.

14.12 and 14.13 (BLANK)

14.14 Scorer's Duties

(a) When targets are scored before their removal from the frame the scorer records the value of each hit on score card while holding the card in such a position that competitor may see score being recorded. Only "X" and Arabic numerals must be used when shot values are entered, misses are marked as "M". While marking score on card the scorer announces each hit value in an audible tone of voice. Example: "Target Number 15, 2 tens, 2 nines and 1 seven." It is the duty of each competitor to watch the marking of score on the score card and to challenge such score immediately if scoring is thought to be incorrect. After each target is scored shot holes are patched or target is changed. (b) When targets are scored after their removal from the target frames, they are removed by the target detail on the command. "CHANGE TARGETS" and given to the Statistical Office for official scoring. **14.15 Score Cards** - Score cards will be prepared by the Statistical Office. Competitors will ordinarily pick up their score cards from the Statistical Office and deliver the correct score card to the scorer before each match. At the conclusion of each relay, Range Officers will take up the score cards and deliver them to the Statistical Office.

14.16 Erasures on Score Cards - Erasures on scorecards are not permitted. If correction is necessary. it must be made and initialed by the Scorer or Range Officer. To make a correction, the Scorer or Range Officer draws a line, or lines, through the incorrect score and places the correct score above.

14.17 to 14.24 BLANK

14.25 Position on Score Card of Refired Scores - Scores (including refired strings) will be placed on the score card in the order in which they are fired.

NRA CONVENTIONAL PISTOL RULES

15. Decision of Ties

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook15.htm

15. DECISION OF TIES

Note: All tie (same numerical score) ranking Rules shall be applied in the order listed below.

15.1 Match - The term "match" as used in this section refers to all individual, team, and aggregate matches.

15.2 Value of 'X' - In all matches where the X is scored, an X is a hit of highest value.

15.3 Single Stage -At any range or stage ties will be ranked by applying the following steps, (a) to (f) inclusive, in the order listed below:

(a) By the greatest number of X's. (b) By the fewest misses. (c) By the fewest hits of lowest value. (d) By the fewest hits of the next lowest value, etc. (e) In slow fire individual matches by inverse order of shots, counting singly from the last shot to the first shot (this will be applied only when targets are being scored after each shot). (f) In matches scored in strings of 5 or 10 shots by the highest ranking score in the last suing, by highest ranking score in the next to last string, etc. (If still a tie, apply Rule 15.10.) **15.4 Multiple Stage** - In matches fired in stages, ties will be ranked by applying the following steps, (a) to (d) inclusive, in the order listed below:

(a) By the greatest number of X's. (b) By the highest ranking score in rapid fire; if still a tie, rank each rapid fire score by applying Rule 15.3. If this does not break the tie, apply Rule 15.4(c). (c) By the highest ranking score in timed fire; if still a tie, rank each timed fire score by applying Rule 15.3. If this does not break the tie, apply Rule 15.4 (d). (d) By the highest ranking score at slow fire; if still a tie, rank each slow fire score by applying Rule 15.3. If this does not break the tie, see Rule 15.10. **15.5 Aggregate Matches** - In aggregate matches, ties will be ranked by applying the following steps, (a) to (d) inclusive, in the order listed below:

(a) By the greatest number of X's. (b) By the highest-ranking total rapid fire score (including both single and multiple stage match rapid fire scores ranked as shown in Rule 15.3). If this does not break the tie, apply Rule 15.5 (c). (c) By the highest ranking total timed fire score (including both single and multiple stage match timed fire scores ranked as shown in Rule 15.3). (d) By the highest ranking total slow fire score (including both single and multiple stage match slow fire scores as shown in rule 15.3). If still a tie, see Rule 15.10. **15.6 Rifle and Pistol** - Competitors having the same numerical score for the entire course of fire in matches which include both rifle and pistol stages, ties will be ranked in the order shown below:

(a) By highest ranking score in the pistol stage. (b) By the highest ranking score in the rifle stage. **15.7 Team Matches** - Ties in team matches will be ranked in the order shown below:

(a) By considering the team score as though it were a single score fired by an individual. The

same precedent applies as indicated above (15.3 to 15.5 inclusive). (b) By the highest individual aggregate score. (c) By the second highest individual aggregate score, etc. (d) By the highest individual score, the second highest individual score, etc., at each stage considered, in the order listed in Rule 15.4. 15.8 Re-Entry Matches - In re-entry matches ties will be ranked in the following order: (a) By considering as a unit the total score fired on all targets comprising the score for record as provided in the tournament program and applying thereto the provisions of Rule 15.3 (a) (b) and (c) above. (b) By outranking the competitor with the lowest score on any one target of those comprising the score for record. (c) If still a tie, by combining all awards to which those tied are entitled and dividing equally among those tied. Lots will be cast for merchandise, medals or other trophies. **15.9 League Ties** - In league type competitions in which the standing of teams is determined by the number of matches won and lost, ties will be decided by a shoot-off over the same course of fire as used during the league season or as set forth in the league program.

15.10 Unbreakable Ties - In case a tie cannot be ranked under the provisions of the above rules of this section, the Match Director will direct that the tie be decided and awards given under one of the following Plans as appears necessary or advisable.

(a) By the firing of a complete or partial score under the original match conditions or at the longest range of the match. (b) By the drawing of lots for merchandise, medals combining any other awards to which those tied may be entitled, and making equal division of awards among those tied.

NRA CONVENTIONAL PISTOL RULES

16. Challenges and Protests

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook16.htm

16. CHALLENGES AND PROTESTS

16.1 Challenges - When a competitor feels that a shot fired has been improperly evaluated or scored the scoring may be challenged. Such challenge must be made immediately upon announcement of the score. No challenge will accepted after the target has been handled by the shooter or removed from the range.

(a) A challenge fee not to exceed $3.00 may be charged to all competitors making challenges. The challenge fee will be collected before making the first re-check of the challenged score. If the competitor's challenge is sustained at any point along the line of re-checks, the challenge fee will be returned. If the challenge is lost, the challenge fee will be included in the general revenue of the tournament. The decision of the Official Referee, Jury or Supervisor will be final in NRA competition. (b) When targets are scored on frames and the scoring of a shot is challenged, the Range Officer will immediately call the Official Referee or Supervisor, who will score the target. If necessary to avoid delaying the match the challenged target will be replaced with a clean target and the match will proceed. The Official Referee, Jury or Supervisor will score the target as soon as possible and notify the competitor. (c) When targets are scored in the Statistical Office, re-check will be made by the Chief Statistical Officer (provided he has not previously scored or checked the target) and the Official Referee, Jury or Supervisor, in that order. (d) The Match Director may re-check any competitor's target by an administrative challenge. Such challenge must be made within the posted. challenge period and checked by the Referee. For further information on the Statistical Office operation, see Rules 11.6 through 11.6.8 inclusive. **16.2 Protests** - A competitor may formally protest:

(a) Any injustice which is felt has been done except the evaluation of a target. which may be challenged as outlined in Rule 16.1. (b) The conditions under which another competitor has been permitted to fire. (c) The equipment which another competitor has been permitted to use. **16.2.1 Authority of Protest Committees** - may be established at National Championships to rule on protests arising from activities at those sites (SEE Rule 11.8). However, the decisions of these special protest committees or juries shall not contravene prior interpretations of the NRA Rules and/or precedents established by the NRA National Protest Committee.

16.3 How to Protest - A protest must be initiated immediately upon the occurrence of the protested incident. Failure to comply with the following procedure will automatically void the protest:

(a) State the complaint orally to the Chief Range (Chief Statistical) Officer. If not satisfied then, (b) State the complaint orally to the Official Referee, Jury Chairman or Supervisor. If not satisfied then, (c) File a formal protest in writing with the Official Referee, Jury Chairman or Supervisor stating all the facts in the case. Such written protest must be filed within I 2 hours of the occurrence of the protested incident. (d) The Official Referee, Jury Chairman or Supervisor will then forward the protest with a complete statement of facts within 48 hours of receipt thereof to the Protest Committee at NRA Headquarters. (e) At the National Championships a protest procedure must begin no later than one hour after the completion of firing for the day. In case of awards, the protest must begin no later than one-half-hour after the completion of the awards ceremony. **16.4 Challenges and Protests in Team Matches** - Must be made by the Team Captain. Team members who believe they have reason to challenge or protest will state the facts to their Team Captain, who will make the official challenge or protest if it is felt such action to be justified.

NRA CONVENTIONAL PISTOL RULES

17. National Records

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook17.htm

17. NATIONAL RECORDS

17.1 Where Scores for National Records Can Be Fired - Scores to be recognized for National Records must be fired in NRA Registered Competition as defined in Rule 1.6, paragraphs (c), (d), (e), and (f). National Records must be approved by the NRA before being declared official. National Records may not be established during the re-entry matches.

17.2 Scores to be Used - Scores must be complete scores for an entire scheduled match. Stage scores or scores for only part of a match will not be used for records.

17.3 Scores for National Individual Records - Such scores must be fired in individual matches. No score fired in a team match will be considered for recognition as an individual record. For recognition as special group records ("Open," "Civilian", "Police," "Service," "Women," "Junior") scores may be fired in either open or restricted matches. Triggers of pistols used in establishing National Records must conform to Rule 9.8.

17.4 Scores for National Team Records - Such scores must be fired in matches where team fire as a unit, and no combination of individual match scores will be considered for recognition as a team record. For recognition as special group records ("Open," "Civilian," "Police," "Service," "Women," "Junior"), all members of the team must be members of the special group concerned teams must be bona fide teams as outlined in Rules 2.11 to 2.18. National Records will not be recognized for "pickup" teams (teams made up of shooters who do not all represent one of the groups outlined in Rules 2.11 to 2.18). Triggers of pistols used in establishing National Records must conform to Rule 9.8.

17.5 Courses of Fire for Which National Records Are Recognized - *(omitted from this abridged version)*

17.6 Co-holder Records – Tie-Breaking Rules, beyond the use of numerical scores including X-count, will not be employed when establishing National Records. Co-holder status will be accorded to individuals or teams when their score equals a National Record.

NRA CONVENTIONAL PISTOL RULES

18. Competitor's Responsibilities

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook18.htm

18. COMPETITOR'S DUTIES AND RESPONSIBILITIES

Note: The following competitors' duties are in addition to those specified elsewhere throughout these Rules.

18.1 Discipline - It is the duty of each competitor to sincerely cooperate with tournament officials in the effort to conduct a safe, efficient tournament. Competitors are expected to promptly call the attention of proper officials to any infraction of rules of safety or good sportsmanship. Failure of a competitor to cooperate in any case arising out of infractions of these rules may result in the sited competitor being considered as an accessory to the offense. (See Rule 9.35).

18.2 Knowledge of Program - It is the competitor's responsibility to be familiar with the program. Officials cannot be held responsible for a competitor's failure to obtain and be familiarized with the program.

18.3 Eligibility - It is the competitor's duty to enter only those events for which eligible and to enter in the proper classification.

18.3.1 Competitors Will Score - Competitors will act as scorers when required to do so by the Match Director or Chief Range Officer, except that competitors will not score their own targets.

18.4 Classification - It is the competitor's duty to have a current Classification Card in possession when using a classification system. Unclassified competitors must obtain their Score Record Book from the Official Referee, Supervisor or Tournament Officials.

18.5 Individual Entries - In individual matches it is the duty of competitors to make their own entries on the forms and in the manner prescribed for that tournament. Errors due to illegibility or the improper filling out of forms are solely the competitor's responsibility. The Statistical Office is not required to accept corrections after the entry closing time.

18.6 Squadding Tickets - It is the competitor's duty to secure squadding tickets for each match (or to consult the squadding bulletin) in ample time to permit reporting at the proper time and place to fire each match. It is not the duty of officials to page competitors in order to get them on the firing line. Competitors upon receipt of squadding tickets, should inspect them for correctness of the competitor's number and non-interference in squadding assignments. Errors should be reported immediately to the Statistical Officer.

18.7 Reporting at Firing Point - Competitors must report at their assigned firing points immediately when the relay is called by the Range Officer. The proper pistol and ammunition

for that particular match must be ready and in safe firing condition. Time will not be allowed for pistol repairs, sight blacking. sight adjustments or search for missing equipment after a relay has been called to the firing line.

18.8 Timing - Time for the firing of a string ~within the official time limit) is the competitor's responsibility. Range Officers will not announce the time during the firing, but, if requested, will give the competitor information as to the remaining time.

18.9 Loading - No competitor will load a pistol except at the firing point and after the command "LOAD" has been given by the Range Officer.

18.10 Cease Firing - When the command "Cease Firing" is given by anyone, all firing will immediately cease and competitors will await further commands or instructions to be given by the Range Officer.

18.11 Checking Scores - It is the duty of competitors to check the shot values and the total score on the score card at the conclusion of each match. Competitors must promptly check the Preliminary Bulletin and call attention to errors within the time specified at that tournament. Failure to check scores within the time limit results in the penalty of losing the privilege of any further challenge. Failure to sign a scorecard before leaving the firing line results in the loss of both the challenge and protest privilege. In team matches, the Team Captain must check and sign score cards. (For procedures on challenges and protests. see Rule 16.)

18.12 Clearing the Firing Point - It is the competitor's duty to leave the firing point promptly at the conclusion of the relay. When leaving the firing point pistols must be unloaded. Cylinders must be open on revolvers, and slides locked back and magazines removed from semiautomatics.

18.13 Checking Bulletin Board - It is the duty of all individual competitors and Team Captains to check Bulletin Board between each match. The Statistical Officer must be immediately notified of apparent errors. Official Bulletins must be checked and the Statistical Officer notified of any discrepancies between the Preliminary and Official Bulletins. Match Director's Bulletins on the Bulletin Board have the same effect as conditions printed in tournament program. It is the duty of competitors to familiarize themselves with all such Match Directors Bulletins.

18.14 Score Cards Must Be Signed - At the conclusion of scoring the scorer will add the value of the shots, place the total on the score card and sign the card. The competitor then checks value of individual shots, the total, and signs the card. If a competitor or Team Captain leaves the firing line without signing the score card, no protest will be allowed. If the competitor or Team Captain desires to challenge, the word "challenge" is written on the score card in place of the Team Captain's signature. Team Captains verify and sign score cards in team matches.

18.15 Responsibility - It shall be the competitor's responsibility:

(a) That all equipment meets all rules and match specifications in any match in which that equipment is to be used. (b) That the competitor's position conforms to the rules. (c) That the

competitor has full knowledge of the rules under which the match is fired. (d) That after due warning of any infraction of existing rules, that the competitor shall understand that a repetition thereof shall be the subject of disqualification for that match or tournament. (e) That when targets are framed by the competitor, it is the competitor' s responsibility to frame the correct target for the specific match and distance.{See Rule 14.7(b)}. (f) To insure that the target is not altered intentionally or with special marks which will be beneficial in any way. Shots fired on such targets will not be scored.

NRA CONVENTIONAL PISTOL RULES

19. National Pistol Classification

By NRA Conventional Pistol Rulebook

Source: http://bullseyepistol.com/rulebook19.htm

19. NATIONAL PISTOL CLASSIFICATION

19.1 Classified Competitors - Are all individuals who are officially classified by the NRA for pistol competition, or who have a record of scores fired over courses of fire used for classification (See 19.4) which have been recorded in a Score Record Book.

19.2 Unclassified Competitor - Is a competitor who does not have a current NRA Pistol Classification, either regular or temporary by Score Record Book (Rule 19.14), nor an "Assigned Classification" (rule 19.6). Such competitor shall complete in the Master Class.

19.3 Tyro Competitors - Are competitors who have not previously fired in organized, civilian, police or military pistol competitions, of the course of fire composing such matches. Any tournament sponsor may include in their program a prize schedule of awards for the Tyro Class.

19.4 Matches Used for Individual classification - Scores to be used for classification and reclassification will be those fired in individual and team matches in both indoor and outdoor NRA Competition as defined in Rule 1.6 (except Postal Matches) over the following courses of fire and under the indicated conditions:

(a) National Match Course. (b) NRA Short Course. (c) National Match Gallery Course (50 feet or 20 yards). (d) Any component part of these courses when fired as single stage events. Scores fired indoors are used to establish indoor classification. Scores fired outdoors are used to establish outdoor classification. Scores from Sanctioned Leagues may be used during the league firing season in Score Record Books (Rule 19.14), but will only be used by the NRA Headquarters office at the end of the league firing season for issue of Official Classification Cards. **19.5 Compilation of Scores for Classification Averages** - Scores fired in complete matches over the above courses, regardless of the caliber of arm used will be combined and averaged to establish a competitor's classification.

19.6 Assigned Classification - A competitor who has an earned classification (a classification obtained through a Score Record Book or an Official NRA Classification Card) for one type of competition in the grouping listed below will be assigned this same classification in any other type in which the competitor is not classified in the same group:

(a) Outdoor Pistol (b) Indoor Pistol (c) Police Combat (d) Action Pistol (e) International Pistol (Free, Air, Center, Rapid Fire or Standard) If a competitor has a classification in more than one type in the list, the higher classification shall be used. In the second tournament in the new type. the Score Record Book is used rather than the assigned classification. **19.7 Lack of Classification Evidence** - It is the competitor's responsibility to have a NRA Official

Classification Card or Score Record Book with required scores for temporary classification (Rules 19.1 and 19.14) and to present this classification evidence when required. Any competitor who cannot present such evidence will fire in the Master Class. A competitor's classification will not change during a tournament. competitors will enter a tournament under their correct classification and fire the entire tournament in that class. Should it be discovered during a tournament that a competitor has entered in a class lower than the current rating, the tournament records will be corrected to show the correct classification for the entire tournament.

19.8 Competing in a Higher Class - Any individual or team may elect, before firing, to compete in a higher classification than the one in which classified. Such individual or team must fire in such higher class throughout the tournament and may not revert to earned classification for any event in that tournament.

When there are insufficient entries in any class to warrant an award in that class according to the match program conditions, the individual or team concerned may be moved by the Tournament Match Director to a higher class provided this change is made prior to the individual or team concerned having commenced firing in the tournament. **19.9 Obsolete Classification and Scores** - All classifications and scores, (including temporary, Rule 19.14) except Master, shall become obsolete if the competitor does not fire in NRA competition at least once during 3 successive calendar years. Master classifications and scores shall become obsolete if the competitor does not fire in NRA competition at least once during 5 successive calendar years. Lifetime Master classifications will not become obsolete.

19.10 Appeals - Any competitor having reason to believe classification is improper may file an appeal with the NRA stating all essential facts. Such appeal will be reviewed by the NRA Protest Committee.

19.11 Protests - Any person who believes that another competitor has been improperly classified may file a protest with the NRA stating all essential facts. Such protests will be reviewed by the NRA Protest Committee. (See Rule 16.2)

19.12 Team Classification - Teams are classified by computing the "team average" based on the classification of each firing member of the team. To compute this "team average" the key in Table No. 1 for the different classes will be used and the team total divided by the number of firing members of the team. Any fractional figure in the team average of one half or more will place the team in the next higher class. The team average will establish classification of the team as a unit, but will not affect in any way the individual classification of the team members.

TABLE NO. I - TEAM Class Key High Master 6
Master .....5 Expert 4
Sharpshooter ..... 3 Marksman 2
Tyro.....1 **19.13 Reporting Scores** - NRA indoor and outdoor competition (see Section 1) sponsors will report to the NRA all individual and fired team

match scores fired over the courses stated in Rule 19.4. Scores fired in individual matches will be reported as aggregate totals and scores from fired team matches will be reported as a separate aggregate total. Scores from all tournaments and sanctioned leagues will be reported by each sponsor no more than 30 days after completion of the tournament firing schedule.

19.14 Score Record Book - (Temporary Classification) - A Score Record Book will be obtained by each unclassified competitor from the Official Referee, Supervisor, Tournament Statistical Officer, or Sanctioned League Secretary at the time the competitor competes in the first tournament. All scores fired will be recorded by the competitor in each NRA competition (except Postal Matches) until such time as an Official NRA Classification Card is received. The competitor will total all scores and divide that total by the number of 10- shot strings represented. The average so obtained will determine the competitor's NRA Classification at that time (see Rule 19.15 for average score for each classification).

Individual and team scores fired by the competitor during ar least one tournament (Rule 1.1), or from the most recent league match (Rule 1.6(h)), must be posted in the Score Record Book to establish a Temporary Classification. The Score Record Book will be presented by the holder at all NRA competition entered until the competitor's Official NRA Classification Card becomes effective. Note: It is the competitor's responsibility to obtain the Score Record Book, enter scores, and present it at each tournament until his Official NRA Classification Card becomes effective. When the NRA Classification Card becomes effective the Score Record Book becomes obsolete. **19.15 Individual Class Averages** - Competitors will be classified as follows and NRA Classification Cards issued accordingly:

TABLE NO. II - INDIVIDUAL, High Master .....97.00 and above
Master .....95.00 to 96.99 Expert ..... 90.00 to 94.99
Sharpshooter ..... 85.00 to 89.99 Marksman ..... Below 85.00 **19.16 Establishing Classification** - A competitor will be officially classified by the NRA when the total score for a minimum of 360 shots has been reported for either indoor or outdoor. However, classification averages will be computed only after the total score for a tournament or league has been posted and, therefore, the average may be based on a greater number of shots, but will not be based upon a lesser number. Total scores so reported to the NRA will be posted to the Classification Record for the competitor concerned. When the scores for the stated minimum of 360 shots (or more if this minimum is reached during the scores of any tournament or league) have been so posted, the average score per 1 O-shot string will be computed. The competitor will be sent an Official NRA Classification Card based on the average so computed and according to the table in Rule 19.15, which classification will become effective the date shown on the card issued by NRA.

19.17 Reclassification - A competitor who has been classified by the NRA will be reclassified as follows:

(a) A record of all completed (see Rule 19.9) NRA Competition scores fired by a classified competitor (except Postal Match) will be maintained by NRA Headquarters. Only scores not

previously used in computing a classification average will be used for reclassification. except as noted in (c) below. (b) A competitor will be considered for reclassification upward when scores for not less than 360 shots, fired subsequent to the last previous consideration for classification or reclassification, have been recorded as prescribed, except that such consideration will not include tournament or league scores until after all scores for the tournament or league competition concerned have been recorded. If the average scores so justifies, an upward classification will be justified. (c) A competitor who believes the classification is too high may file a request with the NRA that classification be lowered. Once a written request is received at NRA an average will be taken using the competitor's most recent reported shots (1,080 shots for outdoor pistol; 720 shots for indoor pistol) posted subsequent to the last classification or reclassification. If the competitor does not have the minimum number shots posted at the time of request, an average will be taken when the minimum number of shots is posted to competitors' record. When the average of such shots places the competitor in a lower class, classification will be lowered accordingly. A competitor who has been reclassified downward and who, by scores fired in NRA sanctioned competition (except NRA Postal Matches), has again earned a former classification, that classification shall become final and the competitor shall retain the earned classification until reclassified into a higher class as outlined in Rule 19.17. (d) A reclassified competitor shall be sent a new classification card which will become effective as of the date shown on the card issued by NRA. **19.21 Lifetime Master** - Competitors who have been certified as Lifetime Masters will retain their Lifetime Master cards and enter competitions in the Master class, except that, (a) No new Lifetime masters will be certified. (b) Lifetime Masters will be reclassified to a higher class, according to the provisions of Rule 19.17(b) and must enter competitions in the higher class. (c) Lifetime Masters may petition NRA to revoke a Lifetime Master card and be reclassified downward according to the provisions of Rule 19.17(c).

IL-MO LEAGUE

Il-Mo Indoor Pistol League

Source: http://bullseyepistol.com/ilmo.htm

The 60th Season of the Il-Mo Indoor Pistol League Is Currently Underway!

Check out the Participating Team Section for Contact Information to Join a Team!

IL-MO LEAGUE

Il-Mo Executive Officers

Source: http://bullseyepistol.com/imadmin.htm

Current Board of League Officers:

2227 Bentley Manor Dr. Manchester, MO 63026 **636-825-6727**

Secretary-Treasurer: **Kirk Yates**

102 Hunters Lake Ct Eureka, MO 63025-1049 **636-587-2858**

Executive Officer: **Karl Schmitt**

203 Dehner Rd, Hillsboro, MO 63050-1936 **636-797-4203**

Further Reading

- Secretary-Treasurer: (mailto:kirk.yates@usa.net)
- Kirk Yates (mailto:kirk.yates@usa.net)
- Executive Officer: (mailto:kschmitt@jcn.net)
- Karl Schmitt (mailto:kschmitt@jcn.net)

IL-MO LEAGUE

League Forms

Source: http://bullseyepistol.com/imforms.htm

Forms for League Match Operations:

Match Sign-up Sheet:	**Sign up sheet for competitors to request their target and relay for a match.**
Match Scorecards:	**Scorecards for recording shot values for each competitor. *(Printed 2 scorecards per page.)***
New Shooter Info:	**Standardized form for collecting information for new team members.**
League Tournament "900" Scorecards:	**Scorecards for use at the 90-shot League Tournaments *(i.e. Mid-Season Tournament)***

IL-MO LEAGUE

Il-Mo History

Source: http://bullseyepistol.com/imhist.htm

The Illinois-Missouri Indoor Bullseye Pistol League was established in 1949. The wise men who founded our organization are:

Max Thompson
Frank Novak Herman Bockstruck Frank Stubits

In the late forties, there were several shooting teams participating in outdoor matches during the summer in the bi-state area. In 1949, employees of the Winchester-Western company thought it would be good idea to organize an indoor winter league to keep everyone sharp over the off-season. Indeed it was! Thus the "Ill-Mo Pistol League" was born. The original teams that participated in that very first season were:

Winchester-Western
Industrial Pistol Club St. Louis Police Piasa Pistol Club St. Louis Revolver Club Small Arms Armory Team

(A complete historiography of our organization, complete with photographs and interviews, is being researched and developed. Check back often as information will be added as it is available.)

IL-MO LEAGUE

Il-Mo News

Source: http://bullseyepistol.com/imnews.htm

A second Piasa team called the "Pirates" now fires its home matches on Wednesday nights. Contact Team Captain, Mike Holland for more information.

Further Reading

- Mike Holland (mailto:mpholl@att.net)

IL-MO LEAGUE

Il-Mo Previous Season Standings

Source: http://bullseyepistol.com/imprev.htm

The Final Team and Individual Results have been archived and are available for the following seasons:

1960-1961
1971-1972
1983-1984
1998-1999
1999-2000
2000-2001
2001-2002
2002-2003
2003-2004
2004-2005
2005-2006
2006-2007

IL-MO LEAGUE

Il-Mo Records

Source: http://bullseyepistol.com/imrecord.htm

If you have knowledge of records that qualify for inclusion here in any of these categories and you can provide official support documentation for confirmation, please contact the league President, John Dreyer.

ILL-MO PISTOL LEAGUE AS OF 12/21/2009
HIGHEST FIRED SCORES ON RECORD

SLOW FIRE	SCORE	DATE
PAUL NOLES	198	12/21/2005
JOHN DREYER	195	03/06/2002
CHRIS CONSTANTINO	193	11/13/2000

TIMED FIRE	SCORE	DATE
PAUL NOLES	200-17	12/21/2005
JOHN DREYER	200-16	10/10/2001
JOHN O'DONNELL	200-14	11/02/2009
CHRIS CONSTANTINO	200-09	02/17/2003

RAPID FIRE	SCORE	DATE
PAUL NOLES	200-18	12/07/2005
JOHN DREYER	200-15	11/14/2001
CHRIS CONSTANTINO	199-14	11/12/2001

AGGREGATE	SCORE	DATE
PAUL NOLES	597-27	12/21/2005
JOHN DREYER	586-28	03/07/2001

JOHN O'DONNELL	586-20	11/02/2009
ALVIN BOXDORFER	586-xx	??/??/1986

IL-MO LEAGUE

Scorekeeping Software

Source: http://bullseyepistol.com/imsoft.htm

Scorekeeping Software Available:	
Match Scorekeeping:	**Excel-based program used at the range for calculating handicaps and match totals.**
Team Administration:	**Excel-based program to track scores to keep indices updated and to print scoresheets.**
Handicap Calculator:	**A small DOS-based program to compute handicap scores**

IL-MO LEAGUE

Il-Mo Current Season Standings

Source: http://bullseyepistol.com/imstand.htm

If you would like the most current update of the standings delivered automatically to your email address every week, you may register for this service by entering the information requested below:

bullseyepistol/orderad
Kirk!
IL-MO League Email Standings

Full Name *

Email Address *

Register

This is a free service provided by Secretary/Treasurer, Kirk Yates

IL-MO LEAGUE

Il-Mo Current Teams

Source: http://bullseyepistol.com/imteams.htm

Piasa Legends Team John O'Donnell, Captain
314-567-1737

Piasa Pirates Team Mike Holland, Captain
314-839-4536

Missouri Gun & Quail Club Team Don Weihl, Captain
618-398-0145

Boeing Employees Rifle & Pistol Club Jon Hopson, Captain
314-535-0448

Further Reading

- John O'Donnell (mailto:johncifco@aol.com)
- Mike Holland (mailto:mpholl@att.net)
- Don Weihl (mailto:dew@greensfelder.com)
- Jon Hopson (mailto:jmhopson@sbcglobal.net)

CAMP PERRY / NATIONAL MATCHES

How to Enter the National Matches

By The Procedure for Getting Information and Registering

Source: http://bullseyepistol.com/perenter.htm

How to Enter for the First Time

If you did not attend in the previous year, you will need to submit your name to be placed on the competitor's list to have entry pack sent to you. If you DID attend last year, you will receive this pack automatically. Telephone the NRA at 1-800-NRA-3888 and ask for the "Competitions Division." Or you can contact them by E-Mail. They will be able to assist you.

The Official Program: Entry Information

This is General Entry Information taken from the 1995 edition of the Official Program of the National Matches. Some of these conditions may be changed for the subsequent National Matches. It is at the very least a preview of the current program. ALL INDIVIDUALS MUST PERSONALLY COMPLETE AND SIGN THEIR OWN ENTRY CARDS. Make certain all entries are indicated correctly. A-1 ENTRIES: a. ENTRIES LIMITED TO RANGE CAPACITY: Pistol:1152. b. The receipt, ONLY, of an entry will be acknowledged by postcard. A-2 ENTRY INSTRUCTIONS: Advance entry must be made, on the appropriate entry card, for any of the National Championship activities. Full entry fees, except for those organizations which have prior arrangements with the Statistical Director, must accompany entry cards. All entries are received on a first come, first serve basis. Fill out the entry card or cards as indicated. Here are some do's and don'ts, which may be helpful to you: a. Do print or type all information. b. Do use your name and address exactly as shown on your NRA Classification Card. c. Do indicate your NRA membership number in the space indicated. (MUST be included as membership is required for NRA Matches, and for proper posting of classification scores.) New members who have not yet received their NRA membership number should write "New Member" on the card. d. Do indicate each event you want to enter. e. Do circle your Regular Awards Category as well as any Special Awards Categories for which you are eligible. f. Do complete the Housing Requedt Card, regardless of where you will be staying. g. Do indicate your telephone numbers and area code so that we may call you if necessary. h. Do indicate your rank (SGT, CPT, etc.) and not your Pay Grade (service personnel only). Police are also required to give rank and name of department. i. Don't use a temporary address on your entry card. ROTC entries showing a military address will not be accepted. j. Don't alter an entry

card in any way. k. Don't submit a photocopied entry card. l. DON'T SEND HOUSING FEES TO CAMP PERRY. ALL HOUSING FEES WILL BE COLLECTED ON ARRIVAL. After completing the entry card, be sure to double check, and sign it. A-3 COMPETITOR ELIGIBILITY: The NRA National Championships are open to all individual NRA Members in good standing and to non-US Citizens who are members in good standing of their respective National Shooting Federations. Memberships are available at Camp Perry. A-4 COMPETITOR CLASSIFICATIONS: a. TEMPORARY CLASSIFICATIONS, RULE 19.14 WILL NOT BE USED. b. Assigned Classifications - Rule 19.6, may be used. c. All competitors who do not hold an NRA Official Classification, either in Conventional Outdoor Pistol, or an Assigned Classifcation according to Rule 19.6 will be placed in the Master Class. d. Competitors may be required to present an Official Classification Card at the Entry Ofice before picking up their Competitor Packet. A-5 COMPETITOR REGULAR AWARD CATEGORIES: a. Except as noted in specific match conditions, competitors will be divided into classes and those classes further divided into categories as follows:(Special Note: A competitor may enter only one Regular Award category. This category will be used for all matches and is seperate from the Special Awards categories used for aggregates and the National Championship.) Pistol- 1. Civilian- Competitors as defined by Pistol Rule 2.2. 2. Service- Pistol Rules 2.5, 2.6, and 2.7. 3. Police- Pistol Rule 2.4. b. Competitors may not change from one Regular Awards category to another during any one phase of the National Championships. c. In National Trophy Pistol and Rifle Matches, a Junior is any person under the age of 18. d. All Junior or collegiate competitors in the Service must compete in the Service category and pay the standard service entry fees. Juniors competing in the Service category will still be eligible for Special Awards in the aggregates and National Championships. e. Special Awards to junior or collegiate competitors are as stated in this program. A-6 SPECIAL AWARD CATEGORY ELIGIBILITY: a. Special Awards are made as specified in match conditions. In order for a competitor to be included in any Special Awards category for which they may be eligible, the appropriate Special Awards category MUST be indicated on the Entry Card. Competitors may be eligible for more than one Special Awards category. Mark all Special Awards categories for which you are eligible at the time of entry. No changes will be accepted once firing begins. b. Special Awards categories are as follows: Pistol Woman, Senior, Junior, Intermediate Junior, Sub-Junior, Collegiate, Police. c. Collegiate competitors who met all academic requirements of Rule 2.8 during the 1994-95 academic year and received a bachelors' degree in the Spring f 1995 are considered to be collegiate competitors for the 1995 NRA National Rifle and Pistol Championships. d. National Trophy Matches Special Awards Categories are indicated on the individual competitor entry and score cards. Competitors must complete score card after firing. e. NRA/USA National Civilian Pistol Team: Eligible competitors must sign the eligibility statement on their entry card if they wish to be considered for selection to the National Team. A-7 ADULT SPONSORS, JUNIOR ENTRIES: All Juniors 17 years and under must be accompanied by an adult sponsor. A-8

SUBSTITUTIONS: No substitutions will be accepted after the final closing dates. Substitutions of a new team member for another entered in the Individual Championships may be made, but a $10.00 cancellation fee will be charged. A $20.00 post entry fee will be charged for each substitution made after the applicable closing date. A "Team" is defined for this purpose as a group of shooters who are all entered together representing one organization, such as an Affiliated Club, State Association, Military Branch, Police Department, etc., and paid for as a group. A-9 POST ENTRIES: Post entries may be accepted after the closing dates for the phase entered, but nt later than June XX for Pistol, provided there is space available. Entries RECEIVED AFTER the closing dates will be charged a $25.00 Post Entry Fee. Squadding of other competitors WILL NOT be changed to accomodate post entries. All inquires concerning post entries must be made to the Statistical Director BY TELEPHONE prior to the time that the competitor leaves home for Camp Perry. A-10 CHANGES: If you have any changes to make in your entry, DO NOT WAIT UNTIL YOU GET TO CAMP PERRY. Phone the Statistical Director at Camp Perrry immediately. The Statistical Office number at Camp Perry is (419)635-4124. Collect calls cannot be accepted. Changes requested after arrival at Camp Perry will be subject to a $10.00 fee. A-11 CANCELLATIONS: In order to be considered for any refund of fees, a written cancellation is required. Telephone cancellations can start the process, but MUST be followed by a confirming letter, veryifying the cancellation. Refunds will not be made until written cancellation is received. No refunds will be made for those individuals who do not show up at Camp Perry and make no attempt to contact the Statistical Office. Mail cancellation letters to:Statistical Director, P.O. Box 700, Camp Perry, Port Clinton, OH 43452. A cancellation fee of $15.00 is assessed against any cancellation. If a cancellation is received at Camp Perry prior to June 29 for pistol, a total refund of fees, minus the $15.00 cancellation fee, will be made. No-shows will not be refunded.

Further Reading

- E-Mail (mailto:compadmin@nrahq.org?subject=Camp Perry Info Packet)

CAMP PERRY / NATIONAL MATCHES

A History of the National Matches

Source: https://www.bullseyepistol.com/perhist.htm

The National Matches are an official function of the United States Government, held by the National Board for the Promotion of Rifle Practice, in conjunction with the National Rifle Association.

U.S. Legislation that Enacted the Matches

> Title 10 United States Code Section 4312
> (a) An annual competition called the "National Matches" and consisting of rifle and pistol matches for a National Trophy, medals and other prizes shall be held as prescribed by the Secretary of the Army. (b) The National Matches are open to members of the Armed Forces, National Guard, Reserve Officers Training Corps, Air Force Reserve Officers Training Corps, Citizens' Military Training Camps, Citizen's Air Training Camps, and rifle clubs and to civilians. (c) A Small Arms Firing School shall be held in connection with the National Matches. (d) Competition for which trophies and medals are provided by the National Rifle Association of America shall be held in connection with the National Matches.

A Historical Look at National Match Ranges

By Joe B. Roberts Jr.

For practical purposes, NRA national championships of the last century and a quarter have been held at one of three ranges, but there have been exceptions.
Shortly after its founding in 1871, the NRA began work on the "Creedmoor Range." The range at Creedmoor (now a part of the Borough of Queens) was built on land owned by the NRA with financial aid from the state of New York, "pick and shovel" help from the U.S. Army Engineers at Fort Totten, and a promise by the Long Island Railroad to provide service. The range was originally intended as a marksmanship training ground for all New York militia organizations. Use of Creedmoor for NRA championships got underway in 1874 when a rifle team made up of NRA members invited the Irish National Team to travel to the United States for an international competition. From then until the late 1880s, when the NRA lost its control over Creedmoor, NRA championships were fired there. Through the 1880s and 90s, interest in marksmanship dwindled and the NRA went into a period of inactivity. The Spanish-American War in 1898 rekindled the flames. A resurgent NRA re-instituted its match program, both national and international, beginning in 1899. Having lost Creedmoor,

however, NRA moved its competitive operation to Sea Girt, New Jersey, and to ranges operated by the New Jersey Rifle and Pistol Association. NRA national championships were held at Sea Girt until 1907 when, combined with matches sponsored by the National Board for the Promotion of Rifle Practice (NBPRP)--"The National Matches"--they were moved to Camp Perry, Ohio.

The National Matches (the National Trophy Individual and Team Rifle and Pistol Matches, the President's Rifle and Pistol Matches, and the National Trophy Infantry Team Rifle Match) and NRA's championships have been at Camp Perry ever since, except:

Between 1914 and 1920, War Department training schemes and U.S. involvement in World War I combined to keep the National Matches generally out of Ohio. In 1914, the War Department, in an attempt to increase training efficiency and encourage greater participation, scheduled four sets of "Divisional Matches." These matches, held at Sea Girt, at the state military camp near Jacksonville, Fla., at Fort Riley, Kan., and near Portland, Ore., were disastrous in terms of attendance. Jacksonville and Sea Girt drew good crowds but the others did not.

In 1915, the National Matches were scheduled for firing at Florida's Black Point Military Reservation, eight miles outside Jacksonville. Jacksonville hosted the 1916 National Matches, too.

No matches were held in 1917, but Camp Perry did host the contests for 1918, and contemporary reports note that attendance was extraordinarily high.

In 1919, the U.S. Navy offered its newly constructed ranges at Caldwell, N.J., for use as a site for the National Matches and for NRA's championships. In 1920, and from that year until World War II interrupted the schedule, Camp Perry was the location for the National Matches. After World War II, the scenario was much like that of the 1914-1920 era.

Between 1941, when smallbore and pistol championships (but no high power championships and no National Matches) were held at Camp Perry, and 1953, when the full National Match schedule returned to the Ohio site, NRA held championships in a variety of locations. NRA smallbore and pistol championships were shot at Camp Perry in 1946 and 1947, and at the Marine Corps base at Quantico, Va., in 1948. In 1949, smallbore matches were held at Camp Dodge, outside Des Moines, Iowa, and the pistol championships were fired at Fort Sheridan, Ill. There were no matches during 1950 the Korean Conflict interfered.

In 1951, smallbore and pistol championships were held near San Francisco and the first high power rifle championships since 1940 were fired at Camp Matthews, north of San Diego. In 1952, pistol and smallbore matches were shot in Jacksonville, Fla., and a high power championship was conducted at Fort Benning, Ga.

In 1953, the National Matches returned to Camp Perry, Ohio. Since then, though the long range matches were shot elsewhere immediately following the cut-off of funds in 1968, NRA's National Championships and the National Matches have been held at Camp Perry.

CAMP PERRY / NATIONAL MATCHES

Lodging, Dining and Entertainment

By A Multitude of Places to Stay and Go at Camp Perry

Source: https://www.bullseyepistol.com/perinfo.htm

Where is Camp Perry?

U.S. Army National Guard Camp Perry is located in North Ohio not far from the beaches of Sandusky Bay of Lake Erie just west of Port Clinton. As you may have guessed, there are plenty of activities and attractions to keep you and your family busy when you are not on the range. In addition to the map and other information included in your NRA competitor packet, the resources below can help you plan your important week.

Housing for the National Matches:

Camp Perry Lodging and Conference Center, Website
For booking huts and other housing on base.(888) 889-7010.

The White Caps Motel, Website
2186 W. Lakeshore Dr.Port Clinton, OH 43452(419) 734-3816

Important Contacts in Port Clinton:

Ottawa County Visitor's Bureau, Website
127 W. Perry Port Clinton, Ohio 43452 (419) 898-0479

Port Clinton Online Travel Guide, Website
Lodging, Restaurants and more on this information site.

Port Clinton Chamber of Commerce, Website
130 Jefferson Street Port Clinton, OH 43452 (419) 735-5503

Put-in-Bay Island Guide Online, Website
(Don't miss the island, it is great!)Put-in-Bay, Ohio 43456

Some Personal Dining Recommendations:

Mon Ami Winery and Restaurant, Website
3845 E. Wine Cellar RdPort Clinton, Ohio 43452(419) 797-4445

Jolly Roger Seafood House, Website
1737 E. Perry St.Port Clinton, Ohio 43452(419) 732-3382

Updated 10/5/2011

Further Reading

- Website (http://www.cplcc.com/)
- Website (http://thewhitecapsmotel.com/)
- Website (http://www.lake-erie.com/)
- Website (http://www.portclinton.org/)
- Website (http://www.portclinton.com/)
- Website (http://www.put-in-bay.com/)
- Website (http://www.monamiwinery.com/)
- Website (http://www.facebook.com/pages/Jolly-Roger-Seafood-House-Port-Clinton-Ohio/87689094430)

CAMP PERRY / NATIONAL MATCHES

Memories of Camp Perry

By Recollections of Past Experiences at Camp Perry

Source: http://bullseyepistol.com/permem.htm

I heard about Camp Perry during my first year of shooting when I asked someone at our range, "Where had everyone gone?" It was the second Week in July 1985.
George A. Andrews, Philadelphia, Pa.
It was my first year there in 1985. The first relay the cannon goes off and the national anthem played on and it was a tear-jerker. I'll never forget it. *Thomas Drumsta, Downers Grove, Ill.*
My favorite memory was when I won the .22 rapid-fire in 1978 with 200-14x. *Harvey LaJiness Jr., Ray City, Mich.*
Meeting and talking to some of the best shooters in America. *C. Sobers, Red Hook, N.Y.*
Some of my favorite memories are of competing in the team matches and looking down the firing line and seeing all the flags of the different clubs and state associations. *Steven Mrelch, Stroudsburg Pa.*
A lasting: memory...having to shoot .22 slow fire, by myself, in front of 50 teams, coaches and friends (400 people) after electing to refire the string due to a crossfire. *Harvie Loomis, Newark Valley, N.Y.*
There were representatives of each branch of service on every single corner of the camp directing traffic. *Mel Treu, St. Louis, Mo.*
I first heard about Camp Perry in 1964, but haven't competed since 1967...but I am starting over. *Daniel B. Wyatt, Winston-Salem, N.C.*
Camp Perry is a vacation site where you can socialize and compete with the best shooters in the country. *Ronald P. Morency, Glendale, Ariz.*
Camp Perry is a place where you need to be prepared for any kind of weather. Extreme heat, subtle chill, high winds, torrential rain, complete calmness, and extreme pleasure can be found there. *Fred Vanderslice, Virginia Beach, Va.*
One year I had functioning problems with my pistol. Ron Zalewski, also a competitor, said "use mine" and gave me his pistol and ammunition so I could compete in the .45 stage. My total score that year was high enough to earn a Place on the NRA/USA National Civilian Pistol Team. I still wear the red jacket...but I call it "Ron's jacket." *T.C. Lewis, Chelsea, Mich.*

CAMP PERRY / NATIONAL MATCHES

Camp Perry Frequently Asked Questions

By Justin Nystrom

Source: http://bullseyepistol.com/perryfaq.htm

Courtesy of Justin Nystrom of the River Bend Gun Club in Dawson County Georgia, here are answers to the most common questions asked by those interested in attending the National Pistol Matches at Camp Perry.

River Bend Gun Club

Before reading anything else, if you are REMOTELY interested in attending the National Matches at Camp Perry, visit this Webpage to obtain entry information.

Why Attend the National Matches at Camp Perry?

The reasons are many: Great camaraderie with your fellow shooter, taking part in a grand tradition, visiting commercial row, Meeting new and interesting people with like interests and values to name just a few.

Where Exactly is Camp Perry?

Camp Perry, geographically speaking is located on the shores of Sandusky Bay, Lake Erie - near the town of Port Clinton, Ohio. Consult your Rand McNally road atlas. (After all, you are a resourceful bullseye shooter!)

When Does the Pistol Phase Take Place?



The Pistol phase is always in the second full week of July. The first two days of Camp Perry are the Small Arms Firing School. This is a quality pistol instruction course given by the US Army Marksmanship Unit of Ft. Benning, GA. It is not required that you attend this part but it is a good, informative class and includes some live fire training. Wednesday will be the preliminary matches. Thursday will be the .22 match, Friday the CF match and Saturday the .45 match. On Sunday are the National Trophy Pistol Matches - (LEG Matches).

Where Do I Stay?

If you don't have hotel reservations already, you are staying on the grounds of Camp Perry. In this respect there are two options: Economy, i.e. the "hut" and Civilized, a module. There are plusses and minuses for each type of accomodation. Huts have slightly fewer creature comforts than a Turkish prison. But, they are only $10 a bunk. As a pistol shooter, you have the honor of being the first person to use the hut in 10 months. Therefore, bring your Pine-Sol, bug bomb, bucket and broom. I stayed in a hut during my first trip to Perry during the great heat wave of 1995. I was poorly prepared for such an experience. Be sure to bring a cooler, box fan, a radio, etc. A piece of carpet and a fridge are a nice touch. I was pared up with three great fellas from Indiana who brought a fridge and a seemingly endless supply of beer (only after shooting, of course!). This helped with the heat if not my scores. Last year's weather was beautiful and I'm sure staying in a hut was just fine. The modules are $22 per person and are air-conditioned and have bathrooms with showers. If you suffer from any heat-related physical ailment - request a module. Modules are not easy to get and are first-come first-served.

How Much Will I Spend?

More than you think possible.

What Is The Weather Like?

It ranges from Hell to Heaven. Expect torrential rain, blistering sun, unseasonably chilly weather, and above all WIND. Don't, whatever you do forget sun block or a rain suit. Also, bring some sort of plastic covering for your gunbox opening. All shooting is done out in the open (no cover).

What Are Some Things To Do There?

You will, no doubt, find things to entertain yourself. Here are a few highlights you may wish to take in:

- Bring your air pistol. There is a semi-indoor airgun range available most of the time at Perry. Shoot in a 60 shot match.
- The lakefront is very nice. Go to Port Clinton. You may wish to go on a charter fishing trip if you are so inclined.
- There is a swap meet at Perry on one of the evenings. Bring any bullseye stuff you want to sell.
- 

  Bring lots of mad money. Commercial Row is a collection of major vendors and gun manufacturers. Clark is always there, so is Champion's Choice among others. REMEMBER: your NRA points are like cash at Perry - don't leave home without them.

- See the NRA trophies, experience the history.
- Visit the Chapel on base. A very peaceful place for reflection, if you are so inclined.
- Bring a bicycle if you have the room. If you are in any fitness at all, it is a great way to get around the flat landscape of the base while not shooting.

References

- geographically
- hotel reservations

Further Reading

- http://www.rbgc.org/ (http://www.rbgc.org/)
- Webpage (http://www.nrahq.org/compete/nm_campperry.asp)

CAMP PERRY / NATIONAL MATCHES

Camp Perry Survival Guide

By Mike LaVoie

Source: https://www.bullseyepistol.com/perrysrv.htm

Having been to Camp Perry a few times myself, I have learned what you need to survive Perry the hard way - from experience! Save yourself the trouble; study Mike LaVoie's excellent observations. You'll be glad you did. -JD

CAMP PERRY NOTES:

- Have your match confirmation in case the police stop you.
- Lock your guns separate from your ammo.(Which must be locked when you drive.)
- Remember to unlock your gunbox before you go to the range.
- Wear light colored clothes, as it can be very hot.
- Bring water, as it is can be hot.
- Bring a sweatshirt, as it may be cold.
- Rain suit and rubbers as it may be raining.
- A C-clamp to hold your gunbox open, as it may be windy.
- A score sheet holder- an enclosed one is nice when it is raining.
- A grease pencil- this is the only pencil that writes when the paper is wet.
- Golf umbrella
- A cart to carry everything. EB Marine in Fairfield sells them at about $43. They fold up, carry a lot and they don't fall over.
- Some people bring a bungee cord to keep the gunbox on the bench during heavy winds.
- Only give the competitor who is scoring you the score sheets for that day, as they may not be there the next day.
- You should also have:
-
    1. Screwdriver
    2. Allen wrenches
    3. Cleaning rod
    4. Shooti



ng diary, pen, pencil

5. Extra batteries for red dot scopes
6. Scoring overlays
7. Scoring plugs
8. Rule book
9. Shooting glasses (indoor and outdoor)
10. Earmuffs (I use earplugs and earmuffs, called double plugging)
11. Magazines
12. One inch surgical tape
13. Garbage bag (When it rains, you place the bag over your box)
14. Wooden or brass rod. (In case the bullet takes a nap in your barrel. Push the bullet out the way it came in.)
15. Spare parts

- I take it back. You don't need a gun box. You need a pickup truck to carry all this stuff around.
- Did I remember AMMO?

NON-SHOOTING EQUIPMENT:

- 

  Clothing should be loose fitting and comfortable. You might need rain gear due to the weather that includes rubbers, the kind that go on your feet. No openings in the shirt for hot brass to enter, which really hurts.
- Shoes should be comfortable and should have flat soles. Never wear new shoes until they are broken in. Never wear sandals, fire ants love sandals.
- Bring: Bug juice, Suntan lotion, Earplugs, Scope, Tools to adjust sights and tighten scopes. Garbage bag to cover box in case of rain, cut out front and rear of bag so you can see through scope. The garbage bag can also make a poncho for you if necessary.
- Bring a towel for the first relay at Camp Perry. Walk to the benches and wipe the morning dew off. It's a nice thing to wipe off the entire bench. Do both the 50 yard and 25 yard bench. The range officers let you. By the time you get to the 25 yard line it usually will be dry.
- 1st relay is near the canon. Wear your earplugs. Prepare for colors. That means the raising of the flag hand over the heart or hat over your heart.
- Wear a cap to keep the sun, rain and brass off.
- Bring a metal screen to place between the shooter on the left and you as their fired cases may hit you. Check the rulebook for the size diameter of the screen. There isn't a rule about how big the screen is, only the size of the holes in it. If the screen is too big they may say something. Most screens are about 2 ft by 2.5 ft. Some people staple the screens to their gunbox. Others use a clamp. I use a spring-loaded clamp.
- Cee Cee makes great earplugs at Camp Perry. I love mine and have had them for years.
- Oh yeah, the most important thing to remember at Camp Perry is to have fun, tell a lot of jokes, make new friends, renew old friendships and bring enough extra money to buy me a beer.

CAMP PERRY / NATIONAL MATCHES

Camp Perry Trivia

By Virgil M. Gordon

Source: https://www.bullseyepistol.com/pertriv.htm



As one enters Camp Perry through the main entrance off Ohio Route 2 you will notice the 2 stone towers on either side of the gate. These are not as old as they look. Both were built as a WPA project in 1937 after the State of Ohio had purchased all or parts of 6 farms as our addition to the military reservation. Both towers were of field-stone construction and had flat roofs. The red brick capalos at the top was added during WWII.

The main entrance road is Niagara Road, named in honor of Commodore Perry's second and winning flagship in the Battle of Lake Erie. When the State Legislature authorized purchase of land for the camp in 1906, they specified it should be named for Oliver Hazard Perry, who won control of Lake Erie by capture of the entire British fleet on September 10, 1813. Legislation may or may not have called for naming roads within the camp to be named for Perry's ship. At present time there are roads named Niagara, Lawrence (Perry's first flagship), Caledonia (captured from British in eastern Lake Erie the previous year), Arid, Scorpioro, Sommers, and Trippe. However, several former Ohio Governors have roads named for them as well. These are: Cox, Davey, Davis, Cooper, Donahey and Harrison Roads. There are at least 4 roads named for men who had long and important connections with the camp. These are: Clem Road, Critchfield Road (this was first called Critchfield Circle), and is a circular one way road; St. Clair Road; and Cartright Trail. Major General James Clem, who was Ohio's Adjutant General, did much to preserve and restore the reservation, which had become badly run down. One of his achievements was the rebuilding of the pier which had been nearly destroyed by Lake Erie's gales. Adjutant General Ammon B. Critchfield founded the camp and was in charge of the early construction of ranges, roads, and buildings. Warrant Officer

Dennis St. Clair served with the permanent personnel of Camp Perry for many years, and his efforts brought about construction and renovation of many buildings and the memorial plaza. Several former members of the 37th Infantry Division were honored by leaving the ranges and the larger permanent buildings named for them. These men were winners of the Medal of Honor during WWI and WWII. The first structure erected after purchase of the original camp may have been the original pier. The roads in 1906 were usually only of dirt or gravel, and usually were impassable, except during dry weather. The lumber, stone, etc. needed for construction of the first buildings were brought from Port Clinton by boat. Another early structure was a large gambrel roofed barn, constructed east of Niagara Road and north of present day Caledonia Road.This was needed at once to house the large number of horses and mules and feed for them. Some of the existing farm houses were used to house many of the construction workers engaged in building of the camp. The barn was razed during WWII. The Mess Hall, old QM Warehouse, and the Baggage Station were the first new buildings put to use for the first National Matches in 1907. The Mess Hall and the Warehouse were both built of precast concrete, which was a new process for building construction at that time. The Post Theater (building No. 2) has had 3 names since its construction in the late 1930s. Since Ohio's Governor then was Martin L. Davey it was named Davey Auditorium. During WWlI it became Army Auditorium.



When the camp reverted to state control after that war it was named Hough Auditorium. Benson Hough had been the Regimented Commander of the 166th Infantry of the Rainbow Division of WWI. He later served as Commander of the 37th Division between the World Wars and as Ohio's Adjutant General.

The small white huts scattered over much of the camp were built during WWII as temporary quarters. Most housed Italian and German POWs. After WWII and again after the Korean War the huts became quarters for units training or conducting firing practice at Perry. At that time they were still covered with black tar paper of the WWII era. Many have been replaced by barracks and modular housing, but most still remain.

RON PORTER FUNDAMENTALS

Fundamentals Series: Stance

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron1.htm

This is the first in a series of articles in which I will attempt to explain to the beginning shooter the application of the fundamentals which make for a good shot or series of shots in conventional bullseye.

STANCE

The main goal of a proper stance is to provide stable and consistent support for the entire body so that it can hold the gun as steady as possible as well as recover from recoil quickly with little movement. The foot positions and body twist should be duplicated naturally, with little effort. It should be comfortable and give a repeatable natural point of aim.

Finding the stance that works for you involves experimentation on your part. Everyone is biomechanically different, so one stance does not fit all. But there are some basics to adhere to as mentioned previously: comfortable, sense of equilibrium, no muscle tension in the legs or torso, arm raises naturally toward the target and 45 recoil is easily handled with no loss of balance.

To find the position that works for you, start by turning somewhere between 45-90 from the target. Keep feet about shoulder width apart. Knees should not be stiffly locked. Back and neck should be straight, shoulders relaxed. Head upright. Your non-shooting arm must be anchored to prevent body sway in recoil...most place their hand in their pocket, but hooking your thumb on you belt loop in front works even better for balance. Their are three twisting points to align you with the target. Use all three instead of just one or two. The three twist zones are hips-torso, shoulder and neck.

The shooting arm must be extended fully with elbow and wrist locked in position (but not with too much tension). To counterbalance the weight of the gun in your extended hand, some backward sway may be necessary so as to keep your center of gravity in line with the center of your body mass. Keep your head level with the sights; your face should be turned as far as possible toward the target without undue strain on the neck muscles.

To find your natural point of aim, close your eyes and assume the positions described above....open them and see where your sights are aimed. Shift your feet to align the sights on the target bull. Now lower your arm, close your eyes and repeat the process until your are properly aligned. Simulate recoil with your arm and see where it falls back naturally. Adjust

again if necessary. Remember, as tension free as possible. You should be able to stand in this position for a long time without any noticeable fatigue or movement.

After a period of time shooting, you may notice that your foot positions may have altered somewhat; this may be your tired body adjusting to find a more comfortable position. It's okay to accept this comfort adjustment. Check your alignment again and if it is off a bit, just use your rear foot to fine tune your point of impact by leaving the heel of the foot stationary and only making very slight (like and inch or two) toe movements left or right.

The stance forms the foundation for the shooter to hold steady and to build on the other fundamentals. All else remaining equal, an inconsistent stance will change your point of impact on the target. Like some other fundamentals, finding the stance that works for you is not given enough attention. But once found and habitualized, a proper stance becomes natural and automatic and leaves you free to concentrate on other things.

Ron Steinbrecher

RON PORTER FUNDAMENTALS

Fundamentals Series: Grip

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron2.htm

This is the second in a series in the explanation of shooting fundamentals.

GRIP

Gripping the pistol correctly and consistently is essential for shooting groups. Orthopaedic grips fitted to your hand will aid in the consistency necessary every time you grasp the pistol. That is your main objective: to grasp the same way every time. The gun should feel comfortable in your hand. The trigger finger should be able to position itself on the trigger so that it can be able to press it firmly straight back without rubbing against the side of the grip.

Here is a step by step method to grasp the pistol:

1. Pick up the pistol with your non-shooting hand and place it firmly in your shooting hand.

2. The pistol is seated high and deep into the web of the hand (that fleshy area between the base of the thumb and index finger). You are trying to make the pistol barrel a natural straight line extension of your arm. If a string traveled from the muzzle to your eye, it would follow a line along the inner portion of your arm. The mainspring housing (or that area of the grips) should catch the meaty portion of your palm below the base of the thumb.

3. The thumb itself is loose and relaxed along the thumb rest (if you have one).

4. The primary gripping pressure is produced by the middle finger.

5. Secondary pressure is provided by the third finger. Also, this finger can produce tiny changes in front sight elevation by increasing/decreasing pressure.

6. The little finger is loose and not exerting any noticeable pressure. Essentially, it is just along for the ride. Be careful...this finger has a lot of leverage, pressure while shooting will bring the muzzle down.

7. The fingertips do not exert pressure on the pistol grip. They are part of the total "package", along with the thumb and little finger, that form the entire grip and help to control the gun in recoil, but are not consciously applying pressure as do the middle and third fingers. If you "let go" with the thumb, little finger, trigger finger and finger tips..the gun should still be held firmly by the middle sections of your middle and third fingers pressing straight back into the lower palm of your hand.

8. The trigger finger must not be in contact with the grip at all.

The trigger finger must be able to flex at the middle knuckle without moving the pistol or varying the pressure of any of the other fingers. This is important and demands a lot of practice. You must be able to pull the trigger without moving the other fingers. The hand is not meant to work this way, so you must train to overcome this tendency.

Total grip pressure is firm but not to the point where your hand starts to shake. Keep the wrist locked. A firm grip and locked wrist have much less play from left to right. If your finger nails are totally white, you are using too much pressure. If you can see a deep impression of the grips in the flesh of your hand, you are overgripping, back off a little (except maybe in hardball rapid fire). Try to keep a consistency in the amount of pressure you use to grip the gun throughout the entire shot process. Inconsistency will definitely change your point of impact.

Spend time working on your application of a good grip. Learn how to easily and consistently acquire it...like putting on a well worn glove. If the grip doesn't quite feel right, start over. Some shooters put memory markers on the grips to let them know that they are holding correctly.

If you feel you lack grip strength, do some exercises with a hard rubber ball. Dumbbell wrist curl exercises working both the inner and top of the forearm will strengthen the finger flexor tendons. Take a single sheet of a newspaper and using just your fingers, wad it up into a tight ball. Make sure you stretch your "tennis elbow" ligament.

Between shots in slow fire, relax your grip (without letting go) to let blood rejuvenate the hand and fingers.

Let me end by repeating:

1. A firm grip gives you a sense of controlling recoil.

2. The repeatability of the way you grasp the gun will give you tighter groups.

3. The trigger finger and nothing else tightens to squeeze the trigger.

Ron Steinbrecher

RON PORTER FUNDAMENTALS

Fundamentals Series: Breath Control

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron3.htm

This is my third article about the different fundamentals needed to become proficient in bullseye.

BREATH CONTROL

To shoot well one must become motionless. Breathing causes motion, so it must stop....but in doing so you are robbing your body of the oxygen it needs to perform well. Lack of oxygen causes muscle fatigue and loss of vision (in fact, the eyes are the first thing in your body to go). Fortunately, these effects are not felt immediately because we have a reserve of oxygen in our system to sustain us for brief periods, and the more aerobically fit you are, the longer you can last.

To insure that you have enough oxygen in your system to last for the 10 to 20 seconds needed for a slow fire shot or timed fire string you need breath control.

Taking two deep breaths (inhale slowly nose, exhale quickly mouth) before a shot or string does two things....lowers your carbon dioxide blood level (which tells the brain when to breathe), thus reducing our urge to breathe and also gives you sufficient oxygen to avoid eye and muscle fatigue.

Then take a third normal breath as you bring your gun on target and exhale half the air to leave your lungs in a "neutral", comfortable state (too much or too little air in the lungs will cause an uneasiness to inhale or exhale to relieve the pressure). Some listers have proposed having a little more air than you need and slowly letting it out as you shoot. Try it and see if you like that method. But remember, your chest must not move.

Since your body is accustomed to pauses between breaths you can stay motionless in this state until you begin to run out of oxygen. Hopefully, you will have finished shooting before that happens. If you run out of air in slowfire, put the gun down and start all over again since you are definitely holding too long and your eyesight is losing focus and your wobble area is increasing. In sustained fire, you can take a quick inhalation of air during recoil to revive yourself (the effect is almost instantaneous for your eyes and muscles).

If you cannot hold your breath for timed fire, even after taking two deep breaths prior, you are not aerobically fit and need to improve your aerobic capacity by walking or jogging exercises. If you are a smoker...well, duh!

This is the easiest fundamental to master since your body "knows" involuntarily how to do it. Just help the process along by remembering to oxygenate first and then leave the lungs and diaphram in a relaxed state while shooting.

I know it seems as if I went into a too long explanation for what amounts to just "take a couple of quick breaths and hold it", but understanding the mechanics of "why" is, in my opinion, a reinforcement in understanding how important subtle things are when you are trying to something as difficult as precision shooting. Plus, I like to "hear myself talk".

Ron Steinbrecher

RON PORTER FUNDAMENTALS

Fundamentals Series: Aiming & Sight Alignment

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron4.htm

This is fourth in a series on fundamentals for beginners,

AIMING AND SIGHT ALIGNMENT

There are different techniques to aiming and sight alignment depending on whether you are using iron sights or red dot scopes.

AIMING involves correct positioning of the sights in relation to the aiming mark on the target. Since the eye very easily centers concentric circles...with dots the aiming mark is the center of the bullseye. It is as simple as that. With iron sights, however, aiming presents three common options. Center and six o'clock holds plus sub-six area aiming. All three are valid aiming marks and the type of shooting and targets used determines which is best. For the less skilled shooters, center hold may be the best option for sustained fire since it allows the brain to accept more readily what it sees. Six o'clock and sub-six aiming are more suitable for slow fire. Sub-six is sometimes referred to as "area aiming" As in center hold it allows the brain to have a more acceptable area in which to pull the trigger. Six o'clock hold is a very precise point and trying to hold it before squeezing the trigger causes hesitation and can lead to poor trigger manipulation by novice shooters.

Make sure you are using your dominate eye for aiming. To find out which eye is dominate, extend your arms, palms forward, then bring your hands together making an opening with your thumbs at the bottom. Center an object in the opening, then close one eye, then the other. The eye that kept the object centered in the opening is your dominate eye. There has been a lot of discussion on the list about cross dominate shooters...refer to the archives for that discussion.

SIGHT ALIGNMENT is one of the most critical of the fundamentals. Minor alignment mistakes make for major variations in you point of bullet impact. For dots, alignment consists of keeping the dot in the center of the scope lens and the bullseye. Again, it is as simple as that. Unfortunately, no one can keep the dot perfectly still and centered, especially while squeezing a trigger.

Iron sights are in perfect alignment when the front sight blade is centered in the rear sight notch and the tops of both are on level plane with each other. Since the eye can only focus clearly on one object at a time, your focus must be on the front sight blade...for it is only here that you can see misalignment....the brain determines if there is equal light on either side of

the front blade and the top of the blade is level with the rear notch opening. You CANNOT do this if your eye focus is on the target or the rear notch. Both of them will be seen in relation to the front sight blade, but slightly fuzzy, that's okay.

The combination of sight alignment and aiming point on the target forms what is known as the "sight picture". Maintaining the sight picture while squeezing the trigger is the secret of successful shooting. Let me write that again...for it is the holy grail of successful pistol shooting. Squeezing the trigger straight back so as to not cause any misalignment or excessive movement will result in a good shot. Everytime. Guaranteed. Unless, of course, your gun and ammo suck.

A couple of clarifications. Don't confuse movement within your aiming area with movement which causes misalignment. If you maintain perfect alignment (no wobble of the dot from center, the front blade centered and level) you can still have side to side and up and down movement of your aiming area as much as the size of the ten ring and still shoot a ten. Jim Lenardson describes it thusly: imagine a pipe with a 3" inside diameter (the size of the ten ring at 50 yds) running perfectly straight from your muzzle to the target. AS LONG AS YOU MAINTAIN PERFECT ALIGNMENT, you can have a 3" movement by the combination of your arm and body sway and still hit the ten ring. But with a very slight misalignment, the path of the bullet is altered and will it hit the side of the pipe before it travels 50 yds.

So you can have three kinds of movement: wobble (misalignment), arm sway and body sway. All three contribute to the point of impact of the bullet, but the wobble is the one that does the most damage toward a good shot.

Ron Steinbrecher

RON PORTER FUNDAMENTALS

Fundamentals Series: Trigger Control

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron5.htm

This is the fifth of my postings on the fundamentals for novice shooters.

TRIGGER CONTROL

The controlled release of the trigger lies at the heart of performing a good shot on target. All of the other fundamentals can be perfectly performed but they mean nothing if the trigger release is faulty, because it will spoil the aim and sight alignment at the crucial moment when the shot is fired. Thus, trigger control is an equally critical fundamental with sight alignment.

Your trigger finger must apply pressure to the trigger in a direct line back toward the aiming eye. For this, the finger must be correctly positioned, with the center of the pad of the first joint (or whatever works for you) touching the trigger. The trigger finger must be clear of the frame and grips so as not to move the pistol as the pad is pulled straight back. ONLY THE TRIGGER FINGER MOVES. Grip pressure of the other fingers must remain static (see the post on "Grip").

Trigger finger pressure must be constant, smooth, and progressive. Once you start applying pressure, continue until the gun fires... do not "stop-start" the trigger for usually only a very skilled shooter can do that successfully (because each stop-start requires the breaking of the trigger's inertia and that causes movement). The "keep the trigger moving" mantra for rapid fire applies to all trigger pulls.

In slow fire, the timing of the hammer fall should practiced so that the release of the shot coincides with the steadiest part of your hold, which is approximately 6-10 seconds after you start the aiming process. In sustained fire, the hammer fall coincides with returning to an acceptable sight picture.

Keep your finger pad centered on the trigger. Too high decreases leverage and the pull feels harder, too low and the shot may fire before you are ready. You must be consistent in your trigger finger placement. Don't be afraid to experiment in training with different placements till you find the one that gives you the steadiest straight back pull.

Trust in your ability to hold and minimize wobble and the eye's ability to center the sights to form a sight picture and pull the trigger smoothly and continuously without trying to "snatch" a shot or jerk the trigger. You can start out with a slow release and with experience and lots of dry fire trigger time, you can increase your trigger speed without additional movement.

With two stage triggers, take up the first stage completely and add slight prepressure to the second stage (the amount of which depends on your "trigger sense" acquired with experience) just before settling in on your aim and alignment. This works well with slow fire. For sustained fire, take up the first stage in recoil recovery and keep the trigger moving. Again, do not "stop-start" as it will destroy your rhythm and cadence (the only exception is if you recover badly and do not have a sight picture which will give you a shot in the bull).

Work toward unconsciously activating the trigger, such that the shot becomes a surprise break to you. This will help in avoiding flinching or anticipated recoil. This is difficult to master, but with practice and patience, it will be achieved. When trigger pull becomes automatic, your brain is free to focus on the other critical fundamental....sight alignment. Just as your eye cannot focus on two objects simultaneously, you brain cannot focus on two separate fundamentals at the same time. All the fundamentals should be done on auto-pilot, but if you have to focus on one, then let it be sight alignment when shooting with iron sights and when you are using a scope, you can let your focus be on trigger pull (since a sight picture is so easily done with a scope).

One of the best ways to develop trigger control, speed and a feel for your particular gun's trigger is to dry fire. I know it is boring compared to live fire, but if you have long periods between matches, it can substitute as practice.

Ron Steinbrecher

RON PORTER FUNDAMENTALS

Fundamentals Series: Follow-Through

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron6.htm

FOLLOW THROUGH

Follow through is an often neglected fundamental; yet, without it, all your good efforts in applying the other fundamentals may be wasted.

Releasing the shot should not be your final effort. You need to do one more thing. It is argued (especially on this list) that at the moment of ignition, the bullet has left the barrel before you could possibly do anything to affect its flight. I believe that also for the simple reason that the nervous system of your body reacts in tenths of a second and the gun fires in thousands of a second. It's no contest. But the problem lies in doing something BEFORE the moment of ignition. Without follow through, a bad habit may develop which is initiated just at or before ignition in which case, the flight of the bullet will be affected by your action or movement.

In all sports the achievement of a quality performance mandates follow through. The pitched baseball, the sprinter keeping form when crossing the finish line, the martial artist breaking a board, the golf shot.....the list of examples is long. What we do is no exception. There can be no let up of form and technique until it is absolutely certain that the act is completed. The danger is in relaxing your grip too early, losing your concentration on the sights, or beginning recoil before it has actually started; all of which will produce a bad shot. To practice follow through, imagine that the bullet is like a rocket leaving the barrel a few seconds after ignition. Try to maintain your form as long as possible to allow the rocket to leave the barrel, keep your focus on the sights or dot and try to call the shot. If you have trouble calling your shot(s)....you are not following through.

This technique of using follow through as a SEAMLESS, CONTINUING FLOW OF ALL THE FUNDAMENTALS will reduce the chance of giving up on the shot before it is fired.

Well, that's it. I hope you will or have been able to improve your score by the application and practice of these fundamentals. I've tried to explain them as best I could. I've already gotten some feedback from one individual who has improved his shooting. That's great.

I encourage any who have questions to direct them to the entire Bullseye-L forum, not just to me, so that all can share in the discussion. That is the very reason Paul Perrone created the forum.

Good Shooting, Ron Steinbrecher.

RON PORTER FUNDAMENTALS

Fundamentals Series: Red Dot Scopes

By Ron Steinbrecher

Source: http://bullseyepistol.com/ron7.htm

As a follow up to my articles on fundamentals, I like to discuss a few other topics.

RED DOT SCOPES

Most participants in matches use the red dot scope. Not that there is anything wrong with using iron sights, it is just that the dot scopes are easier. As our eyes weaken with age, the dot scope allows us to continue to compete at a level comparable with those of sharp vision.

With iron sights, three elements must be aligned to form a sight picture...the target, front sight blade and the rear sight notch. Since our eyes can only focus sharply on one thing at a time, our focus is thus on the front sight blade to ensure perfect alignment.

Red dots, however, are on the same focal plane as the target (infinity) and thus the "eye" has only to look at two things...the target and the dot and they are both in focus when you aim. That is quite an advantage! It allows for faster target acquisition and alignment and because the brain easily aligns concentric circles (the scope lens, the dot and the bullseye), you can concentrate more on trigger squeeze and let the aiming process become more automatic.

Low light levels (relative to the outdoors) at indoor ranges can be a problem with iron sights, but not with dots. Most dots have several brightness intensities to take care of the dimmest indoor ranges to the brightest outdoor sun.

Some things to remember when using scopes: Parallax is a factor even with the best dots (my choice is the Ultradot) so it matters that the dot is centered in the scope when aiming especially when shooting at 50 yds. You can check the amount of parallax your scope has by keeping it rested on the bench and held stationary with the dot centered on the bull. Now moving only your head slightly side to side and up and down, you can see how much the dot moves on the target when it is in different positions in the lens. The smaller the movement= the lesser the parallax= the better the scope. Keeping the dot centered in the lens avoids misaiming.

Always carry a spare battery. Dots will weaken gradually with silver oxide batteries and will fail very quickly with lithium batteries. The batteries last a long time if you can remember to turn off the scope when finished.

Ultradots (and some other brands) offers different dot and lens sizes. The choice is just a matter of personal preference and the style of shooting that you are doing.

The scope and mount will add 6oz. or more to the gun and if you are slide mounting, you may need to increase your powder charge slightly or change your recoil spring to give you reliable functioning.

I'll go out on a limb and say that dot scopes will probably add 10% to your iron sight scores especially if you have older eyes. They take some getting use it, as all changes do, and your scores may initially drop somewhat, but eventually you will be glad you switched. There are those who believe that all shooters should start out with iron sights when learning the fundamentals before transitioning to dots, I never really understood this, except of course if becoming Distinguished is important to you.

Well, that's it for now. When I think of another topic that may be of interest to new shooters, I'll post again.

Ron Steinbrecher

RON PORTER FUNDAMENTALS

About Ron Steinbrecher

By Ron Steinbrecher

Source: http://bullseyepistol.com/ronbio.htm



Ron Steinbrecher is a HIGH MASTER pistol shooter from the great state of Pennsylvania. Apart from holding several individual National Records in conventional pistol, he has set records and won National Titles with the UltraDot team. Ron is one of the best all-round pistol shooters in the entire country and one of the nicest fellows as well.
With the intention of helping new shooters in the sport, Ron wrote a series of articles for his local club newsletter on pistol shooting fundamentals. He graciously took the time to retype and email each of them to the Bullseye-L forum, where they eventually became archived on Ed Hall's website. *Ron contacted me by email to see if I would be interested in including his writings on the Bullseyepistol.com website. After reading his email, the articles were uploaded to this site within an hour! Thank you for the fine work, Ron!*

DAVE SALYER ARTICLES

An Explanation of Area Aiming

By Dave Salyer

Source: http://bullseyepistol.com/salyer2.htm

Question: I am a relatively new pistol shooter who reads everything I can on the art of pistol competition. Please explain the term "area aiming" in ordinary, every day terms to help me get a better grasp of its meaning.

Answer: Area aiming is a term used by better pistol shots to describe an important part of the process of firing a well aimed shot, with confidence that it will end up in or near the center of the bullseye. Aiming area can be described as the area projected on the target by the front sight or scope dot as viewed by the shooter. For example, you point an electronic red dot sight at a target down range and make a mental note of the extreme movements of the dot. Maybe it is moving back and forth and up and down inside an approximate circle the size of the 7-ring. The area of this observed imaginary circle would be your personal aiming area capability on this day.

The 10-ring is so small it is nearly impossible for any shooter to hold his front sight, or red dot inside an area the size of that circle for an entire competitive event. So, the shooter has two choices. He/she can try to fire the pistol as the sight moves into the 10-ring, or place the center of his personal aiming area over the 10-ring and carefully apply pressure to the trigger until the shot leaves the pistol. Area aiming is the more logical choice for just about all pistol shooters, but this requires some explanation. It is sad that so much has been written about the six-o'clock hold, which defines an infinitely small spot. Sounds great but is not practical for off-hand shooting. The reason is that no mortal can hold the pistol absolutely still, or consistently "grab" a shot when the sight is momentarily on the exact desired spot. The good news is that it is not at all necessary to hold the pistol still, even though it is desired. Area aiming means that you just hold the pistol within an arc of movement that is COMFORTABLE and EASY for you! Of course, you want to center your personal aiming AREA about the target center. Shouldn't I always be trying to reduce my aiming area while in competition? NO! Absolutely not during competition! This will cause anxiety and distracts from proper trigger movement resulting in those awful shots that land outside your personal, comfortable aiming area. This is when and where most points are dropped. Remember this. Nobody can consistently shoot a group smaller than his personal minimum arc (or area) of movement. Even the handful of people who can hold ten-ring and follow through at 50 yards will shoot a few nines due to minor trigger control errors and bullet dispersion. These few shooters are the

2670 shooters, because they can control the hammer drop while keeping the front sight in their comfortable area of hold. They don't disturb pistol sight alignment or target picture as they actuate the trigger/hammer mechanism. The saving grace for the rest of us is that the bullets tend to cluster about the center of our personal aiming area. Nature invented statistical distribution to do just that. For example, If a golfer is within reach of a hole and swings comfortably, the ball is more likely to go into the cup than any other single, random point you can pick. In bullseye shooting, with a zeroed pistol the bullet is more likely to go into the x-ring than any other random spot you can pick. So, hold the dot within your personal comfortable area centered about the x-ring. Press the trigger smoothly toward the web of your hand until the bullet has safely hit the target. Then return the dot or front sight to that aiming area immediately. This will help ensure a habit of following through. Let nature do the hard work for you. Can you explain why the shots will tend to cluster in the middle of my aiming area? The front sight or dot will move randomly in your aiming area. As it approaches the outer edge of your "circle", your mind will immediately tend to send it back across the middle toward the other extreme. Kind of like a star pattern of movement. As it crosses the middle of the target, your subconscious (or conscious mind) will not be trying to correct anything. Thus the dot or sight will spend more time in the middle than at the extreme edges of your aiming area. The shot is more likely to break during this time, than at the edge of your area. You must not decide to help the shot fire at this time as you are very likely to change the angle of the pistol, slightly. This is my definition of "grabbing" for a 10 and getting a 6. When should I try to reduce my aiming area? This must all be done in training rather than in competition! This is important! Do most of this after you have improved trigger control by dry-firing and exercising your arm, wrist and hand with a 5 pound weight. Dry-firing, while standing at the 50 yard line, is the best way to do this. You can measure (observe) your improvement by watching the front sight or red dot move around in this area. Shooting does less to improve your hold, compared to dry-firing where you can see what is happening. Fortunately, dry-firing costs less! Unfortunately, It is so boring that few heed this advice religiously. If you want to be a champion, follow this advice! I honestly believe that the basic secrets are in this spiel. I hope this inspires you to become a champion!

DAVE SALYER ARTICLES

Trigger Control and Aiming Area

By Dave Salyer

Source: http://bullseyepistol.com/salyer3.htm

Question: I have guns and ammunition tested over and over on a machine rest and proved to be excellent. Yet, when I fire the pistol off-hand, many of the bullets strike the target somewhere different from where the electronic red dot appeared on the target just before the gun fired. Can you explain why this happens?

Answer: There could be several answers to this question. I will try to comment on the two most common ones. One explanation applies to all pistol shooters but is not well known. The other applies to most, if not all shooters at times.
The second happens because most of us cause some level of unwanted pistol movement as we go through the process of firing a careful shot. This is lack of trigger control. I will not explain this further since it is a simple concept written about in all training materials. (However, it is very hard to achieve with every shot.) I will try to explain the other phenomenon that applies to all pistol shooters. I suspect that the very top shooters understand this either scientifically or intuitively. What you see is not what you get Rather , what you see is what you got! With the help of my good friend Tillman Eddy, inventor of the famous ClearsighT®, I will try to explain what I am talking about. Many of us try very hard to move the front sight or dot to the center of our aiming area and only then apply enough pressure to make the gun fire. Even if we do that with perfect trigger control we will have to be lucky if the bullet strikes the center! You might ask, "Why is that?" No mortal can hold a pistol perfectly still. Some people are more trained and coordinated to approach that goal than the rest of us. The gun is still moving a little or a lot as the gun fires. But we cannot see the target picture at the exact time the gun fires! You say, "But I can call my shots!" I say, "Yes, you can call where it went after the fact." The dot or sight will appear to stop for an instant as the shot breaks. We see this picture in our mind a fraction of a second after it has happened. In Mr. Eddy's own words he provides the scientific explanation of the points I am trying to make: Our consciousness of the surroundings does not live in the present, rather in the near-historical past. For instance: The "seeing process" is described briefly. Light strikes the rods and cones in the back of the eye and a chemical reaction occurs. The nerve impulses from that chemical reaction are transmitted to the optical nerve, also by a series of chemical reactions in the nerve cells. The impulses then arrive in the cortex that allows interpretation in our "mind" (whatever that is).

One notes that these biochemical reactions take a finite and actually rather long time. ***Demonstrable audio reaction time is two to three tenths of a second and (is) faster than visual reaction time.*** In bullseye as well as other shooting disciplines, this concept is extremely important. Quite simply, one living in the past can not have an effect on the present. The present is represented by the "real" (time) sight picture. what we "see" is historical. Since the above is true, once the sight alignment and aiming area have been achieved and positive trigger pressure begins, ego and judgment must cease! ***One ceases to be a "doer" and becomes an "observer".*** The only decision the conscious mind can make is to stop the shot from firing. Mr. Eddy notes that we all have decided to stop a shot and the darn thing goes anyhow We were trying to change history! The unconscious or subconscious is (or should be) trained to move the trigger to the rear and will result in the best result we are physically capable of performing. The conscious mind will judge, worry, fear (failure and success) and cause failure. As the late and great bullseye champion, Allen Fulford said in his training video, "If there is a secret to successful pistol shooting it is making the shot break with the sight inside your personal arc of movement," (superimposed on your target). There is no point in trying to shoot better than we can hold. We should be working on improving our hold (arc of movement) by dry-firing, exercise and focused practice. At the same time let's work on our ability to prevent disturbing this hold until after the pistol finishes its recoil. I hope this helps you improve your scores. If you have comments I would like to hear from you. Please write me in care of THE BULLET.

DAVE SALYER ARTICLES

Master Sustained Fire with Effective Practice

By Dave Salyer

Source: http://bullseyepistol.com/salyer4.htm

Question: In previous columns you have emphasized dry-firing and other means of slow fire related practice. What can I do to improve my timed and rapid scores?

Answer: This is a very good question, especially since two-thirds of bullseye is shooting sustained fire.
The technique is basically the same except aiming and firing time must be compressed to fit into the 10 and 20 seconds allowed for five consecutive shots from one magazine. Remember the basics. Sight alignment and trigger control must be achieved and maintained until the bullet leaves the barrel. Since it is difficult to maintain full concentration on both of these at once, I recommend the following for timed and rapid firing. Put more thought on trigger control than sight alignment. Another reason for this is that the sustained fire targets are closer or bigger. This allows some forgiveness on sight alignment and especially target alignment (with your sights). The other important factor is that we barely have enough time to get five well-aimed shots off in rapid fire - no extra time if we do it right - not enough time if we do it wrong. If we do it wrong, the "well-aimed" part goes out the window. Five shots go downrange "somewhere". Timing is the secret to good sustained fire scores. This means starting on time, firing in a cadence and finishing the fifth shot just before the target turns away. I tried for years in practice to develop this rhythmic cadence by imagining the stationary target was turning toward me and staying faced for an estimated 10 seconds. I never developed the ability to shoot good sustained fire until I was able to practice with a real turning target. I still have to practice regularly with turning targets to maintain the necessary rhythm to get five good shots onto the target without feeling rushed and thinking about time. If you have to think (worry) about time, then you are not thinking on positive, smooth and steady movement of the trigger to the rear. I would recommend that you consider buying and using a portable lightweight, battery-powered turning target mechanism. They come with a single target frame and a control switch you can activate with your non-shooting hand. A programmable timer takes over and causes the target to actuate exactly as the NRA rules recommend and require. The enhanced models can be programmed to meet international and other courses of fire and certain practice modes, such as a "one-shot" drill. Some people use this to learn to break the first shot immediately. If you can achieve this in a real match you

then have 10 seconds to fire four remaining shots. That is much easier than firing five in 10 seconds if you get a late start. I bought a used unit in 1988 from a retiring bullseye shooter who had just recently reached his goal of 2650. When I find time to practice with it my timed and rapid scores are several points higher at the next match. I have found that keeping sustained fire scores up requires regular practice with a turning target to reestablish and maintain cadence and confidence. My unit has been reliable, requiring only a small motorcycle or garden tractor battery replacement every few years. It was manufactured by longtime bullseye shooter and match director Frank Thomas, P. O. Box 3271, Bristol, TN 37625, Phone (423) 761-9725. Reprinted with permission from the January/February, 1997 edition of The Bullet, published by Virginia Shooting Sports Association.

DAVE SALYER ARTICLES

About Dave Salyer

By A Brief Biography of Dave Salyer

Source: http://bullseyepistol.com/salyer6.htm

Dave is an avid master level pistol competitor who has managed to compete in every National Conventional Pistol Championship at Camp Perry, Ohio since his first in 1983. He was a member of the VSSA Pistol Team from 1984 through 1993 before his move to Rock Hill, South Carolina and China. Since then he has been a member of the SC State team, or Greenville Gun Club team. He started bullseye shooting in 1977 in Corpus Christi, Texas.

His other hobby is pistolsmithing the M1911 Colt and its clones to prepare them for match use. He was the original builder of the winning pistol belonging to the 1993 and 1998 National .45 cal Champion, Al Dorman.

Until just recently, Dave was working as an engineer for a major Chemical company in a Joint venture in China. He had spent four of the last six years there. He has now repatriated and has taken early retirement to devote full time to his family and shooting hobbies.

Hopefully, I can talk Dave into sending me a few more of his articles!

Advertise and Support this Tremendous Resource

By The BEST place for marketing targeted for Competitive Pistol Shooters

Source: http://bullseyepistol.com/advertise.htm

TARGETING YOUR COMPETITIVE SHOOTING AUDIENCE

Competitive shooters are definitely a *unique* bunch, with specific needs and particular tastes. They are *passionate* about their equipment and supplies and are typically *loyal* to their suppliers and manufacturers. That presents a huge problem for those trying to tap into selling to the competitive shooting market. The challenge for the marketer of competition products is to find a medium that has much exposure on a national-level that appeals to this narrow market. Chances are good that they're not going to find it, especially in this unusually tough economy.

Order Advertising

THE FALSE PROMISE OF SOCIAL MEDIA

The temptation of exploiting the participants of Facebook with advertising *ad nauseam* was too much to resist for Mark Zuckerburg and company. Targeting their members with tailored ads based on their interests was the sales pitch to advertisers. Unfortunately for the advertisers, other than getting a few "like" nods, it really is not paying off. It turns out that Facebook users are interested in *socializing* while on the site, not purchasing stuff from random advertisers. In fact, most Facebook users find the advertisements a *nuisance*. But for the shooting sports, it doesn't matter anyway. Ads that are in any way firearm-related are *forbidden* from Facebook.

Order Advertising

A TOUGH DECISION TO MAKE

Before you whip out the checkbook and throw money at an advertising campaign, you have to ask yourself some serious questions:

- Is the medium a *worthy* cause? Or am I just lining the pockets of advertising network executives?
- Will I truly be able to control which websites I am endorsing through my ad impressions?

- Are the ads going to drive *real* customers to my business? Or attract only the *bored and curious*?
- Can I really afford up to *$3 per click* for bored and curious surfers who are merely killing time?
- Forget fancy charts and graphs, what will be the *real* return on my ad investment in sales?

Utilizing an automated *middle-man* to handle your Internet advertising might seem quick and effective, but you will lose control of who you endorse. On top of that, you probably won't get the expected bang for your buck in your targeted market: *competitive pistol shooters*. Internet marketing schemes may have changed over time - but *common sense* has remained the same as always.

Order Advertising

WHY BULLSEYEPISTOL.COM IS DIFFERENT

This website was created and is maintained to serve the interests of competitive pistol shooters; not to profit from them. In the early years alone, our praises from the top shooters, our influence in convincing retired shooters to return to competition and our inspiration to get new shooters involved has permanently stamped its value *forever*.

We continue to be that focal point for bullseye pistol shooters. Our home page alone has drawn almost *three million* page views, and its daily average is close to a *thousand*. With over 30 MB in viewable content and another 5 MB in downloadable multimedia content, there is an enormous amount of information on the website. And while the site is interactive and definitely pleasing to the eye, we prefer to error on the side of *simplicity* and continuity to make navigating the website an always familiar experience and to make *page-loads* fast and reliable.

The nature of the site content encourages visitors to return again and again. With a searchable database of over 150 articles and other documents, those seeking the answer to a question or a solution to a problem can find it quickly and easily on the website. New articles are continually being developed not only just in-house, but also by the contributions of top shooters nationwide. Our pages are designed and tested to look good in all of the major Internet browsers, including mobile and tablet applications. Google and other search engines consistently put our website listing at the top for searches related to bullseye pistol shooting.

We believe that we have the proper *mission*, *dedication*, *content* and *traffic* to be worthy of your advertising support.

Order Advertising

WHAT WE CAN OFFER

Our goal is to provide our sponsors with maximum advertising visibility without hindering the navigation of the website. As a result, we elected not to offer advertisements on the very *busy* homepage as we had concerns about how effective they would be. In addition, we have elected to provide sponsors additional confidence by not *rotating* ads - a particular space is allocated to one ad and that is it. In the Internet marketing world, that is known as *"100% ad impressions per page view,"* a real rarity these days. It's like buying an ad sign on the outfield wall of a baseball stadium - it is all yours.

We are offering two monthly sponsor packages. In either package, your *image* ad will appear in a particular page location, on *every* article in the website, every time it is viewed. The two packages, named PREMIUM #1 and #2 respectively are shown on the sample article page below:



Sponsor packages shall be sold on a first-come basis, one calendar month at a time. That means that sponsorship is *very* limited.

If you are genuinely interested in sponsoring the Encyclopedia of Bullseye Pistol for one or several months, continue the process by clicking the button below.

Order Advertising

Thank you for your support,



John Dreyer

John A. Dreyer. Founder and Webmaster

SITE / REFERENCE / GENERAL INDEX

Website Topic Index

Source: http://bullseyepistol.com/alphadex.htm

If you are *new* to the sport or looking for introductory information, first see the following articles; they will cover almost any questions you might have.
Get into Bullseye Shooting by Sherwood Veith, from *Handgunning*
Bullseye Shooting by Charles E. Petty, from the *American Rifleman*

- Accuracy Issues With Dot Sights: by John Dreyer
- Accurize a .45 Pistol, Custom Guns: by Larry Leutenegger
- Adjusting Sights: from USAMU
- Air Pistol: Information About the Sport
- Alcohol, Adverse Effects: from the USAMU
- Alibis In Sustained Fire: from Rulebook, or Common Causes
- Ammunition, Ammo: Selecting .22 Ammo, or Reloading Information
- Analyzing Group Placement: from USAMU
- Arc of Movement, Reducing: from USAMU, or by Bill Blankenship
- Army Marksmanship Unit: Actual Training Guide
- Artificial Group Tighteners: from the USAMU, or from Joe White
- Bad Habits of Shooters: from USAMU
- Bad Shots, Analyzing: from USAMU
- Blankenship, Bill: Writings of the Champion
- Brass, Components for Reloading: General Information
- Breath Control, Breathing: by Bill Blankenship, or from USAMU
- Bullets, Components for Reloading: General Information
- Caffeine, Adverse Effects (Coffee): from the USAMU
- Camp Perry, Lodging and Dining: compiled by John Dreyer
- Camp Perry, Ohio: by Justin Nystrom
- Carrying Cases, Pistol Boxes: by John Dreyer
- Cartes, David: Writings of the Champion
- Charts, for Analyzing Shots: from USAMU
- Checklist, Preparation on Firing Line: from USAMU
- Classification System: from Rulebook

- Cleaning, Firearms: by John Dreyer
- Confidence: by William Joyner, or by Joe White
- Courses of Fire: from Rulebook
- Cowboy Action Shooting, SASS : Information on the Sport
- Crossfires: from Rulebook
- Derr, Greg: Writings of the Champion
- Diet and Health: from USAMU
- Dot Sights: by John Dreyer
- Dreyer, John: Purpose of These Efforts, or Contact Him by Email
- Dry Firing: by Allen Fulford
- Ear Protection/Eye Protection: by John Dreyer
- Electronic Mailing List: for Paul Perrone
- Equipment - Cost Justifying: by Randy Pafford
- Equipment Requirements: by John Dreyer
- Equipment, Sources: Various Links
- Errors, Common: from USAMU
- Etiquette: from the NRA
- Exercise Programs: from USAMU, or by John Dreyer
- Fulford, Allen: Writings of the Late Champion
- Fung, Randall: His Products & Contacting the Craftsman
- Glossary of Shooting Terms: from USAMU
- Grip, Proper: by Bill Blankenship, or from USAMU
- Grips, Anatomical: by John Dreyer
- Groups, Analyzing Placement: from USAMU
- Hardball Pistol: from Rulebook, or from DCM Pamphlet, or from *American Rifleman*
- International Shooting, ISSF, UIT: Information on the Sport
- Internet Mailing List: the Bullseye-L forum
- Internet Resources, Links: Links to Other Sites
- Jerking The Trigger: by USAMU
- Joyner, William: Writings of the Champion
- Leg Match: General Definition, or Service Pistol Definition
- Leutenegger, Larry: Writings of the Armorer
- Lubrication: by Ed Masaki, or by John Dreyer

- Malfunctions During Match: from Rulebook, or Common Causes
- Masaki, Ed: his Pistol Oiling Tips
- Match Formats: from Rulebook
- Matches, Finding Sanctioned: General Information, or Contact NRA Affiliate Organizations
- Mental Aspects: by Don Nygord, or by Bob Rotella, PhD
- Mental Aspects: by William Joyner, or by Joe White, or from USAMU
- Mind Control: by Don Nygord, or by Bob Rotella, PhD
- Mind Control: by William Joyner, or by Joe White, or from USAMU
- Mounting Systems for Dot Sights: by John Dreyer
- National Champions: from the NRA
- National Matches: by Justin Nystrom
- National Records, Conventional Pistol: from the NRA
- National Records: from the NRA
- National Rifle Association, NRA: Official NRA Website
- Nicotine, Adverse Effects (Cigarettes): from the USAMU
- NRA State Affiliates: as Published by the NRA
- Other Disciplines: From Various Sources
- Parallax, Dot Sight: by John Dreyer
- Piasa Rifle and Pistol Club: by John Dreyer
- Pistol Selection - .22 Cal: by John Dreyer
- Pistol Selection - .45 Cal: by John Dreyer
- Pistol Selection - Center Fire: by John Dreyer
- Pistol Teams: Links to Various Websites
- Pistolsmiths Guild: Link to the Guild
- Pistolsmiths, Custom: Links to Pistolsmiths Sites
- Polarizing Filters: by John Dreyer
- Practical Shooting, IPSC, IDPA: from Various Sources
- Practice Habits: by Allen Fulford
- Preparation Period, Three Minute Prep Period: from USAMU
- Range Commands, MP3 File Recordings of: recorded by John Dreyer
- Range Commands: from Rulebook, or Download the Audio File
- Rapid Fire: by David Cartes
- Reloading Data, .45 Caliber: by John Dreyer

- Reloading Data, Hardball DCM Practice: from *American Rifleman*
- Reloading: General Information
- Rifle and Pistol Clubs: Various Links
- Rulebook, Conventional Pistol Rules: The Official NRA Rulebook
- Safety Rules: General Firearm Safety Rules
- Scope, Spotting Scopes: by John Dreyer
- Scopes, Red Dot Sights: by John Dreyer
- Scoring Fired Targets: from Rulebook
- Scoring Targets: The Rules, or Scoring Aids
- Service Pistol, Restrictions: from Rulebook, or from DCM Pamphlet
- Shooting Clubs and Ranges: by John Dreyer
- Shooting Ranges - Locating: General Information, or Contact NRA Affiliate Organizations
- Shot Analysis: from USAMU
- Sight Alignment: from USAMU, or Even More Technical
- Silhouette Shooting: Information on the Sport
- Skid Shots: from Rulebook
- Slow Fire: from USAMU
- Smallbore Rifle Shooting: General Information
- Smith and Wesson: Link to Website, or Model 41
- Software for Match Directors: by John Dreyer
- Stance: from USAMU, or by William Joyner, or by Bill Blankenship
- Stop Watches: by John Dreyer
- Strength and Stamina: from USAMU, or by John Dreyer
- Targets: by John Dreyer
- Tasco Pro Point: Test Information About the Sight
- Team Shooters, Attributes of: by USAMU
- Teams and Organizations: Various Links
- Terminology, Glossary of Shooting: by USAMU
- Tie-Breaking, Rules On: from Rulebook
- Timed Fire: by USAMU
- Tournaments, Types: from Rulebook
- Tournament Scorekeeping Software: Software by John Dreyer
- Training: by Allen Fulford

- Trigger Control: by USAMU
- Ultra-Dot Sight: Test Information About the Sight
- USAMU Training Guide: The Bible of Competitive Training
- Wheel of Misfortune: from USAMU
- Windy Conditions, Shooting In: from USAMU
- Zen In Shooting: compiled by John Dreyer
- Zero, Adjusting Sights: from USAMU

References

- Get into Bullseye Shooting
- Bullseye Shooting
- by John Dreyer
- by Larry Leutenegger
- from USAMU
- Information About the Sport
- from the USAMU
- Common Causes
- Selecting .22 Ammo
- Reloading Information
- from USAMU
- from USAMU
- by Bill Blankenship
- Actual Training Guide
- from the USAMU
- Joe White
- from USAMU
- from USAMU
- Writings of the Champion
- General Information
- by Bill Blankenship
- from USAMU
- General Information
- from the USAMU
- compiled by John Dreyer
- by Justin Nystrom
- by John Dreyer
- Writings of the Champion
- from USAMU

- from USAMU
- by John Dreyer
- by William Joyner
- by Joe White
- Information on the Sport
- Writings of the Champion
- from USAMU
- by John Dreyer
- Purpose of These Efforts
- by Allen Fulford
- by John Dreyer
- by Randy Pafford
- by John Dreyer
- Various Links
- from USAMU
- from the NRA
- from USAMU
- by John Dreyer
- Writings of the Late Champion
- His Products & Contacting the Craftsman
- from USAMU
- by Bill Blankenship
- from USAMU
- by John Dreyer
- from USAMU
- from DCM Pamphlet
- from American Rifleman
- Information on the Sport
- Links to Other Sites
- by USAMU
- Writings of the Champion
- Service Pistol Definition
- Writings of the Armorer
- by Ed Masaki
- by John Dreyer
- Common Causes
- his Pistol Oiling Tips
- General Information
- by Don Nygord
- Bob Rotella, PhD
- by William Joyner

- by Joe White
- from USAMU
- by Don Nygord
- Bob Rotella, PhD
- by William Joyner
- by Joe White
- from USAMU
- by John Dreyer
- by Justin Nystrom
- from the NRA
- from the NRA
- from the USAMU
- From Various Sources
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- by John Dreyer
- by John Dreyer
- by John Dreyer
- Links to Various Websites
- Link to the Guild
- Links to Pistolsmiths Sites
- by John Dreyer
- from Various Sources
- by Allen Fulford
- from USAMU
- recorded by John Dreyer
- Download the Audio File
- by David Cartes
- by John Dreyer
- from American Rifleman
- General Information
- Various Links
- General Firearm Safety Rules
- by John Dreyer
- by John Dreyer
- Scoring Aids
- from DCM Pamphlet
- by John Dreyer
- General Information
- from USAMU
- from USAMU
- Even More Technical

- Information on the Sport
- from USAMU
- General Information
- Link to Website
- Model 41
- by John Dreyer
- from USAMU
- by William Joyner
- by Bill Blankenship
- by John Dreyer
- from USAMU
- by John Dreyer
- by John Dreyer
- Test Information About the Sight
- by USAMU
- Various Links
- by USAMU
- by USAMU
- by Allen Fulford
- by USAMU
- Test Information About the Sight
- The Bible of Competitive Training
- from USAMU
- from USAMU
- compiled by John Dreyer
- from USAMU

Further Reading

- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w10.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w19.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w07.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w14.pdf)
- Contact Him by Email (mailto:jd@bullseyepistol.com)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w03.pdf)
- General Definition (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w01.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w10.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w01.pdf)
- Contact NRA Affiliate Organizations (http://www.nrahq.org/clubs/state.asp)
- from the NRA (http://www.nrahq.org/compete/nat-trophy/tro-008.pdf)
- Official NRA Website (http://www.nra.org/)
- as Published by the NRA (http://www.nrahq.org/clubs/state.asp)

- by John Dreyer (http://www.piasaclub.com/)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w10.pdf)
- The Official NRA Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-index.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w14.pdf)
- The Rules (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w14.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w03.pdf)
- Contact NRA Affiliate Organizations (http://www.nrahq.org/clubs/state.asp)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w14.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w15.pdf)
- from Rulebook (http://www.nrahq.org/compete/RuleBooks/Pistol/pst-w01.pdf)
- Software by John Dreyer (http://tournament.bullseyepistol.com/)

Terms and Conditions

By User Agreement for Accessing Bullseyepistol.com

Source: http://bullseyepistol.com/legal.htm

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SITE / REFERENCE / GENERAL INDEX

Links to Other Sites

By Resources Relevant to Bullseye Pistol

Source: http://bullseyepistol.com/links.htm

CLUBS AND TEAMS

Shooting Clubs and Ranges:
Alton Piasa Rifle and Pistol Club, Illinois
Alaska Interior Marksmanship Committee
Buffalo Target Shooters Association
Camp Perry, Ohio (Site of National Matches)
Canton McKinley Rifle and Pistol Club, Ohio
Cherry Creek Gun Club, Colorado
Colonial Rifle and Pistol Club, Staten Island, NY
Durham Pistol & Rifle Club, North Carolina
Green Valley R & P Club, Missouri
Leni-Lanape Pistol Club, Pennsylvania
McKeesport Sportsmen's Association, Pennsylvania
Metropolitan Rod & Gun Club, New York
Nashua Fish & Game Association, New Hampshire
Newport Rifle Club, Rhode Island
Old Bridge Rifle & Pistol Club, New Jersey
Ottawa Sportsmen's Club, Michigan
Phoenix Rod and Gun Club, Arizona
Tulsa Red Castle Gun Club, Oklahoma
Quantico Shooting Club, Virginia
Woburn Sportsmen's Assoc., Massachusetts
Waukesha Handgun League, Wisconsin
Yellowstone Rifle Club, Montana
Teams and Organizations:
AusPistol: Pistol Competition in Australia
Civilian Marksmanship Program
Illinois-Missouri Indoor Bullseye Pistol League
Illinois State Rifle Association

Les Baer All-Female Shooting Team
Missouri Sport Shooting Association
Ohio State Rifle and Pistol Association
Seattle Police Athletic Association
Tri-County Pistol and Revolver League
US Army Marksmanship Unit
US Army Reserve Shooting Team
US Air Force Shooting Team
US Marine Corps Shooting Team
US Navy Shooting Team
University of the Philippines Rifle and Pistol Team
Utah Precision Marksmanship Society
Washington State Bullseye Pistol Page

INTERNET DIRECTORIES

Internet Firearms Directory
The Gun Index
Gunhoo Shooting Directory

BULLSEYE EQUIPMENT

Pistol Manufacturers:
Browning
Hammerli
High Standard
Ruger
Smith and Wesson
Springfield Armory
Walther
Custom Pistolsmiths:
Mike Curtis Custom
Les Baer Custom
Jim Clark Custom
Volquartsen Custom
Rock River Arms
Travis Strahan's Accu-Lock Pistols

Bob Marvel's Custom Products
American Pistolsmith's Guild
The American Firearms Industry

Dealers:

Larry Carter
Champion's Choice
Brownells Gunsmithing Supplies
Brenzovich Firearms Importers
Champion Shooter's Supply
Scott Pilkington Competition Equipment
Don Nygord Precision

Accessories:

UltraDot Electronic Sights
Aimpoint Electronic Sights
BME Precision Scope Mounts
Randall Fung Custom Grips
The "Patchworm" by 20/20 Concepts
Clearsight Optical
Knapp Scope Mounts for Hammerli
CMM Hammerli-to-1911 Grip Adaptor & Brasscatchers
Pepper Spray

Reloading/Ammunition:

Dillon Precision
UniqueTek
Nosler Bullets
Magnus Bullets
Hornady Bullets
Starline Brass
Eley Limited
Winchester
Alliant Powders
Vihtavuori Oy
CCI Ammunition
ATK, Inc.
Lyman Products

MISCELLANEOUS INFORMATION

The M1911 Pistol Information Page
Gun Web Online Magazine
The Shotgun News Classifieds
Canadian Firearms Information

FIREARM AUCTION WEBSITES

Auction Arms
Gunbroker.com
Guns America

POLITICAL AND RKBA

National Rifle Association
Gun Owners of America
Second Amendment Foundation
A Challenge to Win Back Firearms Culture
Jews for the Preservation of Firearms Ownership
The Firearms Coalition
A Satellite Channel for Firearms?
The Independence Institute

Further Reading

- Alton Piasa Rifle and Pistol Club, Illinois (http://www.piasaclub.com)
- Alaska Interior Marksmanship Committee (http://www.aimcomm.org)
- Buffalo Target Shooters Association (http://www.BuffaloTargetShooters.ca)
- Camp Perry, Ohio (Site of National Matches) (http://www.cpmr-oh.org/)
- Canton McKinley Rifle and Pistol Club, Ohio (http://www.cantonmckinleyriflepistol.org/)
- Cherry Creek Gun Club, Colorado (http://www.cherrycreekgunclub.org/)
- Colonial Rifle and Pistol Club, Staten Island, NY (http://www.colonialrifleandpistolclub.com)

- Durham Pistol & Rifle Club, North Carolina (http://www.dprc.org)
- Green Valley R & P Club, Missouri (http://www.greenvalleyclub.com/)
- Leni-Lanape Pistol Club, Pennsylvania (http://lenilenape.homestead.com/Master.html)
- McKeesport Sportsmen's Association, Pennsylvania (mailto:223845@comcast.net)
- Metropolitan Rod & Gun Club, New York (http://www.metrorgc.org/)
- Nashua Fish & Game Association, New Hampshire (http://www.nfga.org/)
- Newport Rifle Club, Rhode Island (http://www.newportrifleclub.org)
- Old Bridge Rifle & Pistol Club, New Jersey (http://www.obrpc.org/)
- Ottawa Sportsmen's Club, Michigan (http://ottawasportsmen.org/)
- Phoenix Rod and Gun Club, Arizona (http://www.phoenixrodandgun.org/)
- Tulsa Red Castle Gun Club, Oklahoma (http://www.tulsaredcastlegunclub.com/)
- Quantico Shooting Club, Virginia (http://www.quanticoshootingclub.com/)
- Woburn Sportsmen's Assoc., Massachusetts (http://members.aol.com/wsa1933/home.htm)
- Waukesha Handgun League, Wisconsin (http://www.whlshooting.com)
- Yellowstone Rifle Club, Montana (http://www.yrc.org/)
- AusPistol: Pistol Competition in Australia (http://www.auspistol.com.au/)
- Civilian Marksmanship Program (http://www.odcmp.com/)
- Illinois-Missouri Indoor Bullseye Pistol League (http://www.ilmoleague.com)
- Illinois State Rifle Association (http://www.isra.org/)
- Les Baer All-Female Shooting Team (http://www.lesbaerteam.com/)
- Missouri Sport Shooting Association (http://www.missourisportshooting.org/)
- Ohio State Rifle and Pistol Association (http://www.orpa.net/)
- Seattle Police Athletic Association (http://www.spaarange.com/)
- Tri-County Pistol and Revolver League (http://www.tricountypistol.org/)
- US Army Marksmanship Unit (http://www.usamu.com/)
- US Army Reserve Shooting Team (javascript:alert('The%20US%20Army%20Reserve%20Team%20Website%20was%20closed.%20%20If%20you%20receive%20word%20of%20a%20new%20site,%20please%20let%20me%20know!'))
- US Air Force Shooting Team (http://www.airforceshooting.org/)
- US Marine Corps Shooting Team (javascript:alert('The%20USMC%20Team%20Website%20was%20closed.%20%20If%20you%20receive%20word%20of%20a%20new%20site,%20please%20let%20me%20know!'))
- US Navy Shooting Team (http://www.usnst.org/)
- University of the Philippines Rifle and Pistol Team (http://uprpt.cjb.net/)
- Utah Precision Marksmanship Society (http://www.UPMS.org/)
- Washington State Bullseye Pistol Page (http://www.wsrpa.org)
- Internet Firearms Directory (http://www.shootguns.info/)
- The Gun Index (http://www.igun.com/)
- Gunhoo Shooting Directory (http://www.gunsgunsguns.com/gunhoo/)
- Browning (http://browning.com/)
- Hammerli (http://www.hammerli.com/)

- High Standard (http://www.highstandard.com/)
- Ruger (http://www.ruger-firearms.com/)
- Smith and Wesson (http://www.smith-wesson.com/)
- Springfield Armory (http://www.springfield-armory.com/)
- Walther (http://www.waltheramerica.com/)
- Mike Curtis Custom (mailto:mrcurtis@kih.net)
- Les Baer Custom (http://www.lesbaer.com/)
- Jim Clark Custom (http://www.clarkcustomguns.com/)
- Volquartsen Custom (http://www.volquartsen.com/)
- Rock River Arms (http://www.rockriverarms.com/)
- Travis Strahan's Accu-Lock Pistols (http://www.accu-lockpistols.com)
- Bob Marvel's Custom Products (http://www.marvelprod.com)
- American Pistolsmith's Guild (http://www.americanpistol.com/)
- The American Firearms Industry (http://www.amfire.com/)
- Larry Carter (http://www.larrysguns.com/)
- Champion's Choice (http://www.champchoice.com/)
- Brownells Gunsmithing Supplies (http://www.brownells.com/)
- Brenzovich Firearms Importers (http://www.brenzovich.com/)
- Champion Shooter's Supply (http://www.championshooters.com/)
- Scott Pilkington Competition Equipment (http://www.pilkguns.com/)
- Don Nygord Precision (http://www.nygordprecision.com/)
- UltraDot Electronic Sights (http://www.ultradotusa.com/)
- Aimpoint Electronic Sights (http://www.aimpoint.com/)
- BME Precision Scope Mounts (http://www.bme-mounts.com/)
- Randall Fung Custom Grips (http://www.jnb.com/~funggrip/)
- The "Patchworm" by 20/20 Concepts (http://20-20.8m.com/)
- Clearsight Optical (mailto:tleddy@earthlink.net)
- Knapp Scope Mounts for Hammerli (http://www.knappent.com/)
- CMM Hammerli-to-1911 Grip Adaptor & Brasscatchers (http://www.cm-machine.com/)
- Pepper Spray (http://www.buy-pepper-spray-today.com/Pepper-Spray-Gun-s/61.htm)
- Dillon Precision (http://www.dillonprecision.com/)
- UniqueTek (http://www.uniquetek.com/)
- Nosler Bullets (http://www.nosler.com/)
- Magnus Bullets (http://www.magnusbullets.com/)
- Hornady Bullets (http://www.hornady.com/)
- Starline Brass (http://www.starlinebrass.com/)
- Eley Limited (http://www.eley.co.uk/)
- Winchester (http://www.winchester.com/)
- Alliant Powders (http://www.AlliantPowder.com/)
- Vihtavuori Oy (http://www.vihtavuori.fi/)
- CCI Ammunition (http://www.cci-ammunition.com/)

- ATK, Inc (http://www.atk.com/profileAmmunition/civil-ammo/)
- Lyman Products (http://www.lymanproducts.com/)
- The M1911 Pistol Information Page (http://www.m1911.org/)
- Gun Web Online Magazine (http://www.gunweb.com/)
- The Shotgun News Classifieds (http://www.shotgunnews.com/)
- Canadian Firearms Information (http://www.firearmstraining.ca/)
- Auction Arms (http://www.auctionarms.com/)
- Gunbroker.com (http://www.gunbroker.com/)
- Guns America (http://www.gunsamerica.com/)
- National Rifle Association (http://www.nra.org/)
- Gun Owners of America (http://www.gunowners.org/)
- Second Amendment Foundation (http://www.saf.org/)
- A Challenge to Win Back Firearms Culture (http://is-lan.com/challenge/)
- Jews for the Preservation of Firearms Ownership (http://www.jpfo.org/)
- The Firearms Coalition (http://www.nealknox.com/)
- A Satellite Channel for Firearms? (http://www.gunownerstv.com/)
- The Independence Institute (http://i2i.org/)

The NRA's National Records

By For Conventional Bullseye Pistol

Source: http://bullseyepistol.com/records.htm

OUTDOOR CONVENTIONAL PISTOL RECORDS

NRA
NATIONAL RECORDS FOR OUTDOOR PISTOL [15]
AS OF 10-08-98

.22 CALIBER

20 SHOTS SLOW FIRE - 50 YDS

OPEN	F. A, HIGGINSON, USMC	200-12X	04/21/68
	R. J. TOURIGNY JR, HARRISVILLE, RI	200-12x	07/01/84
CIVILIAN	R. J. TOURIGNY JR, HARRISVILLE, RI	200-12X	07/01/84
SERVICE	F. A, HIGGINSON, USMC	200-12x	04/21/68
POLICE	G, DEGIROLAMO, SAN FRANCISCO, CA	198-11X	01/10/71
WOMEN	R, FOX, PARKER, AZ	197-8X	10/18/97

20 SHOTS TIMED FIRE - 25 YDS

OPEN	R. D. WHITACRE, USA	200-20X+5X	02/25/68
CIVILIAN	R. TOURIGNY, CRANSTON, RI	200-19X	08/14/88
	L. HALL, WOODLAND HILLS, CA	200-19X	02/10/91
SERVICE	R. D. WHITACRE, USA	200-20X+5X	02/25/68
POLICE	L. W. DECAMP, W COVINA, CA	200-18X	06/03/60
	J. W. HURST, LOS ANGELES, CA	200-18X	03/17/62
	J. W. HARMON, CHULA VISTA, CA	200-18X	04/20/63
	E. M. BURNETT, ROMEO, MI	200-18X	09/15/73
WOMEN	R. E. FOX, PARKER, AZ	200-17X	10/03/93

20 SHOTS RAPID FIRE - 25 YDS

OPEN	B. D. HARMON, USA	200-20X	06/19/82
CIVILIAN	W. HART, PROVIDENCE, RI	200-19X	10/08/95
SERVICE	B. D. HARMON, USA	200-20X	06/19/82
POLICE	J. W. HURST, LOS ANGELES, CA	200-19X	04/20/63
WOMEN	R. E. FOX, PARKER, AZ	200-16X	10/16/93
JUNIOR	A. SOKOLOWSKI, YEAGERTOWN, PA	200-14X	05/18/97

NATIONAL MATCH COURSE

OPEN	B, D. HARMON, USA	300-24X	10/10/81
	J. E. MEIDINGER, QUANTICO, VA	300-24X	07/05/97
CIVILIAN	L. CARTER, PORTLAND, ME	300-20X	07/23/95
SERVICE	B. D, HARMON, USA	300-24X	10/10/81
	J. E. MEIDINGER, QUANTICO, VA	300-24X	07/05/97
POLICE	L. W. DECAMP, W COVINA, CA	299-19X	01/15/61
WOMEN	B. J. HILE, USA	299-15X	06/20/71
JUNIOR	J. BICKAR, N. CANTON, OH	297-16X	07/05/96

90 SHOTS - SF 50 YDS, TF 25 YDS, RF 25 YDS
& NATIONAL MATCH COURSE

OPEN	F. A. HIGGINSON, USMC	898-62X	04/21/68
	J. MEIDINGER, QUANTICO, VA	898-62X	05/30/98
CIVILIAN	R. J, TOURIGNY JR, HARRISVILLE, RI	897-61X	07/07/85
SERVICE	F. A, HIGGINSON, USMC	898-62X	04/21/68
	J. MEIDINGER, QUANTICO, VA	898-62x	05/30/98
POLICE	E. M. BURNETT, LANSING, MI	895-47X	01/20/68
WOMEN	R. FOX, PARKER, AZ	891-43x	10/18/97
JUNIOR	A. SOKOLOWSKI, YEAGERTOWN, PA	884-55X	05/18/97

NRA

NATIONAL RECORDS FOR OUTDOOR PISTOL [15]

AS OF 10-08-98

CENTER FIRE

20 SHOTS SLOW FIRE - 50 YDS

OPEN	H. L. ANDERSON, USA	200-10X	07/20/75
CIVILIAN	D. R. YOUNG, WINTERBURN, ALBERTA	199-10X	06/20/76
SERVICE	H. L. ANDERSON, USA	200-10X	07/20/75
POLICE	G. KASAT, EAGAN, MN	198-8X	04/26/92
WOMEN	K, S. DYER, USA	195-4X	03/11/78
JUNIOR	B. E. GUIETTE, LA MESA, CA	192-4X	01/16/65

20 SHOTS TIMED FIRE - 25 YDS

OPEN	J. PASCARELLA, NEWTON FALLS, OH	200-20X	08/27/78
CIVILIAN	R. D. KRELSTEIN, MEMPHIS, TN	200-19X	05/16/64
	L. CARTER, PORTLAND, ME	200-19X	09/07/91
SERVICE	J. PASCARELLA, NEWTON FALLS, OH	200-20X	08/27/78
POLICE	E, W. HILDEN, OCEANSIDE, CA	200-19x	10/02/65
WOMEN	K. CHATTERTON, MILLBURN, NJ	200-17X	08/12/92
JlTNIOR	J. C. BICKAR, NORTH CANTON, OH	200-15X	07/16/94

20 SHOTS RAPID FIRE - 25 YDS

OPEN	P. H. COLSON, QUANTICO, VA	200-19X	10/08/89
CIVILIAN	J. H. JOHNSON, LA HABRA, CA	200-17X	10/18/83
SERVICE	P. H. COLSON, QUANTICO, VA	200-19X	10/08/89
POLICE	J, W. HURST, LOS ANGELES, CA	200-17X	09/15/62
WOMEN	G, E. BACKSTROM, HOWUIAM, WA	200-9X	08/13/58
JUNIOR	K. O. SWANSON, BIRMINGHAM, AL	200-10X	10/16/82

NATIONAL MATCH COURSE

OPEN	F. A. HIGGINSON, USMC	300-21X	09/18/76
CIVILIAN	L. CARTER, PORTLAND, ME	300-20X	09/10/95
SERVICE	F. A. HIGGINSON, USMC	300-21X	09/18/76
POLICE	J. W. HURST, LOS ANGELES, CA	299-16X	11/05/67
WOMEN	R. E. FOX, PARKER, AZ	298-15X	06/11/94

JUNIOR	J. BICKAR, N. CANTON, OH	295-18X	08/15/93

90 SHOTS - SF 50 YDS, TF 25 YDS, RF 25 YDS
& NATIONAL MATCH COURSE

OPEN	B. D. HARMON, USA	896-54X	04/04/82
CIVILIAN	D. R, YOUNG, WINTERBURN, ALBERTA	893-55X	06/20/76
SERVICE	B. D. HARMON, USA	896-54X	04/04/82
POLICE	E. M. BURNETT, E LANSING, MI	889-41X	03/11/68
WOMEN	R. E. FOX, PARKER, AZ	886-43X	05/01/94
JUNIOR	B. E. GUIETTE, LA MESA, CA	874-34X	06/05/65

NRA
NATIONAL RECORDS FOR OUTDOOR PISTOL [15]
AS OF 10-08-98

.45 CALIBER

20 SHOTS SLOW FIRE - 50 YDS

OPEN	B. D. HARMON, USA	200-11x	10/30/82
CIVILIAN	R. STEINBRECHER, WYNDMOOR, PA	199-8X	08/23/97
SERVICE	B, D. HARMON, USA	200-1lX	10/30/82
POLICE	E. W. HILDEN, OCEANSIDE, CA	197-8X	02/20/65
WOMEN	G. LIBERTY, USAF	195-9X	08/23/64
JUNIOR	K. O. SWANSON, BIRMINGHAM, AL	192-7X	08/21/82

20 SHOTS TIMED FIRE - 25 YDS

OPEN	D. L. HAMILTON, USN	200-20X+5X	05/23/65
CIVILIAN	R. STEINBRECHER, WYNDMOOR, PA	200-20X+3X	09/27/97
SERVICE	D. L. HAMILTON, USN	200-20X+5X	05/23/65
POLICE	S. L. CLEVENGER, BEAUFORT, SC	200-20X	04/28/96
WOMEN	R. E. FOX, PARKER, AZ	200-17X	07/14/85
JUNIOR	K. O. SWANSON, BIRMINGHAM, AL	200-9X	07/22/80
	J. BICKAR, NORTH CANTON, OH	200-9X	07/03/93

20 SHOTS RAPID FIRE - 25 YDS

OPEN	B. D. HARMON, USA	200-19x	10/23/76
	L. B. CARTER, PORTLAND, MF:	200-19X	05/17/98
CIVILIAN	L. B. CARTER, PORTLAND, ME	200-19X	05/17/98
SERVICE	B. D. HARMON, USA	200-19x	10/23/76
POLICE	E. M. BURNETT, E LANSING, MI	200-15X	06/11/70
	J. L. FARLEY, AMERICUS, GA	200-15X	07/02/72
WOMEN	G. LIBERTY, USAF	199-7X	08/17/63
JUNIOR	J. BICKAR, NORTH CANTON, OH	199-11X	10/04/92

NATIONAL MATCH COURSE

OPEN	B. D. HARMON, USA	300-19x	06/27/81
	A. E. MOODY, QUANTICO, VA	300-19X	06/25/89
CIVILIAN	D. KLING, GILBERT, AZ	300-9x	04/28/96
SERVICE	B. D. HARMON, USA	300-19x	06/27/81
	A. E. MOODY, QUANTICO, VA	300-19X	06/25/89
POLICE	E. W. HILDEN, YUMA, AZ	300-13x	11/05/72
WOMEN	R. E. FOX, USAR	299-16X	12/04/83
JUNIOR	J, C. BICKAR JR., NORTH CANTON, OH	295-10X	07/05/92

90 SHOTS - SF 50 YDS, TF 25 YDS, RF 25 YDS
& NATIONAL MATCH COURSE

OPEN	J. MEIDINGER, QUANTICO, VA	897-49X	05/17/98
CIVILIAN	W. HART, PROVIDENCE, RI	893-57X	06/01/96
SERVICE	J, MEIDINGER, QUANTICO, VA	897-49X	05/17/98
POLICE	E, W. HILDEN, YUMA, AZ	887-42X	09/22/68
WOMEN	R. E. FOX, USAR	884-39X	06/11/94
JUNIOR	A. SOKOLOWSKI, YEAGERTOWN, PA	873-40X	05/18/97

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NATIONAL RECORDS FOR OUTDOOR PISTOL [15]
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3 GUN AGGREGATE

270 SHOTS - SF 50 YDS, TF 25 YDS, RF 25 YDS
& NATIONAL MATCH COURSE CALIBER .22, CENTER FIRE, & .45

OPEN	H. L. ANDERSON, USA	2680-159x	07/24/74
CIVILIAN	C. E. BAXTER, ARGYLE, TX	2677-154X	07/05/87
SERVICE	H. L. ANDERSON, USA	2680-159X	07/24/74
POLICE	S. L. CLEVENGER, BEAUFORT, SC	2659-156X	04/28/96
WOMEN	R. E. FOX, PARKER, AZ	2660-139X	06/11/94
JUNIOR	A. SOKOLOWSKI, YEAGERTOWN, PA	2628-137X	05/18/97

INDOOR CONVENTIONAL PISTOL RECORDS

NRA
NATIONAL RECORDS FOR INDOOR PISTOL [14]
AS OF 06-24-98 PAGE: 1

.22 CALIBER

CATEGORY		SCORE	DATE
	20 SHOTS SLOW FIRE - 50 FEET		
OPEN	F. D. THOMPSON, USA	199	02/14/63
	H. L. ANDERSON, USA	199	01/21/72
CIVILIAN	T, MACMCTRRAIN JR, FT. BENNING, GA	198	01/18/68
	D. GLENN, CLINTON, IL	198	04/21/96
	S, LORENZ, LYNNWOOD, WA	198	03/22/97
SERVICE	F. D. THOMPSON, USA	199	02/14/63
	H. L. ANDERSON, USA	199	01/21/72
POLICE	E. M. BURNETT, HOLT, MI	196	03/20/65
WOMEN	G. E. BACKSTROM, HOQUIAM, WA	194	04/16/55
JUNIOR	P. VALINTAKONIS, WINDSOR, CT	191	01/19/96

20 SHOTS TIMED FIRE - 50 FEET

OPEN	L, CARTER, PORTLAND, ME	200-19X	03/19/95
CIVILIAN	L. CARTER, PORTLAND, ME	200-19X	03/19/95
SERVICE	M. D. BLACK, USA	200-18X	01/19/75
POLICE	J. W. HURST, LOS ANGELES, CA	200-17X	01/28/67
	E. B. HAMB, DETROIT, MI	200-17X	02/22/67
WOMEN	B. J. HILE, USA	199-11X	02/17/74
JUNIOR	R. A, LARUE JR, JEROME, MI	199-10X	03/18/67

20 SHOTS RAPID FIRE - 50 FEET

OPEN	D. L. HAMILTON, USN	200-17X	01/25/70
	R. J. TOURIGNY JR, GLENDALE, RI	200-17X	03/19/72
	J. DORSEY, USMC	200-17X	03/21/76
	R. J. TOYOTA, SPRINGFIRLD, OR	200-17X	03/22/97
CIVILIAN	R. J. TOURIGNY JR, GLENDALE, RI	200-17X	03/19/72
	R, J. TOYOTA, SPRINGFIRLD, OR	200-17X	03/22/97
SERVICE	D. L. HAMILTON, USN	200-17X	01/25/70
	J. DORSEY, USMC	200-17X	03/21/76
POLICE	M. D. HAMILTON, MISSOULA, MT	200-14X	02/12/66
	E. M. BURNETT, LANSING, MI	200-14X	03/31/68
WOMEN	G. P. SCHLERNITZAUER, LOWELL, MI	198-7X	02/25/67
JUNIOR	S. K. COLLINS, SANBORN, NY	199-10X	10/07/79

NRA GALLERY COURSE - 50 FEET

OPEN	L, CARTER, PORTLAND, ME	300-15X	11/20/94
CIVILIAN	L. CARTER, PORTLAND, ME	300-15X	11/20/94
SERVICE	R, M. STINEMAN, USA	300-9X	01/22/64
POLICE	E. M. BURNETT, HOLT, MI	298-10X	04/04/65
WOMEN	S. A. TALBOTT, FORT BENNING, GA	294-16X	02/04/95
JUNIOR	S. K. COLLINS, SANBORN, NY	290-4X	10/01/78

90 SHOTS - SLOW FIRE, TIMED FIRE, RAPID FIRE & NRA GALLERY COURSE - 50 FEET

OPEN	H, L. ANDERSON, USA	896-40X	01/21/72
CIVILIAN	L. CARTER, PORTLAND, ME	893-43X	12/18/94
SERVICE	H. L. ANDERSON, USA	896-40X	01/21/72
POLICE	G. I. DEGIROLAMO, SAN FRANCISCO, CA	886-35X	02/27/75
WOMEN	B. J. HILE, USA	874-30X	02/17/74
JUNIOR	S. K. COLLINS, SANBORN, NY	866-14X	10/01/78

NRA
NATIONAL RECORDS FOR INDOOR PISTOL [14]
AS OF 06-24-98 PAGE: 2

.22 CALIBER

20 SHOTS SLOW FIRE - 25 YDS (NEW B-16 10/1/95)

OPEN	W. HART, PROVIDENCE, RI	198-8x	12/02/95
CIVILIAN	W. HART, PROVIDENCE, RI	198-8X	12/02/95
SERVICE	D. R. YOUNG, MALO, WA	194-4X	11/26/95
POLICE	G. T, SINCLAIR, BOISE, ID	190-3X	01/28/96
WOMEN	D. L, SPENCER, PORTSMOUTH, RI	196-5X	03/02/98
JUNIOR	P. VALINTAKONIS, WINDSOR, CT	177-2X	02/23/97

20 SHOTS TIMED FIRE - 25 YDS

OPEN	D. L. HAMILTON, KINGSTON, MA	200-20X+3X	11/23/86
	M, M. BEEMAN, HOWELL, MI	200-20X+3X	03/28/93
CIVILIAN	D. L. HAMILTON, KINGSTON, MA	200-20X+3X	11/23/86
	M. M. BEEMAN, HOWELL, MI	200-20X+3X	03/28/93
SERVICE	J. MCCARTY, BOLLING AFB, D.C,	200-19X	02/16/92
POLICE	E. M, BURNETT, ROMEO, MI	200-19X	11/07/71
WOMEN	K. CHATTERTON, MILLBURN, NJ	200-14X	12/04/92
JUNIOR	D. J. STARLIPER, DOWNINGTOWN, PA	197-10X	10/25/80

20 SHOTS RAPID FIRE - 25 YDS

OPEN	M, M. BEEMAN, HOWELL, MI	200-20X+4X	04/29/94
CIVILIAN	M. M, BEEMAN, HOWELL, MI	200-20X+4X	04/29/94
SERVICE	S. F. REITER, SAN FRANCISCO, CA	200-17X	10/24/74
	K. R. BUSTER, BILLINGS, MO	200-17X	02/29/75
POLICE	E. M. BURNETT, LANSING, MI	200-18X	11/05/67
WOMEN	K, S. DYER, USA	199-10x	02/19/78
JUNIOR	M. MANCINI, SAN JOSE, CA	197-7X	08/28/84

NRA GALLERY SHORT COURSE - 25 YDS (NEW B-16 10/1/95)

OPEN	G. DERR, MARSHFIELD, MA	300-16X	02/02/98
CIVILIAN	G. DERR, MARSHFIELD, MA	300-16x	02/02/98
SERVICE	D. YOUNG, MALO, WA	296-20x	02/22/98
POLICE	G. T, SINCLAIR, BOISE, ID	299-20X	10/30/94
WOMEN	R. H. NEWMAN, VINELAND, NJ	290-8X	11/19/95

90 SHOTS - SLOW FIRE, TIMED FIRE RAPID FIRE & NRA GALLERY SHORT COURSE - 25 YDS (NEW B-16 10/1/95)

OPEN	W. HART, PROVIDENCE, RI	895-63X	12/02/95
CIVILIAN	W. HART, PROVIDENCE, RI	895-63X	12/02/95
SERVICE	D. R. YOUNG, MALO, WA	888-42X	11/26/95
POLICE	G. T. SINCLAIR, BOISE, ID	886-46X	01/28/96
WOMEN	R. H, NEWMAN, VINELAND, NJ	869-28X	11/19/95
JUNIOR	P. VALINTAKS, WINDSOR, CT	862-33X	02/23/96

Other Popular Disciplines

By Of the Pistol-Related Shooting Sports

Source: http://bullseyepistol.com/other.htm

AIR PISTOL:

Air pistol competition is a sport that is both challenging and greatly popular at both the international and collegiate level. Air pistol also represents a tremendous training opportunity for the Bullseye Pistol shooter as mechanics of follow-through and trigger control are magnified tremendously. The Air Pistol Page

PRACTICAL SHOOTING VARIETIES:

What is Practical Pistol Shooting?

Practical shooting is a sport in which competitors are required to combine accuracy, speed and power to successfully complete many different types of shooting "problems". Competitors use centerfire handguns in large calibers (9mm/.38 special is the minimum allowed) and shoot full-power loads. Fewer points are awarded to competitors using "minor" power loads. These handguns are carried in belt holsters and are accompanied by spare magazines or speedloaders in pouches also attached to the belt. Unlike bullseye or skeet, the events shot in each practical shooting match are different each time - which requires competitors to be diverse in their training. At any given match a shooter may be required to shoot targets 2 meters away in one event, and 50 meters away in the next. Sometimes the targets are paper, sometimes they are steel. Often "no-shoot" penalty targets are placed near "shoot" targets. Points are subtracted from a shooter's score for hitting the "no-shoots". Realistic props are used to simulate a scenario that the shooter must complete. Shooting may be done from freestyle, strong hand, weak hand, prone, or any other imaginable position, depending on the course of fire. Since scoring uses both total points and elapsed time, the shooters strive to find the best combination of accuracy, speed, and power to win.

IPSC (pronounced "ip-sick") was created as an organization in 1976 at Columbia, Missouri, by representatives from nine nations where the sport of "combat" shooting was becoming popular. This became known as the Columbia Conference. The term "practical" went into the name instead of "combat" in deference to public image and Jeff Cooper who was elected the

first President. Jeff's writings and philosophy of "practical pistolcraft" were highly regarded and earned him the title of father of the sport.

As the organization grew, member nations developed their own national sanctioning bodies to administer matches in their own countries, and to hold their own national championships. For the United States, the United States Practical Shooting Association (USPSA) is the sanctioning body. Within the international administrative structure the US is designated as an administrative "Region". The member elected president of USPSA also serves as "Regional Director" (RD) for the US IPSC program.

An important part of USPSA is the National Range Officers Institute (NROI), which sends instructors around the country to conduct training and certification courses for volunteer USPSA/IPSC range officers. Thus insuring that the highest standards of safety and scoring integrity are maintained at official matches, right down to the local club level.
United States Practical Shooting Association
International Practical Shooting Association

What is Defensive Pistol Shooting?

Defensive pistol shooting as a sport is quite simply the use of practical equipment including full charge service ammunition to solve simulated "real world" self-defense scenarios. Shooters competing in Defensive Pistol events are required to use practical handguns and holsters that are truly suitable for self-defense use. No "competition only" equipment is permitted in Defensive Pistol matches since the main goal is to test the skill and ability of an individual, not his or her equipment or gamesmanship.

Prior to the formation of Defensive Pistol, there was no place to compete and hone one's skill with equipment designed for and suitable for self-defense. Other shooting sports are just that, sports that have no relevance to self-defense. Defensive Pistol offers an exciting new forum for practical shooters in which truly practical equipment, techniques and courses of fire are mandated. Prior to Defensive Pistol there was no place at all to compete with common service pistols such as the Beretta, Glock or Sig. Nor was there a shooting sport where your concealed carry holster could also be your match holster without handicap. When you come to a Defensive Pistol match you cannot only use your duty/CCW equipment, you can be completely competitive with it! Other shooting sports have become equipment "races," Defensive Pistol will not. If you're interested in using truly practical pistols, drawn from practical holsters to solve challenging and exciting defensive shooting problems, then Defensive Pistol is the sport for you.

- The goal of Defensive Pistol is to create a level playing field for all competitors to test the skill and ability of the individual, not their equipment or gamesmanship.

- To promote safe and proficient use of guns and equipment suitable for self-defense use.

International Defensive Pistol Association

INTERNATIONAL-STYLE SHOOTING EVENTS:

What is ISSF Shooting?
The International Shooting Sport Federation (ISSF), formerly the UIT is the governing body of the International Amateur Shooting Sports at international and world-wide levels of competition, including World Championships and the Olympic Games. USA Shooting is the national governing body for ISSF-style shooting in the United States by authority of the US Olympic Committee and the ISSF. In the Olympics, the shooting sports continue to draw the third-greatest number of countries and impose participation quotas to keep competitors to a manageable number. One of the greatest surges in participation came in 1984, when separate women's events were first added to the Olympic program. Altogether, there are 17 different ISSF events. International Sport Shooting Federation (Formerly UIT)

SILHOUETTE SHOOTING:

Silhouette shooting involves firing at steel outlines of chickens, pigs, turkeys and sheep from various distances, rifle up to 500 meters, pistol up to 200 meters. Unlike most conventional target games that utilize paper targets and numerical scoring rings, most every shot fired at a metallic silhouette produces an immediate and clearly visible result. Even misses produce a cloud of dust. For a copy of the official rules and regulations governing centerfire or smallbore metallic silhouette shooting, contact: National Rifle Association, Silhouette Department, 1600 Rhode Island Avenue, N.W., Washington, D.C. 20038; Phone: (202) 828-6326 or the National Shooting Sports Foundation, 11 Mile Hill Road, Newtown, CT 06470-2359; Phone: (203) 426-1320, for a copy of the NSSF publication, "Metallic Silhouette." International Handgun Metallic Silhouette Association
National Shooting Sports Foundation

COWBOY ACTION SHOOTING VARIETIES:

Three men are given credit for starting the development of this shooting sport in 1979. They are Harper Criegh, Bill Hahn and Gordon Davis. Little did these gentlemen know that when they formed the Single Action Shooting Society (SASS), their idea of using the firearms of the

American Frontier in competition events likened to those of the old west would be where it's at fifteen years later -- with tens of thousands of shooters participating in hundreds of events around the world every weekend!

This is a game of fantasy gunfighting during the wild and wooley days of Wild Bill Hickock, Jesse James & Wyatt Earp. Not just the men either. Calamity Janes, Annie Oakleys and Belle Stars are right beside them. Hundreds of other colorful figures of the past are seen again at today's shooting ranges having the times of their lives.

Cowboy Action Shooting is basically a three type of gun competition. The firearms are those most typical of the American Frontier from about 1860 to just after the turn of the century.

Shooting scenarios are likened to an events akin to those of the old west. Match themes can come from actual incidents, movie scenes or the match designers imagination.

This shooting sport was designed for all levels of shooters. Targets are generous in size and placed at such proximity that the difficulty or challenge competitors face is the speed they engage a course of fire. However, we all learn soon enough that no target is too big or too close that we can't miss it.

The great success of this activity can be attributed to many reasons. None are more significant than that it is a family oriented, recreational activity and that it is so much FUN. Shooters compete and enjoy the sport at any skill level, it is NOT mandatory to shoot well or to win in order to have FUN. This promotes the values often referred to as the "spirit of the game," and is perhaps best shown by the friends made while sharing a mutual appreciation of what is the legend of America's most colorful and exciting period.

Single-Action Shooting Society
The World Fast Draw Association

Further Reading

- The Air Pistol Page (http://www.pilkguns.com/aphp/aphphome.htm)
- United States Practical Shooting Association (http://www.uspsa.org/)
- International Practical Shooting Association (http://www.ipsc.org)
- International Defensive Pistol Association (http://www.idpa.com/)
- International Sport Shooting Federation (Formerly UIT) (http://www.issf-shooting.org/)
- International Handgun Metallic Silhouette Association (http://ihmsa.org/)
- National Shooting Sports Foundation (http://www.nssf.org/)
- Single-Action Shooting Society (http://www.sassnet.com/)
- The World Fast Draw Association (http://www.gunfighter.com/fastdraw/)

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Technical Data for Pistol Shooters

By Compiled by John Dreyer

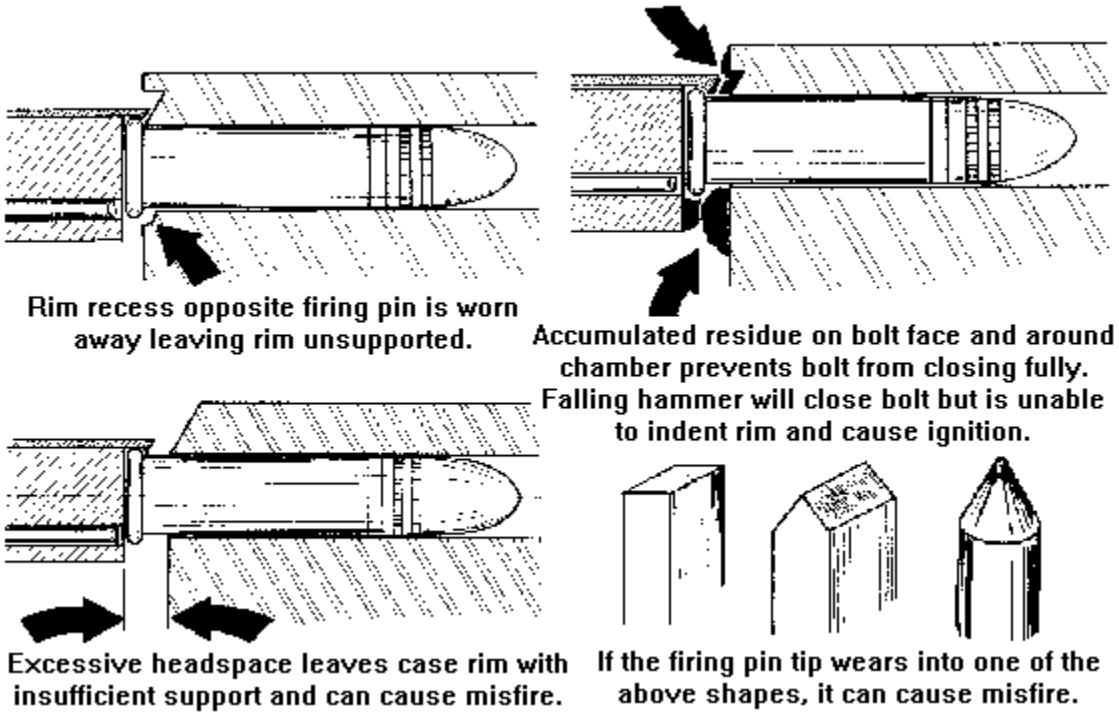
Source: http://bullseyepistol.com/techo.htm

.22 Misfire Causes | .22 Ammunition Selection | Reloading Components

The Causes and Solutions for .22 Misfires

(By the Technical Staff Omark CCI)

With new and properly stored ammunition, the chances that you will experience a legitimate misfire are seldom. But still, misfires do occur, and with the exception of defective primers (as is all too common with Remington ammunition), it is usually a gun problem. There are many more components in the gun than in the ammunition that can cause difficulty. Firing pin wear is a prime offender. At any rate, carefully examine the weapon for the following conditions before condemning the ammunition.



.22 Misfires

Selecting a .22 Ammo

The task of selecting ammo for your .22 pistol is an important one. Regardless of the pistol, the following factors must be considered in the selection process:

1. Accuracy (and consistency from lot to lot)
2. Functioning (some types malfunction in certain guns)
3. Cost (the biggest variable)

If a certain ammo does not function well in your pistol, change to another. It is the only of the three factors of which you have no control. Beyond that, selection is really a matter of how much accuracy you can afford. However, even some of the most inexpensive cartridges still produce satisfactory groups. To read an extensive article about the process of conducting your own ammunition accuracy test, select the following link:
Testing Ammunition Accuracy

Component Manufacturers

In addition to product listings, you will find an enormous quantity of information on reloading at the following places:

Dillon Precision	The maker of the finest progressive presses for the money!
Star Ammunition	Star makes the most accurate swaged lead bullet money can buy.
Nosler Bullets	Manufacurer of popular and accurate jacketed bullets.
Hornady Bullets	Manufacturer of bullets and other components for reloading.
Starline Brass	The Sedalia, Mo. company makes fine handgun brass.
Winchester	Winchester reloading components and reloading guide.
Barnes Bullets	Another well-known U.S. bulletmaker.
Alliant Powders	Alliant (formerly Hercules) manufactures the finest gun powders.
Vihtavuori Oy	Finland's superb manufacturer of gun powders.

Hodgdon Powders	The Shawnee Mission, Kansas manufacturer of gun powders.
IMR Powders	IMR manufactures a full line of gun powders.
Lyman Products	Lyman reloading components and equipment.

References

- .22 Misfire Causes
- .22 Ammunition Selection
- Reloading Components

Further Reading

- Dillon Precision (http://www.dillonprecision.com/)
- Star Ammunition (http://www.411web.com/S/STARAMUNITION/)
- Nosler Bullets (http://www.nosler.com/)
- Hornady Bullets (http://www.hornady.com/)
- Starline Brass (http://www.starlinebrass.com/)
- Winchester (http://www.winchester.com/)
- Barnes Bullets (http://www.itsnet.com/home/bbullets/)
- Alliant Powders (http://www.AlliantPowder.com/)
- Vihtavuori Oy (http://www.vihtavuori.fi/)
- Hodgdon Powders (http://www.hodgdon.com/)
- IMR Powders (http://www.imrpowder.com/)
- Lyman Products (http://www.lymanproducts.com/)

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About this Website...

By The History and Philosophy of this Work

Source: http://bullseyepistol.com/why.htm

MISSION STATEMENT:



My primary objective with this website is to provide the new shooter with any and all information that might help him obtain initial success in this sport. It is unfortunate, but true, that many folks who try out this game of bullseye quickly become frustrated and quit because "it ain't as easy as it looks."
Ultimately, I hope that my efforts will enable more interested shooters to get involved in competition.

AFFILIATIONS:

I have been involved in competitive shooting for many years and currently hold a Master Classification in NRA Conventional Pistol. I am currently or have been formerly associated with the following organizations:



NOTABLE COMMENTS:

Thank you to everyone who has sent me feedback. Especially in the first year, your comments have clearly been the difference in the development of this website! It has been the associations and friendships that I have made over the years via the Internet that have made this project worthwhile. I have received too many emails to include them all, but here are some of them...

"I'm glad to see a page dedicated to bullseye shooting. You couldn't be more right about it being challenging. The best score I have posted to date is 2677, twice. One day I will break the 2680 mark."
- Brian H. Zins, 1996, 1998, 2001, 2002, 2003, 2004, 2005, 2007, 2008, 2010 National Champion

"I just wanted to let you know that it is awsome that someone got the Web Page started. I have enjoyed surfing through it."
- Mario A. Lozoya, 1992 National Champion

"Thanks for the info and all of your hard work ... I along with countless others, appreciate all of the work you do to promote and support our chosen field."
- Kimberly Hobart, Unprecedented 8-time Women's National Champion

"Thank you for offering such a wonderful site!"
- Mark Beeman, has set more Indoor National Records than anyone in history

"You have every right to be proud of your Web Site ... Excellent work, sir!"
- Gil Gilchrist, director of NRA Competitions Division

"Thanks for your excellent web site. It's a valuable resource, and a real service to the sport."
- Michael Dane, NRA Pistol Committee, 2001 Connecticut State Champion

"John, I look forward to meeting you at Perry and personally thanking you for all of your efforts ..."
- Ron Steinbrecher, Pennsylvania's finest

"Bullseyepistol.com was one of the most fantastic sites I've ever visited. The way the whole thing has been systematically arranged and with great links. It was a pleasure staying on it for

more than 45 min."
- Dr. Vivek Jain, National-level air pistol shooter in India

"I just got done looking over your site and am very impressed. We need more people who still care about bullseye shooting. I am a Marine getting ready for the shooting season right now."
- Sgt. Shawn Tweedell, USMC

"Thank you for a really informative and well presented site!"
- Bo Blomqvist, National-level shooter in Sweden

"Neat site!!! I like what I've seen of your site!"
- Renee Tyson, Clark Custom Guns, Inc.

"I never imagined that I could find such a wealth of information on bullseye shooting at one site."
-Bob Abresch

"I am very happy that we have someone like yourself advancing the sport of bullseye shooting. We need more folks like you."
- Barry Vincent

"Thank You for the wonderful home page, links, and dedication that went into producing this site and related links. I go to bed tonight feeling better about where our sport stands and the nature of the people that support it."
- Andris Neimers, Port Moody British Columbia

"What a webpage! Nicely laid out and the information is outstanding. Been shooting NRA Highpower for years, there ain't nothing like this for that sport anywhere I know of. What a wonderful resource."
- Gregory S. Garland, Colorado Springs

WEBSITE HISTORY:

August 17, 1996 is a day that changed my life incredibly. Until that glorious day, I had dismissed the Internet as nonsense and refused to have anything to do with it. I still remember going through the mail that morning, disappointed that all I had received was "junk mail." The entire contents of the mail box was tossed in my office wastebasket. After working several hours at my desk, I glanced over and noticed that one piece of that junk mail had missed the basket. As I stooped to pick it up, my curiosity got the best of me. It was an envelope from Compuserve, marked colorfully "one month free trial." Within an hour of reluctantly opening that package and installing the software, I was surfing the 'net with a big grin on my face. I was hooked. I just HAD to have my own website.

My first webpages were born on August 20, 1996. The page titled "The Competitive Shooting Network" was initially just a set of links compiled from a few days of aggressive surfing. It wasn't until I bought a scanner a week later, that I began to add true "content" to the page. I was determined to produce a unique and professional-looking set of pages, loaded with useful information. After years of compiling information and improving my page designs, I am confident that I have done just that. On October 15, 1998 I purchased the domain name BULLSEYEPISTOL.COM, finally making it easy to find the Competitive Shooting Network. Several weeks later I dropped the old name and renamed it more appropriately, "The Bullseye Pistol Encyclopedia." Traffic has continually increased over the years and the site has seen at least three complete facelifts, the most recent in April of 2003. New material is added often and guest columnists who would like to see their material published are welcome to contact me!

Thank you for your interest in my work!



John A. Dreyer, Webmaster

Zen in Visualization of Fundamentals

By John Dreyer

Source: http://bullseyepistol.com/visualize.htm

This was projected to be my masterpiece of the writings in the Zen category. I spent an enormous amount of time in collecting research materials and in planning to piece the information together in a format that would really dazzle! Unfortunately, the project really kicked my butt and I burnt out on it way back in September of 2001. Years later, as I look over the outline again, I see the potential of the piece, and will resume working on it over the Spring and Summer of 2005.

Thank you for your patience.